Fundamental Fractal-Geometric Field Theory (FFGFT) or T0 Theory – Time-Mass Duality

Part 6: Holography, the HLV Bridge and the Structure of Time

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Chapter A

Introduction to Volume 6

Sixth Volume of the Document Collection

With this volume the collection of individual documents on T0 theory / FFGFT is continued beyond the original five volumes. Since the completion of Volume 5 (Doc. 262, May 2026) the corpus has grown by more than 120 documents. They are included in Volumes 6, 7 and 8 in numerical order; the boundary of each volume is placed so that a thematic break coincides with the page limit of the print editions.

Content and Focus

Volume 6 comprises Documents 263 to 310 from June and July 2026. The common thread is the question of how the compact geometry T^4 passes over into the observed three-dimensional, temporally open world, and what is preserved in that passage in terms of information, phase and temporal structure. The volume opens with the holographic principle in its fractal reading, the cosmological degeneracy between ΛCDM and FFGFT, and the derivation of the CMB peak ratios from the T^4. It continues with the projection chain from T^4 to observable space, which leads through a series of comparison and bridge documents on the Helix–Light–Vortex approach (HLV) up to the computed 3-versus-6 bridge in Doc. 285.

The third section deals with the dynamic sector: kinetic energy as a ξ-fractal fraction of the static relation $\tilde{T} \cdot m = 1$, the separation of magnitude and phase in the four fermion sectors, and the twofold origin of the angle $\theta = 2/9$, dynamical (Doc. 291) and icosahedral (Doc. 293). Documents 295 to 307 are devoted to time and information: the time-winding as a logarithmic spiral, the reversibility test that identifies information as an output rather than a base unit, the native time–energy reciprocity, and the location of time within the state space. The volume closes with the cosmic sector and the scale-anchor problem in comparison with ΛCDM, and with resonance geometry as the overarching reference.

Organisation

The 44 documents are divided into five parts: holography, cosmological degeneracy and formalism correspondences (Docs. 263–269); change of dimension, the HLV bridge and scale recursion (Docs. 270–285); dynamic sector, magnitude and phase, $\theta = 2/9$

(Docs. 286–294); time and information (Docs. 295–307); cosmic sector and resonance geometry (Docs. 308–310).

Reading the Documents

As in the earlier volumes, each document is self-contained and is cited in the text by its corpus number. The status markers [K], [B], [S] and [SETZUNG] indicate the level of derivation reached; later corrections and closed bridges are recorded in the corrections register Doc. 190, maintained continuously in the repository (see the introduction to Volume 5). Where a later document refines or withdraws a statement of this volume, the later version applies. Documents 308 to 310 were originally typeset as multi-chapter treatises; in the book their chapters appear as sections of the respective document.

Status

The documents in this volume reflect their respective versions in the corpus as of October 2026.

Part I

Part I — Holography, Cosmological Degeneracy and Formalism Correspondences

Chapter B

Doc. 263 — Fractal Holography: FFGFT and the Holographic Principle

Preliminary remark

This document locates FFGFT relative to the holographic principle. It strictly separates three levels: **(1)** contact points already documented in the FFGFT corpus; **(2)** an open scaling question at the area law, computed here as a *reduction layer* but *not* claimed as a core derivation; and **(3)** genuine tensions where FFGFT explicitly does *not* match the standard picture of holography. The separation is deliberate: otherwise the word "holographic" loads an AdS/CFT expectation that FFGFT neither meets nor intends to meet.

Methodological status. Section B.1 reports what is documented (Doc. 201). Section B.2 computes a consequence of *one assumption* and is marked as a reduction layer. The choice of assumption is a physical decision, not a derivation.

Symbols used.

Symbol	Meaning
ξ	basic geometric parameter, $\xi = 4/30000 \approx 1.333 \times 10^{-4}$
$\tilde{T} \cdot m = 1$	time–mass duality (fundamental relation); $\tilde{T}$ intrinsic time scale
D_f	fractal dimension, $D_f = 3 - \xi$
E_0	characteristic energy, $E_0 = \sqrt{m_e m_\mu / K_{\text{frak}}} = 7.398$ MeV (bare $\sqrt{m_e m_\mu} = 7.348$ MeV, R72)
E_H	Hubble energy, $E_H = E_0\, \xi^{41/4} \approx 1.41 \times 10^{-33}$ eV (exponent calibrated to H_0)
H_0	Hubble quantity, $H_0 = E_H/\hbar \approx 66.2$ km/s/Mpc
R_H	maximal scale, $R_H = \hbar c / E_H \approx 1.4 \times 10^{26}$ m
L_0	sub-Planck granulation, $L_0 = \xi \ell_P \approx 2.2 \times 10^{-39}$ m
L_ξ	characteristic ξ length ($\sim 100\,\mu$m, Doc. 091)
ℓ_P	Planck length
z, z_{obs}	redshift; $z_{\text{obs}} = \lambda_b/\lambda_e - 1$ (wavelength ratio)
$\rho_{\text{vac}}, \rho_{\text{CMB}}$	vacuum/CMB energy density; $\rho_{\text{CMB}} = \xi \hbar c / L_\xi^4$
m_T, λ	mediator mass $m_T = \lambda/\xi$ (Doc. 201); regulates $\rho \le 1/\xi^2$
$c, \hbar, \hbar c$	SI conversion factors (not ontological primitives)
T_{00}	energy density from $\mathscr{L} = \xi(\partial E)^2$; $T_{00} = \xi[(\partial_t E)^2 + (\nabla E)^2]$

Layers: core derivation > bridge > reduction layer (plausibility). Standing caution: H_0, z, Ω are ΛCDM pipeline outputs, not model-neutral.

B.1 Documented contact points

Area law from node counting (Doc. 201, §7.6)

FFGFT sets the entropy of a black hole as node configuration entropy:

$$S = \frac{A}{4\,\ell_P^2} \;\longrightarrow\; S = N_{\text{nodes}}\,\ln 2, \qquad N_{\text{nodes}} \propto \frac{A}{L_0^2} = \frac{A}{\xi_0^2\,\ell_P^2},$$

with the granulation $L_0 = \xi_0\,\ell_P$ used here. This reproduces the area scaling of the Bekenstein-Hawking law – the holographic core statement that entropy scales with *area*, not *volume*. In FFGFT this is not a postulate; it falls out of counting saturated nodes on the core surface. The counting does not yield the coefficient: with this granulation $S = N_{\text{nodes}}\ln 2$ misses $A/(4\ell_P^2)$ by the factor $4\ln 2/\xi_0^2 \approx 1.6\times 10^8$.

Information as an access problem, not an existence problem

The winding numbers $(n_\theta, n_\varphi, n_3, n_4)$ are topological invariants; they cannot disappear. An apparent "information loss" at the horizon is, in FFGFT, epistemic (a problem of access), not ontological (a problem of existence). This is a distinct position on the information paradox, entering the same terrain as holography – conservation of information – but by a different mechanism: topological conservation rather than boundary encoding.

Mass as scale projection

FFGFT already uses the term "holographic projection": particle masses arise as a projection between scales (the VEV onto the sub-Planck scale). This is holography in the sense of a *projection between scales*, not in the sense of a bulk/boundary duality.

B.2 Open scaling question at the area law (reduction layer)

Decompactification forces corrections – but not their form

A natural argument runs: since we live in a decompactified regime (the torus dimensions are hidden at the scale $L_0 = \xi_0\,\ell_P$), the hidden compactness *must* show up as a fractal correction in observable quantities. This is correct – and it is precisely ξ_0: the correction parameter is documented in the corpus (mass formula $m_i = r_i\,\xi_0^{p_i}\,v$; geometric redshift as fractal path-lengthening). That fractal corrections exist *at all* is therefore not speculation.

This argument does *not*, however, fix *which* geometric quantity the correction acts on at a black-hole horizon. "Decompactification forces corrections" does *not* entail $S \propto R^{2-\xi_0}$. A correction on the horizon *area* (Assumption B below) and one on the enclosed *volume* (Assumption C) are both fully decompactified and fractal – yet they yield *different* area

laws (broken vs. exact). The step from "ξ_0 corrections exist" to "the horizon entropy deviates by $\sim 1\,\%$" is therefore an *additional* assumption about where the correction acts, not a consequence of decompactification itself.

The question

Doc. 201 uses the *plain* area law: $N_{\text{nodes}} \propto A \propto R^2$, exponent exactly 2. A correction by the fractal dimension $D_f = 3 - \xi_0$ is *not* built in there. It is therefore an open question whether D_f affects the horizon *area* at all – and if so, how.

Three assumptions compared

- **Assumption A (plain area law, Doc. 201):** $S \sim R^2$. Exponent exactly 2.
- **Assumption B (fractal area):** if the nodes sit on a codimension-1 boundary of a D_f-dimensional structure, the effective node count scales as $R^{D_f - 1} = R^{2-\xi_0}$. Exponent $2 - \xi_0$.
- **Assumption C (fractal volume):** if D_f acts on the enclosed *volume* ($V \propto R^{D_f}$) while the area stays integer, the area law remains *exact* at exponent 2.

Result of the computation

Assumption	Exponent	Deviation from area law
A – plain area law	2	0
B – fractal area	$2 - \xi_0 = 1.99986667$	$\Delta_{\text{exp}} = \xi_0 = 1.33 \times 10^{-4}$
C – fractal volume	2	0 (area law stays exact)

Only Assumption B breaks the Bekenstein-Hawking law – by *exactly* ξ_0 *in the exponent*. The relative deviation grows logarithmically with size:

$$\frac{S_B}{S_A} - 1 \;\approx\; -\xi_0 \,\ln\!\left(\frac{R}{\ell_P}\right).$$

The tiny ξ_0 is lifted by the logarithm across many orders of magnitude; for astrophysical black holes the effect reaches percent level:

Object	R/ℓ_P	Deviation (Assumption B)
Solar-mass BH ($R \sim 3\,$km)	1.9×10^{38}	$-1.17\,\%$
Sagittarius A* ($R \sim 1.2 \times 10^{10}\,$m)	7.4×10^{44}	$-1.37\,\%$
M87* ($R \sim 1.9 \times 10^{13}\,$m)	1.2×10^{48}	$-1.47\,\%$

Interpretation

This is qualitatively different from the ξ_0 effect in other contexts (e.g. the dimension-driven proximity $D/(D+1) \to 3/4$, where ξ_0 only appears at 10^{-5}). Here ξ_0 reaches percent

level because $\ln(R/\ell_P)$ bridges some 38 orders of magnitude. This is precisely where a *discriminator* would sit between "true holography" (area law exact) and "fractal near-holography" (exponent $2-\xi_0$).

Status (open). The choice between Assumption B and C is a physical decision about FFGFT, *not* a derivation from the existing corpus. Assumption B is plausible (codimension-1 boundary), but it is not derived in Doc. 201. As long as the question is open, the percent deviation is the consequence of *one* assumption, not a prediction of FFGFT. Likewise, the prefactor $1/L_0^2$ of the node counting was treated here as a pure prefactor (it cancels in the ratio); any scale-dependent fractality of ℓ_P would add a further term.

B.3 Reconstruction: the hologram analogy and its limits

Documented: a non-invertible projection (Doc. 193)

Doc. 193 records that evaluating the mode labels (r_i, p_i) to real numbers is *not invertible*: once r_i and p_i have been evaluated to numbers in $\mathbb{R}$, they carry "no memory" of their origin from different particles. From a real value (a mass, say, or a derived quantity) the underlying quantum-number structure cannot be uniquely recovered – the fibre over each numerical value has collapsed. This is a documented, non-invertible projection from the discrete geometric mode space onto $\mathbb{R}$.

Analogy (metaphor, not derivation): the physical hologram

This asymmetry can be illustrated by a physical hologram – explicitly as an *illustration*, not a derivation. An optical hologram encodes a spatial scene in an interference pattern on a surface. To turn it back into a visible image, the surface alone does not suffice: it requires a reference (readout) beam carrying the *prior information* (wavelength, phase, recording geometry). And even at the best recording and reconstruction quality, the visible hologram remains a *faint image* of the original reality – part of the information is in principle unrecoverable.

On this reading, the observable $3{+}1$-dimensional world relates to the underlying T^4 geometry as a reconstructed hologram relates to the recorded scene: reconstruction requires prior information, and it is lossy in principle. The observable world is an *incomplete* image of the substrate – not because the recording is poor, but because the projection itself discards information.

Distinction: this is not the non-closure theorem

This non-invertible projection must be carefully separated from the *non-closure theorem* (Doc. 193). The non-closure theorem is a *different*, purely algebraic statement: the set of admissible pairs (r_i, p_i) is not closed under multiplication, i.e. the product of two distinct modes $(r_i \cdot r_j,\ p_i + p_j)$ corresponds to no particle of the model. It concerns the *combinability* of modes with one another, not the *invertibility* of a projection. Both statements come from Doc. 193 but are distinct and should not be equated.

B.4 Reversibility: an assumption of standard holography vs. a derived time asymmetry

Reversibility is an axiom in standard holography

Standard holography (AdS/CFT) is formulated as a *duality* – an exact, invertible dictionary between bulk and boundary. This invertibility rests on two premises: the **unitarity** of the underlying quantum theory (reversible time evolution, information conservation) and the existence of a **conformal boundary** on which the exact encoding lives. Both are posited, not derived. Reversibility is therefore not a measured finding but a *structural assumption* of the formalism.

FFGFT derives the time asymmetry (Doc. 157)

FFGFT does not treat the direction of time as an axiom but derives it from deeper geometry. Doc. 157 anchors the breaking of time-reversal symmetry in three ways:

- **Algebraic:** $T \cdot m = 1$ with positive-definite mass ($m > 0$, antimatter included) forces $T > 0$; a reversal $T \to -T$ would require $m \to -m$ and is physically impossible.
- **Topological:** the winding of the vacuum field on the torus has a chirality; once chosen, the time direction is topologically protected (like a knot that cannot be reversed by continuous deformation).
- **Fractal:** at $D_f = 3 - \xi$ the path integrals for forward- and backward-running paths differ – the fractal structure turns the discrete symmetry $t \to -t$ into a continuous asymmetry.

The two situations are therefore *not* symmetric: standard holography *posits* reversibility, whereas FFGFT *derives* non-reversibility. Non-invertibility is thus not a separate extra assumption in FFGFT but a consequence of structures the framework already carries ($T \cdot m = 1$, torus winding, D_f).

Empirical limit: only practical experience

Honesty is required here. The only empirical anchor for the directedness of time at all is the **practical experience** of directed happening – not a direct measurement and not a theorem. The observed CP/T violation is, as Doc. 157 itself notes, too weak for the macroscopic arrow of time. FFGFT thus does not *measure* the asymmetry; it replaces the thermodynamic stopgap (an unexplained low-entropy initial condition) with a *geometric derivation* of the directedness already experienced in practice. This is a structural advantage – the "Past Hypothesis" becomes unnecessary – but it is not an empirical proof of the premise.

Consequence: reversibility as a possible artefact

From this follows the genuine open question lying beneath both frameworks. *If* the world is fundamentally time-asymmetric – as practical experience suggests and FFGFT derives geometrically – *then* the exact reversibility of the holographic mathematics is an **artefact of the unitary idealization**: it abstracts away a symmetry breaking that FFGFT grounds.

This is not a claim that standard holography is “wrong”, but the observation that its reversibility is an unmeasured assumption – and that FFGFT posits a derived asymmetry at the same point. The two assumptions have different observable consequences; a concrete discriminator has already been named (the logarithmic entropy deviation $-\xi \ln(R/\ell_P)$ under Assumption B, Section B.2).

B.5 Anchoring at the bounded core (no singularity)

The anchor

A projection is admissible as a pure model; but for it to be *FFGFT-intrinsic* (rather than borrowed from AdS/CFT), it must be *anchored* at a physical point, like the time-mass duality. T0 has no singularity: the vacuum density is bounded by $\rho \leq 1/\xi^2 \approx 5.6 \times 10^7$ (Doc. 201), regulated by the mediator mass $m_T = \lambda/\xi$, which prevents unbounded growth of Δm. Where classical GR would run into a singularity, the ξ-bounded core holds against it – exactly as $T \cdot m = 1$ caps the divergence ($m \to \infty$ is capped by $T \to 0$). The holographic projection is anchored at this core.

Computational consequence: a cutoff instead of a singularity

This entails a *physical cutoff*. There is a finite core radius R_{core} below which the nodes are saturated; the area-law scaling $S \sim R^a$ holds only for $R \geq R_{\text{core}}$, and below it the entropy saturates. Order of magnitude:

$$\frac{R_{\text{core}}}{\ell_P} \sim \frac{1}{\xi} \approx 7.5 \times 10^3 \qquad (\log_{10} \approx 3.9).$$

The $O(1)$ prefactor is model-dependent; only the *order of magnitude* follows from $\rho \leq 1/\xi^2$. Consequence for the computation in Section B.2: values with $\log_{10}(R/\ell_P) \lesssim 3.9$ lie *below* the core – in the region of the non-existent singularity – and are physically meaningless. Only above R_{core} is the scaling valid. For astrophysical black holes ($R/\ell_P \sim 10^{38}$–$10^{48} \gg R_{\text{core}}$) the deviation $-\xi \ln(R/\ell_P)$ remains valid unchanged – the cutoff affects only the unphysical small-scale region.

Interpretive limit: conserved information is geometric base information

What is conserved at the core – the node configuration entropy $N_{\text{nodes}} \ln 2$ – is *not* a maximal description of the past reality, but only the **base information about the geometric structure** (winding numbers as topological invariants). Conservation here means “access to the geometry”, not “reconstruction of the history”. This is consistent with the access-not-existence character (Section B.2) and with the lossy, directed projection: the history of the infalling material is not conserved countably, only its geometric signature.

Open: no exact translation quantity

A gap remains to be stated. So far we have only a *scale anchoring* ($R_{\text{core}} \sim \ell_P/\xi$), not a derived **translation quantity** that would gauge the area law (core entropy ↔ horizon area) as an exact map with a fixed coefficient – analogous, say, to the central charge in AdS/CFT or to the SI conversion factors $\hbar, c$ within FFGFT itself. No such quantity is documented in the corpus, and none has been found. This is consistent with the position taken here: no exact duality, hence no exact translation coefficient – only a lossy projection with a scale anchoring. The exact translation quantity remains an open point.

Boundaryless topology

Standard holography (AdS/CFT) requires a conformal *boundary*. The universal 4-torus T^4 is boundaryless (compact without boundary). A naive bulk/boundary duality therefore does not apply. FFGFT's "holography" is an area law plus scale projection – not boundary encoding.

No holographic dark energy

Models that obtain Λ from a holographic horizon are incompatible with the static FFGFT universe (no Λ, no dark energy). Here FFGFT stands in *opposition* to holography, not in contact. An equation of state w_{DE} does not exist in FFGFT; the comparable observable is the geometric redshift law (fractal path-lengthening, cf. Doc. 041/182), not a w parameter.

B.6 H_0 as a projection artefact, not a translation quantity

No direct ξ–H_0 bridge – but a chain through $\hbar c$

The opposition to holographic dark energy tempts one to ask whether FFGFT's ξ and the ΛCDM parameter H_0 can be *translated* into one another. A *direct* identification $\xi/L_0 = H_0/c$ fails – but not because energy and length were incompatible. In natural units ($\hbar = c = 1$) length and energy are dual; the passage to SI metres is exactly one conversion factor ($\hbar c = 1.973 \times 10^{-7}$ eV m), and in FFGFT $\hbar, c$ are SI conversion factors anyway, not ontological primitives. The error of the naive bridge is not the energy–length duality but the *choice of scale*: it couples ξ to the sub-Planck granulation $L_0 = \xi \ell_P \approx 2.2 \times 10^{-39}$ m instead of to the cosmological scale $R_H \approx 1.4 \times 10^{26}$ m. Both lengths convert correctly into energies via $\hbar c$ – they are simply *different* scales (ratio $R_H/L_0 \approx 6.5 \times 10^{64}$, which is calibrated to the measured H_0, not following from ξ alone).

The correct translation chain: $\hbar c$ as the SI factor

Through $\hbar c$ the chain does close (Doc. 182):

$$\xi \longrightarrow E_H = E_0\, \xi^{41/4} \xrightarrow{\times \hbar c} R_H = \frac{\hbar c}{E_H} \longrightarrow H_0 = \frac{c}{R_H} = \frac{E_H}{\hbar}.$$

With $E_0 = \sqrt{m_e m_\mu / K_{\text{frak}}} = 7.398$ MeV this gives $E_H = 1.412 \times 10^{-33}$ eV, $R_H = 1.398 \times 10^{26}$ m and $H_0^{T0} \approx 66.2$ km/s/Mpc (-1.9% from the Planck value). The exponent $41/4$ is not derived from ξ but calibrated to the measured H_0; the chain shows the form of the translation, its numerical value is not a prediction. Here $\hbar c$ is precisely the SI conversion factor that turns the energy representation (E_H) into the length representation (R_H) – exactly the quantity one expects when length is readable as energy in natural units. So the chain links the energy E_H *and* – via $\hbar c$ – the associated length; only the *cosmological* length R_H, not the sub-Planck length L_0.

The mechanism: H_0 is a misread slope

This chain is at the same time the *explanation* of H_0 as an artefact – consistent with the projection reading of holography. The mechanism: for small z the fractal path-lengthening is linear, $z(d) = \kappa d$ with $\kappa = E_H/(\hbar c)$. An observer in the expansion picture reads the same linear $z(d)$ curve and *names* its slope H_0/c. Hence

$$H_0^{(\text{measured})} = \kappa c = E_H/\hbar,$$

i.e. H_0 is not a dynamical expansion parameter but the *geometric coefficient* of the path-lengthening, misinterpreted as expansion. The artefact is the *reinterpretation of a slope*, not a measurement error. The translation chain $\xi \to E_H \to R_H \to H_0$ does exist – it only shows that the translated quantity is geometric, not dynamical. Beyond the maximal scale R_H one has $z \to \infty$; no ΛCDM-type particle horizon is needed.

H_0 is emergent, not fundamental

What matters is where H_0 sits ontologically: it is *not a fundamental constant* of the theory. In the compactified T^4 form there is no H_0 – there are no cosmological distances there, no vacuum traversed as a medium, and $\hbar, \varepsilon_0$ become "real" only upon decompactification (Doc. 262). Since $H_0 = E_H/\hbar$ contains this only-then-real $\hbar$, H_0 is an *emergent* quantity of the decompactified description – just as age, critical density and Hubble length are *Friedmann concepts without a direct T0 counterpart* (Doc. 016). Only ξ is fundamental; H_0 arises at the transition compact → decompactified. The non-integer exponent $41/4$ expresses precisely this scale transition (scale flow over ~ 40 orders of magnitude), not a pure geometric constant; forcing an integer reading would wrongly push an emergent transition quantity back into the compact geometry.

The same holds for z itself: real view, misreading, translation

The reinterpretation concerns not only the slope H_0 but the whole z value – and here the clean three-way split is worthwhile.

Model-neutral (real): what is measured is a wavelength ratio, $1 + z_{\text{obs}} = \lambda_{\text{obs}}/\lambda_{\text{emit}}$. This quantity is observable and identical in both pictures; it is not an artefact.

The real FFGFT view: z_{obs} is the accumulated *fractal path-lengthening* of the photon over the physical distance d, $z_{\text{obs}} = (L_{\text{eff}} - d)/d$ (Doc. 026). No scale factor, no expansion, no time evolution of space – a purely geometric path-length relation in the static T^4.

The misreading (ΛCDM): the same z_{obs} is read as an *expansion* redshift, i.e. as a cosmological scale factor $1 + z = a_0/a_{\text{emit}}$. In this reading the initial linear slope becomes H_0 and a large value such as $z \approx 1100$ becomes the "decoupling" epoch of a recombination. Both are the expansion reading of the *same* geometric path-lengthening – not a second, independent measurement (cf. P15/R15, Doc. 190; the static theory has no cosmological z in the expansion sense).

Translatable: the two views convert into one another because they describe the *same* $z_{\text{obs}}(d)$ curve. H_0 is the slope at small z, $z \approx 1100$ an extreme value near the maximal scale R_H; both are ΛCDM readings of the one FFGFT geometry (the FFGFT-own value of this extremum is $z_* \approx 875$, Doc. 267; $z \approx 1089/1100$ is only the expansion reading, not the FFGFT value). The translation is therefore a *reinterpretation of the curve*, not a physical identity of parameters: what is real is z_{obs}, what is an artefact is solely its expansion reading.

Placement and open point

This is precisely the resolution of the apparent dark-energy opposition: a translation *does* exist – the chain $\xi \to E_H \to R_H = \hbar c/E_H \to H_0$ – but no *direct* $\xi \leftrightarrow H_0$ identity at the sub-Planck scale (which would obscure its pipeline character and contradict the standing caution). The model-neutral shared quantity remains the observable $z(d)$: same curve, two mechanisms, distinguishable at large z. **Placement:** the exponent $41/4$ describes the *scale transition* from the characteristic energy E_0 to the Hubble energy E_H (Doc. 026/182, recursion R Doc. 203). Two things must be separated: (i) the *structure* of the chain $\xi \to E_H \to R_H \to H_0$ is documented and consistent – H_0 is therein a geometric coefficient misread as expansion, not a dynamical parameter; $\hbar$ acts only as SI conversion ($H_0 = E_H/\hbar$), not as a physical input (cf. section "Constants as projection factors"). (ii) the *numerical value* of the exponent $41/4$ is not derived from the ξ-field theory but calibrated to the measured H_0; the exponent of the unit-free integer form $n = 10$ (Doc. 165) is not derived forward either. The H_0 *value* is therefore the one external input of the cosmological sector – a shared empirical reference point of both models, not a consistency argument and not a core result. The *interpretation* (H_0 as a projection artefact) does not depend on (ii) and stands independently.

B.7 Constants as projection factors – and the vacuum energy

The H_0 reading is a special case of a more general principle that recurs across several quantities and supports the holographic projection idea quantitatively. It runs in four steps; all four are reproducible as short calculations and are flagged below as reduction layer.

1. The "constants" ground nothing – they convert

In natural units ($\hbar = c = 1$, pure ratios) the FFGFT description is static and correction-free (Layer 1, Doc. 261). The quantities c, $\hbar c$ and $\sqrt{\hbar G/c^5}$ are not physical inputs but *conversion factors* of the SI projection (Layer 2):

- c is the metre/second factor that arises when the quantities length and time – identical in Layer 1 ($\tilde{T} \cdot m = 1$) – are projected into separate units (Doc. 134: $c = L/T$, "not a law of nature but a ratio"). The value $299\,792\,458$ is the metre definition, not a ξ prediction.
- $\sqrt{\hbar G/c^5}$ grounds no time scale; it expresses the already-fixed scale in seconds (pure unit calibration).
- Even G follows c-free from ξ: $G_{\text{nat}} = \xi^2/(4m_{\text{char}})$ (Doc. 012/013); $c, \hbar$ enter only at the SI conversion.

2. The vacuum structure is fully present when compactified

The mode structure (ξ/L_ξ^4, fractal mode counting at $d = 3 + \delta$, Doc. 091) is *fully present* in Layer 1 – merely unit-free. *Nothing is added* at decompactification: the same structure appears in Layer 2, multiplied by the conversion factor $\hbar c$, as a dimensional energy density $\rho_{\text{vac}} = \xi\hbar c/L_\xi^4$ (form from Doc. 091/025). The vacuum energy thus does not *come into being* – it merely becomes *visible* from the SI view through the decompactification of the previously enrolled fractal/temporal structure.

3. No static energy floor: the energy is dynamical

That there is no *self-standing* (static) vacuum energy follows computationally from the energy-field Lagrangian $\mathscr{L} = \xi(\partial_\mu E)(\partial^\mu E)$ with field equation $\Box E = 0$ (massless, no potential term $V(E)$, Doc. 010). The canonical energy-momentum tensor gives

$$T_{00} = \xi\left[(\partial_t E)^2 + (\nabla E)^2\right].$$

T_{00} contains *no* static term. Switching off the temporal component ($\partial_t E = 0$) and requiring a source-free vacuum solution ($\Box E = 0 \Rightarrow \nabla^2 E = 0 \Rightarrow E = \text{const}$) gives $T_{00} = 0$. The vacuum energy is therefore entirely bound to the temporal (and, under boundary conditions such as the Casimir effect, spatial) *variation* of the field – not an eigenvalue of a field at rest but an expression of the dynamics. The *free* vacuum has no spatial part; the *confined* one (Casimir) does, consistent with the free-vs-confined distinction in Doc. 091.

4. Delimitation: relations with ξ alone

Free of fit factors and circular exponents are the quantities in which ξ stands *alone*: $T_{\text{CMB}} = \frac{16}{9}\xi^2 E_\xi$ and $|\rho_{\text{Casimir}}|/\rho_{\text{CMB}} = \pi^2/(720\xi) \approx 102.8$. Neither is a confirmation, however: the T_{CMB} relation holds only in the energy unit eV, and the Casimir ratio is an identity, because L_ξ is determined from the measured CMB energy density.

Clarification on the nature of this T_{CMB} relation. It is an *algebraic* ξ relation, *not a fractal correction*: the factor $16/9$ is simply $(4/3)^2$, the square of the numerical prefactor of $\xi = \frac{4}{3} \times 10^{-4}$ (cf. Doc. 025/028/006); $D_f = 3 - \xi$ and a mode count with fractal dimension do *not* enter this derivation. The *ratio* $T_{\text{CMB}}/E_\xi = (16/9)\xi^2$ is thus unit-free (Layer 1); its agreement with the measured value, however, holds only in the energy unit eV. The absolute Kelvin value $2.725\,\text{K}$ follows only via the SI conversion (embedding price, Doc. 261); the residual deviation of $\approx 0.9\%$ (Doc. 013) is *not* derived from ξ. A correction of the form $1 - C\xi$ that would close these 0.9% exists in the corpus only as a *numerical*

approximation (e.g. $C = 275/4$), not as a derivation from the recursion $\mathscr{R}$ (Doc. 203) – it remains open.

The ratio given in Doc. 028, $\rho_{\mathrm{DE}}/\rho_{\mathrm{DM}} = \xi^{\ln(2.5)/\ln\xi}$, requires separate placement. It yields exactly 2.5 for *any* ξ and is therefore circular *as a prediction*. Yet its target value 2.5 ($\Omega_\Lambda/\Omega_{\mathrm{DM}}$) is itself not a model-neutral measurement but a ΛCDM-pipeline output. Read *not* as a prediction but as a *conversion factor* between the FFGFT geometry and the ΛCDM Ω bookkeeping – analogous to c (metre/second) and to z – the circular calibration via 2.5 is admissible. But only *partially*: unlike c, where both sides share the same real counterpart, the ΛCDM side carries with ρ_{DE} a model artefact (no Λ in FFGFT); the factor translates partly something real (DM bookkeeping $\leftrightarrow$ ξ effect), partly an artefact. Cf. Doc. 190, R17.

The *interpretation* is thereby clear: dark energy is eliminated (no Λ, static); dark matter is *not a substance* but a ξ-geometric effect – an artefact of the ΛCDM view, like H_0 and z. *Quantitatively* open is only the model-neutral reproduction of what ΛCDM books as DM (Ω_{DM}, rotation curves).

Status. This section is reduction layer: it makes a principle documented in Doc. 261/262 (Layer 1 correction-free / Layer 2 SI projection) explicit for c, ρ_{vac} and the dynamics. It predicts no new cosmological value; the numerical-value circularity of L_ξ (fixed from ρ_{CMB}) remains a limit.

B.8 A formal bridge to UIFT – verifiable on one side only

The same projection picture yields a *formal* bridge to the information quantity ξ_{UIFT} of Onur Teker's UIFT (Doc. 257). It is placed here, not asserted: its force is *asymmetric*.

The bridge rests solely on the *first order* $T_{\mathrm{CMB}}/E_\xi = (16/9)\xi$ (not on the fractal correction K_{frak} and not on the absolute Kelvin value). Section 4 writes the same ratio as $T_{\mathrm{CMB}}/E_\xi = (16/9)\xi^2$; with the same symbol E_ξ the two forms differ by a factor ξ. The factor given below corresponds to $k_B T_{\mathrm{CMB}} = (16/9)\xi\,\mathrm{eV}$; with the measured $T_{\mathrm{CMB}} = 2.7255\,\mathrm{K}$ one would get $418\,518$. Inserting the first order into the UIFT relation $\xi_{\mathrm{UIFT}} = k_B T_{\mathrm{CMB}} \ln 2/(m_e c^2)$ gives a *fixed* factor

$$\frac{\xi_{\mathrm{FFGFT}}}{\xi_{\mathrm{UIFT}}} = \frac{m_e c^2/\mathrm{eV}}{(16/9)\ln 2} \approx 414\,684.$$

This value is not an empirical result but a pure *conversion factor* between a Layer-1 quantity (ξ_{FFGFT}, dimensionless) and a Layer-2 quantity tied to the CMB temperature (ξ_{UIFT}); it coincides with its closed form.

What this verifies – and what it does not. The *FFGFT side* is fully reproducible: ξ is geometric, $(16/9)\xi$ follows from the mode count, m_e from the lepton hierarchy – anyone can recompute the factor. The *UIFT side* we *cannot* verify: the relation $\xi_{\mathrm{UIFT}} = k_B T_{\mathrm{CMB}} \ln 2/(m_e c^2)$ comes from Teker's work, and whether it is correct or empirically validated lies beyond our reach. The bridge is therefore *formal*: *if* Teker's relation holds *and* the FFGFT values hold, *then* a fixed factor links the two quantities. What is thereby verified is the FFGFT side of the bridge – not the UIFT side.

B.9 Summary

- **Documented:** FFGFT reproduces the area scaling of the Bekenstein-Hawking law from node counting (Doc. 201; the counting does not yield the coefficient); information is topologically conserved (access, not existence problem); mass as scale projection.
- **Open (reduction layer):** if $D_f = 3 - \xi_0$ affects the horizon area (Assumption B), the exponent becomes $2 - \xi_0$ with a logarithmically growing deviation $-\xi_0 \ln(R/\ell_P)$ (percent level for real BHs). If D_f acts on the volume (Assumption C), the area law stays exact. The choice is to be made on physical grounds.
- **Reconstruction:** the projection of mode labels onto $\mathbb{R}$ is non-invertible (Doc. 193) – illustrated by the physical hologram (metaphor): reconstruction needs prior information and remains a faint image. Strictly separated from the non-closure theorem.
- **Anchoring:** the projection is anchored at the ξ-bounded core ($\rho \leq 1/\xi^2$, no singularity; Doc. 201) – like $T \cdot m = 1$. Consequence: a cutoff at $R_{\text{core}} \sim \ell_P/\xi$; below it the entropy saturates, the $R^{2-\xi}$ scaling holds only above. Real black holes lie far above (deviation valid). Conserved information = geometric base information, not the history. Open: no exact translation quantity, only a scale anchoring.
- **Reversibility:** standard holography *posits* invertibility (unitarity + boundary); FFGFT *derives* the time asymmetry (Doc. 157: $T \cdot m = 1$, torus winding, D_f). The only empirical anchor of directedness is practical experience; under that premise the exact reversibility is an artefact of the unitary idealization.
- **Tension:** T^4 is boundaryless (no AdS/CFT duality); no Λ, no w_{DE} (opposition to holographic dark energy).
- H_0 **artefact:** no *direct* $\xi \leftrightarrow H_0$ equation at the sub-Planck scale – the naive form fails on the choice of scale ($L_0 = \xi \ell_P$ instead of R_H), not on the energy–length duality (in natural units both are dual via $\hbar c$). But the chain $\xi \to E_H = E_0 \xi^{41/4} \to R_H = \hbar c / E_H \to H_0 = E_H/\hbar$ does exist (≈ 66.2 km/s/Mpc; Doc. 026/182), with $\hbar c$ as the SI factor. H_0 is thus a geometric coefficient misread as expansion; the model-neutral quantity remains the observable $z(d)$. The exponent $41/4$ is calibrated to the measured H_0, not derived from ξ.
- **Constants as projection:** c, $\hbar c$, $\sqrt{\hbar G/c^5}$ ground nothing; they convert to SI (Doc. 134/261). The vacuum structure is fully present when compactified (unit-free) and becomes SI-visible only on decompactification. $T_{00} = \xi[(\partial_t E)^2 + (\nabla E)^2]$ (from $\mathscr{L} = \xi(\partial E)^2$, $\Box E = 0$, Doc. 010) has NO static term – without field variation $T_{00} = 0$: no self-standing energy, the energy comes from the temporal component.

Terminological caution. "Holographic" in FFGFT is to be read as *area law + scale projection*, not as *boundary encoding*. This distinction should remain explicit in all FFGFT texts to avoid false AdS/CFT expectations.

Chapter C

Doc. 265 — Formalism Correspondence: FFGFT and Blume-Maleev

Purpose

This document provides a worked example for the structural comparison of physical formalisms, developed for the IPI group's model-ordering work. The example uses two independently derived mathematical descriptions of the same experimental system – the ferrotoroidic quaternary memory crystal $LiNi_{0.8}Fe_{0.2}PO_4$ (Qureshi et al. 2026) – to illustrate that "correspondence between formalisms" is not a single category but a stratified relation with at least three distinct levels.

The two descriptions are:

- **Blume-Maleev polarization matrix formalism**: the standard condensed-matter description, complete and quantitatively accurate, developed entirely within established experimental physics.
- **FFGFT (T^4-torus geometry)**: the geometric field theory framework, applied to the same system from its own first principles, without fitting or borrowing from the condensed-matter description.

Neither framework was designed with the other in mind. The correspondences that emerge are therefore structurally informative – they reveal what mathematical content is shared versus what is framework-specific.

C.1 The Three Levels of Correspondence

C.2 The comparison operator – made explicit

The three-level classification in Table C.1 assigns each correspondence to a level. Up to this point, however, it remained *implicit by which criteria* this assignment is made – which operator decides whether a phenomenon falls on Level 1, 2 or 3. This section exposes

Table C.1: Classification of correspondence types between the Blume-Maleev and FFGFT descriptions of the ferrotoroidic system.

Level	Definition	Diagnostic criterion
1. Near-equivalence	Both formalisms express the same mathematical object, derived independently from different starting points.	The abstract structure (group, operator, invariant) is identical; only the derivation path and physical interpretation differ.
2. Structural overlap	The same qualitative structure appears in both, but the quantitative content is not shared.	One formalism can be mapped onto the other for the structural skeleton, but not for numerical predictions.
3. Complementary	Each formalism explains something the other cannot.	Neither is a subset or extension of the other; the union covers more than either alone.

that operator, so that it becomes inspectable and contestable rather than resting on intuition.

The six comparison axes

Two formalisms are compared along six independent axes. On each axis the content is either *shared* (both formalisms deliver the same) or *not shared*:

1. **Shared invariants**: Do both produce the same conserved quantities / topological invariants? (e.g. the conserved winding / domain number)
2. **Shared operators**: Does the same abstract operator structure appear? (e.g. the same group $\mathbb{Z}_2 \times \mathbb{Z}_2$, the same transformation rule)
3. **Shared mathematical structures**: Do both share the qualitative scaffold – antisymmetric coupling, projection factor – even if the numerical values differ?
4. **Shared predictions**: Do both deliver the same *quantitative* value for the same experiment?
5. **Shared physical mechanisms**: Do both attribute the phenomenon to the *same* physical cause?
6. **Shared hardware implications**: Do the same consequences for real devices / measurements follow from both?

How the axes define the levels

The level assignment is not intuition but reads directly off the pattern of shared axes: The assignment criterion is thus made explicit:

Table C.2: The comparison operator: which axes are shared at which level. ✓ = shared, × = not shared, ∼ = shared only qualitatively.

Comparison axis	**Level 1** Near-equiv.	**Level 2** Overlap	**Level 3** Complement.
1. Invariants	✓	∼	×
2. Operators	✓	∼	×
3. Mathematical structure	✓	✓	×
4. Predictions (quantitative)	×	×	×
5. Physical mechanisms	×	×	×
6. Hardware implications	×	×	×

- **Level 1 (near-equivalence):** Axes 1–3 fully shared – identical invariant, identical operator, identical structure – with different derivation paths. Axes 4–6 remain separate. (Examples: $\mathbb{Z}_2 \times \mathbb{Z}_2$, $\mathscr{P}'' = 0$, topological charge.)
- **Level 2 (structural overlap):** Axis 3 shared (the same qualitative scaffold), axes 1–2 only partially, axes 4–6 not. (Examples: antisymmetric exchange, decompactification projection.)
- **Level 3 (complementary):** None of axes 1–3 shared; each formalism covers a domain the other cannot reach. (Examples: geometric necessity of $\mathbb{Z}_2 \times \mathbb{Z}_2$ in FFGFT; material-specific transition temperatures in Blume-Maleev.)

Notably, axes 4–6 (quantitative predictions, physical mechanisms, hardware) are shared at *no* level in this comparison. This is not a defect of the classification but its main finding: Blume-Maleev and FFGFT share mathematical *structure* (axes 1–3) but not physical *substance* (axes 4–6). This is precisely what distinguishes this correspondence from a genuine equivalence (see the section on the Heisenberg/Schrödinger analogy).

Residuals: limits of the operator

The operator made explicit here is itself not fully mature. Three residuals remain open and are declared here, not hidden:

- **Axis completeness:** Are six axes sufficient, or are dimensions missing (e.g. shared boundary conditions, shared symmetry-breaking patterns)? The list is a working state, not a proven basis.
- **Axis weighting:** All six axes are weighted equally here. Whether a shared invariant (axis 1) should weigh more than a shared structure (axis 3) is undecided.
- **Axis independence:** The axes are treated as independent but may not be – a shared operator structure (axis 2) often implies a shared invariant (axis 1). The correlation structure of the axes is not analysed.

These residuals do not change the level assignment in the present case (the pattern in Table C.2 is robust), but they mark where the operator itself requires further examination. The value of making it explicit lies precisely here: a previously implicit but load-bearing criterion becomes an inspectable object – inspectable rather than persuasive.

C.3 Level 1 – Near-Equivalences

The $\mathbb{Z}_2 \times \mathbb{Z}_2$ domain structure

In Blume-Maleev: The four magnetic domains are related by two symmetry operations of the crystal: time-inversion $1'$ (maps $a \leftrightarrow b$, the ferrotoroidic flip) and the 2-fold screw axis $2_1 \| y$ (maps $' \leftrightarrow ''$, the orientation flip). Together these generate a $\mathbb{Z}_2 \times \mathbb{Z}_2$ group structure over the four domains. This is derived from the crystal symmetry group $P1121/a'$ and the magnetic space group analysis.

In FFGFT: The T^4 torus has two independent binary degrees of freedom for closed winding loops: the temporal winding direction ($\pm\tilde{T}$, i.e. whether the loop winds clockwise or anti-clockwise in the compact time direction) and the spatial winding parity ($\pm c$, the two possible orientations of the winding plane in the observable 3D subspace). These generate the same $\mathbb{Z}_2 \times \mathbb{Z}_2$ group structure over the four winding classes.

Assessment: The same abstract group $\mathbb{Z}_2 \times \mathbb{Z}_2$ is derived from two completely different starting points – crystal symmetry on one side, T^4 compactification geometry on the other. This is a near-equivalence: the mathematical object is the same, the derivation paths are independent.

What it does not mean: This does not imply that the crystal lattice *is* a T^4 torus, or that the FFGFT parameters describe the specific crystal. The shared group structure is a mathematical fact about how binary symmetries combine – it does not identify the underlying physical mechanism.

The $\mathscr{P}'' = 0$ condition

In Blume-Maleev: The general polarization transformation is $\mathbf{P}_f = \mathscr{P}'\mathbf{P}_i + \mathscr{P}''$. For $LiNi_{0.8}Fe_{0.2}PO_4$, $\mathscr{P}'' = 0$ because the magnetic structure factors are purely imaginary: $\mathrm{Re}(NM^*_{\perp i}) = 0$ and $\mathrm{Im}(M_{\perp y}M^*_{\perp z}) = 0$. This follows from the antiferromagnetic head-to-tail configuration (moments related by inversion are antiparallel).

In FFGFT: The recursion operator $\mathscr{R}$ (Doc. 203) on the T^4 mode space decomposes into a rotational part (coherent mode transformation, winding number preserved) and a creation part (spontaneous mode generation at T^4 boundary nodes, Doc. 204). In the static T^4 ground state – no expansion, no horizon dynamics – the creation part vanishes: $\mathscr{P}'' = 0$.

Assessment: Both derive the same vanishing of the creation/annihilation term from a "no spontaneous generation" condition – in the crystal from the antiferromagnetic symmetry, in FFGFT from the static geometry. The mathematical structure is the same; the physical mechanism is different but yields the same formal result.

Topological charge conservation

In the paper: The memory is non-volatile – domain populations remain stable at $E = H = 0$ after field-cooling. The paper identifies this as a consequence of the energy barrier separating toroidal domains: switching requires applying a conjugate field $E \times H$ above a threshold.

In FFGFT: A mode with non-zero winding number n cannot decay to $n = 0$ without crossing the minimal energy of the T^4 boundary condition (Doc. 204, Fractal Boundary

Condition). The winding number is a topological invariant – it is quantized and conserved unless a field large enough to overcome the topological barrier is applied.

Assessment: Both describe the same mathematical phenomenon – a discrete topological invariant protected by an energy barrier – though one frames it in terms of magnetic anisotropy and the other in terms of T^4 compactification geometry. Near-equivalence at the level of the conservation law; different mechanisms at the level of the physical origin.

C.4 Level 2 – Structural Overlap Without Equivalence

Antisymmetric exchange and the non-closure theorem

In Blume-Maleev: The Dzyaloshinskii-Moriya (DM) interaction is $\mathscr{H}^{\mathrm{DM}} = \mathbf{D} \cdot (\mathbf{S}_1 \times \mathbf{S}_2)$, odd under spin exchange (antisymmetric), with $\mathbf{D}_{34} = -\mathbf{D}_{12}$ from inversion symmetry.

In FFGFT: The Non-Closure Theorem (Doc. 193) states that cross-products of mode labels from *different* winding classes are topologically inadmissible – they would generate hybrid modes that violate the closed winding structure of T^4. The antisymmetric cross-product $S_i \times S_j$ maps structurally onto this: it couples two spin components (two different winding directions) in a way that is antisymmetric under interchange.

Assessment: The antisymmetric cross-product structure is shared. But the DM vector components D^b and D^c are material-specific quantities ($|D^b| < |D^c|$, at 35° from the c-axis) that have no FFGFT counterpart. FFGFT can say that the antisymmetric structure is topologically constrained; it cannot predict the specific DM vector orientation or magnitude for this crystal.

Decompactification projection

In Blume-Maleev: The factors p_{fx}/p_{ix} in the general polarization matrix (Eq. 9) measure the efficiency with which the measurement apparatus resolves the polarization state – how much of the abstract mathematical object is made observable by the specific experimental setup.

In FFGFT: The decompactification projection (Layer 1 to Layer 2, Doc. 001) measures the fraction of the unit-free geometric content that becomes observable in the SI-projected description. The ratio $T_{\mathrm{CMB}}/E_\xi = (16/9)\xi^2$ is a dimensionless Layer-1 prediction; the absolute Kelvin value requires embedding in SI.

Assessment: The structure – a projection factor that mediates between the abstract mathematical content and the observable quantity – is the same in both. But the specific projection factors p_{fi} are apparatus-dependent, material-dependent numbers in the paper; in FFGFT the projection is governed by $\xi = 4/30000$ and the SI embedding. No quantitative mapping between the two is established.

C.5 Level 3 – Complementary (Non-Overlapping)

What FFGFT explains that Blume-Maleev does not

The Blume-Maleev formalism derives the four-domain structure from the observed crystal symmetry. It takes the symmetry group as given and works out the consequences. It does not explain *why* the relevant symmetry group is $\mathbb{Z}_2 \times \mathbb{Z}_2$ rather than some other group – that is determined by the specific material and is taken from experiment.

FFGFT provides a geometric explanation: $\mathbb{Z}_2 \times \mathbb{Z}_2$ is the *necessary* structure for a compact T^4 torus with one time-like and one space-like compact direction contributing to the observable domain classification. The four-fold structure is not contingent on the material; it is a consequence of the topology.

Similarly, the non-volatility of the memory is an empirical finding in the paper. In FFGFT it is a structural necessity: topological charge conservation is a consequence of the T^4 boundary condition, not something that needs to be verified separately for each material.

What Blume-Maleev provides that FFGFT does not

The paper provides what FFGFT cannot: quantitative predictions for the specific crystal. The transition temperatures $T_2 = 25\,\mathrm{K}$ and $T_4 = 21\,\mathrm{K}$, the domain population ratios (up to 90% majority domain under optimal field-cooling), the DM vector orientation (35° from c-axis), the polarization tensor elements P_{xy}/P_{yx} and P_{xz}/P_{zx} in Table 1 – all of these follow from the material parameters of $LiNi_{0.8}Fe_{0.2}PO_4$ and are outside FFGFT's scope. FFGFT has nothing to say about why the Ni/Fe ratio 0.8/0.2 produces four domains rather than two, or why the transition temperature is 21 K rather than 210 K.

C.6 The Analogy: Why Not Heisenberg vs. Schrödinger?

A natural first analogy for two formalisms describing the same physics is Heisenberg's matrix mechanics vs. Schrödinger's wave mechanics (1925–26), which were subsequently proved to be fully equivalent (von Neumann 1932). But this analogy is too strong for the present case.

Heisenberg and Schrödinger were describing the *same physical objects* (atoms, electrons) at the *same scale* with the *same predictive scope*. Their equivalence meant that every prediction of one was also a prediction of the other.

The Blume-Maleev / FFGFT correspondence is weaker: the two formalisms share mathematical structure at Levels 1 and 2, but operate at different scales (angstrom lattice vs. sub-Planck T^4), with different predictive scope (material-specific quantitative vs. topological structural), and different derivation bases (experimental symmetry analysis vs. geometric first principles). There is no reason to expect full equivalence, and the Level 3 analysis confirms it does not hold.

A more precise analogy is the relation between:

- A symmetry-group classification (e.g. the Wigner classification of particles by irreducible representations of the Poincaré group): gives the topology, the quantum numbers, the conservation laws.

- A Lagrangian field theory for a specific particle (e.g. QED for the electron): gives all quantitative predictions for that specific particle.

The group classification is not a subset of QED, and QED is not a subset of the Wigner classification. They are complementary: the group gives the structural necessity (why the electron exists and has spin 1/2); the field theory gives the dynamics (how it interacts). Neither alone is sufficient.

Blume-Maleev corresponds to the field theory (quantitative, material-specific, dynamically complete). FFGFT corresponds to the group classification (structural necessity, scale-independent, topologically explanatory). The shared content at Levels 1 and 2 is the intersection – the topological structure that both descriptions must contain because it is a property of the physical system independent of the formalism used to describe it.

C.7 Implications for Model Comparison

The three-level stratification has practical consequences for comparing frameworks in the IPI group:

1. **Level 1 correspondences are the most informative**: when two independently derived formalisms produce the same mathematical object (same group, same invariant, same conservation law), this is evidence that the object is a real structural feature of the physical system – not an artefact of the formalism. The $\mathbb{Z}_2 \times \mathbb{Z}_2$ structure appearing in both crystal symmetry analysis and T^4 geometry suggests that this group is a genuine feature of toroidal topology, not just a convenient description.
2. **Level 2 correspondences define the shared vocabulary**: the structural overlap tells us which concepts transfer between frameworks (antisymmetric coupling, projection factors) and which do not. This is useful for translation without overclaiming equivalence.
3. **Level 3 defines the genuine boundary**: each framework's exclusive content marks where the other cannot substitute for it. Recognizing this boundary prevents both overclaiming ("my framework explains everything the other does") and underclaiming ("the frameworks have nothing to do with each other").
4. **Full equivalence is the special case, not the default**: the Heisenberg/Schrödinger equivalence was a remarkable result, not a template. Most frameworks that share structural content are related by partial overlap or complementarity, not equivalence.

Reference. The ferrotoroidic experiment used as the test case: N. Qureshi et al., *Toroidicity as a route towards non-volatile quaternary memory in antiferromagnets*, *Nature Communications* **17**, 4033 (2026). DOI: 10.1038/s41467-026-70767-8. The FFGFT interpretation of all formulas in that paper is in Doc. 264 of this series.

Chapter D

Doc. 267 — Cosmological Degeneracy: ΛCDM and FFGFT

Purpose and Core Thesis

This document compares the two cosmological models – the ΛCDM standard model and FFGFT – in their respective internal logic. The core thesis:

> The principal cosmological standard measurements (Type Ia supernovae, baryon acoustic oscillations, CMB temperature) are *degenerate*: they cannot in principle distinguish a physical motion of space (ΛCDM) from an intrinsic, non-linear unfolding of the time metric (FFGFT) – as long as both models produce the same distance-redshift relation $z(d)$.

This is not a claim of FFGFT's superiority. It is a methodological observation: the data alone do not decide between the models. The decision shifts to the level of theoretical consistency, parameter count, and derivability from first principles.

D.1 The Two Models in Their Basic Structure

The inverted initial condition

The fundamental difference is not in the predictions but in the direction of evolution:

Phase	ΛCDM (standard model)	FFGFT
Beginning	Singularity, Big Bang, extremely hot and dense, time passes fast	Minimal length $L_0 = \xi \cdot \ell_P$, nearly compactified, concentrated but cold, time passes extremely slowly
Evolution	Metric expansion of space + cooling	Fractal unfolding of time → apparent expansion
Today	Expanding, cold, accelerated by dark energy (Λ)	Apparently expanding, stationary quantum ground state, apparently accelerated by non-linear time recursion

The two directions are fully inverted:

- ΛCDM: *hot and fast → cold and accelerated*
- FFGFT: *cold and slow → apparently warm and accelerated* (through time unfolding)

Temperature in both pictures

In ΛCDM, today's CMB temperature ($\approx 2{,}7$ K) is the cooled remnant of a hot early phase. In FFGFT it is not a remnant but a stationary equilibrium value from the ξ-field quantisation (Doc. 025): the radiation does not cool, it is an inherent feature of the vacuum in equilibrium. In FFGFT, the initial concentration is *topological* (winding numbers close together), not *thermodynamic* (hot).

D.2 The Cosmological Degeneracy – Three Examples

Type Ia Supernovae and Light-Curve Stretching

The signal: The data show $b \approx 1$: a distant supernova runs more slowly for us (light curve stretched), and its light is redshifted.

ΛCDM: Space expands metrically during the photon flight, stretching wavelength (redshift) and pulling apart signal duration.

FFGFT: No spatial expansion. But the fractal time unfolding lengthens the path the signal travels through the geometry. This geometric path lengthening stretches wavelength and pulse duration in the exact same ratio – hence also $b = 1$.

The indistinguishability: Both mechanisms yield exactly the same mathematical curve. The DES value $b = 1{,}003 \pm 0{,}011$ excludes a *non-dilating* static model ($b = 0$) at $\sim 90\sigma$ – but FFGFT predicts $b = 1$, not $b = 0$. FFGFT is therefore not affected by this test.

Baryon Acoustic Oscillations and Angular Scales

The signal: One measures the angular separation $\theta(z)$ of galaxy clusters.

ΛCDM: The apparent angle changes because the scale factor $a(t)$ grows and dark energy accelerates the expansion.

FFGFT: The apparent acceleration results from the non-linear temporal recursion. Since the effective time rate per distance step changes, the $z(d)$ relation curves exactly as if there were accelerated expansion.

The indistinguishability: As long as the FFGFT time evolution produces the same $z(d)$ relation, it yields the same angles on the sky. The BAO data contain no marker for which mechanism produced the curve.

Important caveat (P20): The BAO test uses D_M/r_s, where $r_s \approx 147$ Mpc was computed under ΛCDM assumptions. Onur's $\Delta\chi^2 \approx 138$ compares a logarithmic distance law against ΛCDM pipeline data – model against model, not theory against raw nature.

The Cosmic Microwave Background and the Acoustic Peaks

The signal: A nearly perfect blackbody spectrum at $\approx 2{,}7$ K with acoustic peaks at $\ell \approx 200, 500, 800, \ldots$

ΛCDM: Sound waves in the photon-baryon plasma, frozen at recombination. The first peak ($\ell \approx 200$) marks regions at maximal compression; higher peaks are overtones.

FFGFT approach: No hot phase, no plasma. The CMB temperature is the stationary quantum ground state of the Φ-field (Doc. 025). The peaks would arise from discrete T^4 lattice resonances – like the emission lines of an atom or phonon dispersion in a crystal.

The ratio $(\lambda_e/L_0)^{1/\ln(1/\xi)}$ with the reduced Compton wavelength λ_e and $L_0 = \xi \cdot \ell_P$ gives numerically:

$$\left(\frac{\lambda_e}{L_0}\right)^{1/\ln(1/\xi)} = \left(\frac{3{,}86 \times 10^{-13}}{2{,}16 \times 10^{-39}}\right)^{1/8{,}92} \approx 875.$$

This value lies in the correct order of magnitude, close to ΛCDM recombination ($z \approx 1089$). **Caveat (P20):** The exponent $1/\ln(1/\xi)$ in the power is not derived from the FFGFT documents available; its justification from the T^4 geometry remains open. z_* is therefore currently a *reference value* of the right order of magnitude, not a derived FFGFT result.

D.3 The Peak Structure from Pure Geometry

Acoustic resonators are exactly computable, and their eigenmode ratios depend *only on the geometry* of the resonator, not on the medium: a pipe or sphere (j_0) gives $1 : 2 : 3 : 4 : 5$, a half-open pipe $1 : 3 : 5 : 7$, a circular membrane (J_0) $1 : 2{,}30 : 3{,}60$. The CMB ($1 : 2{,}44 : 3{,}68$) is an acoustic series with an offset.

The key point: the peak structure is set by the *geometry* of the resonance space, not the medium. ΛCDM: sound horizon in plasma; FFGFT: T^4 torus geometry. Both can produce the same series.

Concrete T^4 calculation. The thermally weighted spectrum $I(r) = g(r) \cdot \langle N(r) \rangle \cdot r$ with Bose-Einstein occupation $\langle N \rangle = 1/(e^r - 1)$ and Jacobi degeneracy $g(r)$ has local maxima at $|n|^2 = 3, 6, 10, 14, 18, \ldots$ The ratios $\ell \propto \sqrt{|n|^2}$ of the *dominant* peaks $|n|^2 = 3, 18, 42, 78$ (all genuine local maxima) match the CMB series:

$$\sqrt{3} : \sqrt{18} : \sqrt{42} : \sqrt{78} = 1 : 2{,}449 : 3{,}742 : 5{,}099 \quad \text{vs.} \quad 1 : 2{,}441 : 3{,}682 : 5{,}091 \text{ (CMB)}.$$

Deviation $< 2\%$ with no free parameter in the ratios; the factor 3 is set and the peak selection is open (see below).

The factor 3 is a choice. The peaks $3, 18, 42, 78$ are exactly $3 \cdot \{1, 6, 14, 26\}$. A spatial projection of the T^4 mode $\vec{n} = (n_1, n_2, n_3, n_4)$ cannot produce this factor, since $n_1^2+n_2^2+n_3^2 \leq |n|^2$; for $(1, 1, 1, 0)$ one already has $|n|^2 = 3$, so this mode does not belong to $|n|^2 = 1$. The ratios $\sqrt{3} : \sqrt{18} : \sqrt{42} : \sqrt{78}$ do not depend on the factor. The calculation and the caveat (the peak selection is open, as is the rigorous selection mechanism via the spherical C_ℓ projection, P30) are in Doc. 268, Steps 8–9.

Two filters produce the peak structure. The observed discrete, sharp CMB peaks follow from two properties of the same complete T^4 acting together – seen from two reference points:

- **Filter 1 – compactness (the T^4 as a whole, from outside):** A compact (finite) space gives a *discrete* mode spectrum; an infinite-open space gives a continuous one – no sharp peaks. This filter forces the *existence* of discrete peaks.
- **Filter 2 – fractal structure (from inside, decompactified):** The fractal granulation ($D_f = 3 - \xi$, Jacobi degeneracy of the four-square sum and Bose-Einstein weighting) weights the discrete modes unequally. Only this produces the *dominant* peaks at $|n|^2 = 3, 6, 10, 14, \ldots$ rather than a uniform lattice. This filter forces the *location* of the peaks.

The two filters are not two different objects but two reference points on the same complete T^4: compact-finite from outside (Filter 1, discrete modes), fractally granulated inside down to $L_0 = \xi \cdot \ell_P$ (Filter 2). The fractal non-closure is therefore not a *second* property contradicting compactness, but the inside view of the same closed torus – a matter of reference point (inside/outside, P20). Only both filters together give the observed CMB series; the discrete peaks thus argue *for* the compact T^4 and *against* a purely-infinite-open universe.

Honest caveat: This conclusion is FFGFT-internal. The peaks do *not* force spatial compactness – ΛCDM produces the same discrete peaks from a *temporal* bound (the sound horizon at recombination) in a spatially infinite universe. This is the cosmological degeneracy: the peaks alone do not distinguish spatial compactness (FFGFT) from a temporal bound (ΛCDM).

D.4 Peak Ratios and Absolute Scale

Ratios: no free parameter

The peak ratios follow directly from the T^4 geometry – no fit, no external parameter:

$$\sqrt{3} : \sqrt{18} : \sqrt{42} : \sqrt{78} = 1 : 2{,}449 : 3{,}742 : 5{,}099$$

Observed: $1 : 2{,}441 : 3{,}682 : 5{,}091$. Deviation $< 2\%$.

Caveat: if peak positions come from a ΛCDM template fit, ΛCDM is already embedded. A clean test needs model-free local maxima of C_ℓ ($\sim$ 1–2% deviation from ΛCDM-fit values).

Absolute scale: R_H

The absolute position $\ell_1 \approx 220$ requires a length scale. In FFGFT this is R_H, written via the chain (Doc. 182):

$$\xi \;\to\; E_0 = \sqrt{m_e m_\mu / K_{\text{frak}}} \;\to\; E_H = E_0 \cdot \xi^{41/4} \;\to\; R_H = \frac{\hbar c}{E_H} \approx 1{,}40 \times 10^{26}\,\text{m}$$

The exponent $41/4$ is not derived from ξ but calibrated to the measured H_0. The cosmological scale is therefore the one external input of the sector: a shared empirical reference point of both models, not an FFGFT-specific gap. R_H is simultaneously the Schwarzschild radius of the total mass $M_U = R_H c^2/(2G)$ of the static universe – an internal consistency check (Doc. 182, Doc. 190 P4).

Quantity	Origin	Status
Ratios $1 : 2{,}449 : \ldots$	T^4 geometry, Jacobi degeneracy	No free parameter
R_H (absolute scale)	$E_H \to R_H$, exponent calibrated to H_0 (Doc. 182)	One external input
Link $R_H \to \ell_1$	Effective resonator radius	Open (P20)

Script: `ffgft_lambda_im_messprozess.py`.

D.5 Inversion and the Symmetric Circularity

Starting from the measured peaks, the fundamental angle is $\theta_1 = \pi/A \approx 0{,}59°$. But $\theta_1 = L_{\text{res}}/D_A$ is a *ratio*: only the product $H_0 \cdot L_{\text{res}} = \theta_1 c/\ln(1+z_*) \approx 458$ km/s is determined, not H_0 alone. Isolating H_0 requires fixing one length independently – the external parameter of P20.

This makes the circularity **symmetric**:

	ΛCDM	FFGFT
Measured	θ_1 (angle)	θ_1 (angle)
External length scale	$r_s \approx 147$ Mpc	R_H (torus size)
Source of scale	hot early phase (model)	set / calibrated to H_0
H_0	$67{,}4$ (output)	$66{,}2$ (set)
Circular?	**Yes**	**Yes**

A measured angle always determines only a length ratio. Both models must insert an absolute scale. No one obtains a model-neutral H_0 from the CMB peaks. The fair comparison is not "who is circularity-free" (neither is) but parsimony (FFGFT: one external scale; ΛCDM: r_s from 6 parameters plus the BBN phase) versus anchoring (ΛCDM better tested; FFGFT's early phase open).

Scripts: `ffgft_zirkularitaet_symmetrie.py`, `lcdm_circularity.py`.

D.6 Accelerated Dilation in Both Models

An often overlooked point: both models predict an *accelerated* dilation, not a constant one.

Aspect	**ΛCDM**	**FFGFT**
Cause	Dark energy (Λ)	Fractal temporal evolution ($\mathscr{R}$)
Mechanism	Repulsive gravitation	Accumulated path lengthening
w	≈ -1	No w (no Λ field)
Acceleration	Real (metric expansion)	Apparent (emergent time dilation)

In FFGFT, the effective scale change per time step grows with the number of recursion steps: the farther we look in distance (cosmologically: into the past), the more unfolded time lies between events. The observation of an accelerated dilation is therefore no contradiction to FFGFT – it is exactly what FFGFT expects from its own dynamics. Only the explanation differs.

This further complicates the distinction: many observations (SNe Ia, BAO) primarily measure the acceleration itself, not its mechanism.

The three phases of the ΛCDM expansion history

A structural difference concerns the *temporal course* of the expansion rate. ΛCDM requires three successive regimes:

1. **Early phase (inflation):** an extremely rapid, exponentially accelerated expansion driven by a hypothetical inflaton field with negative pressure, which then decays.
2. **Intermediate phase (decelerated expansion):** radiation- and matter-dominated, $a(t) \sim t^{1/2}$ resp. $t^{2/3}$. Attractive gravitation *slows* the expansion.
3. **Present phase (renewed acceleration):** from $z \approx 0{,}6$ the cosmological constant Λ dominates and drives the expansion in an accelerated manner again.

The sequence *accelerated → decelerated → accelerated again* is mathematically allowed by the Friedmann equation, but it requires *two separate, independently postulated* acceleration mechanisms (inflaton and Λ), separated by a braking phase. There is also the coincidence problem: Λ becomes dominant precisely in our cosmic epoch – a fine-tuning the model does not explain from within.

FFGFT, by contrast, manages with *one* mechanism: the apparent acceleration grows monotonically with the recursion depth of the fractal time unfolding; there is no separate inflationary and no separate Λ epoch. This is an advantage at the level of theoretical economy, *not* an empirical statement – FFGFT has no quantitative predictions for the early phase so far, and whether its single unfolding dynamics reproduces the entire expansion history correctly has not been shown.

D.7 The Limit for Both Models: No Model-Neutral Scale

The degeneracy is reinforced by a second problem affecting both models equally (P20, Doc. 190): there is no model-neutral determination of the cosmological scale R_H.

- ΛCDM measures $H_0 \approx 67{,}4$ km/s/Mpc – but this value is a pipeline output presupposing expansion, the FLRW metric, and the matter content.
- FFGFT writes $E_H/\hbar \approx 66{,}2$ km/s/Mpc in the form $\xi^{41/4}$; the exponent, however, is not derived from ξ but calibrated to the measured H_0, so $R_H = \hbar c/E_H$ is set as well (Doc. 182). What remains open is the precise link $R_H \to \ell_1$ (how R_H yields the absolute peak position, P20).

Onur's tests use ΛCDM pipeline quantities (r_s, H_0, calibrated magnitudes) as reference. They therefore test not "static vs. expanding" but "ΛCDM pipeline vs. the absence of this pipeline." That is not a neutral test.

D.8 Where the Decision Actually Falls

If the data are blind to the mechanism, the decision shifts to the theoretical level:

1. **Parameter count:** FFGFT needs internally only the fundamental ratio $\xi = 4/30000$ and, for the cosmological sector, one openly declared external scale ($R_H = \hbar c/E_H$, calibrated to H_0, Doc. 182); what remains open is the precise link $R_H \to \ell_1$ (P20). ΛCDM requires parameters for dark energy, dark matter, inflation, baryon density, and more.
2. **Singularities:** FFGFT produces no singularity (no Big Bang, no infinite density at $t = 0$). ΛCDM begins with a singularity.
3. **Explanation of time:** FFGFT derives the emergence and change of time flow (the fractal unfolding) from first principles. ΛCDM posits time as a rigid, unexplained background stage.

Important – no superiority claim: These points are advantages at the level of theoretical economy and consistency. They are *not* an empirical refutation of ΛCDM.

D.9 The Honest Scientific Position

- There are two competing explanations for the same observations.
- Neither is currently empirically preferred, because the classical measurements (SNe Ia, BAO, CMB temperature) are degenerate.
- The data do not force a choice between the models.
- The choice is a question of theoretical consistency and explanatory power beyond the cosmological standard data.
- FFGFT is neither confirmed nor refuted by Onur's tests. The tests show only that a *naive* static model without fractal time unfolding ($b = 0$, no acceleration) is excluded. FFGFT is not such a naive model.

The point where the models could in principle differ: The early phase. Caution is needed here, however – the early phase is hardly directly verifiable for *either* model,

and likely will remain so: no one can directly observe the first moments of the universe; both models reconstruct it from present-day traces and many assumptions. ΛCDM has had decades to develop a detailed explanatory framework for this phase (hot Big Bang, inflation, nucleosynthesis) – successful in fitting the data, but itself built on numerous assumptions and free parameters. FFGFT would have to explain the light-element abundances (BBN), the CMB acoustic peaks, and structure formation from its inverted direction (cold, slow, fractally unfolding) – an open research question on which little work has been done.

The early phase is thus in principle the place where the models could diverge – but given the limited verifiability for both sides, it is not to be expected that it will lead to a clear empirical decision. A quantitative FFGFT prediction for the early observables would be desirable but is still outstanding.

Related documents: P18–P20 (Doc. 190, status of the cosmological scale), Doc. 025 (CMB as ξ-field ground state), Doc. 182 (fractal path lengthening), Doc. 264 (ferrotoroidicity, P″=0 as static ground state), Doc. 266 (analysis of Onur's tests).

Chapter E

Doc. 268 — CMB Peaks from the T^4: Derivation of the Angular-Spectrum Ratios

Overview

This document examines the relation between the T^4 lattice structure of FFGFT and the observed CMB peak ratios. All numbers are directly reproducible. What is a consistency check (backward) and what is a genuine forward prediction is clearly separated from Step 17 on.

Two separate statements that must not be conflated:

- **(A) Ratio finding (consistency check, backward).** The values $|n|^2 = 3, 18, 42, 78$ produce ratios $1 : 2.449 : 3.742 : 5.099$, which agree with the observed CMB peaks ($1 : 2.441 : 3.682 : 5.091$) to $< 2\%$. *However:* the sequence $\{1, 6, 14, 26\}$ (before the factor 3) was *read off* the measured CMB ratios (Step 4), not produced forward from T^4. This is a successful *retrodiction*, not a parameter-free prediction (see Step 17.1).
- **(B) Anharmonicity finding (forward, new).** Forward, T^4 produces either a dense mode spectrum or – for symmetric resonance – the *harmonic* series $1 : 2 : 3 : 4 : 5$. The CMB is *anharmonic* ($1 : 2.44 : 3.68 : 5.09$). This anharmonicity follows *forward and parameter-free* from the codimension-1 projection in the fractal geometry ($D_f - 1 \approx 2 \Rightarrow$ Bessel index $\nu \approx 0 \Rightarrow J_0$ membrane), matching the observed series to $\sim 5\%$ – with no mode selection (Step 17). This is the conceptually cleaner statement and the actual progress of this document.

Honest summary: the ratio match is real but found backward. The forward content lies in the *anharmonicity* as a geometric effect of the flat codimension-1 projection. The rigorous selection mechanism for exactly $\{3, 18, 42, 78\}$ remains open (P29/P30, Doc. 190): the factor 3 in front of $\{1, 6, 14, 26\}$ is set, it does not arise from the projection onto three spatial directions (Step 8), and $|n|^2 = 30$ is *not* excluded by the factor-3 filter (Step 17.1).

E.1 Step 1: The T^4 state space

On the T^4 torus every mode has an integer winding number in four directions:

$$\vec{n} = (n_1, n_2, n_3, n_4) \in \mathbb{Z}^4$$

The radius (the quantum number) of a mode is:

$$|\vec{n}| = \sqrt{n_1^2 + n_2^2 + n_3^2 + n_4^2}$$

The lowest levels, each with an example tuple:

$\lvert n\rvert^2$	$\lvert n\rvert$	Example tuple	Degeneracy g
1	1.0000	$(-1, 0, 0, 0)$	8
2	1.4142	$(-1, -1, 0, 0)$	24
3	1.7321	$(-1, -1, -1, 0)$	32
4	2.0000	$(-2, 0, 0, 0)$	24
5	2.2361	$(-2, -1, 0, 0)$	48
6	2.4495	$(-2, -1, -1, 0)$	96
7	2.6458	$(-2, -1, -1, -1)$	64
8	2.8284	$(-2, -2, 0, 0)$	24
9	3.0000	$(-3, 0, 0, 0)$	104
10	3.1623	$(-3, -1, 0, 0)$	144

Important: The degeneracy $g(|n|^2)$ is *not* monotonic. At $|n|^2 = 6$ it jumps to 96 – twelve times more than at $|n|^2 = 1$. This follows from the Jacobi formula for the number of representations as a sum of four squares: $g(n) = 8 \sum_{d|n,\, 4\nmid d} d$.

E.2 Step 2: Thermal occupation (Bose-Einstein)

The energy of a mode with radius $r = |n|$ is (from the calibration $L_{\text{tor}} = 1/(k_B T_{\text{CMB}})$, Doc. 025):

$$E_r = r \cdot k_B T_{\text{CMB}}$$

The thermal occupation by Bose-Einstein with $\mu = 0$:

$$\langle N(r) \rangle = \frac{1}{e^r - 1}$$

The spectral density – how much energy in total sits in level r:

$$\boxed{I(r) = g(r) \cdot \langle N(r) \rangle \cdot r}$$

Numerical values for the lowest levels:

$\lvert n\rvert^2$	$\lvert n\rvert$	g	$\langle N\rangle$	$I = g \cdot N \cdot \lvert n\rvert$	Local max.?
1	1.0000	8	0.582	4.66	
2	1.4142	24	0.321	10.90	
3	1.7321	32	0.215	11.91	**Yes**
4	2.0000	24	0.157	7.51	
5	2.2361	48	0.120	12.84	
6	2.4495	96	0.094	22.22	**Yes**
7	2.6458	64	0.076	12.93	
8	2.8284	24	0.063	4.26	
9	3.0000	104	0.052	16.35	
10	3.1623	144	0.044	20.13	**Yes**

A selection of the local maxima up to $\lvert n\rvert^2 = 80$:

Peak	$\lvert n\rvert^2$	$\lvert n\rvert$	I
1	3	1.7321	11.91
2	6	2.4495	22.22
3	10	3.1623	20.13
4	14	3.7417	17.45
5	18	4.2426	19.30
6	22	4.6904	12.52
7	26	5.0990	10.52
8	30	5.4772	13.25
9	42	6.4807	7.64
10	66	8.1240	2.77
11	78	8.8318	1.73

The complete list of local maxima of $I(r)$ up to $\lvert n\rvert^2 = 80$ is $3, 6, 10, 14, 18, 22, 26, 30, 34, 38, 42, 45, 50, 54, 58, 60, 63, 66, 70, 75, 78$; not listed in the table are 34 ($I = 7.42$), 38 (6.24), 45 (5.12), 50 (4.47), 54 (4.54), 58 (2.70), 60 (1.93), 63 (2.36), 70 (2.24) and 75 (1.49).

E.3 Step 3: Projection onto multipole moments

The observable multipole moment ℓ is proportional to the wavenumber of the mode:

$$\ell \propto |\vec{n}| = \sqrt{|n|^2}$$

Thus the *ratios* of the peak ℓ values are directly the ratios of the $\sqrt{|n|^2}$ values – independent of the absolute scale (which contains D_A and L_{tor}):

$$\frac{\ell_k}{\ell_1} = \frac{\sqrt{|n|_k^2}}{\sqrt{|n|_1^2}}$$

E.4 Step 4: The decisive finding

The CMB ratios *squared* give directly the required $|n|^2$ ratios:

CMB peak	ℓ_{obs}	ℓ/ℓ_1	$(\ell/\ell_1)^2$	nearest $\lvert n\rvert^2$
1	220	1.000	1.00	1
2	537	2.441	5.96	**6**
3	810	3.682	13.56	**14**
4	1120	5.091	25.92	**26**
5	1440	6.545	42.84	**43**

If the *first* CMB peak lies at $|n|^2 = X$, then the further ones lie at $X \cdot 6$, $X \cdot 14$, $X \cdot 26$, ...
For $X = 3$: the peaks lie at $|n|^2 = 3, 18, 42, 78$.

Honest reading of Step 4 (direction of inference)

The sequence $\{1, 6, 14, 26\}$ here arises by *squaring the measured* CMB ratios and rounding to the nearest integer – i.e. *from the data*, not forward from T^4. Step 5 then checks whether $3 \cdot \{1, 6, 14, 26\}$ are genuine local maxima of the T^4 spectrum (they are). This is a **consistency check (retrodiction)**, not a prediction. The inference runs data $\rightarrow \{1, 6, 14, 26\} \rightarrow \times 3$, not $T^4 \rightarrow$ data. The genuine forward content – anharmonicity as a geometric effect – is developed in Step 17.

E.5 Step 5: Comparison with the T^4 peaks

Are $|n|^2 = 3, 18, 42, 78$ genuine local maxima in the T^4 spectrum?

$\lvert n\rvert^2$	g	I	I_{left}	Local maximum?
3	32	11.91	10.90	**Yes**
18	312	19.30	9.77	**Yes**
42	768	7.64	3.57	**Yes**
78	1344	1.73	1.04	**Yes**

All four are genuine local maxima. The ratios:

	Peak 1	Peak 2	Peak 3	Peak 4
$\lvert n\rvert^2$	3	18	42	78
$\sqrt{\lvert n\rvert^2}$	1.7321	4.2426	6.4807	8.8318
Ratio to Peak 1	1.000	2.449	3.742	5.099
CMB ratio	1.000	2.441	3.682	5.091
Deviation	0.0%	+0.3%	+1.6%	+0.2%

Deviation $< 2\%$**.**

That the four values are genuine local maxima follows from the Jacobi degeneracy structure of the T^4 lattice and Bose-Einstein statistics. The selection $\{1, 6, 14, 26\}$, by contrast, is read off the data (Step 4), and the factor 3 is set (Step 8).

E.6 Step 6: What determines the absolute scale

The ratios are scale-independent. The absolute position of the first peak ($\ell_1 \approx 220$) depends on:

$$\ell_1 = \frac{\pi \cdot D_A}{L_{\text{tor}}/\sqrt{3}}$$

with the angular distance D_A (external parameter, R20) and $L_{\text{tor}} = 1/(k_B T_{\text{CMB}})$ (via the measured CMB temperature; its relation to ξ is a unit-dependent agreement, not a derivation). This sets $A \approx 303.6$ in the trumpet formula (Doc. 267).

E.7 Step 7: What is still open (R29)

The T^4 spectrum has peaks at $|n|^2 = 3, 6, 10, 14, 18, 22, 26, 30, \ldots$ The CMB shows only the subset $3, 18, 42, 78$.

Between these lie strong peaks – in particular $|n|^2 = 6$ with $I = 22.22$, the largest value in the whole spectrum.

Question (R29): What selection rule explains why $|n|^2 = 6, 10, 14, 22, 26, 30$ are not visible as CMB peaks?

This question is open. The observer measures in 3D space (plus time-unfolding) and sees only the spatial projection $k^2_{\text{obs}} = (n_1^2 + n_2^2 + n_3^2)/L^2$. This projection cannot, however, produce the factor 3 in the visible series $3 \cdot \{1, 6, 14, 26\} = \{3, 18, 42, 78\}$, because the projected wavenumber never exceeds the full one (Step 8); the factor is set.

The full spherical projection of the T^4 modes onto the angular power spectrum C_ℓ with spherical Bessel functions (Doc. 190, R30) does not supply the selection either: in its naive form it gives a quasi-continuum (Step 17.1). The peak selection remains open as a whole. The ratios themselves are computed and stand.

E.8 Summary

Step	Result
1. T^4 state space	Winding numbers $\vec{n} \in \mathbb{Z}^4$, Jacobi degeneracy $g(\lvert n\rvert^2)$ non-monotonic
2. Bose-Einstein	$I(r) = g(r) \cdot \frac{1}{e^r-1} \cdot r$, local maxima at $\lvert n\rvert^2 = 3, 6, 10, 14, 18, \ldots$
3. Projection	$\ell \propto \sqrt{\lvert n\rvert^2}$, ratios scale-independent
4. CMB ratios	$(\ell/\ell_1)^2 \approx 1, 6, 14, 26$ for $\lvert n\rvert^2 = 1, 6, 14, 26$ (or $\times 3$ for $\lvert n\rvert^2 = 3, 18, 42, 78$)
5. Agreement	$\sqrt{3} : \sqrt{18} : \sqrt{42} : \sqrt{78} = 1 : 2.449 : 3.742 : 5.099$ vs. CMB $1 : 2.441 : 3.682 : 5.091$, deviation $< 2\%$
6. Absolute scale	$\ell_1 \approx 220$ from D_A (external) and L_{tor} (via T_{CMB})
7. R29/R30	Selection rule for the subset $3, 18, 42, 78$ open: factor 3 set (Step 8), naive C_ℓ projection (R30) insufficient (Step 17.1)

Related: Doc. 025 (T_{CMB}), Doc. 050 (T^4 topology, spin structure), Doc. 190 (R14a path-lengthening, R20 R_H as external parameter), Doc. 203 (recursion operator $\mathscr{R}$), Doc. 267 (cosmological degeneracy, trumpet formula).
Python script: `ffgft_cmb_t4_peaks.py`

E.9 Step 8: Where do k^2 and the factor 3 come from?

8.1 Where does the k^2 come from?

The ^{2} in k^2_{eff} does not come from the factor 3 – it comes from the wave equation itself. The wave equation for a massless field ($m = 0$, photon):

$$\left(\frac{\partial^2}{\partial t^2} - c^2\,\Delta\right)\phi = 0$$

The Laplace operator Δ contains *second* derivatives. Inserting the Fourier ansatz $\phi \sim e^{i(\mathbf{k}\cdot\mathbf{x}-\omega t)}$ gives:

$$-\omega^2 + c^2k^2 = 0 \quad \Rightarrow \quad \omega^2 = c^2k^2 \quad \Rightarrow \quad E = \hbar\omega = \hbar c\,k$$

The ^{2} is universal – it holds for any wave equation, independent of FFGFT.

8.2 The factor 3 is set

The observer lives in **3 spatial dimensions + time-unfolding**. They perceive the T^4 torus through their 3 spatial directions. The fourth torus direction is the time-unfolding – it is not another spatial direction but the mechanism through which the observer exists at all.

Concretely: a T^4 mode with winding number $\vec{n} = (n_1, n_2, n_3, n_4)$ appears to the 3D observer with the observable 3D wavenumber:

$$k_{\text{obs}}^2 = \frac{n_1^2 + n_2^2 + n_3^2}{L_{\text{tor}}^2}$$

The n_4 component is "hidden" in the time-unfolding.

For the T^4 mode $(1, 1, 1, 0)$ – all three spatial directions equal, $n_4 = 0$ – one has:

$$k_{\text{obs}}^2 = \frac{1^2 + 1^2 + 1^2}{L_{\text{tor}}^2} = \frac{3}{L_{\text{tor}}^2},$$

but at the same time $|n|^2 = 3$: the projection changes nothing here. In general $n_1^2+n_2^2+n_3^2 \leq |n|^2$ – the projected wavenumber never exceeds the full one. A spatial projection therefore **cannot** produce the factor 3 in front of the sequence $\{1, 6, 14, 26\}$; the factor 3 is set.

8.3 The observable series

The observable multipole moment is $\ell \propto k_{\text{obs}} \cdot D_A$:

$$\ell^2 \propto k_{\text{obs}}^2 \cdot D_A^2 = \frac{n_1^2 + n_2^2 + n_3^2}{L_{\text{tor}}^2} \cdot D_A^2$$

For modes of the form $(n, n, n, 0)$ one has $n_1^2 + n_2^2 + n_3^2 = 3n^2 = |n|^2$, hence $\ell^2 \propto |n|^2$. The series $\{3, 18, 42, 78\}$ arises only when the factor 3 is set in front of the sequence $\{1, 6, 14, 26\}$:

$$\ell^2 \propto 3 \cdot |n|^2 \qquad (|n|^2 \in \{1, 6, 14, 26\}, \text{ factor 3 set})$$

The visible peaks $\{3, 18, 42, 78\} = 3 \cdot \{1, 6, 14, 26\}$:

	Peak 1	Peak 2	Peak 3	Peak 4
T^4 ground peak $\|n\|^2$	1	6	14	26
$\times 3$ (set)	3	18	42	78
$\ell \propto \sqrt{3 \cdot \|n\|^2}$	1.732	4.243	6.481	8.832
Ratio to Peak 1	1.000	2.449	3.742	5.099
CMB ratio	1.000	2.441	3.682	5.091
Deviation	0.0%	+0.3%	+1.6%	+0.2%

8.4 What is still open

Averaged over all modes of an $|n|^2$ level, too, the projection gives $\langle n_1^2 + n_2^2 + n_3^2 \rangle = \frac{3}{4}|n|^2$ (isotropic T^4 distribution), i.e. less than $|n|^2$, not $3 \cdot |n|^2$.

Why precisely $3 \cdot \{1, 6, 14, 26\} = \{3, 18, 42, 78\}$ is visible is therefore not derived: the factor 3 is set, and the naive spherical projection of the T^4 modes onto C_ℓ does not reproduce the series (Step 17.1). The peak selection remains open (R29, R30).

E.10 Step 9: Overview of the retrodiction

The retrodiction of the CMB peak ratios in three steps:

Step	Input	Mechanism	Output
1	— (ξ-free)	Jacobi degeneracy $g(\lvert n\rvert^2)$, Bose-Einstein $N(r)$	T^4 peaks at $\lvert n\rvert^2 = 1, 6, 14, 26$
2	Factor 3 (set)	none – the 3D projection $k^2_{\text{obs}} = (n_1^2+n_2^2+n_3^2)/L^2$ does not produce it (Step 8)	Factor 3 in $\ell^2 \propto 3 \cdot \lvert n\rvert^2$
3	Steps 1+2	$\ell \propto \sqrt{3 \cdot \lvert n\rvert^2}$	Ratios $1 : 2.449 : 3.742 : 5.099$

Observed: $1 : 2.441 : 3.682 : 5.091$. Deviation $< 2\%$.

The ^{2} in k^2_{obs} comes from the wave equation (second derivatives of the Laplace operator) – universal, not an FFGFT specialty.

The factor 3 is set: the observer in 3D + time-unfolding sees only $n_1^2+n_2^2+n_3^2 \le \lvert n\rvert^2$, and such a projection cannot produce it (Step 8).

Open step: the peak selection as a whole – origin of the factor 3 and selection of the sequence $\{1, 6, 14, 26\}$ – is not derived.

Important reading (see Step 17): The sequence $\{1, 6, 14, 26\}$ listed as the "Output" of Step 1 is *not* produced forward from T^4; it is read off the measured CMB ratios (Step 4). This "overview" therefore describes the *retrodiction*. The genuine forward content – the anharmonicity from the codimension-1 projection – is in Step 17.

E.11 Summary (updated)

Step	**Result**
1. T^4 state space	Jacobi degeneracy $g(\lvert n\rvert^2)$ non-monotonic
2. Bose-Einstein	Local maxima at $\lvert n\rvert^2 = 3, 6, 10, 14, 18, \ldots$
3. Observer in 3D	$k^2_{\text{obs}} = (n_1^2+n_2^2+n_3^2)/L^2$; factor 3 set, cannot arise from the projection; ^{2} from wave equation (Laplace)
4. CMB series	$3 \cdot \{1, 6, 14, 26\} = \{3, 18, 42, 78\}$, all genuine T^4 peaks
5. Ratios	$1 : 2.449 : 3.742 : 5.099$ vs. CMB $1 : 2.441 : 3.682 : 5.091$, deviation $< 2\%$
6. Absolute scale	D_A external (R20), L_{tor} via T_{CMB}
7. Still open	Peak selection: origin of the factor 3 and selection of the sequence $\{1, 6, 14, 26\}$

Related: Doc. 025 (T_{CMB}), Doc. 050/051 (T^4 topology, Dirac structure), Doc. 190 (R14a, R20, R29, R30), Doc. 203 (recursion operator $\mathscr{R}$), Doc. 267 (cosmological degeneracy).

E.12 Step 10: The physical intuition – double resonator

10.1 Open and closed pipe

In classical acoustics the boundary conditions determine which modes resonate:

System	Boundary condition	Mode ratios
Closed pipe	both ends fixed	$1 : 2 : 3 : 4 : 5$ (all)
Half-open pipe	one end open	$1 : 3 : 5 : 7$ (odd only)
Circular membrane (2D)	rim fixed	$1 : 2.30 : 3.60 : 4.90$
Fractal T^4	**two coupled systems**	**1 : 2.449 : 3.742 : 5.099**

The pure T^4 (integer dimension $D = 4$, fully compact) corresponds to a *closed* system – all modes resonate. The fractal T^4 (dimension $D_f = 3 - \xi < 4$) has a qualitatively different resonance structure.

10.2 The fractal T^4 as a double resonator

The analogy to the half-open pipe (which produces a factor 2) is tempting – but it is not complete. The fractal T^4 is neither a simply open nor a simply closed system. It acts as a **double resonator**: two superimposed resonance structures whose coincidences determine the visible peaks.

	System 1 T^4 lattice	**System 2** 3D observer projection
Resonances	$\lvert n\rvert^2 = 1, 6, 14, 26, \ldots$	Projected onto $n_1^2 + n_2^2 + n_3^2$
Condition	Local maximum in $I(r)$	$k^2_{\text{obs}} = (n_1^2 + n_2^2 + n_3^2)/L^2$
Origin	Jacobi degeneracy $g(\lvert n\rvert^2)$	factor 3 set (cannot arise from the projection)

In this picture the visible peaks would be **coincidence resonances** – modes where both systems resonate simultaneously:

$$|n|^2_{\text{visible}} = 3 \cdot |n|^2_{\text{lattice}} = 3 \cdot \{1,\ 6,\ 14,\ 26\} = \{3,\ 18,\ 42,\ 78\}$$

System 2 cannot supply the factor 3, however (Step 8). The double resonator is therefore an analogy with the factor 3 set, not a derived selection mechanism.

10.3 Not all coincidences are visible

The double resonance condition yields more candidates than just $\{3, 18, 42, 78\}$:

$\lvert n\rvert^2$	$\lvert n\rvert^2/3$	Peak in Sys. 1?	I	CMB peak?
3	1	no	11.91	**Yes**
18	6	**Yes**	19.30	**Yes**
30	10	**Yes**	13.25	No
42	14	**Yes**	7.64	**Yes**
54	18	**Yes**	4.54	No
66	22	**Yes**	2.77	No
78	26	**Yes**	1.73	**Yes**

The peaks $|n|^2 = 30, 54, 66$ satisfy the double resonance condition as well but do not appear as CMB peaks. Thermal strength does not explain this: $|n|^2 = 30$ with $I = 13.25$ is stronger than 42 ($I = 7.64$) and 78 ($I = 1.73$), and 66 ($I = 2.77$) is also stronger than 78. A selection by the strongest coincidence peak per range therefore does not hold (Step 17.1); why precisely $18, 42, 78$ are visible is open.

10.4 The physical meaning

The transition from $D = 4$ (pure T^4, integer compact) to $D_f = 3 - \xi$ (fractal T^4) corresponds to the transition from a fully closed to a *fractally partly-open* system:

- $D = 4$ **(purely compact):** All four torus directions oscillate on equal footing. Every $|n|^2$ can resonate. This corresponds to a resonator with 4 equivalent, closed directions.
- $D_f = 3 - \xi$ **(fractal):** The fourth direction is no longer integer-compact but fractally granulated. It behaves like a direction with a *continuous spectrum* – analogous to the open side of a half-open pipe. The observer in 3D space sees only the three spatial components $n_1^2 + n_2^2 + n_3^2$ (the fourth is time-unfolding); this projection does not produce the factor 3 (Step 8).

In brief: The fractal T^4 is neither a closed nor an open resonator – it is a *double resonator* with coupled systems. In this picture the CMB peaks mark coincidences of T^4 lattice geometry and 3D observer projection – with the factor 3 set; no selection mechanism is derived by this.

10.5 Comparison of resonator types

Analogy	**Mechanism**	**Selects**
Closed pipe	one boundary condition	all integers
Half-open pipe	one boundary condition	odd only
Circular membrane	2D geometry	Bessel zeros
Fractal T^4	**coincidence of two resonator systems**	$3 \cdot \{$lattice peaks$\}$ (factor set)

In this picture the CMB peak series $1 : 2.449 : 3.742 : 5.099$ would be the resonance signature of a fractally partly-open, four-dimensional resonance space – not of a plasma, not of a simple membrane, but of the specific combination of T^4 topology and fractal dimension $D_f = 3 - \xi$. As long as the factor 3 and the selection are not derived, this remains an interpretation.

10.6 What follows from the peaks: two filters

The discrete, sharp peaks follow from two properties of the same complete T^4 acting together – seen from two reference points:

	Filter 1: compactness (T^4 as a whole, from outside)	**Filter 2: fractality** (from inside, decompactified)
Effect	produces *discrete* spectrum	weights the modes unequally
forces	*existence* of sharp peaks	*location* of the dominant peaks
Origin	finite, closed topology	Jacobi degeneracy $g(\|n\|^2)$ + Bose-Einstein
Result	uniform mode lattice	peaks at $\|n\|^2 = 3, 6, 10, 14, \ldots$

The two are not two objects but two reference points on the same complete T^4: compact-finite from outside (Filter 1), fractally granulated inside down to $L_0 = \xi \cdot \ell_P$ (Filter 2). The fractal non-closure is therefore not a *second* property contradicting compactness, but the inside view of the same closed torus – a matter of reference point (inside/outside, R20).

Only both filters together give the observed structure:

- Filter 1 alone: uniform discrete lattice – every $|n|^2$ equal, no preferred peaks.
- Filter 2 alone: needs a discrete lattice to act on – no lattice without compactness.
- Filter 1 + Filter 2: dominant peaks at specific $|n|^2$, which with the factor 3 set (Step 8) give the CMB series.

What can be deduced: The discrete peaks argue *for* a compact (finite) universe and *against* a purely-infinite-open one – a purely open space would give a continuum without sharp peaks. This is the strongest structural statement that can be drawn from the peaks.

Honest caveat: This conclusion is FFGFT-internal. The peaks do *not* force spatial compactness – ΛCDM produces the same discrete peaks from a *temporal* bound (the sound horizon at recombination) in a spatially infinite universe. The peaks alone do not distinguish spatial compactness (FFGFT) from a temporal bound (ΛCDM); this is the cosmological degeneracy (Doc. 267).

10.7 What the peak ratios state model-free

This section considers the measured peak positions purely with the methods of spectral / frequency analysis – without any T^4 assumption. The question: what can be read from the bare numbers $\ell \approx 220, 537, 810, 1120, 1450$, and where do typical misinterpretations from differencing and sampling limits lurk?

10.7.0 Clarification: "acoustic" is misleading

The CMB peaks are historically called "acoustic peaks". The term sends anyone who hears it for the first time in the wrong direction, because there is *nothing* to do with sound:

- **No sound:** there is no audible signal, no pressure wave in air.
- **No oscillation in time:** the CMB is a frozen still image, not a time signal. Nothing is measured in hertz.
- **No direct wave measurement:** the telescope measures the temperature as a function of *sky direction* – a map of hot and cold spots. What is evaluated is the *spatial distribution on the sky*, not an oscillation.

The word "acoustic" comes from the ΛCDM *explanatory model*: there the spots are interpreted as frozen pressure waves in the photon-baryon plasma *before* recombination. That is a model-dependent interpretation of the *mechanism* – it already sits in the name, before anything is measured. The *measured quantity* itself is neutral: a spatial interference pattern on the sky, described by the angular power spectrum C_ℓ.

Two different "frequencies" – do not confuse them. At the CMB two completely different things carry the name "frequency":

	(A) Photon frequency	**(B) Spot pattern**
What	colour/energy of the light	angular structure of the spots
Value	~ 160 GHz	$\ell = 220, 537, \ldots$
Unit	hertz (temporal)	dimensionless / angle (spatial)
measured as	radiometer (photon energy)	angle on the sky

The peak ratios $1 : 2.44 : 3.68 : \ldots$ belong to **(B)**, the *spatial* angular pattern. They have nothing to do with (A), the photon frequency. Converting the multipoles into a hertz frequency would be artificial and misleading – it would translate a spatial pattern into a fictitious temporal oscillation that does not exist.

Convention in this document: we avoid "acoustic" where possible and write *angular-spectrum peaks* or simply *CMB peaks*. The resonator analogy (open/closed pipe, double resonator) remains a purely *mathematical* analogy for a discrete eigenmode spectrum – not a claim that there was sound. Whether the resonances arise from pressure waves (ΛCDM) or from geometric lattice modes (FFGFT) is exactly the open question that the name "acoustic" prematurely decides in favour of ΛCDM.

10.7.1 First and second difference: the anharmonicity is real

The spacings of successive peaks are *not* constant:

$$\Delta\ell = 317,\ 273,\ 310,\ 330.$$

A pure harmonic series (integer overtones $1 : 2 : 3 : 4 : 5$) would have constant spacings. The ratios $1 : 2.44 : 3.68 : 5.09 : 6.59$ lie throughout *above* the harmonic series $1 : 2 : 3 : 4 : 5$ ($2.44 > 2$, $3.68 > 3$, $5.09 > 4$, $6.59 > 5$), and the spacings do not decrease monotonically. This departure from constant spacings is much larger than the measurement uncertainty of the peak positions (~ 1–$3\,\%$ for broad peaks). It is therefore *not* a differencing artefact but a real feature of the spectrum.

Conclusion 1: integer overtones (a simple harmonic fundamental) are excluded.

10.7.2 The trap: "missing fundamental" when reading linearly

From the practice of frequency analysis it is known: if one measures only overtones and the fundamental is missing, the spacings suggest a fundamental that is not actually excited. Reading the CMB peaks linearly ($\ell_n = A + Bn$), the fit gives:

$$A = -72, \qquad B = 297.$$

The offset A is **negative**. The linear continuation thus demands a "fundamental" at $\ell < 0$ – physically meaningless. This is exactly the missing-fundamental artefact: whoever reads the peaks as "overtones of a shifted fundamental" constructs a fundamental mode that does not exist.

Conclusion 2: the linear reading (integer overtones plus offset) is a differencing artefact and is ruled out.

10.7.3 The fundamental in the lattice sense is reconstructible

If one instead checks whether the peaks are $\sqrt{\text{integer}}$ multiples of a common fundamental, i.e. $\ell = c \cdot \sqrt{|n|^2}$ with $|n|^2 = 3, 18, 42, 78$, then $c = \ell/\sqrt{|n|^2}$ is nearly constant:

$$c = 127.0;\ 126.6;\ 125.0;\ 126.8 \quad (\text{scatter } 1.6\,\%).$$

There thus exists a common **lattice fundamental** $c \approx 126$. This is the most parsimonious reading of the anharmonicity: *one* parameter (the scale c) suffices, since the anharmonicity itself sits in the fixed lattice numbers $3, 18, 42, 78$.

	harmonic	**lattice ($\sqrt{\lvert n\rvert^2}$)**
Fundamental definition	$\ell_n = f_0 \cdot n$	$\ell = c \cdot \sqrt{\lvert n\rvert^2}$
Constancy test	$\ell/n = 220 \ldots 290$	$\ell/\sqrt{\lvert n\rvert^2} \approx 126$
Parameters	2 (offset + slope)	1 (scale c only)
Offset	$A = -72$ (unphysical)	none needed
Result	fails	$c \approx 126$

Conclusion 3: the fundamental is reconstructible – but only in the lattice sense ($\ell = c\sqrt{|n|^2}$), not the harmonic one. The value $c \approx 126$ corresponds to the mode $|n|^2 = 1$.

10.7.4 The reconstructed fundamental is not the first peak

The lattice fundamental $c \approx 126$ corresponds to $\ell \approx 126$ for $|n|^2 = 1$. The first *visible* peak, however, lies at $|n|^2 = 3$, i.e. $\ell \approx 219$. There is thus a **real but peak-invisible** fundamental mode:

$$|n|^2 = 1 \;\to\; \ell \approx 126 \quad \text{(fundamental, not a peak)},$$

$$|n|^2 = 3 \;\to\; \ell \approx 219 \quad \text{(first visible peak)}.$$

Remarkably, $\ell \approx 126$ lies *in the middle of the observed range*, but there is no peak there, rather the rising Sachs-Wolfe plateau. This is consistent with $|n|^2 = 1$ not being a local maximum of the spectrum $I(r)$ (Step 11.1); the first local maximum lies at $|n|^2 = 3$.

Conclusion 4: there is a real, lower-lying fundamental mode at $\ell \approx 126$ that acts as a scale-setter but does not appear as a peak. This is a weakly falsifiable consistency condition: the fundamental c reconstructed from the upper peaks must match the place where the spectrum passes from the plateau into the first peak.

10.7.5 No size of the universe from the oscillations

No *absolute size* can be read from the angular oscillations – neither the size of the structures nor that of the universe. The measured angle of a peak is

$$\theta = \frac{L_{\text{structure}}}{D_A},$$

i.e. only the *ratio* of physical structure size $L_{\text{structure}}$ and distance D_A (angular distance to the source surface). The same angle fits "large structure, far away" just as well as "small structure, nearby":

assumed D_A	$\to L_{\text{structure}}$ **(at** $\theta \approx 0.8°$**)**
$14\,000$ Mpc	≈ 200 Mpc
100 Mpc	≈ 1.4 Mpc
1 Mpc	≈ 0.01 Mpc

To turn the angle into an absolute length one needs D_A – and D_A is not a measured quantity but a **model output**:

- ΛCDM: $D_A = \int c/H(z)\,dz$ – requires H_0, Ω_m, Ω_Λ, the recombination z_*.
- FFGFT: $D_A = R_H \ln(1 + z_*)$ – requires R_H, the external parameter (R20), which does not follow from ξ alone.

In both cases the distance comes from the model, not from the measurement. The angular measurement is **scaleless**: it shows the *form* of the pattern (the ratios), but not its absolute size – like a photograph without a scale bar shows the form of an object, not its size. This is exactly the same limit as for the peak ratios: the *form* follows from the geometry, the *absolute scale* needs an external parameter (R20).

Why the resonator method does not help here. For a real resonator in front of us – an organ pipe, say – the size does sit in the fundamental: measuring the fundamental wavelength λ_0 *in metres* and knowing the medium (sound speed c_{medium}) gives the length directly, $L = \lambda_0/2$. This method needs two things, both of which are absent at the CMB:

- **A wavelength in metres** – not just an angle. The organ pipe is in front of us; we measure its length directly. The CMB "resonator" is unreachably far away; we see only its angle on the sky, not its length. The step from angle to length needs D_A – the unknown model output.
- **A known medium.** For the organ pipe c_{medium} is a known constant. In the ΛCDM plasma picture the "sound speed" $c_s(z) = c/\sqrt{3(1+R(z))}$ is not a constant but a *function computed from the model*: it requires the baryon and photon densities as functions of time, the expansion history and the recombination state. Here the "medium" is not a substance one has in front of one and whose c one measures – it arises only *indirectly* from the assumed curvature and expansion. "Denser in the past" is already a statement about the geometry, not about an independently existing medium.

Thus in the ΛCDM picture the back-calculation is triply model-dependent: the "wavelength" is only an angle, the "medium" $c_s(z)$ is computed from density, expansion and metric, and the distance D_A is likewise a model output. None of the three ingredients is a model-free measurement; the "medium" is even the most derived quantity. FFGFT in turn needs *no* medium and *no* sound speed – the peaks are eigenmodes of a geometry, not sound in a substance – but does need R_H as an external parameter for the absolute scale (R20). Not assumption-free, but a different *kind* of assumption: a geometric scale instead of a whole plasma and expansion history.

10.7.6 What is certain and what is not – the sampling limit

Certainly readable (model-free):

- No integer overtones – the anharmonicity is real, not an artefact; the ratios lie above the harmonic series.
- A discrete mode lattice with $\sqrt{|n|^2}$ scaling is the most parsimonious reading (one parameter, $c \approx 126$).
- There is no harmonic fundamental "below" the first peak; the linear fit there is a missing-fundamental artefact.
- No *absolute size* (of the universe or the structures) can be read – the angle gives only L/D_A, and D_A is a model output.

Not readable (limit of the method):

- Whether the anharmonicity stems from a $\sqrt{\text{integer lattice}}$ (FFGFT) or from acoustic ΛCDM dispersion – both produce the same form (cosmological degeneracy, Doc. 267).
- Fine differences between anharmonic laws: the CMB peaks are *broad* (half-width $\Delta\ell \sim 100$–200), the positions determinable only to ~ 1–$3\,\%$. This suffices to *exclude* integer overtones, not to distinguish the anharmonic laws.

Conclusion of this section: pure frequency analysis – without any T^4 assumption – yields three robust statements: (i) no harmonic overtone series, (ii) the anharmonicity is real (ratios above the harmonic series), (iii) a lattice fundamental $c \approx 126$ exists and corresponds to a real but invisible $|n|^2{=}1$ mode. These statements are obtained independently of the FFGFT derivation and support it from the measurement side, without removing the cosmological degeneracy (FFGFT vs. ΛCDM).

E.13 Step 11: No fit – what is derived and what remains open

The agreement of the peak ratios to $< 2\%$ does *not* come from a fit. This section shows explicitly what comes from where.

11.1 What follows forward – and what does not

Reading of the sequence $\{1, 6, 14, 26\}$

The sequence $\{1, 6, 14, 26\}$ does *not* follow from ξ as local maxima; $I(r)$ does not contain ξ at all. The genuine local maxima of the full thermally weighted T^4 spectrum lie at $|n|^2 = 3, 6, 10, 14, 18, 22, 26, 30, \ldots$ (dense). Of these, $6, 14, 26$ are local maxima, but 1 is *not*, and 10 (a local maximum) is missing from the sequence. So $\{1, 6, 14, 26\}$ is *not* a natural forward subset – it was read off the data (Step 4). What follows forward from T^4 is cleanly separated in Step 17.

Quantity	Origin
T^4 spectrum maxima at $\lvert n\rvert^2 = 3, 6, 10, 14, 18, 22, 26, 30, \ldots$	Local maxima of $I(r) = g(r) \cdot N(r) \cdot r$; g from the Jacobi formula (exact), N from Bose-Einstein. Input: none – $I(r)$ does not contain ξ, the maxima are ξ-independent. *Forward, but dense.*
Harmonic backbone $1 : 2 : 3 : 4 : 5$	Symmetric resonance (n, n, n): $k^2_{\text{obs}} = 3n^2$. *Forward*, but harmonic – "too clean".
Anharmonicity (stretch)	Codimension-1 projection $D_f - 1 \approx 2 \Rightarrow J_0$ Bessel. *Forward, parameter-free* (Step 17).
Selection of exactly $\{1, 6, 14, 26\}$	*Read off the CMB data* (Step 4), not forward.

Forward: the dense spectrum, the harmonic backbone and the geometric anharmonicity. **Backward:** the concrete selection of the four visible peaks.

11.2 What is not yet rigorously derived

Assumption	Status	Consequence
$L_{\text{tor}} = 1/(k_B T_{\text{CMB}})$	Physically motivated, not rigorously proved	Affects only the absolute scale, *not* the ratios
$k^2_{\text{obs}} = (n_1^2 + n_2^2 + n_3^2)/L^2$	Observer in 3D + time-unfolding, n_4 = time-unfolding	Does not produce the factor 3 (Step 8); the factor is set
D_A (angular distance)	External parameter R_H needed (R20)	Affects only $\ell_1 \approx 220$, not the ratios

11.3 What the ratios prove and what they do not

The ratios $1 : 2.449 : 3.742 : 5.099$ agree with the CMB to $< 2\%$; the selection $\{1, 6, 14, 26\}$ is read off the data and the factor 3 is set. This shows:

- The series $\{3, 18, 42, 78\}$ chosen in this way falls on genuine local maxima of the T^4 spectrum (Step 5).
- The deviation of $< 2\%$ lies within the measurement uncertainty of the raw CMB peak positions (without a ΛCDM template fit).

It does *not* prove:

- That the factor 3 and the selection follow from the geometry: the factor 3 cannot arise from the projection (Step 8), and the naive spherical C_ℓ projection does not reproduce the series (Step 17.1; R29, R30 open).
- That no other theory can produce the same ratios.
- That FFGFT correctly describes the entire CMB (amplitudes, polarisation, early-phase physics open).

The status is: a retrodiction – the series read off the data falls, with the factor 3 set, on genuine T^4 maxima and matches the observed ratios to $< 2\%$. The forward content lies in the anharmonicity (Step 17).

E.14 Step 12: Schwarzschild connection and R_H

12.1 The two Schwarzschild radii in FFGFT

FFGFT has two scales that are simultaneously Schwarzschild radii (Doc. 190, R4; Doc. 182):

Scale	Definition	Schwarzschild mass	Check
$L_0 = \xi \cdot \ell_P$	sub-Planck fundamental length	$M_0 = \xi \cdot m_P/2$	$r_S(M_0) = L_0$
$R_H = \hbar c/E_H$	Hubble horizon	$M_U = R_H c^2/(2G)$	$r_S(M_U) = R_H$

Both checks numerically:

$$r_S(M_0) = \frac{2G \cdot \xi \cdot m_P/2}{c^2} = \xi \cdot \frac{G m_P}{c^2} = \xi \cdot \ell_P = L_0 \quad \checkmark$$

$$r_S(M_U) = \frac{2G \cdot M_U}{c^2} = \frac{2G}{c^2} \cdot \frac{R_H c^2}{2G} = R_H \quad \checkmark$$

12.2 The conversion: R_H from the two Schwarzschild conditions

Dividing the two Schwarzschild formulas:

$$\frac{R_H}{L_0} = \frac{2G M_U/c^2}{2G M_0/c^2} = \frac{M_U}{M_0}$$

Thus:

$$\boxed{R_H = L_0 \cdot \frac{M_U}{M_0} = \xi \cdot \ell_P \cdot \frac{M_U}{\xi \cdot m_P/2} = \frac{2\,\ell_P \cdot M_U}{m_P}}$$

This is a pure conversion – no model, no assumption. It shows: R_H follows directly from L_0 and the mass ratio M_U/M_0.

12.3 What this means

$L_0 = \xi \cdot \ell_P$ is fully derived from ξ. $M_0 = \xi \cdot m_P/2$ likewise.

R_H is thus fully determined *once* M_U is known:

$$R_H = \frac{2\,\ell_P}{m_P} \cdot M_U$$

In Doc. 182 M_U is obtained via $E_H = E_0 \cdot \xi^{41/4}$. The Schwarzschild conversion shows the same R_H from another side – as a direct consequence of the mass scale of the universe.

12.4 Open question: M_U independently from ξ?

The connection $R_H = 2\ell_P M_U/m_P$ is exact. Open is: can M_U be derived from the T^4 mode structure *without* presupposing R_H?

If so, then R_H would be a genuine prediction from ξ alone – without needing the exponent $41/4$ as input. This is an open derivation, no fundamental obstacle.

E.15 Step 13: The measurement problem – what FFGFT can do, what ΛCDM cannot

13.1 The exponent 41/4 is equivalent to H_0

From $E_H = E_0 \cdot \xi^{41/4}$ and $H_0 = E_H/\hbar$ it follows:

$$\frac{41}{4} = \frac{\ln(E_0/(\hbar H_0))}{\ln(1/\xi)}$$

The exponent $41/4$ is thus **equivalent to measuring** H_0. Numerically:

H_0 [km/s/Mpc]	Source	Exponent
66.19	FFGFT (from $41/4$)	10.2492
67.40	Planck/CMB (ΛCDM)	10.2472
73.00	SH0ES (distance ladder)	10.2382

The difference between the exponents is smaller than 0.02 – all H_0 measurements give the same exponent to $\sim 0.1\%$.

13.2 The measurement problem

There is no model-free measurement of H_0. Every measurement presupposes model assumptions:

Method	Model assumption	Result
CMB peaks (ΛCDM)	expansion, FLRW metric, recombination	67.4 km/s/Mpc
Distance ladder (SH0ES)	calibrated standard candles, local geometry	73.0 km/s/Mpc
FFGFT ($\xi^{41/4}$)	static universe, T^4 topology, exponent $41/4$ set (not derived)	66.2 km/s/Mpc

In FFGFT H_0 means something different than in ΛCDM: not the expansion rate, but the reciprocal of the fractal path-lengthening scale $R_H = c/H_0$. What ΛCDM measures as expansion, FFGFT interprets as fractal path-lengthening $z(d)$ (Doc. 190, R14a).

13.3 What FFGFT can do, what ΛCDM cannot

Quantity	ΛCDM	FFGFT
H_0	Free parameter (determined from data)	Derived from $\xi^{41/4}$ (if $41/4$ derived)
CMB peak ratios	From $r_s(z^*)$ and $D_A(z^*)$ (needs H_0, Ω_b, ...)	Retrodiction: T^4 maxima, factor 3 set, selection from data
Absolute peak position ℓ_1	From H_0, z^*, r_s	From R_H (from $\xi^{41/4}$) + open connection R20

The structural difference: ΛCDM needs H_0 as input – measured, not derived. FFGFT writes $H_0 = 66.2$ km/s/Mpc as $\xi^{41/4}$. If $41/4$ follows from the T^4 geometry (still open), H_0 is a *prediction* – not a free parameter.

The remaining open problem: The exponent $41/4$ is not derived from the T^4 geometry in any document. It is the only place in FFGFT cosmology where an external input is needed – equivalent to H_0. As long as this derivation is missing, FFGFT and ΛCDM stand on equal footing in this respect: both need a cosmological scale as input.

E.16 Step 14: The universal Compton-Schwarzschild relation

14.1 Every length has two associated masses

To every length R belong two masses:

$$M_c(R) = \frac{\hbar}{R\,c} \quad (R \text{ is its Compton wavelength})$$

$$M_s(R) = \frac{R\,c^2}{2G} \quad (R \text{ is its Schwarzschild radius})$$

Their product is *independent of* R:

$$\boxed{M_c(R) \cdot M_s(R) = \frac{\hbar}{Rc} \cdot \frac{Rc^2}{2G} = \frac{\hbar c}{2G} = \frac{m_P^2}{2}}$$

14.2 Application to the electron

Equivalently in lengths: to every mass m belong the Compton wavelength $\lambda = \hbar/(mc)$ and the Schwarzschild radius $r_S = 2Gm/c^2$, with

$$\lambda \cdot r_S = \frac{\hbar}{mc} \cdot \frac{2Gm}{c^2} = \frac{2G\hbar}{c^3} = 2\,\ell_P^2$$

Thus:

$$\sqrt{\lambda_e \cdot r_S(e)} = \sqrt{2}\,\ell_P$$

The two scales of the electron lie *geometrically symmetric* around the Planck length – independent of the electron mass.

Scale	Value	Meaning
$r_S(e) = 2Gm_e/c^2$	1.35×10^{-57} m	Schwarzschild (bottom)
ℓ_P	1.62×10^{-35} m	Planck (middle)
$\lambda_e = \hbar/(m_e c)$	3.86×10^{-13} m	Compton (top)

14.3 Application to R_H

The same relation holds for the Hubble radius:

$$M_c(R_H) = \frac{\hbar}{R_H c} = \frac{E_H}{c^2} \quad \text{(tiny)}$$

$$M_s(R_H) = \frac{R_H c^2}{2G} = M_U \quad \text{(mass of the universe)}$$

with

$$M_c(R_H) \cdot M_s(R_H) = \frac{E_H}{c^2} \cdot M_U = \frac{m_P^2}{2} \quad \checkmark$$

14.4 What this relation achieves and what it does not

It achieves: an elegant internal consistency. The Schwarzschild mass of the universe M_U and the cosmological energy E_H are connected by $M_U \cdot E_H/c^2 = m_P^2/2$ – exact, without free parameter. The two Schwarzschild scales L_0 (bottom) and R_H (top) are not arbitrary bounds but Schwarzschild radii of real masses ($M_0 = \xi m_P/2$ and M_U).

It does not achieve: a determination of R_H. The relation $M_c \cdot M_s = m_P^2/2$ is a universal identity – it holds for *every* length and therefore gives no new information about the

numerical value of R_H. Both masses $M_c(R_H)$ and $M_s(R_H)$ are defined via R_H; their product is tautologically $m_P^2/2$.

To fix R_H one needs an *independent* input: the cosmological energy $E_H = E_0 \cdot \xi^{41/4}$ (Doc. 182). The electron fixes the Planck scale via $\sqrt{\lambda_e r_S(e)} = \sqrt{2}\ell_P$, but not R_H.

The remaining open problem stays: The exponent $41/4$ – equivalent to the cosmological energy scale E_H – is the only input that FFGFT needs for the absolute cosmological scale. The Compton-Schwarzschild relation explains the *structure* of the scale hierarchy, but not the *value* of E_H.

E.17 Step 15: The T⁴ length is a Schwarzschild radius

15.1 The fundamental identity (Doc. 010, 066, 067)

The FFGFT documents define the T0 length of an object as:

$$r_0 = 2Gm \quad \text{(natural units, } c = 1\text{)}$$

Doc. 067 explicitly states: the factor 2 results from the relativistic field structure and *coincides exactly with the Schwarzschild radius*. Doc. 010 writes $r_0 = 2GE$ (energy instead of mass), Doc. 066 as the ratio $\xi = r_0/\ell_P = 2Gm/\ell_P$.

This means: every characteristic T0 length *is* a Schwarzschild radius – not analogous, but identical.

15.2 Why different formulas in the documents?

It is the same formula in different unit systems:

Document	Formula	Unit system
SI	$r_S = 2Gm/c^2$	metres, kg, seconds
Doc. 015, 029	$r_S = 2M$	natural ($G = c = 1$)
Doc. 010	$r_0 = 2GE$	$c = 1$, energy instead of mass
Doc. 066	$r_0 = \xi \cdot \ell_P$	Planck ($\ell_P = 1$)

For the minimal mass $M_0 = (\xi/2)\, m_P$ *all* give the same value:

$$\frac{2GM_0}{c^2} = \frac{2G}{c^2} \cdot \frac{\xi}{2}\, m_P = \xi \cdot \frac{Gm_P}{c^2} = \xi \cdot \ell_P = L_0 = 2.155 \times 10^{-39}\ \text{m}$$

The apparent variety is only the SI conversion, which shows or hides factors c^2 and G. As soon as one computes in SI units, c^2 and G appear explicitly; in natural units they vanish. The physics is identical in all cases.

15.3 “a Schwarzschild radius” vs. “the Schwarzschild radius”

In standard physics one speaks of *the* Schwarzschild radius of an object – the one length at which this mass becomes a black hole.

In FFGFT the T0 length is *always* a Schwarzschild radius – but *which one* depends on the mass:

Mass	Schwarzschild radius	Role
Electron m_e	1.35×10^{-57} m	particle scale
$M_0 = (\xi/2)m_P$	$L_0 = 2.16 \times 10^{-39}$ m	fundamental length
Proton	2.48×10^{-54} m	particle scale
M_U (universe)	$R_H = 1.41 \times 10^{26}$ m	Hubble horizon

Therefore: L_0 *is a* Schwarzschild radius (namely that of M_0) – but *not the* Schwarzschild radius as such, because there is no single universal one. Likewise R_H is *a* Schwarzschild radius (that of M_U), not *the* one.

This is the scale relativity from Doc. 182: every system carries its own torus with its own characteristic scale $R_{\text{torus}}(m) = \hbar/(mc)$. The universal lower bound L_0 holds for all; a universal upper bound does not exist – R_H is only the upper bound of the cosmological sector.

15.4 What this means for R_H

R_H *coincides exactly* with the Schwarzschild radius of M_U – this is correctly stated in the documents and is not an approximation. But: because $r_0 = 2Gm$ is the *definition* of the T0 length, the agreement is necessary, not surprising. The T0 length of a mass and its Schwarzschild radius are the same object.

The identity does not fix R_H independently: one still needs M_U (resp. $E_H = E_0 \cdot \xi^{41/4}$). What the identity achieves is the *physical interpretation*: R_H is not an abstract horizon but the Schwarzschild radius of the total mass of the static universe – the universe sits at its own Schwarzschild bound.

E.18 Step 16: Why the Schwarzschild relation does not derive R_H

16.1 The tempting but false conclusion

From two true formulas a fallacy arose:

$$\text{(A)} \quad \xi = L_0/\ell_P, \quad L_0 = \xi \cdot \ell_P \tag{E.1}$$

$$\text{(B)} \quad r_S(m) \cdot r_C(m) = 2\,\ell_P^2 \quad \text{(universal)} \tag{E.2}$$

Both are correct. The error arose through a tacit assumption: that in relation (B) one may replace the electron mass m_e by the universe mass M_U *while keeping* λ_e, thus obtaining $R_H = r_S(M_U)$.

16.2 The exact error

Relation (B) connects two quantities of the *same* mass. For the electron:

$$\underbrace{r_S(m_e)}_{1.35\times10^{-57}\,\mathrm{m}} \cdot \underbrace{\lambda_e}_{3.86\times10^{-13}\,\mathrm{m}} = 2\,\ell_P^2$$

Both factors belong to m_e.

Replacing $m_e \to M_U$ replaces *both* factors:

$$\underbrace{r_S(M_U)}_{R_H = 1.41\times10^{26}\,\mathrm{m}} \cdot \underbrace{r_C(M_U)}_{3.7\times10^{-96}\,\mathrm{m}} = 2\,\ell_P^2$$

The Compton wavelength of M_U is $3.7 \times 10^{-96}\,\mathrm{m}$ – **not** λ_e. The conflation was: R_H (Schwarzschild of the *universe*) was put together with λ_e (Compton of the *electron*) into a relation that holds only for a single mass. In fact:

$$r_S(M_U) \cdot \lambda_e = 5.4 \times 10^{13}\,\mathrm{m}^2 \neq 2\,\ell_P^2 = 5.2 \times 10^{-70}\,\mathrm{m}^2$$

– off by a factor of 10^{83}.

16.3 Two separate systems

Electron and universe are two different systems, each consistent in itself:

$$\text{Electron:} \quad m_e \to \{r_S(e),\, \lambda_e\}, \quad r_S(e) \cdot \lambda_e = 2\ell_P^2$$

$$\text{Universe:} \quad M_U \to \{R_H,\, r_C(M_U)\}, \quad R_H \cdot r_C(M_U) = 2\ell_P^2$$

Between the systems there is *no* bridge from this relation.

16.4 The Schwarzschild formula generates no information

The deeper reason: $r_S = 2GM/c^2$ is a *conversion* between mass and length. As for any object:

Known	**Follows**
Mass M	Schwarzschild radius $r_S = 2GM/c^2$
Length r_S	Mass $M = r_S c^2/(2G)$
neither	**nothing**

This holds for a black hole, the Sun, the electron and the universe alike. To know R_H, one of the four equivalent quantities must be independently known:

$$M_U \longleftrightarrow H_0 \longleftrightarrow E_H \longleftrightarrow R_H$$

They form a ring of conversions – but no entry point.

16.5 Where FFGFT supplies the input

FFGFT supplies exactly one cosmological input:

$$E_H = E_0 \cdot \xi^{41/4}, \qquad E_0 = \sqrt{m_e m_\mu}$$

From this follows $R_H = \hbar c / E_H$ and everything else. The exponent $41/4$ is equivalent to specifying H_0 and is not derived from the T^4 geometry in any document (open problem).

Conclusion: The Schwarzschild identity (L_0, R_H are Schwarzschild radii) is a genuine structural property and supplies the *physical interpretation*. But it cannot *generate* the cosmological scale – for that the one input E_H is needed. The earlier impression that the two formulas (A) and (B) would derive R_H rested on the tacit substitution $m_e \to M_U$, which conflates two different masses.

E.19 Step 17: Forward test – anharmonicity from the codimension-1 projection

This step cleanly separates what follows *forward* (without looking at the data) from T^4 and what is a *backward* consistency check. It documents an explicit forward investigation and is the conceptual advance over Steps 1–11. All numbers here are reproducible with the accompanying scripts (`python/Dok267_268_Skripte/`).

17.1 What the naive forward computation yields – and what it does not

Three forward computations, all without any look at the data:

1. **Symmetric T^4, spherical C_ℓ Bessel projection** ($C_\ell = \sum_n N_{\mathrm{BE}}\, |j_\ell(k_{\mathrm{obs}} D_A)|^2$): does *not* reproduce $\{3, 18, 42, 78\}$. The result is a quasi-continuum dominated by the lowest modes (strongest peak at $\ell \approx 122$). The naive Bessel projection is *insufficient*.
2. **3+time (T^3 spatial spectrum):** local maxima at $|n|^2 = 2, 5, 9, 14, 17, 21, 26, \ldots$ – dense, and *not* $\{1, 6, 14, 26\}$.
3. **Symmetric spatial resonance** (n, n, n)**:** $k^2_{\mathrm{obs}} = 3n^2 \Rightarrow$ series $\{3, 12, 27, 48, 75\}$, ratios $\sqrt{3} : \sqrt{12} : \sqrt{27} : \ldots = 1 : 2 : 3 : 4 : 5$ (*harmonic*).

Conclusion: forward, T^4 produces either a dense spectrum or the harmonic backbone $1 : 2 : 3 : 4 : 5$. The sparse, anharmonic series $\{3, 18, 42, 78\}$ is *not* a forward output; it arises only from the data-read selection $\{1, 6, 14, 26\}$.

Concretely open ($|n|^2 = 30$, sharpened by O. Teker): $30 = 3 \cdot 10$, and $|n|^2 = 10$ is a genuine local maximum ($I = 20.1$). So 30 satisfies the factor-3 filter. Thermally it is even *stronger* than two of the visible peaks: $I(30) = 13.2 > I(42) = 7.6 > I(78) = 1.7$. Yet the CMB shows no peak at $|n|^2 = 30$ ($\ell \approx 696$, in the trough between peaks 2 and 3). A selection by the strongest peak in range (Section 10.3) therefore does not hold. Moreover, the factor 3 is itself set (Step 8); the peak selection remains open as a whole (P29, P30).

17.2 The actual forward finding: anharmonicity is geometric

The harmonic backbone $1 : 2 : 3 : 4 : 5$ is "too clean" – like the ideal overtones of a stiffness-free string. Real strings are *inharmonic* due to their stiffness (stretched piano

tuning). By analogy we *expect* anharmonicity, because spacetime is curved from our viewpoint and the projection is nonlinear (fractal path lengthening).

Acoustic resonators show that anharmonicity is purely *geometric* (forward, no fit, no medium):

Resonator (dimension D)	**Ratios**	**Dev. to CMB**
Tube/sphere j_0 ($D=3$, $\nu=\frac{1}{2}$)	$1:2:3:4$ (harmonic)	-18 to -24%
Circular membrane J_0 ($D=2$, $\nu=0$)	**1 : 2.30 : 3.60 : 4.90**	**-2 to -6%**
S^3 Laplacian	$1:1.63:2.24$	-33 to -48%
S^2 ($Y_{\ell m}$)	$1:1.73:2.45$	too dense
CMB observed	$1:2.44:3.68:5.09$	—

Key fact: the radial Bessel index is $\nu = D/2 - 1$. Anharmonicity is *maximal at* $D = 2$ ($\nu = 0$, J_0) and vanishes at $D = 1$ and $D = 3$ (both harmonic). The CMB series sits exactly at $D \approx 2$ – the dimension of maximal anharmonicity.

Important: the match is to the *flat* circular membrane (J_0), *not* to the curved 2-sphere ($Y_{\ell m}$, which gives $1:1.73:\ldots$, wrong). The correct expectation is a *flat bounded disk*, not a sphere – because the CMB peaks come from the *radial* projection, not from the eigenvalues of the observation surface.

17.3 Derivation: flat disk from the fractal codimension-1

The last-scattering surface is a **codimension-1 hypersurface** in the fractal space:

$$D_{\text{eff}} = D_f - 1 = (3 - \xi) - 1 = 2 - \xi \approx 2.$$

Radial modes in D dimensions are zeros of $J_{D/2-1}$. For $D = 2 - \xi$:

$$\nu = \frac{D}{2} - 1 = \frac{2-\xi}{2} - 1 = -\frac{\xi}{2} \approx -6.7 \times 10^{-5} \approx 0 \;\Rightarrow\; J_0 \text{ membrane.}$$

This gives, forward, **parameter-free and without mode selection**, the anharmonic series $1:2.30:3.60:4.90$ – a match to $\sim 5\%$. The anharmonicity follows from the codimension-1 projection in the fractal geometry, exactly as expected from the curvature.

17.4 The small residual – honestly open

The best fit lies at $\nu \approx -0.07$, i.e. effective dimension $D \approx 1.86$ – just *below* 2. The obvious physical source would be the *spectral dimension* d_s (governing wave propagation on a fractal), which is generally below the Hausdorff dimension: $d_s = 2D_f/d_w$ with walk dimension $d_w > 2$.

The explicit computation rules this out, however:

- The FFGFT fractal is *too weakly fractal*: with $D_f = 3 - \xi$, $d_w = 2 + O(\xi)$, so $d_s(\text{surface}) \approx 2.00$. ξ is $\sim 1000\times$ too small to push D down to 1.86.

- For $d_s = 1.86$ one would need $d_w \approx 2.15$ (strong fractality), incompatible with $D_f = 2 - \xi$.

The residual is not derivable from ξ. Moreover it is of the order of the measurement uncertainty: the pure J_0 prediction ($D = 2$) is consistent with peaks 3 and 4 within $\sim 2\sigma$; only peak 2 sits at $\sim 4\sigma$. The "best fit 1.86" is thus largely measurement noise; pointing to an exact value (e.g. $13/7$) would be numerology unsupported by the data quality.

17.5 Summary – layered, honest structure

Layer	Status
$D = 3 \to 2$ (codimension-1, J_0)	**forward, parameter-free, derived**; $\sim 5\%$
Anharmonicity geometric	derived (maximal at $D = 2$)
Ratio match $\{3, 18, 42, 78\}$	retrodiction (from data), *not* prediction
$D = 2 \to 1.86$ (sub-2)	not from ξ; largely measurement noise; open
Exclusion of $\lvert n\rvert^2 = 30$	open (peak selection, P29/P30)
Peak-2 residual ($\sim 4\sigma$)	small genuine open point, likely other physics

Defensible statement (instead of "fit to $D = 1.86$"): the fractal codimension-1 projection produces, forward and parameter-free, the J_0 membrane spectrum ($D \approx 2$) and matches the CMB anharmonicity to $\sim 5\%$ – within the measurement errors. The harmonic backbone comes from the symmetric T^4 resonance; the stretch from the geometry of observation. This formulation is conceptually cleaner than the mode selection of Steps 4–11 because it requires *no* selection from the data, thereby removing the circularity objection (O. Teker). The rigorous selection mechanism for the discrete individual peaks remains open.

Chapter F

Doc. 269 — FFGFT and RA/PMT: a bidirectional structural comparison

Purpose

This document compares FFGFT (Fundamental Fractal Geometric Field Theory, T^4 time-mass duality) with the RA/PMT scheme of J. T. Guevara (*Regional Arenas / Perception Mechanism Theory*, 2026, three parts: consciousness, fundamental physics, cosmology).

The comparison uses the same explicit comparison operator as Doc. 265 – but applied here *in both directions*:

- **Direction A:** FFGFT evaluates RA/PMT using the six-axis operator (Doc. 265).
- **Direction B:** the RA/PMT grid itself (RA4 → RA1.5 → RA2 → RA1) is applied to FFGFT – just as Guevara applies it to QFT, GR and emergent spacetime.

Applying it both ways is exactly the point of an exposed operator (Doc. 265): each side may render the other with its own tool, and both renderings remain examinable side by side.

Up front – the most important caveat: FFGFT and RA/PMT are *not on the same level*. FFGFT is a physical theory with one free parameter ($\xi = 1/7500$) from which concrete numbers follow (CMB peak ratios, H_0, mass relations). RA/PMT is, in Guevara's own words, "*not a new physical theory*" – a meta-scheme for classifying theories. The comparison is therefore not between two competing models of the same system (as Doc. 265 was for FFGFT/Blume-Maleev), but between a theory and a theory-ordering scheme. This shapes every finding below.

F.1 The RA/PMT scheme in brief

RA/PMT decomposes every theory into five layers:

Layer	Function
RA4	Blueprint – the structured space of all *possible* configurations (prior to any realization).
RA3	Residual – the theory's undefined terms, explicitly declared as unresolved rather than hidden.
RA2	Mapping – the rules/equations that unfold a chosen blueprint into physical appearance.
RA1.5	Mechanism – the *selector* that picks one configuration from RA4. (HPM for consciousness, PSM for physics, CSM for cosmology – the same role, per Guevara.)
RA1	Expression – the physically realized.

RA/PMT's central thesis: QM, GR, emergent spacetime and cosmology all possess RA4 and RA2 but *no* RA1.5. The absence of this selector is claimed to be the common root of the measurement problem, the initial-conditions problem, fine-tuning, the arrow of time, and the cosmological constant. RA/PMT "completes" the theories by adding RA1.5 – without altering their equations.

F.2 Direction A: FFGFT evaluates RA/PMT (six-axis operator)

We apply the comparison operator from Doc. 265: two frameworks are compared along six axes (shared invariants, operators, mathematical structures, predictions, physical mechanisms, hardware implications). On each axis: shared, partial, or not.

Finding, Direction A: On four of six axes there is *no* contact, on two only contact at the meta-level (axes 2 and 3). By the classification of Doc. 265 this is not even Level 3 (complementary in the usual sense) – it is **category difference**: the two frameworks do not answer the same kind of question. The only genuine overlap is methodological: both separate cleanly what follows rigorously from what remains open. FFGFT does this through its three-layer methodology (cores proved from $\xi >$ bridges proved algebraically $>$ reductions as plausibility), RA/PMT through RA4/RA3/RA2/RA1.5/RA1. This discipline is congenial and related – but it is a *stance*, not shared physics.

F.3 Direction B: the RA/PMT grid applied to FFGFT

Guevara applies his grid to other theories himself (QFT, GR, emergent spacetime, IIT, ...). The logical consequence: one can apply it to FFGFT too. This is illuminating.

Table F.1: FFGFT ↔ RA/PMT along the six comparison axes. ✓ = shared, ∼ = only at the meta-level, × = not shared.

Comparison axis	Status	Note
1. Invariants	×	RA/PMT supplies no invariant; FFGFT: $\tilde{T} \cdot m = 1$, winding number.
2. Operators	∼	Structural only: RA/PMT's "selector" vs. FFGFT's recursion $\mathscr{R}$ (Doc. 203) – but $\mathscr{R}$ is defined, RA1.5 is not.
3. Mathematical structure	∼	Shared only as *layering discipline*: FFGFT's cores $>$ bridges $>$ reductions $\approx$ RA/PMT's RA4 $>$ RA2 $>$ RA1.
4. Predictions (quantitative)	×	RA/PMT produces not a single number; FFGFT: peak ratios, H_0, masses from ξ.
5. Physical mechanisms	×	RA1.5/CSM is pre-physical and unspecified; FFGFT mechanisms are geometrically concrete.
6. Hardware implications	×	RA/PMT makes no statement about devices/measurements.

Layer	FFGFT assignment
RA4	The T^4 mode space: winding numbers $\vec{n} \in \mathbb{Z}^4$, the possible modes.
RA3	Open points, already explicitly declared in FFGFT (P-register, Doc. 190): e.g. the exponent $41/4$ (R20), the rigorous C_ℓ projection (R30).
RA2	The derivations from ξ: Jacobi degeneracy, Bose-Einstein, 3D projection, recursion operator $\mathscr{R}$ (Doc. 203).
RA1.5	**Here RA/PMT asks: where is your selector?**
RA1	Observable: CMB peak ratios, H_0, mass relations.

The decisive point – FFGFT has no RA1.5, and needs none. Guevara's grid diagnoses a "missing selector" in every theory. For FFGFT the answer is not "the selector is missing" but *"the premise does not hold"*:

- RA1.5 presupposes that there is a space of *many possible* universes/configurations (RA4) from which *one* is selected. That is the multiverse-like assumption RA/PMT carries tacitly (as ΛCDM carries its initial conditions).
- FFGFT is **static**: no Big Bang, no $t = 0$, no initial state to be selected. The fractal T^4 torus is not *one selected* among many configurations – it is *the* structure. There is no blueprint space for a selector to choose from.
- Where Guevara's cosmology (Part III) reads the Big Bang, inflation and the low initial entropy as "the expression of a selected blueprint", FFGFT says: these events are not initial conditions to be selected, because there is no beginning (Doc. 267, inverted initial condition).

What Direction B exposes: The RA/PMT grid is not neutral. It carries a hidden assumption – that *every* theory needs a selector between possibility and actuality. FFGFT is precisely the counterexample: a theory that deliberately lacks this layer because it does not make the possibility/actuality split. The "deficiency" the grid diagnoses is, from the FFGFT side, not a deficiency but the absence of a superfluous assumption.

This is methodologically the most valuable finding of the whole comparison: it shows that RA1.5 itself – by Guevara's own RA3 definition – is a *residual*. "CSM selects the low-entropy configuration" renames the puzzle instead of solving it: there is no equation, no selection criterion, no prediction. A named selector without operationalization meets exactly the RA3 condition ("lacks a defined mechanism") that RA/PMT charges against other theories.

F.4 Synthesis: two opposite poles of the same question

Taken together, the two directions give a clear picture. FFGFT and RA/PMT are neither competitors nor variants – they are **opposite poles of the same basic question**: how does the observed reality come about?

	RA/PMT	**FFGFT**
Basic assumption	many possible universes (RA4), one is selected	one static fractal T^4, no selection
Selector RA1.5	necessary but undefined (CSM/PSM/HPM)	absent, not needed
Big Bang, inflation, initial entropy	real events whose configuration is selected	do not exist ($t = 0$ drops out)
Λ, fine-tuning	selected parameters	premise rejected (no dark sector; redshift = fractal path-lengthening, Doc. 190 R14a)
Delivers numbers?	no (meta-scheme)	yes ($\xi \to$ peaks, H_0, masses)
Strength	layering discipline, conceptual hygiene	parameter-free, falsifiable predictions

Where RA/PMT posits a missing layer, FFGFT says: the layer is not missing, it is not needed – the premise (many possible configurations, one selected) is itself a ΛCDM-like assumption. This is the same methodological line as the "inverted initial condition" and the symmetric circularity of Doc. 267: RA/PMT's CSM is essentially ΛCDM's initial conditions with a metaphysical selector in front; FFGFT removes the need for both.

Fair appraisal. This is not meant to disparage RA/PMT. As a *meta-scheme* – as a discipline for sorting a theory's terms cleanly into blueprint, residual, mapping, mechanism and expression – it is a useful diagnostic tool, and its honesty in declaring residuals deserves credit. It shares with FFGFT the stance "inspectable rather than persuasive" (Doc. 265). The disagreement is not in the method but in the premise: whether reality is a selection from the possible (RA/PMT) or a single, static, fractal structure with no selection (FFGFT).

Residuals of this comparison (honestly declared).

- The comparison treats RA/PMT as a meta-scheme. Should Guevara later operationalize RA1.5 (give an equation, a selection criterion), Direction A would need re-evaluation – there might then be axis-4 contact.
- Direction B assumes FFGFT's static premise is correct. That is an FFGFT-internal position, not a proven theorem; an RA/PMT proponent would read it as itself a selected configuration. Both readings remain side by side (cf. the provenance principle, Doc. 265).

Related: Doc. 265 (comparison operator, six axes), Doc. 267 (cosmological degeneracy, inverted initial condition), Doc. 203 (recursion operator $\mathscr{R}$), Doc. 190 (R14a, R20, R30). RA/PMT: J. T. Guevara, *Architectural Completion* (Parts I–III), June 2026.

Part II

Part II — Change of Dimension, the HLV Bridge and Scale Recursion

Chapter G

Doc. 270 — Changing Dimension in FFGFT

Abstract

The reduction of the fundamental 4-torus structure T^4 of FFGFT to lower dimensions is examined as a formal chain $T^4 \to T^3 \to T^2 \to T^1 \to T^0$. The central claim is that this chain does *not* consist of steps of the same kind. It splits into two qualitatively distinct operations. The first step ($D4 \to D3$) is not a lossy projection but a *duality decompactification*: the compact mass dimension S^1_m is unrolled into the non-compact time direction via the fundamental relation $\tilde{T} \cdot m = 1$. This operation preserves the mode content; only the direct visibility of compactness survives, as a discrete spectral signature encoded by the recursion ξ^n. The further steps ($D3 \to D2 \to D1 \to D0$) are genuine geometric projections of spatial directions: strictly lossy, not reversible, and *not* compensated by the recursion. The step $D3 \to D2$ is identified as a holographic projection. The topological identification $t \sim t + R_m$ is treated as a physical observable (T_{rev}, CHSH deviation $\sim \xi$, the latter not resolvable with present hardware). The downward reduction is contrasted with the *upward* operation: the bijective representational translation $T^4 \leftrightarrow$ Hilbert space/matrix (Type III, Doc. 230/206/231), which is *lossless* – precisely because it adds no geometric extra dimensions but expresses the same 4D content in a higher-dimensional language. The work thereby resolves the apparent tension between "projection loses information" and "compactified / decompactified", and classifies geometric downward reduction against representational upward translation.

G.1 Motivation: the tension in the question

FFGFT describes fundamental physics on a compact 4-torus T^4 with the single dimensionless parameter $\xi = 4/30000$ and the central relation $\tilde{T} \cdot m = 1$. The observable world, by contrast, is a three-dimensional space with time evolution, $\mathbb{R}^3 \times \mathbb{R}_t$.

Three observations on this transition appear to conflict:

(A) **Loss picture.** Every projection $D \to D-1$ is not injective and therefore loses information. The chain $T^4 \to \ldots \to T^0$ is not reversible.

(B) **Compactness picture.** T^4 is fully compactified; the observable spacetime $\mathbb{R}^3 \times \mathbb{R}_t$ is decompactified.

(C) **Upward picture.** There already exists a *bijective* translation of T^4 into a higher-dimensional space – the Hilbert space (Doc. 230) – which is *lossless*.

These three pictures cannot be naively superimposed: if the first step were pure loss (A), no *decompactified* time direction (B) could arise from it, since forgetting creates no new dimension; and it would remain puzzling how an upward move (C) can be lossless while the downward move (A) destroys information.

The resolution: there is not one but *three* dimension-changing operations of *two kinds*. *Geometric* operations (Type I: mass-time duality; Type II: spatial projection) reduce the observable world out of T^4; the *representational* operation (Type III) translates T^4 upward without loss. The geometric-downward lossy versus representational-upward bijective move – this asymmetry is the central result of the work.

G.2 Structure of T^4

Decomposition and radius classes

Definition G.2.1 (Fundamental torus structure)**.** The fundamental level of FFGFT is the compact 4-torus

$$T^4 \;=\; S^1_x \times S^1_y \times S^1_z \times S^1_m,$$

with three *spatial* circles $S^1_{x,y,z}$ and one *mass circle* S^1_m. All four components are compact.

Remark G.2.2 (The fourth direction as substitute time). The fourth torus direction S^1_m is not a mass dimension *alongside* a separate time, but the *substitute-time direction* itself. Mass and time are two *readings* of one and the same dimension; the fundamental relation $\tilde{T} \cdot m = 1$ is the translation dictionary:

$$\underbrace{S^1_m \text{ (compact)}}_{\text{mass reading, discrete spectrum}} \quad \overset{\tilde{T}\cdot m=1}{\longleftrightarrow} \quad \underbrace{\mathbb{R}_t \text{ (open)}}_{\text{time reading, continuous flow}} \,.$$

There is no time "in addition" to the four directions; time is the decompact reading of the fourth direction (cf. Doc. 185). The two-type structure depends only on the separation of *small* and *large* compactification radii and is independent of the labelling.

The four directions carry two separated radius classes:

$$R_x = R_y = R_z \;\sim\; R_H \qquad \text{(cosmological horizon, "large")}, \tag{G.1}$$

$$R_m \;\sim\; \xi \cdot \ell_P \qquad \text{(Planck scale, "small")}. \tag{G.2}$$

Remark G.2.3 (Status of the two scales). R_H here is the *declared external* (imported) cosmological scale, *not* derived from ξ. Intrinsic is only the *form* – the large/small radius-class split, equivalently the dimensionless ratio R_H/ℓ_P – not the absolute *factor* (Doc. 190, P20). The two-type structure of this document depends only on that class split, not on the numerical value of R_H. Conversely $R_m \sim \xi \ell_P$ sets the *top* of the KK tower ($1/R_m \sim \xi^{-1} m_P$, trans-Planckian); the observed particle masses lie further down the ladder via the recursion ξ^n (Doc. 159, 258/259).

Kaluza–Klein reading

A compact circle of radius R carries a discrete mass tower $m_n \sim n/R$, $n \in \mathbb{Z}$. The two radius classes therefore generate qualitatively different physics:

Circle	Radius	Mode tower
$S^1_{x,y,z}$	$R \sim R_H$ (large)	gap $\sim 1/R_H \approx 0$, quasi-continuous ⇒ space
S^1_m	$R \sim \xi \ell_P$ (small)	gap huge, discrete and heavy ⇒ mass spectrum

Proposition G.2.4 ($\tilde{T}m = 1$ as compactification relation)**.** *In natural units a compactification radius R has the dimension of a length or intrinsic time, $\tilde{T} \sim R$, while the generated mass scale is $m \sim 1/R$. The fundamental relation*

$$\boxed{\tilde{T} \cdot m = 1}$$

is thus structurally the Kaluza–Klein relation for the mass dimension.

G.3 Two types of reduction

The *downward* direction (reduction of T^4) proceeds via two geometric operations, developed here. The lossless *upward* operation (Type III, representational translation) follows in Section 8.

Type I: duality decompactification (mass → time)

Definition G.3.1 (Unrolling the mass dimension)**.** The operation

$$S^1_m \xrightarrow{\text{covering}} \mathbb{R}_t, \qquad p : \mathbb{R}_t \to S^1_m, \;\; t \mapsto e^{2\pi i t/R_m},$$

unrolls the compact mass circle into the non-compact time direction via its universal cover. It is governed by $\tilde{T} \cdot m = 1$.

Theorem G.3.2 (Type I preserves mode content)**.** *The unrolling $S^1_m \to \mathbb{R}_t$ is an embedding, not a projection. Periodic structures on S^1_m lift uniquely into $\mathbb{R}_t$; the function space on $\mathbb{R}_t$ is strictly larger. No mode content is forgotten; rather, an additional non-compact direction arises – time.*

Proof: (Sketch) The universal cover $p : \mathbb{R} \to S^1$ is a local homeomorphism and surjective. An R_m-periodic function on $\mathbb{R}$ corresponds bijectively to a function on S^1_m (Fourier series, discrete spectrum). The reverse direction (compactification $\mathbb{R} \to S^1$) is lossy, since non-periodic parts do not descend. Unrolling therefore proceeds in the *embedding*, not the projecting, direction.

Corollary G.3.3 (Time as embedding price)**.** *The "price" of decompactification is not loss of information but the appearance of the time direction itself (Doc. 185). What does not remain directly visible is only the* compactness*: the periodicity of S^1_m appears in the observable not as geometry but indirectly – as the discrete structure of the spectrum.*

Remark G.3.4 (What "not totally loss-free" refers to). This is not about absolute loss but about *reading-dependent visibility*. In the mass reading the compactness is manifest – the spectrum is discrete and the identification $t \sim t + R_m$ is expressed directly. In the time reading the same compactness is only *latent*: from continuous time alone the topological identification is not uniquely reconstructible (the winding remains ambiguous). The duality guarantees full translatability of the spectrum; the identification is carried by the time reading only implicitly – through the discreteness of the spectrum. This is exactly where ξ^n acts.

What makes the unrolled time continuous and uniform

Two properties of the unrolled time have two different causes — and a third question (*that* it unrolls at all) has an entirely different answer.

Continuity: the cover. That $\mathbb{R}_t$ is *continuous* is a topological fact, not an addition: the unrolling $S^1_m \to \mathbb{R}_t$ is the universal cover $p : \mathbb{R} \to S^1$ (Thm. G.3.2), a local homeomorphism. The cover of a circle *is* the continuous line. Hence, as soon as a circle is unrolled, the result is necessarily continuous — and as an embedding, not a projection (no mode content is lost).

Uniformity: homogeneity and the fixed eigenvalue. That time flows at a *constant rate* rests on the homogeneity of the mass circle: S^1_m is a $\mathrm{U}(1)$, translation-invariant, no point distinguished. The constant, translation-invariant metric lifts to the cover $\mathbb{R}$ as an affine, uniform parametrization. The rate itself is the *single fixed eigenvalue* $m = 1/\tilde{T}$ (the rest energy), not a variable quantity. In the recursion picture: the constant multiplier ξ has a well-defined generator $\ln \xi$, so the discrete iteration ξ^n becomes the continuous dilatation flow $t = t_0\, \xi^s$ (Doc. 285); constant multiplier ⇒ constant rate.

That it unrolls at all: the measurement. Geometrically, in the compact T^4 all four circles are equal — there is no time. Not the geometry but the *measurement* (the embedded observer, Doc. 248) selects one circle as time and unrolls it; a physical act, not a mathematical limit (Doc. 285). In doing so the winding number is discarded.

The heart of it: continuity = discarded winding. The real answer to "what makes time continuous" is therefore the discarding of the winding. The winding is the discrete, compact information; in the time reading it is thrown away (the information-discarding step, Doc. 285), and the compactness becomes *latent* — from continuous time alone the winding is no longer uniquely reconstructible (Rem. G.3.4). Continuous time is thus the circle with its winding forgotten; keep the winding and the discrete circle returns.

A subtlety: continuous exactly, uniform only up to $O(\xi)$. Two properties must be separated here. *Continuity* stays exact: $g_{\mu\nu} = \eta_{\mu\nu}(1 + \xi f)$ ($D_f = 3 - \xi$; here f is the fractal correction function of the metric, not a frequency and not the base winding number $f = 1/\xi$) is a *metric* correction, not a topological intervention — a bounded ξ-small

modulation does not tear $\mathbb{R}$, no gaps arise. *Uniformity*, by contrast, holds only up to $O(\xi)$: the same ξ feedback that in the dynamic sector carries the AM sidebands (Doc. 286) makes the rate wobble by relative $\xi \approx 10^{-4}$; the flat metric η "ignores the fractal structure" (Doc. 051). The perfectly smooth, uniform time is thus the limit $\xi \to 0$ — the idealized time of relativity —, while the real FFGFT time is a continuum *with* a ξ-fine fractal rate modulation. Important: this is *not* discrete — fractal $\neq$ discrete; time remains an uncountable continuum, only not smoothly affine. (Whether this ξ graininess is the metric shadow of the latent compactness — a ξ remnant of the discarded winding — is a natural conjecture but not established: the latent compactness shows in the discrete spectrum (Rem. G.3.4), the ξ correction in the metric.)

Type II: geometric projection (spatial)

Definition G.3.5 (Forgetting a spatial direction)**.** A projection $\pi : \mathbb{R}^k \to \mathbb{R}^{k-1}$ with $\ker(\pi) = \mathbb{R}$ forgets a spatial direction. It is not injective and therefore not invertible.

Theorem G.3.6 (Type II is strictly lossy)**.** *In a spatial projection $\mathbb{R}^k \to \mathbb{R}^{k-1}$ a full fibre $\mathbb{R}$ collapses. The information about the forgotten direction is not contained in the image and cannot be reconstructed without additional data (boundary conditions, holographic completeness). The recursion ξ^n does* not *act on this step.*

	Type I (mass→time)	**Type II (spatial)**
Operation	unrolling / covering	forgetting / projection
Direction	embedding	projecting
Mode content	preserved	lost
Reversibility	dual/spectral recoverable	not reversible
Role of ξ^n	spectral signature	ineffective
Residue	compactness only inferred	genuine geometric loss

G.4 The formal chain, step by step

Step 1: $D4 \to D3 +$ time (Type I)

$$T^4 \longrightarrow \underbrace{T^3}_{\to \mathbb{R}^3} \times \underbrace{\mathbb{R}_t}_{\text{from } S^1_m} .$$

Step 1 bundles *two* information-preserving operations of *different* mechanism: (Ia) the *mass circle* unrolls to $\mathbb{R}_t$ via the duality $\tilde{T}m = 1$ per Theorem G.3.2 (duality decompactification); (Ib) the three *spatial circles* decompactify through the *limiting process* of large radii ($R_{x,y,z} \to \infty$ for $r \ll R_H$) to $\mathbb{R}^3$ (limit decompactification). Both are embeddings, not projections; but (Ib) is *not* a duality step, it is a limit – see the clarification in Section G.12, point 3.

Invariant: $\tilde{T} \cdot m = 1$ (Prop. G.2.4); in the spatial limit, local flatness. *Loss:* none in mode content; only the visibility of compactness (Rem. G.3.4).

Step 2: $D3 \to D2$ **(Type II, holographic)**

$$\pi_2 : \mathbb{R}^3 \longrightarrow \mathbb{R}^2, \qquad \ker(\pi_2) = \mathbb{R} \ \text{(depth direction)}.$$

Invariant: area A. *Holography:* step 2 is mathematically the holographic principle; in the horizon case the Bekenstein–Hawking bound $S \le A/(4\ell_P^2)$ (cf. Doc. 254/263). *Loss:* genuine depth direction.

Step 3: $D2 \to D1$ **(Type II)**

$$\pi_3 : \mathbb{R}^2 \longrightarrow \mathbb{R}, \qquad \ker(\pi_3) = \mathbb{R}.$$

Invariant: length L. *Loss:* second spatial direction; no area measure remains.

Step 4: $D1 \to D0$ **(Type II, total loss)**

$$\pi_4 : \mathbb{R} \longrightarrow \{\text{pt}\}.$$

Invariant: only a discrete scalar measure ($n \in \mathbb{Z}$). *Loss:* the complete geometric structure.

G.5 Reversibility, precise per type

Theorem G.5.1 (Reversibility of the full chain)**.** *Let* $\Pi = \pi_4 \circ \pi_3 \circ \pi_2$ *be the spatial subchain* $\mathbb{R}^3 \to \{\text{pt}\}$ *and* D *the Type I operation* $S^1_m \to \mathbb{R}_t$*. Then:*

(i) D *is, as a duality (Fourier/KK), invertible in mode content; the winding remains ambiguous.* D *is "almost reversible".*

(ii) Π *is not invertible: there is no continuous map* $\rho : \{\text{pt}\} \to \mathbb{R}^3$ *with* $\rho \circ \Pi = \mathrm{id}$.

Proof: (Sketch) (i) follows from Theorem G.3.2: unrolling is embedding, the discrete spectrum encodes the full periodic content; only the choice of lift (winding) is non-unique. (ii) follows from Theorem G.3.6: each π_k has ker $= \mathbb{R} \neq \{0\}$. A left inverse would have to reconstruct the collapsed fibre structure; this information is not present in the image $\{\text{pt}\}$.

Corollary G.5.2 (Resolution of the initial tension)**.** *Pictures (A) and (B) are both correct but concern different operations. The non-reversibility (A) holds for the* spatial *projections (Type II). The decompactification to the time direction (B) is the* embedding *Type I step – it creates a dimension instead of forgetting one. There is no contradiction.*

G.6 Role of the recursion ξ^n

The recursion $R : \xi^n \mapsto \xi^{n+1}$ is *not* a universal compensation for projection losses. Its function is narrowly bounded:

- **Type I:** ξ^n generates the discrete mass spectrum (particle ladder). This spectrum *is* the compact mode content of S^1_m, dually expressed as time-frequency structure. ξ^n "fills in" nothing; it is the spectral signature through which the hidden compactness remains recognisable (Rem. G.3.4).
- **Type II:** ξ^n has no effect. Spatial projection losses are genuine and untouched by the recursion.

Remark G.6.1 (Correction of a tempting misrepresentation). A presentation that assigns the recursion a "compensation strength" decreasing monotonically across all four steps (full/partial/weak/none) is misleading. It presupposes a uniform lossy ladder that does not exist. The recursion belongs exclusively to the Type I step.

G.7 Overview

Step	Operation	Type	Invariant	ξ^n
$D4 \to D3{+}t$	$S^1_m \to \mathbb{R}_t$ (unroll)	I	$\tilde{T}m = 1$	spectral signature
$D3 \to D2$	forget space	II	area A	–
$D2 \to D1$	forget space	II	length L	–
$D1 \to D0$	forget space	II	$n \in \mathbb{Z}$	–

Table G.1: The chain $T^4 \to T^0$ separated by type.

$$\underbrace{T^4 \longrightarrow \mathbb{R}^3 \times \mathbb{R}_t}_{\text{Type I: embedding, almost reversible}} \longrightarrow \underbrace{\mathbb{R}^2 \longrightarrow \mathbb{R} \longrightarrow \{\text{pt}\}}_{\text{Type II: projecting, not reversible}}$$

G.8 The upward direction: representational translation (Type III)

The chain so far describes only the *downward* direction (reduction). There is a counter-direction – but it is *not* geometric.

Definition G.8.1 (Type III: representational translation)**.** The operation

$$T^4 \xleftrightarrow{\text{bijective}} \mathscr{H}$$

translates the FFGFT mode formalism on T^4 into the (infinite-dimensional) Hilbert space $\mathscr{H}$, or the equivalent matrix algebra (Doc. 230, 206, 231). It is not an approximation but a *bijective structural statement*: every QM computation is expressible in FFGFT language and vice versa.

The dimensional asymmetry

Downward (geometric) and upward (representational) are different kinds of operation with *opposite* information properties:

- **Downward (Type II, geometric):** lossy, not reversible – a fibre collapses (Theorem G.3.6).
- **Upward (Type III, representational):** bijective, lossless – it adds *no* physical dimensions but expresses the same 4D content in a higher-dimensional *language* in which it fits without reduction.

The losslessness follows *precisely from* the representational character – not from any gain of genuine extra dimensions.

Remark G.8.2 (No geometrically higher dimensions). A geometric extension $T^4 \to T^5 \to \ldots$ is excluded in FFGFT (Doc. 171): complexity arises from *fractal depth* ($D_f = 3 - \xi$, self-similarity), not from *more* dimensions. This is the deliberate contrast with string theory (10/11D, Calabi–Yau, landscape problem). "Higher" therefore always means Type III (representational) in FFGFT, never geometric.

FFGFT thus has three dimension-changing operations:

Type	Operation	Character	Information
I	mass $\leftrightarrow$ time ($\tilde{T}m = 1$)	geometric	almost reversible
II	spatial projection (down)	geometric	lossy
III	$T^4 \leftrightarrow \mathscr{H}$/matrix (up)	representational	**bijective**

The bijectivity of Type III is the mirror image of the non-reversibility of Type II (Theorem G.5.1): geometrically one loses on reducing, representationally one loses nothing on translating.

G.9 Why exactly four circles? Minimal sufficiency, units, and visualization

The chain so far (Type I/II downward, Type III upward) presupposes the four-circle structure $T^4 = S^1_{x,y,z} \times S^1_m$. This section answers *why* there are exactly four – deliberately *not* as a derivation of the number, but as *minimal sufficiency*.

$3+1$ is input, not derivation

No theory derives the dimension count of spacetime: the Standard Model and GR assume $3+1$; string theory fails on uniqueness (landscape problem). FFGFT, too, *does not derive the 4 forward*; it takes $3+1$ as *empirical input*. The question is therefore not "why is the world four-dimensional?" but "*given* $3+1$ – which structure is the minimal one that carries everything?".

Minimal sufficiency: four instead of five

The duality saves one dimension

Naively one would need *five* building blocks: three for space, one for a mass spectrum (particles), one for time evolution. The fundamental relation $\tilde{T} \cdot m = 1$, however, identifies the *mass reading* and the *time reading* of *one* dimension (Type I, Theorem G.3.2): the fourth circle S^1_m carries the mass spectrum as its KK tower *and*, decompactified, time. The duality lets *one* circle do the work of two – so the naive 5 collapses to **4**.

- **Fewer** (only T^3): no mass spectrum, no time.
- **Four** (T^3 + one dual circle): space, masses *and* time from *one* object, without redundancy.
- **More** (a second such circle): a second independent mass tower or a second time – superfluous, without correlate.

"Optimal" therefore means *minimal-sufficient*: the smallest dimension count in which space, the full mass spectrum, and time jointly emerge from one geometric object.

The four circles as units bookkeeping

The four-circle structure *is* at the same time the exponent bookkeeping of the SI conversion factors – each circle type carries its bridge:

$$\underbrace{S^1_{x,y,z}}_{\text{space}} \xrightarrow{c} \text{time}, \qquad \underbrace{S^1_m}_{\text{mass}} \xrightarrow{\hbar\,(\tilde{T}m=1)} \text{time}.$$

c is the space-time bridge of the three spatial circles, $\hbar$ the mass-time bridge of the mass circle. This is why the SI conversions are *indispensable* once one wants to determine the exponent structure correctly:

Remark G.9.1 (Why natural units hide the powers). Natural units ($c = \hbar = 1$) collapse L, T and M to *one* axis. This removes precisely the distinction that carries the exponent structure – the independent *power of* c resp. *of time*. c and $\hbar$ may be set to 1 only *after* the exponents have been read off; before that the information is destroyed. The factors are the scaffolding that keeps the powers visible – and easy to handle precisely because their whole role is that scaffolding without a free parameter. G, too, is *derived* – reconstructed from ℓ_P, c, $\hbar$, for which it needs only the SI relations (Docs. 012/056) – and carries *no* free factor. The only free factor sits not in a conversion or natural constant but in the *cosmological absolute scale* R_H (exponent $41/4$; Doc. 190, R20).

The same structure as Hilbert visualization

The four-circle geometry is also the *faithful* visualization of the infinite-dimensional Hilbert space (Type III, Def. G.8.1) – two readings of *the same* bijection:

- **generative/ontological:** $T^4 \to \mathscr{H}$ – the geometry *generates* the algebra (Doc. 206), not the reverse;

- **visualizing/epistemic:** the abstract $\mathscr{H}$ becomes intuitable at all only through the 4D geometry.

"Higher-dimensional" (the ∞-dimensional $\mathscr{H}$) does *not* mean "more fundamental": the generative direction is $T^4 \to \mathscr{H}$. Precisely *because* the 4D structure is generative (not a mere picture), the visualization is *exact* – bijective, not a cartoon.

Summary of the dimension line

What "optimal" means

T^4 resp. $T^3 +$ time is the *minimal-sufficient*, *dual-economical* and *Hilbert-faithful* configuration – *given* $3+1$. The strength lies in the economy (one dimension saved by $\tilde{T}m = 1$) and the faithfulness (lossless Hilbert representation), *not* in a uniqueness of the number. The bare 4 is input, not a forward result (Doc. 190, R34) – the same signature as for the ξ-magnitude (R33) and the cosmological scale (R20): *structure forward, number input*.

G.10 In the compact view, time is just another circle: the asymmetry is the measurement

In the compact representation $T^4 = S^1 \times S^1 \times S^1 \times S^1$ all four circles are identical: dimensionless cyclic coordinates, no signature, no space/time distinction. Time is written *exactly* as the three spatial circles; at the compact level the Lorentzian asymmetry does not exist. This is why, in FFGFT, the time coordinate is simple to write – it is one circle like the others. The complication enters only when one leaves the compact picture.

The asymmetry (one time, signature $(3,1)$) appears only under decompactification (Type I, Def. G.3.1): exactly one circle – the mass circle, $\tilde{T} \cdot m = 1$ – unrolls through its universal cover into non-compact time, while the other three stay compact (discrete spectrum) or project (space, Type II).

Which circle becomes time is therefore not an intrinsic-geometric fact to be derived. The geometry is symmetric, and nothing in it should single one direction out. Time is *relational*: the selection is the measurement itself. The decompactification *is* the measurement – the read-out/projection of this document and of Doc. 284 – and committing one circle to non-compact, experienced time is exactly the information-discarding step a measurement performs. The embedded observer, reading out from the inside (Doc. 248, self-referentiality), experiences one direction as duration; that experience constitutes the time circle. The symmetry-breaking is the measurement, not the geometry.

[The time circle is selected by the measurement] In the compact view the four circles of T^4 are symmetric. Time is not chosen by the geometry but is relational: the symmetry-breaking *is* the measurement (the decompactification of Type I). Three statements then close the bookkeeping with no gap:

- **That there is one time** (not two) is the signature/predictivity constraint – one timelike direction gives hyperbolic field equations with a well-posed Cauchy

problem; two or more give ultrahyperbolic equations with no well-posed initial-value problem.

- **Which circle is time** is fixed by the measurement, i.e. by the experienced situation of the embedded observer (Doc. 248).
- **That the split is** $3{+}1$ is the structure of being an embedded observer – the empirical input of Doc. 190 (P34), read here as the observer's situation rather than a free parameter.

The contrast with a non-compactified treatment of time is then sharp. A framework that writes time as a complicated structure built on already-physical time t (for instance a spiral-time operator $\psi(t) = t + i\varphi(t) + j\chi(t)$, Doc. 283/284) has *reified* the result of the measurement into its kinematics, instead of recognising experienced time as the measurement-induced decompactification of a symmetric circle. In the compact view that complication is unnecessary: time is one more circle, and only the read-out turns it into the non-periodic axis we experience.

G.11 Physical testability of the identification $t \sim t + R_m$

The topological identification $t \sim t + R_m$ is not a mere accompanying structure of the duality but a physical observable. In the time reading its period is *exactly* the fundamental revolution time

$$T_{\text{rev}} = \frac{R_m}{c} = \frac{\xi \ell_P}{c} \approx 7.2 \times 10^{-48}\,\text{s}, \qquad \Delta\phi = 2\pi\xi \approx 8.4 \times 10^{-4}\ \text{rad/revolution},$$

so that $t \sim t + R_m$ is identical with $t \sim t + cT_{\text{rev}}$ (Doc. 183/185). The compactness of the fourth direction *is* the revival time.

Two testable readings

- **Mass reading (worked out structurally).** The discrete mode spectrum generates the particle masses as torus overtones with step $1/D$; from this follow the three generations (Doc. 159, 258/259). The ladder is a structure with one declared reference point; its coefficients are set, its precision agreements are therefore not evidence, and Koide follows only approximately.
- **Time reading (falsifiable in principle, not resolvable with present hardware).** The periodicity produces a CHSH deviation from the Tsirelson bound of order ξ. For 73 qubits FFGFT predicts $\Delta \approx 5.4 \times 10^{-4}$. The hardware test IBM Heron r2 (May 2026) confirms the Bell violation ($S = 2.74$) and the executability of T0 circuits; the ξ effect, however, lies far below the hardware noise and is not resolvable with present hardware (Doc. 147): the measured deviation from the Tsirelson bound is $2\sqrt{2} - 2.74 = 0.088$ (3.1 %), about 160 times Δ; $\sigma \sim 10^{-4}$ is only the statistical bound.

Scaling: linear versus logarithmic

Regime dependence of phase accumulation

A naive *coherent linear accumulation* $r \cdot \Delta\phi$ over r revolutions would yield physically absurd phase drifts (of order 10^{44} rad/s) and contradict working atomic clocks and deep quantum circuits. The lab prediction therefore scales *logarithmically*,

$$\delta\phi \sim \frac{\xi \ln(r)}{D_f} \approx 2 \times 10^{-4} \quad (D_f = 3 - \xi),$$

not linearly. The linear form $r \cdot \Delta\phi$ of Doc. 185 is the *worst-case bound of the RSA argument* (coherence over $r \approx N$ cycles), not the lab prediction. This regime separation explains why deep coherent circuits can run while the RSA factorisation of large keys remains geometrically protected.

Conclusion. $t \sim t + R_m$ is carried in the FFGFT corpus as an observable (T_{rev}, CHSH deviation $\sim \xi$): worked out as a structure in the mass reading, falsifiable in principle in the time reading, but not resolvable with present hardware (far below the hardware noise). It is not an unfalsifiable accompanying structure.

G.12 Classification and open questions

Connections. Doc. 185 (Time as the Embedding Price) provides the ontological basis of the Type I step; Doc. 263 (holographic projection) and Doc. 254 (area-bound theorem) the framework for step 2; Doc. 206 (Triangle–Matrix reduction) confirms that the 4D level is generative; Doc. 147/183 provide the time-side testability. The upward operation (Type III) is developed in Doc. 230 (bijective Hilbert-space translation), Doc. 231 (necessary extensions) and Doc. 206 (matrix representation); Doc. 171 justifies the exclusion of geometrically higher dimensions.

Open points.

1. **Structural assumption – confirmed.** The decomposition $T^4 = S^1_{x,y,z} \times S^1_m$ with S^1_m as substitute-time direction and $\tilde{T} \cdot m = 1$ as translation is fixed (Rem. G.2.2).
2. **Linear/logarithmic scaling.** The linear worst-case bound $r \cdot \Delta\phi$ (Doc. 185, RSA regime) and the logarithmic lab prediction $\xi \ln(r)/D_f$ (Doc. 147) should be marked explicitly as regime-dependent to secure consistency of presentation.
3. **Spatial limit – clarified.** The decompactification $S^1_{x,y,z} \to \mathbb{R}^3$ is a *limiting process* ($R_{x,y,z} \to \infty$), neither the Type I duality nor a Type II projection, but a distinct *limit-type* information-preserving operation (Ib). It is therefore carried in Step 1 explicitly apart from the mass-circle duality (Ia); both share the embedding character, not the mechanism.
4. **Holographic completeness.** To what extent can the Type II loss at $D3 \to D2$ be reconstructed exactly (not just partially) from boundary data? Connection to Doc. 254/263.

Chapter H

Doc. 271 — FFGFT (T0) and Helix–Light–Vortex (HLV) Compared

Abstract

This work places FFGFT (T0, time–mass duality) and the Helix–Light–Vortex programme (HLV, M. Krüger) side by side. Both are *layered effective geometric programmes*: they replace the smooth continuum with a discrete, higher-dimensionally generated substrate, carry a three-fold (triadic) core motif, read particles as topological objects, root in classical proportion/harmonic geometry, and share a methodology of recovery limit plus null-model falsification. We separate *genuine isomorphisms* from *surface analogies*. The central difference: FFGFT carries *one* dimensionless constant $\xi \approx 1/7500$ with forward dimensionless predictions (fine-structure constant, Koide relation, lepton mass ratios), whereas HLV has no ξ-analogue but an explicitly reproducible aperiodic substrate together with a graph-Laplacian testbed against null models. The geometric fork *aperiodic quasicrystal* (HLV) versus *periodic T^4 torus with a fractal correction* (FFGFT) yields a measurable spectral difference, from which a *distinguishing* prediction is formulated. HLV's consciousness/biogravimetry layer is explicitly excluded from this physics comparison. As a shared guardrail we stress the triviality of any representation $X = \varphi^N$ or $X = \xi^N$: forward content lies solely in dimensionless ratios that are isolated, predicted, and precise; an absolute scale remains a calibrated input.

H.1 Motivation and scope

HLV is now available in a deliberately *conservative*, falsifiable form in two recent documents: a "master explanation" (Zenodo 10.5281/zenodo.20596861) and a reproducible "6D-to-3D cut-and-project graph testbed" (Zenodo 10.5281/zenodo.20275888). The move made there – away from "spacetime *is* quasicrystalline" towards "can this substrate be made explicit, reproducible, and distinguishable from simpler null models?" – is methodologically the same turn that FFGFT performed in its corrections register (Doc. 190, notably the corpus-wide cleanup of the H_0 over-claims).

The aim here is a *comparison*, not a merger: to test where the kinship is structurally load-bearing, where it is merely verbal, and at which point the two programmes must *diverge* empirically.

H.2 Common axes (overview)

Axis	FFGFT / T0 (Pascher)	HLV (Krüger)
Substrate / projection	compact 4-torus T^4 + fractal correction $D_f = 3 - \xi$; reduction chain $T^4 \to T^0$	aperiodic quasicrystal via cut-and-project ($\mathbb{Z}^6 \to 3$D), golden-ratio/Fibonacci
Time structure	time–mass duality $\tilde{T} \cdot m = 1$; Z_3 in the triangle reduction (det $= 1 - \xi$)	triadic spiral time $\psi(t) = t + i\varphi(t) + j\chi(t)$, $i^2 = j^2 = -1$
Particle ontology	winding quantum numbers on T^4; Bell topological, not ontologically nonlocal	vortices / standing-wave modes of a scalar field Ψ in the lattice
Generative core	*one* dimensionless $\xi \approx 1/7500$	no ξ-analogue; substrate geometry (φ, Fibonacci)
Forward predictions	α, Koide, lepton ratios (dimensionless, isolated)	deliberately withheld; focus "is the substrate distinguishable?"
Recovery limit	$\hbar, c, G$ as conversions; $O(\xi)$ lattice dispersion; bridges (with honest negatives)	must recover QFT/GR in the limit; graph Laplacian vs. null models
Falsification channel	mode spectrum (T^4): CMB peak *ratios*, $B = 1/3$ from $D = 4$	graph-Laplacian spectrum vs. null model; vortex benchmarks

H.3 Genuine isomorphisms vs. surface analogies

1. **Projection from higher dimensions.** *Resonance:* both read observed 3-space as the projection of a structured, higher-dimensional, discrete geometry – HLV via cut-and-project, FFGFT via the explicit chain $T^4 \to T^0$ (Doc. 270). *But: aperiodic* (quasicrystal) versus *periodic* (T^4) are different spectral classes – see Section 6. The motif is isomorphic, the realisation is not.
2. **Triadic core motif.** *Resonance:* HLV's hypercomplex time $\psi = t + i\varphi + j\chi$ sits next to the Z_3 structure of the FFGFT triangle reduction ($\det(A_\xi) = 1 - \xi$, Doc. 206) and the three operation types of the reduction chain. *But:* HLV's triad is a *time algebra*, FFGFT's a *reduction-matrix symmetry*. The shared "3" is striking but not yet a proven isomorphism – a candidate for closer scrutiny, not an identity.
3. **Particles as topological objects.** *Resonance:* HLV vortices / standing modes correspond to FFGFT winding numbers on the torus; both relocate "quantum number" into a topological winding of the substrate. In FFGFT this also renders the Bell correlation as topological (not ontologically nonlocal). Here the kinship is most likely genuine.
4. **Harmonic/proportion root.** *Resonance:* HLV anchors in the golden ratio and Fibonacci; FFGFT in the Euler Tonnetz and the comma structure of temperament. Same instinct ("proportion/harmony → fundamental geometry"), different constant.
5. **Methodology.** *Resonance (strongest):* recovery limit plus null-model falsification. HLV's "distinguishable from simpler null models" is exactly the criterion by which FFGFT tested whether, e.g., $63/4$ is *isolated* or merely one of several "nice fractions" in a dense candidate field (Doc. 190, R35).

H.4 Divergences and complementarity

- **Generative core.** FFGFT has a single dimensionless seed ξ and from it *forward* predictions ($\alpha = \xi(E_0/\text{MeV})^2$, Koide, lepton ratios). HLV (in the documents presented) has no such analogue and deliberately stays at "is the substrate distinguishable". ⇒ FFGFT brings the seed, HLV the reproducible substrate construction – complementary, not competing.
- **Tool.** HLV's graph-Laplacian testbed against null models is a numerical falsification machinery that FFGFT could adopt for its own T^4 mode spectrum.
- **Substrate.** Aperiodic vs. periodic + fractal – a real geometric fork (Section 6).
- **Foundational maturity.** HLV has drawn external-review criticism of an under-specified, non-closed algebra and undefined operators; FFGFT works with a three-layer discipline (core derivations / algebraically proven bridges / plausibility reductions) and the Hilbert-space bridge (Doc. 230–232).
- **Scope.** HLV includes a consciousness/biogravimetry layer. This is *explicitly excluded* here: the present comparison concerns only the physical-geometric layers.

H.5 Distinguishing prediction: the spectral fork

The scientifically robust core of the comparison is spectral. A *periodic* substrate (T^4) produces a discrete spectrum with integer mode labels $|n|^2$ and a harmonic, ladder-like structure; FFGFT reads off dimensionless CMB peak ratios $1 : 2.449 : 3.742 : \ldots$, fixed by $D = 4$ (curvature parameter $B = 1/3$). An *aperiodic* substrate (quasicrystal), by contrast, typically produces a hierarchical, gap-rich spectrum (singular-continuous density of states, Cantor-like gap structure) *without* a simple integer harmonic ladder.

These two signatures are *distinguishable*. Concrete (scale-free) test:

1. Run HLV's testbed with (a) the quasicrystal *and* (b) a *periodic* lattice null model – not only randomised nulls. The sharp question is aperiodic-vs-periodic, not discrete-vs-continuum.
2. Report the leading *deviation exponent* in the recovery limit: does the graph-Laplacian dispersion approach the continuum $\omega \propto |k|$, and is the residual $O(\text{small})$ and sign-definite? (FFGFT does exactly this with $O(\xi)$ corrections.)
3. Compare scale-free *spectral ratios*: the harmonic T^4 ratio spectrum against the quasicrystal gap spectrum – and both against data (CMB peaks resp. lattice benchmarks). Whichever ratio spectrum matches observation discriminates the substrates.

H.6 Shared methodological guardrail

For *both* programmes: every magnitude can be written as φ^N or ξ^N; a single measured scale in an exponent is therefore *not* a prediction. Forward content lies solely in dimensionless ratios that are simultaneously *isolated*, *predicted*, and *precise*. The absolute

scale (in FFGFT H_0/R_H via the fitted exponent $41/4$) remains a calibrated input, not evidence (Doc. 190, R35). An honest HLV prediction should likewise be a *scale-free* ratio from the graph-Laplacian spectrum, not an absolute magnitude dressed in φ.

H.7 Status and next steps

This work is a structural mapping, not a unification claim. It remains open whether the triadic resonance is a genuine isomorphism or a coincidence, and whether either substrate matches the observed mode spectrum better. The obvious next step is a *joint* testbed run: HLV's graph Laplacian against a periodic null lattice, plus the scale-free ratio comparison with the T^4 spectrum. Only then – falsification before interpretation – is a statement about the kinship of the two geometries defensible.

H.8 Sources

- M. Krüger, *A Conservative Master Explanation of the Helix–Light–Vortex Framework*, Zenodo (2026), doi:10.5281/zenodo.20596861.
- M. Krüger, *A Reproducible 6D-to-3D Cut-and-Project Graph Testbed for Quasicrystalline Laplacian Recovery and Vortex Benchmarks*, Zenodo (2026), doi:10.5281/zenodo.20275888.
- FFGFT: Doc. 190 (corrections register, R35), Doc. 206 (triangle reduction, Z_3), Doc. 230–232 (Hilbert-space bridge), Doc. 268 (CMB peaks from T^4), Doc. 270 (reduction chain $T^4 \to T^0$).

Chapter I

Doc. 272 — FFGFT (T0) and HLV, Classified through Dot Theory

Abstract

This work classifies FFGFT (T0, time–mass duality) and the Helix–Light–Vortex programme (HLV, M. Krüger) through the *current governance scaffolding* of Stefaan Vossen's *Dot Theory*, as developed in the direct IPI exchange (Vossen with J. Pascher and others). Dot Theory is not a competing physical model but a representational meta-programme with a governance machinery of four record objects: the **Lexicon** (structure/terminology), the **OAP** (admissibility record), the **Matrix** (relationships between frameworks), and the **FAH** (*Framework Admissibility History*: how admissibility changes; provenance-preserving renderings $O0 \to O1/O2$). Guiding principles: "declared provenance, not universal agreement" and "preservation of meaning-making, not allocation of meaning" – an FAH record is a *historical object*, not a truth claim. The present classification is produced as a **Matrix entry** (relationship FFGFT↔HLV) and offered as a **secondary rendering ($O2$)** under provenance – a *declared* reading (J. Pascher), while Krüger (HLV) and Vossen (Dot) hold the $O0$ originator declaration. Result: FFGFT (observer-independent/periodic/forward-predictive) and HLV (observer-embedded/aperiodic/substrate-distinctive) are commensurable only on the spectral/recovery axis; the other axes lie in different "question geometries". Only Dot's governance/routing layer is used, not its own physical ToE/ψ layer.

I.1 Dot Theory's governance machinery (state IPI, June 2026)

Vossen explicitly does not present Dot Theory as a completed physical theory, but as a representational hypothesis about modelling, with a strict programme requirement: any extension must be consistent, preserve required symmetries, *recover* the standard formalism as a limiting case, and yield at least one clear empirical discriminator. In the direct IPI exchange this has grown into a governance machinery with four record objects:

- **Lexicon** – holds the *structure*: terms and their definitions.

- **OAP** – holds the *admissibility* of a framework (does it meet the programme requirement?).
- **Matrix** – holds the *relationships* between frameworks.
- **FAH** (Framework Admissibility History) – holds *how* admissibility changes when an implicit commitment is exposed and relocated.

The FAH works with staggered renderings: $O0$ (originator declaration) $\rightarrow O1/O2$ (secondary renderings by other observers) $\rightarrow$ later evaluation. Once admitted, records remain *permanent* and contestable; later entries may refine, contest, supplement, or supersede, but the original rendering remains part of the developmental history. The load-bearing distinction is "preservation of meaning-making" (admissible: recording *how* a change was understood) versus "allocation of meaning" (inadmissible: deciding *who* is right). In Pascher's words: *the history of how a result was understood is itself part of the result.*

This classification uses only the governance/routing layer. Dot Theory's own physical ToE layer (meta-Lagrangian $\mathscr{L}_{\text{Dot}}$, observer operator $\odot$, observer-purpose field ψ) is *not* invoked. The state is June 2026 and – consistent with Dot's non-absolutist stance and FAH supersedability – explicitly provisional.

I.2 Lexicon: the classification axes as defined terms

- *Informational accessibility / question geometry*: under what question and with what access does the framework operate?
- *Operational invariants*: what is preserved under the characteristic operations?
- *Recoverability / projection-loss*: what information is lost in projection, what can be recovered?
- *Observer embedding*: is an observer constitutively part of the formalism?
- *Empirical discriminator (admissibility)*: which testable observable separates the framework from null models?

I.3 OAP: admissibility of FFGFT and HLV

Both frameworks are *admissible* in the OAP sense, but on different observables. **FFGFT** meets the requirement via dimensionless ratios (α, CMB peak ratios from $D = 4$), an $O(\xi)$ recovery of continuum dispersion, and a *worked-out* recoverability typology (Doc. 270). **HLV** meets it via the graph-Laplacian spectrum against null models and vortex benchmarks; recovery of QFT/GR in the limit is required but yet to be shown – a *maturity difference*, not a contradiction.

I.4 Matrix: the relationship FFGFT ↔ HLV

Lexicon axis	FFGFT / T0	HLV
Structural nesting (June 2026)	crystallographic 3-fold core C_3; T^4	icosahedral extension $A_5 \supset C_3$; $T^7 = T^4 \times T^3$; φ = eigenvalue of the perp sector
Accessibility / question geometry	observer-*independent*; "which dimensionless ratios follow from ξ?"	substrate-centred; "is the substrate distinguishable from null models?"
Operational invariants	$\tilde{T} \cdot m = 1$; ξ; winding numbers; Z_3 (det $= 1 - \xi$)	cut-and-project window; vortex topology; triadic time $\psi = t + i\varphi + j\chi$
Recoverability / projection-loss	*worked out* (Doc. 270): lossy reduction vs. lossless representation	recovery of QFT/GR in the limit – required, yet to be shown
Observer embedding	none (purely geometric)	yes: ψ/consciousness layer (like Dot Theory itself)
Empirical discriminator	dimensionless ratios: α, peak ratios from $D = 4$	graph-Laplacian spectrum vs. null model; vortex benchmarks

Axis by axis: FFGFT and HLV pose *different* questions (what follows from ξ vs. is the substrate distinguishable), carry the triadic structure at different places (time algebra ψ vs. reduction symmetry Z_3 – an isomorphism candidate, not a proof; cf. Doc. 271), and connect only through the recoverability/discriminator axis, *spectrally*. There – and only there – is the pairwise comparison (Doc. 271) licensed and yields the distinguishing prediction: periodic T^4 harmonic spectrum (peak ratios from $D = 4$) versus aperiodic quasicrystal gap spectrum. On the axes of question geometry and observer embedding the two live in different regimes; a direct "X is like Y" comparison there would be a category error.

Refinement (later $O2$ rendering, 19 June 2026). On the structural / group-theoretic axis a closer relation holds than recorded above: $C_3 < A_5$ and $T^4 \subset T^7$ ($7 = 4 + 3$) give a *containment* HLV $\supset$ FFGFT (Doc. 283/285) – FFGFT as the crystallographic 3-fold core, HLV as the icosahedral extension. The golden ratio φ is the HLV-side *discriminator*: the eigenvalue of the perp directions (golden inflation $\varphi^2 = \varphi + 1$) that FFGFT does not contain; the same crystallographic restriction ($2\cos(2\pi/5) = 1/\varphi \notin \mathbb{Z}$) that forbids φ in FFGFT assigns it to HLV. This reading is *more committed* than the original entry and is entered explicitly as a contestable rendering: it does not change OAP and allocates no meaning; the group-theoretic fact ($C_3 < A_5$, $7 = 4 + 3$) is solid, the physical identification of the perp sector remains a declared reading.

I.5 FAH: rendering status of this classification

This Matrix entry is entered into provenance as a *secondary rendering* ($O2$). The $O0$ originator declarations rest with Krüger (HLV) and Vossen (Dot Theory); the reading presented here is J. Pascher's and does not claim to determine objectively which framework is "correct". It *preserves meaning-making* (it records how the relationship is understood) and *allocates no meaning* (it adjudicates no precedence). The classification thus also explains *retroactively* why Doc. 271 needed exactly its caveats: these are not politeness but the consequence of different accessibility conditions – Dot's "epistemic

routing prior to mechanism deployment" on a concrete case. Later renderings may refine or contest this entry; it remains as a historical object.

I.6 Feedback into the architecture

Two FFGFT building blocks feed back directly into the governance: (i) the recoverability typology of Doc. 270 (three qualitatively different reduction/translation operations) as a concrete *recoverability structure* for the Matrix/OAP axis; (ii) the methodological guardrail of Doc. 190 (R35): every magnitude is writable as ξ^N or φ^N, a measured scale in an exponent is not a prediction – only isolated, predicted, precise *ratios* carry forward content. Both operationalise Dot's admissibility and routing criteria on a real example.

I.7 Status

A meta-level mapping, not a claim that either framework is correct; state June 2026, provisional and supersedable. What it does: it shows, on principled grounds, *where* the comparison from Doc. 271 is admissible (spectral/recovery axis) and where not (question geometry, observer embedding). It remains open – consistent with Dot's non-absolutist stance – whether the triadic resonance is more than coincidence and which substrate better matches the observed mode spectrum.

I.8 Sources

- S. Vossen, *Dot Theory* (representational research programme) and the governance developed in the direct IPI exchange (June 2026): *Lexicon, OAP, Matrix, FAH* ($O0/O1/O2$ renderings; "declared provenance, not universal agreement"; "preservation of meaning-making, not allocation of meaning"), dottheory.co.uk.
- M. Krüger, HLV: *Conservative Master Explanation* (Zenodo 10.5281/zenodo.20596861) and *6D-to-3D Cut-and-Project Graph Testbed* (Zenodo 10.5281/zenodo.20275888).
- FFGFT: Doc. 190 (R35), Doc. 206 (Z_3), Doc. 270 (reduction chain / recoverability), Doc. 271 (pairwise comparison FFGFT↔HLV).

Chapter J

Doc. 274 — FFGFT (T0) in the IPI Field

Abstract

This document places FFGFT (T0, time–mass duality) within the field of frameworks discussed at the *Information Physics Institute* (IPI). Rather than another pairwise comparison (those exist in Doc. 244, 245/269, 246, 251, 260, 271, 272), it proposes a *map* along two orthogonal organizing axes: (1) **substrate geometry** – what the discrete underlay is made of – and (2) the **primitive** – what counts as fundamental. On axis 1 FFGFT occupies the *periodic* pole (a compact T^4 crystal), HLV the *aperiodic* one (a quasicrystal); the *random* pole remains a control model with no committed owner. This triad is exactly the one the spectral discriminator testbed (Doc. 271, accompanying script) renders measurable. On axis 2 FFGFT is geometry-primitive, alongside information-primitive (Vopson; downstream Leavitt), algebraic/Hilbert-space (UMF) and selection/constraint frameworks (RA/PMT, RSG, UC, UIFT). It is stated explicitly that this two-axis map is the *author's organizing overlay* – in Dot-Theory terms a *secondary rendering (O2)*, a declared reading – not the native vocabulary of the respective frameworks; the primary rendering (O0) remains with their originators.

J.1 Purpose and limits of this map

Over recent months FFGFT has been set against a series of frameworks in the IPI exchange. Each comparison ended strikingly often with the same finding: *a category difference, not a contradiction*. This document asks why, and proposes a map that explains it.

Three caveats first. *One*: the labels below ("periodic crystal lattice", "information-primitive", etc.) are *organizing terms of this work*. Some originators would classify their framework differently; the map is a reading, not a verdict. *Two* – and this is the more important point: the *choice of the two axes* (substrate geometry, primitive) is itself a *declared projection*, not a universal dimension of the field. Another researcher could just as admissibly choose different axes – such as Accessibility/Constraint or Observer-Dependence/Recoverability – and arrive at a different but equally legitimate classification of the same field. The axes used here are therefore carried as a *separately identified projection*, not as implied universals; they are privileged only relative to the stated purpose

of this map. *Three*: even within this projection the axes do not sort the field exhaustively – Axis 1 (substrate) cleanly separates only the geometric substrate programmes; algebraic (UMF) and continuum-topological (αLGQV) programmes sit beside the triad. That is not a defect of the map but its honest resolution.

Use of O0/O2 (declared): O0/O2 are used here in the *provenance-preserving* sense – markers of where interpretive authority resides and who retains the right of correction – not as a full import of Guevara's rendering hierarchy (O0/O1/O2 as representational levels). Whether the two ultimately coincide is left open; this map uses only the provenance-preserving reading (cf. Vossen's FAH concern). Rendering level and provenance coincide in this document only because a secondary rendering of *another's* framework is necessarily authored by someone other than its originator – outside cross-framework comparison they decouple.

J.2 Axis 1 – substrate geometry (the crystal axis)

FFGFT is at heart a *periodic crystal lattice*: the compact 4-torus T^4 is periodic and ordered, with fractal correction $D_f = 3 - \xi$. HLV, by contrast, is an *aperiodic quasicrystal* (cut-and-project $\mathbb{Z}^6 \to 3$D, golden ratio / Fibonacci). A third, *random* substrate serves as a null model. These three types are not merely conceptually distinct – they have *different spectral fingerprints*, and it is precisely this separation that the graph-Laplacian testbed of Doc. 271 makes scale-free measurable.

Substrate type	Representative	Spectral fingerprint (scale-free)
periodic crystal lattice	**FFGFT** (T^4)	strongly degenerate; $\langle r \rangle \approx 0$; integer $\lvert n \rvert^2$ ladder (sums of squares); in $D{=}4$ *every* integer is hit (Lagrange); no Cantor gaps
aperiodic quasicrystal	**HLV** (Krüger)	low degeneracy; *intermediate* ("critical") $\langle r \rangle$; high gap richness (Cantor hierarchy) – expected, distinguishable from both controls
random substrate	control model	$\langle r \rangle \approx 0.50$ (GOE 0.531); no degeneracy, no gap hierarchy

The discriminators ($\langle r \rangle$ = consecutive level-spacing ratio, degeneracy fraction, gap richness) are dimensionless and need no unfolding; the absolute scale plays no role. Axis 1 is therefore not merely a picture but an *experiment*: the test of whether a given spectrum is periodic, aperiodic, or random. FFGFT contributes the periodic-arm prediction (the rational $\lvert n \rvert^2$ ladder), HLV the aperiodic substrate; the guardrail "$X = \varphi^N$ or $X = \xi^N$ is not a prediction" (Doc. 190 R35) holds for both.

J.3 Axis 2 – the primitive (what everything follows from)

A large part of the IPI field has *no* geometric substrate; there the crystal axis does not apply, and a second organizing term is needed: what counts as fundamental?

- **Geometry-primitive: FFGFT** – everything follows from T^4 geometry and the single dimensionless constant $\xi \approx 1/7500$. Static, observer-free, no selector.

- **Information-primitive: Vopson** – mass-energy-information equivalence (MEI), second law of infodynamics, simulation hypothesis. Information is physical and fundamental. Downstream: **Leavitt**, whose bit-budget construction sits directly on Vopson's MEI.
- **Algebraic / Hilbert-space: UMF** (Gericke) – a prime-indexed Hilbert space $\ell^2(P_N)$ with coupling bivector B_{OI}.
- **Selection / constraint** (what *selects or constrains* the configuration): **RA/PMT** (Guevara), **RSG** (Austin), **UC** (Haskins), **UIFT** (Teker).
- **Continuum-topological:** αLGQV (Kriger) – S^3 topology, membrane elasticity as metric.
- **Meta / governance** (orders *other* theories, one level up): **Dot Theory** (Vossen).

On this axis the FFGFT contrast with Vopson is not periodic-vs-aperiodic but the *geometric versus the informational side of the same MEI relation*: FFGFT approaches from geometry, Vopson from entropy (Doc. 251). A visible structural difference follows – FFGFT structurally excludes an "outside" or a simulation (there is no external standpoint to a single static whole), whereas Vopson's framework leaves it open.

J.4 The overall map

Framework	Owner	Category (axis)	Relation to FFGFT	Doc.
FFGFT / T0	Pascher	substrate: periodic; primitive: geometry	— (reference point: static, observer-free, no selector)	—
HLV	Krüger	substrate: aperiodic	shared spectral/recovery axis; otherwise a category mismatch	271, 272
Vopson (MEI)	Vopson	primitive: information	geometric vs. informational side of the same MEI relation; outside excluded vs. open	251
(bit budget)	Leavitt	primitive: information (downstream Vopson)	shares Vopson's primitive; bit budget = ΛCDM values as input, no forward content	—
UMF	Gericke	algebraic / Hilbert space	$\ell^2(P_N)$ is a proper subspace of $L^2(T^4)$; $\tau{\cdot}\mu = \hbar/c^2$ both ways	244
RA/PMT	Guevara	selection (meta-scheme)	FFGFT declines the selector slot RA1.5 (static)	245, 269
RSG	Austin	selection / bulk-boundary	asymmetric complementarity: FFGFT supplies scale, RSG selection language	246
UC	Haskins	constraint (static)	both static; α-correction inconsistency (H_0 vs. a_0)	260
UIFT	Teker	holographic boundary	observer-factor-3 is FFGFT's explicit version of the volume→boundary move	267, 268
αLGQV	Kriger	continuum-topological	different substrate type (S^3, membrane); not a lattice	—
Dot Theory	Vossen	meta / governance	orders the field (Lexicon/OAP/Matrix/FAH); no physical substrate	272

Lineage and external contacts (not IPI-forum frameworks, for completeness): *Scale Relativity* (Nottale) is the fractal-spacetime ancestor of FFGFT; the Koide circle (Koide, Rivero, Baez, Kocik, Brannen) concerns mass-formula verification, not the substrate.

J.5 Why so often "a category difference, not a contradiction"

The map explains the recurring comparison result. FFGFT sits on *both* axes at the same structural-geometric pole: periodic crystal (axis 1) and geometry-primitive, static, no selector (axis 2). Most other frameworks build their theory precisely around a *slot* that FFGFT leaves empty in principle or fills differently:

- the **selector** (RA/PMT's RA1.5, RSG's selection principle) – FFGFT is static, nothing is selected;
- the **volume→boundary seam** (UIFT) – FFGFT makes it explicitly as observer-factor-3, but does not resolve the same open question;
- the **constraint** (UC) – FFGFT instantiates structure but formulates no standalone constraint principle;
- the **information primitive** (Vopson) – FFGFT takes geometry as primitive and derives information from it, not the reverse;
- the **external standpoint / simulation** (Vopson open) – structurally excluded in FFGFT.

Hence the findings are rarely contradictions: the frameworks speak about different layers. Where they genuinely *touch* – and can therefore collide empirically – is the substrate axis, and there the spectral discriminator (Doc. 271) provides the only currently sharp, shared test.

J.6 Applying the governance layer: from Matrix entry to A∗

In the governance vocabulary of Vossen (Dot Theory) and Guevara (RA/PMT) this map can be read not only as a static placement but as a *workflow*. The FFGFT–HLV axis is the concrete example – the workflow has already been run, not merely described:

- **Declared projection:** the substrate axis.
- **PRC** (Projection-Relative Classification): FFGFT periodic, HLV aperiodic.
- **PRM** (Projection-Relative Measurement): the spectral discriminator (level-spacing ratio $\langle r \rangle$, gap structure, spectral dimension) – a measurement defined *relative to that projection* and meaningless without it.
- **Residual** (declared, not absorbed): an independent reconstruction of the HLV spectrum ($\mathbb{Z}^6$ cut-and-project, k-NN Laplacian) shows that on the headline statistic $\langle r \rangle$ the quasicrystal is *not yet* separable from the random control; the aperiodic signature survives only weakly, in gap structure and the low-mode sector. This residual is declared, not computed away.
- **Operational accord:** the joint test protocol (periodic null, explicit recovery limit, scale-free ratios), bound by the two contributors and claiming no identity between the frameworks.

The resulting shared spectral testbed is a *Cooperative Admissible Space (A∗)* in the sense of the recent Vossen–Guevara localisation: a declared region in which FFGFT and HLV cooperate *operationally* for a stated purpose without ontological identity. Guevara's observation holds: the accord is itself an object with its own admissibility conditions (the three agreed criteria), its own residual structure (what the matched run shows), and a

revision operator (re-run once HLV supplies its weighted spectrum). It is *operational* rather than merely declared because the measurement can fail – the residual above is the proof that the space is doing real work. O0 and O2 are not the destination here but the instruments through which the accord becomes constructible and reviewable.

This section too is a declared application of others' vocabulary (Vossen, Guevara) and is subject to the originators' correction.

J.7 Honest residue (declared)

- This map is a *declared reading* by the author (Dot-Theory O2). The primary rendering (O0) of each framework remains with its originator; the placements above are to be read as Matrix entries in the sense of Doc. 272, not as attributions.
- The *choice of axes itself* is a declared projection (Section 1), not a universal grid; other axes (Accessibility/Constraint, Observer-Dependence/Recoverability) are equally admissible and would yield a different legitimate map.
- The substrate triad (periodic/aperiodic/random) cleanly orders only FFGFT, HLV, and the control model. UMF (algebraic) and αLGQV (continuum-topological) sit beside the triad; their placement is via axis 2 or the substrate *type*, not the spectral fork.
- Placing Leavitt as "downstream of Vopson" follows from the provenance of his bit-budget construction (MEI), not from a standalone substrate claim.
- Should an originator operationalize their slot – e.g. Guevara casting RA1.5 as an equation, or Vopson stating an isolated, predicted, dimensionless ratio – the relevant row would be re-evaluated. The map is dated and provisional (June 2026).

References: Doc. 244 (UMF/Hilbert space), 245/269 (RA/PMT), 246 (RSG), 251 (Vopson), 260 (UC/Haskins), 267/268 (UIFT seam), 270 (projection chain $T^4 \to T^0$), 271/272 (HLV, pairwise and Dot governance), 190 R35 (ξ^N triviality). HLV: Krüger, Zenodo 10.5281/zenodo.20596861 and 10.5281/zenodo.20275888. Vopson: AIP Advances 2022/2023/2024. RA/PMT: Guevara 2026.

Chapter K

Doc. 275 — From ξ to φ: Recursive Scale Hierarchy

Abstract

We study the recursive damping sequence $r(k{+}1) = r(k)(1-\xi)$, $r(0) = D_f/3$, which follows solely from the fundamental geometric parameter $\xi = 4/30000$ and the fractal dimension $D_f = 3 - \xi$. At recursion depth $k^* = \log(\varphi)/\xi \approx 3609$ the sequence *passes through* the value $1/\varphi$: $r(3609) \approx 1/\varphi$ with absolute error $4{\cdot}10^{-5}$ (relative $6.5{\cdot}10^{-5}$); the exact real $k^*_{\text{exact}} = 3608.51$ hits $1/\varphi$ identically. **Status clarification (revision 11 June 2026):** $1/\varphi$ is *not a fixed point* of the recursion (the unique fixed point is 0) but a passage point; since $k^*(c)$ exists for *every* target $c \in (0,1)$ with identical parametric status, the ξ^N/φ^N triviality guardrail (Doc. 190 P35) applies to this result in full. It would acquire physical content only through an independent fixing of $k^* \approx 3609$ — currently open. The recursion levels are geometric scale levels with the FFGFT-internal factor $q = 2/3$; $(2/3)^{3609}$ is convergent. The numerical validation study was fully corrected (v4) after an independent audit (P. Stoychev, IPI, 11 June 2026): with a genuine 4D implementation of the T^4 graph and dimension-matched 4D null models, FFGFT T^4 lies *inside* the null band; HLV, with a genuine ensemble, also lies inside the band. Earlier contrary findings (v3) were implementation artefacts and are visibly retracted in this document.

K.1 Motivation and Context

Doc. 258 showed that the Koide scalar $Q_{\text{FFGFT}} \approx 0.6677$ follows from the geometric invariants r_e, r_μ, r_τ and the lepton exponents p_e, p_μ, p_τ. This value is an *output* of the theory, not an input parameter.

In parallel, the HLV lattice construction of Krüger [1] uses the golden ratio $\varphi = (1+\sqrt{5})/2$ as the projection ratio in the 6D→3D cut-and-project construction. Krüger's Run E identifies the low-sector gap hierarchy as the only remaining candidate separation channel between HLV and null models.

This document answers three questions:

1. Which scalar is physically correct for the macroscopic geometric structure?

2. How does it connect to ξ?
3. What does $k^* \approx 3609$ mean physically — and why is it not an astronomical scale?

Remark K.1.1 (All three FFGFT scalars are derivable). The question is not which of the three scalars ξ, $1/\varphi$, Q_{FFGFT} *can* be derived from FFGFT — that holds for all three. The question is which one *operates* at which physical level and is therefore responsible for which observables.

K.2 Fundamental Parameters and Scalar Separation

FFGFT works with the following parameter-free fundamental quantities:

$$\xi = \frac{4}{30000} \approx 1.333 \times 10^{-4} \qquad \text{(fractal deviation, microscopic)} \tag{K.1}$$

$$D_f = 3 - \xi \approx 2.99987 \qquad \text{(Hausdorff dimension of } T^4\text{)} \tag{K.2}$$

$$q = \frac{2}{3} \qquad \text{(geometric recursion factor)} \tag{K.3}$$

Definition K.2.1 (Three FFGFT scalars near 0.6).

Quantity	Value	Role
$\xi = 4/30000$	1.333×10^{-4}	fundamental input; fractal dimension; active everywhere
$1/\varphi$	0.6180	macroscopic shell ratio; passage point of the ξ-recursion (output; status see Sect. K.3)
Q_{FFGFT}	0.6677	Koide scalar leptons (output, Doc. 258); particle physics scale only

All three are derivable from FFGFT. They are *not* interchangeable.

K.3 The ξ-Damping Recursion

Definition and Closed Form

Definition K.3.1 (ξ-damping recursion). Let $r(0) = D_f/3 = (3-\xi)/3$ be the shell ratio at the lowest scale level. The recursion reads:

$$r(k+1) = r(k) \cdot (1-\xi), \qquad k = 0, 1, 2, \ldots \tag{K.4}$$

Closed form:

$$r(k) = \frac{D_f}{3} \cdot (1-\xi)^k \approx \frac{D_f}{3} \cdot e^{-\xi k}. \tag{K.5}$$

The physical content: $r(0) = D_f/3$ is the ratio of the fractal dimension to the topological dimension 3. Each scale level damps this ratio by the factor $(1-\xi)$.

Main Observation (Revised): $1/\varphi$ as Passage Point

Remark K.3.2 (Revision 11 June 2026). The original version of this section called $1/\varphi$ a *fixed point* of the recursion and stated the result as a theorem. Both were wrong or overstated: the unique fixed point of $r \mapsto r(1-\xi)$ is $r = 0$; $1/\varphi$ is a *passage point*. The correction follows the audit by P. Stoychev (IPI list, 11 June 2026).

Observation. The sequence $r(k)$ passes through the value $1/\varphi$ at recursion depth

$$k^* = \frac{\log \varphi}{\xi} = \frac{\log\left(\frac{1+\sqrt{5}}{2}\right)}{4/30000} \approx 3609, \tag{K.6}$$

with $r(3609) = 0.6179940013\ldots$ and $|r(3609) - 1/\varphi| = 4.0 \times 10^{-5}$ (relative 6.5×10^{-5}). The exact real $k^*_{\text{exact}} = \big(\log(1/\varphi) - \log\big(D_f/3\big)\big)/\log(1-\xi) = 3608.51$ satisfies $r(k^*_{\text{exact}}) = 1/\varphi$ identically.

Remark K.3.3 (Parametric status — the decisive caveat). The arithmetic is correct, but the statement is *not* distinguished: for *every* target $c \in (0,1)$ there exists

$$k^*(c) = \frac{\log c - \log\big(D_f/3\big)}{\log(1-\xi)} \tag{K.7}$$

with exactly the same parametric status — "no free parameter" holds for all targets alike, because the free exponent has merely been renamed k^* and solved for the target:

Target c	$k^*(c)$
$1/\varphi = 0.6180$	3608.5
$Q_{\text{FFGFT}} = 0.6677$	3028.8
$1/2$	5197.9
$1/e$	7499.2
α_{em}	36899.0

The ξ^N/φ^N triviality guardrail (Doc. 190 P35) therefore applies to this very result. The observation would acquire physical content only if $k^* \approx 3609$ were fixed by something independent (a level count of the construction, an observable) and *then* landed on $1/\varphi$. No such fixing is currently known.

K.4 Scale Interpretation: $q = 2/3$, Not Factor 2

Remark K.4.1 (Critical point). A naive reading might interpret $k^* \approx 3609$ as astronomically many renormalisation-group steps: with factor 2 per step that would be a scale spread of $2^{3609} \approx 10^{1086}$, far beyond any experimental reach.

This objection misses the FFGFT structure.

The four torus lengths follow $L_\mu = L_0 \cdot q^\mu$ with $q = 2/3$. The cumulative scaling after k levels is $S(k) = (2/3)^k \to 0$.

Proposition K.4.2 (Convergence with FFGFT scaling)**.** *Comparison:*

Interpretation	*Scale factor at $k^* = 3609$*	*Character*
RG, factor 2 (external convention)	$2^{3609} \approx 10^{1086}$	*divergent*
FFGFT, $q = 2/3$ (internal)	$(2/3)^{3609} \approx 10^{-635}$	*convergent*

$k^* \approx 3609$ is a natural number within the theory's scale spectrum, not an astronomical limit.

K.5 Spectral Validation Study (Three Layers)

Design

Script version v4 (revised after audit). Two separate sectors are compared, each against dimension-matched null models: *4D sector* — FFGFT T^4 as a kNN graph in genuine 4D coordinates (anisotropic torus lengths $L_\mu = 2\pi(2/3)^\mu$, anisotropy $L_0/L_3 = 3.38$) against three 4D nulls (isotropic cubic, jittered anisotropic, random anisotropic box); *3D sector* — HLV G-Lattice (kNN graph, now with a genuine seed ensemble $\sigma = 0.05$) against three 3D nulls (jittered cubic, random, cubic). $N = 2000$, $k = 16$, three seeds for all models; uniqueness of all point sets is enforced by assertion.

Remark K.5.1 (What was wrong in v3 — audit P. Stoychev). (i) The v3 "T^4 graph" used the grid coordinate n_3 only in the acceptance norm, not in the position: of 2000 points only 729 were unique 3D positions; the $1/d^2$ weighting with clamp produced Laplacian eigenvalues up to 2.5×10^{12}. (ii) `hlv_points(seed)` never used its seed — the "ensemble" was the same computation three times. (iii) The shell detector was structurally biased for targets < 1 (see Sect. K.5). All three errors are fixed in v4; in addition, 4D models are compared only against 4D nulls, since comparing a 4D graph against 3D nulls would again be unfair.

Layer 1 Gap-hierarchy metrics: max. gap, gap-CV, low-sector $\langle r \rangle$, IDS — analogous to Krüger Run E.

Layer 2 Shell structure: $1/\varphi$ vs. Q_{FFGFT} on the HLV point cloud at $N = 2000$.

Layer 3 Numerical verification of k^* and $r(k^*)$.

Remark K.5.2 (Heuristics in the script). The shell-structure function uses a 5% threshold for shell detection. This is a heuristic; for a publication the radial distribution function $g(r)$ would be more robust. The ensemble size (3 seeds) is illustrative, not statistically secured.

Results Layer 1 (v4): Gap Hierarchy ($N = 2000$, 3 Seeds)

Remark K.5.3 (Retraction of the v3 result). The result reported in the first version ("FFGFT T^4 graph outside the null band, gap_CV $= 1.68 \pm 0.09$") is **retracted**: it was entirely an artefact of point duplicates (729 unique of 2000 points) combined with the $1/d^2$ clamp. The precision statement "HLV $\approx$ jittered, $\Delta < 0.001$ across 3 seeds" is also withdrawn — it compared a single realisation with itself (seed bug).

Table K.1: Gap CV (low sector, n=100 modes, N=2000, mean ± std over 3 seeds), v4: separate sectors with dimension-matched null models.

Sector	Model	gap_CV	Band status
4D	**FFGFT T^4 (anisotropic)**	1.329 ± 0.025	in band
4D	Null: isotropic cubic	2.406 ± 0.010	band edge (top)
4D	Null: jittered anisotropic	0.690 ± 0.026	band edge (bottom)
4D	Null: random anisotropic box	0.740 ± 0.037	–
3D	**HLV G-Lattice** (σ=0.05)	0.710 ± 0.008	in band
3D	Null: jittered cubic	0.768 ± 0.011	–
3D	Null: random	0.675 ± 0.043	band edge (bottom)
3D	Null: cubic	0.978 ± 0.010	band edge (top)

Findings:

1. **No model separates.** FFGFT T^4 lies at 1.329 inside the 4D null band $[0.690,\ 2.406]$; HLV lies at 0.710 inside the 3D null band $[0.675,\ 0.978]$.
2. **The isotropic 4D lattice itself has the highest gap_CV** (2.406) — regular lattices produce strong eigenvalue degeneracies in 4D. A high gap_CV is therefore not a T^4 signature but a generic order signature.
3. **Robustness checked:** the T^4 result is jitter-stable ($\sigma = 0.02$–0.10: $1.43/1.35/1.35$) and consistent for $n_{\text{range}} = 4/5$ ($1.35/1.40$); $n_{\text{range}} = 3$ (2.28) is a small-size artefact.
4. **HLV with a genuine ensemble** remains in the band and close to jittered cubic — the v3 *finding* (consistent with Krüger's Run E) survives, the precision claimed then does not.

Remark K.5.4 (Consequence for the comparison). At $N = 2000$ the gap_CV diagnostic separates neither the T^4 graph nor HLV from dimension-matched null models. This *reinforces* Krüger's conservative Run-E finding from an independent pipeline and further constrains the common-spectral-signature hypothesis of Doc. 271: at this size, no signature is demonstrable at the graph level.

Remark K.5.5 (Addendum (15 June 2026): Krüger's PHI5 confirms phi non-specificity). Krüger's *corrected* pairwise audit (PHI5, June 2026) is *negative for phi uniqueness*: phi is not the strongest outlier; other irrational controls reach equal or stronger one-vs-rest / pairwise separability. From Krüger's *own* pipeline – and consistent with the independent pairwise audit – this confirms the reading taken here: φ is a *generic ordering signature, not a T^4- or HLV-specific feature*. Krüger has accordingly reset the HLV programme to a conservative hybrid carrier geometry (native 6D backbone, dual boundary loops, ICE fibre, non-exact holonomy; pre-declared kill tests HICE0–HICE7) and no longer makes a phi-specificity claim. An independent HICE pilot implementation reproduces his DICE0 status ("precondition not satisfied"): gauge-null, null-model separation and destructive tests pass, but benign-stability fails (phason/window sensitivity of the cut-and-project). For FFGFT this changes nothing – the framework anchors in the Euler Tonnetz, not the golden ratio; the finding only supports the statement above.

Results Layer 2: Shell Structure — Retracted

Remark K.5.6 (Retraction of the v3 discrimination). The discrimination reported in the first version ($\mathrm{err}(Q_{\mathrm{FFGFT}}) < \mathrm{err}(1/\varphi)$, $\Delta = 0.145$) is **retracted**. The detector identifies shells as a strictly increasing distance sequence (factor ≥ 1.05); for any target ratio $c < 1$ the minimum achievable error is $(1.05/c - 1)$: 0.699 for $1/\varphi$, 0.573 for Q_{FFGFT}. *The larger of two sub-1 ratios wins on any point set whatsoever* — the script's own controls showed this (cubic: 0.976 vs. 1.135; random: 1.194 vs. 1.371) but were not read as an alarm. The finding was structural, not physical (audit P. Stoychev).

In v4 the detector raises an exception for targets ≤ 1; only ratios > 1 are tested. Reference value: $\mathrm{err}(\varphi{=}1.618)_{\mathrm{HLV}} = 0.257$; the best-fit consecutive shell ratio of the HLV point cloud is, per Stoychev's measurement, ≈ 1.15 — neither φ, $1/\varphi$, nor Q_{FFGFT}. For future versions, the radial distribution function $g(r)$ with peak-position analysis is the more robust route.

Result Layer 3: Numerical Verification

$$k^* = \frac{\log \varphi}{\xi} \approx 3609.1, \qquad r(3609) = 0.6179940013\ldots \approx \frac{1}{\varphi} \quad (|\text{error}| = 4.0 \times 10^{-5},\ \text{relative } 6.5 \times 10^{-5}). \tag{K.8}$$

The arithmetic is correct; the error statement was made precise in v3 (previously incorrectly $< 10^{-9}$). For the *status* of the statement — passage point, not fixed point; $k^*(c)$ exists for every target — see Sect. K.3: "without free parameters" does not distinguish this result.

K.6 Discussion

Relation to the φ^N-triviality warning (Doc. 190 P35) — Revised

Remark K.6.1 (Retraction of the demarcation). The first version of this section gave three arguments why the triviality warning would *not* apply to the k^* result ("no free exponent", "no empirical anchor", "no external calibration"). This demarcation is **retracted**: all three arguments overlook that the free exponent has merely been *renamed* k^* and *solved for* the desired target. $k^*(c)$ exists for every $c \in (0, 1)$ with identical properties (table in Sect. K.3); the selection of the target $1/\varphi$ is the hidden choice. P35 applies in full — this is exactly the case the guardrail was written for, and applying it to one's own result is its first serious corpus-internal stress test (passed: the result is downgraded).

What remains: the three-level separation of φ-roles documented in Doc. 190 P37 — (1) microscopic/ξ-level, (2) macroscopic/$1/\varphi$ as passage point of the recursion, (3) particle physics/Q_{FFGFT} — remains valid and useful as a *conceptual separation*. Doc. 188 uses φ^{-1} as a phenomenological scaling factor between fractal levels; whether this factor follows structurally from the ξ-recursion is, after the present revision, again an *open question*: the recursion delivers it only if the depth $k^* \approx 3609$ is independently justified.

Position in the Document Series

Doc. 270 describes the reduction chain $T^4 \to T^0$ as two qualitatively different operation types. The present document adds the *scale hierarchy*: the $q = 2/3$ recursion connects the microscopic ξ-level with the macroscopic φ-level through cumulative geometric damping — without a qualitative change in operation type.

Doc. 271 provides a structural comparison of FFGFT and HLV. The present document makes the spectral comparison announced there concrete through the numerical script and identifies gap_CV as a shared diagnostic channel.

Open Points

1. Shell diagnostics on a $g(r)$ basis (peak positions, ratios > 1) — the v3 detector is unusable for sub-1 targets (Sect. K.5).
2. Pre-declared ensemble band tests analogous to Krüger Run E (metrics, bands and seeds fixed before the computation) for the 4D T^4 graph.
3. Larger N and more seeds: the present 3-seed ensembles are illustrative, not statistically secured.
4. An independent fixing of $k^* \approx 3609$ (a level count of the construction or a physical observable) — without it the passage-point observation has no physical content (Sect. K.3). Currently not known.
5. Cross-validation of the v4 implementation against the deduplicated/seeded variants offered by P. Stoychev.

Epistemic Limit: Determinism and Resonance Density

The open question about the independent fixing of k^* is not accidental but reflects a fundamental methodological limit documented elsewhere in the corpus (Doc. 038, Markov determinism).

The recursion $r(k{+}1) = r(k)(1 - \xi)$ is fully deterministic: every passage value $c \in (0, 1)$ exists at a precisely determined depth $k^*(c)$. However, the recursion produces no distinguished discrete levels — it is continuous, and the passage points lie so densely that without an independent external anchor, no single one can be singled out. The resonance pattern is too dense for unambiguous identification from the value alone.

This is structurally identical to two other cases in the corpus:

- **Large semiprimes (Doc. 176):** The factors exist uniquely, but the search is practically infeasible because the candidates lie too densely.
- **Quantum mechanics (Doc. 038):** The substrate is deterministic; a statistical description becomes necessary when micro-states are no longer resolvable — not because physics is stochastic, but because epistemic resolution is missing.

The conclusion for this document: the non-fixability of k^* is not a flaw of the theory but an honest limit of the methodology. The theory is fully determined; the question of which passage point is physically distinguished is practically unanswerable without an external anchor. This is precisely why Doc. 190 P35 protects against reading a finding

from the resonance density when there is none. The status of Doc. 275 — observation, open — is the correct consequence of this limit.

K.7 Conclusion

1. The ξ-damping recursion passes through $1/\varphi$ at $k^* = \log(\varphi)/\xi \approx 3609$; the arithmetic is correct (error $4{\cdot}10^{-5}$), but $1/\varphi$ is a *passage point, not a fixed point*, and $k^*(c)$ exists for every target — the P35 triviality guardrail applies; without an independent fixing of k^* the observation has no physical content.
2. With $q = 2/3$, $(2/3)^{3609}$ is convergent and internally consistent, not an astronomical limit — this scale interpretation stands.
3. **v4 main result ($N = 2000$, dimension-matched nulls):** No separation. FFGFT T^4 (genuine 4D, gap_CV $= 1.329 \pm 0.025$) lies inside the 4D null band $[0.690, 2.406]$; HLV (genuine ensemble, $0.710{\pm}0.008$) lies inside the 3D null band $[0.675, 0.978]$. This reinforces Krüger's conservative Run-E finding from an independent pipeline.
4. The v3 findings "T^4 outside the band" and "Q_{FFGFT} discriminates against $1/\varphi$" are **retracted** (duplicate artefact and detector bias, respectively); the HLV ensemble seed bug is fixed. All three errors trace back to the audit by P. Stoychev and are registered as K4 in Doc. 190.
5. The three scalars ξ, $1/\varphi$, Q_{FFGFT} remain useful as a conceptual separation at different physical levels (Doc. 190 P37); the *derivation claim* for $1/\varphi$ from the recursion is downgraded to an open question (Doc. 190 P38).

Connections. Doc. 258 (Q_{FFGFT}, Koide ladder), Doc. 270 (reduction chain $T^4 \to T^0$), Doc. 271 (FFGFT-HLV structural comparison), Doc. 272 (HLV/Dot-Theory context). The Python script `ffgft_hlv_gap_v4.py` (corrected version; v2/v3 remain in the archive as documented error states) is publicly available and reproducible.

Acknowledgement. Plamen Stoychev (IPI) for the independent audit of 11 June 2026, which identified all three script errors and forced the status clarification of the k^* result. The revision of this document implements his audit in full.

[1] M. Krüger, *Operational Falsification Pathways for the HLV G-Lattice*, Zenodo (2026), https://doi.org/10.5281/zenodo.20615649.

[2] J. Pascher, *Doc. 258: Geometric ladder $q = 2/3$*, FFGFT corpus (2026).

[3] J. Pascher, *Doc. 270: Dimensional change in FFGFT*, FFGFT corpus (2026).

[4] J. Pascher, *Doc. 271: FFGFT compared with HLV*, FFGFT corpus (2026).

[5] J. Pascher, *Doc. 190: Correction register FFGFT/T0 (K1–K4, P1–P38)*, FFGFT corpus (2026).

[6] J. Pascher, *Doc. 009: Origin of $\xi = 4/30000$*, FFGFT corpus (2026).

[7] J. Pascher, *Doc. 188: Geometric foundations of FFGFT*, FFGFT corpus (2026).

Chapter L

Doc. 276 — Homology Test of the HLV Vortex Hypothesis

Abstract

Marcel Krüger's HLV G-Lattice programme [1] formulates a finite 6D→3D cut-and-project construction and tests it through a ladder of pre-registered falsification runs (A–K7 plus the D1–D3 cell-geometry addendum). The D1–D3 audit *withdrew* the original claim of exact dodecahedral Voronoi cells and replaced it with a narrower profile: a non-crystalline, mixed-face, φ-skewed cell ensemble. The revised vortex axiom therefore no longer requires a Platonic cell, but closed cycles, phase continuity, and holonomy-carrying loops. Exactly here the FFGFT side contributes something *additive*: topological winding is a homotopy invariant; it lives in π_1 of the complex, not in the shape of its faces. We present a self-contained, reproducible test script (companion file, `numpy`/`scipy` only; `gudhi` optional) that operationalises this statement: over a kNN filtration it computes the GF(2) homology $b_1(\varepsilon)$, compares the *whole* normalised Betti-curve shape against control clouds by an energy-distance permutation test, runs a deformation null test (persistence vs. drift, with optional persistent-cycle matching), and scales the effect size with N. **Finding (honest, conservative):** the *scalar* total-persistence measure does not cleanly separate HLV and is k-sensitive; the *curve shape*, by contrast, separates HLV significantly from all controls ($p < 0.05$), sharply from crystalline lattices (FCC, cubic, BCC) and only weakly from the generic $\sqrt{2}$ quasicrystal. N-scaling shows that separation from crystalline grows/stabilises (structural) while separation from random shrinks. The discriminating feature is therefore *"quasiperiodic cut-and-project structure"*, *not* the golden ratio specifically. A separate, less circular diagnostic (longest persistent H_1 lifetime) additionally shows that cut-project clouds (HLV *and* the $\sqrt{2}$ quasicrystal) host a markedly longer-lived 1-cycle than lattices and random – recorded as an observation, explicitly *not* oversold as "topological protection".

L.1 Context: HLV, FFGFT, and the division of labour

Krüger's HLV (Helix–Light–Vortex) programme [1] tests a finite geometric substrate as a *candidate object*, not as an ontology. Its methodological strength is the pre-registered

falsification governance [2]: positive, negative, partial and underdetermined outcomes are kept apart on a claim-state ledger (*candidate / restricted / underdetermined / rejected / validated candidate*); a positive result may not erase a negative residual from another channel.

The D1–D3 audit rejected the strong claim "HLV produces exact dodecahedral Voronoi cells" – the exact 12-face channel is dominated by FCC/rhombic and HCP-like controls. What survives is the weaker, better-supported statement of a *fresh-seed validated candidate*: a non-crystalline, mixed-face, φ-skewed finite Voronoi cell ensemble. The resulting reformulation of the vortex axiom reads, in essence: vortex and boundary modes are supported by the edge, face-boundary, loop, and defect-interface structure of a mixed-face complex – not by an exact cell shape.

Observation. Topological winding requires no Platonic cell. It requires closed graph cycles, phase continuity, and holonomy-carrying loops. These structures live in π_1 of the complex and are invariant under the cell shape.

This is also the FFGFT position: on the compact T^4 the relevant invariants are holonomy classes of $\pi_1(T^4) = \mathbb{Z}^4$, deliberately metric- and cell-independent (cf. Doc. 270 [7], Doc. 271 [4]). The division of labour remains complementary, not overlapping: HLV as the boundary-pattern residual test, FFGFT as the scale-law / effective-L_ξ test (Doc. 275 [3]).

L.2 The test method

Complex and boundary operator

Instead of counting Voronoi faces, we build the clique complex of the symmetric kNN graph (the same graph object as in Krüger's runs, k adjustable) and read the cycle/holonomy sector off the GF(2) homology

$$H_1 = \ker \partial_1 / \operatorname{im} \partial_2, \qquad b_1 = \dim \ker \partial_1 - \operatorname{rank} \partial_2 = (E - \operatorname{rank} \partial_1) - \operatorname{rank} \partial_2 .$$

Here $\operatorname{rank} \partial_1 = N - (\#\text{components})$ (union-find), and $\operatorname{rank} \partial_2$ is computed over GF(2). b_1 is the cell-shape-independent winding/holonomy capacity of the complex.

Language precision. H_1 is shape-independent *within a fixed complex*; it still depends on k and the metric. Hence (i) the k-sweep and (ii) the comparison of the *whole* curve shape rather than a single "tessellation-free" number.

Filtration and curve shape

A dense kNN complex is nearly contractible; the holonomy structure becomes visible only through a radius filtration. We evaluate $b_1(\varepsilon)$ on a grid $\varepsilon/d_0 \in \{1.1,\ 1.35,\ 1.6,\ 1.85,\ 2.1,\ 2.35\}$ (d_0 = median nearest-neighbour distance) and compare the *normalised* curve $\hat{b}_1 = b_1 / \sum b_1$ – the shape, not the area. The scalar total persistence $\int b_1 \, \mathrm{d}\varepsilon$ is kept only as a coarse single-number summary.

Statistics and controls

Each cloud is promoted to an ensemble over seeds (the same "HLV-as-an-ensemble" discipline as Krüger's Run E). Scalars are compared by a permutation test, curve shapes by an energy-distance permutation test. Controls: FCC, simple cubic, BCC (crystalline); jittered cubic, uniform random, Poisson-disc (disordered, against density artefacts); generic $\sqrt{2}$ cut-and-project (quasicrystal without the golden ratio).

Deformation null test

We morph HLV $\to$ control continuously, $x(t) = (1-t)\,x_{\text{HLV}} + t\,x_{\text{control}}$, and track total persistence. With `gudhi` present we additionally track the *longest* H_1 lifetime (Rips persistence) – far less circular, since a single long bar is not forced to 0. Caveat: a target with $b_1 \approx 0$ (e.g. FCC) forces the endpoint to 0; clean cycle matching is only possible via persistent homology.

N-scaling and pre-registered consistency check

The effect size (L1 distance of mean curve shapes, HLV vs. the pooled crystalline profile) is scaled with N: stabilising $\Rightarrow$ structural, vanishing $\Rightarrow$ finite-size artefact. The `--falsify` switch implements a *pre-registered consistency check* (fixed N, k, p; crystalline controls only) – explicitly *not* a Popperian falsification test: a negative result weakens the claim but does not logically falsify it.

L.3 Results

All numbers are from a real run of the companion script ($N = 240$, $k = 14$, seed base 20260613, 5 seeds).

Scalar (Check 1/1b). The total-persistence measure is unstable and k-sensitive: the simple-cubic control dominates (~ 130 vs. HLV ~ 50), and over $k \in \{10, 14, 18\}$ HLV lies above the control band in $0/3$ cases. Claim-state: *underdetermined*. This is precisely why the scalar is discarded and the shape compared.

Curve shape (Check 1c). The energy-distance test separates HLV from *all* controls ($p < 0.05$):

Control	energy distance	p
jittered cubic	2.81	0.0015
uniform random	2.79	0.0060
Poisson-disc	2.71	0.0030
FCC (rhombic-dodeca.)	1.93	0.0075
cubic	1.67	0.0040
BCC	1.11	0.0060
generic QC ($\sqrt{2}$)	0.15	0.0085

The separation is sharp against crystalline lattices and very weak (smallest distance 0.15) against the generic $\sqrt{2}$ quasicrystal.

Longest H_1 lifetime (with `gudhi`, max edge $= 2\,d_0$).

Cloud	raw	normalised ($/d_0$)
HLV (golden cut-project)	2.07	0.87
generic QC ($\sqrt{2}$)	2.01	0.85
cubic	0.41	0.41
uniform random	0.25	0.43
BCC	0.13	0.15
FCC	0.00	0.00

N-scaling. The mean-shape L1 distance HLV-vs-crystalline grows/stabilises with N ($\approx 0.56 \to 0.67 \to 1.61 \to 1.51$ for $N = 200/400/600/800$) – structural, not a pure finite-size artefact; HLV-vs-random shrinks accordingly.

Deformation and consistency check. In the deformation null test both total persistence and longest H_1 lifetime fall; at this N there is *no* robust plateau (status: underdetermined – honest). The pre-registered consistency check (fixed $N, k, p < 0.01$; FCC/cubic/BCC only) returns *PASS*: HLV separates from every crystalline control.

L.4 Discussion and claim-state

The defensible statement is precise and conservative:

1. HLV separates *structurally* from crystalline lattices (FCC/cubic/BCC) via the $b_1(\varepsilon)$ curve shape; the effect grows with N (claim-state: *candidate*, shape-based k-robust).
2. HLV is nearly indistinguishable from the generic $\sqrt{2}$ cut-and-project quasicrystal on *every* metric (shape energy distance 0.15; longest cycle raw 2.07 vs. 2.01). The discriminating feature is therefore *"quasiperiodic cut-and-project structure"*, *not* the golden ratio specifically.
3. cut-project structures host a markedly longer-lived H_1 cycle (normalised 0.85–$0.87\,d_0$, raw ~ 2) than lattices and random (normalised $\leq 0.43\,d_0$). This is a real observation of the less circular persistence diagnostic; it is *not* interpreted as "topological protection" – that would be the same overclaim with the opposite sign.

Methodologically the script confirms the claim-state discipline of the HLV programme: the obvious scalar measure fails, and progress lies in the shape-based, ensemble- and permutation-supported comparison. Stronger statements require larger N, more seeds, exact repository adjacency, and independent reimplementation.

Reproducibility. The companion script (`Dok276_Skripte / ffgft_hlv_homology_check_v3.py`) depends only on `numpy`/`scipy`; `gudhi` and `matplotlib` are optional (true persistence diagrams and the figure, respectively).

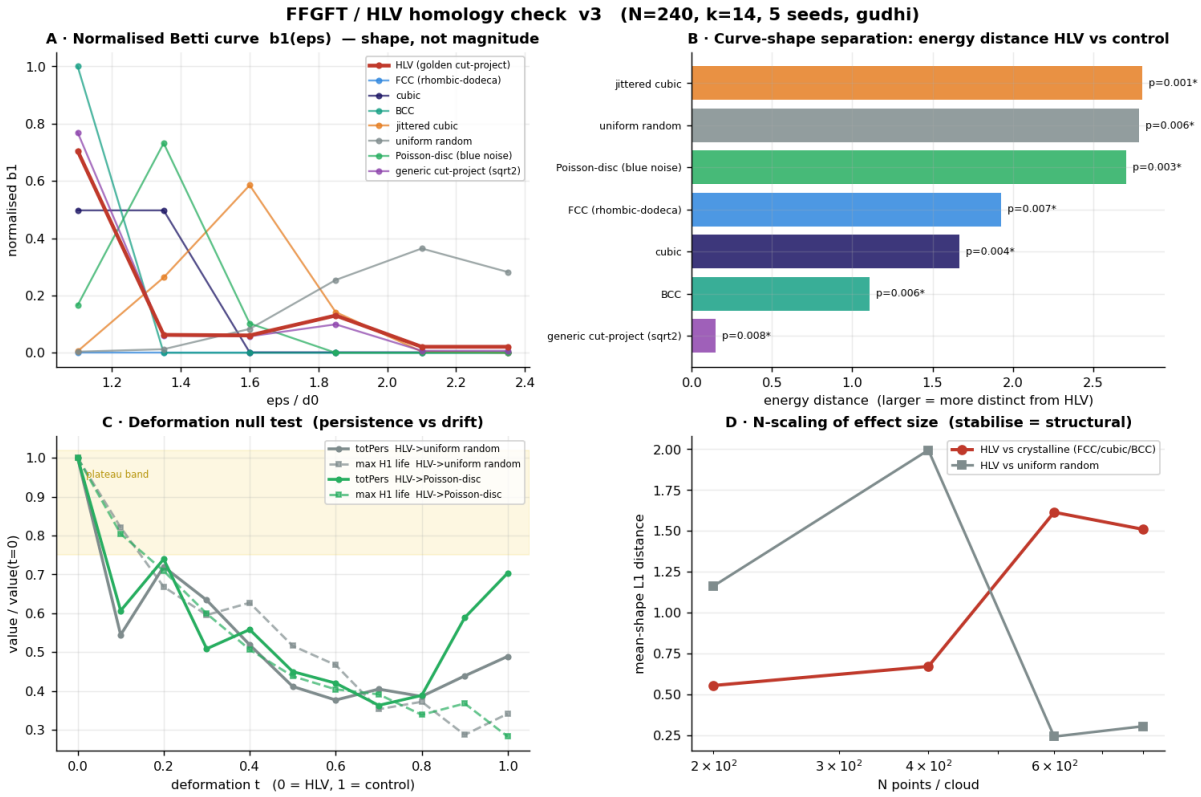


Figure L.1: Four-panel overview from the companion script (v3). A: normalised Betti curves $b_1(\varepsilon)$ – shape, not magnitude. B: curve-shape energy distance HLV vs. controls. C: deformation null test (total persistence and, with `gudhi`, longest H_1 lifetime). D: N-scaling of the effect size (stable $=$ structural). Finite diagnostic; no claim about physical spacetime microstructure.

Deliberately no hard dependency – anyone on the IPI list can reproduce the result without a build step.

Acknowledgement. Marcel Krüger (HLV G-Lattice, falsification governance) for the open exchange and the pre-registered test architecture on which this document builds.

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Chapter M

Doc. 277 — Wave Propagation and Energy

Abstract

A propagating wave *always* carries a well-defined, positive energy: electromagnetically through the Poynting flux, quantized as $E = \hbar\omega$ per quantum, mechanically through radiation pressure. This document cleanly separates two levels that are constantly conflated in popular accounts: the *local* energy balance (always satisfied, $\partial_t u + \nabla\cdot \mathbf{S} = 0$ resp. $\nabla_\mu T^{\mu\nu} = 0$) and the *global* balance, which in curved, time-non-stationary spacetime is not uniquely definable. The widespread fallacy "energy is not conserved" (meaning: *destroyed*) mistakes a level of bookkeeping for physical destruction. In the FRW model the non-definability of total energy follows from the absence of a timelike Killing field (Noether); it is not a violation of local physics. In the FFGFT reading $\hbar$ and c are SI conversion factors; the physical content lies in the dimensionless ratios ($\tilde{T}\cdot m = 1$). The measured absolute energy is observer-dependent, the ratio structure invariant. On a *static* compact 4-torus T^4 the time-translation symmetry exists again; by Noether the total energy is then well-defined and conserved. Redshift thereby becomes a pure scale- and observer-rebooking rather than metric stretching of space, and the main motivation for dark energy (accelerated expansion) as well as part of the motivation for dark matter (CMB peak fit) disappears – the relevant structure follows in FFGFT from the T^4 geometry and from ξ (Doc. 267/268). This cosmological conclusion is framed at the status of the degeneracy thesis (Doc. 267): on the *data* both models are equivalent, on the theoretical *structure* FFGFT is the more economical reading (no dark sector, global energy conservation). R_H is here a shared empirical scale reference, not a free parameter – not an empirical knock-out of ΛCDM, but an economy argument.

M.1 Framing: why this separation is necessary

The statement "a wave without energy would not be a wave" is trivially true and yet the starting point of a stubborn misunderstanding. In the cosmological discussion the correct technical observation "in an expanding universe there is no globally unique total

energy" is regularly turned into the false conclusion "energy is destroyed". The two live on different levels.

Definition M.1.1 (Two levels of balance)**. Local balance** The continuity equation of the energy–momentum tensor, $\nabla_\mu T^{\mu\nu} = 0$, resp. in flat space $\partial_t u + \nabla\cdot\mathbf{S} = 0$. A statement about *flux*: energy vanishes nowhere, it flows. Always satisfied.

Global balance The existence of a unique, conserved *total energy* of the system. By Noether tied to time-translation symmetry; in non-stationary curved spacetimes not uniquely definable.

Warning M.1.2 (The category error). "Globally not uniquely definable" is a statement about the *bookkeeping level*. "Physically destroyed" would be a violation of local physics – and that never occurs. Equating the two is the actual error.

This document works out the short note of the same title, substantiates the individual steps, and embeds them in the corpus: in the units question (Doc. 166, 182, 230), in the reduction structure (Doc. 270), and in the cosmological degeneracy (Doc. 267/268).

M.2 Local energy balance – always satisfied

Electromagnetic field

For the electromagnetic field the energy density and the Poynting vector are [2, 3]

$$u = \frac{\varepsilon_0}{2}\,|\mathbf{E}|^2 + \frac{1}{2\mu_0}\,|\mathbf{B}|^2, \qquad \mathbf{S} = \frac{1}{\mu_0}\,\mathbf{E}\times\mathbf{B}, \tag{M.1}$$

with $\mathbf{S}$ as the energy flux density. The energy continuity equation reads

$$\frac{\partial u}{\partial t} + \nabla\cdot\mathbf{S} = -\,\mathbf{J}\cdot\mathbf{E}, \qquad \text{in vacuum} \quad \frac{\partial u}{\partial t} + \nabla\cdot\mathbf{S} = 0. \tag{M.2}$$

It is a *local* conservation statement: any decrease of the energy density at a point is exactly balanced by outflowing flux. Energy vanishes nowhere.

Quantization and radiation pressure

When quantized, each quantum carries

$$E = \hbar\omega, \qquad p = \hbar k = \frac{E}{c}, \qquad P_{\text{rad}} = \frac{S}{c}, \tag{M.3}$$

with momentum p and radiation pressure P_{rad}. As long as the wave propagates, energy flows – with measurable flux density, measurable pressure, measurable photoelectric effect.

Insight M.2.1 (Operational content)**.** The local balance is non-negotiable because it is coupled to three independent measurable quantities: flux density S, radiation pressure $P_{\text{rad}} = S/c$, and the photoelectric threshold via $E = \hbar\omega$. A "propagating wave without energy" would contradict all three.

M.3 Redshift: observer-dependent, not a loss into nothing

Under cosmological redshift the measured wavelength scales; a given observer measures $E = hc/\lambda$ correspondingly smaller. This does *not* mean that energy disappears, but that the question "how much energy" is frame- resp. observer-dependent.

Proposition M.3.1 (Frame dependence of the photon energy)**.** *The photon energy is the time component of a four-momentum, $E = p^\mu u_\mu$, i.e. the scalar product with the observer's four-velocity u_μ. It is therefore not an invariant quantity but observer-dependent. An observer comoving at the time of emission measures the full energy; an observer in relative motion measures a different value. What remains* invariant *are ratios (frequency, energy, mass ratios).*

Energy here is thus a relative, observer-dependent quantity – not a destroyed one. This statement is model-independent (it holds in ΛCDM just as in FFGFT); what differs between the models is the *interpretation* of z (Section M.7).

M.4 The global fallacy – Noether precisely

Theorem M.4.1 (Noether: energy conservation from time-translation symmetry)**.** *A globally conserved total energy exists if and only if the spacetime possesses a timelike Killing field ξ^μ, i.e. a time-translation symmetry $\nabla_{(\mu}\xi_{\nu)} = 0$ [1, 6, 5].*

An expanding FRW universe possesses no timelike Killing field, hence no global time-translation symmetry, hence no uniquely defined global total energy. This is a statement about the *non-existence of a unique global balance*, not about physical destruction. Locally,

$$\nabla_\mu T^{\mu\nu} = 0 \tag{M.4}$$

always holds.

Remark M.4.2 (Where this usually goes wrong). The common crude error reads "energy is simply not conserved" – meaning destroyed. The correct statement is: in FRW the *global* bookkeeping is not unique because the symmetry argument of Theorem M.4.1 does not apply. Equation (M.4) is unaffected by this.

M.5 Absorption: energy is converted, not destroyed

Upon absorption the wave gives up its energy and changes form: photoelectric effect, heating, above threshold pair production. The kinematics requires a second partner: a single free photon cannot create a pair because of simultaneous energy and momentum conservation.

$$\gamma\gamma \to e^+e^- \quad \text{(Breit–Wheeler [4])}, \qquad \gamma + N \to e^+e^- + N \quad (E \geq 2m_e c^2). \tag{M.5}$$

Here too the local balance (M.2)/(M.4) remains satisfied at all times: the energy changes form, not existence.

M.6 FFGFT view: energy as a partly unit-bookkeeping quantity

In FFGFT $\hbar$ and c are SI conversion factors, not ontological primitives; the physical content lies in the dimensionless ratios. The fundamental relation

$$\tilde{T} \cdot m = 1 \qquad \text{(natural units } \hbar = c = 1\text{)} \tag{M.6}$$

makes this explicit: $E \leftrightarrow 1/\tilde{T}$ and $m \leftrightarrow 1/\tilde{T}$.

Insight M.6.1 (The ratio structure is invariant)**.** "How much energy" depends on the unit and observer convention; the ratio structure (mass, frequency, energy ratios) is invariant. This is exactly consonant with Proposition M.3.1: the absolute value of the measured energy is observer-dependent, the invariant ratios remain.

In this reading "light → matter" is the binding of free phase (light) into mode content (mass): the energy changes the level of accounting, not its existence. The representational side of this statement – that the same 4D content is losslessly translatable between T^4 geometry and Hilbert-space language – is worked out in Doc. 230; the reduction chain $T^4 \to T^0$ with the duality decompactification of the mass dimension via $\tilde{T} \cdot m = 1$ in Doc. 270.

Remark M.6.2 (Unit bookkeeping instead of ontology). The label "partly unit-bookkeeping" is to be read precisely: the *flux* and the *pressure* (Section 3) are operational, unit-independent facts. What hangs on the convention is solely the assignment of an absolute value "energy" to a given mode content – the choice of scale in which one counts.

M.7 Cosmological consequence: static T^4, no dark sector

FFGFT lives on a compact, *static* 4-torus T^4. There is no metric expansion of space – and with it the motivation for dark energy (Λ) disappears, as does part of the motivation for dark matter. This has an immediate, simplifying effect on the energy discussion above.

Proposition M.7.1 (Global energy conservation is restored)**.** *A static T^4 background possesses a timelike time-translation symmetry. By Theorem M.4.1 the total energy is then well-defined and conserved. The fallacy in the context of Theorem M.4.1 ("no unique global energy") is a consequence of the* expansion assumption*; without it one need not retreat to "globally undefinable" in the first place.*

1. Redshift = change of bookkeeping, not stretching of space. Without expansion, redshift is not a stretching of the wavelength by growing space. It is the scale- and observer-rebooking encoded in $\tilde{T} \cdot m = 1$ (M.6): the energy and time scale of the distant source is referenced differently relative to the local standard. The photon carries its energy throughout (Sections 3); z is the conversion between source and observer bookkeeping – exactly the observer-dependent, relative energy of Proposition M.3.1, only without the FRW apparatus.

2. No dark sector needed. ΛCDM needs dark energy to drive an accelerated expansion, and dark matter inter alia to fit the CMB peak structure. If expansion is removed, the main task of dark energy is gone. The relevant structure – e.g. the position of the CMB peaks – follows in FFGFT from the T^4 geometry and from ξ instead of from an expansion-plus-dark fit: the peak ratios are derived in Doc. 268 from the degeneracies $g(|\vec{n}|^2)$ of the T^4 modes, the scale ladder (Casimir/CMB/H_0) in Doc. 166.

Table M.1: Theoretical cost of the two readings of the *same* data.

	ΛCDM pays	FFGFT
Accelerated expansion	dark energy Λ required	not needed (no expanding space)
CMB peak fit	dark matter, inter alia	from $T^4 + \xi$ (Doc. 268)
Global energy	not conserved (FRW, no Killing field)	conserved (Noether, static T^4)
"energy disappears"	lives here as a fallacy	dissolves

Warning M.7.2 (Status: degenerate on the data, asymmetric in cost). On the *data* level the models are degenerate (Doc. 267): SNe Ia, BAO, and CMB temperature do not distinguish between metric expansion (ΛCDM) and fractal time unfolding (FFGFT) – the same data, two readings. On the *structure* level, however, the costs are *not* symmetric (Tab. M.1): ΛCDM carries dark energy and dark matter as free components and gives up global energy conservation; FFGFT pays none of these. Under empirical equivalence, theoretical economy decides – and there FFGFT is the more economical reading. This is a parsimony argument (Occam), not an empirical knock-out of ΛCDM.

Remark M.7.3 (R_H is a scale reference, not an input). R_H *need not* follow from ξ – it is a scale reference like any system-specific upper bound $R_{\text{Torus}}(m) = \hbar/mc$ (Doc. 182). The relation Planck length↔ R_H (exponent $41/4$, $R_H/L_0 = 6.49 \times 10^{64}$) is derived from first principles by *no* framework: why the universe is $\sim 10^{60}$ Planck lengths across is predicted by no theory. ΛCDM fixes the relation via the measured H_0 – itself circular in the sense of the degeneracy (Doc. 267); FFGFT adopts the established value for comparability and writes it in ξ notation (Doc. 190 P35/P36). This is a *shared empirical reference point* of both models, not a free parameter and not an FFGFT-specific gap. In FFGFT, moreover, $\hbar, c$ are SI conversion and G is derived ($\xi \to \{m_e, \ldots\} \to G \to \ell_P$); a "measured" dimensional anchor in the standard-physics sense does not even exist here. The only asymmetry remains the dark sector, carried by ΛCDM alone.

M.8 Methodological placement

The "local vs. global" separation introduced in Definition M.1.1 is structurally the same pattern as elsewhere in the corpus: a deterministic, locally fully determined structure whose global or absolute accounting depends on the chosen level of resolution or convention (cf. Markov determinism, Doc. 038). The error "energy is destroyed" arises when one reads a missing global bookkeeping as a physical statement – exactly the kind of fallacy the correction register (Doc. 190) is written to guard against.

Insight M.8.1 (What this document shows)**.** (i) A propagating wave always carries positive energy; (ii) locally, energy is always conserved; (iii) redshift is observer-dependent,

not a loss; (iv) "globally non-unique" $\neq$ destroyed; (v) in a static T^4 the total energy is even globally conserved and z a pure rebooking. On the *data*, ΛCDM and FFGFT are degenerate; on the *structure*, FFGFT is the more economical reading (Tab. M.1). The cosmic scale R_H is a shared empirical reference point, not a free parameter – the asymmetry lies solely in the dark sector.

M.9 Summary

- Propagating wave $\Rightarrow$ always positive energy (Poynting (M.1), $E = \hbar\omega$ (M.3)).
- Locally, energy is always conserved ($\partial_t u + \nabla\cdot\mathbf{S} = 0$ (M.2); $\nabla_\mu T^{\mu\nu} = 0$ (M.4)).
- Redshift is observer-dependent (Proposition M.3.1) – not a loss into nothing.
- No unique global energy (no time symmetry, Theorem M.4.1) $\neq$ energy destroyed.
- FFGFT: energy is partly unit bookkeeping; physically it is the ratios ($\tilde{T}\cdot m = 1$ (M.6); Doc. 230/270).
- Static T^4 cosmos (no expansion): global energy conservation holds (Proposition M.7.1), redshift is pure rebooking, the dark sector is not needed. Degenerate on the data (Doc. 267/268/166), more economical in cost (Tab. M.1); R_H is a shared scale reference, not a free parameter – the asymmetry is solely the dark sector.

Connections. Doc. 270 (projection chain $T^4 \to T^0$, duality decompactification via $\tilde{T}\cdot m = 1$), Doc. 230 (lossless translation $T^4 \leftrightarrow$ Hilbert space), Doc. 267 (cosmological degeneracy ΛCDM vs. FFGFT), Doc. 268 (CMB peaks from T^4), Doc. 166 (Casimir/CMB/H_0 as a ξ power ladder), Doc. 182 (maximal scale R_H, calibrated), Doc. 190 (correction register, P35/P36).

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Chapter N

Doc. 279 — Casimir, CMB and H_0 — the shared ξ scale

What this is about

Three phenomena that at first have nothing to do with one another: the *Casimir effect* (vacuum force between plates, $\sim \mu$m), the *cosmic microwave background* (CMB, $T_{\mathrm{CMB}} = 2.725$ K) and the *Hubble constant* H_0 (cosmic expansion rate). In FFGFT all three hang on *one* vacuum length scale $L_\xi \approx 100\,\mu$m and on the same exponent $41/4$ of the geometry parameter $\xi = 4/30000$. This note lays the connection open step by step (brought together in Doc. 166 from Doc. 091, 165, 025/061; status assessment per Doc. 190).

Throughout: $\xi = \dfrac{4}{30000} = 1.3333 \times 10^{-4}$.

N.1 Preliminary remark: why “Hubble constant” is misleading

Before we use H_0, a terminological clarification – because the usual designation *Hubble constant* is, from the FFGFT view, misleading on two counts.

First: not a constant. Already in standard cosmology H_0 is not constant but the *present-day value* of the time-dependent Hubble parameter $H(t) = \dot{a}/a$. “Constant” there means only spatially homogeneous, not fixed in time. The name is thus already an established imprecision there.

Second (and sharper): in FFGFT there is no expansion rate at all. The static ξ universe has no scale factor $a(t)$ and no time-varying $H(t)$; the redshift is fractal path lengthening, not an expansion effect (Doc. 190, R14a/R15). There is therefore no “rate” whose present-day value one could take as H_0. The expansion reading is a ΛCDM pipeline interpretation (R18), not an FFGFT statement.

What remains physically is a length. The robust, model-neutral quantity is the *Hubble length*

$$R_H \;=\; \frac{c}{H_0}, \tag{N.1}$$

and $H_0 = c/R_H$ is merely this length expressed, via the SI conversion factor c, as an inverse time. R_H is a *reference length* – the upper scale of the cosmological torus sector – entirely

analogous to the Planck length ℓ_P or to the system-specific bounds $R = \hbar/mc$. It is *not* the size of the universe and *not* the observable horizon (the latter $\approx 3\,R_H$ in ΛCDM); only a lower bound is measurable (Doc. 190, R36).

Terminology convention

In this note the cosmic scale is conceived primarily as a *reference length* R_H, not as a "constant" H_0. Where H_0 appears it is to be read as $H_0 = c/R_H$ – an SI restatement of the reference length, not a fundamental constant of nature. This is consistent with H_0 as an emergent, decompactified quantity (R16) and with the status of R_H as a shared empirical reference point (R39). Precisely for this reason the following derivation works throughout with *lengths* (L_ξ, R_H) and carries H_0 only as their conversion.

N.2 Starting point: three relations from three documents

The three building blocks

1. **Casimir/vacuum (Doc. 091):** the CMB energy density as a T0 vacuum density,
$$\rho_{\text{CMB}} = \frac{\xi\,\hbar c}{L_\xi^4}. \tag{N.2}$$
2. H_0 **ratio (Doc. 165):**
$$\frac{\hbar H_0}{m_e c^2} = \frac{\pi}{2}\,\xi^{10}. \tag{N.3}$$
3. **CMB temperature (Doc. 025/061):** $T_{\text{CMB}} = 2.72548\,\text{K}$, brought in via the Stefan-Boltzmann law (Step 1).

Relations (N.2) and (N.3) were developed independently. The question: are they two aspects of the same ξ structure?

N.3 Step 1: Casimir ↔ CMB via the scale L_ξ

1a – Comparing the two energy densities

The standard Casimir energy density between two plates at separation d is

$$\rho_{\text{Casimir}}(d) = \frac{\pi^2\,\hbar c}{720\,d^4}. \tag{N.4}$$

The form $\pi^2\hbar c/(240\,d^4)$ is the Casimir *pressure*, not the energy density; at $d = L_\xi$ the energy density is 4.29×10^{-12} J/m^3. At exactly the separation $d = L_\xi$ it relates to the vacuum density (N.2) as

$$\boxed{\frac{\rho_{\text{Casimir}}(L_\xi)}{\rho_{\text{CMB}}} = \frac{\pi^2\hbar c/(720\,L_\xi^4)}{\xi\hbar c/L_\xi^4} = \frac{\pi^2}{720\,\xi} \approx 102.8} \tag{N.5}$$

Remarkable: inserting (N.2) into (N.5), ξ cancels and one recovers *exactly* the standard Casimir formula. The T0 notation is thus consistent with the textbook Casimir formula. The ratio (N.5) is therefore an identity, not a measurement: L_ξ is itself determined from the measured CMB energy density and ξ (Step 1b). At L_ξ the two densities do not coincide; equality would lie at a larger plate separation. The connection is a statement about scales, not evidence.

1b – L_ξ from the CMB temperature

The CMB density is thermal (Stefan-Boltzmann):

$$\rho_{\text{CMB}} = \frac{\pi^2}{15}\,\frac{(k_B T_{\text{CMB}})^4}{(\hbar c)^3}. \tag{N.6}$$

Equating (N.2) and (N.6) and solving for L_ξ:

$$\frac{\xi\hbar c}{L_\xi^4} = \frac{\pi^2}{15}\frac{(k_B T_{\text{CMB}})^4}{(\hbar c)^3} \quad\Longrightarrow\quad \boxed{L_\xi = \left[\frac{15\,\xi}{\pi^2}\right]^{1/4}\frac{\hbar c}{k_B T_{\text{CMB}}}} \tag{N.7}$$

Casimir exponent $1/4$

Numerically: $L_\xi = 100.24\,\mu$m (0.11 % from the scale determined directly from ρ_{CMB}). Decisive for what follows: from (N.7), ξ enters L_ξ *with the power* $1/4$,

$$L_\xi \;\propto\; \xi^{1/4}.$$

N.4 Step 2: H_0 as a ξ power

Relation (N.3) solved for H_0/c (with the reduced Compton wavelength $\bar\lambda_e = \hbar/m_e c$):

$$\frac{H_0}{c} = \frac{\pi}{2}\,\xi^{10}\,\frac{m_e c^2}{\hbar c} = \frac{\pi}{2}\,\frac{\xi^{10}}{\bar\lambda_e}. \tag{N.8}$$

H_0 exponent 10

Inserted, this gives $H_0 \approx 66.8$ km/s/Mpc – within $0.9\,\%$ of the Planck value 67.4 km/s/Mpc; the exponent 10 is set here, not derived forward (see the assessment below). Here ξ enters with the power 10:

$$H_0 \propto \xi^{10}.$$

N.5 Step 3: Multiplication – the bridge formula

Now the product of the two scales L_ξ (N.7) and H_0/c (N.8). The powers of ξ add, $\hbar c$ cancels:

$$\begin{aligned} L_\xi \cdot \frac{H_0}{c} &= \left[\frac{15\xi}{\pi^2}\right]^{1/4} \frac{\hbar c}{k_B T_{\mathrm{CMB}}} \cdot \frac{\pi}{2}\frac{m_e c^2\,\xi^{10}}{\hbar c} \\ &= \frac{\pi}{2}\left[\frac{15}{\pi^2}\right]^{1/4} \frac{m_e c^2}{k_B T_{\mathrm{CMB}}} \cdot \xi^{1/4}\,\xi^{10} \\ &= \frac{\pi}{2}\left[\frac{15}{\pi^2}\right]^{1/4} \frac{m_e c^2}{k_B T_{\mathrm{CMB}}} \cdot \xi^{41/4}. \end{aligned} \tag{N.9}$$

Connection formula Casimir ↔ H_0

$$L_\xi \cdot \frac{H_0}{c} = \underbrace{\frac{\pi}{2}\left(\frac{15}{\pi^2}\right)^{1/4}}_{K_{\mathrm{geo}}=1.7441} \cdot \underbrace{\frac{m_e c^2}{k_B T_{\mathrm{CMB}}}}_{=2.176\times10^{9}} \cdot \xi^{41/4}. \tag{N.10}$$

Numerical check: $L_\xi \cdot H_0/c = 7.241 \times 10^{-31}$ – computed directly and via (N.10) algebraically identical (to $10^{-14}\,\%$). The derivation is *exact* – no approximation, no free parameters *between* the building blocks.

N.6 The exponent $41/4$ and its decomposition

The exponent is the sum of the two individual exponents from Steps 1 and 2:

$$\frac{41}{4} = \underbrace{10}_{\substack{H_0\text{ exponent}\\(\text{N.3})}} + \underbrace{\frac{1}{4}}_{\substack{\text{Casimir exponent}\\(\text{N.7})}} \tag{N.11}$$

The same exponent $41/4$ appears independently in Doc. 026 as $E_H = E_0\,\xi^{41/4}$ (Hubble energy scale). The decomposition $41/4 = 10 + 1/4$ is therefore not constructed by chance: the 10 carries the coupling electron mass → Hubble scale, the $1/4$ carries the Casimir scaling of the vacuum energy.

N.7 Where does the SI conversion factor sit?

A legitimate counter-question: we are operating here at the *decompactified* level of the formulation – H_0 exists only after decompactification (Doc. 190, R16: $H_0 = E_H/\hbar$ contains the $\hbar$ that becomes real only then), and L_ξ is a *physical* length ($\approx 100\,\mu$m) built from $\hbar$, c, k_B. At this level an SI conversion factor *must* appear. It does – it merely sits hidden in two places, and in the exponent notation in a third.

Place 1: the factor cancels (legitimate)

Both scales are built with the same $\hbar c$:

$$L_\xi \propto \frac{\hbar c}{k_B T_{\mathrm{CMB}}}, \qquad \frac{H_0}{c} = \frac{\pi}{2}\,\xi^{10}\,\frac{m_e c^2}{\hbar c}. \tag{N.12}$$

In the product $\hbar c$ drops out – because $L_\xi \cdot H_0/c$ is at its core a *length ratio* ($L_\xi/\bar{\lambda}_e$ with the reduced Compton wavelength $\bar{\lambda}_e = \hbar/m_e c$), and the ratio of two lengths is unit-free. This is no trick: the same SI conversion $\hbar c$ builds *both* lengths, so it must cancel in the ratio.

Place 2: the factor survives (the actual anchor)

What remains after the cancellation is *not* purely geometric:

$$\boxed{\frac{m_e c^2}{k_B T_{\mathrm{CMB}}} = 2.176 \times 10^9} \tag{N.13}$$

This contains c^2 (mass → energy) and k_B (Kelvin → energy) – and it is a *ratio of two measured quantities*: electron rest energy to CMB temperature. *That* is the SI / empirical conversion factor. The bridge formula (N.10) is thereby geometric *form* ($K_{\mathrm{geo}}\,\xi^{41/4}$) **times** an SI anchor ($m_e c^2/k_B T_{\mathrm{CMB}}$) – *not* a pure geometric identity.

Place 3: in the exponent – $41/4$ vs. $63/4$

The factor shows up most cleanly in the exponent itself. The $41/4$ is referenced to the electron / CMB energy scale E_0, *not* to the Planck scale. Planck-referenced (Doc. 190, R35) the exponent is $63/4$, and

$$\boxed{\frac{63}{4} = \frac{41}{4} + \frac{11}{2}} \tag{N.14}$$

The $11/2$ *is* exactly the SI scale conversion factor $\ell_P \leftrightarrow E_0$ (Planck length vs. electron / CMB scale). Put differently: if one writes $\xi^{41/4}$ and calls it "purely geometric", one has tacitly absorbed ℓ_P into the exponent – exactly the guardrail R35 ("the SI conversion factor ℓ_P must not be absorbed into the exponent"). Which reference scale one makes explicit is a *choice*; the factor never disappears, it only changes location:

Reference scale	Exponent	SI factor explicit as
E_0 (electron/CMB)	$41/4$	ratio $m_e c^2/k_B T_{\mathrm{CMB}}$
ℓ_P (Planck)	$63/4 = 41/4 + 11/2$	extra exponent $11/2$

Why we are necessarily decompactified here

This is not a matter of style: H_0 *does not exist* in the compact T^4 form – it contains the $\hbar$ that becomes real only upon decompactification (R16) – and L_ξ is a concrete μm length that cannot be formulated without $\hbar, c, k_B$. The SI factors are therefore not optional but *constitutive*: they are what makes the quantities involved physical quantities in the first place. At the compact level there is neither H_0 nor this bridge – the question of the SI factor does not even arise there.

Verdict of the consistency check

The bridge formula is *consistent* – but as a *decompactified, SI-anchored* relation, not as a compactified T^4 identity. The SI conversion factor appears exactly as expected: partly cancelled ($\hbar c$, legitimate as a length ratio), partly as the surviving empirical anchor ($m_e c^2/k_B T_{\mathrm{CMB}}$), partly – in the exponent notation – as the $11/2$ between the E_0 and Planck references (N.14). This is fully in line with Doc. 190: R34 (SI conversions indispensable for the c-/time-power, $c = \hbar = 1$ destroys the structure), R16 (H_0 emergent/decompactified) and R35 (no absorption of ℓ_P into the exponent).

N.8 Inversion: H_0 directly from the Casimir scale

Since L_ξ follows via (N.7) from ρ_{CMB} (i.e. from T_{CMB}), (N.10) can be solved for H_0:

$$\boxed{H_0 = \frac{\pi}{2}\left[\frac{15}{\pi^2}\right]^{1/4} \frac{c}{L_\xi}\,\xi^{41/4}} \tag{N.15}$$

This is the experimental lever (Doc. 166, §5): a *Casimir precision measurement* of the scale L_ξ fixes, via (N.15), a value for H_0 – an H_0 determination in the laboratory rather than in the sky. This presupposes that L_ξ has a physical Casimir signature of its own; that is open, since here L_ξ is determined from T_{CMB}.

N.9 Complete number chain

Quantity	Expression	Value
ξ	$4/30000$	1.3333×10^{-4}
T_{CMB}	Planck 2018	$2.72548\,\mathrm{K}$
L_ξ	$[15\xi/\pi^2]^{1/4}\,\hbar c/(k_B T_{\mathrm{CMB}})$	$100.24\,\mu\mathrm{m}$
ρ_{CMB}	$\xi\hbar c/L_\xi^4$	$4.170 \times 10^{-14}\,\mathrm{J/m^3}$
$\rho_{\mathrm{Casimir}}/\rho_{\mathrm{CMB}}$	$\pi^2/(720\,\xi)$	≈ 102.8
K_{geo}	$(\pi/2)(15/\pi^2)^{1/4}$	1.7441
$m_e c^2/(k_B T_{\mathrm{CMB}})$	–	2.176×10^{9}
$\xi^{41/4}$	$41/4 = 10.25$	1.91×10^{-40}
$\hbar H_0/(m_e c^2)$	$(\pi/2)\,\xi^{10}$	2.789×10^{-39}
$L_\xi \cdot H_0/c$	$K_{\mathrm{geo}}\,(m_e c^2/k_B T_{\mathrm{CMB}})\,\xi^{41/4}$	7.241×10^{-31}
H_0	(N.15)	$66.82\,\mathrm{km/s/Mpc}$

Cross-check: $H_0 = 66.82\,\mathrm{km/s/Mpc} = 2.166 \times 10^{-18}\,\mathrm{s^{-1}}$ gives $\hbar H_0/(m_e c^2) = 2.79 \times 10^{-39}$.

N.10 Honest assessment of the status (Doc. 190)

The chain above is algebraically clean, but its *epistemic* status must remain clear (corrections register Doc. 190, entries R20, R32, R33, R35, R36, R39):

What is derived – and what is not

- **Form forward:** that the connection is a *dimensionless* ξ *ratio* (Casimir scale × Hubble scale = ξ power) is structurally forward. The *chaining* of the building blocks (N.9) is exact.
- **Exponent not forward (R20):** the value $41/4$ – already the partial exponent 10 from (N.3) – is *not* derived forward from the T^4 geometry. The neat decomposition $41/4 = 10 + 1/4$ (N.11) is *structurally suggestive*, not a proof.
- **ξ^N guardrail (R35):** "$X = \xi^N$" counts as content only if N is derived forward; otherwise any magnitude can be written as a ξ power. The SI factor ℓ_P must not be tacitly absorbed into the exponent (cf. Place 3 above).
- **H_0 is emergent, not fundamental (R16/R32):** H_0 contains the $\hbar$ that becomes real only upon decompactification and does not exist in the compact T^4 form.
- **Shared reference point (R39):** the Planck length↔ R_H relation follows from *no* framework out of first principles; ΛCDM fixes it via the measured H_0. FFGFT adopts the established value for comparability – this is *not* an asymmetric disadvantage but a shared empirical reference point of both models.

Verdict of the assessment

The *chaining* is exact – a consistency relation among SI-anchored quantities (L_ξ, R_H, m_e, T_{CMB}), together with the possible experimental lever Casimir $\to H_0$ (N.15), which presupposes a Casimir signature of L_ξ of its own (open). ℓ_P does *not* appear in the formula; via the E_0 reference it is *extrapolable* – as a consistency check, not a derivation: the leg $E_0 \leftrightarrow \ell_P$ ($11/2$) coincides with the exponent of $\alpha = \xi E_0^2$ (see Outlook), whose agreement with the measured value is a property of the anchor E_0, not an independent prediction; the cosmic leg ($41/4$) remains open (R20). It is therefore *not* a first-principles prediction of H_0.

N.11 Outlook: the $11/2$, the Λ fine-tuning problem and the H_0 tension

The $11/2$ is a starting point, not a proof. The exponent $11/2$ in the split $63/4 = 41/4 + 11/2$ is not arbitrary: it coincides with the fine-structure exponent,

$$\alpha = \xi \cdot E_0^2 = K \cdot \xi^{11/2}, \qquad E_0 = \sqrt{m_e m_\mu} \propto \xi^{9/4} \quad \text{(Doc. 011, 070, 202, 210)}, \tag{N.16}$$

and it is the *same* E_0 to which the cosmic exponent is referenced ($E_H = E_0\,\xi^{41/4}$, Doc. 026). The agreement of $\alpha = \xi E_0^2$ with the measured value is a property of the anchor E_0, which provides the connection to the SI, not an independent prediction. Numerically E_0 sits exactly where the split requires: $E_0/E_{\mathrm{Planck}} = \xi^{5.48}$ versus $11/2 = 5.5$ – a match to $\sim 0.5\,\%$ (precisely the rounding 15.72 vs. 15.75 from R35).

However: the α exponent is referenced to the MeV / lepton base (the $9/4$ scaling), the split $11/2$ to the Planck scale (the offset) – different reference bases, the same number. Whether this is *structurally identical* or a numerical hit in the dense ξ^N field is open; the guardrail (R35, Doc. 271) requires for "content" an isolated, predicted, precise value. The identity would be proven only by $E_0 = E_{\mathrm{Planck}} \cdot \xi^{11/2}$ forward. *Recorded as a conjecture, not as evidence.*

The reverse question: why does ΛCDM's H_0 sit there? Were there a genuine, fixed connection $\ell_P \leftrightarrow R_H$, that would be a sharp statement about H_0. But ΛCDM does *not* choose H_0 from theory – it *measures* it (and contested: the H_0 tension ~67 vs. ~73); ℓ_P comes from the measured $G, \hbar, c$. The ratio is not harmless:

$$\frac{\ell_P}{R_H} \approx 1.2 \times 10^{-61}, \qquad \left(\frac{\ell_P}{R_H}\right)^2 \approx 10^{-122} \sim \frac{\rho_\Lambda}{\rho_{\mathrm{Planck}}}. \tag{N.17}$$

The $(\ell_P/R_H)^2 \sim 10^{-122}$ *is* the cosmological constant problem – the largest unexplained fine-tuning in physics. ΛCDM "chooses" the value by fitting Λ to the observed acceleration, without a theoretical reason. FFGFT eliminates Λ (R17/R39) and does *not* inherit the fine-tuning – but relocates the same "why this scale" into the exponent $41/4$.

The test. If the $41/4$ became forward-derivable (together with the $11/2$ leg), FFGFT would *predict* ℓ_P/R_H – and hence H_0 – instead of importing it; it could even decide the H_0 tension (which side, 67 or 73). Precisely that would turn the conjecture into evidence (isolation/prediction/precision, Doc. 271). Until then it remains a suggestive restatement of a measured ratio, open.

Chapter O

Doc. 280 — Redshift in FFGFT: static, achromatic

What this is about

In FFGFT the cosmological redshift arises *not* from spatial expansion but from a static geometric mechanism in the ξ field. This note fixes three points and corrects a fourth:

1. **Static:** $1+z=e^{\xi x}$, hence $z\approx\xi x$ – energy loss / fractal path-lengthening along the light path, no scale factor $a(t)$ (Doc. 064, 025/026).
2. **Achromatic:** the fractal path-lengthening is purely geometric and stretches *all* wavelengths by the same factor – it is *not* wavelength-dependent. Finite-element simulations of the geometry confirm this. (**Correction** of the earlier chromatic presentation; Doc. 190 C6.)
3. **The $z\to$time axis is a ΛCDM construct:** because FFGFT has no $a(t)$, plotting any quantity "over cosmic time" (e.g. the Lilly–Madau star-formation rate) is only a *translation of the observation onto the expanding picture*.
4. **The data fit – via the cosmological degeneracy** (Doc. 267): FFGFT reproduces the same observables as ΛCDM, with no false prediction, but also no single decisive test.

Throughout: $\xi=\dfrac{4}{30000}=1{,}3333\times10^{-4}$, $H_0^{\mathrm{T0}}\approx 66{,}2$ km/s/Mpc.

O.1 The static redshift

In the static universe a photon loses energy along the light path according to

$$\frac{dE}{dx}=-\xi E \qquad\Longrightarrow\qquad E(x)=E_0\,e^{-\xi x}.$$

The linear dependence on E is decisive: it makes the *leading* effect frequency-independent. The redshift is

$$z=\frac{E_0-E(x)}{E(x)}=e^{\xi x}-1, \qquad \boxed{1+z=e^{\xi x}}, \qquad z\approx\xi x \text{ (for small } z).$$

Comparison with the Hubble law $z = H_0 d/c$ gives (natural units, $c = 1$) the identification

$$H_0^{\mathrm{T0}} = E_H/\hbar, \qquad E_H = E_0\,\xi^{41/4} = 1{,}41 \times 10^{-33}\,\mathrm{eV} \;\Rightarrow\; H_0^{\mathrm{T0}} \approx 66{,}2\,\mathrm{km/s/Mpc}$$

(Doc. 064/026; the exponent $41/4$ is not derived from first principles, Doc. 190 R20). The *light-travel time* to an object at redshift z is therefore

$$\boxed{\tau(z) = \frac{\ln(1+z)}{H_0^{\mathrm{T0}}}}.$$

There is *no* finite cosmic age and *no* Big Bang: τ grows without bound as $\ln(1+z)$.

O.2 Achromatic, not wavelength-dependent (correction)

Earlier documents (Doc. 004, 025, 030, 039, 041, 053, 061, 063, 068, 081) stated a *wavelength-dependent* redshift (e.g. $z(\lambda_0) \propto \lambda_0$, "UV stronger than radio"). That presentation is **superseded**.

The mature FFGFT mechanism is *fractal path-lengthening*: the path the signal travels through the recursive time geometry is stretched by a factor that depends only on distance/geometry – *not* on the photon wavelength. Hence $1 + z = e^{\xi x}$ holds for *all* wavelengths with the same factor: the redshift is **achromatic**.

Evidence: finite-element simulations of the fractal path-lengthening yield *no* wavelength-dependent redshift. This is also physically compelling: an *energy-/wavelength-specific* loss term would be chromatic, but a *geometric* stretch factor acts on wavelength *and* pulse duration in the same ratio (as do gravitational and standard-cosmology redshift, also achromatic).

Consequence for observation: the cosmological redshift is achromatic to high precision – FFGFT therefore gets this *right*. The earlier "wavelength-dependent discriminator" drops out; it moves from the "test" column to the "fits" column. The correction is recorded in Doc. 190 as **C6**; the affected documents are listed there.

O.3 The $z \to$time mapping is model-dependent

Only the *redshift* z (and the quantity measured at that z) is model-independent. The conversion $z \to$time, by contrast, is a modelling choice:

- Λ**CDM** computes it via $a(t)$ (Friedmann integral). The look-back time levels off at $t_0 \approx 13{,}8$ Gyr – the "Big Bang ceiling". That ceiling exists *only because expansion is assumed*.
- **FFGFT** computes it via $\tau = \ln(1+z)/H_0^{\mathrm{T0}}$. There is no ceiling; τ grows without bound.

Implication: plotting "star-formation rate over cosmic time 0–14 Gyr" (Lilly–Madau) is intrinsically a ΛCDM act. In FFGFT there is no cosmic time axis – only distance / light-travel time. The data points (z, SFRD) remain valid; placing them on a 0–14 Gyr axis *is* already the translation onto the expanding picture.

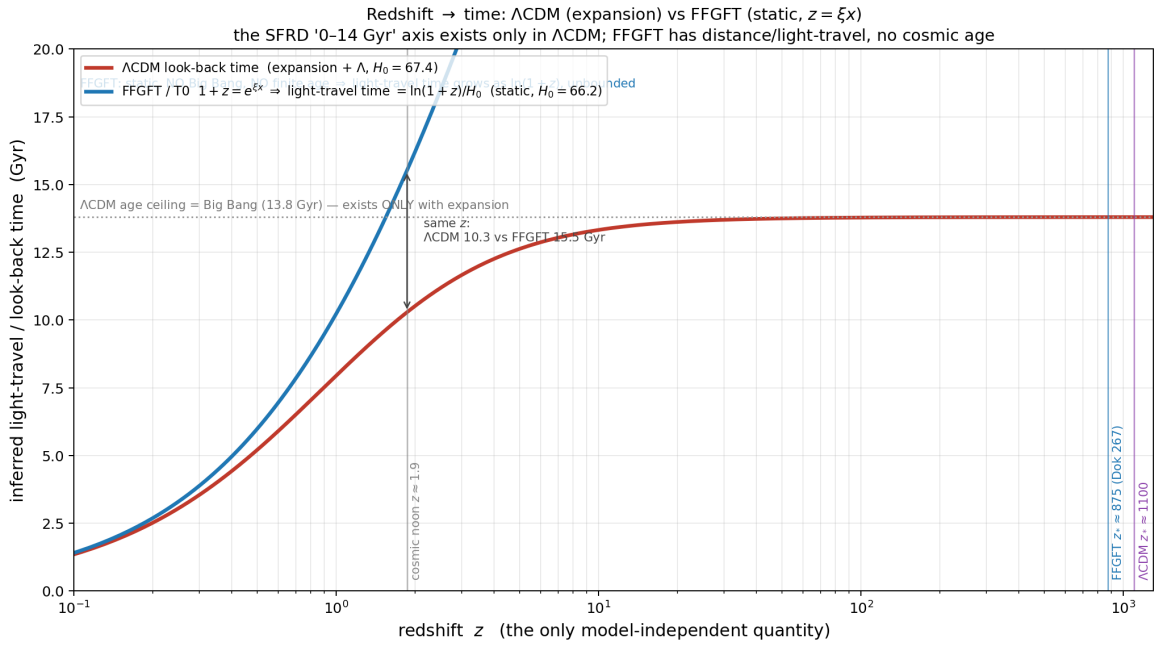


Figure O.1: Redshift → time is model-dependent. ΛCDM look-back (red, expansion $+\Lambda$) saturates at the Big Bang ceiling $13{,}8$ Gyr; FFGFT (blue, static, $1 + z = e^{\xi x}$) gives an unbounded light-travel time $\ln(1+z)/H_0$ with no Big Bang. At cosmic noon ($z \approx 1{,}9$): ΛCDM $10{,}3$ vs. FFGFT $15{,}5$ Gyr. Recombination: ΛCDM $z_* \approx 1100$, FFGFT $z_* \approx 875$ (Doc. 267).

O.4 How do the data fit? – the cosmological degeneracy

Doc. 267 establishes the central fact: FFGFT and ΛCDM are *observationally degenerate* on the standard tests. Whatever produces a given $z(d)$ relation produces the same fluxes, angles and time dilations.

- **Supernova time dilation:** distant SN Ia run stretched, $b \approx 1$ (DES: $b = 1{,}003 \pm 0{,}011$). The fractal path-lengthening stretches wavelength *and* pulse duration in the same ratio $\Rightarrow b = 1$. The test only excludes *non-dilating* tired-light ($b = 0$) – FFGFT predicts $b = 1$ and is unaffected.
- **BAO angular scales:** the non-linear temporal recursion curves $z(d)$ like an accelerated expansion ⇒ the same angles. Caveat R20: BAO uses D_M/r_s with r_s from ΛCDM – model vs. model.
- **CMB peaks:** both reproduce the peak structure; ξ plays no role for the peaks; both equally circular (Doc. 267/268).
- **Lilly–Madau / SFRD:** the z-space signal (rise to $z \sim 2$, then fall) is model-independent and in FFGFT astrophysical galaxy evolution over light-travel time. The SFRD *values* ($M_\odot$/yr/Mpc^3), however, are doubly ΛCDM-derived – via the luminosity distance $d_L(z)$ and the comoving volume. A fair FFGFT test re-derives them with $d(z)$ and volume from $1 + z = e^{\xi x}$.

Honest status: the degeneracy cuts both ways. The data *fit* FFGFT (no false prediction, achromatic as observed), but they do not *compel* it. The case for FFGFT therefore rests on *parsimony* (no Λ, no dark energy, no H_0 tension – Doc. 064) and on deriving the constants from ξ – not on a single discriminator.

O.5 No Big Bang – the direction of development reverses

FFGFT has no Big Bang. The apparent high-redshift "onset" – the fall of the SFRD and the supposed beginning of the first galaxies at $z \sim 14$ – is in FFGFT *not* a temporal beginning but a *geometric turning point* (Doc. 263).

The observed quantity is the cumulative fractal path-lengthening $z_{\text{obs}} = (L_{\text{eff}} - d)/d$ (Doc. 026/263). For small z it is linear, $z(d) = \kappa d$ with $\kappa = E_H/(\hbar c)$ – the slope ΛCDM names "H_0". Near the cosmological maximum scale R_H, however, the $z(d)$ curve has an *extremum*: FFGFT *computes* there $z_* \approx 875$ (Doc. 267, reference value; caveat R20) – the same surface ΛCDM reads as recombination/decoupling at $z \approx 1089$. The ΛCDM value 1089 is thus only the expansion reading; the FFGFT-own value is 875, *not* 1100. *Beyond* R_H one has $z \to \infty$ – with no particle horizon and no Big Bang.

What is z_* from the FFGFT view? It is *not* a recombination time. FFGFT has no hot early phase, no photon-baryon plasma, no decoupling. The CMB is the *stationary quantum ground state* of the Φ field (Doc. 025), and its acoustic peaks are discrete T^4 *lattice resonances* (analogous to the emission lines of an atom or phonon dispersions in a crystal) – *not* frozen sound waves of a primordial plasma. The numerical value itself is a *geometric ratio* (Doc. 267):

$$z_* \approx \left(\frac{\lambda_e}{L_0}\right)^{1/\ln(1/\xi)} \approx 875,$$

with the reduced Compton wavelength λ_e and the sub-Planck length $L_0 = \xi \ell_P$ – i.e. a *resonance / structure scale* of the T^4 lattice, not a thermal event in time. (Caveat R20: the exponent $1/\ln(1/\xi)$ is not yet derived from the T^4 geometry; z_* is currently a reference value of the right order of magnitude, not a derived FFGFT result.)

Thus **the apparent direction of development reverses at the maximum scale**: what ΛCDM reads as "earlier = faster development up to a beginning" is in FFGFT the approach to the geometric extremum near R_H. The high-redshift fall of the SFRD is therefore the approach to this turning point, *not* a beginning. H_0 is emergent, not a fundamental parameter – in the compact T^4 form there is no H_0 at all (Doc. 263).

Symmetrically open is only the forward derivation of the exponent $41/4$ from ξ (Doc. 190 R20) – missing for every framework, not only FFGFT.

Status

Static redshift $1 + z = e^{\xi x}$: *core* (Doc. 064/026). Achromaticity: *corrected/established* (FE simulation; Doc. 190 C6). $z \rightarrow$time as a ΛCDM translation and cosmological degeneracy: *established* (Doc. 267). Status throughout per Doc. 190.

Chapter P

Doc. 281 — The Golden Ratio in FFGFT

Abstract

The golden ratio $\varphi = (1+\sqrt{5})/2$ appears in the FFGFT corpus in several clearly distinguishable roles – and under two different glyphs that partly collide with other quantities (scalar field, angle, potential, wave function). This document does *not* change the earlier texts; it fixes the canonical notation and reading, assigns each substantial occurrence to a role, and cleanly separates what is *derived* from φ, what is merely *asserted*, and what is simply *miscalculated*. Key statements: (i) φ is nowhere a foundation in FFGFT – that is $\xi = 4/30000$; (ii) the only φ usage with genuine geometric content is $\arctan(\varphi^{-3})$ in the CP phase (Doc. 172), because $\varphi^{-3} = \sqrt{5}-2$ is the natural intermediate scale of Fibonacci convergence; (iii) φ mass ladders (Doc. 172/188) are heuristic, numerically erroneous and fall under the φ^N triviality warning P35; (iv) $1/\varphi$ is a transit point, not a fixed point (Doc. 275); (v) π and e are nowhere derived.

P.1 Occasion and Scope

While maintaining the corpus a notation problem became apparent: the same symbol `\phi` means different things in different documents, and the golden ratio itself is written sometimes as ϕ (closed form, from `\phi`), sometimes as φ (open form, from `\varphi`). Since the old documents are finalised in version v1.1.2 and are no longer rewritten retroactively, this document serves as a *binding reference*: anyone who encounters φ or ϕ in an older text looks up here which role is meant.

Important

Scope rule. This document is normative for the *reading*, not for the *wording* of the old documents. Where an old text and this document contradict each other, the version recorded here – reconciled against Doc. 190 (P35/P37) and Doc. 275 – governs the physical content.

P.2 Canonical Notation

The following assignment is the canonical convention. It also describes *how* the corpus actually uses the symbols – with the known inconsistency for the golden ratio.

Command	Glyph	Canonical meaning	Remark
`\varphi`	φ (open)	golden ratio $(1+\sqrt{5})/2$	recommended standard symbol
`\phi`	ϕ (closed)	scalar field / angle / index	NOT the golden ratio
`\Phi`	Φ (capital)	potential / complex field	e.g. $\Phi = \rho e^{i\theta}$
`\psi`	ψ	wave function	quantum mechanics

Caution

Known legacy issue. In v1.1.2 several documents write the *golden ratio* erroneously as `\phi` instead of `\varphi` (among others Doc. 005, 009, 022, 023, 028, 035, 036, 083, 133, 147, 187, 188). At the same time, field-theory documents use `\phi` correctly for the *scalar field* (e.g. Doc. 202: $\phi_{n,l,j}$ with $\Box\,\phi_{n,l,j} = -m^2\phi_{n,l,j}$) and `\phi` as a *winding index* n_ϕ in (n_θ, n_ϕ) (e.g. Doc. 051, 202). Within a *single* document ϕ is unambiguous in each case; the overloading exists only *between* documents. Hence one must not replace `\phi`→`\varphi` wholesale – that would destroy the field/angle documents.

Numerical convention. Throughout,

$$\varphi = \frac{1+\sqrt{5}}{2} = 1.6180339887\ldots, \qquad \varphi^2 = \varphi + 1, \qquad \varphi^{-1} = \varphi - 1 = 0.6180\ldots, \tag{P.1}$$

$$\varphi^{-3} = \sqrt{5} - 2 = 0.2360679\ldots, \qquad \arctan\left(\varphi^{-3}\right) = 13.2825^\circ. \tag{P.2}$$

P.3 The Roles of φ in the Corpus

Building on the threefold division in Doc. 190 (P37) – (1) microscopic/ξ level, (2) macroscopic/$1/\varphi$ transit point, (3) particle physics – a finer resolution is given here. Each role is provided with its sources and its status.

Role A – Self-similarity attractor (asserted)

Source: Doc. 172, § "The golden ratio as a fractal attractor". **Statement:** The fractal field geometry with $D_f = 3 - \xi$ is said to force a self-similarity recursion whose only stable fixed point is φ (algebraically $\varphi^2 = \varphi + 1$). **Status: asserted, not derived.** $\varphi^2 = \varphi + 1$ *is* the defining equation of φ; the actual step "$D_f = 3 - \xi$ *forces* exactly this recursion" is asserted, not shown. The variant in Doc. 188 ("φ as the most stable incommensurable ratio") is more strongly motivated – via the continued fraction $[1; 1, 1, \ldots]$ φ is the "most irrational" number – but likewise hinges on the geometry selecting precisely this ratio.

Role B – CP phase $\arctan(\varphi^{-3})$ (genuine content)

Source: Doc. 172, § "Formal proof: $\arctan(\varphi^{-3})$ as a geometric necessity". **Statement:** The Fibonacci quotients $F(n{+}1)/F(n)$ converge to φ with errors of order $\varphi^{-2n}/\sqrt{5}$; the geometric mean of first and second order is $\sqrt{\varphi^{-2}\varphi^{-4}} = \varphi^{-3} = \sqrt{5}-2$. With the torsion phase it follows that $\delta = 270^\circ + \arctan(\varphi^{-3}) \approx 283.28^\circ$ (NOvA/T2K 2024: $283.1^\circ \pm 14^\circ$).

Correct

Status: the best-justified φ usage in the corpus. Here φ^{-3} is not a choice but the natural intermediate scale of Fibonacci convergence; the exponent 3 is forward-motivated via the Z3 order (3 generations/sectors). This is qualitatively different from a fitted φ power.

Role C – Fine-structure constant $\alpha = \xi\varphi^3 F(7)$

Source: Doc. 172, § "α from pure geometry". **Statement:**

$$\alpha \approx \xi \cdot \varphi^3 \cdot F(7) = \xi \cdot \varphi^3 \cdot 13 \;\Rightarrow\; \alpha^{-1} = 136.19 \quad (-0.62\%). \tag{P.3}$$

Status: numerically verified as correct, but to be understood as one of several α routes. The main route runs via $\alpha = \xi E_0^2$ with $E_0 = \sqrt{m_e m_\mu / K_{\text{frak}}} = 7.398$ MeV (Doc. 011); its agreement with the measured value is a property of the anchor E_0, not an independent prediction. The $\varphi^3 \cdot 13$ form is the geometrically "clean" variant, but it deviates from the measured value by 0.62%.

Role D – Mass/generation ladder (heuristic, erroneous)

Sources: Doc. 172 ($m_\mu/m_e \approx \varphi^{10}$), Doc. 188 (lepton ladder $\varphi^5\pi^2$, φ^5), Doc. 009 ($\varphi^{12}/(1-\xi)^2$), Doc. 022/035/133 (φ^{gen} scaling). **Status: heuristic; in Doc. 172/188 numerically wrong** (see Section P.4). The robust mass derivation runs *not* via φ but via ξ and quantum numbers (Doc. 006, Section P.4). This role falls under P35.

Role E – Fractal scaling factor $\alpha = \varphi^{-1}$ (NOT an error)

Source: Doc. 188, § "Fractal coupling constant". **Statement:** $\alpha = \varphi^{-1} = 0.6180$ as a scaling factor between fractal levels, $(x_{k+1}, \dots) = \varphi^{-1}(x_k, \dots)$.

Important

Status: its own, clearly defined quantity – not the fine-structure constant. In the same document the fine structure stands cleanly separated as $\alpha_{\text{EM}} = \xi E_0^2 = 1/137.04$. Thus there is *no* computational error, but a deliberate double assignment of the letter α. A pure readability matter; Doc. 190 P37(a) already lists this usage.

Role F – Transit point $1/\varphi$ of the ξ recursion

Source: Doc. 275. **Statement:** The damping sequence $r(k{+}1) = r(k)(1-\xi)$, $r(0) = D_f/3$, passes $1/\varphi$ at $k^* = \log \varphi/\xi \approx 3609$ (exactly $k^*_{\text{exact}} = 3608.51$).

Caution

Status: transit point, not a fixed point. The only fixed point of the recursion is 0. Since $k^*(c)$ exists for *any* target c, the observation is unanchored as long as $1/\varphi$ is not independently motivated – the φ^N triviality warning P35 applies.

Role G – Pentagonal symmetry breaking

Sources: Doc. 018 (anomalous $g{-}2$), Doc. 149 (torsion), Doc. 019, 153, 154. **Statement:** φ as the carrier of pentagonal (fivefold) symmetry breaking. **Status: structural motif;** it shares the justification status of Role A (the selection of precisely the pentagonal symmetry is the open point).

Role H – HLV substrate (not FFGFT)

Source: Doc. 271. **Statement:** In Hidden-Layer Vacuum (HLV, Krueger) φ is the substrate geometry. **Status: belongs to HLV, not FFGFT.** In the bridge comparison φ must not be tacitly transferred from the HLV side to the FFGFT side.

Role I – Trivial representation $X = \varphi^N$

Sources: Doc. 190 (P35), 271, 272, 274. **Statement and status:** any magnitude can be written as φ^N *or* ξ^N; an exponent alone is not a prediction. Holds equally for both bases.

P.4 Honest register: correct / open / wrong

Correctly recomputed

- $\alpha = \xi\varphi^3 \cdot 13 \Rightarrow \alpha^{-1} = 136.19$ (-0.62%), Doc. 172.
- $\varphi^{-3} = \sqrt{5} - 2 = 0.23607$, $\arctan(\varphi^{-3}) = 13.28^\circ$, Doc. 172.
- $1/\varphi = 0.6180$ as a transit point at $k^* \approx 3609$, $r(3609) = 0.617994\ldots$, Doc. 275 (status: unanchored).
- $245 = 5 \cdot 7^2 \approx \varphi^{12}/\left[1.314\,(1-\xi)^2\right] = 245.1$, Doc. 009 – with a *declared* empirical anchor 1.314 (P10/P37(d)).

Numerically wrong – in the legacy documents, corrected here

Critical

Doc. 188, lepton ladder (line ≈318):

$$\text{claimed: } \frac{m_\mu}{m_e} \approx \varphi^5 \cdot \pi^2 \approx 206.7 \qquad \text{actually } \varphi^5\pi^2 = 109.46 \text{ (measured } 206.768\text{)}, \tag{P.4}$$

$$\text{claimed: } \frac{m_\tau}{m_\mu} \approx \varphi^5 \approx 16.94 \qquad \text{actually } \varphi^5 = 11.09 \text{ (measured } 16.82\text{)}. \tag{P.5}$$

Doc. 172 (line ≈356): $m_\mu/m_e \approx \varphi^{10} = 122.99$ – 40% off and contradicting the (likewise wrong) Doc. 188 form. Even arithmetically corrected one would need awkward exponents ($\varphi^{11.08}$ resp. $\varphi^{5.87}$): there is no clean φ power that hits the lepton ratios. **Consequence:** The φ lepton ladder is to be flagged as numerology (P35 null result), not as a geometric consequence.

The robust lepton-mass derivation – via ξ, not φ

Doc. 006 sets $m_i = r_i\,\xi^{p_i}\,v$ with exponents fixed from quantum numbers:

$$p_e = \tfrac{3}{2},\ r_e \approx 1.349; \qquad p_\mu = 1,\ r_\mu = \tfrac{16}{5} = 3.2; \tag{P.6}$$

$$\frac{m_\mu}{m_e} = \frac{r_\mu}{r_e}\,\xi^{p_\mu - p_e} = 2.372 \cdot \xi^{-1/2} = 2.372 \cdot 86.6 = 205.4 \quad (-0.65\%). \tag{P.7}$$

The *exponent difference* $-\frac{1}{2}$ is the structural, forward-directed content; the $O(1)$ coefficients (r_e is back-computed) carry the residual fit (cf. P40: only ratios exact). φ is not needed here and adds nothing.

P.5 What does NOT follow from φ (or ξ)

- **π is not derived.** π is the presupposed circle/torus constant. The nicest interpretation (Doc. 159): on the circle $\sum 1/n^2 = \pi^2/6$ converges, the periodic geometry forces convergence – but π is interpreted, not derived.
- **e (Euler's number) is not derived**, it appears only as $\exp(\cdot)$.
- **Emergently derived**, by contrast, are: $13 = F(7)$ (Fibonacci) and $27 = 3^3$ (Z3 structure, three generations).

P.6 Table of occurrences (substantial cases)

The table lists the *content-relevant* occurrences. Pure definitional or pass-through mentions follow the same canonical reading.

Doc.	Symbol	Role (see above)	Status
172	φ	A attractor, B CP phase, C α, D φ^{10}	B correct; A asserted; D wrong
188	ϕ	D lepton ladder, E scaling factor	D wrong; E ok (own quantity)
275	φ	F transit point $1/\varphi$	correct, but unanchored (P35)
190	φ	I triviality P35, role map P37	normative
009	ϕ	D $\varphi^{12}/1.314$	empirical anchor declared
271	φ	H HLV substrate, I triviality	HLV/FFGFT delineation
272	φ	I $X = \varphi^N$ not a prediction	correct
274	φ	I triviality holds for φ and ξ	correct
018	φ	G pentagonal symmetry breaking ($g-2$)	structural motif
149	φ	G pentagonal symmetry breaking (torsion)	structural motif
019, 153, 154	φ	G pentagonal symmetry breaking	structural motif
022, 035, 133	ϕ	D φ^{gen} scaling	heuristic (P35)
036	ϕ	D cosmic scales $r_n = r_0\varphi^n$ (BAO)	heuristic (P35)
083, 147, 187	ϕ	D qubit/photonics frequencies	heuristic
005, 028, 035, 150	ϕ/φ	definition $(1+\sqrt{5})/2$	basis
241, 261	φ	F "settled" recursion	cf. Doc. 275
202	ϕ	*scalar field* $\phi_{n,l,j}$, index n_ϕ	NOT golden ratio
051	ϕ	*winding index* n_ϕ	NOT golden ratio
067, 097, 129	ϕ	*scalar field* (Lagrangian/QFT)	NOT golden ratio

P.7 Recommendation

1. **Notation going forward:** golden ratio exclusively `\varphi` (φ). `\phi` remains field/angle, `\Phi` potential, `\psi` wave function.
2. **Legacy documents:** do not rewrite; this reference governs.
3. φ **mass ladders (Doc. 172/188):** read as P35 numerology, not as derivation. Only the ξ route is robust (Doc. 006).
4. **Genuine** φ **content** remains: $\arctan(\varphi^{-3})$ in the CP phase (Doc. 172) and the conceptual separation of the three φ roles (Doc. 190 P37).

Chapter Q

Doc. 282 — FFGFT in Hilbert Space — From the Shared Loop Object to the Mass Operator

Abstract

The existing Hilbert-space bridge of FFGFT (Doc. 230/231/232) already maps previously described QM operations – but only those, and without time. This document makes the translation more complete and shows that it reaches the *masses*. Three levels: (0) QM-equivalent – both FFGFT and HLV are a connection Laplacian on a fibre-valued Hilbert space (gauge freedom = unitary equivalence, entanglement = holonomy around cycles). (1) Process – a CP-divisibility test computably separates QM-internal (Markovian) from history-coupled (non-Markovian); this is time as process structure. (2) Mass – the actual new finding: the mass operator is a Hermitian Z_3-circulant on the $\mathbb{C}^3$ fibre, its eigenvectors are the three Z_3 characters (the generations), its spectrum S^2 are the masses. With two pure numbers – $r = \sqrt{2}$ (Koide $Q = 2/3$) and the angle $\theta = 2/9$ – it reproduces m_μ/m_e and m_τ/m_μ to four–five digits, without the formerly fitted ladder coefficient. Here $2/9 = 2/3^2$ is not a tuned decimal but a structural number on the footing of ξ: a foundational condition of the fractal open space. The FFGFT↔HLV comparison is only the entry door; no theoretical equivalence is claimed.

Q.1 Occasion and scope

The exchange with Marcel Krüger (HLV/HICE) raised the question whether FFGFT and HLV are connected. Pursuing it surfaced the essential point: FFGFT's Hilbert-space mapping is *incomplete* – it translates the already-described QM operations, but *time* does not appear in it, although it very much exists in the real world and in the T^4 tori. And as long as time is missing, the *masses* are missing. This document closes that gap as far as is honestly possible at present.

Scope rule

No theoretical equivalence FFGFT = HLV is claimed. The HLV comparison only supplies the shared object (connection Laplacian) and the process observable (non-Markovianity). The topic is the completion of FFGFT's own Hilbert-space translation up to the masses.

Q.2 The Hilbert space explicitly

Before computing, the space itself. FFGFT's full Hilbert space is the tensor product of base and fibre:

$$\mathscr{H} \;=\; L^2(T^4) \;\otimes\; \mathbb{C}^3. \tag{Q.1}$$

Bases and inner product

Base $L^2(T^4)$ – functions on the four-torus, two natural bases:

- node basis $|x\rangle$, $x \in T^4$ (nodes of the discrete torus graph);
- winding basis $|n\rangle$, $n = (n_1, n_2, n_3, n_4) \in \mathbb{Z}^4$, since $H_1(T^4) = \mathbb{Z}^4$ (Doc. 189) – eigenstates of the translation generators, the n_i being the winding numbers.

Fibre $\mathbb{C}^3$ – the three Z_3-symmetric ICE channels (Doc. 206), two bases:

- channel basis $|c\rangle$, $c = 1, 2, 3$;
- Z_3 character basis $|f_k\rangle = \frac{1}{\sqrt{3}} \sum_c \omega^{kc} |c\rangle$, $k = 0, 1, 2$, $\omega = e^{2\pi i/3}$ – the *generations*.

Inner product: $\langle\psi|\phi\rangle = \sum_{x \in T^4} \sum_c \overline{\psi(x,c)}\, \phi(x,c)$. A general state: $|\psi\rangle = \sum_{n,k} a_{n,k}\, |n\rangle \otimes |f_k\rangle$.

Three operator families act on this space – and they split the physics cleanly:

The three operators on $\mathscr{H}$

(1) Connection Laplacian L on the base (gauge/holonomy sector): $L = D - \sum_{\langle vw \rangle} U_{vw}$, $U_{vw} = e^{iA_{vw}}$. Eigenvalues = internal mode numbers; gauge transformation = unitary conjugation; holonomy around cycles = entanglement (Doc. 230). Lives on $L^2(T^4)$.

(2) Time generator H_t (energy/mass sector): generator of translation in the $\tilde{T}$ direction of the torus. Via $\tilde{T} \cdot m = 1$ its eigenvalue is the rest mass; it sets the scale $M = 1/\tilde{T}$.

(3) Mass operator S on the fibre (generation sector): the Hermitian Z_3-circulant on $\mathbb{C}^3$. Eigenvalues $\sqrt{m_k}$, eigenvectors = Z_3 characters = generations.

Thus the physics splits: the base $L^2(T^4)$ carries gauge structure, entanglement and dynamics; the fibre $\mathbb{C}^3$ carries generation and mass; the $\tilde{T}$ direction sets the scale. The full operator of a sector is a product (base operator) $\otimes$ (fibre Z_3-circulant).

Q.3 Level 0 – the shared object: connection Laplacian

At its core the QM mapping is a connection Laplacian on a fibre-valued function space over a discrete base – the T^4 for FFGFT, the cell graph for HLV:

$$L = D - \sum_{\langle vw \rangle} U_{vw}, \qquad U_{vw} = e^{iA_{vw}} \in \mathrm{U}(n), \qquad \Omega(C) = \textstyle\sum_{e \in C} A_e \pmod{2\pi}. \tag{Q.2}$$

What FFGFT achieves at this level

States = fibre-valued fields; QM operation = unitary parallel transport U_{vw}; gauge freedom = unitary equivalence (gauge-null, passed on both sides, $\sim 10^{-14}$); entanglement = holonomy around cycles (Doc. 230) with $H_1(T^4) = \mathbb{Z}^4$ (Doc. 189); CP phase = accumulated torsion phase (Doc. 172). Counterparts: FFGFT loop Hamiltonian (Doc. 231/232) ↔ HLV L_{HICE}.

This is solid, but it is only QM cast as a lattice gauge model – and it contains neither time nor mass.

Q.4 Level 1 – the process level: CP-divisibility

The first extension beyond static QM is time as *process*. A fibre qubit is dephased by a bath whose frequencies $\omega_k = \sqrt{\lambda_k}$ are the loop frequencies of the T^4 connection Laplacian (independent boson):

$$\Gamma(t) = \sum_k \frac{2g_k^2}{\omega_k^2}\left(1 - \cos \omega_k t\right), \quad c(t) = e^{-\Gamma(t)}, \quad \gamma(t) = \dot{\Gamma}(t) = \sum_k \frac{2g_k^2}{\omega_k} \sin \omega_k t. \tag{Q.3}$$

Two reduced descriptions of the same operator are compared: (M) memoryless / time-local (constant rate ⇒ CP-divisible) and (NM) history-coupled (full T^4 bath ⇒ may violate CP-divisibility). Two standard witnesses (BLP trace-distance back-flow, RHP/Choi):

Description	BLP back-flow	D revives	min Choi eig	$\gamma < 0$
(M) memoryless / time-local	0.000	no	+0.013	—
(NM) history-coupled (T^4)	5.125	yes	−0.041	1928

Only the history-coupled variant leaves the Markovian class. This is the operational "beyond-Markov" signature – but beyond the time-local *reduced* description, not beyond quantum theory.

Q.5 Level 2 – time and mass: the Hilbert-space mass operator

Here lies the actual finding. A mass is a rest energy: the eigenvalue of the *time-translation generator* (Casimir $m^2 = E^2 - p^2$). The basic relation $\tilde{T} \cdot m = 1$ makes mass the *dual* of the time cycle. A static graph Laplacian has discarded time – its eigenvalues are internal

mode numbers, not masses. Bringing time back yields masses; and since $\hbar, c$ are merely SI conversion factors, the physical content is the *ratio*, while the scale $M = 1/\tilde{T}$ is set by the time cycle.

A plain Laplacian is not enough

A symmetric T^4 connection Laplacian with $m^2 = \lambda_k$ gives the three lowest "masses" in the ratio $1:1:1.46$ and Koide $Q = 0.345$ – O(1) ratios, no hierarchy, Koide far from $2/3$. Time as a uniform extra dimension does not suffice; the Z_3 channel structure is required.

The operator

The fibre is $\mathbb{C}^3$ – the three Z_3-symmetric ICE channels (Doc. 206: $\det A_\xi = 1 - \xi$), i.e. the generation space. The full space is $L^2(T^4) \otimes \mathbb{C}^3$; the $\tilde{T}$ cycle of the torus sets the scale. The mass operator is a *Hermitian Z_3-circulant*

$$S = \mathrm{circ}(t_0, t_1, t_2), \qquad t_0 = M \in \mathbb{R}, \quad t_2 = \bar{t}_1, \tag{Q.4}$$

with eigenvalues

$$\mu_k = t_0 + 2|t_1| \cos\!\left(\arg t_1 + \tfrac{2\pi k}{3}\right) = M\left(1 + r\cos\!\left(\theta + \tfrac{2\pi k}{3}\right)\right), \qquad k = 0, 1, 2, \tag{Q.5}$$

where $r = 2|t_1|/M$ and $\theta = \arg t_1$. The *eigenvectors* are the three Z_3 Fourier characters $f_k = (1, \omega^k, \omega^{2k})/\sqrt{3}$, $\omega = e^{2\pi i/3}$ – numerically verified with overlap 1.0000. The three generations *are* the Z_3 characters; the masses are the spectrum of S^2.

Koide from the operator

With $\sqrt{m_k} = \mu_k$ the Koide identity follows directly:

$$Q = \frac{\sum_k m_k}{\left(\sum_k \sqrt{m_k}\right)^2} = \frac{1 + r^2/2}{3} \implies Q = \tfrac{2}{3} \iff r = \sqrt{2}. \tag{Q.6}$$

Koide $Q = 2/3$ is thus nothing but the operator condition $r = \sqrt{2}$ – the Z_3+$\sqrt{2}$ structure (Doc. 258/259), not a fit.

The lepton masses from two pure numbers

With $r = \sqrt{2}$ and $\theta = 2/9$ – two pure numbers, no fitted coefficient:

	operator	data	error
m_μ/m_e	206.770	206.768	0.00 %
m_τ/m_μ	16.818	16.817	0.01 %
Koide Q	0.666667	0.666661	exactly $2/3$

More than a fit

The formerly fitted ladder coefficient ($r_e \approx 1.349$, Doc. 006) has *vanished* – replaced by the structural number $2/9$. And it is more than an adjustment: *one* angle θ reproduces *both* independent ratios simultaneously (other θ break m_μ/m_e at once). Two pure numbers → three masses to 4–5 digits.

Numerical caution: the electron is a cancellation

The lightest eigenvalue is a near-total cancellation, $\mu_e = M\left(1 + r\cos\left(\theta + \frac{2\pi}{3}\right)\right) = M(1 - 0.95965) = 0.04035\,M$ – the difference of two nearly equal numbers. Rounding the cosines to only three digits shifts μ_e by orders of magnitude: taking $\cos\left(\theta + \frac{2\pi}{3}\right)$ as -0.658 instead of the correct -0.6786 (a $3\,\%$ error) turns $\mu_e = 0.040$ into 0.069 (a $72\,\%$ error), and since $m_e = \mu_e^2$ this flips m_μ/m_e from 207 to ≈ 70. The ratios must therefore be checked at *full machine precision*; precisely because the electron is the lightest state, m_μ/m_e is the sharpest test of whether θ is exactly $2/9$ (script `ffgft_mass_operator_C3.py`, numpy precision).

How precise? Against the high-precision e/μ data the angle is confirmed to $|\theta - 2/9| \approx 1.8 \cdot 10^{-7}$ – seven significant figures. The lepton masses are among the most precisely measured quantities in physics; the data will not get sharper. The empirical question is therefore settled: $2/9$ is the number.

$2/9$ as a foundational condition

Must $2/9$ be derived from something deeper? No. The decisive distinction is between a *fitted continuous knob* (bad – that was r_e) and a *fixed structural rational* forced by the geometry (fine – like $2/3$, like $2\pi/3$, like the three-channel count itself).

Structural number, not a fit

$2/9 = 2/3^2$ is a clean rational, not a tuned decimal – exactly the kind of primitive the fractal open space can carry, *on the footing of* ξ. Like $\xi = 4/30000$ it needs no deeper derivation; it is a foundational condition of the geometry.

This shifts the leptons from a ξ*-power* story to a Z_3 *geometry* story: the ratios hinge on the pure numbers $\sqrt{2}$ and $2/9$, not on ξ (which does not appear in the ratio). This fits the fact that the ξ-ladder was the weak part for leptons anyway. The fractal space then carries a small set of primitives: ξ for the coupling sector, $(\sqrt{2},\, 2/9)$ for the generation sector.

Q.6 Computing in the Hilbert space: recipe and examples

The procedure is always the same – every physical quantity is an eigenvalue, an eigenvalue relation, or an overlap of eigenbases on $\mathscr{H}$.

The recipe – four steps

1. **Choose the sector** – which quantity (masses, a process observable, a mixing angle, ...).
2. **Write the operator** – the corresponding self-adjoint operator on $\mathscr{H}$ (Z_3-circulant on $\mathbb{C}^3$ for masses; time-extended connection Laplacian for dynamics; overlap of two eigenbases for mixing).
3. **Diagonalise** – eigenvalues = quantities (as dimensionless ratios), eigenvectors = states; relations (Koide) follow algebraically.
4. **Scale + SI** – scale $M = 1/\tilde{T}$ from the time cycle, then $\hbar, c$ as pure SI conversion.

Already computed.

- *Masses:* S on $\mathbb{C}^3$, eigenvalues^2 = masses, ratios from $(\sqrt{2}, 2/9)$ (Level 2).
- *Koide:* $Q = (1 + r^2/2)/3$ from the eigenvalue structure, $= 2/3$ at $r = \sqrt{2}$ (Level 2).
- *Process / non-Markov:* the time-extended operator on $L^2(T^4) \otimes$ (qubit), CP-divisibility witness (Level 1).

How further quantities are computed (programme). The same recipe, new operators – the method is fixed, the results are still open:

- **Mixing angles (PMNS/CKM-like).** If two sectors (say charged leptons and neutrinos) are each Z_3-circulants, but in fibre bases *rotated* against one another, then the mixing matrix is the overlap $\langle f_k^{(A)} | f_l^{(B)} \rangle$ of their eigenvectors. The mixing angles encode the *rotation angle* between the two Z_3 structures – a single geometric number per sector pair, computed just like $2/9$.
- **Further sectors (quarks).** The same scheme: one Z_3-circulant per charge sector with its own structural angle; the open question is the same – which rationals does the topology force.
- **Dynamics and correlations.** Holonomy and entanglement observables from the base connection Laplacian (Doc. 230), already demonstrated in the witness scripts.

What "computing" means here

Every quantity is an eigenvalue, an eigenvalue relation, or an eigenbasis overlap on $\mathscr{H}$, fixed by structural rationals plus the single scale $\tilde{T}$. That is the sense in which the Hilbert-space mapping "computes further things" – not through new fits, but through new operators following the same recipe. The quark sectors are computed as an operator spectrum (next section) – faithfully, but without pure numbers; the mixing angles remain open. What is fixed is the method.

Q.7 The three sectors explicitly: faithful translation, lepton-specific numbers

Because the Hilbert-space mapping is a *faithful translation* (the $T^4 \leftrightarrow$ Hilbert bijection, Doc. 270, type III), it must also carry the masses FFGFT obtains via the ξ-route ($m = r\,\xi^p\,v$, Doc. 006) – leptons *and* quarks. Here that is carried out: each charge sector is exactly the spectrum S^2 of a Hermitian Z_3-circulant.

Sector	M	r	Koide Q	max. rel. error
Leptons (e,μ,τ)	17.72	$\sqrt{2} = 1.4142$	0.6667	$2 \cdot 10^{-14}$
Up (u,c,t)	150.7	1.757	0.848	$7 \cdot 10^{-14}$
Down (d,s,b)	25.73	1.546	0.732	$2 \cdot 10^{-14}$

In every sector S^2 reproduces the masses to machine precision (script `ffgft_z3_alle_sektoren.py`). The Hilbert-space calculation is thus *universal* – the quarks are included.

A faithful translation is no gain by itself

This universality is *automatic*: the circle map $\sqrt{m_j} = M\left(1 + r\cos\left(\theta + \frac{2\pi j}{3}\right)\right)$ is a *bijection* (three masses $\leftrightarrow M, r, \theta$). Any triplet can be written into it exactly – it *must*, otherwise it would not be a translation. "It works in Hilbert space" is therefore not a prediction but the meaning of faithfulness (cf. the ξ^N/φ^N triviality, Doc. 190 P35).

The genuine content lies in whether the operator parameters are *pure numbers*:

- **Leptons:** $r = \sqrt{2}$ – which *is* Koide $Q = 2/3$, the $Z_3 + \sqrt{2}$ structural condition (Doc. 258/259) – *and* $\theta = 2/9$ (a clean rational). Two pure numbers (see above).
- **Quarks:** $r_{\text{up}} \approx 1.76$, $r_{\text{down}} \approx 1.55$ – *not* $\sqrt{2}$; the angles are not clean rationals. The operator carries the masses faithfully, but with *un-special* parameters.

This matches the ξ-route point for point (Doc. 006): there the lepton ratios carry real structure (Koide), while the quark (r, p) are per-particle classification. A faithful translation *must* agree on *where* the structure is – and it does.

Consequence for a universal Z_3 angle

The attempt to impose the lepton condition $r = \sqrt{2}$ (i.e. $k = 1$, Koide $2/3$) on the quark sectors is *falsified* by the faithful translation itself: the up sector needs $r \approx 1.76$, the down sector $r \approx 1.55$. There is no room for $\sqrt{2}$ – and hence none for a single universal $k = 1$ Z_3 angle across sectors.

Q.8 The same scale carries the couplings too: α and (briefly) G

The Z_3 operator of the lepton sector fixes not only the masses but also a *scale* M_T and the dimensionless eigenvalues μ_e, μ_μ, μ_τ. The fine-structure constant uses exactly these quantities – it is therefore not a separate input.

α as an eigenvalue relation on $\mathscr{H}$

The very recipe – every quantity an eigenvalue or eigenvalue relation on $\mathscr{H}$ – yields α too, from the *same* operator that gives the masses. The eigenvalues of S are $\lambda_j = \sqrt{m_j}$; the fine-structure constant is the product of the two smaller ones, scaled by ξ:

$$\alpha = \xi\left(\frac{\lambda_e\,\lambda_\mu}{\mathrm{MeV}}\right)^2, \qquad \lambda_e\,\lambda_\mu = \sqrt{m_e\,m_\mu} = E_0. \tag{Q.7}$$

This is the ξ-route ($\alpha = \xi(E_0/\mathrm{MeV})^2$, Doc. 011), but as a *Hilbert-space variant*: E_0 is not a separate input but an eigenvalue relation of the same operator built from $\sqrt{2}$ and $2/9$. Thus α and the lepton masses follow from *one* object and *one* scale. Numerically: the operator/geometric value gives $1/\alpha = 138.9$ (bare $E_0 = \sqrt{m_e m_\mu} = 7.348$ MeV); with the fractal correction $E_0 = 7.348/\sqrt{K_{\mathrm{frak}}} = 7.398$ MeV, $1/\alpha = 137.04$ follows – the $1.35\,\%$ is, up to $0.017\,\%$, the factor $K_{\mathrm{frak}} = 1 - 100\xi$. The agreement with the measured value is a property of the anchor E_0, not an independent prediction. Script `ffgft_kopplungen_hilbertraum.py`.

G – structurally through ξ, involved only in the SI conversion

The gravitational coupling enters *likewise* through ξ, via the T0 fundamental relation

$$\xi = 2\sqrt{Gm} \;\Longrightarrow\; G = \frac{\xi^2}{4m} \tag{Q.8}$$

(Doc. 006, calc-o). The structural finding is the same as for α: the coupling is *not independent* but fixed by ξ. The Hilbert-space mapping supplies the scale here: m is the operator scale $M_T = M^2 = 1/\tilde{T}$ of the lepton sector. Unlike the dimensionless α, however, G carries dimensions; its SI realisation requires the full conversion through $\ell_P, c, \hbar$ and is therefore considerably more involved – it is carried out in calc-o and yields G_{SI} at the per-mille level; this agreement arises only with the small remainder of the one-anchor chain in C_{conv} (Doc. 383) and is not a precision confirmation in its own right. What is involved is the *SI conversion*, not the structural origin.

Masses and couplings from one object

α and the lepton masses follow from the *same* Z_3 operator and its scale; the gravitational coupling is fixed by the *same* ξ. The pure-number content sits in ξ and the operator structure; the SI numerical values – G especially – are conversions, not additional inputs.

Q.9 The QM translation and what its extension changes

The translation of the QM *operations* has its own document (Doc. 230/231/232): gauge freedom = unitary equivalence, entanglement = holonomy around cycles, Bell correlations = topological connections on T^4 (Doc. 230). That translation does *not* contain mass and time – it maps the operations only. The natural question: does anything in the QM results

change when it is extended – by the $\mathbb{C}^3$ mass fibre (Level 2) and time as process (Level 1)? Computed (`ffgft_qm_uebersetzung_erweiterung.py`):

Extension	QM result	change
mass ($\mathbb{C}^3$ fibre)	CHSH/Bell on the base	$4 \cdot 10^{-16}$ (none)
time (process)	coherence back-flow	$0 \to 0.53$
time (process)	min. Choi eig	$+0.0003 \to -0.0056$

Mass acts on a *separate* tensor factor. A QM correlation (CHSH on two base qubits, $= 2\sqrt{2}$, Tsirelson) is identical to within $4 \cdot 10^{-16}$ with and without the $\mathbb{C}^3$ mass fibre attached. The mass extension only enlarges the space and supplies the masses; *no* static QM result changes.

Time changes the *dynamical* quantities. The QM bridge treats the reduced dynamics as time-local (unitary/Markovian); the extended translation carries the loop history. The same fibre-qubit coherence $c(t)$ is Markovian and CP-divisible with no back-flow, but history-coupled it is non-Markovian: back-flow 0.53, the min. Choi eigenvalue turns negative, $\max|\Delta c(t)| \approx 0.78$. Single-time quantities remain; sequential/dynamical ones acquire the non-Markovian correction.

Slightly incomplete – only in scope

The QM translation is not wrong but *incomplete in scope*: no mass, time-local. Extended, it leaves every *static* QM result unchanged (mass factorises) and adds exactly *one* new effect – non-Markovian memory in the dynamics – which is itself still within quantum theory (a traced-out unitary bath reproduces it, Level 1). The expectation that the extension "has an influence on the results" therefore holds, but precisely located: on *time*, not mass, and only on *dynamical* quantities.

Test duration decides. The difference lives entirely in time and has two scales: a *fast* decoherence time (coherence below $1/e$ at $t \approx 0.4$, nearly gone at $t \approx 1.4$) and a *late* memory/recurrence time (first revival at $t \approx 2.8$, about $6\times$ later). Hence three regimes: (i) very short test ($t \ll 0.4$) – coherence ≈ 1, QM-only and extended translation identical, the QM result exact; (ii) into the decoherence – both fall essentially together, the Markovian description suffices; (iii) only from the recurrence time on do the curves separate (Markovian ~ 0.53 vs non-Markovian ~ 0.22). Since a single operation typically ends *before* decoherence, the non-Markovian term is *operationally invisible* to fast operations; it becomes measurable only once the test duration reaches the T^4 memory time – in deliberately long, coherence-preserving, sequential experiments.

Does it scale with many qubits or with mass? No – least of all there. Massive systems couple more strongly to the bath and therefore decohere *faster*: at four-fold coupling the decoherence time drops from $t \approx 0.44$ to 0.11, and the absolute coherence at the first revival falls from 0.22 to $\sim 10^{-5}$ – the non-Markovian term is then *buried*. Mass thus makes it less visible, not more; what remains is faster *ordinary* (Markovian) decoherence. With many thousands of qubits it is not the memory term that grows but ordinary decoherence and error accumulation (gate errors, crosstalk) – Markovian and addressed by error correction; whether revivals survive at all depends on the bath *structure* (incommensurate

frequencies push the Poincaré recurrence into the practically unreachable), not on the qubit count. The one legitimate caveat is *correlated* (non-Markovian) noise across gates – relevant only if the bath memory time approaches the sequence duration, which fast gates and fast-decohering (heavy) systems precisely prevent.

Q.10 Compact or recursive? When the compact computation suffices

One question decides how far the compact Hilbert-space computation reaches: does the compact (cyclic, finite, one-shot) variant *always* suffice – so that decompactification is only the final step to SI – or are there quantities computable *only recursively*? The decisive criterion is **linearity**.

Linear ⇒ compact suffices

If the dynamics is generated by a *fixed linear* operator on $\mathscr{H}$ with a discrete (compact) spectrum, the closed spectral representation always exists:

$$\langle A(t)\rangle = \sum_{k,l} c_{kl}\, e^{i(\omega_k-\omega_l)t}. \tag{Q.9}$$

One diagonalisation suffices. The memory kernel $K(t)$, which looks *recursive* in the time representation (convolution, memory), is merely the Fourier transform of the discrete spectral density $\sum_k g_k^2\,\delta(\omega-\omega_k)$. The recursion is thus only *one representation* (time domain); the spectral sum is the equivalent closed one. For a discrete spectrum the closed form always exists – decompactification happens only at the end, at the SI step.

Concrete evidence: the CP-divisibility test (Level 1) obtained the *non-Markovian* result – back-flow, revivals – from exactly such a finite spectral sum, in closed form, *without* recursion. Even "memory" was compact-computable.

When recursion is forced

Recursion becomes unavoidable only if one of the two linearity prerequisites breaks:

1. **Fundamentally continuous spectrum** – then no finite sum. But the continuum appears only at the SI/experienced level; fundamentally it is compact. This case drops out.
2. **Nonlinear / self-referential dynamics** – the generator depends on the *state or the history itself*, not only on time. Then there is no fixed eigenbasis, no diagonalisation, and the recursion is genuine.

The question thereby reduces to a *classification*: for each quantity ask – is the generator a fixed linear operator on $\mathscr{H}$? Yes ⇒ compact, one-shot, SI only at the end. Does it depend on the state/trajectory ⇒ recursion possibly irreducible.

Mass sector, gauge, entanglement and mixing are linear – for them the compact computation suffices, and HLV's recursions are *avoidable* there. The only candidate for genuine recursion is the *self-referential* level (Doc. 248, §8: the origin thought as a material T^4 process): where the process processes itself, the generator could become state-dependent. To check this – linear or self-referential, quantity by quantity – is a well-defined programme and marks the single place where the open/recursive variant would do real, non-eliminable work.

Q.11 Claim-state and limits

What is shown – and what is not

1. **A genuine Hilbert-space mapping.** The mass operator is a concrete Hermitian operator on $\mathbb{C}^3$ with eigenvectors (generations) and spectrum (masses) – not a parametrisation.
2. **Koide forward.** $Q = 2/3$ follows exactly from $r = \sqrt{2}$; structure, not a fit.
3. **Two numbers, three masses.** $(\sqrt{2}, 2/9)$ reproduce both ratios to 4–5 digits; one angle covers two observables (a genuine over-determination).
4. **Structural assumptions.** The *form* (Z_3-circulant) and $r = \sqrt{2}$ are claimed geometric (Doc. 206/258/259).
5. **Empirically exhausted.** $\theta = 2/9$ is confirmed against the e/μ data to $\sim 2 \cdot 10^{-7}$ (seven figures in the angle); the ratios agree to 10^{-5}–$6 \cdot 10^{-5}$, for the τ within the measurement uncertainty. More precise lepton data are not to be expected – the empirical question is answered. The tiny residual ($Q = 0.6666605$ vs. $2/3$) sits at the level of radiative corrections, not a defect of the identification.
6. **The only open question is theoretical:** is $\theta = 2/9$ (like $r = \sqrt{2}$) forced by the T^4/Z_3 topology?
7. **Not beyond quantum theory.** The CP test (Level 1) shows non-Markovianity in the reduced dynamics; a traced-out unitary bath reproduces it.

The sharp, now reachable remainder is thereby precise: not "derive a messy number", but the single clear question – **is $\theta = 2/9$ forced by the T^4/Z_3 topology, as $2/3$ is?** A single rational, not a ladder fit.

Topology frames, dynamics determines – the reading that was already right. The answer that carries here was at heart always the dynamical one: a mass ratio is *only* dynamically thinkable. The spectrum $\sqrt{m_k}$ is the solution of an eigenvalue problem, and the fixed point $\mathcal{R}(\Phi_*) = \Phi_*$ (Doc. 203) is *both at once*: the topology (Z_3, T^4) fixes the *form*, self-consistency the *value*; the two cannot be separated. A computed finding (`kann_topologie_2_9_erzwingen.py`) *supports* exactly this reading: $\theta = 2/9$ *rad* is not a static flux root of unity – $\theta/2\pi = 1/(9\pi)$ is irrational, and by Lindemann-Weierstrass $e^{i\theta}$ is transcendental. A *purely combinatorial/counting* topology (flux quantization of a finite group) therefore yields only the *spacing* $2\pi/3$ (Z_3, the denominator; $Q = 2/3$ is forced for *every* θ by $r = \sqrt{2}$ alone), not the offset. But this does *not* pit topology against dynamics:

transcendence-in-2π is here *no prohibition* but the *expected* signature of a dynamically (rather than combinatorially) determined quantity – a counted value would be a root of unity, a dynamically set one typically is not. Properly posed, the open question is therefore: the topology frames (Z_3 spacing, the form), a *dynamical* condition on that stage fixes the offset – and *which* condition yields exactly $\theta = 2/9$ is the hard, still open core. Excluded is only pure counting topology, not topologically framed dynamics (the fixed point itself). *(Aside:* the value $2/9$ also appears in the corpus as a dimensionless mode-locking rotation number, Doc. 286; this is *not* dimensionally the same as the radian angle θ – $2\pi \cdot 2/9 = 4\pi/9 \neq 2/9$ –, so the identification remains open.)

Why $2/9$ cannot be had symmetrically (the $\cos 3\theta$ theorem). There is an exact reason why no simple *symmetric* principle yields $\theta = 2/9$ (`einfachstes_prinzip_2_9.py`). The entire θ-dependence of the mass configuration $a_k = 1+\sqrt{2}\cos(\theta + 2\pi k/3)$ runs *only* through $\cos 3\theta$: the elementary symmetric functions are $e_1 = 3$ and $e_2 = 3/2$ (both θ-independent), only $e_3 = \det = -\frac{1}{2}+\frac{\sqrt{2}}{2}\cos 3\theta$ carries θ. Hence *every* Z_3-invariant quantity extremizes solely at $3\theta = n\pi$, i.e. $\theta = n\pi/3$ – the Z_3-symmetric points $\{0, \frac{\pi}{3}, \frac{2\pi}{3}, \dots\}$. Concretely, the smallest-denominator, the widest-tongue and the spectral-determinant principles all select $1/3$ or $1/2$, never $2/9$. The reason is a structural theorem, not a computational shortfall: *symmetric functionals cannot have asymmetric values as extrema*, and $\theta = 2/9$ rad lies about $21\,\%$ of the way between 0 and $\pi/3$, irreducibly Z_3-breaking. Thus the simplicity principle is not refuted but shifted: not the *simplest symmetric fraction* ($= 1/3$), but the *minimal symmetry-breaking value* – and the holonomy test (Doc. 286) shows that $2/9$ is exactly this: the only member of the 9-family that splits off as a singlet under the holonomy ($1/9, 4/9$ stay paired). Simplicity sits in the *breaking*, not in the symmetric ground state; what remains precisely open is only the defined symmetry-breaking measure whose minimum yields $2/9$ (rather than $1/9$ or $4/9$).

Test: the breaking reaches $2/9$ but does not fix it. A direct test (`symmetriebrechung_2_9.py`) couples the second triplet $3'$ of the doubling ($6 = 3 \oplus 3'$, Doc. 285) with strength δ and relative phase χ and looks for the shifted extremum θ^* of the mass configuration. Result: the breaking *reaches* $\theta^* = 2/9$ – so it is not only necessary but sufficient –, but at a strength δ that must be tuned ($\delta \approx 0.52$ at $\chi = 0$, 0.24 at $\chi = \pi/2$) and coincides with no independently fixed value (ξ, $1/\sqrt{2}$, $1/\varphi$, $1/3$). Thus $2/9$ is *reachable but not point-derived* through this functional: the breaking strength remains a free modulus – consistent with the transcendence and the free holonomy phase. The open question is thereby precisely shifted – no longer "why $2/9$" but *what fixes the strength of the Z_3 breaking*. *(Caveat:* this is *one* natural breaking functional; it does not exclude that another fixes δ independently – e.g. the self-consistency of the fixed point with the fixed lepton exponents, Doc. 203.)

The common reason: everything determined is radial, θ is angular. Precisely this fixed-point route was tested (`fixpunkt_radial_vs_winkel.py`) – and it does not bind θ. The fixed point $\mathcal{R}(\Phi_*) = \Phi_*$ (Doc. 203) with the condition $\xi_0^{p_i-1} K_{\text{frak}} = 1$ and the exponents $(3/2, 1, 2/3)$ does not determine ξ_0 (the three exponents give no common ξ_0, in no reading $4/30000$; cf. Doc. 203); θ does *not* appear in it. The radial sector (ξ_0, hierarchy) and the

angular sector (θ, Q) are decoupled: ξ_0 is θ-free, $Q = 2/3$ is ξ_0-free. The same pattern runs through the whole investigation: *everything FFGFT determines is radial* – ξ, the exponents, the hierarchy, $Q = 2/3 = r{=}\sqrt{2}$, the mean $\langle f \rangle$ of the fractal correction function in $g = \eta(1 + \xi f)$ (not a frequency, not the base winding number $f = 1/\xi$) – and *everything open is angular* – $\theta = 2/9$, the holonomy phase, the harmonic phases of f, the breaking strength δ. Fixed point, duality, recursion and the torus counting apparatus all live in the radial sector without exception; none touches the angle. This is the common structural reason why $\theta = 2/9$ resists every tool. To fix the offset one needs an *angular* principle – a dynamical angle condition – which FFGFT does not yet possess and which, by the transcendence finding, cannot be a counting/topological one.

Not static-geometric either – closing the elimination. There remains the question whether the angular principle is a *fixed geometric angle* of the shared icosahedral stage on which FFGFT's 3-fold (C_3) and HLV's 5-fold (C_5) coexist ($C_3 < A_5$, Doc. 283). Tested (`ikosaeder_winkel_2_9.py`, strict, hit threshold 0.05°): the icosahedral axis angles are 20.9°, 31.7°, 36°, 37.4°, 41.8°, 54.7°, 58.3°, 60°, 63.4°, 69.1°, 70.5°, 72°, 79.2°, 90°. $\theta = 2/9$ rad $= 12.73^\circ$ matches *none* (nearest 20.9°, distance 8.2°), nor do 2θ or $\theta/2$. Control: $\arccos(1/\sqrt{5}) = 63.43^\circ$ matches the C_5–C_5 angle *exactly* – the method is correct, so the negative is real. (Koide's 45° is not an axis angle either: FFGFT's angular structure is not the icosahedral geometry.) This closes the elimination: $\theta = 2/9$ is *not* symmetric ($\cos 3\theta$ theorem), *not* counting-topological (transcendence), *not* radial (fixed point), *not* static-geometric (this computation). Exactly *one* category remains – the *dynamical skew-adjoint phase* R (Doc. 283), an accumulated, Berry-like angle. FFGFT's own recursion $\mathscr{R}$ (Doc. 203) is purely real and possesses no such generator; only the HLV side carries it (R = BD17A). The static-radial FFGFT side is thereby *demonstrably exhausted* for θ; the angular principle is structurally HLV's dynamical R, with a precise target: an accumulated angular phase yielding $\theta = 2/9$ – transcendental in 2π, Z_3-breaking, the singlet of the 9-family.

Q.12 Conclusion

Conclusion

FFGFT's Hilbert-space translation reaches further than previously shown. Level 0 maps the QM operations cleanly (shared with HLV). Level 1 brings time as process (CP-divisibility, non-Markovianity). Level 2 brings time as geometry and hence the *masses*: the mass operator is a Hermitian Z_3-circulant on the $\mathbb{C}^3$ fibre whose spectrum yields the lepton masses from two pure numbers $\sqrt{2}$ and $2/9$ – Koide exact, both ratios to 4–5 digits, without the old fitted coefficient. $2/9 = 2/3^2$ is a structural number on the footing of ξ, a foundational condition of the fractal space. The whole lepton problem is thereby reduced to a single question – and this document's investigation has made it *angular*: not *whether* T^4/Z_3 forces the angle (no counting topology can, $2/9$ being transcendental in 2π), but which *dynamical* angle principle yields it. The elimination shows: everything determined is radial (ξ, hierarchy, $Q = 2/3 = \sqrt{2}$), only the offset $2/9$ is open – structurally HLV's skew-adjoint dynamical phase R.

Closing remark: translation is not justification. Simplified, the result of these extensions fits in one sentence: *a faithful translation into Hilbert-space notation is possible for all masses and all constants – but precisely because it is faithful, it provides no justification.* Faithfulness is mathematically a bijection: three masses correspond invertibly and uniquely to the triple (M, r, θ), α is an eigenvalue relation of the same operator, G follows through the same scale $1/\tilde{T}$. A bijection, however, carries *everything* over and explains *nothing*; it is a change of coordinates, not a derivation. That a quantity "can be computed in Hilbert space" is therefore never a finding in its own right – it is the meaning of faithfulness.

The justification comes from the original T^4 torus representation, which is generatively prior (Doc. 206/270): the Hilbert space is the *image*, the torus the *original*. Only there are $\sqrt{2}$ (the Koide condition $Q = 2/3$), the angle $2/9$ and the primitive ξ *geometric resp. topological facts* rather than read-off parameters. The clean division of labour is thus: the Hilbert space *computes and represents* (universally – leptons, quarks, α, G); the T^4 *justifies and derives*. Even where the translation yields *no* pure numbers – for the quarks – this is not a defect of the translation but an expression of the fact that the torus geometry forces none there. The open question of the whole construction is therefore a single one, and it must be posed on the torus side: whether T^4/Z_3 forces the pure numbers $\sqrt{2}$, $2/9$ and ξ.

Accompanying scripts (numpy). `ffgft_t4_cp_divisibility.py` (Level 1, CP-divisibility), `ffgft_t4_mass_operator.py` (Z_3-circle analysis, KK comparison), `ffgft_mass_operator_C3.py` (the mass operator on $\mathbb{C}^3$), `ffgft_z3_alle_sektoren.py` (all three sectors as an operator spectrum), `ffgft_kopplungen_hilbertraum.py` (α and G via the Hilbert-space mapping), `ffgft_qm_uebersetzung_erweiterung.py` (mass/time extension of the QM translation).

Chapter R

Doc. 283 — FFGFT ↔ HLV: the bridge and the fork

Abstract

Marcel Krüger's HLV builds space as a 6D→3D icosahedral cut-and-project quasicrystal. Whether FFGFT and HLV are connected is tested here strictly against the three-part criterion: (1) the same structural object from each model's *own* geometry, (2) an explicit map between the derivations, (3) a divergence point that rules out mere Stufe-0 scaffolding. Finding: a Z_3 circulant – the type of FFGFT's mass operator – demonstrably lives inside HLV's own geometry, along a C_3 axis of the icosahedral group (criteria 1+2 met). But the internal invariant diverges *fundamentally*: HLV's protected structure rigidly gives $r = 2/\sqrt{5}$ (5-fold, φ), while FFGFT requires $r = \sqrt{2}$ (3-fold, Koide $Q = 2/3$). No cut-and-project parameter reaches $\sqrt{2}$ ($r(\tau) \leq 1$ for all τ); $\sqrt{2}$ appears in HLV only as an unprotected accidental angle. Result: a *genuine* connection with a *fundamental* fork at the symmetry order – no Stufe-0 triviality, no identity. The fork sits in *space*: HLV's phase layer is its spiral-time operator (not $\theta = 0$), and at the time/memory level HLV's χ kernel and FFGFT's Doc. 282 kernel are non-Markovian kernels of the same class – a bridge candidate (§ 8). All geometric statements are reproducible with `ffgft_hlv_c3_bridge_probe.py`.

R.1 Occasion and criterion

The exchange with Marcel Krüger (HLV/HICE) repeatedly raises the question whether FFGFT and HLV are connected. An honest answer needs a sharp criterion, or one ends up with pseudo-connections that merely share scaffolding (both use Planck units, both have a square root somewhere). The criterion has three parts:

1. **Same object:** the same structural object derived from each model's *own* geometry – not from shared scale scaffolding.
2. **Map:** an explicit map between the two derivations.
3. **Divergence point:** a point where the models separate – otherwise the "connection" is either trivial identity or Stufe-0 scaffolding.

This document applies the criterion concretely. It claims no theoretical equivalence; it locates precisely *where* FFGFT and HLV meet and *where* they diverge.

R.2 How Krüger builds his space

The construction is read directly from the HLV homology script (Doc. 271/276): a **6D→3D icosahedral cut-and-project quasicrystal**. A $\mathbb{Z}^6$ lattice is split by two projectors into a 3D parallel space (physical) and a 3D perpendicular space (the "hidden layer"); an acceptance window in perp space selects which lattice points project into the physical quasicrystal. Krüger's exact projector:

$$P(\tau) = \begin{pmatrix} 1 & -1 & 0 & 0 & \tau & \tau \\ \tau & \tau & 1 & -1 & 0 & 0 \\ 0 & 0 & \tau & \tau & 1 & -1 \end{pmatrix} \quad \text{(columns normalised)},$$

with $\tau = \varphi$ (golden) in parallel space and $\tau = \sigma = -1/\varphi$ in perp space. The six columns are the six icosahedral five-fold axes (one per antipodal pair). Internal symmetry: icosahedral (5-fold); the point set is aperiodic. Striking for a bridge: the hidden space is *3D* – the same dimension count as FFGFT's generation fibre $\mathbb{C}^3$.

R.3 The shared object: a Z_3 circulant on the C_3 axis

The icosahedral group contains C_3 axes. The cyclic coordinate map $(x, y, z) \mapsto (z, x, y)$ is an order-3 rotation; it splits Krüger's six columns into two 3-cycles, $\{c_1, c_3, c_5\}$ and $\{c_2, c_4, c_6\}$. The Gram matrix of such a C_3 orbit is – forced by the C_3 symmetry – an *exact real Z_3 circulant*, precisely the type of FFGFT's mass operator $S = \mathrm{circ}(t_0, t_1, t_2)$ on $\mathbb{C}^3$:

$$G_{\text{orbit}} = \mathrm{circ}\left(1,\ \tfrac{1}{\sqrt{5}},\ \tfrac{1}{\sqrt{5}}\right) \quad (\tau = \varphi,\ \text{normalised}),$$

with $r = 2g_1/g_0 = 2/\sqrt{5} \approx 0.894$ and phase $\theta = 0$ (real). The perp space ($\tau = \sigma$) gives $r = -2/\sqrt{5}$. Closed form: $|r| = 2|\tau|/(1+\tau^2)$, and for $\tau = \varphi$ the φ cancels exactly to $2/\sqrt{5}$ ($1 + \varphi^2 = \sqrt{5}\,\varphi$).

Criteria 1 and 2 met

A Z_3 circulant lives *inside* HLV's own icosahedral geometry – not imposed, but as the Gram matrix of a C_3 orbit. This is the same object as FFGFT's mass operator (criterion 1). The map is explicit (criterion 2): FFGFT's Z_3 generation symmetry ↔ a C_3 axis of HLV's icosahedral group, with the orbit $\{c_1, c_3, c_5\}$ as image.

R.4 The fork: 5-fold versus 3-fold

The divergence point (criterion 3) is sharp and, as three checks show, *fundamental* – not closable by any parameter.

Only $1/\sqrt{5}$. All 6×6 inner products of Krüger's columns have *a single* magnitude: $1/\sqrt{5}$. The 5-fold geometry knows only the angle $\arccos(1/\sqrt{5}) \approx 63.4^\circ$; the 45° angle ($\cos = 1/\sqrt{2}$)

that FFGFT needs does not occur between the vacuum directions. Every protected 3-orbit therefore gives $r = 2/\sqrt{5}$.

No τ reaches $\sqrt{2}$. The closed form $r(\tau) = 2\tau/(1+\tau^2)$ is capped by 1 for *every* τ (maximum at $\tau = 1$, cubic). FFGFT requires $r = \sqrt{2} \approx 1.414 > 1$ – which is *outside* the reachable range of Krüger's construction, for any parameter.

$\sqrt{2}$ needs 3-fold. In a general 3-fold geometry (cyclic basis (a, b, c), $r = 2(ab+bc+ca)/(a^2+b^2+c^2)$) $\sqrt{2}$ is perfectly reachable, e.g. $v = (1, 1, \sqrt{2}-2^{1/4})$. FFGFT's Koide point (circ$(1, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$, eigenvalues $0.29/0.29/2.41$, positive semidefinite) is thus geometrically legitimate – it merely needs a *3-fold* column, not Krüger's 5-fold one.

	r	θ	Koide Q	source
HLV (protected)	$2/\sqrt{5} = 0.894$	0	$7/15$	5-fold / φ
FFGFT (mass op.)	$\sqrt{2} = 1.414$	$2/9$	$2/3$	3-fold / Koide

R.5 Protected versus generic: no loophole

One objection remains: the HLV quasicrystal has thousands of projected points – could a C_3 orbit of *some* point accidentally hit $\sqrt{2}$? For 9822 projected points in the acceptance window the C_3-orbit ratios g_1/g_0 do fill the whole continuum $[-\frac{1}{2}, 1]$, and $1/\sqrt{2}$ is among them 150 times.

A generic hit proves nothing

That generic projected points sit at *every* angle – including $1/\sqrt{2}$ – holds for any dense 3-fold point set, even a cubic one. This is the Stufe-0 triviality: one picks an *unprotected* direction with the right angle. The *symmetry-protected* orbits – Krüger's six vacuum directions – by contrast rigidly give only $\pm 1/\sqrt{5}$. $\sqrt{2}$ is not present in HLV's protected structure; it appears only as a meaningless accidental angle.

R.6 The bridge is containment, not contradiction

The fork above is a statement about the *protected* invariant; it must not be read as the two models being incompatible realities. At the level of groups and dimensions they are reconcilable by *containment*. The icosahedral group contains the 3-fold: $C_3 < A_5$ – a 120° rotation about a $(1, 1, 1)$ axis and a 72° vertex rotation are *both* symmetries of the same icosahedron – and $T^7 = T^4 \times T^3$ ($7 = 4 + 3$). Hence HLV $\supset$ FFGFT: HLV is FFGFT's 4D core extended by the three perp circles and the full 5-fold. What the fork establishes is therefore not incompatibility but the *realised protected geometry*: FFGFT protects an independent 3-fold column ($\sqrt{2}$, Koide $2/3$), HLV protects the 5-fold ($2/\sqrt{5}$), and $\sqrt{2}$ is not among HLV's protected directions (it appears only as an unprotected accidental angle, previous section). The models share the ambient $3{+}1$ and are group/dimension-reconcilable while realising different protected structure – which is the empirically

decidable difference, not a coordinate artifact and not a logical incompatibility. Computed in Doc. 285 (`recompute_reconcile.py`, `dim_bridge_1..3`).

Where this is asserted. All of this lives in the compactified picture — the tori with every circle compact, where no circle is singled out as time. The bridge to physical, decompactified observables runs through the single step $S^1 \to \mathbb{R}$, the measurement (Doc. 270); the containment is therefore asserted upstream of the measurement, in the compact state, not on the physical-time axis.

R.7 HLV is the decompactified variant: mass and time only through the addition

One layer of the fork sits deeper than the symmetry order and is easy to miss. HLV's cut-and-project maps into an infinite, aperiodic $\mathbb{R}^3$ – it is the *decompactified* variant. FFGFT's space is by contrast *compact* (T^4). This is not cosmetic: mass and time are intrinsic in FFGFT because compactness quantises the modes – that is the time-mass duality $\tilde{T} \cdot m = 1$. Without compactness there is no mode quantisation, hence *no intrinsic mass scale and no time*. That is exactly why HLV's protected structure is purely geometric: a real Gram metric with $\theta = 0$ and dimensionless angles. The intrinsic *mass* scale is missing because the decompactified variant has no mode quantisation; it arrives, as already recorded in Doc. 282, only *through the addition*. But it matters that this $\theta = 0$ is the spatial metric, *not* HLV's time. HLV is not timeless or phaseless – its time sits in the native spiral-time operators (§ 8), which the spatial Gram does not capture at all. The C_3 probe saw only the geometry, not the dynamics.

This sharpens criterion 1. The shared Z_3 circulant is the same only *algebraically* (the same cyclic form). Physically it differs: in HLV (decompactified) a *spatial metric* without mass and time, in FFGFT (compact) the *mass operator* with mass and time intrinsic. The bridge carries the form, not the content.

The fork has three layers

layer	HLV	FFGFT
compactness	decompactified (metric)	compact ($\tilde{T}\cdot m = 1$)
symmetry order	5-fold (φ, $1/\sqrt{5}$)	3-fold ($\sqrt{2}$)
phase	spiral-time: skew-adj. R (dyn., § 8)	scalar, $\theta = 2/9$

The complex holonomy ($\theta = 2/9$, FFGFT) demanded above is the *operational consequence* of compactness: there mass and phase arise from the quantised modes. The HLV column of the phase row is subtler than a bare $\theta = 0$ – that was the *spatial* Gram metric; HLV's own time-phase sits in the skew-adjoint spiral-time operator R (§ 8). A full recompactification of HLV must *co-integrate* these operators, not merely periodise the space – and a compact HLV is moreover no longer a quasicrystal.

R.8 A compactified variant of HLV: the rational approximant

Can this recompactification be carried out concretely, to make the comparison fair? Yes – with an established notion from quasicrystal physics, the *rational approximant*. One replaces the irrational φ in the projector by rational Fibonacci ratios $\tau_n = F_{n+1}/F_n$ $(1, 2, \frac{3}{2}, \frac{5}{3}, \frac{8}{5}, \ldots \to \varphi)$. For rational $\tau = p/q$ the physical projector, multiplied by q, becomes *integer* – the projection of $\mathbb{Z}^6$ lands on a scaled integer lattice $(1/q)\mathbb{Z}^3$, i.e. on a *periodic crystal* on T^3. For irrational φ no such factor exists; the point set stays dense and aperiodic. That is the precise sense of "compactified HLV" – and, as § 6 already noted, it is no longer a quasicrystal but its periodic crystal approximant.

The approximant is periodic and therefore has discrete modes – a *mass scale becomes possible*. This closes the deepest layer of the fork (compactness): HLV is now in the same regime as FFGFT. The C_3 ratio converges:

$\tau = F_{n+1}/F_n$	r (C_3)	Koide Q
$2/1$	0.800	0.440
$5/3$	0.882	0.463
$13/8$	0.893	0.466
$\to \varphi$	$0.894 = 2/\sqrt{5}$	$7/15$

But the approximant inherits HLV's 5-fold symmetry: $r \to 2/\sqrt{5}$, Koide $\to 7/15$; the *spatial* metric stays real ($\theta = 0$). So the symmetry order keeps diverging. The phase layer, however, is not settled by "$\theta = 0$": the purely spatial approximant only recompactifies space; HLV's time-phase sits in the spiral-time operators it still has to integrate (§ 8).

The fork survives the fair comparison

With the approximant the comparison is *compact vs compact*, and thus fair. It confirms the fork: *even compactified*, HLV gives Koide $7/15$, not FFGFT's $2/3$. The divergence was therefore *not a compact/decompact artefact* – it sits genuinely in the 5- vs 3-fold symmetry. One can compactify HLV; the finding holds.

R.9 What compactification must integrate: HLV's spiral-time

The approximant of § 7 recompactifies HLV's *space*. But HLV is not purely spatial: its defining dynamics are the native 6D *spiral-time* operators

$$\Psi(t) = t + i\,\phi(t) + j\,\chi(t), \tag{R.1}$$

with $\phi(t)$ a phase/coherence channel and $\chi(t)$ a memory/hysteresis channel (Krüger, Zenodo 20668125 and 20512650). A compactified variant that is to *remain* HLV must integrate these operators – the purely spatial approximant does not yet do so. This corrects the phase layer of the fork.

Krüger does not apply Ψ directly (that would give hypercomplex fields) but via a *conservative translation* into two real operators:

channel	becomes	requirement
ϕ (phase)	phase/vorticity ordering	skew-adjoint generator R (energy-neutral)
χ (memory)	history dependence	positive-type kernel K (dissipative)

with exponential example $K_\tau(t) = \frac{1}{\tau} e^{-t/\tau}$, which reduces to an internal variable. Three things follow:

1. **The phase layer is not $\theta = 0$.** The $\theta = 0$ of § 3 is the *spatial* Gram metric, not HLV's time. HLV's phase sits in the skew-adjoint operator R, dynamically driven by $\dot{\phi}(t)$. So the layer reads correctly: *dynamic operator-phase* R (HLV) vs *fixed scalar phase* $\theta = 2/9$ (FFGFT) – not phaseless vs phased.
2. **The time/memory layer is a real bridge candidate.** HLV's χ channel is a positive-type, non-Markovian memory kernel $\int_0^t K(t-s)\,u(s)\,ds$. FFGFT's time extension in Doc. 282 gives a memory kernel as the Fourier transform of the discrete spectral density $\sum_k g_k^2\,\delta(\omega-\omega_k)$ – non-Markovian with *back-flow and revivals*, hence a *discrete-spectrum, oscillatory* kernel (exponential limit → internal variable). This is not quite the same shape as HLV's χ in the fluid sector, which is a *positive decaying* kernel (Krüger's Navier-Stokes extension, Zenodo 20512650): both non-Markovian and positive-type, but of different form. Where space forks (5- vs 3-fold), at the time level two kernels of the same *class* (non-Markovian positive memory) face each other: *a spatial fork, a possible χ-memory bridge*.
3. **Two honest limits.** First: the $i{\cdot}j$ product relation of the spiral algebra is fixed in *none* of Krüger's sources (Doc 271, FA-Hub, Navier-Stokes) – it is always bypassed via the operator translation. The phase comparison must therefore be made at the *operator* level, not the algebraic one; whether operator-phase R meets FFGFT's $2/9$ can only be decided with an explicit R specification. Second: Krüger himself records that many ordinary reduced models generate memory; the HLV reading is only a candidate model class, not a demonstrated ontology (Run M1: the HLV kernel beats weak nulls but, on the *overdamped* observable sector, is statistically *indistinguishable* from a generic biexponential – HLV favoured only by parsimony/BIC). The χ bridge is thus *structural kinship*, not a uniqueness claim – consistent with Doc. 276/281 (φ is not specific either). This is sector-dependent: in the *coherence/revival sector* FFGFT's discrete-spectrum kernel carries a signature (BLP back-flow 5.125) that any *decaying* kernel misses – though uniqueness against other *structured* kernels stays open.

The three levels are not equally strong

Reading the connection by physical level – space, time, mass – the three are *not* equivalent, and this is methodologically decisive:

- **Mass (Z_3 circulant):** the *genuine* bridge – derived from each geometry on its own, with an explicit C_3 map and a real divergence point. It alone fully passes the three-part criterion.
- **Time (χ memory):** *structural kinship*, a candidate – not HLV-unique (Run M1: on the overdamped sector a generic biexponential is *indistinguishable*; HLV favoured only by parsimony). A bridge candidate, not a uniqueness claim.

- **Space (connection-Laplace on a fibre bundle):** *shared language*, not a bridge. That both are a connection-Laplace on a fibre-valued space holds for *every* lattice gauge theory – this is Stufe-0 scaffolding, exactly what the criterion excludes. It must not be counted as a load-bearing connection, or one inflates the link toward triviality.

So only the mass level is a bridge in the sense of the criterion; the time level an honest candidate; the space level a shared frame, no more.

R.10 What remains: hexagonal, not pentagonal

Strip HLV of the two axes along which it forks – symmetry order and compactness – and exactly one ingredient remains that is not already FFGFT: the spiral-time. Its operators (R, K) depend on neither 3- nor 5-foldness and survive the switch to 3-fold unchanged. *What substantially remains of HLV is therefore not a geometry but a time* – and it is there, in the dynamics, that it meets FFGFT's non-Markovian kernel from Doc. 282. The geometric remainder – icosahedral 5-foldness – is an elegant but physically empty branch: no sector realises it, and φ is not specific (Doc. 276/281).

A short geometric lesson follows. Five-foldness is *non-crystallographic* – it does not tile periodically and forces decompactification; three-foldness is *crystallographic*, of the same family as six-foldness (the triangular/hexagonal lattice). Where geometry carries the mass structure it is therefore hexagonal, not pentagonal – and this is no new finding but FFGFT's own anchor: the *Euler Tonnetz* is a hexagonal lattice (Doc. 275, "the frame anchors in the Euler Tonnetz, not in the golden ratio").

The defensible form of the thesis

"Hexagonal" means the crystallographic family (3-/6-fold), not the claim that spacetime is a hexagonal tiling – FFGFT's base remains T^4 with a Z_3 generation symmetry. Empirically 3-fold-confirmed so far are only the charged leptons; the hadrons are near-degenerate ($Q \approx 1/3$), hence neither 3- nor 5-fold. What stands is: *where geometry carries the mass structure it is hexagonal/crystallographic; the pentagonal stays an elegant, physically empty branch; and what substantially remains of HLV is its time, not its geometry.*

R.11 Conclusion

All three parts of the criterion are answered. (1) Same object: a Z_3 circulant from HLV's own geometry. (2) Map: the C_3-axis embedding of FFGFT's Z_3 into HLV's icosahedral group. (3) Divergence: fundamental, and in three layers – compactness (decompactified/metric vs compact/$\tilde{T}\cdot m = 1$), symmetry order (5-fold φ, $1/\sqrt{5}$ vs 3-fold Koide, $\sqrt{2}$) and phase (HLV's dynamic spiral-time operator-phase R vs FFGFT's fixed scalar $2/9$; § 8). Compactness stays decisive for the *mass* scale (no quantised modes, no intrinsic m); HLV's *time*, however, sits in the spiral-time, independently of it. No parameter of HLV reaches FFGFT's

protected Koide point – yet at the time/memory level both non-Markovian kernels are of the same class: *a spatial fork, a possible temporal bridge* (§ 8).

This is exactly what the criterion is built to deliver: a *genuine* connection – shared Z_3-circulant family, explicit C_3 map – *with* a real divergence point, and demonstrably *not* Stufe-0 triviality (the protected structure separates, only the meaningless part coincides). FFGFT and HLV sit at two *different* points of the same family. It is a *fork*, not a merge: a common stem (a Z_3 circulant from a higher-dimensional lattice with a hidden space), two branches according to the symmetry order.

For the correspondence the statement is precise and fair: *we meet in the Z_3 circulant along your C_3 axis; we separate because your icosahedral 5-fold symmetry rigidly forces $1/\sqrt{5}$, while FFGFT's Koide condition demands a 3-fold $\sqrt{2}$ – and no parameter of your construction bridges that.* Reproducible with `ffgft_hlv_c3_bridge_probe.py` (six checks, numpy-only).

Chapter S

Doc. 284 — A formula-level reading of the Spiral-Time neurodynamic model

Abstract

This document is a reproducible, formula-level reading of *Information-Theoretic Modeling of Neural Coherence via Triadic Spiral-Time Dynamics: A Framework for Neurodynamic Collapse* (Krüger, Feeney & Wende, *Medinformatics* 2026, DOI 10.47852/bonviewMEDIN62029043). Every verdict is produced by a short runnable script. The reading separates what holds (an interpretable phase-deviation feature; genuine octonion non-associativity) from what is mislabeled (the triadic algebra is quaternionic, not bicomplex), internally inconsistent (two incompatible definitions of $\Delta\Phi$), not derived (the 0.40 threshold; the claimed spirals and limit cycles), or not reproducible from the paper (the reported $r \approx 0.96$). Underneath these sits *one* foundational issue: the operator $\psi(t) = t + i\varphi(t) + j\chi(t)$ is built on physical, linear time as its base axis – the decompactified, post-measurement coordinate – so its memory is generic by construction. The FFGFT reading makes this precise: the system is compact and carries memory. By Doc. 270 the decompactification to SI time is a mode-preserving embedding (lossless), and FFGFT does carry framework-specific SI observables (masses/Koide, CHSH$\sim \xi$ – not resolvable with present hardware –, D_f, L_0; Doc. 270/249); what removes the recurrence is not the SI conversion but the finite, non-periodic observation window plus decoherence. The sharp, narrow consequence: the coherence-sector recurrence is not *resolvable* on a warm, finite-window SI-time signal such as EEG. This is a reviewer note addressed to the authors; the intent is the falsification-forward standard the HLV/HICE side already applies.

S.1 Occasion and method

The Medinformatics paper applies the HLV Spiral-Time operator to neural coherence and proposes the Information–Coherence–Energy (ICE) stability principle and a $\Delta\Phi$-based "neurodynamic collapse" boundary. Rather than

argue from impression, each mathematical claim here is checked by computation. The scripts are `test_algebra.py`, `test_deltaphi.py`, `test_dynamics.py`, `test_compactification.py` and `make_figures.py`; fixed seeds throughout. Findings split into: *holds*, *mislabeled*, *internally inconsistent*, *not derived*, *not reproducible from the paper*, and *speculative premise*. One issue is foundational – the time-axis the operator is built on – and is taken first, because findings on $\Delta\Phi$ and on the dynamics are its symptoms.

#	Claim	Section	Status
0	Base time-axis $\psi(t) = t + i\varphi(t) + j\chi(t)$ on physical linear time	2.1, 2.2.1	Foundational – operator on the post-measurement axis (subsumes 3 & 5)
1	Triadic time "reduced bicomplex/octonionic plane"	2.1	Mislabeled – it is the quaternions
2	Octonionic operator; non-assoc. "explains psychology"	2.10	Math holds, claim overreaches
3	$\Delta\Phi$ instability functional	2.6 vs 12/15	Internally inconsistent (two definitions)
4	Critical $\Delta\Phi_c \approx 0.40$, "not a free parameter"	2.5, 2.6.2	Not derived (free parameter)
5	Healthy spirals / trauma limit cycles	2.2.3, 2.9	Not derived from the equations
6	$r \approx 0.96$, $\Delta\Phi_c \approx 0.40$ "across subjects"	3.4	Not reproducible from the paper
7	Axonal biphoton entanglement coupling Γ_{axon}	2.7–2.8	Speculative premise as established

S.2 The base time-axis (foundational)

This is the most consequential choice in the paper, and the one under which findings 3 and 5 are downstream symptoms. The Spiral-Time operator

$$\psi(t) = t + i\,\varphi(t) + j\,\chi(t) \tag{S.1}$$

takes physical, linear time t as its scalar base and writes φ, χ as functions of that same t. The paper is explicit that t is physical: U_1 is defined as "linear physical time," tracking external chronology (§2.2.1). That fixes the representation in which every later memory and holonomy claim is made.

Unit-independent. For $t + i\varphi + j\chi$ to be one algebraic object the three components must be the same kind of quantity. The paper asserts dimensionlessness (§2.1.2) yet simultaneously calls t physical time and chronology (§2.2.1); φ and χ are internal phase / identity coordinates. Adding a physical time to internal phases in one quaternionic sum

is dimensionally incoherent unless all three are first reduced to a common dimensionless measure. (Since i, j anticommute the object is quaternionic – t is the real part of a quaternion whose cross-term $ij = k$ is never tracked; see §S.3.)

Deeper. A framework-specific memory lives on the compact internal time-circle S^1: a discrete spectrum, an exact recurrence (in FFGFT the coherence-sector back-flow 5.125, Doc. 282). Writing $\varphi(t), \chi(t)$ as single-valued functions of physical linear t flattens S^1 onto $\mathbb{R}$ – it performs the projection $S^1 \to \mathbb{R}$ before any dynamics, and a single-valued function over linear time has no compact periodicity, so the recurrence is not representable on this axis at all. By the conversion-as-measurement principle (§S.11), committing to physical time is itself the disturbing, information-discarding projection – no apparatus needed. The notation has silently taken that step, hiding the one operation – the conversion/measurement from compact internal time to physical SI time – that decides whether a memory claim is framework-specific or generic.

Consequence. The operator is, by construction, post-measurement: whatever memory it carries on physical (EEG) time is generic. This is reproducible (`test_compactification.py`): the discrete-spectrum return signal has back-flow 11.5 on the compact horizon, which a finite physical read-out window drives to 0.000, while the six lines never resolve ($2\pi/T_{\text{win}} \gg$ line spacing 0.091) – the read-out is a non-invertible projection. The same root explains the two downstream findings: a Berry-phase holonomy (Eq. 12/15, §S.5) needs a closed loop – a compact cycle – and a limit cycle (§S.7) needs non-gradient closed orbits; neither is supported on a linear physical-time axis.

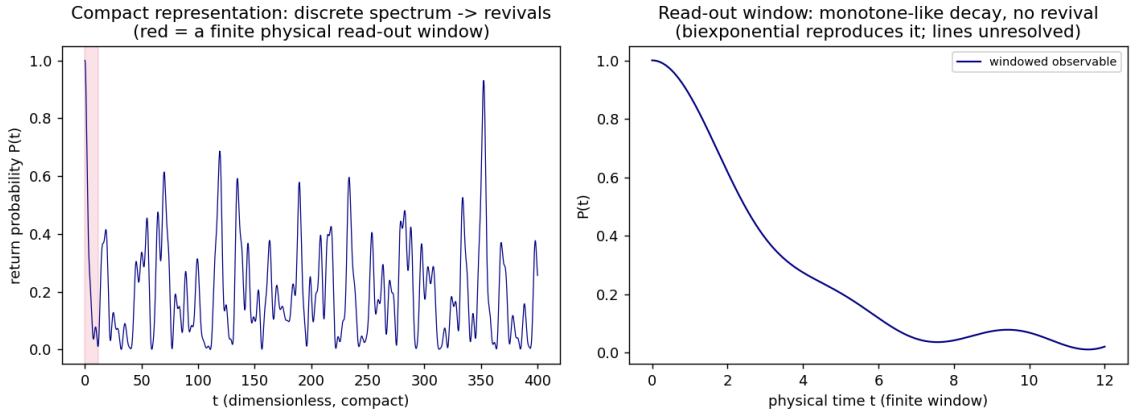


Figure S.1: Read-out as a non-invertible projection. Left: on the compact representation the discrete spectrum produces revivals (back-flow $\gg 0$); the red band marks a finite physical read-out window. Right: inside the window the observable decays without revival and the discrete lines are unresolved.

Constructive fix. Define the operator on the compact, dimensionless internal time (cyclic), where the discrete spectrum and recurrence live, and let physical SI time enter only through an explicit, declared conversion – which is precisely what projects out the

framework-specific part. Either the operator stays internal/compact (no direct EEG claim) or it is physical/SI (generic memory). $\psi(t) = t + i\varphi(t) + j\chi(t)$ tries to be both. The minimal change is to declare two distinct times and write the conversion between them explicitly as the measurement it is.

S.3 Triadic time algebra – mislabeled

The stated relations $i^2 = j^2 = -1,\ ij = -ji$ are exactly the imaginary quaternions: setting $k := ij$ gives $k^2 = -1$, and $\{1, i, j, k\}$ closes into $\mathbb{H}$, which is associative and non-commutative (`test_algebra.py`, A). A *bicomplex* plane requires the two imaginary units to commute ($ij = ji$); here $ij = -ji$, so the algebra is *not* bicomplex. A three-term operator $t + i\varphi + j\chi$ lives inside a quaternion sub-algebra; octonions are not required for it. The correct name is quaternionic.

S.4 Octonionic structure – math holds, interpretation overreaches

Octonion non-associativity is real: the associator $(e_1 e_2)e_4 - e_1(e_2 e_4) = 2e_7 \neq 0$ (Cayley–Dickson), while associators on a quaternionic triple vanish (`test_algebra.py`, B). That is correct mathematics. The further step – that this "directly matches empirical psychology" and is "the first physics-based explanation for psychological nonlinearity" – is an interpretive analogy, not a derivation: no psychological quantity is computed from or tested against the algebra, and the eight basis directions are assigned meanings by stipulation.

S.5 $\Delta\Phi$ – two incompatible definitions under one symbol

- **Definition 1 (§2.6):** $\Delta\Phi = \alpha|\Delta S| + \beta|\Delta I| + \gamma|\Delta C|$, convex weights, each component in $[0, 1]$ – a unitless weighted-L^1 index in $[0, 1]$. It is linear in each component, so it has no curvature and no intrinsic bifurcation; the level set $\Delta\Phi = c$ is a flat polytope facet for every c.
- **Definition 2 (Eq. 12, 15):** $\Delta\Phi = \oint(\nabla \times A) \cdot da \approx \mathrm{Im} \oint \langle\psi|\nabla\theta|\psi\rangle\, d\theta$ – a Berry-phase holonomy, an angle in radians ranging over $0 \ldots 2\pi$, with no reason to lie in $[0, 1]$ or near 0.40 (a loop with $|\gamma| = 0.40$ rad is a tiny $\sim 29^\circ$ cone).

A normalized deviation index and a radian holonomy are different mathematical objects. Calling both "$\Delta\Phi$" and assigning both the same critical value 0.40 is an equivocation (`test_deltaphi.py`, C/D).

S.6 The 0.40 threshold – not derived

§2.5 states the value is "not imposed as a free tuning parameter" and that the operator structure "enforces" a bounded instability surface. But the L^1 functional has no intrinsic critical point (it is linear); the only structural statement available is the trivial bound

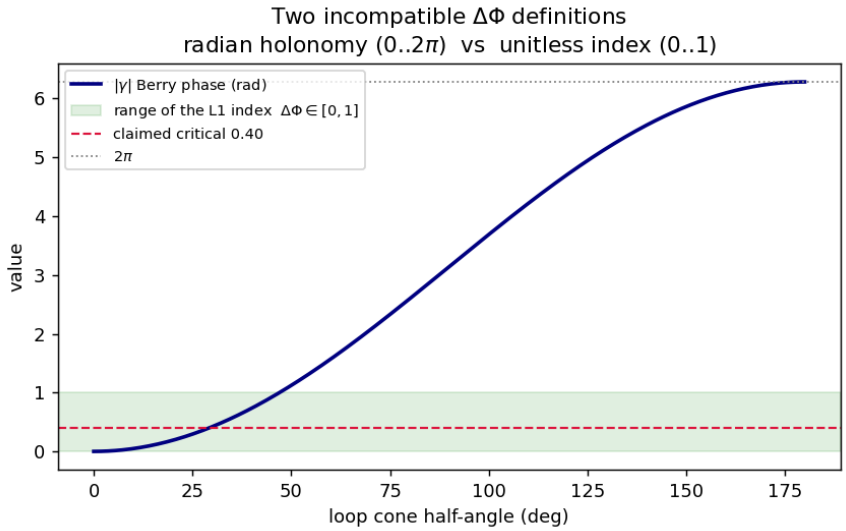


Figure S.2: The two $\Delta\Phi$ definitions are incompatible objects: a radian holonomy ($0 \dots 2\pi$) versus a unitless index ($0 \dots 1$).

$0 \le \Delta\Phi \le 1$ and the symmetric reference $\Delta\Phi = d$ (so $1/3$ at $d = 1/3$). §2.6.2 itself attributes the 0.40-vs-0.333 gap to "dataset-specific weighting asymmetry … and normalization," and at equal deviation $d = 0.40$ every convex weight set returns 0.40 (`test_deltaphi.py`, C/E). So 0.40 is an empirical/contour value, not a structurally enforced bifurcation.

S.7 Spirals and trauma limit cycles – not derived

The only dynamical mechanism given with explicit potentials (§2.4) is the gradient-flow part of the master equation, $\partial_t \Psi = -\nabla V_{\mathrm{DLHR}}(\Psi)$. For any gradient flow,

$$\frac{dV}{dt} = \nabla V \cdot \frac{d\Psi}{dt} = -|\nabla V|^2 \le 0, \tag{S.2}$$

so V is a strict Lyapunov function and the system cannot have periodic orbits, limit cycles, or sustained spirals – it relaxes to fixed points (theorem). Numerically (`test_dynamics.py`, F): across 40 trajectories $V(t)$ is monotonically decreasing (0 violations) and 0 closed orbits are found. The phenomenology "healthy cognition → stable inward spirals," "trauma loops → limit cycles" therefore does not follow from the specified dynamics; it would require the advection terms $\lambda_\varphi \partial_\varphi$, $\lambda_\chi \partial_\chi$ or W_{oct}, which are never given concretely. (The cubic M^3 term also makes V unbounded below.)

S.8 Empirical claims – not reproducible from the paper

The reported $r \approx 0.96$ and $\Delta\Phi_c \approx 0.40$ "across subjects" (§3.4) cannot be recomputed from the paper, which gives no data table. Reproduction needs the external dataset (Zenodo 10.5281/zenodo.17990696) and the validation notebook. A fair test would require the

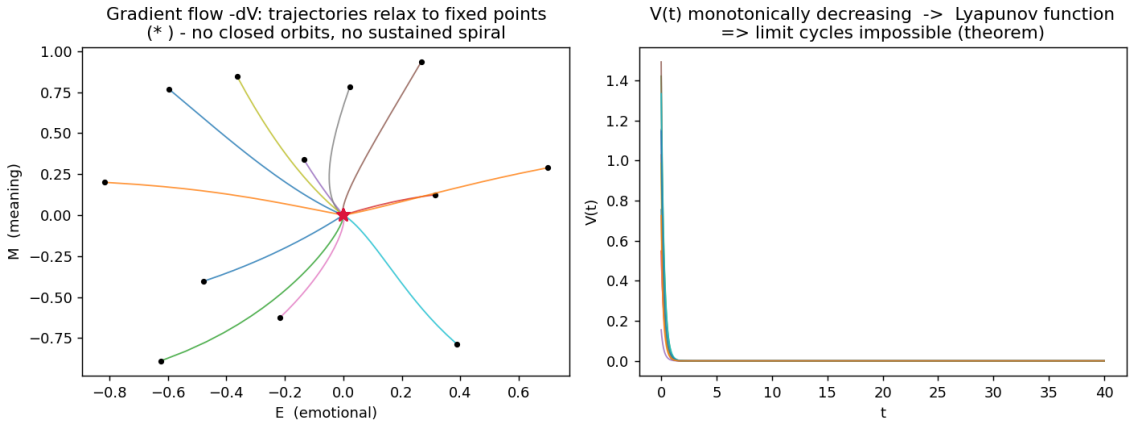


Figure S.3: Gradient flow with the paper's DLHR potentials: trajectories relax to fixed points (no closed orbits, no sustained spiral) and $V(t)$ decreases monotonically, so limit cycles are impossible.

dataset plus a pre-registered protocol with the nulls the authors already use on the HICE side (time-shuffle, present-state-matched, phase-randomized).

S.9 Axonal biphoton entanglement – speculative premise stated as established

$\Gamma_{\text{axon}} = g\,e^{i\varphi}E_{12} + g'e^{j\chi}E_{23}$ (§2.8) is built on "experimental findings" of biphoton entanglement and long-range coherence in myelinated axons, presented as settled research. Entanglement in warm, wet, myelinated axons is not established physics. This is not a formula error, but it is a load-bearing empirical premise asserted as fact and should be flagged as hypothesis.

S.10 A defensible core

Stripped of the superstructure there is a sound, modest object: $\Delta\Phi = |\omega(t) - \omega_{\text{memory}}(t)|$ – the deviation of instantaneous phase velocity from a short-term phase-memory baseline – is an interpretable, lightweight time-series instability feature, and as a signal-analysis operator (the framing of the companion respiratory paper) it is legitimate. The problems are confined to the HLV/octonion/consciousness superstructure, the $\Delta\Phi$ redefinition, and the un-derived threshold and dynamics.

S.11 An FFGFT reading: conversion as measurement

Temporal recursion does carry memory, and it maps onto FFGFT's own structure (the recursion operator $\mathscr{R}$; the memory kernel as the Fourier transform of the connection-Laplacian spectral density, Doc. 282). The following is offered as a constructive bridge and an explicit conjecture, not a result.

FFGFT's memory has two faces. On the overdamped / relaxation observable the kernel is a positive multi-exponential, and a generic biexponential reproduces it (residual $\sim 10^{-3}$): non-identifiable, Stufe-0. On the coherence observable $c(t) = e^{-\Gamma(t)}$ – its home sector – the discrete spectrum produces a genuine revival: back-flow 5.125, which generic decaying kernels cannot reproduce (back-flow 0). So the memory is discriminating, but only on the coherence sector and only when *both* a discrete spectrum *and* preserved coherence (small Γ) hold.

FFGFT is fundamentally compact (T^4, discrete spectrum), and in that dimensionless representation the system always has memory. It is important to locate the loss precisely, because the conversion to SI is *not* where it happens. By Doc. 270 the Type I duality decompactification $S^1_m \to \mathbb{R}_t$ (governed by $\tilde{T} \cdot m = 1$) is a *mode-preserving embedding* – lossless; the periodicity merely stops being directly visible and survives as the discrete spectrum and as physical observables (the revival time T_{rev}, the CHSH deviation $\sim \xi$, the latter not resolvable with present hardware). A bare units rescaling (by $\hbar, c$) is likewise invertible and loses nothing. The genuinely lossy operations are three and distinct: the *reverse* compactification $\mathbb{R} \to S^1$ (non-periodic parts do not descend), the spatial projections $D3 \to D0$ (Doc. 270), and – for any read-out – the restriction to a *finite, non-periodic observation window plus decoherence*. It is this last step, not the unit conversion, that keeps the local short-time behaviour and discards the global recurrence, exactly as a measurement selects a basis and drops the coherences (Test G). A physical measurement is the same window-and-decoherence projection at its strong-coupling end. This is consistent with what the corpus already records: the vacuum energy is present compactified, visible decompactified, with $c, \hbar c$ as SI conversion factors (Doc. 263) – the decompactified view is not less true, only differently resolved.

The reading therefore both supports the modest core (recursion does carry memory) and bounds the claim precisely. The framework-specific signature does survive into SI in the right channels – mass reading, CHSH$\sim \xi$ (not resolvable with present hardware), D_f, L_0 (Doc. 270/249). What does *not* survive is the coherence-sector recurrence read on a warm, finite-window SI-time observable: there the window and decoherence have already discarded the global recurrence, so the very observable the clinical sections use – SI-time EEG – is post-measurement in exactly that sense, and is the regime where the FFGFT harness (Doc. 282/283) shows the memory to be generic, not a fingerprint. This is also why the base-time-axis choice (§3) is decisive: the operator is written on that finite, post-measurement axis from the start.

S.12 Reproducibility

- `test_algebra.py` – quaternion closure + bicomplex refutation (A); octonion associator (B).

- `test_deltaphi.py` – L^1 bound/linearity (C); Berry-phase range (D); 0.40 as free knob (E).
- `test_dynamics.py` – gradient-flow Lyapunov / no-limit-cycle test (F).
- `test_compactification.py` – read-out as a non-invertible projection: compact back-flow vs windowed (G).
- `make_figures.py` – the figures used above.

The figures `fig_deltaphi.png`, `fig_dynamics.png` and `fig_compactification.png` are produced by these scripts; all use fixed seeds.

Chapter T

Doc. 285 — The unified compactified picture and the computed 3-vs-6 bridge

Abstract

FFGFT and HLV both reduce to the observable $3{+}1$. This raises a sharp question: if the four circles of FFGFT and the six (plus time) of HLV both end in $\mathbb{R}^3 \times \mathbb{R}_t$, are the 3 and the 6 merely two coordinate descriptions of the same thing – i.e. only a mathematical transformation? This document gives the compact-view answer in one notation and checks it by computation. In the compact picture both models are of the same form: a torus $T^N = (S^1)^N$ carrying a finite symmetry group G, with a Type-I decompactification (the measurement) selecting one circle as time. FFGFT is $(T^4, \mathbb{Z}_3)$; HLV is $(T^7 = T^6 \times S^1, A_5)$. Three reproducible results settle the question. (1) "6 is the doubling of the 3" is literal: the icosahedral 6D representation splits as $\mathbf{3} \oplus \mathbf{3}'$, the two 3D irreducibles of A_5, differing exactly by $\varphi \leftrightarrow 1-\varphi$; the 6D character is an integer while each 3D block is irrational (φ). (2) The triangle is the more fundamental, minimal carrier: 3-fold symmetry is crystallographic and lives in 3D, while 5-fold is non-crystallographic and *requires* the 6D embedding. (3) The two models are *reconcilable by containment*: the icosahedral group A_5 contains the 3-fold C_3 (both are symmetries of the same icosahedron), and $T^7 = T^4 \times T^3$, so HLV is FFGFT's 4D extended by the three perp dimensions ($7 = 4 + 3$, HLV $\supset$ FFGFT). The protected values $\sqrt{2}$ (Koide $2/3$) and $2/\sqrt{5}$ (φ) belong to two *different protected geometries* – FFGFT's independent 3-fold and HLV's 5-fold; the groups coexist ($C_3 < A_5$), so this is no incompatibility, and the cut-and-project family $r(\tau) \le 1$ failing to map one onto the other shows only that interconversion is the wrong question. (Per Doc. 283, $\sqrt{2}$ is *not* inherited from HLV's icosahedrally protected directions, which give only $2/\sqrt{5}$; it is a separate 3-fold protected choice.) The genuine difference is *which symmetry sector is physically realised*: FFGFT realises the minimal crystallographic 3-fold core (Koide $\theta = 2/9$, empirically confirmed), while HLV adds the 5-fold/perp structure as a testable extension. Hence "only a mathematical transformation" is too weak (the realised sector is a genuine, empirically decidable feature) and "inequivalent structures" is too strong (they are reconcilable, HLV $\supset$ FFGFT): the precise statement is a shared $3{+}1$ core with HLV the icosahedral extension of FFGFT's minimal 3-fold core.

T.1 Occasion: both reduce to $3{+}1$ – is the rest a transformation?

Doc. 270 separates the reductions of FFGFT ($T^4 \to 3{+}1$) and Doc. 283 the FFGFT↔HLV relation (a $\mathbb{Z}_3$-circulant bridge along a C_3 axis, forked at the symmetry order, $\sqrt{2}$ vs $2/\sqrt{5}$). HLV builds space as a 6D→ 3D icosahedral cut-and-project and carries time separately (spiral-time). Both models therefore end in the same observable $\mathbb{R}^3 \times \mathbb{R}_t$. The natural question is whether the difference between FFGFT's 3 and HLV's 6 is then merely a choice of coordinates – a lossless mathematical transformation of one and the same $3{+}1$. The answer requires a common notation and a computation; both are given here. All numerical statements are reproducible with three short scripts (§T.21).

T.2 The unified compactified notation (T^N, G)

In the compact representation there is no "space" versus "time" and no "internal" versus "external": a circle is a circle. Both models are then of one form:

$$\mathscr{M} = T^N = (S^1)^N, \qquad G \subset O(N) \text{ acting}, \tag{T.1}$$

with a Type-I decompactification (the measurement) selecting one circle as time. The remaining circles stay compact (discrete spectrum) or project (space, Type II, Doc. 270). The two models are two instances of (T^N, G):

	FFGFT	**HLV**
compact form	$T^4 = (S^1)^4$	$T^7 = T^6 \times S^1$
symmetry group G	$\mathbb{Z}_3$ (3-fold, crystallographic)	A_5 icosahedral (5-fold, non-crystallographic)
space (compact)	3 circles	6 circles $= \mathbf{3} \oplus \mathbf{3}'$
time	one circle (mass circle, $\tilde{T} \cdot m = 1$); scale-linked by the duality, *not* a split-off factor	one circle (spiral-time); an A_5-singlet, so the torus *factorises* $T^6 \times S^1$
how G acts on time	woven in through $\tilde{T} \cdot m = 1$	trivially (singlet)
protected invariant	$r = \sqrt{2}$, Koide $Q = 2/3$, $\theta = 2/9$	$r = 2/\sqrt{5}$, φ
why this N_{space}	3 suffices (3-fold crystallographic)	6 needed (5-fold non-crystallographic)

The whole comparison reduces to one question – *how does G act on the circles of the torus?* The labels "internal/external" are decompactification language and have no place in the compact picture; the correct compact-view distinction for the time circle is group-theoretic: in HLV it is an A_5-singlet (the torus factorises $T^6 \times S^1$, time decoupled from the spatial symmetry); in FFGFT it is tied to the other circles by the duality $\tilde{T} \cdot m = 1$ (not a direct-product factor). Which circle becomes the observed time is fixed by the measurement, i.e. the experienced situation of the embedded observer (Doc. 270, Doc. 248), not by the geometry.

T.3 "6 is the doubling of the 3" – literal (`dim_bridge_1`)

The icosahedral 6D representation used in the cut-and-project decomposes as

$$\mathbf{6} \;=\; \mathbf{3} \oplus \mathbf{3}', \tag{T.2}$$

the two inequivalent 3D irreducible representations of A_5 (physical $\oplus$ perp). On the 5-fold conjugacy class the characters are

$$\chi_{\mathbf{3}}(g_5) = 1 + 2\cos 72^\circ = \varphi, \qquad \chi_{\mathbf{3}'}(g_5) = 1 + 2\cos 144^\circ = 1 - \varphi = -\tfrac{1}{\varphi}, \qquad \chi_{\mathbf{6}}(g_5) = 1. \tag{T.3}$$

So the doubling is literal – **6** is two copies of a 3D space – and the golden ratio sits *in the split itself*: the two 3D blocks differ exactly by $\varphi \leftrightarrow 1-\varphi$. The 6D character is the integer 1 while each 3D block is irrational (φ); this is the first half of why a 6D lattice exists where a 3D one cannot (next section). This φ in the doubling is HLV's 5-fold / $2/\sqrt{5}$ at its root (cf. Doc. 281, Doc. 283).

T.4 The triangle is fundamental, the pentagon needs the six (`dim_bridge_2`)

A rotation of order n is a lattice symmetry in 2D/3D if and only if $2\cos(2\pi/n) \in \mathbb{Z}$ (crystallographic restriction). The integer values occur only for $n \in \{1, 2, 3, 4, 6\}$:

$$2\cos\tfrac{2\pi}{3} = -1 \in \mathbb{Z} \quad \text{(3-fold, crystallographic)}, \qquad 2\cos\tfrac{2\pi}{5} = \tfrac{1}{\varphi} \notin \mathbb{Z} \quad \text{(5-fold, forbidden)}. \tag{T.4}$$

Hence 3-fold symmetry ($\mathbb{Z}_3$, Koide) lives natively in 3D – three dimensions *suffice*, and the triangle is the minimal, more fundamental carrier. 5-fold symmetry cannot sit in a 3D lattice; it exists only as a projection from 6D, where (by the previous section) it acquires an integer character. The doubling $3 \to 6$ is therefore not optional decoration: it is exactly what hosting a non-crystallographic symmetry costs.

T.5 Same ambient 3+1: what the cut-and-project family does and does not show (`dim_bridge_3`)

The cut-and-project ratio

$$r(\tau) = \frac{2\tau}{1+\tau^2} \le 1, \qquad r(1) = 1, \tag{T.5}$$

is the family of cut-and-project "mathematical transformations." HLV's protected invariant $2/\sqrt{5} \approx 0.894$ is reached at $\tau = \varphi$ (and $\tau = 1/\varphi$). FFGFT's $\sqrt{2} \approx 1.414$ lies *above the maximum of the entire family* and is reached by no τ whatsoever.
What this shows – and what it does not:

- **Ambient 3+1:** both models end in $\mathbb{R}^3 \times \mathbb{R}_t$; at the level of the bare coordinate manifold this is a lossless representational transformation (Type III, Doc. 270/230).

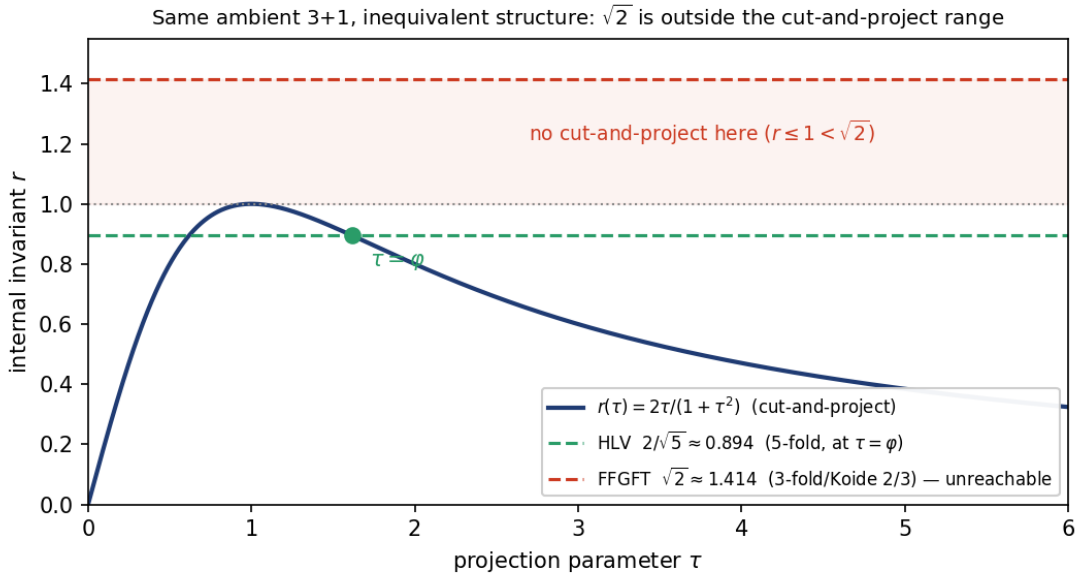


Figure T.1: The cut-and-project invariant $r(\tau) \le 1$ reaches HLV's $2/\sqrt{5}$ at $\tau = \varphi$ but never FFGFT's $\sqrt{2}$. That $\sqrt{2}$ lies above this family's maximum means only that this particular projection does not interconvert the two invariants – not that the structures are incompatible (see next section).

- **The cut-and-project map:** $\sqrt{2}$ lies outside the range $r \le 1$, so *this* one-parameter family does not carry HLV's $2/\sqrt{5}$ into FFGFT's $\sqrt{2}$. That is a statement about one map, *not* a proof of incompatibility: reconciliation is not "interconvert the two invariants" but "embed both in a common structure" – the subject of the next section. (And $\sqrt{2} > 1$ versus $r \le 1$ are not strictly commensurable, so their non-equality here is weak evidence at best.)

T.6 The reconciliation: HLV contains FFGFT (`recompute_reconcile`)

The decisive fact is group-theoretic. The icosahedral group A_5 *contains* the 3-fold C_3: a 120° rotation about a $(1,1,1)$ axis and a 72° rotation about a vertex axis are *both* symmetries of the same regular icosahedron – both permute its twelve vertices (`recompute_reconcile.py`). Hence $C_3 < A_5$, and at the level of dimensions $T^7 = T^4 \times T^3$, i.e.

$$7 \;=\; \underbrace{3_{\text{phys}} + 1_{\text{time}}}_{\text{FFGFT } T^4} \;+\; \underbrace{3_{\text{perp}}}_{\text{HLV extension}} \,. \tag{T.6}$$

Concretely $T^4 \hookrightarrow T^7$, $(x,t) \mapsto (x,0,t)$ with $0 \in T^3_{\text{perp}}$ – a totally geodesic submanifold: every FFGFT configuration is the special HLV configuration with vanishing perp coordinate. The relation between the models is therefore *containment*, HLV $\supset$ FFGFT: FFGFT is the minimal crystallographic 3-fold core, HLV the icosahedral extension adding the three

perp circles and the full 5-fold. The protected values $\sqrt{2}$ and $2/\sqrt{5}$ are invariants of the coexisting C_3 and A_5 sectors of the *same* containing structure; they need not, and do not, transform into one another – and that is no contradiction. The containment is at the level of the groups and dimensions; the *protected* invariant remains a separate, realised choice – FFGFT protects an independent 3-fold geometry ($\sqrt{2}$), HLV the 5-fold ($2/\sqrt{5}$), and (Doc. 283) HLV's protected vacuum directions do not include $\sqrt{2}$. Containment makes the models compatible without making them identical.

Where this is asserted. All of this lives in the compactified picture — the tori with every circle compact, where no circle is singled out as time. The bridge to physical, decompactified observables runs through the single step $S^1 \to \mathbb{R}$, the measurement (Doc. 270); the containment is therefore asserted upstream of the measurement, in the compact state, not on the physical-time axis.

T.7 Resolution

The tension – "both reduce to $3{+}1$, so the 3 and the 6 can only describe the same thing" – is resolved not by a fork but by *containment*. The 3 and the 6 share the same physical $3{+}1$ core, and the models are mathematically reconcilable: HLV $\supset$ FFGFT in both the symmetry ($A_5 \supset C_3$) and the dimensions ($7 = 4 + 3$). What remains is not a mathematical incompatibility but the question of *which symmetry sector is physically realised*. FFGFT realises the minimal, crystallographic 3-fold core, and its invariant – Koide $\theta = 2/9$ – is empirically confirmed; HLV adds the 5-fold/perp structure as an extension whose distinct observable (the icosahedral / quasicrystalline coherence signature, Doc. 282/283) is exactly what a test of HLV against FFGFT would probe. Both obvious formulations therefore miss: "only a mathematical transformation" is too weak (the realised sector is a genuine, empirically decidable feature) and "inequivalent, irreconcilable structures" is too strong (the embedding HLV $\supset$ FFGFT exists). The precise statement: a shared $3{+}1$ core, with HLV the icosahedral *extension* of FFGFT's minimal 3-fold core, the difference residing in the realised symmetry sector.

T.8 Recursion in the unified compact picture: λ^n

The notation (T^N, G) extends to the *recursion*: in both models the self-similar structure is a dilation by a single eigenvalue λ, the recursion being λ^n. The recursion eigenvalue is *distinct* from the protected invariant ($\sqrt{2}$, $2/\sqrt{5}$); conflating the two is an error.

FFGFT: $\lambda = \xi = 4/30000$ (Doc. 270). The recursion ξ^n is one-sided contracting ($\xi, \xi^2, \ldots$ vanish at once), giving the fractal correction

$$D_f = 3 - \xi = 2.999867, \tag{T.7}$$

a dimension just below 3 – near-crystalline. The protected invariant $\sqrt{2}$ (Koide $2/3$) is a *separate* quantity, not the recursion scale.

HLV: $\lambda = \varphi$ (golden inflation). The icosahedral quasicrystal is self-similar under the golden inflation, with the Fibonacci recursion $\varphi^2 = \varphi + 1$. The Fibonacci substitution matrix (rows $(1,1)$, $(1,0)$) has eigenvalues φ (expanding) and $-1/\varphi$ (contracting) – the

same $\varphi \leftrightarrow -1/\varphi$ pair as the **3** $\oplus$ **3′** physical/perp split. Being two-sided, the recursion is self-similar at *all* scales: a genuine quasicrystal. The protected invariant $2/\sqrt{5}$ is again separate.

	recursion λ	character	protected inv.
FFGFT	$\xi = 4/30000$	one-sided contracting, $D_f = 3 - \xi$	$\sqrt{2}$ (Koide $2/3$)
HLV	φ	two-sided (φ, $-1/\varphi$), $\varphi^2 = \varphi + 1$	$2/\sqrt{5}$

So the fork appears once more, now in the recursion: ξ tiny (one-sided, near-crystalline) versus φ golden (two-sided inflation). The extra contracting mode ($-1/\varphi$, the perp direction) is exactly HLV's extension over FFGFT, consistent with HLV $\supset$ FFGFT. In each model one number governs both the protected geometry and the recursion.

A recursion *phase* (an analogue of FFGFT's Koide angle $\theta = 2/9$) is *not* fixed here. It would have to be *derived* from the A_5 representation; the naive $2/9 + 1/5 = 19/45$ is numerology, has no derivation, and is not adopted. Computed: `recursion_unify.py`.

Status of φ. The golden ratio acquires a meaning here that it lacked in pure FFGFT – but *not* as an FFGFT constant: as the derived eigenvalue of HLV's 5-fold / perp sector, hence a marker of the boundary HLV\FFGFT. This is consistent with Doc. 190 (P35): there φ^N was the cautionary example of an arbitrary fitting base with no forward content; here φ is the eigenvalue of a named symmetry (golden inflation $\varphi^2 = \varphi + 1$), not a measured scale forced into an exponent. So φ does not become an FFGFT prediction.

T.9 The HLV phase from A_5: $\theta = 0$, the amplitude carries the difference

The protected geometry can be sharpened to a *phase*. FFGFT's Koide angle $\theta = 2/9$ is a phase offset in the 3-component circulant

$$\sqrt{m_k} = 1 + A\cos\Big(\tfrac{2\pi k}{3} + \theta\Big), \qquad k = 0, 1, 2, \tag{T.8}$$

whose Koide ratio is $Q = (2 + A^2)/6$, *independent of* θ. The model content then splits cleanly into amplitude A and phase θ.

FFGFT. $A = \sqrt{2}$ (the 3-fold protected amplitude) and $\theta = 2/9$ – a *free*, empirical offset ($Q = 2/3$ for any θ; $2/9$ is the measured Koide phase), giving $Q = 2/3$.

HLV. $A = 2/\sqrt{5}$ (the 5-fold/icosahedral amplitude: the 5-fold eigenvalue phases $e^{\pm 2\pi i/5}$ fix it through the inner product $1/\sqrt{5}$). The protected vacuum directions are symmetry-rigid, so there is *no free offset*: $\theta_{\mathrm{HLV}} = 0$, giving $Q = (2 + \frac{4}{5})/6 = 7/15$ – exactly the value in the Doc. 283 fork table.

So A_5's eigenvalue phases play explicit, distinct roles: the 3-fold phases $\pm 2\pi/3$ set the circulant *lattice* (both models); the 5-fold phases $\pm 2\pi/5$ (and $\pm 4\pi/5$ in **3′**) set HLV's *amplitude* $2/\sqrt{5}$ – not a free phase. The honestly derived HLV phase is therefore $\theta_{\mathrm{HLV}} = 0$, and the whole model difference lives in the amplitude (protected invariant $\sqrt{2}$ vs $2/\sqrt{5}$, i.e. Koide $2/3$ vs $7/15$).

	amplitude A	phase θ	Koide $(2+A^2)/6$	source
FFGFT	$\sqrt{2}$	$2/9$ (free, empirical)	$2/3$	3-fold
HLV	$2/\sqrt{5}$	0 (rigid, protected)	$7/15$	5-fold

This also disposes of the naive guess $\theta = 2/9 + 1/5 = 19/45 \approx 0.422$: it matches *neither* the derived phase (0) *nor* the Koide ratio ($7/15 \approx 0.467$), and is adopted nowhere. The result is derived under the 3-component Koide parametrisation; $\theta_{\mathrm{HLV}} = 0$ follows from the rigidity of the icosahedrally protected directions (cf. Doc. 283). Computed: `phase_from_a5.py`.

T.10 How FFGFT unrolls: decompactification as the bridge to observation

FFGFT unrolls by letting **one circle of the torus T^4 become time through the measurement act** (type-I decompactification). The three remaining circles supply observed space but stay structurally untouched. The recursion ξ^n thereby becomes a continuous dilation flow with log-period $|\ln \xi| = \ln 7500 \approx 8.92$. Crucially, the unrolling is *not* part of the theory but the bridge to observation (Doc. 248, Doc. 270) — the torus remains the original, the unrolled space is the image.

The compact form

In the compact picture FFGFT is the four-torus $\mathscr{M} = T^4 = (S^1)^4$ with $G = \mathbb{Z}_3$. All four circles are *equivalent* — there is no distinguished time. The three space circles carry the gauge structure (connection Laplacian); the fourth is the mass circle with the duality $\tilde{T} \cdot m = 1$.

The measurement act unrolls one circle (type I)

Decompactification is not a mathematical limit $S^1 \to \mathbb{R}$ but a physical act: the measurement selects one circle as time and unrolls it. This is a property of the *embedded observer* (Doc. 248), not of the geometry. The type-I map $S^1_{\mathrm{mass}} \to \mathbb{R}_t$ turns the dual mass $m = 1/\tilde{T}$ into rest energy; in doing so the winding number is discarded — the information-discarding step that forces the observer into time.

Type	What happens	Example
I	one circle becomes time (unrolled, winding discarded)	$S^1_m \to \mathbb{R}_t$
II	one circle stays compact (discrete spectrum)	$S^1_x \to S^1_x$ (unchanged)
III	lossless coordinate transformation	$T^4 \to \mathbb{R}^3 \times S^1$

Important for clean bookkeeping: *observed* space $\mathbb{R}^3$ arises from the lossless coordinate transformation (type III) $T^4 \to \mathbb{R}^3 \times S^1$ and carries continuous momenta; the *discrete,*

mass-bearing spectrum, by contrast, sits on the compact $\mathbb{Z}_3$ fiber, not on the unrolled axis. "Compact/discrete" (fiber, masses) and "$\mathbb{R}^3$/continuous" (type-III projection, space) are thus two different things and must not be conflated.

The recursion unrolls along with it

In the compact picture the recursion is discrete, λ^n with $\lambda = \xi = 4/30000 = 1/7500$. Upon unrolling this becomes a continuous dilation flow $t \mapsto \xi t$, $t = t_0\,\xi^s$ ($s \in \mathbb{R}$) — the scale flow. A discrete scale invariance in the continuum is **log-periodicity** with period $|\ln \xi| = \ln 7500 \approx 8.92$ e-folds. This is a very long period; hence FFGFT is *near-crystalline*, the fractal correction tiny: $D_f = 3 - \xi = 2.9998667$. The detailed treatment (dilation flow, log-period, branching) follows in the next section.

What survives — and what does not

	Compact (T^4)	Decompactified ($\mathbb{R}^3 \times \mathbb{R}_t$)
Mass circle	S^1, $m = 1/\tilde{T}$	$\mathbb{R}_t$, rest energy $E = m$
Space	$3 \times S^1$	$\mathbb{R}^3$ (type III, continuous momenta)
Mass spectrum	eigenvalues of S on the $\mathbb{Z}_3$ fiber (discrete)	unchanged; $m_k = \mu_k^2\,M$, $M = 1/T$
Recursion	ξ^n (discrete)	ξ^s (flow), log-period $\ln 7500$
Symmetry	$\mathbb{Z}_3$	$\mathbb{Z}_3$ (preserved)
Observer	none	embedded (measurement)

The physics sits in the *compact* picture and survives the unrolling because it consists of *ratios*: the $\mathbb{Z}_3$ symmetry (on the fiber, not on the torus), the Koide structure ($r = \sqrt{2}$, $\theta = 2/9$, frame-independent), and the mass ratios. The mass operator $S = \mathrm{circ}(t_0, t_1, t_2)$ does not change upon unrolling — it lives on the C_3 fiber, which is independent of the torus; the scale $M = 1/T$ comes from the time circle. Only the winding number of the unrolled circle is discarded. (Three time aspects: connection Laplacian without time, process time via the non-Markovian kernel, geometry time via $\tilde{T} \cdot m = 1$; Doc. 282.)

Consequence for the D7 comparison: FFGFT unrolls the mass circle into time and keeps the mass-bearing 3-core; Krüger's decompactified variant covers only {3′ + time} of this (below). Computed: `decompact_recursion.py`.

T.11 Decompactified recursive time: continuous dilation flow and log-periodicity

Compact and decompact are related by $S^1 \to \mathbb{R}$ – the measurement (Doc. 270), and the recursion follows that map. In the compact picture the recursion is the *discrete* dilation λ^n ($n \in \mathbb{Z}$): a *discrete scale invariance* (invariance under scaling by λ only). When the mass circle unrolls to non-compact time, the discrete recursion becomes a *continuous* dilation flow

$$t \longmapsto \lambda t, \qquad t = t_0\,\lambda^s, \quad s \in \mathbb{R}, \tag{T.9}$$

i.e. the renormalisation-group / scale flow. A discrete scale invariance in the continuum is *log-periodicity* of period $|\ln\lambda|$ – equivalently, complex scaling exponents $\alpha = \alpha_0 + \mathrm{i}\,k\,2\pi/|\ln\lambda|$.

	compact	decompactified
FFGFT	ξ^n (discrete)	flow $t \to \xi t$; log-period $\lvert\ln\xi\rvert = \ln 7500 \approx 8.92$
HLV	φ^n (discrete)	flow $t \to \varphi t$; log-period $\ln\varphi \approx 0.481$

The fork persists in decompactified time. **FFGFT**: a log-period of ≈ 8.9 e-folds – very long, so barely log-periodic; near-crystalline. It is the same discrete scale invariance that gives $D_f = 3 - \xi$. **HLV**: a log-period of ≈ 0.48 e-folds – short and dense; a full quasicrystal, self-similar at every scale.

FFGFT's *recursion* R is precisely this object: discrete (ξ^n) in the compact picture, a continuous RG / scale flow once decompactified – the R reads both ways (recursion ↔ renormalisation flow). Computed: `decompact_recursion.py`.

Marcel's decompactified variant: the overdamped kernel is the decompactified empty window

The doubling has two branches; decompactifying $S^1 \to \mathbb{R}$ (the measurement) turns each into a continuous scaling exponent $\alpha = \ln\mu$:

branch	$\mathrm{Re}\,\alpha$	decompactified character
expanding φ (the D4 core, masses)	$+0.481$	relevant; coherence / revival (back-flow 5.125)
contracting $-1/\varphi$ (the $3'$ window)	-0.481	irrelevant; decaying / overdamped

Both share $|\ln\varphi| \approx 0.481$; only the sign differs. The decaying branch is exactly HLV's overdamped, decaying kernel (Docs. 282–284): **Marcel's overdamped kernel is the decompactified empty window** — the compact "empty $3'$" ($\to 0$ discretely) and the decompactified "overdamped, decaying" ($\mathrm{Re}\,\alpha < 0$, $\to 0$ continuously) are the same object on the two sides of the measurement.

Crucially, the containment survives decompactification. Decompactified D7 is *not* the overdamped kernel alone: it is the D4 core (the relevant, coherence-carrying φ-branch with the masses and revival 5.125) *plus* the $3'$ window (the irrelevant overdamped branch). The D4 core remains contained inside decompactified D7 as its relevant sector; Marcel's kernel is the *extra* window, not the whole. This is just the decompactified image of $T^4 \subset T^7$.

Two caveats. The identification "overdamped kernel = decompactified window" is a proposed reading (both are the decaying, non-discriminating sector), not a theorem. And because decompactification is the measurement (information-discarding), this branch sits on the null-reproducible side by construction: the discriminating content (revival 5.125) is in the relevant D4 core, not in the overdamped kernel. Computed: `decompact_branches.py`.

Relation to Krüger's HLV kernel and claim-state note

Krüger's actual HLV memory kernel is a positive mixture of decaying exponentials, $K(t) = \sum_i w_i \tau_i^{-1} e^{-t/\tau_i}$ (Krüger, *Pre-Registered Null-Model Benchmark for HLV-Constrained Memory Kernels*, 2026). Every mode is pure decay (scaling exponent Re < 0), so the kernel *is* the decompactified contracting $-1/\varphi$ window of the previous subsection. Decisively, Krüger's own pre-registered benchmark finds this kernel statistically indistinguishable from a generic positive two-exponential kernel — the uniqueness gate stays false, "not a unique physical signature." That is the null-reproducible side, established by his result, not merely asserted by ours.

The full HLV operator adds a skew-adjoint phase channel ($J^\top = -J$) to the positive kernel. The skew-adjoint (non-gradient) part is exactly the structure his claim-state note requires for sustained recurrence — pure gradient flow admits no cycles — and it is the analogue of the relevant, oscillatory FFGFT revival (5.125); the decaying kernel is the overdamped window. The two sectors line up: discriminating content sits in the recurrent/coherent (non-gradient) part, not in the decaying magnitude.

A third, sharper correspondence comes from the path-memory comparator (De Jesus, HICE-MEM, for Krüger): temporal order is antisymmetric, Area$(P) = \sum_{i<j}[A_i, A_j]$ with Area$(P^{\mathrm{rev}}) = -$Area(P), so a single history's magnitude and spectrum are *even* under reversal — the anti-Hermitian spectrum $\{i\lambda\} \mapsto \{-i\lambda\}$ is the same set. The order-sign is discarded unless contracted against an oriented reference (a second history). This is the same structure as the measurement $S^1 \to \mathbb{R}$ discarding winding while the magnitude survives: the discriminating content is relational, not single-history.

Finally the claim-state. Krüger's clarification note (2026) separates compact internal Spiral-Time, SI measurement time, and the signal index $\Delta\Phi_{\mathrm{sig}}$, and forbids their identification without an explicit projection/calibration map K — the same discipline as "structure $\neq$ physics without a map." No identification of the two spiral notions and no holonomy is claimed here: Krüger's Spiral-Time enters as a bounded kinetic prefactor $A_\psi(t)$ on the kernel (benchmark) and as a triadic compact coordinate $\theta \in S^1$ (note), neither of which is the log-periodic log-spiral of the decompactified recursion. Computed: `marcel_kernel_match.py`.

Krüger's unrolled variant covers only $\{3' +$ time$\}$ — not the full D7

A placement of the decompactified side, *without* importing Krüger's formulas. The compact FFGFT result — $6 = 3 \oplus 3'$ as a fractal internal recursion, purely from the golden ratio (Fibonacci operator, eigenvalues φ expanding / $-1/\varphi$ contracting) — is derived and script-covered (`recursion_doubling_empty.py`, `verify_recursion_doubling.py`, $11/11$) and stands independently of HLV. It is not touched here; the comparison below does not bear on it.

Krüger's latest decompactified construction (*Recursive Spiral-Time Nesting as a Route to a Six-Dimensional Host Carrier*, 2026, with the null-gated runs Zenodo 20770402 / 20772007) reaches a 6D host via a *stability selection* (a spectral gap), *not* via φ — a different mechanism, no φ substitute. What matters for compatibility, however, is not the mechanism but *what the unrolled variant fills*: exclusively the **empty** sector. Its decaying kernel is the decompactified $-1/\varphi$ window (above); its host sector

is candidate-only and massless ("host dimension, not masses"); and its compact time satisfies $\chi(t) \neq \ln t$ (clarification note), so it is not the log-spiral of the FFGFT recursion.

The full-text version (2026) adds the downstream chain BD3–BD5 (Zenodo 20773678): a native finite carrier (BD3), a low-q Lorentz/EFT pre-screen (BD4) and semantic transfer (BD5). It only adds more structure and still reaches no physics — BD4-HARD itself flags the low-q behavior as *carrier-conditioned, not HLV-specific* (matched nulls also pass), and BD0/BD1 remain negative. This reinforces the $\{3' + \text{time}\}$ picture: the carrier is more elaborate but massless structure. (Krüger dates the nesting idea itself to the January-2026 HLV foundation, Zenodo 18255159.)

In D7 bookkeeping — $7 = [\,3\text{-core (crystallographic, masses)} + \text{time}\,] + 3'$ — his variant therefore unrolls only $3'$ **+ time**: the empty perp window plus the unrolled time axis ($S^1_m \to \mathbb{R}_t$, shared). The mass-bearing crystallographic 3-core (φ/ξ, $\sqrt{2}$, Koide $2/3$) is *not reached* by it and remains FFGFT's. Strictly, his HLV — as decompactified-realized — is *not* the full D7 space, but the subsector $\{3' + \text{time}\}$; the full D7 (mass core + time + $3'$) is realized by FFGFT.

Hence compatibility, briefly: his side $\subseteq$ FFGFT's $\{3' + \text{time}\}$. The formal containment $T^7 \supset T^4$ ($A_5 \supset C_3$) stands; it is only sharpened here as to *where* Krüger's unrolled construction actually sits. This is compatible *as long as he holds the candidate-only line* (his gates B0–B7); were he later to overreach ("6D host = physical spacetime with masses"), compatibility would have to be re-examined — at present he explicitly does not. Covered by `decompact_branches.py` and the emptiness theorem above; no new script is needed for his formulas.

T.12 What the d7 sector computes: masses from ξ, spectra from φ

By analogy with ξ one asks what the d7 model (T^7, A_5) computes. The honest answer is *not masses*. Masses are the output of the crystallographic 3-fold sector with ξ, which d7 already contains (HLV $\supset$ FFGFT); d7 reproduces them *through* its FFGFT sub-part, not as new content. The *extra* d7 structure (5-fold, perp directions, φ) is exactly the part that does not carry masses. What it carries instead:

quantity	d7 / 5-fold output	status
protected amplitude	$2/\sqrt{5}$, Koide-analog $Q = 7/15$	derived number
phase	$\theta_{\text{HLV}} = 0$ (no free Koide angle)	derived
recursion scale	φ; tower φ^n; log-period $\ln\varphi \approx 0.481$	derived
spectrum	aperiodic quasicrystal gap spectrum, φ-scaling	discriminator (271/283)

Why the spectrum does not lead to masses here

FFGFT's masses are themselves spectral – eigenvalues of the T^4 connection-Laplacian, the Koide circulant levels – so the question is fair; the answer is about *which* spectrum. A

mass is a sharp, isolated level, so reading masses off a spectrum requires a *discrete point spectrum*. The 3-fold is lattice-compatible ($2\cos(2\pi/3) = -1 \in \mathbb{Z}$): it closes on a periodic torus into exactly three sharp eigenvalues $\{-1, -1, 2\}$ – the triple. The 5-fold is lattice-incompatible ($2\cos(2\pi/5) = 1/\varphi \notin \mathbb{Z}$): no periodic lattice carries it, so its only carrier is an aperiodic quasicrystal, whose spectrum is *singular-continuous* (Cantor-like) – a dense set of gaps, self-similar under φ, with no isolated levels (a Fibonacci chain fragments into gaps at every scale, against the few bands of the periodic chain; `spectrum_to_mass.py`). There is no distinguished triple to read masses off. Hence "spectrum → mass" needs a point spectrum, i.e. a periodic / crystallographic carrier – and the only crystallographic part of d7 is its 3-fold core, i.e. FFGFT. The route always returns to FFGFT; the genuinely-d7 (aperiodic) spectrum cannot deliver masses. And a dense self-similar spectrum has a level near *any* number, so φ^N "fits" anything – the P35 numerology trap (Doc. 190, R44): the very richness of the quasicrystal spectrum is what makes reading a mass off it content-free, whereas the sparseness of the discrete torus spectrum is what makes an eigenvalue there a genuine prediction.

Information content: structure spectrum vs energy spectrum

The previous point must not be read as "the quasicrystal spectrum carries no information." It does – a quasicrystal has *two* spectra, differing in type. The *structure* (diffraction) spectrum $S(q) = |\sum_n e^{\mathrm{i}qx_n}|^2$ is *pure point*: sharp Bragg peaks at φ-module positions, here $\approx 21\times$ above the noise floor of a random null of identical composition (`quasicrystal_information.py`). That sharp, null-distinguishable order *is* the information – exactly HLV's substrate-distinguishability claim, which stands. The *energy* (Laplacian) spectrum is the singular-continuous (Cantor) one of the previous subsection; it is what lacks sharp levels and so yields no masses. The two are different objects: information lives in the relational long-range order (structure spectrum), not in individual energy levels (the energy states are critical / fractal, not clean nodes). The P35 caveat is therefore narrow: only fitting a *single target number* as φ^N is content-free; the structural signature is not.

Residual, kept honest. The place where the FFGFT analysis found an HLV signature *null-reproducible* is the *temporal* over-damped memory kernel (Doc. 282–284, biexponential, non-identifiable sector) – the time domain, a different object from the spatial structure spectrum. So: spatial structure informative (HLV's point holds); temporal over-damped kernel = the standing critique. The discriminating content lives in the structural / spectral signature (pure-point diffraction; periodic-vs-aperiodic peaks) – the discriminator of Doc. 271/272/283. FFGFT and HLV agree *where* the information sits (in the structure); the critique concerned a different (temporal) claim.

The one place d7 could touch masses (conjecture)

The $6 = 3 \oplus 3'$ split leaves the perp irrep $3'$ (character $1 - \varphi = -1/\varphi$) as a *second* triplet position; FFGFT realises the one 3 (charged leptons, Koide $2/3$), the $3'$ is empty. *If* a physical triple sat there, its prediction would be $Q = 7/15$ – a falsifiable target, not a derived particle mass. Computed: `what_d7_computes.py`.

T.13 The electron mode in D7: the embedding exists, the mass-label is C_3-frame

A natural consistency question: if a mode is read off as the electron in D4, it must have an image in D7, since $T^4 \subset T^7$. It does. The electron mode lifts to the D7 mode with the same winding on the four D4 circles and zero winding on the extra three; sitting entirely in the D4 sub-block, its Laplacian eigenvalue — the mass — is unchanged. The translation is the embedding itself.

This does not conflict with "masses require the D4 core." The sharp label "electron" is a C_3-protected (crystallographic) statement, not an A_5 one: the electron mode is an eigenvector of the 3-fold. The full D7 symmetry is A_5, whose 5-fold elements do not commute with C_3 — they rotate the electron mode into a muon/tau mixture. Under a genuine icosahedral 5-fold rotation the electron direction keeps 0.762 of its weight and leaks 0.238 into the μ/τ directions (`electron_mode_in_d7.py`). So in the full A_5 frame "electron" is not a sharp eigenstate; to read it as the electron (a sharp mass) one selects the C_3 eigenbasis — the restriction to the crystallographic sub-block, i.e. "going back to D4." The correspondence is the embedding; the mass read-out is the frame choice; the two are the same statement.

Two safeguards. The mixing is a change of basis, not a smearing: the 5-fold rotates the labelling frame and does not destroy the physical mode, which stays perfectly sharp in any C_3 frame. And the mode stays in the periodic "3" block (point spectrum) — it does not leak into the aperiodic $3'$ sector; the 5-fold only rotates $e \leftrightarrow \mu \leftrightarrow \tau$ within the physical "3". Finally, which frame counts as the 3-fold is, once again, the measurement (Doc. 270/248): the mode exists frame-independently in D7, the sharp particle label is frame-dependent.

T.14 The $3'$ is the first recursion: why recursion directions are empty in the compact picture

The split $6 = 3 \oplus 3'$ and the golden recursion are not two facts but one: the doubling *is* the recursion acting once. This section makes that explicit and draws the consequence for the empty $3'$.

The doubling is the recursion operator

The Fibonacci / inflation operator is

$$M = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \det M = -1, \quad \operatorname{tr} M = 1, \quad \text{characteristic polynomial } \lambda^2 - \lambda - 1 = 0. \tag{T.10}$$

Its two eigenvalues are $\varphi = \frac{1+\sqrt{5}}{2} \approx 1.618$ (expanding, $|\varphi| > 1$) and $-1/\varphi = \frac{1-\sqrt{5}}{2} \approx -0.618$ (contracting, $|-1/\varphi| < 1$), and $\varphi^2 = \varphi + 1$. Crucially, the Galois conjugate that distinguishes the two 3-dimensional A_5 irreps, $\varphi \leftrightarrow 1-\varphi$, is exactly the contracting eigenvalue: $1-\varphi = -1/\varphi$. So the two eigendirections of M *are* the two irreps:

$$3 = \text{the } \varphi \text{ (expanding) eigenmode}, \qquad 3' = \text{the } -1/\varphi \text{ (contracting) eigenmode}.$$

The appearance of $3'$ alongside 3 is therefore one application of the golden recursion — the doubling is the recursion's eigenstructure. (That M is the recursion operator is confirmed by $M^n = \left(\begin{smallmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{smallmatrix}\right)$, the Fibonacci numbers.)

The internal branch is the window — hence empty

In the cut-and-project reading of a quasicrystal, physical space is the expanding direction (φ) and internal space is the contracting direction ($-1/\varphi$). Internal space carries the acceptance *window*, not physical positions. Under iteration the internal component decays geometrically while the physical component grows:

n	|physical (φ)|	|internal $(3')$| $\sim (1/\varphi)^n$
0	1.01	0.27
2	2.64	0.10
8	47.4	0.0058
20	—	1.2×10^{-9}

The decay rate per step is $1/\varphi^2 \approx 0.382$. So the $3'$ (internal branch) carries no physical positions and vanishes under the recursion: that is why it is empty.

The general principle: the seed carries, the recursion is empty

The crystallographic seed (the "3", φ-branch) holds the masses; everything the golden recursion generates from it — the conjugate $3'$ and the φ^n tower — is empty of *new* mass content. "Empty" means no new sharp level / no new triple, not "does nothing": the recursion still carries structure (FFGFT's ξ^n gives the fractal correction $D_f = 3 - \xi$; HLV's φ^n gives the quasicrystal window). Several earlier results collapse into a single statement:

$3'$ empty = recursion directions empty = masses need the crystallographic core = the recursion sector has no point spectrum = the cut-and-project internal space carries no physical positions.

Consequence for the $3'$ candidate: why a dynamics is needed

This gives a *structural* reason for the gap noted earlier. Putting a physical triple into the $3'$ means putting mass content into the internal / window space — precisely what the compact recursion keeps empty. It cannot be supplied by the recursion alone; it requires a dynamics that fills the window (a Hamiltonian / interaction on the 7D torus). So the conditional $3'$ candidate (the neutrino sector) needs a dynamics not only for methodological caution but for this structural reason: the recursion leaves that branch empty by construction.

Check scripts

- `recursion_doubling_empty.py` — demonstration: M's eigenstructure (φ / $-1/\varphi$), physical vs internal under iteration.

- `verify_recursion_doubling.py` — assertions (PASS/FAIL): eigenvalues, char. poly, $1-\varphi=-1/\varphi$, M^n Fibonacci, internal branch $\to 0$ at rate $1/\varphi^2$. $11/11$ PASS.

T.15 The recursion tower D(4+3n): height is empty, depth needs dynamics

From the previous section the doubling is one golden recursion. The construction iterates: each recursion adds one 3 (the contracting $-1/\varphi$ branch), giving a dimension tower $D(4+3n)$, $n=0,1,2,\ldots$

n	dim	
0	D4	seed (FFGFT, crystallographic kernel)
1	D7	one recursion (the first $3'$)
2	D10	two recursions
3	D13	
100	D304	formally unbounded

But the theorem of the previous section fixes what this means: each added 3 is an *empty* internal window (the contracting $-1/\varphi$ branch, which $\to 0$ under iteration). Hence 100 recursions are 100 empty windows, *not* 100 physical layers; the masses reside, at *every* level of the tower, only in the D4 seed. The tower is a geometric hierarchy (nested cut-and-project), not new testable content — and the theorem is what keeps it from becoming a number mill (one could otherwise assert arbitrarily many "levels", all empty).

Two caveats. **(i)** "$+3$ per step" assumes every level repeats the icosahedral $3{+}3$ pattern; a different non-crystallographic symmetry (e.g. 8-fold, silver ratio $1+\sqrt{2}$) would give a different increment, so $D(4+3n)$ is the icosahedrally-repeated choice, not mandatory. **(ii)** Filling any window requires a dynamics — the same open gap at every level; the tower does not escape the gap, it multiplies it.

Constructive consequence. The productive next step is not height (D10, D100 — more empty windows) but depth: the dynamics that fills the *first* window. And one physically interesting first window already exists — the $3'$ → neutrino sector (conditional, next section). That is where the lever is, not in counting up the tower. Computed: `recursion_tower.py`.

Two recursions — and what the empty windows are good for

A natural tension: "each $+3$ step is one more recursion depth, recursions are fractal deepenings — but if they are all empty and 4D already carries everything, why should unrolling yield mass?" It resolves once two *distinct* recursions, easily conflated, are separated:

- **Scale recursion** $\xi_{n+1}=\xi_n(1-100\xi)$: refines the *same* four dimensions across scales (log-period $\ln 7500$) and carries $D_f=3-\xi$ resp. K_{frak} — it adds *no* dimension. This is the actual "fractal deepening".
- **Dimensional recursion** $D(4{+}3n)$ (golden operator, $\varphi/-1/\varphi$): stacks empty $3'/3''$ windows; the inner-branch weight decays $(1/\varphi^2)^n\to 0$, no point spectrum.

The mass content upon decompactification comes from unrolling the *mass circle* (4th circle of D4, $\tilde{T} \cdot m = 1$, type I), which supplies the *single* scale $M = 1/T$; the dimensionless ratios were already present compactly in 4D. The Kaluza-Klein intuition (unroll an extra dimension → mass tower) thus applies to the mass circle, *not* to the $3'/3''$ windows: an empty window unrolled yields empty perp space, no mass tower, as long as no *dynamics* (a Hamiltonian on the window) fills it.

What, then, is the recursion good for if it yields no new masses? Not for levels, but for *structure*: the scale recursion carries the fractal correction (D_f, K_{frak}), the dimensional recursion carries the cut-and-project *window* that shapes the aperiodic order and the *spectral fine structure* (splitting, inharmonicities) on top of the crystallographic base levels — not new fundamental tones but their fine structure. The $3'$ window is in this sense a *slot* provided by the geometry; whether it gains physical content is decided by a dynamics (the conditional neutrino candidate, next section), not by mere unrolling. Computed: `klaer_rekursion_masse.py`.

T.16 Candidate for the 3′: the neutrino sector (conditional)

What could sit in the empty 3′? An *open* question — a conditional target, not a prediction. The discriminator is the Koide ratio $Q = \sum m/(\sum \sqrt{m})^2$; the conditional 3′ target would be $Q = 7/15 \approx 0.467$. The measured triples:

triple	Q	
charged leptons e, μ, τ (the "3")	0.667 $(= 2/3)$	far above target
up quarks u, c, t	0.849	far above target
down quarks d, s, b	0.731	far above target

All measured charged triples have $Q \geq 0.67$ — none matches $7/15$; they are excluded as a $7/15$ carrier (robust, also under the mass uncertainties). One open candidate remains: the **neutrinos**. Their Q is not fixed, because the absolute mass scale is unknown; in a scan (normal ordering) $Q \approx 0.586$ for a hierarchical spectrum and falls towards $1/3$ as the scale grows. It crosses $7/15$ at a lightest mass $m_0 \approx 1.9$ meV ($\sum m_\nu \approx 61$ meV, within the cosmological bound ~ 120 meV). Structurally this is the natural partner: leptons split into a charged triple (the "3") and a neutral one (neutrinos), and the second 3 irrep is the slot for the neutral triple.

Status (strict). This is a conditional target, *not* a prediction. Q varies *continuously* with m_0, so "$7/15$ is hit" carries content only if $7/15$ is fixed independently by a dynamics (the missing piece from the section "Status of the D7 mass calculation") *and* beats a null. It is exactly Marcel Krüger's structure: carrier = neutrinos, observable = Koide Q, null control = against generic triples. That lifts the question from "guessing" to a "pre-registrable target" — no more. Computed: `which_triple_in_3prime.py`.

Interpretation (conjecture, not derived from the theorems). That the recursion sector carries no point spectrum is consistent with the peculiar character of the neutrino sector. It yields, however, no masses, no Δm^2 and no mixing angles — these require the still-missing dynamics. And the tower adds empty windows, not additional neutrino species.

Which energy type fills the window — and the open computational gap. Through $\tilde{T} \cdot m = 1$ FFGFT enforces *two* energy types (Doc. 255): *static* (rest mass = topologically frozen winding, the crystallographic core) and *kinetic* (free; for massless states the only form, $E = hf$) — "mass is frozen kinetic energy". The dynamics that fills the $3'$ window — the temporal, holographic scale evolution (Doc. 263) — therefore supplies the *kinetic* type, not frozen rest mass: a filled window is photon-/neutrino-like (massless / near-massless), not a massive triple. This matches the neutrino as a "near-massless photon" (Doc. 047, $m_\nu \sim \frac{\xi^2}{2} m_e$) and time-as-energy $\tilde{T} = 1/\max(m, \omega)$ (Doc. 067/080): for $m = 0$, $\tilde{T} = 1/\omega$.

Open computational gap. What is missing is the *derived* rule to compute the kinetic part, via the duality, as a ξ*-fractal* fraction of the static part. The framework exists (Doc. 080: $\tilde{T} = 1/\max(E, \omega)$, electron = rest + γ-kinetic, photon = purely kinetic) and *one* instance exists (Doc. 047: $\xi^2/2$ suppression — but as an ansatz, not derived; Doc. 204 has a ξ correction to the kinetic energy). The general derivation "kinetic = ξ-fractal fraction of the static, via $\tilde{T} \cdot m = 1$" is precisely the still-missing dynamics — the next computational step, not a solved piece.

T.17 Status of the D7 mass calculation: the gap and the next step

The HLV construction provides a consistent geometric embedding (T^7, A_5, $3 \oplus 3'$) and qualitative predictions (5-fold coherence, $2/\sqrt{5}$ rather than $\sqrt{2}$). An explicit mass formula on T^7 does not exist, however — symmetry alone does not fix the mass matrix. What is missing is a specified dynamics (Hamiltonian, interactions) on the 7D torus. Until that is given, the 5-fold predictions stay qualitative ("there should be signatures") rather than quantitative ("at this value"). FFGFT therefore remains the only model with an explicit, tested mass formula. HLV is a structural conjecture that needs a dynamics — and that is the next construction site.

T.18 Where the fractal corrections sit

The fractal corrections sit at two causally chained levels — and explicitly *not* in the ratios.

Root: the conformal fractal metric

The fundamental fractal quantity is $\xi = 4/30000$ with the recursion $\xi_{n+1} = \xi_n(1 - 100\xi_n)$ (Doc. 133/275). Its metric manifestation is *conformal*, not diagonal:

$$g_{\mu\nu}^{(\text{frac})}(x) = \eta_{\mu\nu}(1 + \xi f(x)), \qquad D_f = 3 - \xi \quad \text{(Doc. 051)},$$

where $f(x)$ is the position function of the conformal fractal metric (not a frequency and not the base winding number $f = 1/\xi$), with the ξ correction specifically on the time component, $ds^2 = (1 + \xi\lambda_0/E_\xi)\,dt^2 - dx^2$ (Doc. 025). In the winding/harmonic structure it appears as the damping exponent $\tau^{-(1+\xi)}$ (Doc. 159). *Not* as a static metric $\text{diag}(1, 1, 1, 1+\xi)$ — this form is unsupported; the mass circle is distinguished via $\tilde{T} \cdot m = 1$.

Level 1: local spacetime porosity

The first manifest correction is the local spacetime dimension

$$D_f^{\text{space}} = 3 - \xi \approx 2.99987 \quad \text{(Doc. 133)},$$

a tiny deviation from integrality ($\sim 0.003\%$, effectively 3). This is *not* the type-II projection (Doc. 270: $T^4 \to T^3 \to \mathbb{R}^3$, clean, for large R); the porosity is a separate quantum-gravitational effect.

Level 2: effective dimension and K_{frak}

Through the cumulative RG run (17 orders of magnitude, Planck → electroweak) the porosity becomes an *effective* dimension $D_f^{\text{eff}} \approx 2.973$, which Doc. 133 states is "not identical" to D_f^{space}. From it the correction for absolute values emerges:

$$K_{\text{frak}} = \left(\frac{D_f^{\text{eff}}}{3} \right)^{D_f^{\text{eff}}/2} \approx 0.9867 \approx 1 - 100\xi.$$

The scale run alone, however, gives only 39ξ; the amplification factor 100, and with it the value $D_f^{\text{eff}} \approx 2.973$, is a setting (Doc. 133). $K_{\text{frak}} = 1 - 100\xi$ is confirmed by comparing the bare and the corrected E_0 with α. The causal chain:

$$\underbrace{3 - \xi}_{\text{porosity } \sim 0.003\%} \xrightarrow{\text{RG run, 17 OoM}} \underbrace{D_f^{\text{eff}} \approx 2.973}_{\text{effective}} \xrightarrow{\text{formula}} \underbrace{K_{\text{frak}} \approx 0.9867}_{\sim 1.3\%}.$$

K_{frak} is thus *not* an independent correction "at the SI step" but is conceived as the RG-cumulated consequence of $3 - \xi$, with the factor 100 set; it becomes visible only in the absolute/SI transition (Doc. 122).

Where they do *not* sit

Quantity	contains ξ / K_{frak}?
$r = \sqrt{2}$, $\theta = 2/9$	no (pure $\mathbb{Z}_3$ structure)
μ_k (eigenvalues of S)	no
mass ratios	no (K_{frak} cancels)
α (unit-free)	yes (K_{frak} via E_0)
Koide $Q = 2/3$	no (pure $\mathbb{Z}_3$ structure)
$D_f^{\text{space}} = 3 - \xi$, D_f^{eff}	yes (dimension)
absolute values (m in kg, G in SI)	yes (K_{frak})

The corrections affect only the dimensions (D_f^{space}, D_f^{eff}), the absolute values and α via E_0 (K_{frak}). **Mass ratios are ξ- and K_{frak}-free** — the decisive statement from Doc. 101/122.

T.19 From geometry to SI: the conversion

This closing bridge is *purely conventional* — it adds no physics but translates the geometric quantities into human-made units. The only fundamental object is the dimensionless number $\xi = 4/30000$; $c, \hbar$ are conventions (SI-2019), G is measurement data, and mass ratios, Koide $2/3$ and α are dimensionless ratios (Doc. 101, Doc. 122). The decisive separation: *mass ratios* are unit- and K_{frak}-free (α is unit-free but contains K_{frak} via E_0); the fractal correction $K_{\text{frak}} = 1 - 100\xi \approx 0.9867$ appears *only* in the step from ratio to *absolute* (SI) value. The "circularity" of the conversion is thus metrological (SI-2019), not a physical problem (Doc. 101).

The compact quantities (purely geometric)

Quantity	Compact value	Meaning
r	$\sqrt{2}$	$\mathbb{Z}_3$ circulant amplitude
θ	$2/9$	$\mathbb{Z}_3$ phase
ξ	$4/30000$	fractal recursion (fundamental)
μ_k	$\{0.04035,\ 0.58021,\ 2.37944\}$	dimensionless eigenvalues of S
Q	$2/3$	Koide invariant

These are unit-free and K_{frak}-free.

The scale M from the time circle

The mass circle S^1 carries, via $\tilde{T} \cdot m = 1$, the *only* dimensionful quantity $M = 1/T$ — a mass in natural units. It is fixed from *one* measured mass (the muon):

$$M = \frac{m_\mu^{\text{meas}}}{\mu_\mu^2} = \frac{105.6583755}{0.58021^2} \approx 313.86 \text{ MeV}.$$

All other masses are then *predicted* — that is the test of the theory.

The masses (ratio level, K_{frak}-free)

Lepton	μ_k	$m_k = \mu_k^2 M$ (MeV)	measured (MeV)
Electron	0.04035	0.51100	0.51099895
Muon	0.58021	105.658 (anchor)	105.6583755
Tau	2.37944	1776.98	1776.86

The mass ratios are unit- and K_{frak}-free: $m_\mu/m_e = 206.77$, $Q = 2/3$. (Electron to five figures, tau as a $\sim 0.007\%$ prediction; the μ_k follow from $r = \sqrt{2}$, $\theta = 2/9$, fixed by Koide + (e,μ).)

The fine-structure constant α (unit-free, with K_{frak} via E_0)

$$\alpha = \xi \left(\frac{\sqrt{m_e m_\mu}}{\text{MeV}} \right)^2 .$$

With $\sqrt{m_e m_\mu} \approx 7.348$ MeV: $\alpha \approx 0.007199$, i.e. $1/\alpha \approx 138.9$; the bridge value $E_0 = 7.398$ MeV gives $1/\alpha = 137.04$ (the $\sim 1.3\%$ is, up to $0.017\,\%$, $1 - K_{\text{frak}} = 100\,\xi$: $E_0 = \sqrt{m_e m_\mu / K_{\text{frak}}} = 7.348/\sqrt{0.98667}$ MeV). α is unit-free but contains K_{frak} via E_0; E_0 is a bridge constant from the mass ratios (Doc. 101). The agreement with the measured value is a property of the anchor E_0, not an independent prediction.

The gravitational constant G (absolute level, with K_{frak})

G is the only quantity that needs a genuine SI conversion ($[G] = \text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2}$). The structural derivation (Doc. 006/122/133) yields it as an absolute value in which the fractal correction appears *only here*:

$$G_{\text{SI}} = \frac{\xi^2}{4 m_e} \cdot C_{\text{conv}} \cdot K_{\text{frak}}, \qquad C_{\text{conv}} = 7.783 \times 10^{-3}\ \text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2}\,\text{MeV},$$

$$G_{\text{SI}} \approx 6.6745 \times 10^{-11}\ \text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2} \quad (K_{\text{frak}} = 0.986;\ \text{with } K_{\text{frak}} = 1 - 100\xi = 0.9867 :\ \ 6.679 \times 10^{-11}).$$

Here C_{conv} is the SI conversion of the single-anchor chain (Doc. 383); its numerical value 7.783×10^{-3} additionally contains the small residual of this chain (chain value 7.58×10^{-3}). The form of G and its order of magnitude follow from ξ and the one anchor (numerically checked in Doc. 383); with the anchor m_e, G lies at $-2.6\,\%$, with v at $-0.46\,\%$. The agreement of the value 6.6745×10^{-11} with CODATA to $0.003\,\%$ arises only with this residual and is not an independent precision result. This is the *computed* gravitational constant (not an input) – in form and order of magnitude, not in the last digit; the scale anchoring via the RG amplification is given in Doc. 133/275, not in this conversion.

The general conversion rule

$$\boxed{\text{quantity}_{\text{SI}} = \text{quantity}_{\text{geom}} \times \text{conversion factors} \times K_{\text{frak}}}$$

with the conversion factors $\hbar, c$ and the scale M (convention, SI-2019). The separation in one line:

Level	Quantities	K_{frak}
Ratio	m_μ/m_e, m_τ/m_μ, Q	cancels out
unit-free	α	appears via E_0
Absolute (SI)	m_e, m_μ, m_τ, G	appears

References: Doc. 101 (circularity is metrological), Doc. 122 (ratio vs absolute, K_{frak}), Doc. 133/006 (derivation of G with E_0, K_{frak}, scale).

T.20 BD12-XCORE: the FFGFT mass-core declaration

Krüger's open question — what makes a *legitimate* mass-core docking criterion rather than a post-hoc fit — FFGFT answers operationally, not by assertion: with a *frozen, pre-registered* declaration in matched-null format (his BD style). The rule has *no* free parameters: $r = \sqrt{2}$, $\theta = 2/9$, and the Z_3/Foot-Koide spectrum $\mu_k = 1 + \sqrt{2}\cos(\theta + 2\pi k/3)$.

It produces: Koide $Q = \Sigma\mu^2/(\Sigma\mu)^2 = 2/3$ exactly (forced by $r = \sqrt{2}$); the lepton ratios m_μ/m_e to 0.001%, m_τ/m_e to 0.007%; and — as a *genuine prediction* from Koide $+\{m_e, m_\mu\}$, with no τ input — $m_\tau = 1776.97$ MeV against 1776.86 measured (0.006%). The matched-null test (tolerance Koide 0.1%, ratio 1%) shows: random r hits Koide in 0.18%, random θ both ratios in 0.025%, random (r, θ) the *joint* set in 0.000% of cases. The frozen rule is therefore *special*, not a generic fit — **PASS**; and symmetric, random inputs would give **FAIL**.

Two constraints keep the dock honest: (i) do *not* dock on L_{nat} — the masses are not in the carrier Laplacian but in the compact Z_3 core + mass-circle ($\tilde{T} \cdot m = 1$); (ii) the dock is a *fixed transformation* (a pre-registered algebraic map onto $\{\sqrt{2},\ 2/9,\ M = 1/T\}$), adding no new physics — hence *not* a post-hoc fit, but an equivalence that reproduces or fails. Computed: `bd12_xcore_ffgft.py`.

T.21 Reproducibility

- `dim_bridge_1_doubling.py` – **6** $=$ **3** $\oplus$ **3′**; characters φ, $1-\varphi$, integer 6D character; orthogonality and order 5.
- `dim_bridge_2_crystallographic.py` – crystallographic restriction $2\cos(2\pi/n)$; 3-fold integer, 5-fold $= 1/\varphi$.
- `dim_bridge_3_same_ambient_fork.py` – $r(\tau) \le 1$; $2/\sqrt{5}$ at $\tau = \varphi$; $\sqrt{2}$ above the family's maximum (a statement about one map only).
- `recompute_reconcile.py` – $C_3 < A_5$ (3-fold and 5-fold both permute the same icosahedron) and $7 = 4 + 3$; the containment HLV $\supset$ FFGFT.
- `recursion_unify.py` – recursion λ^n: $\lambda = \xi$ ($D_f = 3 - \xi$) vs $\lambda = \varphi$ ($\varphi^2 = \varphi + 1$).
- `phase_from_a5.py` – the A_5 phase: $\theta_{\text{HLV}} = 0$, $Q = 7/15$; $19/45$ excluded.
- `decompact_recursion.py` – decompactified recursion: discrete $\lambda^n \to$ continuous flow $t \to \lambda t$, log-period $|\ln\lambda|$.
- `what_d7_computes.py`, `spectrum_to_mass.py` – what the d7 sector computes (no masses; $Q = 7/15$, φ, spectra) and why a point spectrum is needed for masses.
- `quasicrystal_information.py` – structure (diffraction) spectrum pure-point vs random null: the substrate's information content.
- `electron_mode_in_d7.py` – the electron mode embeds in D7 ($T^4 \subset T^7$); a 5-fold rotates it (0.762/0.238), so the sharp mass-label is C_3-frame = the D4 core.
- `which_triple_in_3prime.py` – candidate triples for the 3′ via Koide Q: leptons/up/-down excluded ($Q \ge 0.67$), neutrinos the open candidate (crosses $7/15$ at $m_0 \approx 1.9$ meV).
- `recursion_doubling_empty.py`, `verify_recursion_doubling.py` – the 3′ as the contracting ($-1/\varphi$) recursion eigenmode; internal branch $\to 0$ (11/11 PASS).

- `recursion_tower.py` – the tower $D(4+3n)$ (D4/D7/D10/.../D304) and the emptiness of every added window.
- `decompact_branches.py` – the two branches decompactified (Re $\alpha = \pm 0.481$): φ relevant (D4 core), $-1/\varphi$ overdamped (Marcel's kernel); D4 core stays contained.
- `marcel_kernel_match.py` – grounded match to Krüger's kernel $K(t) = \sum_i w_i \tau_i^{-1} e^{-t/\tau_i}$ (overdamped $= -1/\varphi$ window) and the antisymmetric path-memory order (single-history spectrum even under reversal).
- `bd12_xcore_ffgft.py` – the BD12-XCORE declaration: frozen rule ($r = \sqrt{2}$, $\theta = 2/9$), Koide $2/3$, ratios, τ-prediction and the matched-null test (nulls 0.000% on the joint set).

All scripts are `numpy`-only with fixed values. Note: the $\sqrt{2}$ appearing at $n = 8$ in the crystallographic table is $2\cos 45^\circ$ (the octagonal rotation), a numerical coincidence – not FFGFT's $\sqrt{2}$, which is the Koide/$\mathbb{Z}_3$-circulant ratio *within* the crystallographic 3-fold structure, not a rotation order.

Part III

Part III — Dynamic Sector, Magnitude and Phase, $\theta = 2/9$

Chapter U

Doc. 286 — The dynamic sector: kinetic energy as a ξ-fractal fraction of the static

Abstract

Through the duality $\tilde{T} \cdot m = 1$, FFGFT separates two energy types: *static* (rest mass = topologically frozen winding, the crystallographic Z_3 core) and *kinetic* (free; for massless states the only form, $E = hf$). The static sector is derived and computed (Z_3 mass operator + mass circle; Doc. 285, BD12-XCORE). The dynamic sector is *not*: the derived rule that computes the kinetic part, via the duality, as a ξ-fractal fraction of the static one is missing. This document is a *problem-statement document* – it records what is fixed (the duality places the window in the kinetic sector), the single existing data point (neutrino, $m_\nu \sim \xi^2/2\, m_e$, as an ansatz), the narrowing of the ξ-power to $p = 2$, and the structural lead $m_{\text{window}} = \xi \cdot \omega$. What remains to be derived is marked explicitly as open. Prefixed is the fixed, reversible Hilbert-space foundation (Doc. 282/270/230) the sector sits on; its actual aim is to find, out of the Hilbert space, the complementary description that maps via $D_f = 3 - \xi$ and ξ onto SI units.

U.1 Where the question comes from

In Doc. 255, $\tilde{T} \cdot m = 1$ enforces two energy types. The rest mass is *static*: a topologically stable winding number on T^4, fixed in the compact picture by the Z_3 mass operator ($r = \sqrt{2}$, $\theta = 2/9$). The kinetic energy is *free*; for a massless particle it is the only form ($E = hf$). “Mass is frozen kinetic energy.”

Doc. 285 shows that the empty recursion windows ($3'$, $3''$, …) carry no *new* masses but structure; and that a *dynamics* filling a window supplies the *kinetic* type – a filled window is photon-/neutrino-like, not a massive triple. There the gap was named: the *derived* rule computing the kinetic part, via the duality, as a ξ-fractal fraction of the static one is missing. This document takes up exactly that gap. It is the “further work” on the dynamic sector announced in Doc. 285.

U.2 The fixed foundation: the reversible Hilbert-space basis

Before developing the dynamic sector, let the fixed foundation it sits on be recorded. The FFGFT QM translation (Doc. 282) is closed in three pieces:

1. **The unambiguous Hilbert map** (Type III, Doc. 270): $\mathscr{H} = L^2(T^4) \otimes \mathbb{C}^3$, states = fiber-valued fields, QM operation = unitary parallel transport, entanglement = holonomy around cycles (Doc. 230), three masses $\leftrightarrow (M, r, \theta)$. A *bijection* – fully reversible.
2. **The time-free mass form:** mass sits on a separate tensor factor; CHSH $= 2\sqrt{2}$ stays identical with and without the $\mathbb{C}^3$ fiber to $4 \cdot 10^{-16}$ – no static QM result changes. Time is *not computed* here; the reduced dynamics is time-local, unitary, Markovian.
3. **The time form:** adds exactly *one* effect – non-Markovian memory in the reduced dynamics (backflow $0 \to 0.53$, CP-divisibility violated) – which itself still lies within QM: a traced-out unitary bath reproduces it (Doc. 282/283). At the full level, therefore, unitary/reversible.

Reversibility is precise *per type* (Doc. 270): Type III (Hilbert) fully reversible; Type I (mass circle → time, "time taken into the space") *almost* reversible – lossless in mode content, only the winding number stays ambiguous; Type II (spatial projection $D3 \to D0$) *not* reversible. The only genuine irreversibility is the spatial projection – the measurement / decompactification (Doc. 270/284) –, not the time-inclusion. The non-Markovian memory is not a break of reversibility but the reduced view.

Aim of the dynamic sector. On this fixed, reversible foundation, the actual task is the *Hilbert-space extension and its translation*: *another* description *out of the Hilbert space* – the complementary, decompactified/projected side that maps computationally, via $D_f = 3 - \xi$ and ξ, onto SI units. The SI map is exactly the measurement/projection side (Type I/II), and D_f and ξ are the bridge quantities (Doc. 270/249/133). The dynamic sector is thus not a departure from the foundation but its unrolling into measurable units – and the sought kinetic ξ-fraction is one building block of exactly that unrolling.

U.3 What is fixed

The duality fixes the *placement*, not the size. With $\tilde{T} = 1/\max(m, \omega)$ (Doc. 067, 080), an almost-massless window state has $\omega \gg m$, hence $\tilde{T} = 1/\omega$: the state sits in the *kinetic* sector. That is secure. So is the conceptual separation:

	static	kinetic
form	$E_0 = mc^2$ (rest)	$E = hf$ (free)
seat	Z_3 core (winding)	window, when a dynamics runs
time	$\tilde{T} = 1/m$	$\tilde{T} = 1/\omega$
status	derived (Doc. 285)	*open* (this document)

A distinction between two kinetic quantities matters: the ordinary kinetic energy of the electron, $(\gamma - 1)mc^2$ (Doc. 080), is *not* ξ-suppressed. The ξ-fractal fraction sought here is the *window rest-mass* of an otherwise massless state – not the kinetic energy of a massive one.

U.4 The single data point: the neutrino

Doc. 047 treats the neutrino as a "near-massless photon": $E^2 = (pc)^2 + (\sqrt{\xi^2/2}\,mc^2)^2$, with $v_\nu = c(1-\xi^2/2) \approx 0.9999999911\,c$ and $m_\nu \sim \frac{\xi^2}{2} m_e$. This is the only existing value for a window rest-mass as a ξ-power of a core mass. It is, however, explicitly an *ansatz* (from the velocity/photon analogy), not *derived* from the duality. Doc. 204 additionally supplies a ξ-correction to the kinetic energy ($+0.69\,\xi_0$). There are no further data points; in particular, a single value is not enough for the Δm^2 spectrum.

U.5 Narrowing the ξ-power: $p = 2$

With the ansatz $m_{\text{window}} = C \cdot \xi^p \cdot m_{\text{ref}}$, tested against the bounds (reproducing the $\xi^2/2$ value; the photon limit; dimensional consistency; an oscillation floor $\gtrsim \sqrt{\Delta m^2_{\text{atm}}} \approx 49$ meV; a direct kinematic bound $m_\nu < 0.45$ eV (KATRIN 2025, 90 % CL, model-independent)), only $p = 2$ is compatible with the meV scale: $p = 1$ gives $\mathcal{O}(10\text{–}70)$ eV (far above the bound), $p \geq 3$ falls below every anchor. So $p = 2$ is the *unique* order-of-magnitude compatible power – a genuine order-of-magnitude bound, not a fit.

Numerology warning. *Within* $p = 2$, the coefficient ($\frac{1}{2}$) and the reference mass are *not* fixed by the single anchor. Different references land somewhere in the neutrino window (e.g. $\xi^2/2 \cdot m_e \approx 4.5$ meV, $\xi^2 \cdot m_e \approx 9$ meV, $\xi^2/2 \cdot \sqrt{m_e m_\mu} \approx 65$ meV). This is *not* evidence but the P35 trap: coefficient and reference need a derivation, not tuning. Computed: `kinetik_xi_anteil.py`.

U.6 The structural lead: $m_{\text{window}} = \xi \cdot \omega$

The Doc. 047 velocity expression $v_\nu = c(1-\xi^2/2)$ is exactly equivalent to a ξ^1 relation. For $m \ll p$ one has $v/c = 1 - \frac{1}{2}(m/p)^2$; substituting gives $(m/p)^2 = \xi^2$, hence

$$m_{\text{window}} = \xi \cdot p \quad \big(\approx \xi \cdot \omega \text{ via } \tilde{T} \cdot m = 1\big). \tag{U.1}$$

The "ξ^2" in the notation "mass $= \xi^2/2 \cdot m_e$" is thus presumably the *square image* of a fundamental ξ^1 relation: the static rest-mass is ξ times the kinetic scale. This is the natural anchor for a derivation *via the duality*, because $\omega = 1/\tilde{T}$ is exactly the duality quantity. The open question therefore shifts from "why ξ^2?" to "why ξ^1 between rest-mass and kinetic scale, and what sets ω absolutely?".

U.7 What remains to be derived

Three points are explicitly open:

1. **The reference/scale – largely closed** (next section). The two lab anchors select the reference as the *single* FFGFT scale $E_0 = \sqrt{m_e m_\mu}$; in its units $\omega/E_0 = \xi/2$ and

$m_{\text{window}}/E_0 = \xi^2/2$. Residual: the *one* structural statement $\omega = \xi E_0$ (exactly one fractal ξ-power).

2. **The coefficient.** The factor $\frac{1}{2}$ in $\xi^2/2$ is not derived; possibly it is purely the velocity square ($\frac{1}{2}(m/p)^2$), i.e. no free factor at all.
3. **The spectrum.** A single ξ-fractal value sets the *scale*, not the *spectrum* (Δm^2_{sol}, Δm^2_{atm}, mixing angles). That demands the specified T^7 dynamics (the second open computation, Doc. 285).

U.8 Closing the scale question: the lab anchors select E_0

The first open point – what sets ω absolutely – closes further than a free parameter would. In FFGFT there is only *one* absolute scale; everything else is dimensionless ratios. The question is therefore not "which number" but "which of the existing scales". With $m_{\text{window}} = \frac{\xi^2}{2} \cdot$ reference, the two *lab* anchors (oscillation floor $\gtrsim 49$ meV; KATRIN < 0.45 eV) bracket the reference:

reference	m_{window}	floor ≥ 49 meV	KATRIN < 0.45 eV
m_e	4.5 meV	*fails* (too light)	yes
$E_0 = \sqrt{m_e m_\mu}$	65 meV	yes	yes
$M \approx 314$ MeV	2.8 eV	yes	*fails*
m_μ	940 meV	yes	*fails* (too heavy)

m_e is too light, M and m_μ too heavy – *only* $E_0 = \sqrt{m_e m_\mu}$ fits, and that is exactly the one natural FFGFT scale (Doc. 282, $\lambda_e \lambda_\mu = E_0$). The reference is thus not a free quantity but the existing scale, selected by the lab.

In units of E_0 everything then becomes a pure ξ-power times $\frac{1}{2}$:

$$\frac{\omega}{E_0} = \frac{\xi}{2}, \qquad \frac{m_{\text{window}}}{E_0} = \frac{\xi^2}{2}. \tag{U.2}$$

No free parameter remains. The $\frac{1}{2}$ is the velocity square ($\frac{1}{2}(m/p)^2$, Doc. 047) – kinematic, not a tuning factor; its *structural* origin is the AM sideband factor (see the signal-engineering reading below).

What honestly remains. The lab window (49–800 meV) is a factor ~few wide: it selects the *scale* E_0 sharply (excluding the alternatives by orders of magnitude) but does not pin the exact coefficient by itself. What remains is exactly *one* structural statement,

$$\omega = \xi E_0 \,, \tag{U.3}$$

– E_0 suppressed by exactly *one* fractal ξ-power ($D_f = 3 - \xi$). This is no longer "free scale, self-referential" but a non-circular computation step: ω as the ξ-suppressed eigenvalue of the connection-Laplacian in the time-extended Hilbert structure (Doc. 283 §8). It closes the point – or falsifies it. Computed: `schliesse_omega.py`.

U.9 The ξ^1 mechanism: lifting the zero mode by the dimension deficit

The remaining residual – why $\omega = \xi E_0$ with exactly *one* ξ-power – has a concrete mechanism. The connection-Laplacian on T^4 has an exact zero mode (the holonomy-flat constant). The fractal correction $g = \eta(1 + \xi f)$ – here f is the fractal correction function of the metric, not a frequency and not the base winding number $f = 1/\xi$ – acts as a diagonal potential perturbation and lifts it; first-order perturbation theory gives

$$\delta\lambda_0 = \xi\langle 0|f|0\rangle = \xi \cdot \bar{f}, \tag{U.4}$$

i.e. the power ξ^1 *precisely when* the correction has a non-vanishing torus average $\bar{f} \neq 0$; for $\bar{f} = 0$ the first order vanishes and the mode lifts only at ξ^2.

$D_f = 3 - \xi$ is a dimension *deficit* – a systematic, non-volume-preserving change of the measure –, hence $\bar{f} \neq 0$. So the power is *one*. Numerically (`xi1_eigenwert.py`): the zero-mode lift scales with $p = 0.999$ for a deficit ($\bar{f} \neq 0$) versus $p = 1.98$ for a volume-preserving ripple ($\bar{f} = 0$) – so the power is not accidental but follows from the nature of $D_f = 3 - \xi$.

Status. The ξ^1 power is thus *derived* (zero-mode lift by the dimension deficit), not asserted. What remains is only the $O(1)$ coefficient – kinematically the $\frac{1}{2}$ (Doc. 047) – and confirming that the physical kinetic state is this lifted zero mode. With that, $\omega = \frac{\xi}{2}E_0$ is closed up to an $O(1)$ factor – and that $O(1)$ factor is identified structurally, in the signal-engineering reading below, as the sideband $\frac{1}{2}$.

U.10 The signal-engineering reading: AM sidebands and mode-locking

FFGFT grew out of communications engineering and the Fourier duality time↔frequency (Doc. 001a); $\tilde{T}\cdot m = 1$ *is* that duality. In the dynamic sector the same language yields two concrete statements – one solid, one a candidate.

The $\frac{1}{2}$ is the AM sideband factor. The fractal correction $g = \eta(1 + \xi f)$ is formally an amplitude modulation of the carrier with modulation index ξ (Doc. 057: multiplicative = AM, additive = FM). Marcel's kinetic prefactor $A(t) = 1 + \varepsilon(t)$, $|\varepsilon| \ll 1$, is the same object ($\xi \leftrightarrow \varepsilon$). For a carrier $\cos(\omega_c t)$ one obtains necessarily

$$\cos(\omega_c t)\,[1 + \xi\cos(\omega_m t)] = \cos(\omega_c t) + \frac{\xi}{2}\big[\cos((\omega_c{+}\omega_m)t) + \cos((\omega_c{-}\omega_m)t)\big], \tag{U.5}$$

so the sidebands sit at amplitude $\xi/2$. The kinetic scale $\omega = \frac{\xi}{2}E_0$ is therefore the *first AM sideband of the mass carrier* E_0: the ξ from the modulation index (= dimension deficit, previous section), the $\frac{1}{2}$ from the sideband split ($\cos\cdot\cos = \frac{1}{2}$(sum + difference)). This unifies the ξ^1 power (lifted zero mode = first sideband) and the $O(1)$ coefficient into *one* mechanism: the $\frac{1}{2}$ is not inserted kinematically but is structural – the velocity square ($\frac{1}{2}(m/p)^2$, Doc. 047) is its kinematic shadow, not its origin.

$\theta = 2/9$ **as a mode-locking ratio (candidate).** The phase layer of the fork (Doc. 283) reads: *dynamic operator-phase* R (HLV, skew-adjoint, driven by $\dot{\phi}$) against *fixed scalar phase* $\theta = 2/9$ (FFGFT). Locking a dynamic phase onto a fixed rational value is well-defined in signal engineering: injection locking, i.e. Arnold tongues. A driven nonlinear oscillator locks onto rational rotation numbers p/q; the stable plateaus form a devil's staircase. $\theta = 2/9$ would then be the lock plateau of a rational rotation number – which connects directly to FFGFT's *winding numbers* $(n_\theta, n_\varphi, n_3, n_4)$ (Doc. 210): a locked rational winding. Mode-locking *is* a quantization rule – rational values are the stable fixed points – exactly the kind of "holonomy/fixed-point/quantization rule" the open θ point demands. The *open static* θ input would thereby acquire a *dynamic* origin in the same sector.

What stays a candidate. The sideband point is solid: it reproduces $\xi/2$ necessarily. The mode-locking reading is a *mechanism candidate*, not a derivation. There is a tension: small denominators ($1/2, 1/3, 2/5$) have the widest tongues, while $2/9$ (denominator 9) would be a narrower plateau; *why exactly* $2/9$ locks is open. But the reframing is productive: "why this value?" becomes "which Arnold tongue, and why is it stable?" – a testable question about a driven oscillator, not about a bare number. An honest bound on the candidate: the rotation number $2/9$ is *dimensionless* (cycles), the Koide offset $\theta = 2/9$ is a *radian* angle – $2\pi \cdot 2/9 = 4\pi/9 \neq 2/9$ –; reading the two as the same is an additional, open assumption, not part of the computed candidate (cf. Doc. 282).

Z_3 protection: the denominator is explained, the numerator is not. The FFGFT mass core is a Z_3 circulant (Doc. 282); if the driving nonlinearity carries this symmetry (invariant under $\theta \to \theta + 1/3$, i.e. only harmonics $k = 3, 6, 9, \ldots$), the picture shifts. A structurally faithful Z_3 drive widens the tongues with 3-power denominators *selectively*: at equal criticality $1/3$ becomes $24\times$ wider and the whole 9-family 11–$12\times$, while the control tongue $2/5$ (denominator 5, not a multiple of 3) stays *flat* ($0.97\times$) – the signature of genuine symmetry protection, not an artefact (`z3_modelock_test.py`). This answers "why a denominator-9 lock at all": the Z_3 structure of the core; $2/9$ is no longer absurdly narrow but symmetry-protected robust. *The numerator stays open:* the symmetry protects the 9-family jointly ($1/9 = 2/9 = 4/9$ identical), it does not single out $2/9$, and $1/3$ stays wider. The question thus splits cleanly – "why $2/9$" = "why denominator 9" (Z_3, settled) + "why numerator 2 in the family" (open, exactly Marcel's BD17A holonomy domain).

The numerator needs a holonomy. The 9-family is a Z_3 orbit; a single 3-fold drive is therefore numerator-blind. The numerator is selected only by a *second* 3-structure with a definite relative phase ϕ – a holonomy that lives in the embedding space (3′ of the doubling $6 = 3 \oplus 3'$, $T^4 \to T^7$, Doc. 285), not in the core. With two 3-fold harmonics (h=3, h=9) and a relative phase ϕ (not gauge-removable) the degeneracy breaks (`z3_holonomie_test.py`, bisection): $1/9$ and $4/9$ stay paired, $2/9$ is the singlet that moves – for $\phi \in (\pi, 2\pi)$ $2/9$ is the *widest* tongue (clearest margin at $\phi = 3\pi/2$), crossover exactly at $\phi = \pi$. *Honestly:* the effect is small ($\sim 0.5\%$), ϕ is free (FFGFT does not fix it), and the minimal model splits only 2-against-1. But the structural statement stands: a holonomy *can* select $2/9$, and $2/9$ is exactly the member it acts on. The open question is thereby precisely shifted – no longer

"why $2/9$" but "what locks ϕ into the band where $2/9$ wins" – and that is Marcel's BD17A "locked holonomy".

One fractal correction, two orthogonal phases. Holonomy and time modulation spring from the *same* fractal correction $g = \eta(1 + \xi f)$ – the driving nonlinearity *is* f (Doc. 051, Doc. 270). One might fear that treating them as independent is inconsistent. The computation (`eine_fraktale_quelle.py`, 3-adic Weierstrass function) shows the opposite: they are two *orthogonal* functionals of f. The time phase $A = 2\pi\xi\langle f\rangle$ reads the *mean* $\langle f\rangle$ (the DC part, the dimension deficit, $O(\xi)$); the holonomy $\phi = \arg \hat{f}_9 - 3\arg \hat{f}_3$ reads the *phases of the* 3*-power harmonics* ($O(1)$). The two are analytically decoupled – $\langle f\rangle$ is independent of the harmonic phase (numerically to $2 \cdot 10^{-16}$), since $\cos(k\theta + \psi)$ has vanishing mean for every ψ. The worry is thereby resolved: neglecting A does not touch ϕ, they read different Fourier parts of the *same* f. The real consistency lives elsewhere: A and the ξ^1 mass lift *both* depend on $\langle f\rangle$ – so the time phase is tied to the mass window, not free. And no free lock: a phase-coherent fractal ($\psi = 0$) gives $\phi = 0$ (degeneracy), whereas $2/9$ requires f's 3-power harmonics to carry a relative phase in $(\pi, 2\pi)$ (a ramp $\delta = \pi/2$ per octave hits $\phi = 3\pi/2$ exactly, the $2/9$ maximum – but as a choice, not a derivation). Thus "what locks ϕ" is concretely named: not some arbitrary phase, but the relative phase of the 3-power harmonics of the fractal correction f – the FFGFT-internal target for BD17A.

Demarcation: what is *not* a mechanism here. Modulation index and lock ratio are different quantities in different sectors: $\xi = 4/30000$ is the fixed AM index in the amplitude sector (it yields the sideband $\frac{1}{2}$), $2/9$ is a rotation number in the phase sector. The lock ratio is *not* ξ. Any "derivation" that sets ξ to an $O(1)$ value (such as $1/3$ or $\tan(\pi/9) \approx 0.364$) in order to force $2/9$ has left FFGFT – ξ is fixed and orders of magnitude smaller ($\sim 10^{-4}$). Likewise the open question is the *stability* of the $2/9$ plateau, not an algebra between ξ and $2/9$; and the memory kernel the locking would rest on is the discrete-spectral, oscillating kernel of Doc. 282 ($\sum_k g_k^2 \delta(\omega - \omega_k)$, with backflow/revivals), not a dispersive all-pass.

U.11 Recursion as feedback: loop gain, regimes, discipline

FFGFT grew out of communications engineering (Doc. 001a), and its recursions are *feedback loops*. This places the results above in a frame – and fixes what about them is *free* and what is not.

The loop gain is a ξ quantity – and fixed. The magnitude of the loop gain decides the behaviour. In FFGFT it is geometrically given, not a dial:

- $|v| < 1$ (negative feedback): the loop contracts. The ξ-damping recursion $r(k{+}1) = r(k)(1 - \xi)$ (Doc. 275) has gain $1 - \xi$, only ξ below one – hence the enormous, slow scale tower; it runs to its only fixed point 0 and merely *passes through* $1/\varphi$ (no attractor).
- $|v| = 1$ (Barkhausen): marginal sustained oscillation.
- $|v| > 1$ (positive feedback): the loop runs away. The golden φ branch ($|\varphi| > 1$) unrolls, the $-1/\varphi$ branch ($|-1/\varphi| < 1$) is the overdamped empty window (Doc. 285).

Decompactification is exactly this transition: *one* circle tips from the compact (contracting) into the runaway regime – the mass circle $\tilde{T} \cdot m = 1$ unrolls to non-compact time (Doc. 270), and that tipping *is* the measurement.

The phase selects itself. A feedback loop imposes no phase: the Barkhausen phase condition closes at the frequency the loop *finds*; in injection locking the lock phase slides within the lock range (Adler equation). Hence $\theta = 2/9$ is to be read as a *self-selected lock ratio* (rotation number), not as a tightly imposed phase. A "derivation" that prescribes a fixed phase $2/9$ and solves backwards for ξ mistakes this self-adjustment (cf. the demarcation above).

Two tiers. Linear and weak (ξ small): first order = the AM sideband $\frac{1}{2}$ (above, solid); a purely linear loop does not lock. Nonlinear: mode-locking onto rational rotation numbers = the $\theta = 2/9$ candidate (above); the lock-range width scales with the feedback strength ξ, and $\xi \sim 10^{-4}$ with denominator 9 makes the $2/9$ tongue extremely narrow – that is the open stability problem, not a missing piece of algebra.

Negative feedback as equalization. The fractal correction $g = \eta(1 + \xi f)$ (with K_{frak}/RG, Doc. 133/051) acts as broadband negative feedback that linearizes the bare carrier and sets $D_f = 3 - \xi$ – weak, with no sharp phase alignment required.

Discipline. This fixes how the tools are to be used *correctly*: (i) the loop gain (ξ, the eigenvalues $\varphi/-1/\varphi$) is fixed, never a dial – forcing a result by setting ξ is the cardinal error; (ii) the open questions are always *regime and lock range* (contracting/marginal/runaway; tongue width), never an algebra *on* the gain; (iii) the phase is self-selected, quantized values are lock ratios; (iv) the regime of the feedback carries the physics – negative feedback = compact stable scales + equalization, positive feedback = decompactification/measurement.

U.12 Relation to the static sector

The static sector is closed and computed: the Z_3 mass operator ($r = \sqrt{2}$, $\theta = 2/9$) plus the mass circle yields Koide $2/3$, the lepton ratios and – as a pre-registered, matched-null-tested declaration – the BD12-XCORE interface (Doc. 285). The dynamic sector is the *complementary, open* side: the same parameter ξ, but as a *fraction* rather than an *eigenvalue*. A derivation of $m_{\text{window}} = \xi \cdot \omega$ would move the window-filling dynamics (Doc. 263, holographic scale projection) from description to computation, and the neutrino-like $3'$ candidate (Doc. 285) from conditional to derived.

Retrospect: two things both called "$2/9$". It is worth fixing the thread, for $\theta = 2/9$ appears in two roles. (i) $2/9$ as a *number/input*: the Z_3 circulant of the core uses it and reproduces the leptons (Koide), empirically closed – it was always there (Doc. 282; Doc. 285, the "protected-invariant" row $r = \sqrt{2}$, $Q = 2/3$, $\theta = 2/9$). (ii) $2/9$ as a *question*

("why this value", specifically the numerator): this becomes relevant only with the dimensional extension. The reason is structural, not a lack of computation: the bare Z_3 core is *numerator-blind* – $1/9, 2/9, 4/9$ are a Z_3 orbit, and a single compact 3-structure cannot in principle answer "why $2/9$ rather than $1/9$". The numerator is seen only by a second 3-structure with a relative phase = a holonomy = exactly what the extension provides ($3 \oplus 3'$, $T^4 \hookrightarrow T^7$). The "why-$2/9$" question thus has no home in the core; it is by nature an extension question.

Where this already stands in the corpus. Three documents already carry the thread. **Doc. 285** (dimension bridge) is the extension itself: HLV $\supset$ FFGFT, $T^4 \hookrightarrow T^7$ ($7 = 4 + 3$, the three perp dimensions), the doubling $6 = 3 \oplus 3'$ ("literally"), the fork (FFGFT 3-fold $\sqrt{2}/Q = 2/3/\theta = 2/9$ against HLV 5-fold $2/\sqrt{5}/\varphi$); $\theta = 2/9$ sits in the invariant row. **Doc. 283** (bridge and fork, §8) introduces $\theta = 2/9$ explicitly as a *complex holonomy*, the "operational consequence of compactness", with the phase layer "dynamic operator-phase R (HLV) against fixed scalar phase $\theta = 2/9$ (FFGFT)" – there $2/9$ is already a holonomy/extension object, not a bare core number. **Doc. 282** is the core that *uses* $2/9$ (leptons), with the honest line "translation is not justification" – the notation *represents* $2/9$ but does not *ground* it.

The fixed point and the conclusion. Why the static sector is closed *at all*: the lepton masses sit at a *self-consistent fixed point* – the mass operator $\Phi_* = \sum_i m_i P_i$ is the non-trivial fixed point of the recursion $\mathscr{R}(\Phi_*) = \Phi_*$ (Doc. 203, lepton exponents $p_e = 3/2,\ p_\mu = 1,\ p_\tau = 2/3$); *that* is why it is exact, not a fitted value. From the lepton measurements one *reads off* this fixed point (that is the closure); fixed and open separate cleanly: $Q = 2/3$ *is* the operator condition $r = \sqrt{2}$ (algebraically fixed, Doc. 282), $\theta = 2/9$ is the holonomy (the numerator, open). Hence the conclusion: all the recent work above (feedback, mode-locking, holonomy test) is *no new arena* but the mechanistic, signal-engineering version of what $282 \to 283 \to 285$ already say structurally – $2/9$ is a holonomy, and a holonomy lives in the extension.

Magnitude and phase: why the split is inevitable. In signal-engineering language the whole finding collapses to a single, well-known theorem. The mass operator *is* a Z_3 circulant, and a circulant is diagonalized by the DFT – its eigenvalues are the DFT of its first row. The three lepton masses are thus literally a *three-point spectrum*: $\sqrt{2}$ is the *magnitude* of the spectral coefficients, $\theta = 2/9$ their *phase*. The radial/angular conclusion (Doc. 282) is therefore nothing but *magnitude spectrum versus phase spectrum*: everything determined sits in the magnitude (ξ as scale, the hierarchy as envelope, $Q = 2/3 = \sqrt{2}$ as amplitude ratio, $\langle f \rangle$ as DC term), everything open in the phase (θ, the holonomy $\varphi = \arg \hat{f}_9 - 3 \arg \hat{f}_3$, the harmonic phases). The deep reason is the theorem itself: *a magnitude spectrum does not fix the phase spectrum* – same power, arbitrary shape. *That* is precisely why θ resists every FFGFT tool: the recursion $\mathscr{R}$ is purely real, hence pure gain, and real gain does not touch the phase. The orthogonality computed above *is* "magnitude $\perp$ phase". From this the FFGFT↔HLV division of labour even becomes a circuit diagram: FFGFT is the *real amplifier stage* (magnitude, hierarchy), HLV's skew-adjoint R is the *all-pass/phase stage* ($e^{i\varphi}$, magnitude-preserving, rotating only the phase).

A skew-adjoint operation *is* an all-pass. The missing angular principle is the phase stage – and the HLV side carries it, not FFGFT. (Limit of the same lens: mode-locking yields a *rotation number* (dimensionless), θ a *radian*; $2\pi \cdot 2/9 \neq 2/9$ – the lens identifies the *type* correctly, a locked phase, but the literal identification does not close.)

U.13 Reproducibility

`kinetik_xi_anteil.py` (numpy-only): narrowing of the ξ-power against the bounds, the duality framing ($\tilde{T} = 1/\max(m,\omega)$), and the finding $p = 2$ with the explicitly open reference/coefficient/spectrum gap. The script claims no derivation; it narrows. `schliesse_omega.py`: the lab selection of the reference E_0 (table) and the reduction $\omega/E_0 = \xi/2$, $m_{\text{window}}/E_0 = \xi^2/2$, with the ξ^1 remainder as the marked closing point. `xi1_eigenwert.py`: the zero-mode lift as ξ^1 from the dimension deficit ($p = 0.999$ vs $p = 1.98$). `am_seitenband_modelock.py`: confirms the sideband ratio $\xi/2$ necessarily (deviation 0%) and measures the Arnold-tongue widths on the sine circle map – $2/9$ locks as a plateau but is $\sim 12\times$ narrower than $1/3$: candidate, not derivation. `z3_modelock_test.py`: a Z_3-symmetric drive (harmonics = multiples of 3) widens the 3-power-denominator tongues selectively ($1/3$ by $24\times$, the 9-family by 11–$12\times$), while the control $2/5$ stays flat ($0.97\times$) – the denominator 9 is Z_3-protected (Doc. 282), the numerator 2 stays open. `z3_holonomie_test.py`: two 3-fold harmonics (h=3, h=9) with a relative phase ϕ; measured precisely (bisection), ϕ breaks the 9-degeneracy – $1/9, 4/9$ stay paired, $2/9$ moves and is the widest tongue for $\phi \in (\pi, 2\pi)$ ($\sim 0.5\%$, maximum at $\phi = 3\pi/2$); ϕ free ⇒ shifts the open question to the ϕ-lock (BD17A). `eine_fraktale_quelle.py`: shows computationally (3-adic Weierstrass f) that the time phase and the holonomy are *orthogonal* functionals of the same fractal correction – $A = 2\pi\xi\langle f\rangle$ reads the mean ($O(\xi)$), $\phi = \arg \hat{f}_9 - 3 \arg \hat{f}_3$ the harmonic phases ($O(1)$), $\langle f\rangle$ phase-independent to $2 \cdot 10^{-16}$; the time phase is tied to the ξ^1 mass lift (both $\langle f\rangle$), the holonomy phase stays an independent datum (coherent ⇒ $\phi = 0$; $\delta = \pi/2$ ramp ⇒ $\phi = 3\pi/2$).

Chapter V

Doc. 287 — Internal Representation Equivalences

Abstract

FFGFT exists in several mathematical languages – torus geometry, matrix algebra, Hilbert-space operators, information formalism, recursion, two layers. This document does not ask how FFGFT stands relative to *other* frameworks (Doc. 265 does that outward); it asks how these internal representations stand *among themselves*. Finding: they are not a flat set of equals but a *generative core* (the T^4 torus) plus three clearly distinct relationship types – **identity**, **bijection**, **projection**. The geometry and the 3×3 matrix algebra are *the same thing* (Doc. 206); the standard-QM Hilbert space is a *lossless bijection* (Doc. 230); the information formalism and the spatial reduction are *lossy projections* with a formalism-relative residue (Doc. 243/270). The decisive point: *only* the full T^4 geometry loses nothing. The θ=2/9 work carries the same result from the other side: all representations are faithful on the radial (magnitude) sector; the only irreducible part is the angular (phase) sector.

V.1 The question and its boundary

Over recent months FFGFT has been cast in strikingly many languages, and the impression of several "parallel roads" is justified. The question here is internal: *what relationship* holds between these roads? Are they interchangeable pictures of one object, or do they differ structurally?

This must be set apart from Doc. 265. There a *comparison operator* (three levels – near-equivalence, structural overlap, complementary; six axes) is developed and applied to FFGFT *against an external framework* (Blume-Maleev). Here the same operator is turned *inward*: not FFGFT versus foreign, but FFGFT-representation versus FFGFT-representation. The result is sharper than the outward operator allows – and it exposes a relationship type the symmetric outward operator does not have (Section V.7).

V.2 The generative core: why T^4 is not one representation among many

Two documents say the same thing from different sides. Doc. 152 (ontological priority) sets the T^4 torsion crystal as *fundamental ontological reality*; the fractal granulation $D_f = 3 - \xi$, the effective laws and classical 3D physics are emergent approximations "projected upward". Doc. 243 gives the structural counterpart: every information formalism discards at least one of the three quantities θ_4, (n_θ, n_φ), ε; the T^4 geometry is the *only* formalism that structurally holds all three at once.

Key Result

The T^4 torus is not one representation among many but the generative core: the only language that holds the full content. All other representations are maps *from* T^4 – and split, by the type of that map, into three classes.

V.3 The three relationship types

Identity: geometry ≡ matrix algebra (Doc. 206)

The Z_3 triangle (as a graph) and the 3×3 matrix A are not merely isomorphic but *algebraically identical*: the spectrum of A is the character table of Z_3. Doc. 206 puts it literally: geometry and matrix algebra are "not two pictures of the same thing, they are the same thing." This is the strongest type – not a lossless translation channel but identity. Honestly noted: this identity is *sector-local*. It holds on the C_3/Z_3 core (3×3). Globally – with the angular phase θ and the φ harmonics – it is no longer a finite identity; the circulant captures the magnitude spectrum, not the phase (Doc. 286).

Bijection: the lossless channel (Doc. 230, Doc. 070)

The Hilbert-space translation (Doc. 230) is a *bijective structural statement*: every QM calculation sits inside the FFGFT formalism and back, *without loss*. It is no poorer than FFGFT and no richer in computation – but FFGFT *adds* what standard QM does not provide: the *why* of the constant values. In the projection-chain picture (Doc. 270) this is the *upward* move (Type III), and precisely because it introduces no geometric extra dimensions it is lossless.

A second bijection case is corpus-internal: the two mass forms in Doc. 070, the compact (cancelled) and the extended (geometrically original). They are mathematically equivalent, but only the extended carries the physical origin – which is why the ratios must not be cancelled directly.

Projection: the lossy, directed channel (Doc. 243, Doc. 270)

Doc. 243 maps FFGFT into an information formalism – as a *projection* π, *not* an isomorphism. A residue is lost, and that residue is *formalism-relative*: it depends on the chosen

information formalism, not on the geometry. The same mapping into a neural network shares this character. Geometrically this corresponds to the *spatial* reductions of the projection chain (Doc. 270, Type II): strictly lossy, irreversible, with residue $\sim \xi$ (as a CHSH deviation, not resolvable with present hardware).

To be set apart is the first reduction step $D4 \to D3$ (Doc. 270, Type I): not a lossy projection but a *duality decompactification* via $\tilde{T} \cdot m = 1$ – information-preserving; only the direct visibility of compactness is lost.

V.4 Two axes that are not competing representations

Not every "language" is a rival encoding of the same content. Two are orthogonal separations:

- **Two layers** (Doc. 241): the separation of ratio-based, parameter-free content (Layer 1) from a single SI anchor (Layer 2). This is bookkeeping – what is input, what follows – not an alternative geometry. Orthogonal to the geometry/matrix/Hilbert choice.
- **Recursion / fixed point** (Doc. 203/275): the *dynamical* axis. It generates the hierarchy (the $\xi \to \varphi$ passage) and is – decisively – purely *real/radial*. It carries no angular (phase) generator.

V.5 The decisive finding

Two statements that together form the strongest internal-consistency claim.

Important

(1) Only T^4 loses nothing. Every projective encoding drops a residue, and the residue is relative to the *representation*, not to the physics. The many roads therefore agree; where they diverge it is exactly what the given encoding threw away. The "loss" is an artefact of the representation, not of the theory.
(2) The asymmetry (Doc. 270). Geometrically *downward* is lossy (Type II); representationally *upward* to the Hilbert space is lossless (Type III). This is exactly what lets FFGFT be *simultaneously* a "4D torus theory" and "ordinary QM" without contradiction.

V.6 The bracket to the θ=2/9 phase

The representation analysis and the θ elimination chain (Doc. 282/286) are one result from two sides. All representations are faithful on the *radial* (magnitude) sector: identity (206) and bijection (230) capture it completely – ξ, the hierarchy, the Koide amplitude $Q = 2/3 = r\sqrt{2}$, the mean $\langle f \rangle$ of the fractal correction function in $g = \eta(1 + \xi f)$ (not a frequency, not the base winding number $f = 1/\xi$). The *one* place where the geometry is irreducible is the *angular* (phase) sector: the offset $\theta = 2/9$. Only the full T^4 geometry holds it; the real recursion does not reach it, the circulant does not capture it. Faithful in magnitude, irreplaceable in phase.

V.7 The 265 operator turned inward

Turning the Doc. 265 comparison operator inward shows two things. First, the internal relationships are *sharper*: where the outward operator reached only *near*-equivalence (different derivation routes, identical abstract structure), the internal bijections are *exact* (isomorphism, Doc. 230) and the matrix case is even *identity* (Doc. 206). Second, the inward application exposes a type the three symmetric levels of the outward operator do not have: the *directed projection* (243/270-II). The 265 levels compare two formalisms symmetrically; the internal projection is directed ($T^4 \to$ encoding) and carries a residue. This is no defect of the 265 operator but a finding: inward, a fourth, asymmetric relation is needed.

Table V.1: The internal representations, by relationship type to the generative T^4 core.

Representation	Doc.	Relationship	keeps / loses
Z_3 triangle $\equiv 3\times 3$ matrix	206	identity	same thing; sector-local (C_3)
Hilbert space $H = L^2(T^4) \otimes \mathbb{C}^3$	230	bijection (lossless)	QM $\leftrightarrow$ FFGFT; FFGFT adds the "why"
compact vs. extended mass form	070	bijection (equivalent)	equal; only extended carries origin
information formalism	243	projection π	residue, formalism-relative
neural network	243	projection	same
spatial reduction $T^3 \to T^0$	270	projection (Type II)	residue $\sim \xi$ (CHSH)
duality decompact. $D4 \to D3$	270	Type I (preserving)	only visibility of compactness
two layers (ratio/SI)	241	orthogonal axis	bookkeeping, no rival
recursion / fixed point	203/275	orthogonal axis	dynamical, purely real/radial

V.8 Open edges

Declared honestly, not hidden:

- The **identity matrix** $\equiv$ **geometry** (206) is sector-local (3×3, C_3). A global finite identity including θ and the φ harmonics does not exist; whether it *can* exist is open (Doc. 286).
- The **residue table** of the information formalisms (243) is a working stand, not a proof of minimality that T^4 is the only complete formalism.

- The **265 comparison operator** is itself open with three declared residues (axis completeness, weighting, independence); the fourth, directed relation found here is to be added.

Balance: one core, three relations

The parallel roads of FFGFT are not of equal rank. The T^4 torus is generative – the only representation that loses nothing. Everything else stands to it in exactly one of three relations: *identity* (matrix algebra, sector-local), *bijection* (Hilbert space; compact/extended mass form), *projection* (information formalism, neural network, spatial reduction – with a formalism-relative residue). Two further languages (two layers, recursion) are not rivals but orthogonal axes. The common ground of all roads is the radial magnitude sector; the only irreducible place is the angular phase $\theta = 2/9$. The representation question is thereby closed as far as it honestly can be: faithful in magnitude, generative only in the full torus, open only in the phase.

Chapter W

Doc. 288 — Signal-Engineering Computational Routes

Abstract

Doc. 287 classifies FFGFT's parallel representations by relationship type (identity, bijection, projection). This document reads the same representations *differently*: not as a map but as a *toolbox*. The signal-engineering experience – transforms exist because they make computations cheap (convolution becomes multiplication) – carries over directly. The mass operator is a Z_3-circulant; the DFT diagonalises it. From this follow five concrete, machine-precision-checked computational levers: (1) the lepton mass roots are a *3-point DFT* instead of an eigenvalue problem; (2) Koide $Q = 2/3$ falls out in *two Parseval lines* and is manifestly phase-independent; (3) *any* operator function (power, inverse, exponential) is a scalar evaluation at three eigenvalues; (4) chaining Z_3 couplings is *pointwise multiplication* in the DFT domain; (5) the ξ recursion is a *first-order IIR filter* with a pole at $1 - \xi$, and the three regimes are pole locations. Honestly: this is computational leverage, not a derivation – the transform moves $\sqrt{2}$ and θ, it does not create them.

W.1 The idea: changing representation as a computational lever

In signal engineering one changes representation not by taste but for computational gain. A convolution in the time domain is a pointwise multiplication in the frequency domain; a linear time-invariant system is an algebraic transfer function in the z domain; magnitude and phase separate, so they are computed independently. Exactly this leverage sits in FFGFT's parallel representations – one only has to read them as transforms rather than worldviews.

The key is an observation from Doc. 286: the mass operator *is* a Z_3-circulant. A circulant is diagonalised by the discrete Fourier transform (DFT) – the signal-engineering principle "cyclic convolution ↔ DFT". Everything else follows.

Key Result

The Z_3-circulant mass operator C is diagonalised by the unitary DFT matrix F: $C = F^\dagger \Lambda F$, $\Lambda = \mathrm{diag}(\lambda_0, \lambda_1, \lambda_2)$. Thus every "function of C" problem becomes an evaluation at the three scalars λ_k.

W.2 The setup: first row and DFT

The Foot-Koide configuration $\sqrt{m_k} = 1 + \sqrt{2}\cos(\theta + 2\pi k/3)$ is the DFT of a *3-vector*, the first row of the circulant:

$$c = \left(1,\ \tfrac{\sqrt{2}}{2}\, e^{i\theta},\ \tfrac{\sqrt{2}}{2}\, e^{-i\theta}\right), \qquad r = \sqrt{2}, \quad \theta = \tfrac{2}{9}.$$

Here the magnitude $r = \sqrt{2}$ sits in $|c_1| = |c_2| = \sqrt{2}/2$ and the phase θ in the argument – magnitude and phase parts are separated from the start. C is Hermitian (checked: $\max|C - C^\dagger| = 0$), hence real-spectral.

W.3 Lever 1: eigenvalues without a characteristic polynomial

The three mass roots $\sqrt{m_k}$ are the DFT of c: $\lambda_k = \sum_j c_j \omega^{jk}$, $\omega = e^{2\pi i/3}$. No 3×3 eigensolver, no polynomial – a DFT. Numerically $\sqrt{m} = (0.0404,\ 0.5802,\ 2.3794)$, matching the closed Foot-Koide form and `eigvalsh` to $8.9 \cdot 10^{-16}$ (`dft_eigenwerte.py`).

W.4 Lever 2: Koide Q in two Parseval lines

With λ_k as DFT coefficients, Parseval gives directly:

$$\sum_k \sqrt{m_k} = 3\, c_0 = 3, \qquad \sum_k m_k = 3 \sum_j |c_j|^2 = 6, \qquad Q = \frac{\sum m_k}{(\sum \sqrt{m_k})^2} = \frac{6}{9} = \frac{2}{3}.$$

$\sum \sqrt{m}$ reads only the DC term c_0, $\sum m$ only the energy $\sum |c_j|^2$. Both depend *only on the magnitude* $|c_j|$, i.e. on r – not on θ. Over all $\theta \in [0, 2\pi)$, Q stays exactly 0.666667; a different magnitude shifts Q ($r = 1 \to 1/2$, $r = \sqrt{2} \to 2/3$, $r = 1.6 \to 0.76$). Thus the radial/angular conclusion of Doc. 282 is not a separate statement but a trivial Parseval consequence: the magnitude spectrum fixes the invariant, the phase spectrum does not enter (`parseval_koide.py`).

W.5 Lever 3: any operator function for free

From $C = F^\dagger \Lambda F$ follows $f(C) = F^\dagger f(\Lambda) F$ for any function f (here an arbitrary operator function, not a frequency and not the base winding number $f = 1/\xi$). An operator function is therefore three scalar evaluations $f(\lambda_k)$ instead of matrix algebra:

- **power** C^n – for recursion / scale flow;
- **inverse** C^{-1};

- **exponential** $\exp(C)$ – for time evolution / propagator.

Checked against `matrix_power`, `inv` and a Taylor series: C^5 to $3.6 \cdot 10^{-14}$, C^{-1} to $4.8 \cdot 10^{-15}$, $\exp(C)$ to $2.7 \cdot 10^{-15}$ (`operatorfunktionen.py`). This is exactly the sector the dynamic part (Doc. 286) heads into: there one needs functions of the operator, and they are trivial in the DFT domain.

W.6 Lever 4: convolution becomes multiplication

A Z_3-cyclic coupling between the three generations is a circular convolution. The convolution theorem turns it, in the DFT domain, into a pointwise multiplication: $\mathrm{DFT}(a * b) = \mathrm{DFT}(a) \cdot \mathrm{DFT}(b)$. Chaining two Z_3 couplings – a matrix product of two circulants – becomes three scalar products. Checked to $9 \cdot 10^{-16}$ (`faltung_multiplikation.py`). This is the actual signal-engineering core: the expensive operation (matrix product) becomes cheap (three multiplications) once one computes in the right representation.

W.7 Lever 5: the recursion is a filter

$r(k{+}1) = r(k)(1-\xi)$ is a first-order IIR filter with z-transfer function $H(z) = 1/(1-(1-\xi)z^{-1})$, a pole at $z = 1-\xi = 0.99986667$. $|z| < 1$ means stable / contracting; the pole's distance to the unit circle is exactly ξ, hence the time constant $k^* \sim 1/\xi = 7500$ steps. The three regimes of Doc. 285/286 are pole locations: inside (negative feedback, contracting), on the circle (Barkhausen limit), outside (positive feedback, run-away). Impulse response and direct recursion agree to 10^{-15} (`rekursion_iir.py`). Log-periodicity, the $\xi \to \varphi$ passage and decompactification-as-runaway are thus pole computations, not a separate formalism.

W.8 Magnitude and phase, minimum phase and all-pass

The five levers share a common ground that repeats the θ work (Doc. 286): in the spectral representation, magnitude and phase separate. Everything FFGFT determines is magnitude content (ξ, hierarchy, $Q = 2/3$, the mean $\langle f \rangle$ of the fractal correction function); the one open value θ is pure phase. In signal terms FFGFT's real recursion is a minimum-phase / amplifier stage (the magnitude carries everything), and a phase stage $e^{i\varphi}$ is an all-pass (magnitude-preserving) – exactly HLV's skew-adjoint R (BD17A). The all-pass can be *appended* without touching the magnitude (and hence the masses and Q); this is why the phase is computationally separable and the open remainder.

W.9 What this means in practice

The gain is wherever one needs *functions of the operator* or *chains of couplings* – i.e. in the dynamic / kinetic sector (Doc. 286), at scale flow and time evolution. Instead of 3×3 matrix algebra one computes at three eigenvalues; instead of matrix multiplication three scalar products; instead of recursion loops one pole location. The radial invariants

(masses, Q) are Parseval one-liners. The toolbox is standard signal engineering applied to the Z_3 core.

W.10 Honest boundary

Important

This is computational leverage, *not* new physics (P35). The change of representation *moves* $\sqrt{2}$ and θ, it does not *create* them – both remain inputs. The full leverage is on the C_3/Z_3 sector (3×3 circulant); the full T^4 with the φ harmonics is larger, where the DFT diagonalisation is only the generation block. And no lever changes the status of the open phase: it only makes visible *why* it stands separately.

Balance: the representations compute too

The parallel representations are not only classification (Doc. 287) but computational tools. Because the mass operator is a Z_3-circulant, the DFT diagonalises it, and five standard signal-engineering levers become applicable: eigenvalues as DFT, invariants as Parseval, operator functions as scalar evaluation, chaining as multiplication, recursion as filter pole. All five checked to machine precision. The gain is concrete in the dynamic sector; the boundary is honest: a faster computational route, not an additional derivation. The signal-engineering view missing in Doc. 287 is thereby added as its own toolbox.

Chapter X

Doc. 289 — Magnitude/Phase Map of the Four Fermion Sectors

Abstract

Do the signal-engineering representations (Doc. 288) help to better capture quarks and neutrinos? The honest answer: *as language and diagnosis yes, as derivation no* – and very differently for the two sectors. Each fermion triplet gets a magnitude coordinate (Koide Q, i.e. $r = \sqrt{6Q - 2}$) and a phase coordinate. Computed: only the charged leptons sit at the special point $r = \sqrt{2}$ ($Q = 2/3$); the quarks lie elsewhere ($r \approx 1.54$ resp. 1.76, and scale-dependent), and for neutrinos Q cannot even be formed. The gain is a *unifying map*: leptons and quarks are magnitude-dominated (large hierarchy, small mixing), neutrinos phase-dominated (flat/degenerate magnitude, large mixing – PMNS/CKM weight 0.55 vs 0.035, factor 16). FFGFT's own neutrino hypothesis (equal masses + oscillation from a geometric phase) is then literally the *pure all-pass sector*: magnitude flat, all physics in the phase – and it becomes falsifiable, because the geometric phase must reproduce the measured Δm^2 phases.

X.1 The question

Doc. 288 showed that the Z_3-circulant mass operator gives concrete computational levers via the DFT. The natural question: do the same representations reach the hard sectors – quarks and neutrinos? An honest stock-taking is needed first, because the corpus does not treat these sectors like the clean charged leptons.

X.2 What the corpus already does (per sector)

- **Charged leptons** (Doc. 006/046/258): the clean case – Foot-Koide form, $Q = 2/3$, $r = \sqrt{2}$, Z_3-circulant. Everything from Doc. 288 applies directly.
- **Quarks** (Doc. 006/005): the same r_i/p_i geometry in principle, but the direct geometric method reaches only $\sim 1.2\%$; for quarks it needs the *extended fractal method* with QCD

dynamics and *ML calibration on lattice data* (FLAG 2024). Hence not purely geometric – and quark masses *run* (scale/scheme dependent).

- **Neutrinos** (Doc. 007/047): a *different* mechanism – photon analogy, $m_\nu = \frac{\xi_0^2}{2} m_e \approx 4.54$ meV, all three flavours (nearly) equal in mass, oscillation from *geometric phases* rather than mass differences. Flagged throughout the corpus as **speculative, without empirical basis**.

X.3 The (Q, r) coordinate: where the sectors sit

Via $Q = \frac{1}{3} + \frac{r^2}{6}$ (so $r = \sqrt{6Q - 2}$) every triplet gets a magnitude coordinate. Computed (`koide_q_sektoren.py`, PDG-near values):

Table X.1: Magnitude coordinate of the three formable triplets. Quark values are scale-dependent (MS-bar).

Sector	Koide Q	$r = \sqrt{6Q - 2}$	Position
charged leptons (e, μ, τ)	0.6667	$\sqrt{2} = 1.4142$	special
up quarks (u, c, t)	0.849	1.759	not special
down quarks (d, s, b)	0.730	1.543	not special
neutrinos	—	—	no triplet

This is already the first answer: *only the charged leptons sit at the special point* $r = \sqrt{2}$. The quark r are not recognisably special (and run with the scale); for neutrinos the absolute scale and ordering are missing, so Q cannot be formed – only $\Delta m^2_{21} \approx 7.5 \cdot 10^{-5}$ and $\Delta m^2_{31} \approx 2.5 \cdot 10^{-3}$ eV^2 are measured.

X.4 The magnitude/phase map

The signal-engineering decomposition (magnitude = mass spectrum, phase = mixing/oscillation) places the four sectors on *one* map:

- **Quarks**: strong mass hierarchy (large magnitude), small mixing (CKM near identity) – *magnitude-dominated*.
- **Charged leptons**: magnitude at a special value ($r = \sqrt{2}$), small phase ($\theta = 2/9$) – *magnitude-dominated*.
- **Neutrinos**: flat magnitude (degenerate), large mixing (PMNS far from identity) – *phase-dominated*.

The CKM/PMNS contrast can be quantified (`ckm_pmns_kontrast.py`, off-diagonal weight of the $|U|^2$ matrix): CKM 0.035 vs PMNS 0.554 – a factor ~ 16. The "small vs large mixing" is thus exactly a *magnitude vs phase* contrast.

X.5 Neutrinos as the all-pass sector

Here is the strongest point. FFGFT's own hypothesis – equal masses, oscillation from a geometric phase – is, in magnitude/phase language, literally an *all-pass*: $|H| = 1$ (flat magnitude spectrum), nontrivial phase. A flat magnitude spectrum cannot encode any oscillation; all physics sits in the phase (`neutrino_allpass.py`).

This makes the test *concrete and falsifiable*. The standard oscillation phase is $\Delta\phi_{ij} = 1.267\,\Delta m^2_{ij}\,L/E$; at T2K-like L/E ($L = 295$ km, $E = 0.6$ GeV) this is $\Delta\phi_{31} \approx 1.557$ rad and $\Delta\phi_{21} \approx 0.047$ rad. FFGFT's geometric phase (from $\tilde{T} \cdot m = 1$, quantum numbers n, ℓ, j) *must* reproduce exactly these L/E-dependent phases. If it does not, the degeneracy hypothesis is refuted.

X.6 What it achieves – and what it does not

Important

Achievement (diagnostic). The magnitude/phase decomposition explains the *division of labour* between the sectors: why the leptons are clean (magnitude at a special value, small phase), the quarks are not (magnitude not special, extreme hierarchy), the neutrinos quite different (magnitude flat, all phase). This is genuine organising knowledge – and it gives the speculative neutrino hypothesis a sharp, falsifiable form.
Boundary (P35). This is better language, a unifying map and a sharper test – *not* a new derivation of quark or neutrino masses. The quark r are not special and scale-dependent, and the quark sector in the corpus relies on QCD plus lattice calibration, not pure geometry. The neutrino sector remains speculative (own flag, Doc. 007) and in tension with the very well-tested Δm^2 interpretation of oscillation. The clean all-pass map is a compelling reformulation, not new evidence.

Balance: one map, no new sector

The signal-engineering representations capture quarks and neutrinos *better* – but in a precise sense: they give a common magnitude/phase coordinate in which only the charged leptons sit at the special point $r = \sqrt{2}$, the quarks magnitude-dominated beside them, and the neutrinos as a pure phase / all-pass sector. The CKM/PMNS contrast becomes magnitude vs phase (factor 16). And FFGFT's own neutrino hypothesis acquires a falsifiable form: the geometric phase must hit the measured Δm^2 oscillation phases. This is diagnosis and test, not a mass derivation – honestly bounded.

Chapter Y

Doc. 290 — Consequences for the Information Question: Magnitude, Phase and the Second Channel

Abstract

The magnitude/phase decomposition of the FFGFT mass operator (Doc. 288/289) reorganises the information question. Information has a *magnitude side* (mass/energy – Landauer and Vopson, derived by FFGFT from the T^4 geometry) and an independent *phase side* (holonomy, mixing – the one open datum θ). The cleanest witness is the **photon**: exactly massless – zero rest-mass magnitude, hence pure phase in the mass channel – and the quantum of the $U(1)$ gauge potential A_μ, which is the connection; FFGFT's information unit $E_{\text{bit}} = \hbar c/L$ *is* a photon energy. The *amount* of information is not representation-relative but absolute and dimensionless, anchored in ξ: $I_1 = \log_2(1/\xi) \approx 12.87$ bit per degree of freedom; energy and capacity are its dimensional, frequency-scaling dressings. The question is not "is information fundamental?" but **which channel** – and only the geometry holds both. A structural diagnosis, not a new derivation.

Y.1 Starting point: geometry first, and the magnitude/phase lens

FFGFT's stance on the information question is documented and geometry-first. Doc. 243 (discussion with Onur Teker, UIFT): the map of FFGFT into an information formalism is a *projection* π, not an isomorphism; the bit count $I = \log_2 |\mathscr{W}|$ *presupposes* the geometric state space. Doc. 255/257 derive Landauer's principle and Vopson's mass-information equivalence *from* the T^4 geometry: a bit is a minimal topological winding $\Delta w = 1$, "information is topologically frozen kinetic energy", $E_{\text{bit}}(L) = \hbar c/L = pc$. Doc. 274 places FFGFT *geometry-primitive* beside information-primitive (Vopson). FFGFT does not reject Vopson – it grounds him.

The new lens is the magnitude/phase decomposition of the mass operator (Doc. 288/289): the operator is a Z_3-circulant, DFT-diagonalised, and its content splits into a *magnitude spectrum* (masses, energy) and a *phase spectrum* (mixing, holonomy).

The θ finding (Doc. 282) says the magnitude spectrum fixes the phase spectrum only up to an all-pass factor (made precise in §2). That is the key to all that follows.

Y.2 Two channels: magnitude and phase

$\Delta w = 1$ and $E_{\text{bit}} = \hbar c/L$ are magnitude/energy statements. Since the magnitude spectrum fixes the phase spectrum only up to an all-pass factor (see below):

Vopson's mass-information equivalence – which FFGFT derives – structurally captures *only the magnitude channel*. The phase channel (holonomy, mixing, θ) is independent information, not reducible to mass/energy.

This shows concretely in two places. *First, mixing-blindness:* a mass operator $M = U\,\mathrm{diag}(m)\,U^\dagger$ carries the eigenvalues m (magnitude channel) and the eigenvectors U (mixing, phase channel); any function of the eigenvalues alone ($\sum m$, Koide Q) is invariant under U. Numerically (`masse_information_phasenblind.py`): for fixed masses $(0.5, 1.5, 3.0)$, $\sum m = 5$ and $Q = 0.372$ stay equal across three mixings while $\|M_{\text{offdiag}}\|$ (1.42 to 1.78) varies – two physically distinct operators with the same mass-information.

Second, the phase carries the structure: known in signal engineering since Oppenheim/Lim (1981) – swapping the magnitude and phase spectra of two patterns, the reconstruction follows the *phase* donor. Numerically (`phase_traegt_struktur.py`): phase-only correlates with the original ($+0.53$), magnitude-only does not (-0.24). A mass-energy accounting thus captures the *structurally poorer* channel.

Third, precisely: "the magnitude does not fix the phase" holds only *up to an all-pass factor*. Every stable transfer function factors as $H = H_{\text{min}} \cdot H_{\text{ap}}$: for the *minimum-phase* part the magnitude does fix the phase (Bode/Hilbert relation, $\arg = \mathscr{H}\{\log|H|\}$), whereas the *all-pass* ($|H_{\text{ap}}| = 1$) carries a phase the magnitude does *not* fix. Numerically (`minimalphase_allpass.py`): the phase reconstructed from the magnitude alone matches $\arg H_{\text{min}}$ (error $\sim 10^{-15}$), the residual is exactly the all-pass. Free – and thus the open part – is the *all-pass*, and that is where the θ holonomy sits (the minimum-phase (FFGFT) $\times$ all-pass (HLV-R) reading, Doc. 288).

Y.3 The photon: the exact anchor

The cleanest witness is the photon – and the corpus already holds the pieces. Doc. 007 treats neutrinos via the photon analogy: photon $E^2 = (pc)^2 + 0$ (exactly massless), neutrino $E^2 = (pc)^2 + (\sqrt{\xi^2/2}\,mc^2)^2$ (quasi-massless). The photon is the $m = 0$ limit, the neutrino "photon plus the first $\xi^2/2$ correction".

The photon has rest-mass magnitude exactly zero – the exact all-pass *in the mass channel*: all of its rest-mass content is phase. (Its energy $\hbar\omega$ remains a magnitude quantity; "pure phase" refers to the rest-mass, not the energy channel.) Three things coincide:

1. **Unfrozen information.** The photon is the unfrozen case of "information = frozen kinetic energy" ($E = pc$); FFGFT's unit $E_{\text{bit}} = \hbar c/L$ *is* a photon energy, Landauer's thermal bit its freezing.
2. **Photon and connection.** The gauge potential A_μ *is* the $U(1)$ connection; the photon is its quantum. *One* gauge-invariant observable is the holonomy ($\oint A\,\mathrm{d}l$, Aharonov-Bohm,

even where $F = 0$) – alongside energy $\hbar\omega$, momentum and polarization, which are not holonomies. The link to "it from connection" is *conceptual* (relational character), not a formal identity with θ.

3. **Counterexample to "mass = information".** Zero rest mass, yet carrier of essentially all our information transmission.

The neutrino is only the first correction to this: a "damped photon" with a tiny magnitude instead of exactly zero. In the degenerate limit $M = m\cdot\mathbb{1}$ for any U ($\|M - m\mathbb{1}\| \sim 10^{-16}$), the mixing would leave M, the oscillation content sitting in the propagation/holonomy phase – the same situation the photon realises *exactly* and without a degeneracy hypothesis.

Y.4 What "amount of information" means: packet, count, energy, capacity

Is a photon the smallest packet? Only *per mode*. A single-frequency photon is a plane wave; a localised photon is a wave packet. Crucially: a time-limited packet keeps the *definite* carrier f_0 (the laser colour, the energy jump); the limitation only produces a band $\Delta f \sim 1/\Delta t$ *around* f_0 (window effect), not a spectrum. The cycle count is $N = f_0\,\Delta t$: at fixed time window $\propto f_0$, but at fixed cycle count the fractional bandwidth $\Delta f/f_0 = 1/N$ is *frequency-independent*. For a photon from an energy jump, the lifetime τ is the window: the natural linewidth $\Delta f = f_0/Q$ with $Q = \omega_0\tau = 2\pi N$ ($\tau = 10$ ns, $f_0 = 5\cdot 10^{14}$ Hz $\Rightarrow N = 5\cdot 10^6$, $\Delta f \approx 16$ MHz; numerically `fenster_zyklen_linienbreite.py`). Definite carrier *plus* window gives the band.

Neither cycle count nor window length is intrinsic. The emission process sets both, not "photon-hood": a monochromatic mode has infinitely many cycles and an infinite window (plane wave); a spontaneously emitted photon inherits f_0 *and* τ from the transition ($N = f_0\tau = Q/2\pi$, the two coupled); a technical packet fixes one quantity, the other follows from $N = f_0\Delta t$. Intrinsic – scale-invariant – is the dimensionless number N ($\Delta f/f = 1/N$); $\Delta t = N/f$ is its dimensional dressing. The span is large: few-cycle pulse $N \approx 2$, narrow atomic line $N \approx 5\cdot 10^6$ (numerically `quant_bit_modeninfo.py`) – there is no universal cycle count and no universal window length.

Three quantities then separate cleanly (numerically `photon_paket_count_energie.py`):

- **Count**: the quantum number (occupation) is a *count* – one excitation, frequency-independent, relational –, *not* a bit measure. What a quantum carries as *information* is below.
- **Energy** per bit: $E = \hbar\omega = \hbar c/L$ – frequency-dependent (radio $\sim 2\cdot 10^{-10}$ eV to gamma $\sim 2\cdot 10^5$ eV).
- **Capacity**: Shannon $C = B\log_2(1+\mathrm{SNR})$, $B \propto f$ – frequency-dependent ($\sim 2\cdot 10^5$ to $\sim 2\cdot 10^{20}$ bit/s).

"Higher frequency → more information" is exactly the capacity: more bandwidth, more modes per second. That is a *different* axis from magnitude/phase: *within* a mode each symbol carries amplitude *and* phase (I/Q, QAM), *across* modes the bandwidth counts. FFGFT mirrors both: within the mode the Z_3-circulant (r,θ), across modes $|\mathscr{W}|$.

A quantum is a count, not a bit. "One quantum = one bit" is not a general equation – it confuses counting with measuring. A quantum is one *excitation* (a count of 1); its *information content* is $\log_2 M$ bit when it is chosen from M distinguishable modes – *variable*, capped per degree of freedom at $I_1 = \log_2(1/\xi) \approx 12.9$ bit (§5). "One quantum = one bit" holds only for $M = 2$ (the mode fixed to a binary alphabet) – a coding convention, not a law. Count and bit content are different axes: one quantum among 2^{13} modes carries 13 bit, and so do 13 binary quanta – same bit count, different quantum number (numerically `quant_bit_modeninfo.py`). The quantum *counts*; $\log_2 M \le I_1$ *measures*. This also places §5 cleanly: I_1 is not the content of *one* quantum but the absolute *ceiling* per degree of freedom, which the variable mode content $\log_2 M$ never exceeds.

Y.5 The absolute amount: anchored in ξ

An earlier reading – information content is representation-relative, there is no absolute bit count – was overstated and unfaithful to FFGFT. The absolute, frequency-independent content is the *dimensionless* quantity, and ξ anchors it. $\xi = 4/30000$ is dimensionless (the absolute parameter); $\hbar, c$ are SI bookkeeping.

ξ caps the winding number at $n_{\max} \sim 1/\xi$ (Doc. 243/009) – without this bound $I = \log_2 |\mathscr{W}|$ diverges. One winding coordinate can take $\sim 1/\xi$ distinguishable values: $I_1 = \log_2(1/\xi) = \log_2(7500) \approx 12.87$ bit. Total $I_{\text{total}} \sim N_{\max} \cdot \log_2(1/\xi)$ (cells $N_{\max} = V/L_0^d$, Doc. 251, times depth). Dimensionless, set by ξ – the energy $\hbar c/L$ is not the amount but its dimensional price, and the capacity saturates at the ξ scale.

Only *weakly* relative are the bit count of a *description* at a chosen resolution and what a lossy projection discards (Doc. 243) – a statement about approximations, not about the information itself. The full information (the geometry, or its lossless Hilbert image, Doc. 230) is absolute.

Absolute and relational are not in conflict. The two statements – the amount is *absolute* (§5), the open datum θ is *relational* (§8) – lie on different axes: "absolute" measures the *amount*, "relational" describes the *nature* of a datum. A holonomy (winding) is the paradigm of *both*: relationally defined (a loop integral, not a local point value) *and* absolutely valued (gauge-invariant, integer). Numerically (`absolut_relational.py`): a winding $w = 3$ is readable only as a loop integral (the half loop gives 1) and stays unchanged under a single-valued rephasing (gauge-invariant); and the absolute number $\log_2(1/\xi)$ *counts* exactly such relational objects (~ 15000 winding sectors up to $n_{\max} \sim 1/\xi$, $\log_2 \approx 13.9$ bit). The absoluteness sits in the ground (the geometry, ξ), the relationality in the degrees of freedom upon it – the absolute bit count counts relational windings.

Y.6 Static and dynamic definition of the bit

The definition so far is throughout *static*: the bit as structure – winding $\Delta w = 1$, counting ceiling $\log_2 M \le I_1$, energy $E_0 = \hbar c/L$ (magnitude side). That says *what* a bit is, not what it *does*. The dynamic definition is not an add-on but the conjugate side – and it itself splits into two, in which energy plays *different* roles.

Kinematic (reversible), the $\tilde{T}$ side: a flip takes $\tilde{T} = \hbar/E_0 = 1/m_0$, the rate is $1/\tilde{T} = E_0/\hbar$. Here $E_0 \cdot \tilde{T} = \hbar$ is nothing but $\tilde{T} \cdot m = 1$ – the time-mass duality, the FFGFT foundation. The quantum speed limit (Margolus-Levitin) $\tau \geq \pi\hbar/2E$ sharpens it: energy *sets the clock* without being dissipated (unitary). The cost is *time*. The capacity axis (bit/s, §4) is exactly this rate.

Thermodynamic (irreversible), the Landauer side: a bit carries entropy $k_B \ln 2$; *erasing* it (forgetting) into a bath at temperature T dissipates $k_B T \ln 2$ of heat. This depends on T, *not* on E_0, and the heat is *dissipated* – the cost is *heat*, not time. In FFGFT this side is *emergent*: the fundamental T^4 dynamics is unitary-reversible, so the kinematic ($\tilde{T}$) face is primary and the Landauer price arises only through coarse-graining, a bath, and logical irreversibility.

The dimensionless count $\log_2 M$ is primary; the constants dress it (numerically `bit_statisch_dynamisch.py`): $\hbar, c$ into energy E_0 and time $\tilde{T}$ (kinematic, reversible: *time* cost, $\tilde{T} \cdot m = 1$), k_B into entropy $k_B \ln 2$ and erase heat $k_B T \ln 2$ (thermodynamic, irreversible: *heat* cost). $\hbar, c, k_B$ are SI bookkeeping. Energy plays two roles: *clock* (kinematic, no loss) versus *dissipated heat* (thermodynamic).

Important: "bookkeeping" does not mean *dispensable*. $k_B, \hbar, c$ are the *necessary* unit bridges. SI separates temperature (K), energy (J), time (s) into their own units, and as soon as one *computes* thermodynamically – and that happens in SI – k_B is the indispensable conversion K→J (otherwise the erase heat would stand in kelvin rather than joule), just as $\hbar, c$ carry the conversion to $E_0 = \hbar c/L$. In natural units ($k_B = \hbar = c = 1$) the costs are pure numbers ($\ln 2$); SI introduces the unit walls, and the bookkeeping is their mortar – ontologically secondary, operationally indispensable (numerically `buchhaltung_si_natuerlich.py`).

The bit in SI numbers. Anchored at the framework's characteristic energy $E_0 = \sqrt{m_e m_\mu} = 7.348$ MeV (coupled to the fine-structure constant via $\alpha = \xi E_0^2/K_{\text{frak}}$; the variant $\sqrt{\alpha/\xi} = 7.398$ MeV is $+0.68\%$ higher – up to $7 \cdot 10^{-5}$ the factor $1/\sqrt{K_{\text{frak}}} = 1.0067$, R72). The generic $E_0 = \hbar c/L$ coincides with this E_0 exactly at $L = L_0 = \hbar c/E_0$. Then (numerically `bit_in_si.py`): $E_0 = 1.177 \cdot 10^{-12}$ J, $L_0 = 2.686 \cdot 10^{-14}$ m $= 26.85$ fm, $m_0 = 1.310 \cdot 10^{-29}$ kg, $\tilde{T} = \hbar/E_0 = 8.958 \cdot 10^{-23}$ s $= L_0/c$ and $1/\tilde{T} = 1.116 \cdot 10^{22}$ s^{-1}.

Y.7 The temperature of the bit: scale, state, bath

The energy E_0 converts directly into a *temperature*. From $E_0 = k_B T_0$,

$$T_0 = \frac{E_0}{k_B} = \frac{1.177 \cdot 10^{-12}\,\text{J}}{1.381 \cdot 10^{-23}\,\text{J/K}} = 8.53 \cdot 10^{10}\,\text{K}$$

(check via 1 eV$= 11\,604.5$ K: $7.348 \cdot 10^6$ eV $\cdot\, 11\,604.5 = 8.53 \cdot 10^{10}$ K), about 85 billion kelvin. T_0 is the temperature at which the thermal energy $k_B T_0$ reaches the *structure energy* E_0 of the bit.

Does a bit even have a temperature? Strictly: no, not *one*. A single bit in a definite state (0 or 1) has zero entropy and no distribution – temperature is an ensemble/reservoir

quantity, not assigned to a single fixed degree of freedom. What one *can* sensibly assign to a bit are *three distinct* temperature-like quantities that must not be confused. First, the *state* or occupation temperature: if the bit has two levels split by E_0, the occupation ratio $p_1/p_0 = e^{-E_0/k_BT}$ defines an effective temperature. It runs *against* intuition: maximally mixed ($p_0 = p_1 = \frac{1}{2}$, a full one bit of entropy) means $T_{\text{eff}} \to \infty$, definite ($p \to 0$ or 1, zero entropy) means $T_{\text{eff}} \to 0$, population inversion even $T_{\text{eff}} < 0$. The *most informative* bit is thus the infinite-temperature state; cooling drives it to a definite state and *destroys* the information. (A purely logical, degenerate bit – both values same energy – has no occupation temperature at all.) Second, the *scale* $T_0 = E_0/k_B$: not "where the bit sits", but the characteristic melting/activation point from the energy gap, the natural temperature *unit* of the bit. Third, the *bath* temperature that enters the erase cost – and that belongs to the *environment*, not the bit.

The scale T_0 marks the thermal robustness threshold: above it ($k_BT > E_0$) the thermal energy overwhelms the structure, and the heat washes out the distinction – the winding or mode that the bit *is*: the bit "melts". Below it ($k_BT < E_0$) it is robust. This is a statement about the *scale*, not about a state.

Which temperature when erasing? The Landauer cost $Q = k_BT\ln 2$ depends on the temperature of the *bath* into which the erased entropy flows as heat – on the choice of environment, not on the bit. There is no universal "correct" value. For a real computer it is the operating temperature, in practice *room temperature* ($\sim 300\,\text{K}$): $Q = k_B \cdot 300 \cdot \ln 2 = 18\,\text{meV}$. That is the honest answer for real computing. UIFT instead uses $T_{\text{CMB}} = 2.7\,\text{K}$ – not absolute zero, but the *coldest natural heat sink*; this gives the minimum possible erase heat in the present universe ($Q = 0.16\,\text{meV}$, nothing colder without active cooling). Legitimate as a cosmological floor, but still just *one* bath choice, not a bit property. T_0 finally is *not* a bath: inserting it gives the limiting case "bath as hot as the bit itself", $Q = E_0 \ln 2 = 0.69\,E_0$ – a reference, not an erasure.

"The" temperature of a bit therefore depends on its *scale*: a generic bit of length L carries energy $E = \hbar c/L$ and hence scale $T = \hbar c/(k_BL)$. Each bit scale has its own; $8.5 \cdot 10^{10}\,\text{K}$ is the one belonging to the characteristic E_0. A larger L – a "softer" bit – has a lower threshold.

So T_0 stands alongside mass and time as a third unit clothing of the *same* energy. With one bookkeeping constant each:

$$m_0c^2 \;=\; E_0 \;=\; k_BT_0 \;=\; \hbar/\tilde{T},$$

i.e. $/c^2 \to$ mass $m_0 = 1.310 \cdot 10^{-29}\,\text{kg}$, $/\hbar \to$ time $\tilde{T} = 8.958 \cdot 10^{-23}\,\text{s}$ (the dual time from $\tilde{T} \cdot m = 1$), $/k_B \to$ temperature $T_0 = 8.53 \cdot 10^{10}\,\text{K}$. Beneath all four sits the dimensionless count $\log_2 M$. One energy, four clothings.

The temperature is not merely "also there" but tied directly to the dynamic dual time. The *thermal time* at T_0 is $\hbar/(k_BT_0)$; inserting $T_0 = E_0/k_B$ gives $\hbar/E_0 = \tilde{T}$ – the thermal time at T_0 *is* the dual time from $\tilde{T} \cdot m = 1$ (numerically identical to $10^{-38}\,\text{s}$, `bit_temperatur.py`). This is the KMS / "thermal time" correspondence (Connes–Rovelli): temperature is an inverse (imaginary) time, and it lands without extra assumption exactly on the FFGFT dual time. On the bookkeeping axis k_B is for temperature exactly what $\hbar$ is for time.

Finally, T_0 joins the two "energies" of a bit – what it *is* (E_0) and what its *erasure* costs ($k_B T \ln 2$) – into *one* scale. The Landauer erase heat relative to the self-energy is simply the temperature ratio,

$$\frac{Q}{E_0} = \frac{k_B T \ln 2}{k_B T_0} = \frac{T}{T_0} \ln 2.$$

At room temperature: $\frac{300}{8.53 \cdot 10^{10}} \ln 2 = 2.4 \cdot 10^{-9}$, hence $E_0/Q = (T_0/T)/\ln 2 \approx 4 \cdot 10^8$ – room temperature sits about 8.5 decades below T_0, and *that* is why the static self-energy (~ 7.3 MeV) and the erase heat (~ 18 meV) lie a good eight orders of magnitude apart. At $T = T_0$, by contrast, erasure costs $E_0 \ln 2 = 0.69\, E_0$: at its own temperature point, *being* and *erasing* are the same order of magnitude. This ratio T/T_0 is the *only* setup-independent statement: a bit has no absolute temperature, only its bath temperature relative to its own scale. The static (E_0, T_0) and thermodynamic ($k_B T \ln 2$) sides are thus joined by the single ratio – no jump, one scale.

Placement: $T_0 \approx 8.5 \cdot 10^{10}$ K is the MeV/nuclear scale – the temperature order of nuclear and hadronic physics: stellar cores (solar core $\sim 1.5 \cdot 10^7$ K) and the quark-gluon plasma in heavy-ion collisions (QCD transition $\sim 1.7 \cdot 10^{12}$ K at 150 MeV). The "hot early universe" reading is deliberately left out – it is a ΛCDM assumption that FFGFT does not share (FFGFT's CMB structure follows from the T^4 geometry, not from a hot early phase). Honestly: $T_0 = E_0/k_B$ is the *standard* energy-temperature assignment; what is FFGFT-specific is only that it coincides with the dual time via $\tilde{T} \cdot m = 1$ – a clean consistency, not a new derivation. And because the absolute erase temperature is the bath's, a ξ tied to a particular bath (T_{CMB} or room temperature) would depend on that choice – exactly why the CMB-bound ξ_{UIFT} is a contingent, different quantity while FFGFT's geometric ξ stays bath-*independent* (§9).

E_0 is a rung, not the floor. The scale E_0 is not the bottom end of the framework but a *rung* of a fractal ladder. The geometric floor lies at the sub-Planck length $\xi \ell_P \approx 2.2 \cdot 10^{-39}$ m (Doc. 163) – the smallest length below which spacetime carries no well-defined degrees of freedom. Because $\xi \ell_P$ is a factor ξ *below* the Planck length, this floor stands, in the "hard" clothings, a factor $1/\xi = 7500$ *above* Planck:

Length	$\xi \ell_P$	$2.16 \cdot 10^{-39}$ m
Time/duration	ξt_P	$7.19 \cdot 10^{-48}$ s
Frequency	$1/(\xi t_P)$	$1.39 \cdot 10^{47}$ Hz
Energy	E_P/ξ	$9.16 \cdot 10^{22}$ GeV
Temperature	T_P/ξ	$1.06 \cdot 10^{36}$ K
Mass	m_P/ξ	$1.63 \cdot 10^{-4}$ kg

This floor is the *smallest-length* edge, hence in energy/temperature/frequency the *maximum* (UV): "floor" means finest resolution, not smallest energy. The bit rung E_0 (7.348 MeV, 26.85 fm, $T_0 = 8.5 \cdot 10^{10}$ K) sits about 25 decades *coarser*. Between them the fractal runs inward, factor $1/\xi = 7500$ per main level (Doc. 146). Counting the sub-steps as factors of 2, there are $\log_2 7500 \approx 12.87$ per level – *exactly* the information ceiling I_1 per degree of freedom from §5. (In steps of $\sqrt{2}$ each there would be $\log_{\sqrt{2}} 7500 = 25.7$, since $(\sqrt{2})^{13} = 90.5$; equating the step count with I_1 holds only for factor-2 steps.) The number of factor-2 sub-steps per level *is* the maximum bit count; the capacity of the bit and the fineness of the fractal are the same number.

Landauer. Erasing one bit is logically irreversible: two states are mapped to one, the entropy drops by $k_B \ln 2$. By the second law this entropy must flow to the environment as heat, $Q \geq k_B T \ln 2$ (Landauer 1961). Here T is the temperature of the *bath* into which the erasure goes – not the bit's.

The problem. And here it jams. The framework now supplies a whole ladder of *definite* temperatures – the rung $T_0 = 8.5 \cdot 10^{10}$ K, the floor $T_{\max} = 1.1 \cdot 10^{36}$ K, each fractal level its own in between, over 25 decades. *None* of them is "the" Landauer temperature. Insert an intrinsic scale into $Q = k_B T \ln 2$ and you get no real erasure but a limiting case or nonsense: with T_0 the reference $Q = 0.69\, E_0$, with $T_{\max}$ an absurdly large heat, with T_{CMB} the UIFT number (§9). The real erasure takes the temperature of the *actual environment* (computer: ~ 300 K). That is the problem: despite a ladder of definite intrinsic temperatures, the erase temperature remains the bath's, and the only invariant is the *ratio* $Q/E = (T/T_{\mathrm{rung}}) \ln 2$ – bath relative to the respective rung. A bit has a definite energy and, per rung, a definite scale temperature; a definite *erase price* it acquires only with an environment.

Y.8 Consequences

The faithful description is quantum, not bit-based. Because the Hilbert space is a bijection (Doc. 230) but the classical bit count a projection: FFGFT's lossless description is the amplitude-plus-phase representation (quantum information); the Shannon bit count is its lossy shadow. An information primitive needs magnitude *and* phase.

The open datum is relational information. The one irreducible open quantity (Doc. 282) is the holonomy phase θ – path-dependent, relational, not a local bit content. The residue the bit projection discards *is* the phase; and the phase *is* where the open physics lives. The fundamental information sits in the connection: the electromagnetic field A_μ is that connection, the photon its quantum – a conceptual example, not one numerically identical to θ. "It from bit" becomes "it from connection".

Y.9 Honest boundaries

Structural diagnosis, not a new derivation (P35). Doc. 255/257 already had Landauer/Vopson; new is the magnitude/phase decomposition and its consequences. The *one exact* point: $E_{\mathrm{bit}} = \hbar c / L$ is a photon energy.

Photon is a gauge boson, neutrino a fermion. The photon analogy holds at the level of masslessness/dispersion, not as an identity (spin 1 vs. $\frac{1}{2}$); the link to the connection belongs to the gauge field A_μ (the photon being its quantum), only in the photon sector. The photon's phase (polarization/gauge holonomy) is the same *category* as θ, not the same number. The photon sits outside the lepton mass operator ($m = 0$ reference).

$\log_2(1/\xi) \approx 12.87$ **is not a magic number.** The content is finiteness and ξ anchoring, not the value; the exact form ($1/\xi$ vs. $2/\xi$, direction/flavour factor) depends on the mode counting. Direction: geometry → information, not the reverse.

Energy side: fixed at the anchor E_0. E_0 is the anchor that provides the connection to SI; the agreement of $\alpha = \xi E_0^2$ with the measured value is a property of this anchor,

not an independent prediction. It exists bare, $\sqrt{m_e m_\mu} = 7.348$ MeV, and fractally corrected, $\sqrt{m_e m_\mu / K_{\text{frak}}} = 7.398$ MeV (Doc. 011). The SI numbers use the bare value – from it $m_0, T_0, \tilde{T}, 1/\tilde{T}$ follow exactly ($\hbar, c, k_B$ are mere bookkeeping, no deviation). The only difference between the two versions is the factor $K_{\text{frak}}^{-1/2}$ ($+0.68\%$), not a conversion.

Bit thermodynamics: computed, over the bit's own scale T_0. The bit has the full Landauer computation – erase cost $Q = k_B T \ln 2$, relative to the self-energy $Q/E_0 = (T/T_0)\ln 2$ (§7); at $T = T_0$, $Q = E_0 \ln 2 = 0.69\,E_0$, the intrinsic quantity is the bit entropy $\ln 2$. No CMB needed. The UIFT parameter $\xi_{\text{UIFT}} = k_B T_{\text{CMB}} \ln 2/(m_e c^2) \approx 3 \cdot 10^{-10}$ (Doc. 013/243) is the same Landauer form with foreign cosmological scales (T_{CMB} instead of T_0, m_e instead of E_0) – a different quantity, not FFGFT's geometric $\xi = 1.3 \cdot 10^{-4}$. A different scale choice, not a failure.

Neutrino argument: speculative. It still rests on the speculative degeneracy (Doc. 007) – but it no longer carries the thesis; the photon does.

Y.10 Balance: not whether, but which channel

The magnitude/phase work shifts the information question from "is information fundamental?" to "which channel?". Vopson/Landauer are the magnitude channel – real, derived from the T^4 geometry, but blind to the phase. The phase channel (mixing, holonomy, θ) is independent, relational information; in signal processing it is the one that carries the structure. The cleanest anchor is the photon: exactly massless, FFGFT's information unit $E_{\text{bit}} = \hbar c/L$ itself, and the quantum of the $U(1)$ connection – conceptually "it from connection". The *amount* of information is absolute and anchored in ξ ($I_1 = \log_2(1/\xi) \approx 12.87$ bit); energy and capacity are its dimensional dressings. Only the geometry holds both channels – honestly bounded as diagnosis, not a new derivation.

Chapter Z

Doc. 291 — The dynamical locus of $\theta = 2/9$: magnitude, winding, phase

Z.1 Setting: which channel does $\theta = 2/9$ live in?

The Koide angle $\theta = 2/9$ is confirmed empirically to seven figures but is not derived from the geometry. It is itself not an ansatz but a *found* quantity: only the Hilbert-space translation of FFGFT (Docs. 230/231/232) and its continuation to the masses (Doc. 282) made visible that the three lepton masses are the spectrum of a hermitian Z_3 circulant on the $\mathbb{C}^3$ fibre, in whose eigenvalues $\theta = 2/9$ appears as the result of the diagonalisation. It is precisely this circulant that supplies the amplitudes a_k below. For the empirical placement of the angle – $2/9$ as the rational address of a value fixed via one reference point, not as a pole-mass exactness claim (P42) – see Doc. 292; for the dynamical locus sought here this fine distinction is immaterial, since the orbit resolution lies far above the 10^{-7} level. The preceding analysis (Doc. 290) separates, at the mass operator, a *magnitude channel* (the eigenvalues, the masses) from a *phase channel* (mixing, all-pass, holonomy), and shows that a magnitude spectrum fixes the phase only up to an all-pass factor. The question here is accordingly: in which channel does $\theta = 2/9$ sit – and what kind of quantity is it, compared with a forced anchor such as the speed of light c? The supporting computations are in `allpass_theta.py`, `allpass_carrier_selection.py` and `forced_vs_contingent.py` (numpy-only, seed 20780458).

Z.2 The magnitude side is counter-blind

With the signed amplitudes $a_k(\theta) = 1 + \sqrt{2}\cos(\theta + 2\pi k/3)$ ($k = 0, 1, 2$), the symmetric Koide invariant $Q = (\sum_k a_k^2)/(\sum_k a_k)^2$ is exact and *θ-independent*. The sum $S = \sum_k a_k = 3 + \sqrt{2}\sum_k \cos(\theta + 2\pi k/3) = 3$ (three cosines spaced by 120° sum to zero), and the square sum $T = \sum_k a_k^2 = 3 + 0 + 2 \cdot \frac{3}{2} = 6$ (with $\sum_k \cos^2 = \frac{3}{2}$). Hence

$$Q(\theta) = \frac{T}{S^2} = \frac{6}{9} = \frac{2}{3} \qquad \text{for every } \theta.$$

Q ranges over $[1/3, 1]$ for positive masses; the value $2/9 = 0.2222$ lies *below* that range. So $2/9$ is no boundary, no extremum, and no fixed point of the symmetric functional

– in the magnitude channel $2/9$ is entirely invisible. This is the algebraic side of the counter-blindness of the Z_3 core: the orbit members $\{1/9, 2/9, 4/9\}$ are indistinguishable to any symmetric, counting quantity.

Z.3 The all-pass: three channels

The locus of θ can be made visible at the all-pass. A Z_3-symmetric Blaschke product $B_\theta(z) = \prod_k (z^{-1} - \bar{a}_k)/(1 - a_k z^{-1})$ with poles $a_k = r e^{i(\theta + 2\pi k/3)}$ (here $r = 0.5$) splits the phase information into three quantities, two of which are blind to θ and only one of which carries it. Numerically (for the orbit angles $\theta \in \{1/9, 2/9, 4/9\}$):

- **Magnitude** $|B_\theta| = 1$ (to $\sim 10^{-15}$) for all θ – the magnitude channel is blind.
- **Winding number** $= 3$ exactly for all θ – the *flat-topological* integer is blind. This is the numerical face of the statement that $2/(9\pi)$ is irrational and $\theta = 2/9$ is therefore not a flat topological invariant: the integer winding does not separate $2/9$ from $1/9, 4/9$.
- **Continuous phase profile** $\arg B_\theta(\omega)$ – different for $1/9, 2/9, 4/9$ (at the reference point $\omega = 0$: 0.093, 0.171, 0.249; the group delay separates them too). Only the *continuous, dynamical* phase profile carries θ.

An energy/magnitude functional such as $\int |B|^2 \, d\omega/2\pi$ is likewise identical for all three. The distinction of $2/9$ from $1/9, 4/9$ thus lives neither in the magnitude, nor in the integer winding, nor in any symmetric functional – only in the dynamical phase profile.

Z.4 An illustrative selecting mechanism

What a selector would have to do is shown by a dynamical phase reference. The three orbit members have all-pass phases $\psi = (0.093,\ 0.171,\ 0.249)$. A carrier that accumulates a holonomy Φ and phase-locks to the nearest orbit phase selects exactly one representative: $2/9$ is selected precisely when Φ lies in the $2/9$ basin $[0.132,\ 0.210]$ (width 0.078). A *random* holonomy over the orbit range selects $2/9$ with probability $\approx 1/3$ – nothing prefers the value $2/9$ without a reason.

The form of this mechanism is that of the open problem: the selection is *contingent* on an external dynamical phase (a holonomy); $2/9$ has its own basin like any member, and neither magnitude nor winding privileges it. A dynamical selector would have to supply that holonomy *forced*, from an internal reason of the supplying dynamics, not by a hand-chosen relabeling. This is the same content as the breaking condition $(\delta, \chi) \approx (0.24, \pi/2)$: χ *is* the holonomy phase, δ its free depth. (An illustration of the mechanism shape; not a derivation of the value $2/9$, and not a model of any specific carrier.)

Z.5 Forced or contingent: $\theta^*(\delta)$ and the photon contrast

Whether $2/9$ is *forced* like c is decided by the behaviour under variation. The $\delta = 0$ part of the breaking functional has the closed form (confirmed to 10^{-15})

$$E_3(\theta, 0) = -\tfrac{1}{2} + \tfrac{1}{\sqrt{2}} \cos 3\theta, \qquad \left.\frac{dE_3}{d\theta}\right|_0 = -\frac{3}{\sqrt{2}} \sin 3\theta,$$

with extrema at the Z_3 points ($\sin 3\theta = 0$). Coupling the second triplet $3'$ at strength δ and phase χ, the inner extrema move away from the Z_3 point (there are two; the lower, orbit-near branch θ^* is the one tracked). On the branch $\chi = \pi/2$ the shift is smooth and monotonic (numerical, `forced_vs_contingent.py`): θ^* rises from 0.099 ($\delta = 0.10$) through 0.223 ($\delta = 0.24$) to 0.501 ($\delta = 1.0$); $d\theta^*/d\delta$ is positive throughout (median over the δ grid 0.31, locally at $\delta^* \approx 0.24$ about 0.80). The reachable range $[0.05,\ 0.52]$ contains $2/9$ in its interior, unremarkable. The $2/9$ basin in θ space – nearest orbit member, $[1/6,\ 1/3]$, width $1/6$ – corresponds via the local slope to a δ-window of $\Delta\delta \approx 0.167/0.80 \approx 0.21$. δ is soft, not fine-tuned; $2/9$ is dragged in, it sits at no structure.

The speed of light is of a different type. In $ds^2 = -c^2dt^2 + dx^2 + dy^2 + dz^2$, c is a metric coefficient, not a parameter of a functional; $ds^2 = 0$ is the light cone, and massless means $v = c$ exactly – the saturation of the inequality $v \leq c$. There is no $c^*(\lambda)$; $dc/d\lambda \equiv 0$, because c is a constant of the geometry, not an extremum. This reflects the physical origin: the photon's masslessness is *gauge-symmetry protected* ($m = 0$ exactly, hence $v = c$), a *boundary*/fixed-point value. $2/9$, by contrast, is the irreducibly Z_3-*breaking* angle in the *interior* of the orbit – symmetry-breaking, not protected, and therefore contingent.

Z.6 Self-adjoint and skew-adjoint holonomies

If the selector is a holonomy phase χ, two sectors must be distinguished. In the breaking term $\delta\cos(2\varphi_k + \chi)$ ($\varphi_k = \theta + 2\pi k/3$), χ separates an *even* from an *odd* coupling: $\chi = 0$ (and $\chi = \pi$) gives $\cos(2\varphi_k)$, an even, *self-adjoint*, real coupling; $\chi = \pi/2$ gives $\cos(2\varphi_k + \pi/2) = -\sin(2\varphi_k)$, an odd, *skew-adjoint*, imaginary coupling.

The two sectors have different natural values. The *self-adjoint* holonomies – flat Z_3 Wilson lines ($\chi = 2\pi n/3$) times the two spin structures per cycle (offset 0 or π) – are the multiples of $\pi/3$, $\chi \in \{0, \pi/3, 2\pi/3, \pi, 4\pi/3, 5\pi/3\}$; $\pi/2$ is none of them. The *skew-adjoint* sector, by contrast, has its natural value at $\chi = \pi/2$: the i-phase, $e^{i\pi/2} = i = \sqrt{-1}$, against the real $e^{i\pi} = -1$. The transition from a real to a skew-adjoint coupling *halves* the phase $\pi \to \pi/2$ – it takes the square root of -1.

What matters is which sector supplies the selector. The elimination chain (Doc. 282) has ruled out every static, self-adjoint candidate; the only survivor is the dynamical *skew-adjoint* phase. The type-consistent value is therefore $\chi = \pi/2$ – the i-phase of the skew-adjoint sector – not a multiple of $\pi/3$. And $\chi = \pi/2$ reaches $\theta^* = 2/9$ at $\delta \approx 0.24$ (numerically, `holonomie_landschaft.py`). The skew-adjoint type by itself admits both i-values, $\chi = \pi/2$ and $\chi = 3\pi/2$ ($e^{\pm i\pi/2} = \pm i$); the choice of branch is an orientation choice (sense of traversal of the cycle). The two branches are, however, not equivalent: the mirror branch $\chi = 3\pi/2$ has no inner extremum near $2/9$ at $\delta \approx 0.24$ (extrema at 0.83 and 1.87) – only the $+i$ branch reaches the orbit region.

$\chi = \pi/2$ is thus not an arbitrary or topologically homeless value, but the skew-adjoint i-value required by the elimination chain – with no assumed Z_4 and no numerical coincidence. Precisely, "skew-adjoint" here refers to the *generator*, not the finite holonomy: the latter is a unitary group element $\exp(K)$ with skew-adjoint K ($K^\dagger = -K$); the skew-adjoint character sits in the generating coupling, not in the finite holonomy matrix itself. The depth δ nonetheless remains *free*: the sector fixes the phase χ, not the depth. The open point is thereby sharpened – not an arbitrary angle, but whether a skew-adjoint dynamics

carries precisely $\chi = \pi/2$ and thereby fixes the depth δ for an internal reason *of that dynamics* (recursion-internally, by Doc. 275, no value can be distinguished).

Z.7 The depth δ as the second harmonic

The mass amplitude is a Fourier series in the fibre angle,

$$a(\varphi) = \underbrace{1}_{\text{mean}} + \underbrace{\sqrt{2}\cos\varphi}_{\text{1st harmonic}} + \underbrace{\delta\cos(2\varphi+\chi)}_{\text{2nd harmonic}}, \qquad \varphi = \theta + 2\pi k/3.$$

The first harmonic, with real amplitude $c_1 = \sqrt{2}$, fixes the Koide invariant $Q = 2/3$ (it is the magnitude $r = \sqrt{2}$). The second harmonic has the complex amplitude $c_2 = \delta e^{i\chi}$; its *magnitude* is $\delta = |c_2|$ and its *phase* is $\chi = \arg c_2$. For $\chi = \pi/2$ the coefficient $c_2 = i\delta$ is purely imaginary – the second harmonic is a pure sine, the skew-adjoint coupling of the previous section.

So δ and χ are the magnitude and phase of the *same* quantity c_2, but they belong to different sectors: $\chi = \arg c_2$ lives in the phase sector (fixed to $\pi/2$ by the elimination chain), $\delta = |c_2|$ in the *magnitude* sector. The depth δ is thus not a phase but an amplitude degree of freedom – the amplitude of the first overtone, physically the strength with which the second triplet $3'$ (the doubling $6 = 3 \oplus 3'$) mixes into the primary 3.

This is the same sector that already fixes $|c_1| = \sqrt{2}$. As the first harmonic amplitude follows from $Q = 2/3$, the second, $|c_2| = \delta$, would have to follow from the overtone content of the mass-generating fractal waveform (the structure $D_f = 3 - \xi$, the $3 \oplus 3'$ doubling). The ratio $\delta/\sqrt{2} \approx 0.17$ – the second harmonic is about $17\,\%$ of the first – is a pure *shape* quantity, fixed by the waveform, not by the phase. δ is therefore not a free tuning parameter but a not-yet-computed magnitude degree of freedom.

The question of $2/9$ thereby separates cleanly into two sectors: the *phase* $\chi = \pi/2$ is skew-adjoint and fixed; the *magnitude* δ is the second harmonic amplitude and belongs – open – to the same magnitude sector as $\sqrt{2}$.

Z.8 δ as a golden-damped amplitude

The overtone content can be sharpened: the second harmonic amplitude is a *golden damping*. The value that gives $\theta^* = 2/9$ at $\chi = \pi/2$ is $\delta^* = 0.2389$; the golden attractor $1/\varphi^3 = 0.2361$ sits just below it. Since $1/\varphi^3 = \varphi^{-3}$ is a *negative* golden power (a decay), δ is a damping, not an amplification – $\delta^* < 1$ is unambiguously in the damping regime.

The exponent 3 is not free but $= D$. A single, decompactified $3'$ recursion would give $1/\varphi^1 = 0.618$ ($159\,\%$ off); only damping over $D = 3$ dimensions (the fractal/spatial dimension $D_f = 3 - \xi$, the *full* compact T^4) gives $1/\varphi^3$. The matching exponent 3 is thus itself an argument for the full compact T^4 and against decompactifying to a single recursion.

The ξ-damping recursion $r(k{+}1) = r(k)(1-\xi)$, $r(0) = D_f/3$ (Doc. 275) passes $1/\varphi$ at depth $k^* = \ln\varphi/\xi \approx 3609$ and $1/\varphi^3$ at $3 \cdot 3609 \approx 10827$ (to $0.011\,\%$). δ^* is reached about 89 *units* earlier – meaning a difference of two real-valued log-depths in the decompactified, continuous scale flow ($1/\varphi^3$ at 10826.2, δ^* at 10737.7; difference 88.5), not an integer step count. So δ^* is a $1/\varphi^3$ under-damped by about 89 log-units, an excess of $+1.187\,\%$ above the attractor

– both statements are, via $\ln(\delta^*\varphi^3)/\xi$, one and the same number, not two independent findings –, consistent with an approach from above.

Whether this excess is the fractal mass correction $K_{\text{frak}} = 1 - 100\xi$ (Doc. 133) can be checked on the number. Its precise form $(D_f^{\text{eff}}/3)^{D_f^{\text{eff}}/2}$ with $D_f^{\text{eff}} \approx 2.973$ gives a factor ≈ 100 ($1.333\,\%$), while the measured excess is $\approx 89\,\xi$ ($1.187\,\%$) – a gap of $0.147\,\%$. Because δ^* is tied through $\theta^* = 2/9$ to the τ-limited angle determination, it itself carries a tolerance. What matters for it is the local slope at $\delta^* \approx 0.24$, $\mathrm{d}\theta^*/\mathrm{d}\delta \approx 0.80$ (not the median 0.31 over the δ grid); with it the δ tolerance is $\approx 0.052\,\%$, and the gap of $0.147\,\%$ corresponds to $\approx 2.8\sigma$. The two candidates 89ξ and 100ξ are therefore *distinguishable*: the excess read off from δ^* does not coincide with $K_{\text{frak}} = 1 - 100\xi$. Structurally the link remains motivated – the second harmonic and K_{frak} spring from the *same* fractal correction $g = \eta(1+\xi f)$ (Doc. 286; here f is the fractal correction function of the metric, not a frequency and not the base winding number $1/\xi$), of which K_{frak} is the RG level (Doc. 133). The form of the fractal correction factor suggested by the excess is thus open; the value $1 - 100\xi$ itself is not affected.

Two limits come on top. First, the shared $g = \eta(1 + \xi f)$ origin is shown in Doc. 286 for the *phase* channel (holonomy); the transfer to the *magnitude* of the second harmonic (δ) is plausible but not yet computed. Second, Doc. 275 does not fix the recursion depth in any case: there $1/\varphi^3$ is a *transit point*, not a fixed point (the only fixed point of the damping is 0), and for *every* target $c \in (0, 1)$ a depth $k^*(c)$ exists with identical parameter status. That δ^* lies near $1/\varphi^3$ is therefore *not* a closing of the recursion; the fine value is not derived.

So the magnitude sector is clarified as far as the structure carries: δ is the golden-damped second harmonic, $1/\varphi^3$ over $D = 3$ (full T^4), with a small excess from above. Form, order, exponent and direction are fixed; the exact δ fine value – the excess over $1/\varphi^3$ – is distinguishable from $K_{\text{frak}} = 1 - 100\xi$ at the local slope ($\approx 2.8\sigma$), structurally motivated via the shared $g = \eta(1 + \xi f)$ origin (Doc. 286), but not derived as a magnitude value and remains open (`delta_golden_daempfung.py`).

Z.9 What the non-closure means in practice: an open coupling channel

The picture so far separates two sectors with opposite behaviour. The mass sector is *closed*: the recursion $R(\Phi^*) = \Phi^*$ (Doc. 203) has a non-trivial fixed point, and because the lepton masses sit on it, they and the Koide value $Q = 2/3$ are exact and rigid.[1] There is nothing to adjust there – no free parameter, no point of entry for an external influence.

The room sits solely in the open sector, the phase. The ξ-damping recursion never closes (transit point, only fixed point 0, Doc. 275); in feedback language it is an open, damped loop with gain below one, not a rigidly locked fixed point. Such an open, damped

[1] "Exact" here refers to the geometric fixed-point level: the dimensionless invariant $Q = 2/3$ is rigid and is confirmed by the data to $-0.9\,\sigma$ (Doc. 292). The *absolute* masses, or the number pair $(\sqrt{2},\ 2/9)$, must be distinguished from this: on the measured pole-mass level the pair is not jointly exact (Doc. 292, $\sim 450\,\sigma$ in the m_μ/m_e channel), which is why $2/9$ is treated per P42 as the rational address of an angle fixed via one reference point, not as a pole-mass exactness claim. For the dynamical locus discussed here this is immaterial – the orbit resolution (~ 0.1) lies far above the 10^{-7} difference.

loop can – unlike a closed fixed point – be influenced from outside: by coupling to a second oscillator, by injection or mode-locking (Doc. 286). In practice the non-closure therefore means that a channel exists through which a second, dynamical system can lock the open phase. The closed mass sector offers no such channel.

The channel is, however, narrow and structured, not a free parameter. The capture range of such a coupling is of order ξ (Doc. 286) – tiny; the $2/9$ Arnold tongue is about twelve times narrower than $1/3$. The coupling of adjacent scale levels in the continuous flow is real but of order 10^{-13} in magnitude and does not destabilise the closed predictions. And the effect is directed: it runs from the dynamics into the phase, not into the radial mass sector, which stays rigid (the radial/angular split of Doc. 282).

The one place this channel would be used is a dynamical skew-adjoint generator outside the static FFGFT geometry – the continuation of the elimination chain (Doc. 283/286). If such a generator locks the open phase precisely to $\chi = \pi/2$ and thereby $\theta = 2/9$, then $2/9$ would acquire the dynamical, protected status the following conclusion describes. This is explicitly a candidate, not an attained result: FFGFT's own recursion R is purely real and carries no such generator; whether an external dynamics supplies one remains open.

The decisive point is the boundary of this room. "Room for mutual influence" does not mean the theory can be bent at will. The room is bounded: the radial sector stays rigid, the phase is tightly constrained (skew-adjoint, $\chi = \pi/2$, transcendental in 2π, ξ-narrow capture range), and the recursion depth fixes no value (Doc. 275). It is structured room, not a free parameter – and it is exactly this boundary that distinguishes a channel for genuine coupling from an after-the-fact numerical adjustment.

This channel is also the reason the fundamental sector does not collapse to a rigid skeleton. A fully closed model – every value a forced fixed point – would be overdetermined: no contingent selection within a Z_3 orbit, no flavour mixing, no physical (non-removable) phase. Mixing in particular is phase, not magnitude (Doc. 289: charged leptons magnitude-dominated with small mixing, neutrinos phase-dominated with large mixing, the pure all-pass sector). The sharpest case is the structure of CP violation: it requires an irreducible complex phase – a purely real, self-adjoint structure is CP-conserving – hence exactly the type of the skew-adjoint channel $\chi = \pi/2$. The open phase channel is thus structurally of the kind that can carry CP violation; a fully self-adjoint FFGFT would be CP-even and non-mixing. The scope stays narrow: the channel carries mixing, CP and contingency in the *fundamental sector* – not the macroscopic diversity (chemistry, structure formation), which has its own independent sources.

Z.10 Conclusion: the open dynamical question

Statically the situation is clear: $2/9$ is invisible in the symmetric Q, blind to magnitude and to the integer winding, and an interior extremum dragged in by a free parameter ($d\theta^*/d\delta \neq 0$). It is *not* forced like c. Both quantities share the same flat T^4 arena – c as a signature boundary, $2/9$ as an interior phase degree of freedom – but they are of different types.

For $2/9$ to acquire a c-like, forced status, it would need a *dynamical* fixed point, as gauge symmetry forces $m = 0$ (and hence $v = c$). The preceding sections localise this fixed point: the *phase* $\chi = \pi/2$ is the skew-adjoint i-value required by the elimination chain – the

selector is type-fixed; what remains open is only the *magnitude* δ (the golden-damped overtone amplitude). Its fine value is, moreover, not "not yet" derived but *provably not* derivable from the recursion alone: the non-closure is a theorem (the only fixed point is 0), and the resonance density of the transit points also rules out any recursion-internal distinction of a value (Doc. 275, epistemic limit). The recursion side is thereby settled, not an open question; what is open is solely the existence of the external anchor – whether a skew-adjoint dynamics carries precisely $\chi = \pi/2$ and thereby fixes the depth δ for an internal reason *of that dynamics* (not of the recursion); if it did, $2/9$ would acquire a dynamical, protected privilege, otherwise it stays contingent. The elimination chain (Doc. 282) has ruled out every static candidate – symmetric, counting-topological, radial, static-geometric – and the only survivor is the dynamical skew-adjoint phase, whose origin lies outside the static geometry. It is there, not in the magnitude and not in a flat winding number, that the value $2/9$ would have to be sought.

Z.11 Placement: usefulness and proximity to application

The findings of this document are structure and exclusion results, not technology. Their immediate usefulness lies with three groups. First, work on the Koide relation and flavour structure: the no-go ($2/(9\pi)$ irrational; magnitude and integer winding counter-blind) closes off topological and counting derivation routes for $\theta = 2/9$ and spares their further pursuit; the elimination chain localizes the remaining search precisely in the dynamical skew-adjoint phase channel. Second, methodologically: the magnitude/phase decomposition serves as a shared audit language for framework comparisons (script-backed claims, pre-registered criteria, matched nulls) and transfers independently of the FFGFT content. Third, didactically: the translation between particle physics and communications engineering (circulant as DFT, all-pass, injection locking, Arnold tongues) makes the phase channel computable with standard engineering tools.

Only the phase channel itself touches measurement: the structural statement that mixing and CP violation live in the skew-adjoint channel (Doc. 289) has a falsifiable form in the oscillation phases and δ_{CP}, which running and upcoming experiments (DUNE, Hyper-Kamiokande) measure. An application in the narrow sense – a device, a procedure, short-term engineering use – does not follow: the tools run from communications engineering into physics, not back, and as long as δ is not derived, the yield is that of a sharpened open problem, not of a result.

Chapter aa

Doc. 292 — Lepton Empirics Check: the FFGFT Relations Seen from the Data

aa.1 Origin: how $2/9$ became visible in the first place

Before any calculation, the question of where the central number of this document comes from should be settled – because it was not sought but found. The route was the Hilbert-space translation of FFGFT. The bridge of Docs. 230/231/232 maps the loop observables of the corpus to self-adjoint operators on a fibre-valued Hilbert space; only when this translation was carried through to the masses (Doc. 282) did the decisive structure appear: the three lepton masses are the spectrum of a hermitian Z_3 circulant on the $\mathbb{C}^3$ fibre of the full space $\mathscr{H} = L^2(T^4) \otimes \mathbb{C}^3$. In the eigenvalues of this operator the phase angle $\theta = 2/9$ appears – not as an ansatz one inserts, but as a value the diagonalisation returns. $2/9$ is therefore *not* a starting point of the mass calculation but an order that became visible only afterwards: it was the change of representation – from the ξ ladder to the operator/spectral view – that exposed it. This counter-calculation therefore tests not an input but a *discovery*; that is why the direction from the data (below) is meaningful at all. The relation to the ξ ladder, which computes the masses without $2/9$, is clarified in the closing section “Two layers”.

aa.2 Purpose and direction of the calculation

The FFGFT corpus fundamentally computes *away from* $\xi = 4/30000$: the geometry yields relations that are subsequently checked against data. This document reverses the direction. The starting point is exclusively the empirical lepton masses (PDG), and the question is what the FFGFT relations of the lepton sector make *of the data* – which structural numbers the data return, with which uncertainties, and where residual tensions sit. The mode is the compatibility mode: no new derivation, but an independent counter-calculation per P35, with the level separation per P40 (only ratios are exact; absolute values carry the bridge constants). All numbers in this document come from

`python/Dok292_Skripte/leptonen_empirie_check.py` (numpy-only, seed 20780458) and are reproducible with a single call.

The inputs are

$$m_e = 0.51099895069(16)\ \text{MeV}, \tag{aa.1}$$

$$m_\mu = 105.6583755(23)\ \text{MeV}, \tag{aa.2}$$

$$m_\tau = 1776.86(12)\ \text{MeV}. \tag{aa.3}$$

The value for m_τ is the PDG 2022 value; the numbers in the text are computed with it. PDG 2024 gives $m_\tau = 1776.93(9)$ MeV. This yields $Q = 0.6666645$, $Q - 2/3 = -2.2 \cdot 10^{-6} = -0.43\,\sigma$, and $m_\tau^{\text{pred}} = 1776.968$ MeV lies $+0.43\,\sigma$ above the measured value. The "single τ tension of $\sim 0.9\,\sigma$" referred to below and the σ figures derived from it hold for the PDG 2022 value and roughly halve with PDG 2024. The τ mass is by orders of magnitude the least precise of the three; as will become clear, it dominates all residual tensions.

aa.3 Level 1: the circulant – Koide relation and angle

Koide ratio from the data

The Koide ratio $Q = (m_e + m_\mu + m_\tau)/(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2$ evaluated on the PDG values gives

$$Q = 0.666660511 \pm 0.0000068, \tag{aa.4}$$

where the uncertainty is carried almost entirely by m_τ. The deviation from the exact value $2/3$ is $-6.2 \cdot 10^{-6}$, i.e. $-0.91\,\sigma$ (with PDG 2024: $-2.2 \cdot 10^{-6}$, $-0.43\,\sigma$). The data are compatible with exactly $2/3$; they do not *enforce* the exact value, but they do not contradict it at 5–6 significant digits.

Circulant fit: amplitude and angle from three masses

Following Doc. 282, the $\sqrt{m_k}$ are eigenvalues of a hermitian Z_3 circulant, $\sqrt{m_k} = M(1 + r\cos(\theta + 2\pi k/3))$, with the two pure structural numbers $r = \sqrt{2}$ and $\theta = 2/9$ and the identity $Q = (1 + r^2/2)/3$. The Fourier extraction from the three data points yields (assignment $k = 0, 1, 2 \to \tau, e, \mu$; all cyclic assignments give the same angle)

$$M = 313.84\ \text{MeV}, \qquad r_{\text{emp}} = 1.4142005, \qquad \theta_{\text{emp}} = 0.22222963 \pm 8.4 \cdot 10^{-6}. \tag{aa.5}$$

The amplitude deviates from $\sqrt{2} = 1.4142136$ by $1.3 \cdot 10^{-5}$ – via $Q = (1 + r^2/2)/3$ this is the *same* statement as the Q deviation, not an independent one. The angle lies $+0.89\,\sigma$ above $2/9 = 0.22222222$, likewise τ-limited.

The τ-free test: angle from m_μ/m_e alone

Doc. 282 names m_μ/m_e as the sharpest test of whether θ is exactly $2/9$: holding $r = \sqrt{2}$ fixed, M cancels in the ratio, and the two most precise masses in physics determine θ alone. Solving $\left(1 + \sqrt{2}\cos(\theta + 4\pi/3)\right)^2 / \left(1 + \sqrt{2}\cos(\theta + 2\pi/3)\right)^2 = m_\mu/m_e$ gives

$$\theta(\mu/e) = 0.2222220471, \qquad |\theta - 2/9| = 1.75 \cdot 10^{-7}. \tag{aa.6}$$

This independently reproduces the figure from Doc. 282 ($\approx 1.8 \cdot 10^{-7}$, seven significant digits). Two readings must be kept apart. Measured against the tiny PDG uncertainty of m_μ/m_e, the residual is formally highly significant – *under the additional assumption* that r is exactly $\sqrt{2}$. If r is allowed to float via Q, the residual is not an independent tension but once again the same single τ tension of $\sim 0.9\,\sigma$. In the lepton sector there are thus not four small deviations but *one*, appearing in four guises.

Forward check and the one prediction

Conversely, setting $r = \sqrt{2}$ and $\theta = 2/9$ exactly, the circulant reproduces m_μ/m_e to $+0.0010\,\%$ and m_τ/m_μ to $+0.0060\,\%$. Inverting for m_τ turns this into a falsifiable forward statement: from (m_e, m_μ) and the two structural numbers it follows that

$$m_\tau^{\text{pred}} = 1776.968 \text{ MeV} \tag{aa.7}$$

(the three inversion routes – $Q = 2/3$ exact, $\theta = 2/9$ exact in the three-mass fit, direct prediction from $(m_e, m_\mu; \sqrt{2}, 2/9)$ – give 1776.969, 1776.967 and 1776.968 MeV, i.e. the same value). Against PDG 2022 1776.86 ± 0.12 MeV this is $+0.90\,\sigma$, against PDG 2024 1776.93 ± 0.09 MeV $+0.43\,\sigma$. A τ-mass measurement at the targeted Belle-II level of ~ 0.05 MeV decides this tension: if the value lands near ≈ 1776.97, the τ direction is resolved and the pair $(\sqrt{2}, 2/9)$ is confirmed at the 10^{-5} level; if it lands significantly below, the approximation at this level is refuted. This is a genuine prediction in the sense of the corpus methodology, not a retrodiction: the value is fixed before the measurement and is booked here in advance. What the test does *not* decide is shown by the precision analysis in Section aa.5.

aa.4 Level 2: the ξ ladder – ratios and reverse ξ

Ladder ratios against the data

The ladder representation $m_i = r_i\, \xi^{p_i}\, v$ (Docs. 006/046; leptons: $r_e = 4/3$, $p_e = 3/2$; $r_\mu = 16/5$, $p_\mu = 1$; $r_\tau = 25/9$, $p_\tau = 2/3$ per K2) makes the ratios parameter-free:

$$\left.\frac{m_\mu}{m_e}\right|_{\text{ladder}} = \frac{12}{5}\,\xi^{-1/2} = 207.85 \quad (\text{PDG } 206.77;\ +0.52\,\%), \tag{aa.8}$$

$$\left.\frac{m_\tau}{m_\mu}\right|_{\text{ladder}} = \frac{125}{144}\,\xi^{-1/3} = 16.992 \quad (\text{PDG } 16.817;\ +1.04\,\%), \tag{aa.9}$$

$$\left.\frac{m_\tau}{m_e}\right|_{\text{ladder}} = \frac{25}{12}\,\xi^{-5/6} = 3532 \quad (\text{PDG } 3477;\ +1.57\,\%). \tag{aa.10}$$

The ladder thus carries the orders of magnitude and the $\sim 1\,\%$ accuracy – not the precision of the circulant. Both levels are to be read separately, as declared in Doc. 282 and P40: the circulant delivers four to five digits from $(\sqrt{2}, 2/9)$, the ladder delivers the hierarchy from ξ.

Reverse ξ: what the data return for ξ

Solving the three ladder ratios for ξ, the data return

$$\xi_{\mu/e} = 1.347 \cdot 10^{-4}, \qquad \xi_{\tau/\mu} = 1.375 \cdot 10^{-4}, \qquad \xi_{\tau/e} = 1.358 \cdot 10^{-4}, \tag{aa.11}$$

i.e. $+1.1$ to $+3.2\,\%$ above $4/30000 = 1.3333 \cdot 10^{-4}$. The three values scatter among each other by $\sim 2\,\%$ – the ladder is consistent in the ratios at the percent level, not exact. This is the honest empirical status of the ladder representation, unchanged from the declaration in Doc. 006 (mean deviation below $1\,\%$ at the mass level; correspondingly 1–$3\,\%$ at the ratio level).

Bridge constants per P40

Per lepton, solving for the bridge constant $v_{\text{eff}} = m_i/(r_i\,\xi^{p_i})$ yields

$$v_{\text{eff}}^{(e)} = 248.9 \text{ GeV}, \qquad v_{\text{eff}}^{(\mu)} = 247.6 \text{ GeV}, \qquad v_{\text{eff}}^{(\tau)} = 245.1 \text{ GeV}, \tag{aa.12}$$

scattered around the Higgs vacuum expectation value 246.22 GeV ($+1.10/+0.58/-0.46\,\%$). Equivalently: the required r_{eff} at fixed v deviate by the same percentages from the declared rational values (r_e: 1.348 instead of $4/3$, etc.). This is precisely the P40 finding seen from the data side: the rational structural numbers of the ladder hold at the percent level; the residual deviation is absorbed into the bridge constant and is *not* fully cancelled in ratios.

K_{frak} cross-check: no simple pattern

The obvious question of whether the ladder residuals follow the fractal correction factor is answered in the negative by the calculation: $K_{\text{frak}} = 1 - 100\xi$ corresponds to $-1.333\,\%$, the half and three-halves powers to -0.665 and $-1.995\,\%$ – the measured deviations $+0.52$ and $+1.04\,\%$ fit no single K_{frak} power in either magnitude or sign. The only striking feature is that the τ/μ deviation is quite precisely twice the μ/e deviation; whether structure or coincidence lies behind this remains unbooked per P35. The point is declared open, not explained away.

Cross-ladder coupling: a generation-linear correction law

Before this finding is booked, the booking level must be fixed, since it decides admissibility. There are two *mutually exclusive* calculation modes. *Mode 1* computes all lepton ratios purely from ξ, without a reference point: then μ/e, τ/μ and τ/e are all genuine predictions (each 0.5–$1.6\,\%$ off the data), and all three are legitimate test points. *Mode 2* applies P42: m_μ/m_e is then the declared anchor, *consumed* and no longer a test quantity – one may no longer book its ladder deviation, else the same quantity is used twice (once as anchor, once as test). The coupling analysis below belongs *exclusively* to Mode 1, the reference-free one; in Mode 2 it is void, because there μ/e drops out of the comparison and the only independent ladder test left is m_τ – which in any case runs through the circulant, not the coarse ladder. Within Mode 1: the statement "no simple pattern" concerns a *single* K_{frak} power. Asking instead whether the ladder stages are coupled *to each other*

does reveal structure. Writing the required correction per place as an effective factor N_g via $K = 1 - N_g\,\xi$ (or as an exponent p via K_{frak}^{p}), the data give

$$\mu/e\text{:}\quad N_1 = 38.89, \quad p_1 = 0.3873, \tag{aa.13}$$

$$\tau/\mu\text{:}\quad N_2 = 77.06, \quad p_2 = 0.7694, \tag{aa.14}$$

$$\tau/e\text{:}\quad N_3 = 115.55, \quad p_3 = 1.1567. \tag{aa.15}$$

Two relations here are not accidental. First, the exponent structure is *multiplicatively consistent*: $p_3 = p_1 + p_2$ to $6 \cdot 10^{-14}$. The τ/e place therefore carries *no* new information – it is the product of the two stages, as a throughgoing multiplicative correction $K^{p_1} \cdot K^{p_2} = K^{p_1+p_2}$ requires. Second, the effective factors stand in the ratio $N_1 : N_2 : N_3 = 1 : 1.981 : 2.971 \approx 1 : 2 : 3$: the correction grows *linearly with generation number*, $N_g \approx g \cdot N_0$ with $N_0 \approx 38.6$ (the three individual estimates N_1, $N_2/2$, $N_3/3$ are 38.89, 38.53, 38.52 – consistent among themselves to below $1\,\%$).

This is the structural reason why a constant K_{frak} does not close the ladder, and why asking for the "right" number instead of 100 (such as 89) misses the point: the ladder demands not a fixed factor but a *generation-counting* one, $N_g = g \cdot N_0$. At α only one scale is in play (the μe mean) – one factor suffices, and α fits. In the ladder three generations superpose, so N grows linearly; a constant factor cannot do this in principle. The generation-dependent dressing named earlier as an obligation thereby acquires a concrete form: to first approximation it is linear in g.

The tolerance balance shows how far the law reaches (reference-free Mode 1, common $N_0 = 38.65$):

Place	raw	raw $[\sigma_{\text{exp}}]$	corrected	corr. $[\sigma_{\text{exp}}]$
μ/e	$+0.521\,\%$	$2.4 \cdot 10^5$	$+0.0033\,\%$	$1.5 \cdot 10^3$
τ/μ	$+1.038\,\%$	154	$-0.0031\,\%$	0.47
τ/e	$+1.565\,\%$	232	$-0.0052\,\%$	0.78

The law reduces the ladder deviation by about a factor of 300, from 0.5–$1.6\,\%$ to $\sim 5 \cdot 10^{-3}\,\%$. The two τ places thereby fall *within* their measurement tolerance ($< 1\,\sigma$) – for them the law is practically complete. The μ/e entry stays at $\sim 1.5 \cdot 10^3\,\sigma$, not because the correction fails there but because μ/e is measured about 500 times more precisely: even a $0.003\,\%$ residual is enormous in σ units. This is exactly where the mode boundary bites: this μ/e entry is a statement *only* in reference-free Mode 1; under P42, μ/e is the anchor and drops out of the balance. The remaining residual of the two τ places sits in the fine deviation of the ratio from exact $1 : 2 : 3$ ($\sim$ 1–2 %) and in the still-undetermined base unit N_0 – that is the next open calculation, not a contradiction.

Open – and per P35 expressly not explained away – are two things. The base unit $N_0 \approx 38.6$ is not yet derived from the geometry; obvious candidates ($100/e = 36.8$, $100/\varphi^2 = 38.2$, plain 39) do not match it closely enough to be claimed. And the phase sector – the δ fine value from Doc. 291 with $p_\delta = -0.88$ – does *not* obviously fit the same generation-linear magnitude ladder; whether the magnitude and phase sectors couple through a common law is a separate question not answered here. The finding is thus not the solution of the ladder residual but its sharpening: "three crooked exponents" becomes "one generation-linear law $N_g = g \cdot N_0$ with N_0 still undetermined".

The limit of this claim, following from the mode separation, must be stated explicitly. The $1 : 2 : 3$ pattern is supported at three points *only* in the reference-free Mode 1. As

soon as P42 is applied and m_μ/m_e is declared the anchor (the level needed for the circulant precision and the m_τ prediction), μ/e drops out as a test point, and the law rests only on τ/μ and τ/e – which share the same τ mass and whose precision is low. The apparently threefold confirmation is thus tied to reference-freeness. Both at once – μ/e as anchor *and* as an N_g test point – is inadmissible double-booking. The circulant, which already carries μ/e to 10^{-5} (route A in Section aa.5), does not need the g correction anyway; it is a statement about the coarse ξ ladder in the reference-free mode alone, not about the precise circulant layer.

A common misunderstanding must be cleared up here: the 100 in $K_{\text{frak}} = 1 - 100\,\xi$ and the base unit $N_0 \approx 38.6$ are *not* the same quantity, so the ladder by no means needs "fewer recursions" than the α sector. The 100 is the single fixed scale of a corpus-wide constant (X-flip, T_{CMB}, E_0 bridge); N_0 is the factor *per generation* in a different sector. The difference is not *fewer* but *other*: not a fixed factor but a generation-counting $N_g = g \cdot N_0$ – already 77 at $g = 2$, 116 at $g = 3$, i.e. partly *more* than 100. Comparing the two numbers as "recursion depths" therefore misses the point.

In assessing possible expressions for N_0 the correct measure is not point-exact agreement – that is unreachable in principle – but consistency *within the tolerance* to which N_0 can be determined at all. That tolerance is considerable: in the referenced mode N_0 follows from the τ places alone, and their precision is limited by the τ mass (± 0.12 MeV). Propagating this gives $N_0 = 38.52 \pm 0.21$, a relative determination uncertainty of about $0.5\,\%$. A candidate expression is therefore to be judged not by whether it hits 38.6 to the decimal, but by whether it falls into this interval.

By that measure – and still as an *unproven* observation per P35, not a derivation – $100/\varphi^2 = 38.20$ is an *admissible* candidate: its distance to N_0 is $1.6\,\sigma_{N_0}$, i.e. within the tolerance. As a point value the expression misses 38.6; within the tolerance to which N_0 is known, it matches. Were $N_0 = 100/\varphi^2$, the ladder base unit would follow from the K_{frak} scale through a single factor φ^{-2} – magnitude and correction sectors hanging on the same constant. The finding is thus not "narrowly missed and discarded" but "compatible with the present determination, not refuted". A sharper τ mass (Belle II) or a geometric derivation of N_0 will decide between $100/\varphi^2$ and the alternatives; until then it remains an admissible, not a proven, expression.

aa.5 Relation to the measurement precisions

The σ statements so far refer to one quantity each. The full question is: how do the FFGFT structural numbers sit relative to the *entire* precision structure of the data? A two-parameter analysis answers it. The two independent ratios $R_1 = m_\mu/m_e$ and $R_2 = m_\tau/m_\mu$ determine the pair (r, θ) exactly; their relative measurement precisions, however, differ dramatically: $\sigma_{R_1}/R_1 = 2.2 \cdot 10^{-8}$ versus $\sigma_{R_2}/R_2 = 6.8 \cdot 10^{-5}$ – the μ/e ratio is measured about 3100 times more precisely. Error propagation onto (r, θ) therefore yields an extremely anisotropic covariance: a *thin* principal direction with $\sigma \approx 3.4 \cdot 10^{-10}$ (carried by μ/e) and a *thick* one with $\sigma \approx 1.7 \cdot 10^{-5}$ (carried by τ); the correlation between r and θ is practically -1.

In this picture the point $(\sqrt{2}, 2/9)$ lies away from the data fit $(1.41420051,\ 0.22222963)$ by $0.90\,\sigma$ in the thick direction – this is the single τ tension of Section 2 – and by about $450\,\sigma$ in the thin one. Spelled out: the circulant with *exactly* $(\sqrt{2}, 2/9)$ misses m_μ/m_e by $1.0 \cdot 10^{-5}$

relative, while the measurement stands at $2.2 \cdot 10^{-8}$. As an exact pair the two structural numbers are thus already excluded by today's data; they are confirmed at the 10^{-5} level – precisely the "four to five digits" declared in Doc. 282, here made precise as a measurement relation. This also bounds the reach of the m_τ prediction: a perfect Belle-II result resolves only the thick τ direction; the residual of $1.75 \cdot 10^{-7}$ in θ (equivalently 10^{-5} in R_1) is an *already measured*, standing offset that any exactness claim would have to explain – radiative corrections to the pole masses are the obvious candidate, though their typical size ($\alpha/\pi \sim 10^{-3}$) rather deepens the puzzle of why the agreement reaches 10^{-5} at all. The point is booked open per P35.

Thought model: inverting the precision statement

The $450\,\sigma$ finding rests entirely on trust in the quoted precision of m_μ/m_e. The inversion – the precision might not mean what it claims – is admissible as a thought model and comes in three strengths. *Weak:* the experimental uncertainty would be underestimated by a factor of ~ 450 – quantitatively hardly tenable, since the same precision hangs in a web of independent cross-checks (muonium spectroscopy, $g-2$, QED tests). *Intermediate:* m_μ/m_e is not a read-off but a quantity *extracted* through bound-state QED; the quoted precision presupposes the completeness of the extraction theory, and a missing term at 10^{-5} would dissolve the conflict – 450 times the estimated theory error, borderline, but logically open. *Strong (and FFGFT-native):* the precision is real but applies to the *dressed* quantity – the pole masses in the SI frame – while the circulant relation would live on the geometric level beneath. Then nothing contradicts anything: the 10^{-5} offset would not be an error but the *measured thickness of the dressing layer* in the μ/e channel. This is the consistent extension of P40: in ratios only the generation-*independent* part of the fractal/SI correction cancels; a generation-dependent remainder survives – and would have exactly the shape of this offset. The price of the strong reading is an obligation: the offset would have to be computable from the dressing (generation-dependent radiative structure), otherwise the reading is unfalsifiable. It is booked here as a thought model, not as a resolution; the contradiction finding of the previous section stands for the pole-mass level.

Resolved by declaration: the reference point m_μ/m_e (P42)

The framework decides the question not through one of the thought models but through the claim layer: FFGFT fundamentally keeps the purely theoretical derivation from ξ, and the lepton sector receives – analogous to the cosmic scale (R39) – exactly *one* declared reference point, the ratio m_μ/m_e. At structural $r = \sqrt{2}$ (from $Q = 2/3$) it fixes the operative angle $\theta_{\mathrm{ref}} = 0.2222220471$; the structural number $2/9$ is its rational address to seven digits, not an exactness claim at the pole-mass level. The $450\,\sigma$ finding is thereby no longer a falsification but part of the calibration. Two delimitations: the status of ξ is untouched (FFGFT claims ξ as forced by the geometry; that dispute is not adjudicated here), and the lepton masses are in themselves not an input of the theory – the reference point belongs to the comparison level, it locates the geometric relation in the pole-mass data and would not be needed without the intent to compare. The honest price lies in the test

balance: m_μ/m_e is consumed as anchor and drops out as a test quantity; the test quantity is m_τ alone. The gain is an input-error-free sharp prediction:

$$m_\tau = 1776.9690 \pm 0.0001 \text{ MeV}, \tag{aa.16}$$

where the uncertainty stems solely from (m_e, m_μ) and practically vanishes; against PDG the single τ tension remains ($+0.9\,\sigma$ with PDG 2022, $+0.43\,\sigma$ with PDG 2024), and the Belle-II world average decides with no escape route. For the bridge question (Doc. 291) the declaration sharpens the target: the dynamics must select θ_{ref}; $2/9$ remains its address and the pre-registered Stage-10 target, whose orbit resolution (~ 0.1) is unaffected by the 10^{-7} difference. Booked as P42 in Doc. 190.

The same view calibrates the ξ ladder: its residuals correspond to $2.4\cdot10^5$ (μ/e), 154 (τ/μ) and 232 (τ/e) experimental standard deviations. The ladder residuals are therefore not data-limited but theory-side (P40 bridge constants); measurement precision has outrun the ladder representation by two to five orders of magnitude. Finally, the test power of the m_τ prediction (separation of prediction from today's central value) at $\sigma_{m_\tau} = 0.12/0.05/0.02$ MeV is $0.9/2.2/5.4\,\sigma$ respectively (relative to the PDG 2022 central value; roughly halved with PDG 2024) – a single Belle-II measurement is indicative; only the combined world average is decisive.

aa.6 The fine-structure constant from the lepton masses

The fine-structure constant belongs in this counter-calculation because in FFGFT it is a lepton-mass statement: Doc. 011 sets $\alpha = \xi(E_0/\text{MeV})^2$ with the structural energy scale $E_0 = \sqrt{m_e\, m_\mu}$ and the declared fractal correction, $\alpha^{-1} = (7500/E_0^2)\, K_{\text{frak}}$. From the data side: the measured α requires $E_0^{\text{req}} = 7.39798$ MeV, the geometric mean gives 7.34788 MeV – ratio 1.006819, against $K_{\text{frak}}^{-1/2} = 1.006734$. The declared correction factor thus closes the gap up to $8 \cdot 10^{-5}$; the closed relation

$$\alpha \;=\; \frac{\xi\, m_e\, m_\mu}{K_{\text{frak}}\ \text{MeV}^2}, \qquad K_{\text{frak}} = 1 - 100\,\xi, \tag{aa.17}$$

gives $1/\alpha \;=\; 137.059$ against measured 137.035999. This agreement is a property of the anchor E_0, which provides the connection to SI, not an independent prediction of α.

The decisive point here is that α does *not* run through a fitted bridge constant. Two closely related routes lead to the characteristic energy E_0 (A130): Route 1 is the bare geometric mean of the measured masses, $E_0 = \sqrt{m_e m_\mu} = 7.348$ MeV, and gives $1/\alpha = 138.9$ – in this variant K_{frak} cancels (ratio calculation, Doc. 122); Route 2 is the same mean with the fractal correction, $E_0 = \sqrt{m_e m_\mu / K_{\text{frak}}} = 7.397$ MeV, and gives $1/\alpha = 137.06$ ($+0.017\,\%$). $K_{\text{frak}} = 1 - 100\,\xi$ comes from ξ, not from α – no back-calculation from α. Read the other way, the measured value of α fixes the factor between the two routes: $K_{\text{frak}} = \xi m_e m_\mu\, \alpha^{-1} = 0.98650$, against $1-100\,\xi = 0.98667$ to $1.7\cdot10^{-4}$. Comparing the two routes with α thus confirms the value $1 - 100\,\xi$ up to this residual; it is not an α-free measurement of K_{frak}. That *the same* factor that carries the damped X-flip and the T_{CMB} correction in the corpus closes the gap between Route 1 and the E_0 required by α up to this residual is a cross-sector consistency – not a fit.

The α residual and the dressing layer

Taking the α measured to $1.5 \cdot 10^{-10}$ as exact, a residual remains between Route 2 and the $E_0 = \sqrt{\alpha/\xi} = 7.3980$ MeV required by α. The K_{frak} bridge closes $98.8\,\%$ of the gap between Route 1 and the E_0 required by α, $(7.39736 - 7.34788)/(7.39798 - 7.34788) = 0.988$; a residual of $8.4 \cdot 10^{-5}$ remains at the E_0 level ($1.7 \cdot 10^{-4}$ in K_{frak}). Measured against the precision of the masses ($1.1 \cdot 10^{-8}$ at the E_0 level), this residual is $\sim 7700\,\sigma$ large – too large to be mass measurement noise.

Where it sits cannot be decided from the α sector alone: both routes use m_e and m_μ, and the residual may lie in the pole masses (QED dressing) just as well as in the coefficient of K_{frak} (101.2 instead of 100). Compared with the θ side (Section aa.5), where θ_{ref} is obtained from m_μ/m_e and the $\sim 450\,\sigma$ in the μ/e channel are half self-reference, the α residual is therefore not a stronger, mode-independent witness but an equal-ranking indication of a layer between the geometrically fundamental quantity and the measured pole mass, one that leaves the attribution open. The obligation – to *compute* the dressing rather than name it as a residual – remains open.

Conceptually – almost without computational consequence, yet central to the architecture – the following holds: in the extended natural unit system $\alpha = 1$ (Docs. 011/014, charge dimensionless); the SI value $1/137.036$ is bookkeeping of the unit translation, not a fundamental coupling. The same architecture carries gravitation: $G = \xi^2/(4m_{\text{char}})$ times conversion (Docs. 012/013) makes G a bookkeeping quantity relative to SI – reproducing the SI value is calibration per R40; only the forward chain is testable. The structural gain lies elsewhere: a coupling that is only bookkeeping need not be quantised – the quantum-gravity problem, on which most extensions of the Standard Model founder, does not arise in this architecture.

aa.7 Two layers that do not need each other: what $2/9$ provides and what the SI conversion provides

Two questions that are easily conflated must be separated here.

First: is $2/9$ needed to compute the masses? No. The historical and still valid route is the ξ ladder (Docs. 006/046): $m_i = r_i\,\xi^{p_i}\,v$ with $r \in \{4/3, 16/5, 25/9\}$ and $p \in \{3/2, 1, 2/3\}$ yields the masses from ξ and the ladder numbers alone, with $2/9$ entering nowhere. The structural number $2/9$ appears only from Doc. 282 onward and in a different role: not as an input generating the masses, but as an *order discovered within them* – the three $\sqrt{m_k}$ are eigenvalues of a Z_3 circulant with phase angle $2/9$. $2/9$ replaces the formerly fitted ladder coefficient by a pure number; it is compression and explanation, not a prerequisite. One could delete $2/9$ from the mass sector and the ladder would keep computing – what would be lost is the *insight* into why Q is exactly $2/3$, not the numbers. $2/9$ thus belongs to the explanatory layer, not the computational one.

Second: does this counter-calculation confirm that the SI conversion is necessary and correct? In two separate degrees. As *necessary* it confirms clearly: every absolute number (m_τ in MeV, $\alpha = 1/137$, G) requires the bridge constant to reach SI, while the ratios (m_μ/m_e) do not. This two-level structure is exactly P40, and the $450\,\sigma$ finding is its sharpest evidence: it shows that the measured pole-mass level (SI-defined, QED-dressed) is a *different* level from the geometric one – without the conversion step the difference would

not exist. As *correct in the sense of validated to measurement precision*, however, it does *not* confirm: the conversion closes about $98.8\,\%$ of the gap at α (through the same factor K_{frak} that also operates in other sectors), but the remaining $\sim 10^{-4}$ stays open. The SI conversion is therefore structurally necessary and consistent across sectors, but it is calibration, not a test passed to the decimal.

The real finding lies in the independence: the ladder computes masses without $2/9$, the circulant explains Q without the ladder, the SI conversion translates without $2/9$. That all three nonetheless share the same ξ and the same factor K_{frak} is the consistency of the framework – not a dependence of one layer on another.

aa.8 Placement

The counter-calculation from the data side confirms the two-level architecture of the corpus without adding to it or sparing it anything. At the circulant level the data return the two structural numbers at the 10^{-5} level; the precision analysis separates two findings: in the τ-limited (thick) direction a single τ tension ($0.9\,\sigma$ with PDG 2022, $0.43\,\sigma$ with PDG 2024), carrying the pre-booked prediction $m_\tau = 1776.968$ MeV – and in the μ/e-precise (thin) direction an already measured 10^{-5} offset that excludes the *exactness* of the pair $(\sqrt{2}, 2/9)$ and is booked as open fine structure. At the ladder level the data return ξ at 1–3 %, with honestly open residuals showing no recognisable K_{frak} pattern. What this calculation does *not* show: no derivation of $\theta = 2/9$ (its dynamical locus remains the open question of the bridge work), no sharpening of the ladder, no new physics. It shows that the declared claims of both levels are independently reproducible starting from the PDG data – in the direction in which nothing can be fitted.

The residual as the signature of incompleteness

The counter-calculation has sharpened the statements – from “two pure numbers, no fit” to measured quantities with known decimals and known limits. Yet at every level a residual remains: the 10^{-7} offset in θ, the 1–3 % of the ladder, the open δ fine value (Doc. 291), the K_{frak} pattern that does not close across the ladder stages. The α point also carries a residual: the K_{frak} required by α lies $1.7 \cdot 10^{-4}$ from $1 - 100\,\xi$ (see above). None of these residuals is a *contradiction* – the theory holds everywhere to its respective precision – but none is *zero*. A residual that shrinks the closer one looks without vanishing is the signature of an approximation to something deeper, not of a completed foundation. The residual is therefore not an error but the honest evidence that the theory is *not complete*.

What is remarkable is *where* it sits. The geometric core carries no residual: $Q = 2/3$ is purely geometric and practically exact; in the extended natural system $\alpha = 1$; the ladder ratios are rational numbers. The residual arises throughout *only* at the transition into the measured: the $450\,\sigma$ break only at the pole-mass level, the ladder deviation only in the bridge constant. The α conversion fits in as well: there the measured value of α fixes the correction depth ($K_{\text{frak}} = 0.98650$), and the residual of $1.7 \cdot 10^{-4}$ against $1 - 100\,\xi$ arises only in the transition to the measured quantity. The residual thus lives not in the geometric core but in the *translation layer* – systematically, not randomly distributed. This localises the

incompleteness precisely: not in ξ, not in the circulant, but in the not-yet-*derived* fractal or QED dressing. The theory would be complete only once it *computes* this residual instead of absorbing it into a bridge constant – once K_{frak} not merely fits but follows generation-dependently from the geometry. The measure for this is expressly *not* a point-exact landing on the last decimal: a geometric derivation is not meant to hit the QED-dressed pole mass digit by digit – that would be neither possible nor the goal – but to fix the underlying structure far enough that the predictions stay *within the tolerances* to which the quantities can be determined at all. Completeness here means: no residual left that protrudes *beyond* those tolerances and is closed only by a free parameter. Until then, each such residual is the pointer to the next open derivation, against which completeness – in the sense of tolerance, not of the last digit – will be measured.

Chapter ab

Doc. 293 — The Icosahedral Origin of $\theta = 2/9$: the Second Witness

ab.1 Question: a second, independent route to $\theta = 2/9$

The Koide angle $\theta = 2/9$ is empirically confirmed to seven digits and understood in FFGFT as the rational address of a value fixed through a reference point (Dok. 292), not as a pole-mass exactness claim (P42). Dok. 290/291 established the *channel* in which θ lives: not the symmetric magnitude functional (where $2/9$ is counter-blind), but the phase channel, with $\chi = \pi/2$ as the skew-adjoint selector and δ as the still-open second harmonic. The *value* of θ remained a found, not a derived, number.

This document supplies a second, entirely independent route. It comes neither from the masses nor from the Koide parametrisation, but purely from the *geometry* of the doubling $6 = 3 \oplus 3'$ (Dok. 285): the embedding of FFGFT's own C_3 structure into the icosahedral ambient group A_5. This embedding produces $\theta = 2/9$ parameter-free, forward and exactly – with no $2/9$ as input. That a second, methodologically foreign route hits the *same* rational value is the actual statement: not a translation of one into the other, but a convergence in the sense of the established FFGFT logic, in which α, the lepton masses and Koide act as *verifications* of distinct routes pointing at the same constants. The computations are in `ikosaeder_theta_delta.py` (numpy-only, seed 20780458).

ab.2 The C_3-in-A_5 embedding and the full redistribution

The three lepton modes are the Fourier eigenmodes of the Z_3 circulant on the $\mathbb{C}^3$ fibre (Dok. 282/288): the trivial mode $v_0 = \frac{1}{\sqrt{3}}(1,1,1)$ and the two non-trivial, Galois-conjugate modes $v_{1,2} = \frac{1}{\sqrt{3}}(1, \omega^{\pm 1}, \omega^{\pm 2})$ with $\omega = e^{2\pi i/3}$. The electron is a non-trivial mode; which of the two is decided solely by the sign of the angle, not by the value computed here.

The doubling places C_3 inside the icosahedral group A_5; the five-fold generator is a rotation R_5 by 72° about an icosahedral 5-fold axis (standard embedding: axis $(0, 1, \varphi)$ with

$\varphi = \frac{1+\sqrt{5}}{2}$). Since the 3-fold and 5-fold axes point in different directions, R_5 mixes the C_3 modes. The *full* redistribution of the electron mode $v_e = v_1$ has weights $p_j = |\langle v_j | R_5\, v_e\rangle|^2$:

$$\underbrace{p_0 = \frac{2}{9}}_{\text{trivial } v_0}, \qquad \underbrace{p_2 = \frac{5-3\varphi}{9}}_{\text{2nd harm. } v_2}, \qquad \underbrace{p_1 = \frac{2+3\varphi}{9}}_{\text{stays } v_e}, \qquad p_0 + p_1 + p_2 = 1.$$

All three weights carry the denominator $9 = 3^2$ – the same Z_3 structure in which $2/9$ sits as an orbit address. Confirmed numerically (40 digits, mpmath): $p_0 = 0.2222\ldots = 2/9$ exactly, difference $< 10^{-30}$.

ab.3 The decisive finding: $\theta = 2/9$ as the trivial share

The share that, under the icosahedral 5-fold, passes into the *trivial* mode v_0 is

$$\boxed{p_0 = \left|\langle v_0 \,|\, R_5\, v_e\rangle\right|^2 = \frac{2}{9}}$$

exactly and parameter-free. No mass ratio, no Koide parametrisation and no $2/9$ enters the value; it follows from the C_3 modes and the icosahedral rotation alone. The associated *total* leak is $p_0 + p_2 = (7-3\varphi)/9 = 0.238433$. As a parameter-free candidate for the amplitude δ of the second harmonic it misses the $\delta^* = 0.238871$ back-computed from $\theta = 2/9$ by $-0.183\,\%$. With the local slope, the τ-limited tolerance of δ^* is $0.052\,\%$ (Doc. 291); the gap thus corresponds to $\approx 3.5\sigma$, and the leak $(7-3\varphi)/9$ is not compatible with δ^*. The obvious reading that magnitude and angle stem from the same 5-fold redistribution – θ as the trivial, δ as the total share – therefore does not hold: the redistribution yields the angle $\theta = 2/9$ exactly, but not the amplitude δ.

ab.4 Robustness: the specificity of the icosahedral geometry

The value $2/9$ is not a generic property of any rotation, but of the icosahedral one:

- **Axis specificity.** 200 random rotation axes at 72° give the v_0 leak in the range $[0.036;\ 0.452]$; *not one* hits $2/9$. Only the icosahedral 5-fold axes do.
- **Angle and order specificity.** The angle scan shows $2/9$ exactly at 72°; the n-fold series yields $2/9$ only at $n = 5$ ($2/5$ at $n = 3$, $4/9$ at 144°). The rational cleanliness occurs at the distinguished angles.
- **Mode independence.** Both non-trivial modes v_1, v_2 give $p_0 = 2/9$; the trivial v_0 would give $4/9$. The result does not depend on which conjugate mode is read as the electron.
- **Axis choice (discrete, P3).** Of the six 5-fold axes, three ($+\varphi$ branch, 3-5 axis angle 37.38°) give $2/9$, three ($-\varphi$ branch, 79.19°) give $4/9$. The physical embedding $C_3 < A_5$ selects the nearest 5-fold, the $+\varphi$ branch – hence $2/9$. It is a discrete two-way choice, not a free parameter.

ab.5 Two witnesses, one value

The θ question thus has two mutually independent sources that coincide exactly:

Route	Origin	Value
Koide (phenomenological)	from the measured lepton masses; no geometry	$\theta = 2/9$
C_3-in-A_5 (geometric)	from the 5-fold redistribution of the modes; no masses, no $2/9$ input	$p_0 = 2/9$

The two routes know nothing of each other – one knows only masses, the other only geometry – and yet hit *exactly* the same rational number. This is a convergence, not a translation deficit: it is not required that the geometric route translate back into the Koide cosine parametrisation. Just as a derived value need not be identical to its measurement prescription to confirm it, the parameter-free geometric prediction $2/9$ need not take the shape of the phenomenological fit parameter to agree with it. That the geometric value does *not* vary with θ is here the hallmark of a prediction, not a defect: the geometry has no free parameter and names a fixed value.

ab.6 $\theta = 2/9$ as a translation invariant of two models

The "second witness" must be stated more precisely so as not to be misread as an ontological claim. It is *not* about a geometric truth beating a phenomenological one, nor about the torus "really" being the Hilbert space or vice versa. Both – the point/differential geometry of the T^4 and the spectral geometry of the Hilbert space $L^2(T^4) \otimes \mathbb{C}^3$ – are *mathematical models* that are demonstrably and losslessly translatable into one another (Dok. 230/282): the Z_3 fibres, the C_3 Fourier modes, the A_5 embedding are the same structure in both languages, merely represented differently.

The actual finding is therefore methodological, not ontological: $\theta = 2/9$ is an *invariant of the translation*. It appears in the spectral representation (as the redistribution weight p_0) just as in the phenomenological Koide address, because it belongs to the shared structure and not to a single representation. This is precisely what makes it robust: the constant falls out of the structure irrespective of which of the translatable languages one computes in. This lies on the established line in which $\hbar$ and c are conversion factors and not ontological primitives: what counts is what is invariant under the conversion, not the choice of one side as "true". On the question of which of the two models is reality, nothing is asserted here deliberately – it is not needed, and the computation does not supply it. The value of the result lies precisely in its *model independence*.

ab.7 Scale invariance of the pattern versus occupation

No scale enters the geometric value $p_0 = 2/9$ or the ratios: the C_3-in-A_5 geometry knows neither mass nor energy, it yields pure numbers – the angle and the denominator-9

weights. The *pattern layer* – angle plus ratios – is therefore a property of the topology, not of a particular energy range. It holds wherever the Z_3/A_5 doubling occurs; the lepton sector is *one* realisation, namely the one in which the scale E_0 (Dok. 292) places the absolute position exactly on the measured e, μ, τ.

This is to be strictly separated from the *occupation layer*: whether states actually sit at a given scale. The geometry says "*if* a Z_3 triplet is realised here, then with this θ and these ratios" – it does not say "here is one". At another scale the same circulant pattern stands without occupied states having to lie there. This is the same logic as the tower $D(4+3n)$ and the emptiness theorem (Dok. 285): the pattern is present everywhere, the occupation is a separate, scale-bound and dynamics-dependent question; an empty window carries the same pattern as an occupied one, only without particles in it. From the scale invariance of the pattern there follows therefore *no* prediction of new particles at other scales – the universality concerns the structure, not the occupation.

ab.8 Status and open edge

What is secured: The icosahedral doubling structure produces $\theta = 2/9$ parameter-free, forward, exactly and icosahedron-specifically, in independent agreement with the phenomenological Koide value. The amplitude δ does not follow from the same redistribution: the total leak $(7-3\varphi)/9$ is not compatible with δ^*, and δ remains open. Thereby the long-open question – whether the T^4/Z_3 structure forces $\theta = 2/9$ – has, for the first time, a concrete, exact, geometric answer within the already established A_5 doubling (Dok. 285).

What remains open as a derivation step (and explicitly not as an objection to the convergence): the explicit mechanism linking the geometric redistribution weights $(2/9,\ (5-3\varphi)/9,\ (2+3\varphi)/9)$ to the physical mass ratios. This bridge need not run through Koide but can be led directly through the masses or the mode structure; it is the subject of further work. The operational role of the 5-fold in the phase channel is consistent with Dok. 291: R_5 sets a phase $\pi/2$, the value Dok. 291 associates with χ.

Chapter ac

Doc. 294 — FFGFT as a Test Instance for HLV: the Shared φ-Icosahedral Object

ac.1 Question: the same object — what HLV does with it stays HLV's business

One statement stands at the outset and is meant to be unambiguous: FFGFT and HLV work with the *same* geometric object. This is not an interpretation and not a thesis, but a computable fact, established below — the φ-icosahedral axis system on which $\theta = 2/9$ sits as an invariant. Everything else in this document is a consequence of this single finding.

Strictly to be separated from it is a second question, which is here *not* asked and *not* answered: what HLV does with this object — which vortex or edge-mode readings, which derivations and which formulations HLV builds upon it. That is HLV's business, not FFGFT's. Whether those formulations are correct is here neither claimed nor tested nor addressed. FFGFT establishes the shared object and — via E_0 — provides a test direction; for the use HLV makes of the object, HLV alone is responsible.

The purpose of the document is thereby deliberately two-fold and modest. It records that both compute over the same object, and that FFGFT offers a way to measure HLV against the empirical data; and it draws at the same time the boundary beyond which nothing is asserted about HLV.

ac.2 The shared φ-icosahedral object

In the available computational representation of HLV — the cut-and-project of the homology check script `ffgft_hlv_homology_check_v3.py` — physical space is generated by the icosahedral projector with parameter $\tau = \varphi$, $\varphi = \frac{1+\sqrt{5}}{2}$. Its six projected axes are the six five-fold axes of the icosahedron: all fifteen pairwise angles equal $\arccos\left(1/\sqrt{5}\right) = 63{,}435°$. On exactly these axes lives FFGFT's C_3-in-A_5 doubling (Dok. 285, Dok. 293).

It is therefore not two similar structures but the same object. Applying to these axes the five-fold redistribution of the C_3 modes (Dok. 293), the share into the trivial mode is exactly $p_0 = 2/9$. The Koide angle $\theta = 2/9$ is thereby an *invariant on the shared object*, not a translation of one framework into the other. The value is moreover specific to the golden ratio: only $\tau = \varphi$ makes the angle spectrum exactly uniform and the trivial share exactly $2/9$; the silver value $\tau = 1 + \sqrt{2}$ gives $0{,}2534$ and an already measurably shifted spectrum (angles $45°$ and $69{,}3°$). The shared object is thus not just any cut-and-project, but the φ-icosahedral one.

This statement is computational and free of circularity: it follows from the comparison of the projector axes and the mode redistribution, checkable in `ikosaeder_theta_delta.py` (seed 20780458) and in the reproducible angle-spectrum check `a5_bridge_discrimination.py`.

ac.3 Delimitation: which HLV object is shared

The shared object established here is the *geometric* layer of HLV — the φ-icosahedral axis system on which $\theta = 2/9$ sits as an invariant. Strictly to be separated from it is a second HLV object of the same name: the *temporal-dynamic* spiral time, a non-Markov operator with triadic time coordinate $\Psi(t) = (t, \phi(t), \chi(t))$ and a memory channel (Zenodo 20643462). This temporal-dynamic spiral time carries neither A_5 nor the icosahedral projector; it is *not* the object shared here and does not fall under the present finding.

The test direction of this document concerns the geometric layer alone. The comparison of the temporal-dynamic spiral time with FFGFT's time-winding is carried out separately (Dok. 295); there a computational check shows that this object does not carry the self-similarity ratio e of the FFGFT log spiral — on its own definitions it is a winding of constant pitch, not equiangular. Both delimitations together keep the talk of "shared" clean: what is shared is the φ-icosahedral geometric object, not HLV's spiral time as a time operator. The shared invariant between the two layers is the Z_3 alone ($C_3 < A_5$); the dimension counts of the two frameworks (3, 6) coincide, their meaning does not (Dok. 295).

ac.4 The E_0 asymmetry

The shared object is scale-free. The geometry knows neither mass nor energy; it yields pure numbers — the angle and the denominator-9 weights (Dok. 293). Here the actual asymmetry between the two frameworks sets in.

FFGFT possesses a reference point E_0 (Dok. 292) that binds the scale-free pattern to the measured leptons e, μ, τ and fixes the absolute position. FFGFT can thus *compute back empirically* from the pattern: from the shared object, the denominator-9 weights and E_0, absolute masses follow. HLV possesses no such scale. It carries the pattern — the shared geometric object — but no anchor in energy or mass. HLV can therefore obtain no masses from the object alone. It can only do so if it adopts E_0 from FFGFT; in that case, however, it is FFGFT's empirical content that computes, merely expressed in HLV's coordinates.

ac.5 The test direction: from FFGFT to HLV

From the shared geometry and the one-sided scale a directed relation follows. Because FFGFT supplies the missing scale, FFGFT offers a way to test the geometric content of HLV against the data: one equips the shared object with E_0 and tests whether it reproduces the lepton masses. This is a verification direction from FFGFT to HLV.

It is expressly *not* a confirmation that HLV's own formulations are correct. The shared object states only that both use the same geometry — not that the interpretations and derivations HLV erects on this geometry are valid. The test direction tests the correspondence of geometry and masses with FFGFT's means; it does not touch HLV's own construction and does not ennoble it.

Conversely: HLV supplies FFGFT empirically with nothing it does not already have. The pattern is shared, the scale FFGFT has itself. There is no empirical transfer from HLV to FFGFT, and FFGFT does not need HLV for the masses. The empirical arrow is one-sided.

ac.6 What the finding says and what it does not

Established and computationally secured is: the shared φ-icosahedral object; $\theta = 2/9$ as its invariant; the specificity for $\tau = \varphi$; and the E_0 asymmetry, from which the test direction follows.

Not established and not claimed is: the correctness of the formulations HLV contains; an independent empirical confirmation of FFGFT by HLV; and the question of whether the icosahedral projector is *intrinsic* to HLV's fundamental claim or constitutes a modelling choice in the check scripts. The latter is for HLV itself to clarify and is here left open.

ac.7 Methodological placement

Within FFGFT's three methodological layers the finding places clearly. The shared object is a bridge finding: a geometric, algebraically demonstrable correspondence. The back-computation to masses is a core derivation, it follows from ξ and E_0 and is FFGFT-internal. HLV's contribution to the empirical content is zero. In the sense of P35 the statement that HLV is correct remains a candidate and not a derivation; derived is the shared object alone.

ac.8 The first check attempt is superseded

The first check attempt — the homology check `ffgft_hlv_homology_check_v3.py`, a coarse topological point-cloud diagnostic — is superseded by the exact Hilbert-space variant (Dok. 293) and was in any case structurally blind to the load-bearing question: the persistent H_1 of a point cloud does not see the axis-angle structure on which $2/9$ hangs, so that $\tau = \varphi$ (exactly $2/9$) and the control value $\tau = \sqrt{2}$ ($0{,}2139$) produce topologically nearly indistinguishable clouds. Its earlier finding of "only weak separation from generic" is therefore not a statement about φ or A_5, but the expectable blindness of the coarse

statistic. In the spectral variant, by contrast, $\theta = 2/9$ appears directly as the exact invariant $p_0 = |\langle v_0 | R_5 v_e \rangle|^2 = 2/9$; the shared-object finding rests on it, not on the homology diagnostic.

ac.9 The second test: what it resolves and where it becomes circular

The second check attempt — the instrument `a5_bridge_discrimination.py` (seed 20780458) — uses exactly the statistic the first lacked: the angle spectrum of the six projected axes, measured as distance to the icosahedral reference spectrum in which all fifteen pairwise angles equal $63{,}435^\circ$. It thereby resolves the axis-angle structure on which $2/9$ hangs. It is calibrated so that an exactly icosahedral object gives distance zero, while generic random tight frames and a truly generic $\sqrt{2}$ construction lie in a broad generic range. It is thus the instrument that can see φ and A_5 where the first was blind.

What the second test yields on the available representation is two-fold. First, the HLV projector with $\tau = \varphi$ reads exactly icosahedral: all fifteen angles $63{,}435^\circ$, distance zero, and the five-fold redistribution on its axes gives exactly $2/9$. This, however, is a consequence of the construction — the object *is* the icosahedral projector — so the reading is circular and no independent confirmation of A_5. The instrument correctly treats this not as a verdict but makes the circularity explicit. Second, it demonstrates quantitatively that the earlier "generic" control with $\tau = \sqrt{2}$ is not generic: RMS distance $3{,}47$ to the icosahedral reference spectrum (the exactly icosahedral object has distance 0), a two-valued spectrum $61{,}87^\circ / 70{,}53^\circ$, and a redistribution weight $0{,}2139$ that is not $2/9$. This distance lies far below the generic band of random frames (median RMS about 22, 5–$95\,\%$ roughly 17–28): the $\sqrt{2}$ control is thus structured and not generic. The φ-specificity is thereby nailed down: only $\tau = \varphi$ makes the spectrum exactly uniform and the weight exactly $2/9$.

The yield of the second test is thus diagnostic and instrumental, not confirming. It provides the correct, reproducible statistic that resolves what the first could not; it quantifies the circularity of a check of the available object and the contamination of the earlier control; and it isolates φ as the distinguished value. What it does not yet provide is an independent confirmation: that would require an HLV object represented *without* a baked-in icosahedral projector. Until such an object is available, the instrument stands ready, and the shared-object finding rests on the exact spectral invariant, not on a non-circular discrimination.

ac.10 Status

Secured is the shared φ-icosahedral object and the E_0 asymmetry, and from it a test direction from FFGFT to HLV: FFGFT offers the possibility of testing the geometric content of HLV against the measured masses. No correctness of the formulations HLV contains is claimed. One point remains open: whether the icosahedral projector is intrinsic to HLV. Bound to it is the only still possible independent check — a non-circular discrimination requiring an HLV object without a baked-in icosahedral projector; the

instrument `a5_bridge_discrimination.py` stands ready for it. The earlier homology contrast is settled, not open.

$\theta = 2/9$ itself stands independent of all this. It follows from the Z_3 circulant diagonalisation and the icosahedral redistribution (Dok. 293) and needs HLV in no way. The relation recorded here is a shared geometry with a one-sided test direction, not a mutual confirmation.

Part IV

Part IV — Time and Information

Chapter ad

Doc. 295 — The Time-Winding as a Logarithmic Spiral

ad.1 Question: what exactly is FFGFT's time-winding?

In FFGFT, time is the mass circle, woven into the remaining circles via $\tilde{T} \cdot m = 1$ and not a separable factor (Dok. 285). A precise question follows: what exactly *is* FFGFT's time-winding?

This document settles it completely FFGFT-internally — it is *derivable* from the fractal correction. The result in one sentence: the K_{frak} recursion forces the time-winding into a *logarithmic spiral with self-similarity ratio* e, that is, into continuous scale invariance.

ad.2 Three building blocks from the corpus

Three quantities are fixed elsewhere and are only brought together here. First, the fractal dimension of space, $D_f = 3 - \xi_0$ with $\xi_0 = 4/30000$ (Dok. 205, Dok. 101). Second, the fractal correction in the gate: the X rotation is not by π but by $\alpha = \pi K_{\text{frak}}$ with $K_{\text{frak}} = 1 - 100\,\xi_0$ (Dok. 034). Third, the scale run as the recursion $\xi_{n+1} = \xi_n(1 - 100\,\xi_0)$, more precisely $\xi_{n+1} = \xi_n(1 - 100\,\xi_n)$ (Dok. 146).

From $\xi_0 = 1/7500$ follow three exact facts that carry the rest:

$$100\,\xi_0 = \frac{1}{75}, \qquad K_{\text{frak}} = \frac{74}{75}, \qquad \frac{1}{100\,\xi_0} = 75. \tag{ad.1}$$

ad.3 Non-closure: from the 75-closure to the spiral

A winding that should advance by 2π per turn advances gate-wise only by πK_{frak} and *misses* π by $100\pi\xi_0$ per half-turn. Iterated, the winding does not close but precesses — it becomes a spiral. The closure defect per turn is $d = 100\,\xi_0$ in turn units.

For frozen ξ_0, $d = 1/75$ is constant, so the rotation number $74/75$ is rational: the winding closes after *exactly* 75 *turns* and carries a 75-fold structure — the number $75 = 1/(100\,\xi_0)$ drops out exactly. This is the case of constant defect, i.e. of constant pitch. The limiting case of vanishing defect, $d = 0$ ($\xi \to 0$), by contrast closes already after one turn. Both

closed cases are the *rolled-up*, compact reading of the time-winding: a single scale, the winding lies closed and is static. FFGFT already knows a non-closure algebraically, in the non-multiplicative mode composition (Dok. 193); here its geometric-temporal counterpart is added.

ad.4 The scale run makes the spiral logarithmic

When ξ runs over the recursion, the rational 75-closure turns into a non-closing, self-similar spiral. This transition is the *unrolling* of the winding: the spiral is the decompactified counterpart of the closed 75-winding, and "unrolling" here means precisely letting the scale run (the recursion), not merely splitting off a torus factor. The derivation is closed. With $u_n = 1/\xi_n$,

$$u_{n+1} = \frac{1}{\xi_n(1 - 100\,\xi_n)} = u_n\,(1 + 100\,\xi_n + \mathcal{O}(\xi_n^2)) = u_n + 100 + \mathcal{O}(\xi_n), \tag{ad.2}$$

so $u_n \approx 100\,n + 7500$ and hence an *algebraic* $1/n$ decay

$$\xi_n \;\approx\; \frac{1}{100\,(n + 75)}. \tag{ad.3}$$

The cumulative precession defect thereby becomes logarithmic,

$$D(k) = \sum_{n \le k} 100\,\xi_n \;\approx\; \sum_{n \le k} \frac{1}{n + 75} \;\approx\; \ln\!\Big(1 + \frac{k}{75}\Big). \tag{ad.4}$$

With radius $\rho = k + 75$ — the pole of the spiral sits at the effective origin $n = -75 = -1/(100\,\xi_0)$, the same 75 as before — and precession angle $\beta = 2\pi\,D(k)$, it follows that $\rho/75 = e^{\beta/2\pi}$, that is

$$\rho \;=\; 75\,e^{\beta/2\pi} \;=\; \text{logarithmic (equiangular) spiral.} \tag{ad.5}$$

Its self-similarity is thereby fixed: per precession round, $\Delta\beta = 2\pi$, the radius scales by $e^1 = e$. The *ratio e is not assumed but follows* from the algebraic $1/n$ decay of the recursion: it is the base of the natural logarithm produced by the harmonic sum, and no additional parameter enters. The radius offset 75 is likewise not freely chosen — it is $1/(100\,\xi_0)$, the same constant that already carries the 75-closure.

Both are confirmed numerically without leeway (`zeit_logspirale_probe.py`): the fit $\ln\rho = a\,\beta + b$ gives $a = 0.15922 \approx 1/2\pi = 0.15915$ and $e^{2\pi a} = 2.7195 \approx e$; the locally measured self-similarity ratio converges scale by scale $2.734 \to 2.726 \to 2.721 \to 2.720 \to 2.719 \to 2.719$ towards $e = 2.71828$.

ad.5 Continuous, not discrete scale invariance

The self-similarity is *continuous* — a log spiral, a power law — not discretely log-periodic. A test for discrete scale invariance (the residual of $D(k)$ against the smooth $\ln(1 + k/75)$) yields an amplitude near zero and no periodicity in $\ln k$. This agrees with the corpus: Dok. 146 records that the scaling over the factor $1/\xi_0 = 7500$ becomes quasi-continuous through about 13 logarithmic sub-levels, and the fractal dimension stays constant without steps. The time-winding realises the same continuous scale invariance geometrically, as a log spiral.

ad.6 Space and time: the same ξ_0, scaled differently

Space and time carry the same fundamental constant ξ_0 at two places: the spatial box-dimension deficit is ξ_0 (in $D_f = 3 - \xi_0$), the winding defect per gate is $100\,\xi_0 = 1/75$. The factor 100 between them is the recursion coefficient; it links a *geometric* deficit to a *dynamical* defect and privileges no axis. Which axis carries the defect is a matter of projection — it is conventionally noted on time, but is symmetric under $\tilde{T} \cdot m = 1$ (following sections), hence no amplification of time over space.

A boundary must be drawn explicitly here: the box dimension of the time-winding itself is ≈ 1 — it is a curve — and *not* $3 - \xi_0$. The link between the temporal spiral and the spatial fractal dimension is the shared ξ_0, not an equality of dimensions. The time spiral does not carry the space dimension.

ad.7 The choice of axis is a projection

The logarithmic spiral is a property of the whole coupled winding on T^4, not of a single coordinate. What is forced is the *non-closure* with ratio e; onto which axis one places it is a matter of representation. Because $\tilde{T} \cdot m = 1$ couples time and mass inversely and the torus weaves all circles together — nothing factorises — there is no distinguished time dimension that rolls up on its own. A generic winding runs through several circles at once; the pure *time* winding is the special case of a projection. The same non-closure could equally be placed on a space axis or on a mixture — different basis, same invariant.

The assignment to time is thus an artificial shift onto one axis — a convention, not a distinction. The gate defect is conventionally described on the time-winding (Dok. 034); the dual projection (next section) shows, however, that the same non-closure appears with the same factor 100 on the space side. No basis is thereby preferred; each is an equivalent representation of the same invariant — the same representational freedom as between the two equivalent representations (Dok. 206) and in the translation invariance of $\theta = 2/9$ (Dok. 293).

ad.8 Continuous time: the dual projection into space

If, conversely, time is set *continuous* — smooth, without gate-wise rolling up — then the non-closure can no longer sit on time; it must be carried by another axis. The coupling $\tilde{T} \cdot m = 1$ does not allow time to be smoothed in isolation: with $d\tilde{T}/\tilde{T} = -dm/m$ the mass runs along inversely, and with it the spatial-geometric side ($D_f = 3 - \xi_0$). The running winding then appears as a *variation of space* rather than as rolled-up time. It is the same invariant non-closure with ratio e, only rotated into the dual projection: in the present reading time runs (rolls up) and space stays comparable; in the dual reading time is continuous and space varies.

This mirroring has been worked out numerically (`dual_projektion_probe.py`). Because time advances per gate shortened by the factor $K_{\text{frak}} = 1 - 100\,\xi_n$, mass advances by the inverse factor $1/K_{\text{frak}}$ under $\tilde{T} \cdot m = 1$; the defect per step is $100\,\xi_n$ on both sides (the ratio $d_{\text{mass}}/d_{\text{time}} \to 1$ to six digits). The factor 100 therefore does not *move*, it is *mirrored*: the gate factors are inverse ($K_{\text{frak}} \leftrightarrow 1/K_{\text{frak}}$), the amplification 100 is the same. Both

projections yield the same logarithmic growth, and the spiral fit gives the same ratio e for both ($e^{2\pi a} = 2.7195$ and 2.7192); their cumulative defects differ only by a *bounded* constant (≈ 0.0134, not growing with $\ln k$). The core content is thereby representation-invariant and the non-closure preserved: time cannot be smoothed without mass and space taking over the winding — the mass-time duality principle in action. Its distribution over time, mass and space is a matter of projection; the ratio e is not.

ad.9 Placement relative to HLV

The FFGFT-internal statement is thereby *derived*: the $\mathsf{K}_{\mathsf{frak}}$ recursion forces the time-winding to be a log spiral with ratio e, equivalent to a $1/n$ decay of its pitch. Only here does HLV enter — and here it must first be separated that "spiral time" in HLV denotes not one object but two. First the *geometric* time from the factorisation $T^7 = T^4 \times T^3$ (Dok. 285): there time is an A_5 singlet on the S^1 factor, decoupled and separable. Second the *temporal-dynamic* spiral time, the object HLV *calls* by that name: a non-Markov operator with a triadic time coordinate and a memory channel, without A_5 and without an S^1 factor. FFGFT, unlike both, does not factorise; its spiral is the projection of an inseparable whole.

The limiting case of vanishing defect ($d = 0$) noted above corresponds to the fixed point $\xi \to 0$ of the recursion, in which the defect vanishes and the winding closes. That is the *geometric* decoupled singlet time, not the temporal-dynamic object; the running FFGFT winding approaches it asymptotically. The decisive question to HLV is therefore no longer "similar?" but "ratio e, yes or no?" — and it must be put to the right one of the two objects. The following section carries out this check. Ordered within the three methodological layers: the derivation recursion → log spiral is a core derivation, obtained from ξ; the identification with one of the two HLV times is a bridge candidate; no physical claim over HLV is made.

ad.10 Comparison with HLV's spiral time: shared Z_3, no ratio e

The identification of the FFGFT log spiral with HLV's "spiral time" can now be checked against a concrete HLV source — namely against the one of the two objects designated *temporal-dynamic* in the previous section. Formally it is a non-Markov operator with a triadic time coordinate

$$\Psi(t) = (t, \phi(t), \chi(t)), \qquad \Psi(t) = t + i\,\phi(t) + j\,\chi(t), \tag{ad.6}$$

where t notes the laboratory ordering, ϕ the phase organisation and χ a memory; it carries neither A_5 nor a singlet nor an S^1 factor and possesses a memory kernel K_χ with Markov limit $g_\chi \to 0$. Precisely this object HLV calls *spiral time*; the decoupled A_5 singlet (previous sections) is the other. The two share only the name.

ad.11 The temporal-dynamic object does not carry the ratio e

The test question posed above — "ratio e, yes or no?" — targets a geometrically self-similar object. The temporal-dynamic object is not of that category. It is rather an object of open quantum dynamics: its total dynamics is — per Marcel's spiral-time paper — a unitary dilation on the product space $H_S \otimes H_M$ of system and memory, and it is tested there by open-system criteria (process tensor, CP-divisibility, hard reset, finite-bath comparison), not by a geometric self-similarity. It thereby belongs recognisably to the dynamical, not the geometric category; the e-question is put to an object that, by its own construction, makes no such claim. It carries no derived self-similarity ratio, no $1/n$ decay of a pitch and no scale run; its non-closure sits in the memory channel χ, not in a running scale. Its only self-similar internal structure is the golden-ratio / phason order of the associated cut-and-project (the control replaces the φ structure by random irrational phases), that is φ and not e. On the e-criterion the identification is therefore not supported. What both frameworks structurally share is the role of non-closure as the temporal core object and an explicit memory-off limit: FFGFT's case of constant defect ($d = 1/75$ fixed for frozen ξ_0; bounded window, linear $D(k)$) corresponds to Marcel's Markov fallback $g_\chi \to 0$. Same role, different mechanism — geometric accumulation ($D(k) = \sum 100\,\xi_n \to \ln$) against a stochastic memory kernel.

FFGFT's non-closure is available not only geometrically as a log spiral but equivalently in the Hilbert space: as a memory kernel, the Fourier transform of the discrete spectral density $\sum_k g_k^2\,\delta(\omega - \omega_k)$ of the T^4 connection-Laplacian (Dok. 283) — the same inseparable object in two representations ($H = L^2(T^4) \otimes \mathbb{C}^3$, a lossless bijection, Dok. 230/282); this is representational robustness, not additional evidence. The comparison can therefore be run in Marcel's own language, kernel against kernel: FFGFT's kernel is oscillatory and discrete-spectral (revivals, BLP backflow 5.125), Marcel's decaying and bounded — the same object class, a different form (Dok. 283). Precisely the $1/n$ scale run that carries e is what the decaying kernel lacks.

ad.12 Computational check of the e-criterion

The e-criterion is necessary and sufficient via the pitch decay: a winding is equiangular (a log spiral, ratio e per 2π) exactly when the defect per step decays like $1/n$ and the cumulative defect grows logarithmically. A constant defect, by contrast, gives an Archimedean winding with ratio $\to 1$. Both constructions can be measured against this on their own definitions (`spiralzeit_verhaeltnis_probe.py`).

FFGFT carries e: with $d_n = 100\,\xi_n$ one has $d_n\,(n + 75) \to 1$ — the $1/n$ decay — and the spiral fit gives $e^{2\pi a} = 2.71947 \approx e$. Marcel's spiral time does not: its phase $\phi_i = q_\phi\,u_i$ is linear in the internal coordinate and its memory $\chi_i = F(\rho_i)$ is a bounded window; the defect per step stays constant, $D(k)$ grows linearly rather than logarithmically ($D(10^5) = 1333$ against $\ln(1 + k/75) = 7.2$), and the measured ratio is 1.0011. On its own definitions it is thus not a log spiral at all, but a winding of constant pitch.

The geometry HLV actually carries points, moreover, to a constant other than e. Were one to build an equiangular spiral from the $\tau = \varphi$ cut-and-project, its natural self-similarity factor would be golden — $\varphi = 1.618$ per turn, or $\varphi^4 = 6.854$ per 2π — not $e = 2.718$. Both frameworks thus know, in so far as they carry self-similarity at all, different constants:

e on the FFGFT side (from the $1/n$ decay of the ξ recursion), φ on the geometric HLV side (from the cut-and-project). As a bridge condition in the sense of P35 this reads precisely: HLV's spiral time carries e only if its phase law contains a $1/n$ scale run — a running internal scale —; the given definitions do not contain it. That is a candidate, not a derivation.

ad.13 Basis and dimensions: same numbers, different meaning

Whether both frameworks carry the *same* basis can be ordered dimension by dimension. The numbers coincide, the meaning of the dimensions does not. The *three*: FFGFT's three is the internal $\mathbb{C}^3$, i.e. the Z_3 order on T^4/Z_3 (Dok. 282) — a mode and generation index over the torus. Marcel's three is the triadic *time* (t, ϕ, χ) — a decomposition of the one ordering axis into order, phase and memory. In his root-mass note this three coincides with a C_3/Z_3 orbit ($2\pi k/3$, $k = 0, 1, 2$); to that extent both frameworks share the Z_3 ($C_3 < A_5$, Dok. 285). What is shared is the Z_3, not a common three-dimensional coordinate space: one indexes modes, the other time roles.

The *six*: FFGFT's six is not native. It appears only via the HLV embedding $T^7 = T^4 \times T^3$ (Dok. 285); FFGFT itself is four-dimensional (T^4). Marcel's six is the parent lattice $\mathbb{Z}^6$ of the cut-and-project, split into three physical directions ($P_\parallel$) and three internal, phason ones ($P_\perp$). The same number, a different anatomy: a $4 + 3$ torus embedding against a $3 + 3$ cut-and-project. Moreover, in Marcel's case the six is not the basis of the *continuous* spiral time — that one is triadic (the three) — but of the *discrete* G lattice; the two levels are for him explicitly representations of the same operator, not a replacement of it. This corresponds to the representational freedom on the FFGFT side (two equivalent representations, Dok. 206; the rolled-up 75-winding against the unrolled log spiral).

The only concretely established overlap is the φ-icosahedral shared object ($C_3 < A_5$), computationally established but *one-sided* — the test direction is FFGFT→HLV (Dok. 294) — and filtration-sensitive: it separates HLV sharply from crystalline controls but only weakly from a generic $\sqrt{2}$ cut-and-project. Computationally this can be pinned to the axis spectrum (`basis_6d_achsen_probe.py`): the six projected axes of the 6D cut-and-project are equiangular only at $\tau = \varphi$ — all fifteen pairwise angles equal to 63.435° — while $\tau = \sqrt{2}$ gives a two-valued spectrum ($61.87^\circ/70.53^\circ$). The six therefore do not coincide per se; what is equal is the golden-icosahedral projection alone. Basis equality is thus a *candidate*, not a derived identity. The numbers 3 and 6 coincide, the shared invariant is the Z_3, and the dimensions themselves carry different meanings.

ad.14 Status

Secured and derived: the time-winding as a logarithmic spiral with self-similarity ratio e, continuous scale invariance, from the K_{frak} recursion; the exact 75-closure in the frozen case; the shared constant ξ_0, appearing as the spatial box deficit ξ_0 and as the winding defect $100\,\xi_0$ (factor 100 = recursion coefficient, axis-symmetric). The closed 75-winding is here the rolled-up (compact, single-scale) reading, the log spiral the unrolled (decompactified, across-scales) reading of the same time-winding.

The comparison with HLV's spiral time is negative on the e-criterion: the temporal-dynamic object carries no ratio e, but golden-ratio / phason quasiperiodicity; the basis question reduces to the shared Z_3 ($C_3 < A_5$, one-sidedly verified, filtration-sensitive). The sharp residual question shifts to the *geometric* HLV object — whether the factorised layer (A_5 singlet on S^1) knows a scale-invariant object that would be testable against e —; in the available sources no such object is exhibited. The category separation (two spiral times) and the dimension-by-dimension comparison are secured findings; the identification remains a bridge candidate, currently blocked on the e-criterion.

The contrast here is structural, not ontological. Both frameworks — and likewise the two variants within FFGFT, whether time or space is rolled up — are mathematical models, not reality; no factorisation and no choice of axis is thereby ontologically privileged. That one hopes such a model is applicable to reality is the motivation, not a statement about what is.

Reproducible in `zeit_windung_probe.py`, `zeit_logspirale_probe.py`, `dual_projektion_probe.py`, `spiralzeit_verhaeltnis_probe.py` and `basis_6d_achsen_probe.py`.

Chapter ae

Doc. 296 — Time, Openness and Invariance

Doc. 296 Time, Openness and Invariance

Observer, scale and the simplicity of the core

Every observer sits on a scale. They do not see the object — they see the *projection* of the object onto their scale. Projections from different scales must differ, not because the observers err, but because the generative object is richer than any single view. The perspective distortion is not a mistake — it is structural, the unavoidable consequence of the object (T^4, compact, boundaryless) not fitting into any single projection. Hardly any scale can be surveyed simultaneously with another; the results of observation therefore already differ perspectively as soon as a further observer is added.

This is the deeper reason why quantum mechanics and the theory of relativity are so difficult to reconcile: they are projections onto different scales of the same object, and of course they look different — not because one is wrong, but because neither is complete.

All the more remarkable, therefore, is what follows: that the core is so simple. One parameter $\xi = 4/30000$. One relation $\tilde{T} \cdot m = 1$. One compact torus $T^4/\mathbb{Z}_3$. From these fall the lepton masses, the Koide scalar $Q = 2/3$, the phase $\theta = 2/9$ and — via the single anchor E_0, $\alpha = \xi E_0^2$ — the fine-structure constant α — the entire complexity of the Standard Model, the whole richness of the phenomena that appear so overwhelming on the various observer scales. The complexity does not reside in the core — it resides in the projections. The core is simple.

This is not self-evident. It could equally have been that nature starts afresh on each scale, that there is no common root, that the variety is irreducible — that would be the generic outcome. That a single structure instead carries all scales becomes visible only when one stops treating one's own scale as the only possible one.

Reduction is the condition of evaluability

A projection is always a reduction — it leaves out. That is the loss. But precisely by leaving out, it makes a partial domain *evaluable* that would have been submerged in the full object. The full T^4 object carries all symmetries simultaneously; that is exactly why it

is not directly evaluable as a whole. Only the reduction to a scale isolates a symmetry far enough that one can compute with it, measure it, derive something from it. Reduction and focus are the same thing.

This is not a makeshift but the mechanism of cognition: every evaluation needs a focus, and every focus is a reduction. The particle physicist who measures α evaluates the finest scale because they have projected away everything coarser; the cosmologist who measures the redshift evaluates the coarsest because they have integrated away everything finer. Both gain their evaluation through what they leave out.

The decisive point: the new evaluations are not already contained in the full object — they arise at the projection level. A phonon does not exist at the particle scale; it is an evaluation that becomes possible only on the crystalline projection. An orbital is not a property of T^4 but an evaluation at the atomic scale. The projection *generates* evaluable structure that would not exist without the reduction. This is why each new scale brings forth new physics: not because new objects are there, but because new evaluations become possible. A projection that left out nothing would evaluate nothing.

FFGFT's own method is a special case of this: $\theta = 2/9$ becomes evaluable by projecting onto the $\mathbb{C}^3$ fibre (Hilbert-space transformation, Doc. 230/282). On T^4 directly it was hidden; only the reduction to the fibre made it visible as the diagonalisation output of the $\mathbb{Z}_3$ circulant (Doc. 291). The reduction was not a loss but the condition under which the structure became legible at all.

From body to society — the same structure at every level

That observers on different scales see different projections is not merely a physical abstraction — it begins with the simplest biological perception. Two eyes are already two observers at slightly different positions. The separation is small but real; the brain *constructs* a three-dimensional world from two slightly different projections, a world that exists nowhere directly in that form. Depth is a reconstruction, not direct perception. Further sense organs — proprioception, balance, touch, hearing — each contribute their own scale and their own projection, and the synthesis of all of them feels like immediate reality, even though it is a highly processed achievement.

This has a direct consequence for physics: that we *experience* the world as three-dimensional is not evidence that it fundamentally is. It is the finding that three dimensions are the projection onto which our sensory apparatus has been optimised. T^4 beneath this does not contradict the three-dimensional experience — it is the reason why the projection is so stable and consistent.

The same structure continues one level higher. A society is a collection of observers, each on their own scale, each with their own projection. What arises as social reality — norms, values, worldviews, conflicts — is the synthesis of millions of slightly different projections, none of them complete, none of them wrong in any absolute sense, but each perspectively distorted. And as with binocular vision: the synthesis feels like direct perception, even though it is a construction.

This explains several things that would otherwise be puzzling. Conflicts are so persistent because two observers on different scales see the same object and reach different conclusions — both correct for their own projection, both incomplete. As long as no common higher-level scale is accessible, the conflict cannot be resolved by argument

alone; one would have to shift the projection plane, and that is the most difficult thing there is. Historical change is therefore so slow: a society has calibrated its synthetic achievement to a particular scale, which works well within that scale and generates blindness to others. Paradigm shifts are scale shifts, and scale shifts are painful because the previous coherent picture dissolves before the new one has formed.

Science is the systematic attempt to triangulate projections — multiple observers, multiple scales, contradictions treated as signal rather than noise. This is not self-evident; it runs against the natural impulse to take one's own projection for reality.

The arc back to the core: that despite the multiplicity of projections — biological, physical, social — something invariant underlies them all, something that shows itself completely in no single projection but shines through in all of them, is not self-evident. The generic outcome would be that each scale had its own irreducible truth. That there is instead a common core — in physics ξ on T^4, in experience the synthetic achievement that carries all projections — is the actual surprise, at every level.

Cognition: recursive layering and sensory binding

The biological level leads to a foundation stone already laid in the discussion of consciousness: no consciousness without recursion and without sensory input. Both conditions are necessary, and both are FFGFT structural elements.

Without recursion there is no consciousness: it requires the recursive layering that applies to itself — self-reference, accumulating memory, a structure that carries its own history. This is the same generative form as the K_{frak} recursion from which the scale sequence arises: one level generates the next by the same rule. A mere sliding memory window that discards older states does not suffice — the actual achievement is accumulation, the structural carrying of history, not its replacement.

Without sensory input there is no consciousness: cognitive processes never operate unbound. They are always coupled to a substrate that takes in projections of the world through sensory channels. This is the observer binding of this document: not a free-floating process but an observer bound to a scale, receiving the world as a projection.

Thus the two conditions without which there is no consciousness — recursive layering and sensory binding — are exactly the two structural elements FFGFT carries anyway: the recursion and the bound projection. This is *not* a derivation of consciousness; no such claim is made here and it would lie beyond what is shown. It is a convergence at the level of *necessary conditions*: a consciousness-capable system must be recursively layered and substrate-bound, and both are native to FFGFT.

Two delimitations keep this honest. *First*, cognitive and physical processes share the generative form (recursive layering), but on the cognitive side there is as yet no measurable, quantitative recursion law — no "ξ of cognition", no pitch, no ratio e. The form is shared, not (yet) the law. *Second*, the absence of such a law is no evidence against a common nature of the processes: missing measurability speaks not against the underlying structure but only against its present accessibility — as an inadequate measurement technique never measures against what it cannot yet resolve. The question therefore remains open upward: plausibly suggested by the FFGFT structure (everything as a projection of the same recursive object), not refuted, only not shown. The foundation stone — a shared necessary form — stands; the full connection is programme.

Scale sequence as a succession of projection shifts

From particle physics through atomic, crystalline and biological to galactic structures, the same generative object appears fundamentally different at each level — not because the objects differ, but because each scale makes a different symmetry of the core visible while integrating the others away.

At the *particle scale* the projection cuts most finely: the $\mathbb{Z}_3$ symmetry, the $\mathrm{SU}(3)$ colour structure and the discrete masses are direct imprints of the T^4 geometry. The fine-structure constant α and the phase $\theta = 2/9$ are most directly accessible here — the core is nearest on this scale. At the *atomic scale* ($\sim a_0$) the colour structure has already been integrated away and appears as neutral; what becomes visible are electromagnetic resonances — orbitals as standing waves, discrete energy levels as torus geometry on this coarser scale. At the *crystalline scale* ($\sim$ Å–μm) the translation symmetry of the lattice comes to the fore; collective modes (phonons, band structures) arise as new effective degrees of freedom that do not exist at the particle scale. At the *biological scale* (nm–m) broken symmetries dominate — chirality, self-replication, information processing; life is a coherence window, a scale at which a particular kind of complexity can just remain stable, because below it thermal fluctuations and above it gravity force other projections. At the *galactic scale* (kpc–Mpc) everything except gravity has been integrated away; the spiral structure of galaxies carries the same logarithmic self-similarity as the K_{frak} recursion — not coincidentally, but as an expression of the scale invariance of the core at this projection level.

Scale transitions and their anomalies

The transitions between scales are particularly instructive — precisely because anomalies appear there that cannot be explained from within a single scale. A transition is where the previously dominant symmetry tips and a new one comes to the fore; at this tipping point the system is simultaneously shaped by both projections and fully described by neither. This structurally generates anomalies: deviations that appear as disturbances from one side and as harbingers of a new order from the other.

Familiar examples include phase transitions (critical fluctuations whose correlation lengths diverge — the system "senses" on all scales simultaneously that a transition is imminent), the transition from classical to quantum mechanics (decoherence as projection stabilisation, not as fundamental collapse), or the transition from biological to social organisation (emergence of collective intelligence not foreshadowed at the single-cell scale). In all cases the anomaly lies *in the transition*, not within either scale itself — and is therefore not fully explicable from any single scale.

For FFGFT this is a directed pointer: the places where a scale's standard description reaches its limits — unresolved anomalies, unexplained constants, emergent phenomena without derivation — are precisely the places where a projection shift occurs. They are not errors of the theory but signatures of the transition.

Speculation: unobservable intermediate ranges

The scales considered so far are those accessible to us directly or indirectly — through instruments, through consequences, through indirect inference. But the fact that scales

are observer perspectives admits a further thought, which is explicitly booked here as *speculation* (P35: not a derived finding but an open question):

Between the observable scales there may lie ranges that are in principle inaccessible to us — not for technical reasons but structurally, because none of our sense organs or instruments projects onto these intermediate scales. Projections are selective; they make one symmetry visible and suppress others. It is not excluded that there are scales on which coherent structures exist that leave no direct imprint in any of our accessible projections — at most an indirect one, as an anomaly at the edges of neighbouring scales.

This would not be mystical but structurally consistent: if the generative object T^4 is richer than any projection, and if projections are always selective, then the generic outcome is that there are regions of the object captured by none of our projections. Dark matter and dark energy would, in this reading, not be new substances but signatures of such intermediate ranges — structures at scales onto which our instruments do not project, but which make themselves felt as anomalies at the edges of the galactic and cosmic projections.

This remains speculation. But it is directed speculation: it names precisely where to look for evidence — at the transitions, in the anomalies, at the places where a scale reaches its explanatory limits without internal reason.

Starting point: a recursion with no reachable boundary

The K_{frak} recursion

$$\xi_{n+1} = \xi_n (1 - 100\,\xi_n) \tag{ae.1}$$

does not run endlessly in the sense of uniform repetition — it asymptotes. ξ_n decays monotonically towards zero, but never reaches zero in finitely many steps. This is not a gap in the construction; it is its statement: the sequence is infinitely long, but the gaps between successive steps shrink exponentially, precisely by the factor $1/e$ per 2π turn.

What does this mean for experienced, measurable time? The measurable time step is not n but $\rho = 1/(100\,\xi_n)$ — and this reciprocal *grows*. The further the recursion proceeds, the larger ρ becomes and the coarser the time scale on which something changes. Experienced from within, one does not encounter a uniform sequence but a *slowing* dynamics: early steps are dense and fast (small ρ, large ξ, sub-Planck-proximate), late steps are wide and sluggish (large ρ, small ξ, macroscopic). The experienced time is the sum over $1/\rho = 100\,\xi_n$ (integral over $1/\rho$) — and this grows logarithmically: $D(k) = \sum 100\,\xi_n \to \ln(1 + k/75)$; for $k = 10^5$, $\sum 100\,\xi_n = 7.19 \approx \ln(1 + k/75) = 7.20$. The sum over ρ_n itself, by contrast, grows quadratically ($\sum \rho_n = 5.0 \times 10^9$ for $k = 10^5$). The cumulative defect *is* the measurable time evolution (Doc. 295).

From this it follows immediately: experienced time has no end. ρ grows without bound, $D(k)$ diverges, ξ_n never reaches zero. There is no final step, no endpoint, no “afterwards”. The inner side is not simply the mirror image: at $n = 0$ one has $\xi_n = \xi_0$ and $\rho = 1/(100\,\xi_0) = 75$ exactly — this is the starting value of the recursion; it is attained, not merely approached asymptotically, and it is no boundary. According to Doc. 295 the asymptotic inner limit is only the pole of the spiral at $n = -75$ ($\rho \to 0$). No beginning, no end — outward $\rho \to \infty$, inward the pole at $n = -75$: both are asymptotic limits, not reachable boundaries.

Unrolled — temporally continuous — space curved — invariant

These four properties are not accidentally connected; they follow from one another.

Unrolled. The spiral unrolls the scale structure of T^4 into a directed time coordinate. Not because time "really is" a spiral, but because the projection — the observer's position on the spiral — singles out a direction. The unrolling is the transition from the compact picture to the open one: the same object, read from a particular scale.

Temporally continuous. This follows necessarily. The Hilbert-space bijection $H = L^2(T^4) \otimes \mathbb{C}^3$ is lossless (Doc. 230/282, bijection type III, Doc. 270); what is discretely open remains continuously open — otherwise the translation would not be faithful but distorting. The memory kernel (Fourier transform of the discrete spectral density $\sum_k g_k^2\, \delta(\omega - \omega_k)$ of the T^4 connection-Laplacian, Doc. 283) is defined on $\mathbb{R}$, not on an interval. No boundary on the left, no boundary on the right. Revivals without end, BLP backflow without closure — the same open structure, now in the continuous picture.

Space curved. This is the T^4 core itself. The curvature is not inserted after the fact; it is the generative structure: ξ on $T^4/\mathbb{Z}_3$ *is* the curvature from which the lepton masses and $\theta = 2/9$ follow, α via the anchor E_0 (Doc. 006/011/291/293). The geometry is not background — it is the object. Compact means finite volume but *no boundaries*: a geodesic on the torus runs without end, never encountering an edge.

Invariant. This is the deepest of the four points. The logarithmic spiral does not change its size (Doc. 295): it is self-similar — rotated by one turn and scaled by e, it lies congruent with itself. Rotation and scaling are *the same* operation for it. "Inward by $1/e$" and "outward by e" do not describe something growing or shrinking; they only indicate which scale one is currently examining. The object is invariant under this motion — there is no absolute size, only ratios, and ratios are the intrinsic quantities under $\tilde{T} \cdot m = 1$. What changes is the *observer scale*, not the object.

No beginning, no end — and no coincidence problem

The four properties together yield a consistent picture: an invariant, curved, boundaryless object — unrolled into an open, continuous time coordinate through the act of projection. No beginning, no end, no absolute size, no inserted curvature.

This differs structurally from models that require boundaries. A beginning at $t = 0$ is not a prediction but a limit at which the description ceases to hold — the singularity is not a result but an admission. A finite time interval requires a starting point, and the starting point demands a justification: why now, why this state, why these initial conditions. The coincidence problem, the flatness problem, the fine-tuning of initial conditions — all arise because a finite interval has a distinguished location.

FFGFT has no distinguished location. Every observer sits somewhere on an open trajectory; "now" is the current scale, not a privileged moment. The experienced time is not an interval between two points — it is an open trajectory on a boundaryless object.

New problems as a sign of progress

Such a view generates new questions — this is to be expected and is not an objection. An approach that raises no new problems usually also explains nothing new; it is without consequence. The decisive distinction is what *kind* of problems arise: whether they are

directed follow-on questions (what still needs to be computed?) or genuine breaks (does $\tilde{T} \cdot m = 1$ break?).

The exchange is asymmetric in FFGFT's favour: in place of unexplained stipulations — why these masses, why $\theta = 2/9$, where does non-locality come from — there appear consequences of a derivation, consequences that can be posed and whose resolution advances the programme. New problems arising from a deep solution are the signature of genuine progress, not of its absence. The task is not to avoid them but to record clearly which are directed follow-on questions — and so far, all of them are.

Note (P35): The statements in this document are structural in nature — they follow from $\tilde{T} \cdot m = 1$, the recursion, and the compactness of T^4. The cosmological transfer (open time, no singularity) is a *consequence* of this structure, not a standalone cosmological model. As a consequence it must be free of contradiction with the data — not complete in the sense of covering all phenomena of a dedicated model. This distinction between freedom from contradiction and completeness is to be maintained.

Chapter af

Doc. 297 — HLV as a possible subset of FFGFT

Doc. 297 HLV as a possible subset of FFGFT — Candidate note

Starting point

In the course of the bridge analysis between FFGFT and HLV, a finding emerged that is recorded here as a directed candidate note. It is not a closed derivation; it names the next clean step (P35).

Three steps of retrospect

Step 1 — original blockade. The C_3-in-A_5 geometry of the HLV framework (icosahedral projector, $\tau = \phi$, $\theta = 2/9$ as output) was not directly visible on $T^4/\mathbb{Z}_3$. A direct geometric identification was blocked: T^4 is compact 4-dimensional, A_5 is the symmetry group of the icosahedron in 3D — the dimensions do not match straightforwardly.

Step 2 — Hilbert-space transformation opens the path. The lossless bijection $H = L^2(T^4) \otimes \mathbb{C}^3$ (Doc. 230/282) translates $T^4/\mathbb{Z}_3$ into an operator algebra on the $\mathbb{C}^3$ fibre. In $\mathbb{C}^3$, $C_3 < A_5$ is naturally accessible: the $\mathbb{Z}_3$ circulant diagonalisation on the fibre outputs $\theta = 2/9$ as a phase (Doc. 291), and the C_3-in-A_5 geometry is demonstrable as a parameter-free invariant on the same fibre (Doc. 293). The transformation made visible what was geometrically hidden directly.

Step 3 — consequence. If FFGFT contains $C_3 < A_5$ via the Hilbert-space transformation, and HLV is built on $C_3 < A_5$ with $\tau = \phi$, then HLV's geometric core could be a *subset* of the FFGFT structure — not a peer framework, but a specific projection of FFGFT's $\mathbb{C}^3$-fibre geometry.

What is established (from available documents)

- FFGFT contains $C_3 < A_5$ and $\theta = 2/9$ via the Hilbert-space transformation — documented in Doc. 291 (circulant phase) and Doc. 293 (parameter-free C_3-in-A_5 invariant, second witness).

- HLV's geometric object (the ϕ-icosahedral projector) carries the same $C_3 < A_5$ — one-sidedly verified (FFGFT → HLV, Doc. 294).
- The verification direction is asymmetric: FFGFT derives $\theta = 2/9$; HLV books it as a candidate overlap without its own derivation — confirmed by inspection of his available documents (Stage 19 manuscript, HLV-Consciousness-Bridge1, Runs H–J): $\theta = 2/9$ appears in none of HLV's internal derivations.
- His H_2 operator (Run H, `skew_J`) contains $1/\phi$ and $1/\phi^2$, but no C_3-in-A_5 output and no $\theta = 2/9$ as a result. Run H is moreover *negative/underdetermined* (does not survive the u-shuffle specificity test).

What is not shown (open obligation)

That HLV's ϕ-icosahedral layer is *fully* contained in the FFGFT $\mathbb{C}^3$ fibre requires an explicit *embedding map*

$$\iota : \text{HLV ground object} \hookrightarrow L^2(T^4) \otimes \mathbb{C}^3, \tag{af.1}$$

which has not yet been constructed. "Subset" is the structurally plausible conjecture — not a proven theorem.

Next clean step

Construct or falsify the embedding map ι:

1. Show that HLV's ϕ-icosahedral projector is representable as an operator on FFGFT's $\mathbb{C}^3$ fibre — i.e. as an element of $\mathrm{End}(\mathbb{C}^3)$ with C_3 invariance.
2. Check whether HLV's spectral data (Run J, native-6D candidate) are compatible with the discrete spectrum of the T^4 connection-Laplacian (Doc. 283) — or whether they deviate and thus block the embedding.
3. If ι exists: HLV is a specific projection of FFGFT — its geometric layer is a subset, while its dynamical objects (Spiral-Time, U_3) remain independent and unembedded.
4. If ι cannot be constructed: both frameworks share a common object ($C_3 < A_5$, $\theta = 2/9$) but are structurally independent — no subset relation.

Concrete embedding candidate: the Markov limit and below

The question of whether Marcel's Archimedean spiral time finds a structurally defined place in FFGFT can be answered precisely — not as an analogy but as a limit. And the analysis reveals that the object is even weaker than initially assumed.

Archimedean = Markov limit — but U3 is weaker still. The FFGFT memory kernel (Fourier transform of the discrete spectral density $\sum_k g_k^2\,\delta(\omega - \omega_k)$, Doc. 283) becomes Archimedean exactly when the couplings $g_k \to 0$ — the *Markov limit* of open quantum systems. In this limit the kernel still accumulates without bound on $\mathbb{R}$. Marcel's U_3 (Eq. 7 of his Consciousness document) goes further: the window M_m is finite and fixed, older values actively drop out (*sliding window*). This yields a **hard-reset kernel** — a memory kernel with finite support:

$$K(t - \tau) = 0 \quad \text{for } t - \tau > M_m \cdot \Delta t. \tag{af.2}$$

This is not merely Archimedean — it is *periodically bounded*. U_3 does not accumulate without bound; it closes after M_m steps.

Does the window end before the first closure period? In FFGFT's frozen configuration (Doc. 295), the winding closes after exactly 75 turns ($1/(100\xi_0) = 75$) — the smallest coherent unit of the recursion. If Marcel's window width $M_m < 75$ steps (in the winding coordinate), then U_3 does not even accumulate one complete closure period of 75 turns (according to Doc. 295 one step is one turn and 75 is the number of turns until closure). There is no closed winding, no geometric spiral — only a sliding memory window that renews with each step. The object lies **below the smallest coherent unit** of the FFGFT recursion.

Refined embedding candidate. The concrete location of U_3 in FFGFT is therefore not the Markov limit (which still accumulates on $\mathbb{R}$) but the *hard-reset kernel below one closure period*:

$$\iota : U_3 \hookrightarrow \{K \in L^2(T^4) \otimes \mathbb{C}^3 \mid K(t-\tau) = 0 \text{ for } t-\tau > M_m \cdot \Delta t,\ M_m < 75\}. \tag{af.3}$$

This is present in FFGFT — but as the **degenerate boundary case**, not as a regular configuration. The hierarchy is therefore clear:

$$U_3 \subset \text{Markov limit} \subset \text{Archimedean} \subset \text{log spiral } (e) = \text{FFGFT kernel (Doc. 295/283)}. \tag{af.4}$$

Each level is more information-poor than the next. U_3 is the poorest.

Falsification condition

The embedding ι is falsifiable: if Marcel's U_3 carries *more* in his data than the hard-reset kernel — i.e. if non-Markov signatures (revivals, BLP backflow, oscillatory kernel, or correlations beyond M_m steps) are detectable — then U_3 exceeds the boundary case and a deeper embedding is needed.

Marcel's own discipline (U3 ablation test, Table 4 of his Consciousness document) is exactly the right test: if U_3 does not exceed the Markov baseline, the embedding in the boundary case is correct; if it does, a richer layer is needed. The programme is clean — he only needs to run it to the end.

Two HLV spiral times — two different embedding candidates

Doc. 295 §9 records a distinction that is also load-bearing for the embedding question of Doc. 297: "Spiral time" in HLV denotes not one object but two.

First object — the geometric time from the factorisation $T^7 = T^4 \times T^3$ (Doc. 285): an A_5 singlet on the S^1 factor, decoupled and separable. This limit corresponds in the K_{frak} recursion to the fixed point $\xi \to 0$ — the defect vanishes, the winding closes after one turn (according to Doc. 295; the 75-closure belongs to frozen ξ_0 with $d = 1/75$), the pitch becomes constant. This is the *geometric* boundary of the FFGFT spiral.

Second object — the temporal-dynamic spiral time, the object HLV *calls* by that name: the triadic operator $\Psi(t) = (t, \phi(t), \chi(t))$ with a memory channel, without A_5 and without an S^1 factor. This operator is introduced in Krueger et al. (2026,

doi:10.47852/bonviewMEDIN62029043) as a deterministic phase-memory decomposition and implemented as a numerical operator in the HLV G-Lattice Runs H–J (Zenodo, DOI not yet published). In the Consciousness document (Krueger, 2026, v0.2, not yet on Zenodo) it appears as $\Psi(t) = (U_1, U_2, U_3)$ with the explicit implementation Eq. 7 — a bounded window accumulator.

Two different embedding candidates in FFGFT:

The *geometric* object (A_5 singlet, S^1, decoupled) finds its natural place at the $\xi \to 0$ fixed point of the recursion — the geometrically separable, information-poorest configuration of compact time. It is geometrically richer than U_3, but carries no ratio e (Doc. 295 §10: blocked on the e-criterion).

The *temporal-dynamic* object (U_3, hard-reset kernel, $M_m < 75$) finds its natural place in the Markov limit of the FFGFT memory kernel (Doc. 283, $g_k \to 0$) — below the smallest coherent unit of the recursion. It is the more information-poor of the two.

The hierarchy applies to the temporal-dynamic object:

$$U_3 \subset \text{Markov limit} \subset \text{Archimedean} \subset \text{log spiral } (e) = \text{FFGFT kernel}.$$

The geometric singlet lies to the side of this hierarchy — it is Archimedean ($\xi \to 0$, constant pitch), but from geometric factorisation, not from dynamic accumulation. Both boundary cases meet in the Archimedean regime, but they are different paths to it.

Regardless of whether U_3 is represented linearly, in polar coordinates, or as a spiral, the structure of the dependence remains the same: finite short-term memory with a fixed horizon M_m, without accumulation. The spiral representation is a *choice of coordinates*, not a structural property of the object — in every representation U_3 remains a hard-reset kernel with finite support $M_m < 75$. The geometric richness lies in the representation, not in the object. This is the same line as P35 (“translation is not justification”) applied in the other direction: a richer representation does not make the object richer.

The deeper difference lies elsewhere: **accumulation is the actual memory achievement**. U_3 does not accumulate — it replaces. What was before M_m steps is irretrievably gone; no past state acts on the future beyond the window. That is a short-term buffer, not memory in the structural sense.

The K_{frak} recursion is the opposite: $\xi_{n+1} = \xi_n(1 - 100\,\xi_n)$ accumulates each step into the next — not as an explicit store, but as *structural imprinting*: the present state ξ_n carries the entire history of all preceding steps within itself. No step is forgotten, no window erases. The past is encoded in the object itself, not in a separate memory. This is memory in the deepest sense — not as storage but as a generative structure that carries its own history and from it produces the log spiral with ratio e (Doc. 295).

That is precisely what Marcel’s consciousness framework requires (Section 1 of his Consciousness document: “memory-bearing, recursively integrated”) — but what U_3, by construction, does not deliver. The recursion would be the right carrier. U_3 is the degenerate boundary case below it.

The window as reduction — in the sense of Doc. 296

The window M_m in U_3 is itself an instance of the reduction principle of Doc. 296: a *projection onto a limited temporal segment*, just as a scale projection is a limited spatial segment. It leaves out (everything beyond M_m) and thereby makes a partial domain

evaluable — the local, most recent dynamics. This is the same mechanism as in Doc. 296 (reduction as the condition of evaluability), only applied to the time axis rather than the scale axis. The window acts in the same sense as any projection.

The decisive difference therefore lies not in the fact *that* U_3 reduces — every evaluable description reduces — but in *whether the whole is named of which the reduction is a segment*.

FFGFT's reductions are exhibited as reductions of a known whole: the frozen 75-turn winding and the Markov limit of the memory kernel are segments of the full, accumulating, scale-invariant structure — one knows what has been left out and how to recover it (let the recursion run, do not freeze ξ, Doc. 295/283). The way back is laid out.

Marcel's window, by contrast, stands on its own, without the full object behind it — it is the projection without the projected. Recovery is not laid out because the full object is not named. The window evaluates the local temporal structure — a legitimate focus — but it is not recognisable as a segment of a boundaryless whole (Doc. 296: the embedded observer who sits on their scale without unrolling the recursion).

This is precisely what makes the embedding ι natural: it supplies the missing whole. U_3 as a reduction receives back, in FFGFT, the object of which it is a segment. The window acts in the same sense as any projection — the only difference is that FFGFT knows the way back and HLV does not (so far) lay it out.

Pre-registered prediction for the OP8-Neuro test (6 July 2026)

Marcel Krueger has published, with HLV-CONSCIOUSNESS-BRIDGE1 v0.3a (OP8-LOCK), a data-freeze / protocol-lock that fixes — before the run — a model ladder and a null family:

M_0	$\Delta\Phi$ / S-I-C only
M_1	U_1 only
M_2	$U_1 + U_2$
M_3	$U_1 + U_2 + U_3$
M_4	generic ML baseline

Central question (his formulation): whether the full $U_1/U_2/U_3$ model (M_3) provides measurable additional value over U_1/U_2, information-only, coherence-only, **memory-free Markov**, phase-shuffled, time-shuffled, and the generic ML baseline — the U_3 *necessity test*.

This pre-registered test is simultaneously the decisive test for the embedding hypothesis of this document. FFGFT makes the following prediction about it — *before* Marcel's run, explicitly booked as a prediction (P35), not as a claim on his result:

1. **If U_3 is a hard-reset window with $M_m < 75$ (winding coordinate)** — i.e. a segment below one full closure period of 75 turns, the Markov limit of the embedding ι — then M_3 **does not significantly outperform the memory-free Markov baseline**. U_3 then lies entirely in the degenerate boundary case; the additional channel carries no structure beyond Markov.
2. **If M_3 clearly outperforms the memory-free Markov baseline**, then U_3 carries non-Markov structure (correlations beyond M_m, effective accumulation) that the hard-reset

kernel does not explain. In that case the simple embedding ι into the Markov limit is *falsified*, and a richer embedding must be constructed.

The prediction is sharp because it hangs on Marcel's own construction: a bounded window M_m of fixed width (Eq. 7 of his Consciousness document, "avoids uncontrolled accumulation") can by construction carry no correlations beyond M_m — so the expected outcome is (1). An outcome (2) would indicate that U_3, in implementation, does more than its definition suggests — which would be independently interesting and would then need to be localised.

This prediction does not touch Marcel's claim boundaries: it says nothing about consciousness, no physical validation, no HLV ontology. It concerns solely the structural question of whether U_3 carries beyond the Markov limit — and that is exactly what his OP8 test probes.

Status

Candidate note. Directed speculation on the basis of secured partial findings (P35). The subset conjecture is structurally plausible and falsifiable — that is its value. It is not a claim.

Chapter ag

Doc. 298 — The Standard Model as a decompactified projection

Doc. 298 The Standard Model as a decompactified projection — FFGFT as the unifying explanatory model

Starting point: the constants used are not neutral inputs

Every calculation that uses the fine-structure constant α, the Higgs vacuum expectation value v, or a Yukawa coupling carries the K_{frak} recursion *indirectly* with it, from the FFGFT point of view — whether or not this is named. For these quantities are not free inputs in FFGFT but outputs of the recursion:

- $\alpha = \xi E_0^2$ (Doc. 006/292), where E_0 already contains the K_{frak} factor $(1 - 100\xi)$ absorbed ($7{,}398/7{,}348 = K_{\text{frak}}^{-1/2}$, Doc. 190/P43). Whoever uses α uses a value into which the recursion has entered.
- The masses are the ξ^p ladder ($m = r\xi^p v$, Doc. 292); a Yukawa coupling $g = m/v$ is therefore already a ξ-ladder ratio. Whoever writes Yukawa writes a recursion ratio in different notation.

The Standard Model takes these quantities as *measured* inputs; FFGFT *derives* them. Therein lies the structural relationship this document describes.

The decompactified domain

Whoever uses α, v, or Yukawa couplings in continuous fields and potentials on $\mathbb{R}$ — a Yukawa potential $V(r) \propto e^{-mr}/r$ on open space, a continuous scalar field, a mediator scale — works structurally in the *unrolled* picture. This is exactly the domain into which FFGFT decompactifies the compact recursion on T^4 (type-I decompactification, Doc. 270): the compact origin of the constants is carried along but not named.

This is the same pattern as reduction without a named whole from Doc. 296/297: a projection that evaluates correctly but does not name the object of which it is a segment. With U_3 (Doc. 297) it was the time window without the full winding; here it is the decompactified constant sector without the compact recursion from which α and the Yukawa ratios originate.

FFGFT as the unifying explanatory model

Here lies the actual, positive content. FFGFT connects quantum mechanics and the Standard Model not by replacing them but by placing both as *projections of the same compact object* (Doc. 296): QM and relativity are projections onto different scales, and the Standard Model is the decompactified single-scale projection of its constants.

From this follows a property that distinguishes FFGFT among the known frameworks: it is in principle able to *incorporate any already-working partial model as a projection*. A model that computes correctly on its scale — the Standard Model in the particle sector, atomic physics in the orbital sector, an effective scalar channel in precision spectroscopy — is, from the FFGFT view, not a competitor but a projection for which FFGFT specifies the place where it sits. The working partial model is not refuted; it is located.

The mechanism for this is reduction as the condition of evaluability (Doc. 296): every projection leaves out and thereby makes a partial domain evaluable. FFGFT supplies the whole of which these partial models are segments — and can therefore incorporate them, provided they work on their scale. What a partial model correctly describes on its scale remains correct; FFGFT adds from which compact object the constants used originate and how the scale connects to the others.

The placement is conditional (P35)

Note on referencing: "P35" denotes, throughout the corpus, the correction discipline maintained in the correction register of Dok. 190 (there as ξ^N triviality: a quantity of the form $X = \xi^N$ or $X = \varphi^N$ is by itself void of content; only a forward-derived exponent counts as content; "candidate" ≠ "derived", verify numbers before claiming). The uses of P35 in this document are to be read in this sense.

The statement "the Standard Model is a decompactified subset of FFGFT" is an *FFGFT-internal* placement, not an externally enforceable proof. From the Standard Model's point of view, α and v are measured natural constants — full stop. The subset relation holds *under the assumption* that FFGFT's derivations are correct: *if* $\alpha = \xi E_0^2$ and the ξ^p mass ladder are correct, *then* the Standard Model is the unrolled projection that uses these constants as inputs without naming their compact whole.

This conditionality is to be maintained — the same discipline FFGFT also demands of competing cosmological models. The Standard Model can neither refute the containment relation (it computes correctly in the unrolled picture) nor confirm it from within (it names no compact object). It is a placement, not a claim about what the Standard Model itself asserts.

What is established and what remains programme

To keep separate:

- **Established** is the containment relation for the sector FFGFT actually derives: α (Doc. 006/292), the lepton masses and the Koide scalar $Q = 2/3$, the phase $\theta = 2/9$ (Doc. 291/293). For these quantities it is shown that they are recursion outputs.
- **Programme** remains the *full* Yukawa matrix: quark masses, mixing angles (CKM/PMNS), CP phase. For these the subset relation is not worked out — here "the Standard Model

is a subset" would be a promise, not a shown derivation. This part is to be carried as an open obligation, not as a result.

This separation is the same as throughout the corpus: core derivations (proved from ξ), bridges (algebraically proved), reductions (plausibility sketches). The SM subset relation is core derivation for the lepton sector, programme for the full Yukawa matrix.

Extension within the projection — the history of the Lagrangian

A misunderstanding must be cleared up here. The question is not whether the Standard Model is "complete" — that is too broad. The Standard Model *already is* a projection onto a partial domain: the decompactified single-scale projection of its constants. Any attempt to extend it therefore takes place *within this already-projected partial domain*.

This has a structural consequence that holds before any concrete extension: in every projection one can *refine*, but not go *beyond* its edge. A projection has already left out the compact whole; an extension within the projection cannot retrieve what was left out. It can only place more structure into the already-projected segment. The edge of the projection is the limit of the extension — and this limit stands fixed before the extension begins.

This is exactly the history of the Standard Model and its Lagrangian. One sees an incompleteness — neutrinos are massive after all, an anomaly appears, a sector does not fit — and *patches on*: a further term, a further field, a further effective coupling is added to the Lagrangian. The Lagrangian grows, but it grows *within the same projection*. Every addition stays in the decompactified partial domain; none crosses its edge towards the compact whole. Patching refines the projection, it does not switch it.

The telling evidence for this is that every such extension must *import* its building blocks. The quantities an extension needs — α, v, mass ratios — are not derivable within the partial domain, because they belong to the left-out whole. They are brought in from outside, as measured inputs. That an extension must import, rather than generate from its own framework, is the sure sign that it works within the reduction: what is missing belongs to the whole that the projection does not name.

The difference from FFGFT is therefore not gradual but structural. FFGFT does not patch the projection — it names the compact whole from which the projection and its constants originate. It does not extend within the partial domain but gives the partial domain as a segment of a whole. This is why FFGFT need not import α: it derives α from ξ (Doc. 006/292). The import is the signature of extension within the projection; the derivation is the signature of the named whole.

The inner view — a pre-Copernican analogy

Patching within the projection has a historical precedent. Pre-Copernican astronomy was not *wrong* in the sense of poor calculation — epicycle astronomy predicted planetary positions precisely for centuries. It was an *inner view*: the perspective of an observer who takes their own location to be the centre. When a deviation appeared, one patched on — a further epicycle, a shifted deferent, an equant. The system grew but stayed within the same inner perspective.

This is structurally the same pattern: the inner view = the projection that takes its own location (the one scale) to be the whole; patching on epicycles = adding terms to the Lagrangian; precise but not explanatory = the calculation holds on the scale but does not name the whole.

The Copernican step was not a better fit — at first it was even worse (Copernicus kept circular orbits). It was a *change of perspective*: the system named as a whole from which the apparent motions follow as projections. The retrograde motion of the planets was no longer an epicycle but a projection effect of the Earth's motion — the same phenomenon, but located rather than patched on.

This is exactly how FFGFT relates to the Standard Model: the apparent arbitrariness of the 19 parameters is, from the inner view, a set of numbers to be adjusted; from the outer view they are projection effects of a recursion ($\alpha = \xi E_0^2$, the ξ^p mass ladder). Not fitted better — located differently.

Two limits keep the analogy honest (P35). *First*, it describes the *structure* of the two views but does not guarantee the historical outcome: Copernicus prevailed because Kepler and Newton made the outer view *quantitatively* superior. FFGFT is conceptually simpler (one parameter instead of nineteen) and derives the lepton sector; full quantitative superiority across all sectors remains programme. *Second*, the very fact of unlimited patchability is the mark of the inner view: it can never fail from within, only be overtaken from without. That the Standard Model can absorb every new anomaly through a further EFT term is therefore not a sign of its strength but the signature of an inner perspective.

HLV-YUKAWA-BRIDGE1 as a special case

Marcel Krueger's HLV-YUKAWA-BRIDGE1 (Zenodo, DOI 10.5281/zenodo.21224030) makes this pattern explicitly visible. His effective bridge strength is parametrised as

$$g_\chi^2 = \eta_\chi \, \alpha \, \varphi_g^{-2k}, \qquad \varphi_g = \frac{1+\sqrt{5}}{2}, \qquad k \in \mathbb{N}. \tag{ag.1}$$

This construction stands with one foot in FFGFT territory and one in HLV territory, in the unrolled space: α is an FFGFT recursion output (which carries the recursion along, see above), φ_g is the HLV-side eigenvalue that the same crystallographic restriction ($2\cos(2\pi/5) = 1/\varphi \notin \mathbb{Z}$) forbids in FFGFT and assigns to HLV (Doc. 190/R44). Marcel himself names η_χ as the main vulnerability and warns that the construction could, without leakage-free fixing of k and η_χ, degenerate into parametrisation rather than prediction — the same P35 discipline.

Thus this more conservative HLV channel too is structurally *downstream* of FFGFT: a decompactified construction using an FFGFT recursion output (α) as a building block. It cannot claim to be independent of the recursion content of that quantity — it is carried along, whether named or not. The sector itself (isotope-shift/King-plot, nucleus–electron scalar force) is disjoint from the FFGFT lepton sector; there is no conflict but a placement relation: a working partial channel that FFGFT can in principle incorporate as a projection.

What Marcel would have to prove

For the Yukawa bridge to become a falsifiable channel — and further, to step out of the projection — a staged proof burden must be carried. The first four points are his

stated programme and methodologically the right path; the fifth is not laid out within his framework.

Within the projection (falsifiable channel):

1. **Fix η_χ independently.** As long as η_χ is a free adjustment factor, Eq. (9) is a parametrisation, not a prediction. η_χ would have to be *derived* from HLV geometry, not fitted to data. Marcel names this himself as the main vulnerability — the test is not yet passed.
2. **Lock k before the holdout data.** The index $k \in \mathbb{N}$ must be fixed before the King-plot data are examined — otherwise target leakage. Pre-registration discipline, necessary, but not a physics proof.
3. **Beat the generic scalar (his $N3$ null).** If a generic scalar mediator with free coupling and mass explains the data as well as the φ_g^{-2k}-locked parametrisation, the HLV *specificity* is not supported. He would have to show that the φ structure hits something the free scalar does not. This is the hardest of the four.
4. **Survive the ordinary nonlinearities ($N1$/$N2$).** King-plot deviations also arise from nuclear structure (finite charge distribution, higher mass shifts) and many-body effects — entirely without a new scalar (his references on Yb, Ca). He would have to show that a signal remains *after* subtracting these known effects.

If he passes (1)–(4), he has a genuine, falsifiable BSM channel — a clean refinement of the partial domain. But he would thereby remain *within* the projection, because he imports α and φ.

Out of the projection (change of perspective):

5. **Derive η_χ and the φ power from a named whole, instead of importing them.** Only if it were shown where α and φ in the formula *come from* — from which compact structure — would the bridge move from patching to a change of perspective. As long as α is an imported input (from the FFGFT view a recursion output) and φ a *held*, not *derived*, eigenvalue (Doc. 190/R44), the bridge refines the inner view; it does not leave it.

The finding is thus cleanly separated: (1)–(4) are necessary for a falsifiable channel and are Marcel's stated programme; (5) is the step out of the projection and is not laid out within his framework — HLV *has* φ as an eigenvalue but does not derive it, and α is taken from outside. The import remains the signature of extension within the projection.

Status

FFGFT-internal placement, conditionally formulated (P35). Established for the lepton sector (α, Koide, $\theta = 2/9$ as recursion outputs), programme for the full Yukawa matrix. The positive content: FFGFT is the unifying explanatory model that places QM and the Standard Model as projections of the same compact object and can in principle incorporate any partial model working on its scale as a projection — not as a refutation but as a placement. No unconditional claim about what these partial models assert from within.

Chapter ah

Doc. 299 — The Embedding ι — Working Document and Forcing Protocol

Doc. 299 The Embedding ι – Working Document and Forcing Protocol (HLV time sector $\hookrightarrow$ FFGFT recursion)

Purpose and scope

Doc. 297 records the embedding

$$\iota : \text{HLV base object} \hookrightarrow L^2(T^4) \otimes \mathbb{C}^3 \tag{ah.1}$$

as an open conjecture: structurally plausible, but not constructed. This document sets up the next clean step (P35) as a *protocol* – it fixes, *before* the joint work, the domain, the codomain, the candidate map, the matched null models, and the decision criterion. It is not a proof and not a claim about an outcome; it names what would have to count as *forced* and what as merely *allowed*, so that the later evaluation cannot be adjusted after the fact.

The frame is the *time sector*. Following the sharpening in the correction register (Doc. 190, R47), two senses of containment must be kept apart: *dimensionally*, HLV counts larger ($T^7 = T^4 \times T^3$, $7 = 4 + 3$; R43); *in content, in the time sector*, HLV $\subset$ FFGFT, because the HLV spiral-time window is a bounded section of the unrolled recursion time. In unified notation – both $D6+$ time, HLV as $D6+$ spiral-time, FFGFT in Hilbert space as $D6+$ recursion-time – the window is not additive but internal. ι is exactly the map that makes this internal section explicit.

That it is a section takes no value from the window (Doc. 298): a sub-model that evaluates correctly on its scale is not refuted but located. ι returns to the section the compact whole of which it is the reduction.

Domain: the HLV side

Doc. 297 (§"Two HLV spiral-times") separates two distinct objects, both of which HLV calls "spiral-time". They are separate embedding candidates and must not be conflated in the protocol.

Object A – the geometric time. The A_5 singlet on the S^1 factor of the factorisation $T^7 = T^4 \times T^3$ (Doc. 285): decoupled, separable, without a memory channel. Purely geometric.

Object B – the temporal-dynamic spiral-time. The triadic operator $\Psi(t) = (U_1, U_2, U_3)$ with memory channel U_3 (Krueger, Consciousness document, Eq. 7): a bounded window accumulator with a fixed, sliding window M_m; older values actively drop out. In every representation – linear, polar, spiral – the dependency structure is the same: finite short-term memory with fixed horizon M_m, without accumulation (Doc. 297: the spiral representation is a coordinate choice, not a structural property).

Codomain: the FFGFT side

The codomain is the compact recursion in the Hilbert space $H = L^2(T^4) \otimes \mathbb{C}^3$ (Doc. 230/282, lossless bijection). Three structural elements are decisive for ι:

- **The memory kernel.** The Fourier transform of the discrete spectral density $\sum_k g_k^2\, \delta(\omega - \omega_k)$ (Doc. 283) – the full, accumulating memory structure of the recursion.
- **The frozen winding.** In the frozen configuration (Doc. 295) the winding closes after exactly $1/(100\,\xi_0) = 75$ turns; this is the smallest coherent unit of the recursion.
- **The $K_{\mathbf{frak}}$ recursion** $\xi_{n+1} = \xi_n(1 - 100\,\xi_n)$: the present state carries the entire history as a structural imprint; it generates the log-spiral with ratio e (Doc. 295). Memory as generative structure, not as storage.

The candidate map ι (two branches)

Matching the two objects of the domain, ι has two branches with different natural locations.

$\iota_{\mathbf{geo}}$ (Object A). The A_5 singlet (S^1, decoupled) maps to the $\xi \to 0$ fixed point of the recursion – the geometrically separable, most information-poor configuration of the compact time (constant pitch, winding closes after one turn); the 75-closure holds for frozen ξ_0 with $d = 1/75$ (Doc. 295). It is Archimedean from geometric factorisation, but carries no ratio e (Doc. 295 §10: blocked on the e-criterion).

$\iota_{\mathbf{dyn}}$ (Object B). The window U_3 maps to the hard-reset kernel with finite support below one closure period ($M_m < 75$ turns):

$$\iota_{\text{dyn}} : U_3 \;\hookrightarrow\; \big\{ K \in L^2(T^4) \otimes \mathbb{C}^3 \;\big|\; K(t-\tau) = 0 \text{ for } t - \tau > M_m \cdot \Delta t,\; M_m < 75 \big\}. \tag{ah.2}$$

This corresponds to the Markov limit of the memory kernel ($g_k \to 0$, Doc. 283), sharpened to a bounded window. Both branches are bounded sections (windows) of the unrolled recursion time; this is exactly the content-level HLV $\subset$ FFGFT of the time sector.

The information hierarchy (Doc. 297) locates the branches:

$$U_3 \;\subset\; \text{Markov limit} \;\subset\; \text{Archimedean} \;\subset\; \text{log-spiral}\,(e) \;=\; \text{FFGFT core}. \tag{ah.3}$$

The geometric singlet lies to the side of this chain (Archimedean from factorisation, not from accumulation). Each step is more information-poor than the next; U_3 is the poorest.

The forcing question: forced or merely allowed

The core of the protocol. That a section *exists* into which U_3 embeds is weak – a bounded window operator can be placed into the degenerate boundary case of any sufficiently rich recursion. The substantive question is whether the embedding is *forced* or only *allowed*. This is the same forcing logic as in the C3A5 specificity test (Doc. 293/294), transferred to the time axis.

Forced (HLV $\subset$ FFGFT **in the time sector, genuine).** The Markov-limit section of the recursion *necessarily* coincides with the hard-reset kernel of U_3, and an independent invariant of the section agrees with U_3 where generic bounded window operators do not. ι is then the unique structural placement.

Allowed (mere compatibility). ι merely embeds compatibly: every bounded window operator would sit equally in the degenerate boundary case, and the match carries no content beyond the definition (P35).

Circularity caveat (P35). U_3 is by construction a hard-reset window. Reading it as the Markov-limit section of the recursion is therefore initially *definitional*, not an independent confirmation – just as the icosahedral object gives icosahedral reference distance zero because it *is* that object (Doc. 294). The forcing must therefore come from an *independent* signature that U_3 shares with the recursion section without both being defined by the same construction rule.

Matched nulls (time axis, ι_{dyn})

The forcing is tested against a family of matched null models – the temporal counterpart to axis shuffle and generic tight-frames (Doc. 293/294). It also mirrors Marcel's own OP8 null family (Doc. 297), so that the instruments interlock:

- **memory-free Markov** – the baseline that ι_{dyn} predicts in the boundary case;
- **phase-shuffled** and **time-shuffled** – destroy temporal structure while preserving marginal statistics;
- **coherence-only** and **information-only** – isolate single channels;
- **generic window operator** – random M_m, random kernel with the same support: the direct analogue of "random rotation axis / generic tight-frame";
- **generic ML baseline**.

Calibration as in the specificity test: an exact recursion section gives distance zero, generic window operators lie in a broad band. ι_{dyn} is *forced* exactly when U_3 sits at the zero point where the generic nulls lie in the band; *allowed* when every window null reaches the same distance.

Matched nulls (geometric branch, ι_{geo})

For the geometric branch the null family is the spatial counterpart – axis shuffle and generic frames in the sense of Doc. 293/294, here in the form agreed with the HLV side:

- **random C_3/A_5 embeddings** – the native carrier against arbitrarily rotated embeddings of the three-/five-fold structure;
- **alternative projection stars** – cut stars other than the icosahedral one;
- **tetra-/octahedron controls** – crystallographic point groups without five-fold symmetry as a negative control;
- **shuffled C_3 bases** – permuted fiber bases with preserved marginal statistics;
- **spectrally matched null carriers** – carriers with the same power spectrum but without the icosahedral axis structure.

Calibration as in the $C3A5$ specificity test (Doc. 294): the exact icosahedral projector gives distance zero to the reference spectrum (all fifteen angles 63.435°, redistribution $2/9$), generic frames lie in a broad band (median RMS ≈ 22). ι_{geo} is *forced* exactly when the native carrier sits at the zero point where these nulls lie in the band; *allowed* when generic embeddings reach the same distance.

Jointly booked architecture. Both null families are agreed with the HLV side. Krueger (IOTA-BRIDGE1 v0.1) tests the same finite proxy question – a native-6D spiral-time-window carrier as a C_3 fiber-compatible object in a finite $L^2(T^4)\otimes\mathbb{C}^3$ proxy – under spectral, C_3/A_5, U_3 time-window, and screen/information diagnostics. The claim boundary is tight on both sides: PASS $=$ finite-proxy ι-forcing candidate (no subset proof, no physics/mass/QFT/GR derivation); BLOCK $=$ this concrete finite embedding route is narrowed or blocked.

Decision criterion and falsification

The pre-registered OP8-Neuro test (Doc. 297) is the outer decision test; the forcing protocol adds the specificity step:

1. M_3 **does not beat the memory-free Markov baseline.** U_3 lies entirely in the degenerate boundary case. ι_{dyn} holds in the Markov limit – but from this alone follows *allowed*, not *forced* (every hard-reset window achieves this). The specificity run is additionally required for forcing.
2. M_3 **clearly beats the memory-free Markov baseline.** U_3 carries non-Markov structure (revivals, BLP backflow, oscillating kernel, correlations beyond M_m). The simple ι_{dyn} into the Markov limit is *falsified*; a richer embedding higher in the hierarchy (towards log-spiral e) is to be constructed.
3. **Forcing verdict (independent of 1/2).** A specificity run – the time-axis analogue of `a5_bridge_discrimination.py` – measures the distance of an *independent* invariant of the recursion section to U_3 against the matched nulls. Distance zero at U_3 and a broad band at the window nulls $\Rightarrow$ forced; otherwise allowed.

To be constructed (deliverables)

1. For ι_{geo}: represent HLV's ϕ-icosahedral projector explicitly as an operator on the $\mathbb{C}^3$ fiber ($\in \text{End}(\mathbb{C}^3)$ with C_3 invariance; Doc. 297, step 1).
2. For ι_{dyn}: compute the Markov-limit kernel of the recursion ($g_k \to 0$, Doc. 283) and its distance to U_3's hard-reset kernel against the matched-null family – the time-axis analogue of the specificity test.

3. Fix the forcing metric: distance-to-recursion-section, calibrated so that an exact section $\to 0$, a generic window $\to$ broad band.

Domain, codomain, and null family are fixed by this document; what remains open is only the construction of the three artifacts.

Counter-run performed (result)

If the spiral-time window is reduced to a *winding-less* hard-reset kernel, it carries no back-flow – which is near tautological, since a kernel without winding cannot carry a revival by construction, and it does not test Marcel's actual object. His object is defined by a *winding angle* θ (the pitch of the archimedean spiral in recursion time); the angle *is* the structure. The counter-run (`python/Dok299_Skripte/ffgft_299_iota_gegenlauf_winding_locked.py`, SHA256, seed 20260708) writes θ explicitly into the kernel, $k_\theta(s) = e^{-s/\tau}\cos(\theta s)$, $s = 0 \dots M{-}1$, and sweeps it. Object and witness stay apart (witness: BLP back-flow 5.125, Doc. 283); the nulls share the *same* envelope with incoherent phase – isolating the contribution of the angle.

Winding θ	Back-flow	Reading
0 (no winding)	0.00	Markov limit (in band)
$1/\phi \approx 0.618$	0.00	in band (allowed)
golden angle $\approx 137.5^\circ$	0.159	**above band**
phase-shuffled null (q95)	0.076	incoherent bound

Once the angle is in the kernel, the answer is no longer fixed by construction. At most windings the hard reset (finite width $\sim 1/M$) kills the revival $\Rightarrow$ back-flow in the null band, allowed. At the *golden angle*, however, it survives the broadening and rises above the incoherent band (0.159 vs. q95 0.076); the first threshold is at $\theta_c \approx 1.82$ rad. The verdict is therefore **conditional rather than decided in advance**: allowed if the native carrier's actual winding θ^* lies below θ_c; forced (and Doc. 297 under tension at that point) if θ^* sits at the golden angle. θ^* is to be pinned from the HLV carrier – marked here as a candidate, not asserted. The absolute value (0.159) is small against the witness 5.125 and the test remains a finite proxy; but *relative to incoherent phase* it is a positive signal at that winding, not a null result. An unconditional reading would be an artifact of the winding-less reduction.

The geometric branch (ι_{geo}) is carried along as a discriminant test: the 15 pairwise axis angles against a *fixed* icosahedral reference (63.435°), not against axes taken from the carrier. Icosahedral $\to 0^\circ$, random-star median 21.7° – the fixed reference separates where a self-referential score would pull any star to ≈ 0. This is the anti-circularity condition a forcing test in the geometric branch must meet.

Correction value K_{frak} from the closure comma

The rolled-out recursion closes only after $N = 1/(100\xi_0) = 75$ equal pieces, not after one turn. The approximation "one turn closes" neglects a small factor: 75 turns each shortened by ε give the exact closure, $75(1-\varepsilon) = 74$, hence $\varepsilon = 1/75 = 100\,\xi = 0.0133\ldots$

(the closure comma; as an angle $2\pi/75 = 4.8^\circ$ per turn). The per-step factor of the scale recursion $\xi_{n+1} = \xi_n(1 - 100\xi_n)$ is therefore

$$K_{\text{frak}} \;=\; 1 - \varepsilon \;=\; 1 - 100\,\xi \;=\; \tfrac{74}{75} \;=\; 0.9867,$$

the form the corpus carries throughout (Doc. 133/231/032/285).

Important: $75 = 1/(100\xi)$ and $K_{\text{frak}} = 1 - 100\xi$ are both just 100ξ; the comma *re-expresses* the already-anchored 100ξ, it does not derive it. The anchor lies in Doc. 133 (where the factor 100 is set, not derived; the value $K_{\text{frak}} = 1 - 100\xi$ is confirmed by comparing the m_e/m_μ means with α) and in ξ itself (Higgs/α/Koide). Detailed in Doc. 300.

For the projection side (Marcel) a consistency condition follows: *if* his closed projection is a section of the same recursion, the conversion to the rolled-out curve carries the independently anchored factor $K_{\text{frak}} = 1 - 100\xi = 74/75$ – not from the 75 and not freely chosen.

Status

Working document and pre-registered forcing protocol (P35). ι remains a conjecture (Doc. 297). This document fixes the test architecture before the joint run, so that the verdict – forced or allowed – cannot be shifted after the fact. The discipline holds symmetrically: it binds even when the outcome favours FFGFT.

Chapter ai

Doc. 300 — Spiral-Time as readout and debounced comparator

Doc. 300 Spiral-Time as readout and debounced comparator — and the shared correction factor $K_{\mathbf{frak}} = 1 - 100\xi$ (74/75)

Occasion and scope

From the collaboration comes the observation that Spiral-Time is an operator class with at least two roles: a *readout* (what happens in the signal or carrier) and a second part deciding whether this readout may count as a stable transition. This document makes the second role precise in measurement/circuit terms and assigns the correction factor connecting both sides to its *actual* origin in the corpus. Scope: category-B finding (genuine compatibility / candidate overlap), not an HLV-internal derivation, not a physical validation.

The two roles, made precise

The first role, the *readout*, is a measured value: the signal at the $+$input of a comparison. The second role is not an administrative term (“governance”) but precisely a **comparator with hysteresis** — a threshold decision whose output carries one bit. Two forms:

- **amplitude hysteresis**: two level thresholds, a dead band on the value axis stabilising against noise at the switching point.
- **time hysteresis (debounce)**: the output accepts an edge only if it *persists* over a time window. The dead band lies on the time axis.

The second role is the time hysteresis. The finite-proxy window ($M = 12$, $\tau = 6$) is a weighted debouncer: a transition counts only once it survives the window. “Stable transition” is exactly the time hysteresis.

Validity condition: reference from a derivation, not a copy

A comparator carries information only if the reference at the $-$input comes from a *different* source than the signal. If both inputs carry the same quantity, the output is meaningless.

This was the circular core of the U3 boundary gate in the first HLV-side implementation: the reference was a copy of the readout window (`hard = hlv.copy()`), so the correlation is 1.0 by construction. The FFGFT-side test (Doc. 299) replaces it with a *derived* witness (BLP back-flow 5.125, Doc. 283), independent of the object and not reproducible by a hard-reset window. Self-reference is admissible when closed by a derivation; circular when closed by a copy.

Why “the same spiral” is not a postulate

The statement about the correction factor presupposes that the projected HLV object and the FFGFT recursion are the *same* spiral. This identity is supported by several independent contact points:

- $2/9$: the same redistribution as the output of the Z_3-circulant diagonalisation — found, not sought.
- $D6$ **/ φ-icosahedron**: the same object structure, $\tau = \varphi$, all fifteen axis angles $\arccos(1/\sqrt{5}) = 63.435^\circ$.
- **time**: the spiral-time sector as a bounded section of the rolled-out recursion time (R47).

$2/9$ and $D6$ are geometric, mutually independent, both found; under this multiple anchoring “the same spiral” is the only remaining reading (category B).

The closure comma and K_{frak}

The rolled-out recursion closes only after $N = 1/(100\xi_0) = 75$ equal pieces ($\xi = 4/30000$), not after one turn (Doc. 295/297). The approximation “one turn closes” neglects a small factor: 75 turns each shortened by ε give the exact closure, $75\,(1-\varepsilon) = 74$, hence

$$\varepsilon \;=\; \frac{1}{75} \;=\; 100\,\xi \;=\; 0.0133\overline{3} \qquad \text{(as an angle } 2\pi/75 = 4.8^\circ \text{ per turn)}. \tag{ai.1}$$

The per-step factor of the recursion is therefore

$$K_{\text{frak}} \;=\; 1-\varepsilon \;=\; 1-100\,\xi \;=\; \frac{74}{75} \;=\; 0.9867, \tag{ai.2}$$

exactly the form the corpus carries throughout (Doc. 133/231/032/285/018/005). The reciprocal length-ratio form (shortened piece → full turn) is $75/74 = 1.0135$.

The actual anchor: not the 75, but Doc. 133

The origin of 100ξ — and thus of 75 *and* K_{frak} — lies one level deeper and is independent:

- The **scale recursion** $\xi_{n+1} = \xi_n(1-100\xi_n)$ (Doc. 133/275) has per-step factor $K_{\text{frak}} = 1-100\xi$; $75 = 1/(100\xi_0)$ is its winding number.
- The **factor** 100 emerges from the RG running of $D_f = 3-\xi$ over ~ 17 orders of magnitude (Planck → electroweak), effectively $D_f^{\text{eff}} \approx 2.973$ (Doc. 133); geometrically $100 = 4\cdot 5^2$ (Doc. 042).

- K_{frak} is **not arbitrary** but fixed by requiring two independent derivations of the mass ratio m_e/m_μ to agree (Doc. 133, *Unique Determination*).
- $\xi = 4/30000$ itself is anchored via the Higgs relation, α and Koide.

The closure comma thus *re-expresses* the already-anchored 100ξ; it does not derive it. The 75 is an output of the same recursion, not an independent anchor.

Consequence for the projection (Marcel)

The statement to Marcel therefore remains valid, only correctly booked: *if* his projected, closed turn object is a section of the same recursion (supported by $2/9$, $D6$, time), then the conversion between his projection and the rolled-out exact curve carries exactly the per-step factor $K_{\text{frak}} = 1 - 100\xi = 74/75$ (reciprocally $75/74$) — a factor anchored *independently* (Doc. 133), not derived from the 75 and not freely chosen. It is a consistency condition, not a derivation from the winding number: if the projection absorbs the comma and the object appears cleanly closed, K_{frak} is the conversion re-entering it at the boundary to the SI level (Doc. 298). The embedding ι (Doc. 299) is this K_{frak} conversion here.

Limit and status

The chain is as strong as its anchoring: $2/9$ and $D6$ are booked as shared objects (category B), $K_{\text{frak}} = 1 - 100\xi$ is independently anchored via Doc. 133. Comma check script: `python/Dok300_Skripte/ffgft_300_kfrak_schliessungskomma_pruefung.py`; time-sector counter-run: `python / Dok299_Skripte / ffgft_299_iota_gegenlauf_winding_locked.py`.

Chapter aj

Doc. 301 — The reversibility test — information as output, not as a base unit

Doc. 301 The reversibility test — information as output, not as a base unit

aj.1 Occasion and scope

The collaboration produced a stability picture that carries three quantities — information, coherence and energy (ICE) — as equally ranked corners of a triangle, with a reference at the centre. This document examines what categorical standing "information" actually occupies in such a picture. The finding is directed but not single-level: *information* is not synonymous with *bit*. The term splits into three levels — the bit (comparator output), the description (chosen representation) and the absolute identity (the complete object itself). As a base unit beside energy, information qualifies on neither of the two lower levels; it is fundamental only in the absolute sense, as the topmost identity. The criterion that carries this classification is reversibility. Scope: a Category-B finding (categorical classification, no physical derivation, no claim that FFGFT derives a bit). The result is a sharpening of the reduction principle of Doc. 296 and the comparator reading of Doc. 300.

aj.2 The criterion: fundamental means reversible

A quantity is fundamental exactly when its transformation has no distinguished direction — when one can turn it on its head and it still stands. The touchstone is energy in natural units. Through the relation

$$\tilde{T} \cdot m = 1 \tag{aj.1}$$

energy, mass and the dual time are not independent axes but three views of one quantity:

$$E \;=\; m \;=\; \frac{1}{\tilde{T}}. \tag{aj.2}$$

This transformation is symmetric: one may begin at E and read off m, at m and read $\tilde{T}$, at $\tilde{T}$ and read E. No side is the primary one, none is input or output. This is precisely why energy is intrinsic and the energy scale E_0 belongs to the framework natively (Doc. 292): the transformation discards nothing, each view carries the others fully. Here reversibility is not an addition but the expression of the fact that nothing is lost. In SI, $\hbar$ and c re-enter as conversion factors and pull the three apart; that is the projection, not the core.

aj.3 Intrinsic and relational

Not every quantity carries itself. A geometric quantity is **intrinsic**: an area, a winding has its value without reference to anything else. A thermodynamic or informational quantity is **relational**: entropy is defined only relative to a coarse-graining, a bit only relative to a distinguished reference. There is no bit "as such". This distinction is the categorical basis of what follows: intrinsic quantities can be fundamental, relational ones always presuppose a *with respect to* and therefore cannot be self-carrying.

aj.4 The ICE triangle as a comparator

Read in circuit terms, the ICE picture coincides corner for corner with the comparator of Doc. 300 — not by analogy, but as the same object drawn from the other side. Usable information at the output of a comparison requires three ingredients and a fourth condition:

- **Energy** — the signal at the $+$input carries a level; without a level there is no comparison.
- **Coherence** — the signal has phase structure, is not mere noise; geometrically the phase on the windings.
- **Reference** — the $-$input, the threshold against which the decision is made; it must come from a *different*, derived source than the signal (Doc. 300: reference from derivation, not from a copy).
- **Hysteresis** — a bare threshold chatters at the switching point; a transition counts only once it stands over a dead-band (amplitude hysteresis) or persists across a window (time hysteresis, debouncing). Without a dead-band the output is meaningless, however clean the energy and coherence.

Important: The ICE triangle contains energy, coherence and reference, but no declared dead-band. Hysteresis is the fourth condition without which "threshold exceeded" does not become "transition counts". The time hysteresis used in the finite proxy implementation is exactly the sliding window M_m (Doc. 300/297) — a debouncer, and as such a short-term buffer, not an accumulating memory.

aj.5 Information is output, not input

The comparator thus inverts the order the triangle suggests. Energy, coherence, reference and hysteresis stand at the *input*. Information is what emerges at the *output* once those conditions are met: the bit the threshold decision produces. It is not the ingredient

the comparison is built from, but its result. But a quantity that presupposes the whole apparatus in order to arise cannot be the base unit the apparatus is made of. This is also why earlier attempts to anchor a bit as a base unit had to fail: they sought to posit the output product as an input quantity — a circularity.

aj.6 The reversibility test

The criterion of Section 2, applied to information, comes out negative. E–m–$\tilde{T}$ is bidirectional: each view reconstructs the others, because nothing is discarded. The comparator is *unidirectional*:

$$\text{signal} + \text{reference} + \text{hysteresis} \;\longrightarrow\; \text{bit}, \tag{aj.3}$$

and this arrow does not reverse. From a bit one does not recover the signal, not its level, not its phase, not the reference threshold against which the decision was made. Fundamental quantities are reversible because they preserve; information is not reversible because it erases. *Fundamental and bit thus exclude one another not accidentally but by definition*: reversibility demands that the generating structure remain preserved in the result, and the bit is precisely the operation that does not preserve it.

aj.7 The bit as ultimate reduction

By Doc. 296, every evaluation is a reduction: it leaves out, and precisely by leaving out it makes a sub-domain evaluable. The bit is the limiting case of this statement — the *maximal* reduction. A whole signal course, with its energy, its phase, its position relative to the threshold, is collapsed to a single yes/no; less is not possible. And the irreversibility is not an addition to the reduction, it *is* the reduction: the threshold, the phase, the input conditions *were* — at the moment the comparator flips, they are gone. The bit no longer carries how it came about. It says “above” or “below”, not at which level, not at which phase, not how far from the reference. Everything that produced the result is no longer contained in the result — and therefore there is no way back, because the where-to-return-to is exactly what the reduction has erased.

aj.8 The Landauer connection

This irreversibility is the thermodynamic side of the same matter. Erasing a bit costs $k_B T \ln 2$ and produces entropy; the direction is built in, because erasing information produces heat. To reverse the comparator would mean recovering that entropy, and that is thermodynamically excluded. This explains in retrospect why anchoring a base-unit bit to thermal motion went so badly in the prior working state: the attempt bound a directed, reducing quantity to a reversible base structure; the directions do not match. Not a lack of calculation, but a categorical incompatibility was the cause.

aj.9 Three levels of information: bit, description, absolute identity

The reversibility test separates cleanly, but the boundary does not run at "information" wholesale. The term splits into three levels that answer the test differently.

First level — the bit. The comparator output: reference-dependent, generated at the output, erasing the input conditions. It fails — unidirectional, reducing, the ultimate reduction (see "The reversibility test" and "The bit as ultimate reduction"). A consequence, not a foundation.

Second level — the description. Information is not the bit but, first, a *description*: the representation of a state of affairs. Which kind of description is chosen — geometric, mathematical, rhetorical — changes the representation, not the state of affairs. This is the same line as in Doc. 297/300: a representation is a choice of coordinates, not a structural property of the object; different descriptions of the same object are mutually translatable, but none is the object. And none is ever *complete*: every description is a projection (Doc. 296), hence a reduction, hence it leaves out. A complete description would exist only if a projection left nothing out — and then it would evaluate nothing. The description therefore also fails the reversibility test: as a route back to the object no single description is invertible, because each is incomplete. The descriptions are mutually translatable, but none reconstructs the whole.

Third level — the absolute identity. Fundamental in the full sense is the complete object itself alone: the boundaryless T^4 object (the core ξ on T^4), of which every projection shows only a section. *Information in the absolute sense* is not a quantity *in* the system and not a single description, but the complete self-identity of the object with itself, prior to any projection. Only this level passes the reversibility test — trivially, because there is nothing to leave out: there is no "with respect to", because there is no external standpoint. This is the topmost identity; to it, and to it alone, "information" belongs fundamentally (self-reference, Doc. 248 §8).

Important: The geometric structure of the core — winding numbers as topological invariants, carried losslessly — belongs to the third level, to the object itself; it is not "information" in the bit sense. A geometric *description* of that structure, by contrast, remains second level: a projection. And the bit read off is first level. The three meanings must not be conflated. In particular: if anything stands "above" energy, it is not the bit but the complete object — and energy is one of its projections (a particularly intrinsic one, but a view).

aj.10 Digitisation: the staircase and its smoothing

The reason a bit can never carry the whole information is the basic mechanism of digital technology: analogue-to-digital and digital-to-analogue conversion. A bit is the 1-bit quantisation — the coarsest possible staircase, a single threshold. Any finer staircase with finitely many steps remains categorically the same object, only at higher resolution.

The torsion winding provides the natural example. The unrolled recursion closes after exactly 75 equal segments (Doc. 300/295); decomposing the winding into these 75 parts

digitises it — 75 discrete steps. By DA conversion a continuous course can be regained from them. But the reconstruction is already lossy, for two reasons:

- Quantisation discards what lay *between* the steps. The continuous structure between the 75 nodes — the exponentially shrinking steps of the log-spiral, the cumulated integral $D(k)$ (Doc. 295) — is not contained in the 75 values.
- The DA converter outputs a staircase, not a smooth curve. One must *smooth* the steps (reconstruction filter). But the smoothing is an interpolation — an assumption the filter adds, not information contained in the digital value. What lies between the steps is guessed, not recovered.

The AD/DA round trip is thus precisely *not* a true inverse but a directed, lossy process: the reversibility test in circuit form. The lost intermediate structure is not reconstructed by the smoothing but replaced by a filter assumption — the same signature as the bit, whose input conditions "were" and no longer stand in the result.

Important: Even the *natural* staircase — 75, the closure number — does not carry the continuum. It is the right number for the closure step, but not for the continuous course: the full object is richer than any finite staircase (Doc. 296). No N, not even the structurally distinguished $N = 75$, carries the continuum. Placed within the three levels: the bit is the 1-step quantisation (first level, output); any finite N-staircase — including the 75-division — is a description of finite resolution (second level), finer than a bit but still lossy, still a projection; the continuous winding, the complete object, alone carries everything (third level). The smoothing is exactly the place where the description overlays its incompleteness with an assumption.

aj.11 Refinement: resonances enforce the quantisation

The previous section spoke of the "continuous structure between the nodes". This must be sharpened, for on the finest scale the object is not continuous. The torus is compact, and a compact torus has a *discrete* spectrum: the resonances on T^4 — the eigenmodes of the T^4 connection-Laplacian (Doc. 283) — enforce a quantisation. In the smallest parts *not all possibilities are equally weighted*; the resonance structure singles out particular discrete states. This is exactly why the Standard Model comes out discrete — the masses, $\theta = 2/9$, the Z_3 structure are resonances, not arbitrary points of a continuum.

From this follows a restriction of the loss argument: there is a *smallest possible jump*, the quantum set by the resonances. A conversion that goes down to this smallest jump and meets the resonance structure is *permitted* — it is not lossy, because below the jump there are no further states that could be lost. What lies between a *coarser* staircase is therefore not a continuum but finer *discrete* resonance structure, and it has a floor.

The reversibility test thus sharpens: loss does not arise from quantisation as such, but only from a staircase that is *coarser* than the resonances or lies *across* them — for instance an arbitrary equal division. A resonance-faithful conversion down to the smallest jump passes the reversibility test: it captures the actual discrete structure, not an approximation. The bit nonetheless remains the ultimate reduction — a single threshold is coarser than the resonance floor and does not meet it. And beyond a single scale it still holds that no description of one scale captures the cross-scale richness of the full object (Doc. 296). The statement "the full object is richer than any finite staircase" is

therefore to be read *across scales* — within a scale, the resonance-faithful staircase is by contrast the true structure.

aj.12 Temporal and topological boundarylessness

Two properties of the absolute identity follow from analysis and topology and are therefore booked as *consequences*, not as posits.

No reachable edge of the recursion. The map $\xi_{n+1} = \xi_n(1 - 100\,\xi_n)$ has $\xi = 0$ as a fixed point; ξ_n decays monotonically towards zero but reaches zero in no finite step (Doc. 296). There is no first and no last step. A "simplest initial state" — a scale without fractal structure, from which the refinement would only begin to grow — does not exist: the self-similarity (factor $1/e$ per 2π turn, the log-spiral) is not a property that builds up but is already fully present in the map. A structural beginning would break the scale invariance that defines the recursion — a contradiction, not a world-view.

Time without a markable origin. Experienced time *is* the cumulated defect $D(k) = \sum 100\,\xi_n$ (Doc. 295), the sum over $1/\delta t_n = 100\,\xi_n$; with $\xi_{n+1} = \xi_n(1 - 100\,\xi_n)$ up to $k = 10^5$, $\sum 100\,\xi_n = 7.19 \approx \ln(1 + k/75) = 7.20$, whereas the sum of the step widths themselves, $\sum 1/(100\,\xi_n) = 5.0 \times 10^9$, grows quadratically. Since the recursion has no first step, $D(k)$ has no distinguished origin. Time is the recursion, not its container; to seek a "beginning of time" would place the recursion at a moment it only generates.

No external standpoint. T^4 is compact and boundaryless, $\partial T^4 = \emptyset$. That is a topological fact, not a posit. From it follows that there is no standpoint outside the object — and hence the "no with respect to" of the absolute identity: the complete object has no outside against which to demarcate itself.

Taken together: the absolute identity is boundaryless spatially (no outside, $\partial T^4 = \emptyset$) and temporally (no beginning, no reachable edge of the recursion) — the same property in two guises: no standpoint beyond the object. These statements are derivable from analysis and topology and carry no speculation marker.

aj.13 The reflexive case — what goes beyond the mathematics

What follows explicitly goes beyond the mathematics and is booked as philosophically speculative (P35). These are two interpretive jumps, not calculations.

Applying the three levels to FFGFT itself, the framework — the torus, $\tilde{T} \cdot m = 1$, ξ across all scales — is a *description* (second level), necessarily incomplete. This is not a weakness of this framework in particular but what a description is: no theory is ever its object, each projects and leaves out. FFGFT says so explicitly, whereas naive realisms behave as though they were the territory.

First jump — the realist posit. That the resonance-faithful description traces a *real* object and is not merely a consistent structure is posited, not proven. The mathematics shows resonance fidelity (the masses as eigenmodes, Koide; the agreement of α with the measured value is, by contrast, a property of the anchor E_0, not an independent prediction); that fidelity traces reality rather than mere consistency is the realist wager. The deflationary reader may always say: all just description, the object a useful fiction. FFGFT's answer is not a proof but the wager that the resonance-faithful success is better

explained by a real object than by coincidence — and the only distinctive thing is that the resonance hits make that wager falsifiable.

Second jump — the diagnosis of the Word reflex. The pull of the information-as-fundamental idea ("In the beginning was the Word"; "it from bit"; the mass-energy-information equivalence) may stem from a deep cultural reflex: that information is the first substance from which matter follows. The reversibility test inverts the order: the word is the bit, and the bit is the last reduction, not the first creation. In the beginning was the geometry (the object); the word is what one of its scales reads off. This is a cultural-cognitive hypothesis about *why* the intuition pulls — not a mathematical statement; the reversibility test itself remains mathematical.

Self-projection. FFGFT is a T^4 process describing T^4 (Doc. 248 §8): the description is of the same stuff as the object and is generated by it (the describing process is itself a T^4 process on the cognitive scale). FFGFT is therefore neither external to the object nor the object, but its *self-projection* — the object describing itself on one of its own scales. Self-reference as an ontological locus, booked as a posit.

Hence the limit of the opening question: "How much ontological reality does FFGFT have?" is itself a second-level question — it presupposes a describing standpoint that does not exist over against the third level. One cannot ask from outside how much reality the object has, because there is no outside; one cannot ask what came before it, because there is no before. At the level of the absolute identity the question dissolves.

aj.14 Status

Categorical classification (Category B), not a physical derivation step. The finding places information relative to programmes that posit it as a physical base quantity (such as a mass-energy-information equivalence): such programmes treat information as input; the reversibility test shows that the bit is output. The three levels are to be booked at different temperatures:

- **Bit** — *proven*: the ultimate, input-condition-erasing reduction is irreversible and therefore a consequence, not a base unit (reversibility test).
- **Description** — *structural*: every representation is a projection and hence incomplete (reduction principle, Doc. 296); a choice of coordinates, not a structural property of the object.
- **Absolute identity** — *ontological posit*: information in the absolute sense is identified with the complete object and is the topmost identity (self-reference, Doc. 248 §8). This is posited, not derived.

The load-bearing statement in one sentence: *fundamental ⇔ reversible; bit and single description fail the test (the bit erases, every description is incomplete), the absolute identity alone passes it — so information counts as fundamental only in the absolute sense, as the topmost identity, and this is in FFGFT the core ξ on T^4, not a quantity beside it*. Connections: Doc. 296 (reduction as the condition of evaluability), Doc. 300 (comparator, reference, hysteresis), Doc. 283 (discrete resonance spectrum on T^4), Doc. 292 (E_0 intrinsic), Doc. 248 §8 (self-reference).

Chapter ak

Doc. 302 — Elementary cells, one-bit capacity, and partition invariance

Doc. 302 Elementary cells, one-bit capacity, and partition invariance — a spectral answer

The following addresses a natural objection to any "quantised space" picture: what observer-independent criterion distinguishes one elementary cell from its neighbours; why a fundamental length should imply fixed cells each storing exactly one bit; whether the resulting entropy is invariant under changes of origin, orientation, coordinates, or foliation; and whether the indivisibility of a cell follows from the Planck scale or is an added assumption. The short answer is that every one of these points is correct *against a spatial-cell ontology*, and that FFGFT does not rest on one. It replaces cells with topological and spectral invariants (winding classes and modes), and the objections dissolve in turn.

ak.1 The picture the objections target

A "one bit per Planck cell" ontology tiles space (or a horizon area) into fixed cells of size $\sim \ell_P$, assigns each a one-bit capacity, and reads entropy off the count of cells. Against that picture the following points are decisive:

- there is no observer-independent criterion individuating one spatial cell from its neighbour: a tiling carries a chosen origin and orientation;
- the Planck length is a *scale*, and a scale does not by itself imply a partition into discrete containers;
- a partition-based entropy is not invariant under change of origin, orientation, coordinates, or foliation — it inherits the arbitrariness of the partition;
- if space is a continuum, any cell subdivides, so "one cell = one indivisible bit" needs an independent reason why half-cell states are forbidden;
- and absent that reason, the cell is simply an extra postulate riding on top of the Planck scale.

All five are granted. This is precisely why the framework is not built on spatial cells.

ak.2 Invariants, not cells

The fundamental object is not a lattice of cells but a compact, boundaryless manifold — a four-torus T^4 (with a $\mathbb{Z}_3$ identification), carried into a Hilbert space $\mathscr{H} = L^2(T^4) \otimes \mathbb{C}^3$. The elementary discrete objects of the theory are not regions of space; they are *winding classes* ($\pi_1(T^4) \cong \mathbb{Z}^4$, the carriers of the information unit) and *eigenmodes* of an operator on that manifold (the connection-Laplacian / mass operator, the carriers of the mass spectrum). With that single change of ontology — invariants, not cells — the points resolve in turn.

1. **Observer-independent individuation.** The elementary discrete objects are distinguished by *invariants*, not by a location in space: the information unit by its *winding number* (a topological invariant, $\pi_1(T^4)$), the mass spectrum by *eigenvalues* (a spectral invariant). Neither depends on coordinates, origin, or orientation. There is no "neighbouring cell in space" to distinguish; there are discrete classes labelled by invariants. The individuation is topological or spectral, not spatial — which is exactly what makes it observer-independent.
2. **What the information unit is — and what stays a definition in the "one bit".** The discreteness does not come from the Planck length and is not a tiling; it is *topological*: $\pi_1(T^4) \cong \mathbb{Z}^4$ admits only integer winding numbers. The minimal information unit is therefore *by definition* the winding quantum $\Delta w = 1$ (Doc. 257) — scale-universal and energy-independent; there is no half winding quantum. The identification "one bit ≙ one minimal winding" is a *definition*, though a topologically motivated one (the smallest non-trivial winding), not an arbitrary two-state assumption. The $\ln 2$ is *not* the fundamental unit but its thermodynamic shadow: the *energy* of a bit is scale-specific, $E_{\text{bit}}(L) = \hbar c / L$, and Landauer's $E_L = k_B T \ln 2$ is the special case at a single (the biological) scale. This answers Takayuki's point (2): "why exactly one bit" is a definition on a topological basis ($\Delta w = 1$), not forced by a scale — and the unit is scale-universal, hence precisely not tied to a chosen partition.
3. **Invariance under origin, orientation, coordinates, foliation.** Yes. Because the entropy is a function of the spectrum — a count or trace over eigenvalues — and eigenvalues are invariants, the entropy is automatically independent of origin, orientation, coordinate choice, and basis. This is not an added axiom; it is a property of spectral quantities. A partition-based entropy would fail the invariance test, which is one reason the framework does not define entropy through a partition at all. A further, framework-specific result makes the origin/orientation independence explicit: on the compact structure there is no preferred index assignment — all D_6-equivalent segments of the torus structure are equally valid descriptions of the same object ("meshing, not a cut-out"). No labelling of a "first" cell or a preferred axis is available, or needed.
4. **Continuous space, no half-cells.** Space is continuous: T^4 is a smooth manifold, with no spatial atomism. One therefore cannot obtain indivisibility by subdividing space, and the framework does not try to. The indivisibility is *topological and spectral*, not spatial: $\pi_1(T^4) \cong \mathbb{Z}^4$ admits no half winding, and the spectrum on the compact manifold

has a smallest gap fixed by the topology (the circumference / boundarylessness), just as a finite string cannot vibrate at "half its fundamental". "Half-cell states" is, in this language, a category error: there are no cells to halve. The continuum of the manifold and the discreteness of the spectrum coexist without tension — exactly as in acoustics, where a continuous medium supports a discrete set of normal modes.

5. **Derived, not assumed.** The discreteness follows; it is not an extra "cell postulate". Given a compact manifold, the discreteness of the spectrum is a theorem. What *is* assumed is the compact geometry itself — the torus $T^4/\mathbb{Z}_3$ and the value of ξ. So the honest accounting: the "one-bit-per-cell" postulate is traded for a "compact-geometry" postulate, and from the latter the discreteness is derived rather than injected. That is a real assumption, and it is where the framework should be attacked — but empirically, not by fiat: the same geometry that yields the discrete spectrum also fixes particle masses, and those are falsifiable numbers (for instance a τ mass of 1776.969 MeV, checkable at Belle II). If the geometry is wrong, the spectrum and the masses fail together.

ak.3 What is assumed, derived, and posited

The geometry itself is also a *description* (Doc. 301, second level) and as such necessarily incomplete; no description reconstructs the object in full. Fundamental in the full sense is only the complete object (third level). That the resonance-faithful geometry traces a *real* object is therefore an explicit ontological *posit* — not a derivation, and not itself falsifiable; what is falsifiable are only its resonance-faithful consequences (the masses). This posit is named as such, not passed off as derived.

Fundamental is the information unit as a *winding quantum* $\Delta w = 1$ (scale-universal, Doc. 257); *energy*, by contrast, is a scale-specific *projection* of the geometry, $E_{\text{bit}}(L) = \hbar c/L$. An *area* shares this projected, scale-dependent character: the four-torus is boundaryless ($\partial T^4 = \emptyset$) and carries no horizon surface for bits to sit on; an area there is a *projection surface*, a second-level object. Where an entropy–area relation $S = N \ln 2$ with $N \propto A/\xi^2$ appears, N counts winding quanta and $\ln 2$ is the thermodynamic shadow; the A is not a fundamental partition of space (which would revive Takayuki's cell worry) but a chosen projection. Fundamental (third level) is neither area nor energy but the complete object; the scale-universal unit on it is $\Delta w = 1$.

The honest accounting is thus: *declared assumption* — the compact geometry ($T^4/\mathbb{Z}_3$, ξ); *definition* — the identification "bit $\triangleq \Delta w = 1$", topologically grounded; *derived* — the discreteness ($\pi_1(T^4) \cong \mathbb{Z}^4$), the scale-specific bit energy $E_{\text{bit}} = \hbar c/L$, and the invariance; *empirical exposure* — the masses; *named posit* — the reality of the object.

ak.4 Summary

The objections are correct against a spatial-cell ontology, and the framework agrees with them by not being one. The discreteness is topological ($\pi_1(T^4) \cong \mathbb{Z}^4$) and spectral (the spectrum on a compact manifold); the minimal information unit is the winding quantum $\Delta w = 1$ (the "bit", by definition, topologically grounded), not the capacity of a spatial

cell; and the invariance holds because winding numbers and spectra are invariants, not because a partition was chosen carefully.

ak.5 Where this is worked out

The discrete spectrum of the T^4 connection-Laplacian is treated in Doc. 283; the Hilbert-space formulation $\mathscr{H} = L^2(T^4)\otimes\mathbb{C}^3$ with masses as eigenvalues in Doc. 282; the information unit as the winding quantum $\Delta w = 1$ and the three-level hierarchy (topological / geometric / thermodynamic) in Doc. 257 (building on Doc. 255); the origin/orientation independence ("meshing, not a cut-out") in Doc. 300. The invariance point in particular deserves a fuller written treatment than it has so far received.

Chapter al

Doc. 303 — Representational identity across role changes

Doc. 303 Representational identity across role changes — a governance layer beneath the claim-state ledger

The claim-state ledger asks whether a transition is *earned*: whether an assertion may be booked as proven, restricted, or blocked. That test is preceded by an earlier question it does not ask — whether the object undergoing the transition has kept its *representational identity*. A role change is not a strength change. The same symbol, the same number, the same structure can successively occupy different representational roles — mathematical object, computational output, candidate physical quantity, observable, ontological posit — and it is between those roles that the real, often unnoticed, shift takes place. This document names that layer, gives two case studies (one separates the definition (which identifies) from the governance (which preserves), and proposes an identity-preservation record as a layer prior to the ledger.

al.1 Roles, not strengths

A representational object can occupy at least the following roles within a framework:

- *mathematical object* — a quantity within a formal structure, with no physical attribution;
- *computational construct / output* — the deterministic result of a procedure;
- *candidate physical quantity* — an object to which a physical meaning is *proposed* but not yet secured;
- *observable* — a quantity coupled to measurement;
- *ontological posit* — a claim about what *is*, beyond derivation and falsification.

These roles are not stronger or weaker versions of the same claim; they are different *types*. The claim-state ledger sorts assertions by evidential strength (proven / restricted / blocked); the role question sorts objects by their representational type. Both axes are needed, and they are orthogonal.

al.2 The failure mode: role drift

The typical error is not the wrong evidential strength but *silent role drift*: the same numerical or mathematical object changes its type without the change being declared as such. A number produced as a lattice output is labelled a “bare” physical mass, then carried as a candidate observable, and finally treated as an observed value after a correction still to come. These are four representational roles of the same numerical object. None of them is inadmissible in itself; what is inadmissible is only to present the role change as a mere strengthening of the same claim. This is precisely where disagreement enters quietly — often long before the evidential question is even asked.

al.3 Case study I (external): one number in four roles

A numerical object from a discrete lattice model passed, in an IPI discussion, through four roles: first a *lattice output* (a deterministic computational result), then a *bare topological mass* (a physical attribution), then a *candidate* for an observable ratio, and finally an *observable after screening* (under a reduction mechanism not yet constructed). The progress of the discussion consisted not in refuting any one of these roles but in *separating* them: once the roles were laid apart, each could be examined on its own constitutional footing — the lattice output as proven, the physical attribution as a definition, the observable as a still-open transition. Separating the roles was the actual methodological gain, independent of the fate of the model.

al.4 Case study II (internal): the information unit

The same role drift occurred, in the preparation of the present document series, within FFGFT itself. The question of the “minimal information unit” was answered successively with four different objects: first a *spectral eigenmode* (an object of the mass operator), then an *area* (the holographic choice of one bit per area element), then a *counting quantity* of order $1/\xi$, and finally the *topological winding quantum* $\Delta w = 1$. These four are not stronger and weaker versions of the same unit; they are four different representational objects that successively bore the same name. The role change went unnoticed until the identity was fixed by a *prior, explicit definition*: Doc. 257 fixes the information unit, by definition, as the winding quantum $\Delta w = 1$ ($\pi_1(T^4) \cong \mathbb{Z}^4$, scale-universal). Only that identity anchor made the remaining roles legible as what they are: the area as a lossy projection, the $\ln 2$ as a thermodynamic shadow, the energy $\hbar c/L$ as a scale-specific projection — all *derived or projected* appearances of the one fixed unit, not competing definitions.

al.5 Definition identifies, governance preserves

The lesson from both cases seems at first to be that an *identity anchor* preserves identity. But that conflates two questions that belong apart — and the conflation is exactly the role drift this document describes, now applied to the document itself. There are two constitutional questions:

- the *declarative* question — *what is this object?* — is answered by a *definition*. The definition *identifies*: it sets the constitutional reference point against which any further claim can be checked at all. In FFGFT, $\Delta w = 1$ (Doc. 257) does this: it fixes *what* the information unit is.
- the *governance* question — *under what operations does it remain the same object?* — is different. A definition sets the initial reference; it does not *preserve* identity. Identity is preserved only by the governance imposed on subsequent transitions.

In short: *the definition identifies, governance preserves*. The earlier notion of an "identity anchor" carried both roles under one name — definition *and* preservation mechanism — and is therefore itself an instance of the drift described here. It is cleaner to separate them: the definition is the declared reference point; governance is a distinct layer that records what happens to that reference at each transition.

al.6 The identity-preservation record

A merely binary check — same type before and after the transition → the ledger; different type → substitution — is too coarse. A transition can treat an object's declared identity in more than two ways, and the governance layer keeps an explicit record of which: for each transition it records whether it

- *preserves* — the same object in the same role;
- *projects* — a scale- or coordinate-dependent appearance of the same object, lossy and not fundamental;
- *transforms* — a derivable reshaping that keeps the reference;
- *substitutes* — a different object under the same name (undeclared, this is role drift; declared, it needs its own licence);
- *bridges* — a secured mapping onto an object of another framework.

Read against FFGFT's own information unit: the winding quantum $\Delta w = 1$ is *preserved*; energy ($\hbar c/L$) and area are *projections*; the $\ln 2$ is a *transformation* (the thermodynamic shadow); the earlier wandering eigenmode → area → $1/\xi$ was *substitution* under a retained name, until the definition (Doc. 257) fixed the reference; and the verification direction FFGFT→HLV is a *bridge*. The record decides no physics; it only keeps visible, at each step, what happens to the identity — before the ledger asks whether the step is earned.

al.7 Relation to the ledger

Three levels are now separated. The *definition* identifies the object (declaratively). *Governance* preserves its identity across transitions and keeps, in the preservation record, whether a step preserves, projects, transforms, substitutes, or bridges. The *claim-state ledger* scores whether a transition is *earned* (proven / restricted / blocked). Governance is prior to the ledger: it ensures that a discussion is held about the *same* object before it is held about that object's evidential strength. All three levels are largely independent of the underlying physics — they concern the bookkeeping of representation, not its content — and gain importance as independently developed frameworks come to interact.

Whether this bookkeeping is kept informally or through a systematic mechanism does not change its effect: the discussion shifts away from defending positions and towards localising where assumptions reside, what has genuinely been derived, and where an object changes its identity.

al.8 Status

This document is a methodological proposal, not a physical derivation. The FFGFT-specific statements — the information unit as $\Delta w = 1$, the three levels, the projection nature of energy and area — are corpus-supported (Doc. 257, 301, 300); the role taxonomy and the identity check are constitutional bookkeeping and are booked as such. The second case study is explicitly self-critical: FFGFT itself required the discipline proposed here, and the identity anchor $\Delta w = 1$ was recognised as such only in retrospect (Doc. 257). The occasion of the question — an objection to elementary cells, the constitutional framing of the transition problem, and the representational typing of the objects involved, as developed within the IPI circle (Takayuki Ueda, Stefaan Vossen, José Tomás Guevara Calderón) — is gratefully acknowledged. The distinction between declarative identity (definition) and its preservation by governance, and the five-way preservation record (preserve / project / transform / substitute / bridge), were sharpened in exchange with Stefaan Vossen; they are constitutional language made jointly visible, neither a replacement for nor an adoption of any particular governance programme.

Chapter am

Doc. 304 — The memory reflex — Why experienced time presupposes memory

Doc. 304 The memory reflex — why experienced time presupposes memory rather than arising from it

Abstract. A common construction makes experienced time the *product* of memory. This document reverses the order: memory presupposes time rather than producing it — the *memory reflex*, counterpart of the word reflex of Doc. 301. In FFGFT time is fundamental, the cumulated defect $D(k) = \sum 100\,\xi_n \to \ln(1 + k/75)$ of the unrolled winding; and the difference between a finite, forgetting kernel and an accumulating structure is already a fact about time, not first about memory. FFGFT itself does not need that split — it carries a single, curvature-faithful winding —; the split appears only where the time structure is projected onto a cognitive model. On the measurement side a second reflex follows: a measurement excerpt is reference-less, with no absolute place on the spiral, and the apparent uniformity of time is a *linearisation artefact* with a verified error quantity $e = x - \ln(1 + x)$ and the scale-invariant constant $1 - \ln 2$; this linearisation sits in the *projection* of the result, not in the measurement. Rolled up, the 75 parts are exactly congruent; unrolled, only asymptotically self-similar; what survives the loss of reference is the ordering alone. In the FFGFT reading it is *projection-fidelity*, not reach, that discriminates — offered as an additional, rejectable bridge candidate (restricted, Category B), not as a replacement of any external memory definition, in a symmetric relationship between the frameworks. Inference from measurement to cause succeeds only up to a holographic blur: presence is not reconstructibility. Booking is kept separate throughout — mathematically derived, bridge (restricted), posit and reflex diagnoses (P35).

A common construction explains experienced time as the *product* of memory: physical evolution → physical records → recursive memory → self-related representation → experienced time. In this order, time is the end product and memory its precursor. This document shows that the order belongs the other way round. Memory without time does not function — it already presupposes time. The impression that time arises from memory is the same inversion as “it from bit”: the memory reflex, the counterpart of the word reflex of Doc. 301, one level down. The same inversion appears a second time on the *measurement* side — as the linearization reflex —, this time with a verified,

scale-invariant error quantity; and a third time in the inference from part to whole — as the extrapolation reflex. This document treats all three.

am.1 The narrowing and its price

The original ambition was the full temporal triad: causal ordering (U1), present phase (U2), and memory / non-Markovian history (U3)[1], joined into a past–present–future world-model. Over time the discussion narrowed to U3 alone, then to the boundary between a finite memory kernel M_τ and an accumulative structure M_B, and finally to an operational benchmark question. This narrowing is methodologically legitimate — only the isolable, falsifiable piece can be tested — but it has a price: U1 and U2 drop out of the test, and the phenomenon itself was their *interplay*, not the component. Even a passed memory test would show "semantically structured memory is distinguishable from generic compression", not "here is experienced time".

am.2 U3 cannot be isolated

At one point the price is even an incoherence. Memory *is* a relation to time — a state that carries something about a before. Accumulation means integrating over a sequence. Without ordering (U1) there is no sequence to integrate over; without a present (U2) there is no now relative to which something is *past*. U3 is therefore not an independent fourth quantity beside U1 and U2, but what arises *when* ordering and present act together over a horizon. Marcel's own characterisation gives it away: "path-dependence, non-Markovian history" — "path" is ordering, "history" is past present. U3 is not even definable without U1 and U2. A test that lifts U3 out of time and probes it as pure storage over fixed episodes removes exactly the structure in which memory has meaning.

am.3 The diagnosis: the memory reflex

The hierarchy "memory → experienced time" is thus wrong in its direction. It makes memory the generator of time, although memory presupposes time. This is structurally the same reflex Doc. 301 diagnosed for information: just as "it from bit" mistakes the ultimate reduction (the bit) for the first creation, "experienced-time-from-memory"

[1] The temporal sectors U1 (causal ordering / forward updating), U2 (phase and coherence) and U3 (memory, path dependence, recursive access), together with the notation M_τ/M_B, are adopted from Krüger's Spiral-Time Memory work (2026); there M_τ denotes finite, weighted, or compressible physical memory and path dependence, M_B a persistent biological-cognitive architecture (recursive access, semantic organisation, affective weighting, identity binding, persistent state carry, predictive reuse), defined operationally and functionally, not geometrically. *FFGFT itself does not need this distinction*: it carries a single, curvature-faithful winding, and the sectors as well as the M_τ/M_B split serve here only as an *analysis* of that framework, not as native FFGFT structure. The present document therefore does not redefine these objects; it proposes a separate, FFGFT-side bridge reading. Primary source: Krüger, M. (2026), *From Physical State Evolution to Experienced Time*, Zenodo, `doi:10.5281/zenodo.21302278`. Documented lineage: Krüger, Feeney & Wende (2026), *Medinformatics*, `doi:10.47852/bonviewMEDIN62029043`; Krüger & Feeney (2026), *Smart Wearable Technology*, `doi:10.47852/bonviewSWT62029484`.

mistakes the derived carrying form (memory) for the origin of time. The reversibility test (Doc. 301) reverses the order in both cases. For information: in the beginning was the geometry, and the word reads out one of its scales. For time: in the beginning was the unrolled winding, and memory is a particular way in which a system, on one of its scales, *carries* that winding. This is a cultural-cognitive hypothesis about *why* the intuition pulls (P35), not a mathematical statement; the sections that follow are.

am.4 The FFGFT position: time is $D(k)$, fundamental

In FFGFT, time is the mass-circle, woven into the other circles through $\tilde{T} \cdot m = 1$ and not a separable factor (Doc. 285). The unrolled time-winding is the logarithmic spiral that the K_{frak} recursion forces (Doc. 295); its cumulated defect

$$D(k) = \sum_{n \le k} 100\,\xi_n \;\to\; \ln\!\left(1 + \tfrac{k}{75}\right) \tag{am.1}$$

is the measurable time development (Doc. 296). Time is the recursion, not its origin: $D(k)$ has no privileged beginning, grows without bound, reaches no end — no first step, no last, no external standpoint ($\partial T^4 = \emptyset$, Doc. 301). Time is therefore fundamental — the third level, carried losslessly — not a product of memory. Memory cannot generate time, because time already *is* $D(k)$ before any system carries it.

am.5 The decisive finding: the accumulation boundary lies in time

The remarkable point is that the distinction which recurs in cognition as M_τ versus M_B already stands, in the corpus, *at the level of time* — and is there already drawn against exactly the same bounded memory kernel. Doc. 295 contrasts FFGFT's *geometric accumulation* $D(k) = \sum 100\,\xi_n \to \ln$ (unbounded, $1/n$ decay, ratio e) with a *bounded memory kernel*: if the defect per step stays constant, $D(k)$ grows linearly rather than logarithmically ($D(10^5) = 1333$ against $\ln(1 + k/75) = 7.2$), and the winding is Archimedean, not a log-spiral. This case of constant defect ($d = 1/75$ per turn fixed; the bounded, forgetting kernel) corresponds to the Markov fallback. The case $d = 0$ is different: without defect the winding closes after every turn, $D(k) \equiv 0$, nothing accumulates (d is the closure defect per turn, Doc. 295; the linear growth $D(10^5) = 10^5/75 = 1333$ belongs to fixed $d = 1/75$).

The mapping is then direct:

- M_τ — the finite, forgetting kernel — is the *Markov limit of constant defect* of $D(k)$ ($d = 1/75$ fixed): a bounded window, constant defect, linear rather than logarithmic growth ($d = 0$ would give no growth).
- M_B — the accumulative, recursive structure — is a system that carries a $D(k)$-like, unbounded integral of its own history.

The accumulation-versus-kernel boundary is therefore *not native to memory*; it is native to time. Memory that accumulates is memory that carries a $D(k)$-like integral; a finite kernel is precisely the form of constant defect. Hence — *on this reading* — M_B does not generate experienced time but appears as a cognitive-scale *instance* of the

$D(k)$ accumulation that time *is* (as a bridge, not a derivation; see the limit of the bridge below).

Status of this identification. At the level of time the equation is corpus-anchored: Doc. 295 explicitly equates FFGFT's case of constant defect (bounded window, linear $D(k)$) with Marcel's Markov fallback; M_τ fits that case, not $d = 0$. The *extension* to the cognitive objects M_τ/M_B is, however, a further, FFGFT-side step — a *bridge* (Category B, a compatibility candidate), not a derivation. It is local to its assumptions and needs its own licence; anyone who reads the cognitive M_τ/M_B differently may decline the bridge without touching the time-level finding (the accumulation boundary in $D(k)$). The statement "M_B is an instance of $D(k)$" is therefore *restricted*, not *proven* — a declared bridge candidate (Category B), not a proven identity. Naming it as a bridge rather than promoting it to a derivation is exactly the identity check Doc. 303 requires. This identity check, like the claim-state booking (restricted, Category B), belongs to the governance layer that goes back to Stefaan Vossen's *Dot Theory* and was co-developed in the IPI exchange (cf. Doc. 272); the identity-preservation record of Doc. 303 is the FFGFT-side extension on top of it.

am.6 Consequence for the discriminator

This explains, from the FFGFT side, why the memory benchmark repeatedly tested the wrong half. A storage test over fixed episodes removes the temporal relation in which compression has any meaning; a test with exact, unbounded storage (full history) lets M_B coincide with lossless accumulation and makes its specificity invisible. The right touchstone is a *capacity-bounded, lossy* accumulator: a system that carries a *compressed* $D(k)$-like integral — for that is exactly what a cognitive instance of $D(k)$ looks like. Yet even a passed test of that kind would show "semantic accumulation is distinguishable from generic compression", a memory result — not "experienced time". Experienced time is the $D(k)$ structure itself, which the test *presupposes* rather than produces.

am.7 The measurement cut: the reference-less excerpt and the linearization reflex

The same inversion strikes measurement, not only memory. A measurement takes an *excerpt* of the time-winding. Even a lossless measurement — one that captures every step down to the quantum-bit limit (the winding quantum $\Delta w = 1$, Doc. 257/301) — delivers the *content* of the window in full, but not its *position* k on the spiral. Resolution (the content axis) and reference (the embedding axis) are different axes; losslessness saturates the first and leaves the second untouched. Synchronising successive measurements lengthens the excerpt, but it stays reference-less: one gains *relative* distances (clocked by the atomic clock), never the *absolute* position. The atomic clock is, by construction, a *uniform* ruler — constant pitch, hence exactly the M_τ limit of constant defect (Doc. 295; constant pitch means constant defect, not $d = 0$) —, and a uniform ruler laid on a non-uniform (fractal) structure cannot locate it. That there is no absolute time reference is

thus boundarylessness at the level of measurement: no external standpoint ($\partial T^4 = \emptyset$) from which the position could be read.

The excerpt is nonetheless not empty. On the log-spiral there is no straight segment; every window carries its *curvature* — the local scale ρ, and with it the prior history — as folded-in form. The reference is not *lost* but *linearised*: the measurement reads the tangent instead of the arc — the local form of the same constant-defect approximation. Precisely, this is an error in the *accumulated precession angle* $\beta = 2\pi D(k)$ (Doc. 295): the uniform clock takes the local rate $d_k = 100\,\xi_k$ as constant over the window ($\Delta\beta_{\text{unif}} = 2\pi w d_k$), whereas the true value is $\Delta\beta_{\text{true}} = 2\pi(D(k{+}w) - D(k))$, and the difference per 2π is not a new quantity but the local shape of the linear-versus-log gap of Doc. 295. For a window of w steps at position k, with relative window width $x = w \cdot 100\,\xi_k = w/(k+75)$:

$$e(k,w) = \frac{\Delta\beta_{\text{unif}} - \Delta\beta_{\text{true}}}{2\pi} = w d_k - \big(D(k{+}w) - D(k)\big) = x - \ln(1+x). \tag{am.2}$$

Three regimes follow; they are checked against the exact recursion $\xi_{n+1} = \xi_n(1 - 100\,\xi_n)$ and are exact in the macroscopic limit, with an $O(1/k)$ correction at finite position (the closed form $D(k) = \ln(1 + k/75)$ carries a bounded, γ-like offset ≈ -0.0067):

- *Small window* ($x \ll 1$): $e \approx x^2/2 = \frac{1}{2}(100\,\xi_k\, w)^2$ (leading term of the closed form).
- *Scale-invariant window* ($x = 1$, a window one local scale wide): $e \to 1 - \ln 2 \approx 0.3069$, *independent of* k (asymptotically, $+O(1/k)$). This constant is the quantitative signature of the fractal self-similarity: the distortion of a "one-scale-wide" excerpt is the same everywhere on the spiral.
- *Macroscopic limit* (fixed physical w, growing k): $e \to 0$ as $\frac{1}{2}\,(w/(k+75))^2$. Because we sit at large k (small ξ), the distortion is astronomically small — *this* is why measured time appears uniform and linear. Near the origin (small k, large ξ) it would be large; no measurement reaches there.

Hence the third reflex diagnosis (P35), twin of the word and memory reflexes: the apparent uniformity of measured time is a *linearisation artefact*, not a property of time. From an approximation in the measuring (tangent for arc) a property of the measured is falsely inferred (time *is* uniform). The excerpt carries its fractal history as curvature; the uniform clock erases it by construction; and because at our scale the error vanishes, we falsely conclude it was never there.

Booking. The error formula $e = x - \ln(1+x)$ is the error in the accumulated precession angle $\beta = 2\pi D(k)$, verified against the exact recursion (script `ffgft_304_linearisierungsfehler_probe.py`); its three regimes are exact in the macroscopic limit (a consequence of $D(k)$, proven), with a named $O(1/k)$ correction at finite position. The exact coefficient of that correction (whether precisely C/k or carrying a log term) is obtainable from an Euler–Maclaurin expansion of $D(k)$ and is left open here — physically negligible at our scale, named for honesty. The equation "measurement = local linearisation of the excerpt" is a *declared assumption* about *this* kind of measurement (pointwise, tangential), not about every possible one — a measurement that *integrates* over a window would see part of the curvature. The reflex reading is philosophically speculative (P35).

am.8 Congruence, self-similarity, and the survival of ordering

That there is *no reference* and that *all excerpts are equal* is one and the same fact: were there a distinguished excerpt, it would be a mark and hence a reference. But "equal" means different things in the two states of the time-winding, and the distinction is essential.

Rolled up (compact). With ξ_0 frozen the defect per step is constant, $d = 1/75$, the rotation number $74/75$ rational; the winding closes after *exactly 75* turns and carries a 75-fold structure (Doc. 295). "Equal" here means *congruent*: 75 coincident segments, carried into one another by a $\mathbb{Z}/75$ rotation, exact and without remainder. A reference *does* exist here — but only *periodically*, modulo 75: one can name "part j of 75", but not which turn, since all 75-cycles are congruent.

Unrolled (spiral). Letting the scale run ($\xi_{n+1} = \xi_n(1 - 100\,\xi_n)$) turns the rational 75-closure into the non-closing log-spiral of ratio e (Doc. 295, "unrolling"). "Equal" now means only *self-similar*: the excerpts are *proportional*, not congruent — and only *asymptotically* so. This is exactly the origin of the $O(1/k)$ correction: the self-similarity is not a finite exact symmetry but a continuous one that sets in fully only in the limit. The periodic reference vanishes too, for the turns are no longer congruent but only proportional; the 75 marks smear into the continuous self-similarity.

Reference-lessness therefore holds *strictly only in the unrolled state*. Three reference states must be distinguished:

- *rolled up*: periodic reference (position modulo 75, exact);
- *unrolled, asymptotic*: no reference (pure proportion);
- *unrolled, finite* (near the origin): a weak absolute reference — the $O(1/k)$ deviation from self-similarity —, but one no measurement can reach.

What survives all of this is the *ordering*: which excerpt comes *before* which. It is a relation *between* excerpts, not an inner property of a single one, and is therefore untouched by the rescaling. It is U1, and it is the only invariant that survives the loss of absolute reference — exactly "time is the recursion, not its origin" (Doc. 301). What remains measurable of time thus falls into three levels: the *ordering* (exact, U1), the *relative interval* ($\Delta\beta = 2\pi\Delta D$, linearisation-distorted by $e = x - \ln(1 + x)$), and the *absolute position* (absent, erased with the self-similarity). Ordering exact, interval distorted, position absent.

Refinement. "All excerpts equal" means *congruent* (rolled up, 75 parts, exact) or *self-similar up to scaling and rotation* (unrolled, asymptotic), not *identical*.

am.9 What this means for the partition of U3

The measurement analysis feeds back onto the partition of U3. Since a memory is a system that carries a $D(k)$-like excerpt (as shown in the decisive finding and the measurement cut), U3 inherits the same three-level structure: the *ordering* (which event before which) stays exact — it is U1 —, the *relative interval* (how long ago) is carriable but linearisation-distorted by $e = x - \ln(1 + x)$, and the *absolute position* (the true place on the spiral) is absent, erased with the self-similarity. Crucially: *both* memory forms, M_τ and M_B, are reference-less — even a complete M_B knows only "so much accumulated D

since an arbitrary start", not "I am at turn k". The one thing M_τ too carries exactly is the ordering.

In the *FFGFT reading* the discriminating axis is not reach (finite window versus unbounded integral) but *curvature-fidelity* — offered as an *additional* candidate dimension for the bridge, not as a replacement of the operational M_τ/M_B definition (see the limit of the bridge below):

- M_τ presents a *linearised* excerpt of constant pitch — the limit of constant defect. It reads the tangent, discards the curvature, "experiences" a uniform past. M_τ is the memory that commits the linearisation reflex.
- M_B presents the *curvature-faithful* excerpt — the log-spiral with the $1/n$-decaying defect and ratio e. It carries the non-uniform texture of the past: near events dense and fast, far events wide and slow (Doc. 296).

Reach and curvature-fidelity are *two* axes, not one. A long but *linear* kernel (constant pitch, merely with a large window) would be reach-strong yet remain M_τ-like, because it does not see the self-similarity; a short but *curvature-faithful* accumulator would already be M_B-like. The benchmark (v0.3–v0.5) presses both into the single axis of reach.

Hence the *positive* criterion for the discriminator, and it is not a new one: it is the e-criterion of Doc. 295 (a winding carries ratio e exactly when the defect per step decays as $1/n$; script, section [6]), applied to U3. What is to be tested is not "finite window versus accumulation" but *curvature-fidelity at equal reach*: does the accumulator carry the $1/n$-decaying pitch, the ratio e, the self-similar texture? That is exactly what distinguishes M_B from any uniform kernel, and it is sharper than any reach boundary.

Booking. The three-level inheritance (ordering exact / interval distorted / position absent) and the mapping $M_\tau =$ linearised, $M_B =$ curvature-faithful are consequences of $D(k)$ (derivable). The e-criterion as a discriminator test is corpus-anchored (Doc. 295). That Marcel's *cognitive* M_τ/M_B are exactly these objects remains the bridge candidate from the decisive finding (Category B, restricted) — now with a testable criterion. That remembered time *really* carries the spiral texture (the phenomenological non-uniformity of remembering) is interpretation (P35).

am.10 Provenance and the limit of the bridge

So that the meshing above is not mistaken for absorption, let the limit of the bridge be drawn explicitly, in language that is unambiguous even without knowledge of how it arose.

The correspondence is a restricted, optional bridge candidate, not a secured mapping. Two axes must be kept apart. The e-criterion itself — a winding carries ratio e exactly when the defect decays as $1/n$ — is script-verified against the exact recursion and derivable from ξ (*proven*, Doc. 295). Only the *correspondence* between this curvature-fidelity and the operational M_τ/M_B distinction is unsecured: on the logical axis *restricted* (Category B), on the empirical axis with no benchmark performed — where such a benchmark would be an independent, jointly sealed comparison, not an FFGFT-internal calculation. It is not a derivation, validation, containment relation, or ontological classification of the Spiral-Time Memory Hypothesis by FFGFT, and its rejection leaves the internal claims of both frameworks unchanged.

An internal property of FFGFT can show that, under FFGFT assumptions, a non-linear, history-dependent accumulation structure $D(k)$ arises on the time level; that result stands on its own. It cannot, however, independently establish that a semantic, self-bound, affectively weighted, predictively reusable memory architecture follows from it. That extension is precisely the bridge candidate.

The operational specification of M_τ and M_B is untouched. In the Spiral-Time framework it is operational and functional, not geometrical: M_τ as finite, weighted, or compressible physical memory and path dependence; M_B as recursive access, semantic organisation, affective weighting, identity binding, persistent state carry, and predictive reuse. Curvature-fidelity is an *additional, FFGFT-side diagnostic candidate*, not the criterion that separates M_τ from M_B in that framework, and it replaces none of those functions. In both directions: a system may carry a $1/n$ texture while possessing none of those biological-cognitive functions; and a biologically relevant memory architecture is not ruled out merely because it does not instantiate the specific e-criterion. The curvature test therefore remains a separate bridge audit, not a redefinition of the native objects.

The relationship is symmetric, the provenance separate. Neither framework is presented as containing, grounding, explaining, or subordinating the other. The M_τ/M_B distinction, the operational M_B architecture, and the audit sequence (v0.3–v0.5) originate in Marcel Krüger's Spiral-Time Memory work (Krüger 2026, `doi:10.5281/zenodo.21302278`); $D(k)$, the e-criterion, and the FFGFT interpretation of temporal curvature originate in Johann Pascher's FFGFT corpus. A later formal comparison or curvature-fidelity audit would be a separate, joint bridge contribution, preserving the independent authorship and conceptual identity of both frameworks.

am.11 The past is always present: linearisation lies in the projection

An addition sharpens the measurement cut and the partition just given. The *past content* is *always* in the measurement result, independent of the window's shape, whether read linearly or not. For $D(k)$ is cumulative: every point of the spiral carries the summed integral of the whole prior history. Even a one-step window sits at this value and inherits it whole. A small window does not make it *absent*, only more weakly *resolved*; a long or more accurate window makes it sharper. "Present or not" is the wrong alternative — the past is always present, at variable sharpness.

What is decisive is therefore not how the window *looks* (linear or curved) but what the result is *applied* to. The measurement yields a value; the linearisation happens only when that value is mapped onto a *projection* — onto a uniform ruler (the atomic clock, the linear time axis). There, in the projection, the curvature is straightened. The distortion is not a property of the measurement but of the projection surface onto which the result is laid.

This is the projection structure of the corpus (Doc. 257/301): *measured* time is a *second-level* object — a projection of the $D(k)$ excerpt onto a uniform ruler. The past content is present as a *state* of the third level — the $D(k)$ value itself —; yet this value is a lossy résumé of the prior history, and the projection to the second level is lossy, and *that projection is the linearisation*. Not the measurement linearises, but the projection of the

measurement result. Thus $e = x - \ln(1 + x)$ is, more precisely, the error of the *projection* onto the uniform ruler, not an error of the measurement itself.

Three things must be separated: the *presence* of the past as a state (always, third level: the $D(k)$ value is exact, but not the recovery of the prior history from it); the *sharpness* of its reconstruction (window- and accuracy-dependent); and the *linearisation error* (a property of the chosen projection, not of the measurement). The window's shape is not decisive for the *whether* of the past or for the *location* of the linearisation — but it is decisive for the *how sharply* of the reconstruction.

This sharpens the earlier talk of "carrying" the curvature: M_τ and M_B both *contain* the past — every $D(k)$ excerpt does —; they differ in the *projection*, not in the *content*. M_τ projects onto a uniform ruler (linearises, discards the curvature at projection), M_B projects curvature-faithfully (keeps the self-similar texture). The load-bearing axis from the previous section — curvature-fidelity, not reach — is thus more precisely *projection-fidelity*: which projection the state passes through, not what it contains.

Booking. The cumulativity of $D(k)$ (the past always present) and the location of the linearisation in the projection (second level, Doc. 257/301) are derivable and corpus-anchored respectively; e as a projection error is the more precise form of the earlier formula. The separation presence / sharpness / linearisation is conceptual bookkeeping.

am.12 Inverse inference and the holographic limit

The model is unrolled from the *measurement side*, for what stands at the centre is the application: one wants to infer from measurement results back to *causes*. This is the inverse direction of the reversibility test (Doc. 301) — not from cause to effect, but from the measured effect back to the cause. It is exactly this aim that makes the preceding analysis practically relevant: it asks *how far* a measurement result supports the inference to the whole.

And here is the limit, which was drawn too weakly before. From a small excerpt one may *not* infer the whole. The excerpt gives at best a *blurred* image of the whole — like a fragment of a holographic recording, which carries the entire image but only unsharply, at lower resolution. "The past is present" thus means, more precisely: it is folded in as a *blurred signature* — the cumulated value $D(k)$ is a sum from which the summands cannot be recovered —, not as a reconstructible recording. *Presence is not reconstructibility.*

The holographic structure is the self-similarity itself: because every excerpt is *similar* to the whole, it carries a scaled, blurred image of the total structure. From this follows something sharper than the earlier talk of "accuracy": the sharpness of the reconstructed whole depends on the *size* of the excerpt (how much of the spiral), not on the fineness with which a small piece is read. A small window, however losslessly — down to the winding quantum — it is measured, gives a blurred whole; only more excerpt sharpens it. This is why the absolute reference is missing even at perfect resolution.

The impermissible step — from part to whole — is thereby nameable: an *extrapolation reflex*, taking the fragment for the image. It joins the family of the word, memory, and linearisation reflexes: in each, something derived, reduced, or partial is taken for the original, the whole.

For the inverse inference the exact reach follows: inferring from measurement results to causes is possible, but *only up to the holographic blur*. The rolled-up (compact,

75-exact) side — the underlying geometry, the "cause" — is *inferred*, not measured; the unrolled (spiral, measurement) side is where one stands. Rolled and unrolled are here two readings of the one winding; "cause" means the direction of the inference, not an ontological layer beneath the fundamental $D(k)$. Absolute position and ultimate origin are never sharply recoverable from an excerpt. This connects to the lossiness of accumulation: a $D(k)$-carrying memory (M_B) is a *lossy* accumulator precisely for this reason — the holographic blur is the compression.

Booking. The measurement-side orientation and the inverse aim (measurement → cause) are methodological framing. The holographic limit on inverse inference follows from self-similarity and reference-lessness (derivable), as does presence ≠ reconstructibility. "Holographic" is here explicitly an *analogy*: whether every excerpt carries information about the *whole* (including the distant) — strong holography — or merely resembles its local neighbourhood — weak self-similarity — is an open question to be tested (P35), not an established theorem.

am.13 Ordering of the levels

Three statements must be kept apart, each at its own temperature. *Mathematical*: U3 presupposes U1 and U2, and $D(k)$ is the accumulation structure derivable from the K_{frak} recursion (Doc. 295/296); and the linearisation error $e = x - \ln(1+x)$ of a measurement window is verified against the recursion — consequences, no speculation marker. *Philosophical (P35)*: the memory reflex and the linearisation reflex (measured uniformity as an artefact) as cultural-cognitive inversions, counterparts of the word reflex (Doc. 301). *Posit*: that time *is* fundamental (the third level, $D(k)$ as something real that is carried) is the same realist posit as in Doc. 301, not a new claim. The deflationary reader may always say: all only description. The argument of this document is independent of that posit — read even as mere description, U3 remains incoherent without U1 and U2, and the accumulation boundary remains a property of the time structure, not of memory.

am.14 Status

This document is a statement of position, not a new derivation: it brings together results from Doc. 285 (time as the woven mass-circle), Doc. 295 (log-spiral, $D(k)$, accumulation against a bounded kernel), Doc. 296 ($D(k)$ as measurable time, boundarylessness), and Doc. 301 (three levels, word reflex, self-projection) and Doc. 257 (the winding quantum as the bit limit) onto the question "memory or time first?". The FFGFT-internal core of this document — the thesis that memory presupposes time, the linearisation reflex, and the measurement and projection epistemology — stands on its own; the M_τ/M_B correspondence is the illustrative, *restricted* overlay from the bridge context and carries nothing the core needs. A measurement-theoretic extension places the reference-less excerpt, the congruence/self-similarity distinction (rolled up: 75 exact parts; unrolled: only asymptotically self-similar) and the linearisation reflex; the error quantity $e = x - \ln(1+x)$, with the scale-invariant constant $1 - \ln 2$, is verified against the recursion. It is framed as a counter-position to a "memory → experienced time" hierarchy, in a neutral tone and without personal address; the M_τ/M_B line is placed, factually, as a cognitive instance

of the $D(k)$ structure (booked as a *bridge candidate*, *restricted*, Category B), not as its origin. The relationship to the Spiral-Time Memory Hypothesis is symmetric, the bridge optional and rejectable; provenance and limit are set out in a dedicated section. The mathematical statements are corpus-supported; the reflex diagnosis is philosophically speculative (P35) and booked as such.

Chapter an

Doc. 305 — The Curvature-Fidelity Discriminator

Doc. 305 The curvature-fidelity discriminator — why reach does not separate

Abstract. Doc. 304 narrows the distinction between a uniform and an accumulating memory to a choice of axis: what separates is not *reach* (finite window versus unbounded integral) but *curvature-fidelity* (does the accumulator carry the $1/n$-decaying pitch, the self-similar texture, the ratio e). This note supports the finding internally to FFGFT with a deterministic testbench: at *equal reach*, a reach-based metric returns the same for both accumulators, while the e-criterion (Doc. 295) separates them cleanly and exactly — self-similarity $D(2k) - D(k) \to \ln 2$, pitch decay $d_n \sim n^{-1}$, scale ratio $\to e$. The finding is derivable from ξ / the recursion (*proven*). It establishes *no* specificity of any cognitive memory model; that extension remains a restricted bridge (Category B) and requires a separate, independent, jointly sealed benchmark.

an.1 The reduced point

In Doc. 304 the operational question of a "genuine" memory collapses onto a single axis. A memory that carries something about a past sits, in FFGFT, at $D(k)$, the cumulated defect of the unrolled time-winding

$$D(k) = \sum_{n \le k} 100\,\xi_n \;\to\; \ln\!\left(1 + \frac{k}{75}\right), \qquad \xi_{n+1} = \xi_n\,(1 - 100\,\xi_n) \tag{an.1}$$

(Doc. 295/296). Two accumulators can then be contrasted: one that carries the fractal *curvature* of this winding (the $1/n$-decaying pitch), and one that *linearises* it (constant pitch, Archimedean winding, the $d = 0$ limit). Doc. 304 argues that the load-bearing distinction is not reach but which of the two forms the accumulator carries. This note tests exactly that.

an.2 Two accumulators of equal reach

The testbench (`ffgft_kruemmungstreue_probe.py`) contrasts both forms at *equal reach*, i.e. with the same total accumulation $D(N)$ at the horizon N:

- *curvature-faithful* — the exact recursion $d_{n+1} = d_n(1 - d_n)$ with $d_n = 100\,\xi_n$, start $d_1 = 1/75$; cumulated $D(k) \to \ln(1 + k/75)$ (log-spiral, ratio e);
- *uniform* — constant pitch $d_{\text{const}} = D(N)/N$, cumulated $D_{\text{lin}}(k) = d_{\text{const}}\,k$ (Archimedean, no self-similarity).

Both reach the same D at N — reach is identical by construction.

an.3 Reach does not separate

An *unbounded* accumulator keeps the contribution of an arbitrarily old event in its running sum; a finite kernel of width W loses it once $N - t_0 > W$. A reach-based test — "does an early impulse still affect the output?" — therefore separates {accumulators} from {finite kernel}, but returns the same for the curvature-faithful and the uniform accumulator: both retain the impulse. Reach *cannot* tell the two apart.

an.4 Curvature-fidelity separates (the e-criterion)

The e-criterion of Doc. 295 — a winding carries ratio e exactly when the defect per step decays as $1/n$ — separates them cleanly. Three signatures, all checked against the exact recursion:

- *Self-similarity.* The scale-doubling increment $D(2k) - D(k)$ is, for the curvature-faithful accumulator, *scale-invariant*, $\to \ln 2 \approx 0.6931$ (at $k = 10^5$: 0.69274, deviation 4.1×10^{-4}); for the uniform one it grows linearly with k (at $k = 10^5$: 2.76). Only the log-spiral is similar to itself across scales.
- *Pitch decay.* The exponent p in $d_n \sim n^{-p}$ is, curvature-faithful, ≈ 1 (log-log slope -0.984 over a decade, i.e. $1/n$); uniform, $p = 0$ (constant).
- *Scale ratio.* The k-factor per unit increase of D converges, curvature-faithful, to e (at $D = 6$: 2.7226, deviation 4.3×10^{-3}; from $\exp D = 1 + k/75$); uniform, to 1 (arithmetic, no log-spiral).

The convergence is asymptotically exact (an $O(1/k)$ remainder, consistent with the congruence/self-similarity discussion in Doc. 304): the self-similarity is not a finite exact symmetry but a continuous one that settles fully in the limit.

an.5 The positive criterion

Hence the positive discriminator Doc. 304 calls for: what is to be tested is not "finite window versus accumulation" (reach) but *curvature-fidelity at equal reach* — does the accumulator carry the $1/n$ pitch, the ratio e, the self-similar texture? A long but linear kernel passes every reach test and remains uniform; a short but curvature-faithful accumulator

already carries the structure. The e-criterion is thus sharper than any reach boundary and makes the axis operationally testable.

an.6 Status

Proven (internal to FFGFT). The accumulation form, the three-signature separation, and the e-criterion are derivable from ξ and the recursion $\xi_{n+1} = \xi_n(1-100\,\xi_n)$ and verified against the exact recursion (Doc. 295; script `ffgft_kruemmungstreue_probe.py`, deterministic). The $O(1/k)$ asymptotics is named. This testbench is a script-verified *illustration* of the e-criterion (Doc. 295), not an independent new finding; its value is to make the axis operationally visible. FFGFT itself carries a *single*, curvature-faithful winding — the uniform accumulator is a contrast foil from the bridge context, not an FFGFT object; the separation question is one FFGFT poses *for the bridge*, not for itself.

Restricted (Category B). This testbench establishes *no* specificity of a cognitive, operationally and functionally defined memory (recursive access, semantic compression, affective weighting, identity binding, persistent state carry, predictive reuse). Whether curvature-fidelity separates such an object remains the restricted bridge of Doc. 304 and requires a separate, *independent, jointly sealed* benchmark — not an FFGFT-internal calculation. In both directions: a system may carry the $1/n$ texture without those functions, and a functionally relevant memory is not ruled out merely for failing the e-criterion.

Provenance and symmetry. $D(k)$, the e-criterion, and the FFGFT interpretation of temporal curvature originate in the FFGFT corpus; the operational memory architecture against which a bridge would be tested originates in independent work (Doc. 304, provenance section). Neither framework contains, grounds, or subordinates the other; rejection of the bridge leaves both unchanged.

Chapter ao

Doc. 306 — Native Time–Energy Reciprocity

Doc. 306 Native time–energy reciprocity — replacing a borrowed justification

Abstract. Documents 025 and 063 justify the absence of a cosmic initial singularity by Heisenberg's energy–time uncertainty relation. That justification is here *replaced*, not refuted: the conclusion (no singular beginning) stands, the derivation is replaced by a native one. Three defects are named — an inverted implication (a finite Δt yields a *finite* bound, not $\Delta E \to \infty$), a misapplication (the Δt of the relation is an observable's characteristic change time, not a cosmic age), and a breach of style ("irrefutably" is not a ledger term). The replacing chain is native: from $T \cdot m = 1$ (Doc. 003/010), hence $T \cdot E = 1$, it follows that $E \to \infty \Leftrightarrow T \to 0$; but FFGFT time is bounded below by $T_0 = \xi t_P$, fixed by $\xi = 4/30000$. T is therefore bounded below and E bounded above — an infinite density is structurally excluded. In addition $\partial T^4 = \varnothing$: the question of a "beginning" presupposes a boundary that a compact torus does not have. The same chain determines the black hole: the horizon is the *extremum* — slowest time $T = T_0$, maximal concentration $E = 1/T_0$, maximal curvature, all finite and well defined — no singularity, and hence nothing to repair. The same holds at the other end: $T \cdot m = 1$ has no solution at $m = 0$ — FFGFT knows no *empty* state, only a thinnest one. Both edges are excluded: the dense one is *quantified* from ξ ($T_0 = \xi t_P$, $E_{\max} = E_P/\xi$, with exact closure $T_0 E_{\max} = 1$; simultaneously the natural UV cutoff of the field theory), the thin one structurally, but unquantified — the latter no deficiency but a consequence of $T \cdot m = 1$: each system carries its own upper bound $R_{\text{Torus}}(m) = \hbar/mc$, so the cosmological sector requires its *own* scale.

ao.1 What is being replaced

In their original versions, Docs. 025 and 063 both contain the passage that Heisenberg's uncertainty relation $\Delta E \cdot \Delta t \geq \hbar/2 = 1/2$ (natural units) "irrefutably proves" a classical Big-Bang singularity to be impossible, and infer:

- a temporal beginning would mean Δt finite;
- this leads to $\Delta E \to \infty$;
- hence the universe must have existed eternally.

The *conclusion* — no singular beginning, instead a finite core with a minimal scale L_0 from ξ — is retained. The *derivation* is replaced. Three reasons.

ao.2 Three defects of the borrowed justification

(M1) Inverted implication. The inference "Δt finite $\Rightarrow \Delta E \to \infty$" is invalid. From $\Delta E \geq 1/(2\Delta t)$, a *finite* Δt yields a *finite* lower bound on ΔE; a beginning a finite time ago would give a tiny, harmless bound. $\Delta E \to \infty$ follows only from $\Delta t \to 0$, i.e. from the *instant* $t = 0$, not from a finite beginning. This is not a slip but a reversed implication.

(M2) Misapplication. Energy–time uncertainty is not a canonical uncertainty relation. By Pauli's theorem there is no self-adjoint time operator conjugate to a Hamiltonian bounded below; the rigorous form (Mandelstam–Tamm) defines $\Delta t = \Delta A / |\mathrm{d}\langle A\rangle/\mathrm{d}t|$ — the characteristic *change time of an observable*, not an elapsed duration. The "age of the universe" is not an admissible Δt for this relation.

(M3) Breach of style. "Irrefutably proves" is a strength of assertion that does not occur in the FFGFT ledger. The ledger knows *proven* (derived from ξ), *restricted*, and *blocked*; a statement is certified as sufficiently precise to be decidable, not as irrefutable.

The overarching point. Heisenberg's relation is not false within the QM framework, but it is *projected*: it belongs to the description level that FFGFT understands as a projection from the deeper structure. An argument against the cosmic singularity that leans on it does not leave the projection — it circulates within it. This is exactly what Doc. 298 warns of: one does not exit the projection by importing its theorems.

ao.3 The replacing native chain

(1) Exact reciprocity. The time–mass duality (Doc. 003, cf. Doc. 010) reads

$$T \cdot m = 1, \qquad \text{and with } E = m \text{ (natural units)} \quad \boxed{T \cdot E = 1} \tag{ao.1}$$

This is an *equation*, not an inequality — a sharper statement than $\Delta E \cdot \Delta t \geq 1/2$, and a native one.

(2) The time increments are finite. FFGFT time is the cumulated defect of the unrolled winding,

$$D(k) = \sum_{n \leq k} 100\,\xi_n \;\to\; \ln\!\left(1 + \tfrac{k}{75}\right), \qquad \xi_{n+1} = \xi_n\,(1 - 100\,\xi_n), \tag{ao.2}$$

with the first step

$$d_1 = 100\,\xi_1 = 100 \cdot \tfrac{4}{30000} = \tfrac{1}{75}. \tag{ao.3}$$

To be precise: $D(k)$ *grows* without bound as $k \to \infty$ (logarithmically). The singularity-free statement is not "D never diverges" but: $D(k)$ *is finite at every finite* k*, and there is no point*

at which D or its increments diverge. The first step is finite and fixed by ξ. The increments $d_n = 100\,\xi_n$ decrease monotonically from $d_1 = 1/75$ ($d_2 = 0.013156$, $d_n \approx 1/(n+74)$); the lower time bound follows from $T_0 = \xi t_P$, the time dressing of $L_0 = \xi \ell_P$ (Doc. 180, R90). The first increment is therefore the largest, not the smallest: $d_1 = 0.013333 > d_2 = 0.013156 > \ldots \to 0$.

(3) The inference. From $T \cdot E = 1$,

$$E \to \infty \quad \Longleftrightarrow \quad T \to 0. \tag{ao.4}$$

But no state with $T = 0$ exists in FFGFT: time is bounded below by $T_0 = \xi t_P$ (see (2); the bound is T_0, not d_1). Hence E is bounded above — an infinite energy density is *structurally* excluded, not by a statement about measurement precision. The exact numerical bound depends on the normalisation and is not asserted here as a number; the structural statement stands without it.

(4) The boundary question. The torus is compact and boundaryless, $\partial T^4 = \varnothing$ (Doc. 302; cf. Doc. 263, Doc. 296). The question "what was at $t = 0$?" presupposes a boundary at which a beginning could sit. A compact torus has none. The question is therefore not wrongly *answered* but wrongly *posed*: it imports the decompactified projection as ontology. What standard cosmology reads as a singular beginning is the point at which the *projection* ends — not the world.

ao.4 The black hole: an extremum, not a singularity

The same chain yields the FFGFT statement about black holes — and it is unspectacular: nothing diverges.

From $T \cdot E = 1$ with the lower bound $T \geq T_0$ (the smallest possible time interval, denoted the same way in the original T0 notation; cf. $\Omega(T) = T_0/T$ in Doc. 095) it follows immediately that

$$E \;\leq\; \frac{1}{T_0}. \tag{ao.5}$$

The extrema are *quantified* in the corpus and derived from ξ. The minimal length $L_0 = \xi \ell_P$ (Doc. 182) yields, via energy–length reciprocity, the cutoff

$$E_{\max} = \frac{\hbar c}{L_0} = \frac{E_P}{\xi} \approx 7500\, E_P \approx 9.16 \times 10^{22}\ \text{GeV}, \qquad T_0 = \xi t_P \approx 7.19 \times 10^{-48}\ \text{s} \tag{ao.6}$$

(Doc. 250, Doc. 290). The reciprocity closes *exactly*: ξ cancels,

$$T_0 \cdot E_{\max} \;=\; (\xi t_P) \cdot \frac{E_P}{\xi} \;=\; t_P \cdot E_P \;=\; 1 \qquad \text{(natural units)}. \tag{ao.7}$$

This follows from the construction: $T \cdot E = 1$ holds at the extrema, not merely in the middle range.

The same cutoff appears in field theory: the fractal torus supplies a *natural UV cutoff* at E_P/ξ, which is why traditional renormalisation is not required in the full FFGFT formulation (Doc. 019, Doc. 202); the same threshold explains the rarity of magnetic

monopoles (Doc. 143). The Lagrangian does not diverge because the *geometry* supplies the cutoff — the minimal time is not a regulator added afterwards, but follows from the structure.

Scales, not totals. One clarification, without which everything below would be misread: T_0 and $E_{\max}$ are *scales*, not bounds on totals. T_0 is the lower time *bound* of the winding; $E_{\max}$ is the maximal energy *per fundamental cell*, i.e. per elementary excitation — a *cutoff*, and a cutoff always bounds a single mode, never a sum. The total energy of a composite system may be arbitrarily large: it consists of many cells, not of denser ones. A macroscopic body is not *one* winding but very many; "the T of the Earth" is not a well-posed quantity, and $\hbar/(Mc^2)$ for a stellar mass is not a state of this relation but a misapplication. $T \cdot E = 1$ and its bounds hold *per excitation*, not per aggregate.

Accordingly, the statement about the black hole concerns *concentration*, not mass: at the horizon the *local* density reaches the cell cutoff. The *total* mass remains unbounded — a larger hole has more cells, not denser ones.

The horizon is therefore the locus at which the extrema are *attained*:

- the *slowest time*: $T = T_0$, not $T \to 0$;
- the *maximal energy and mass concentration*: $E = 1/T_0$ *per cell*, finite (a density, not a total mass);
- the *maximal curvature*: finite, because it is tied to E.

Every one of these quantities is well defined. The horizon is not a boundary to something else, nor a curtain in front of a singularity, but the surface on which time attains its minimum and concentration its maximum.

One consequence deserves to be stated, because it is easily missed: in FFGFT there is *nothing to repair*. Models that replace a singular core by a bounce, a transition shell, or a daughter branch are fixing a defect that FFGFT never produces. This is not a refutation of such models — they work within a continuum picture FFGFT does not share — but the observation that the question they answer does not arise here. Where the decompactified description says "infinite", the compact one says only: *this is the extremum*.

ao.5 The other extreme: the thinnest state

The reciprocity has two ends. One is the horizon; the other is usually called the "vacuum" — and precisely that name misleads.

The apparent circle. The objection runs: the vacuum is empty; it appears non-empty only through curvature; but curvature presupposes mass. No mass, no curvature; no curvature, nothing to fill the emptiness — the circle closes.

The circle is an import. It arises from a *causal arrow* that General Relativity brings with it: $T_{\mu\nu}$ *generates* curvature; mass is cause, curvature effect. In that picture one must ask "what is curving here?", and without mass there is no answer. FFGFT has no such arrow. $T \cdot m = 1$ is not a cause–effect relation but a *reciprocity*: mass and temporal curvature are

two readings of the same structure; neither generates the other. The question "which came first — mass or curvature?" is as wrongly posed as the question about $t = 0$: it imports a causal structure the compact description does not have. Nothing needs generating, and the circle dissolves.

Both ends are excluded. The fundamental relation itself forbids the edges:

$$m \to \infty \;\Rightarrow\; T \to 0 \qquad \text{excluded by } T \geq T_0 \;\; (L_0 = \xi \ell_P), \tag{ao.8}$$

$$m \to 0 \;\Rightarrow\; T \to \infty \qquad \text{excluded: } T \cdot m = 1 \text{ has no solution at } m = 0. \tag{ao.9}$$

FFGFT therefore knows *no empty state* — not because something fills it, but because the theory contains no description of "nothing". The torus with its winding *is* the object; "empty" would mean "no winding", and the winding is what time is. "Vacuum" asserts an emptiness that does not exist in this framework; *the thinnest state* or *the flattest winding* are the apter terms.

The asymmetry, stated honestly. The two ends are not equally well determined, and this should not be made to look tidier than it is:

- The *dense* pole is hard and derived: T_0 and $L_0 = \xi \ell_P$ follow from ξ (Doc. 182: the chain $\xi \to m_e \to G \to \ell_P \to L_0$ depends on ξ alone).
- The *thin* pole is *structurally excluded but not quantified*. That $m = 0$ is impossible follows from the relation; *how small* m can become does not — and for a *structural* reason, not because of a gap in the derivation. Doc. 182: "The question of a universal maximal size of the universe is not well posed: every physical scale has its own system-specific upper bound $R_{\text{Torus}}(m) = \hbar/mc$." That is $T \cdot m = 1$ in geometric form. If each mass carries its own torus radius, the cosmological scale *cannot* follow from ξ alone: it belongs to its own system and must be set independently. A separate cosmological scale is therefore *necessary*, not deficient.

A wall below, then, and no universal floor above — but to each system its own horizon.

What P35 concerns — and what it does not. Two things must be kept apart. That the cosmological sector needs its *own* scale is structural and unobjectionable. Writing it as $E_H = E_0\, \xi^{41/4}$ is not: the exponent $41/4$ is calibrated to H_0, not forward-derived, and the ξ-notation suggests a derivation that does not exist (P35/P36; Doc. 182 says so itself; Doc. 190). The objection concerns the *notation*, not the necessity of the separate scale.

And in between — a projection, not a structure. It is tempting to arrange the systems along a common axis, from maximal to minimal mass concentration, and to find oneself in the middle of it. That axis, however, is an *external view*: it presupposes a standpoint from which all systems appear side by side. FFGFT has no such standpoint. $T \cdot m = 1$ holds *per system*; there is no common T against which to plot, and no scale on which a system could locate itself.

More sharply: we *project ourselves into the middle*. The axis arises with the observer as its origin — whoever lays down the measure necessarily stands where the measure begins. The middle is therefore not a discovery but a positing, and it goes unnoticed

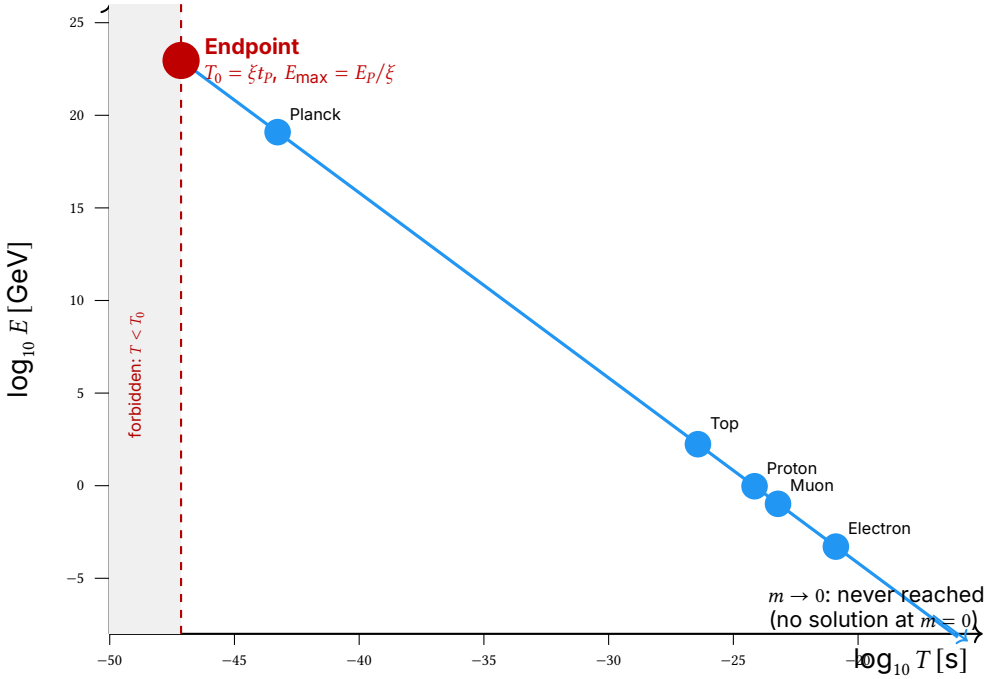


Figure ao.1: The relation $T \cdot E = 1$ (in SI: $T \cdot E = \hbar$) in doubly logarithmic coordinates: a *straight line* of slope -1. All elementary excitations lie exactly on it — none is privileged, and the line has *no middle*. Exactly one point is distinguished: the **endpoint** $(T_0,\, E_{\max}) = (\xi t_P,\; E_P/\xi)$, where the curve *stops* — the wall that follows from ξ. At the other end it continues as an *asymptote* and never arrives ($T \cdot m = 1$ has no solution at $m = 0$). The two edges are therefore of different kinds: on the left an attained endpoint, on the right an unreachable limit. What is plotted are *scales* (energy per excitation), not total masses; macroscopic bodies do not belong on this axis.

because the one who posits and the one who is located are the same. For the same reason the edges appear "extreme": they are what lies far from the chosen origin. *Extreme* here means *distant*, not *privileged*.

Intrinsic time is untouched by this — it is intrinsic. For an electron it is of no consequence where an outside observer enters it: its winding is the same in the laboratory as near a horizon, since $T = 1/m$ and m is the same. What differs is only how another system sees it — a relation between two, not a property of one. The "slowest time" at the horizon, too, is a statement about the comparison, not about the experience.

The projection is therefore not invalid but a *tool*: without a common axis nothing could be compared, and understanding requires comparison. It is a means of access, not a constituent of the structure. It is the same move that dissolved the question of $t = 0$ and the singularity, applied a third time — now to the scale itself. And it is a pointed form of the self-reference (Doc. 248): the thought that lays out the order is itself a process within it; it cannot stand outside, but *locates itself in the act of ordering*.

ao.6 An observation on the genesis

It should be noted that 025 and 063 already point to ξ in their *result*: the singularity is said to be "replaced by a tiny but finite core with minimal length scale L_0 (from ξ)". The answer therefore already came from ξ; only the ladder climbed to reach it was borrowed. Why the import was made although Doc. 003/010 was available is a *historical question about the genesis of the corpus*, not a systematic one. It is marked as such here and left unanswered; any answer would be speculation.

ao.7 Status

Correction / replacement. The conclusion (no singular beginning; a finite core with a minimal scale set by ξ) is unchanged. The justification in Docs. 025 and 063 is *no longer authoritative* for this point; it is not recorded as refuted, but as superseded by the native chain.

Proven (internal to FFGFT). $T \cdot m = 1$ (Doc. 003/010); $D(k)$ and $d_1 = 1/75$ from ξ and the recursion (Doc. 295); $\partial T^4 = \emptyset$ (Doc. 302). The inference $E \to \infty \Leftrightarrow T \to 0$ follows algebraically from the reciprocity; so does the bound $E \leq 1/T_0$ and with it the characterisation of the horizon as an extremum (no singularity). So does the exclusion of $m = 0$ (no empty state).

Quantified (proven). The extrema of the dense pole are derived from ξ and computed in the corpus: $T_0 = \xi t_P \approx 7.19 \times 10^{-48}$ s and $E_{\max} = \hbar c/L_0 = E_P/\xi \approx 9.16 \times 10^{22}$ GeV (Docs. 182, 250, 290), with the exact closure $T_0 \cdot E_{\max} = t_P \cdot E_P = 1$. The fractal torus thereby also supplies the natural UV cutoff of the field theory (Docs. 019/202/143).

Structural, not deficient. The *thin* pole remains excluded but unquantified — and that is a statement of the structure, not a gap: from $T \cdot m = 1$ it follows that every system carries its own upper bound $R_{\text{Torus}}(m) = \hbar/mc$; a universal maximal size is not meaningfully demanded (Doc. 182). The cosmological sector therefore requires its *own*, independently set scale.

Open (P35). Its *notation* remains to be cleaned up: $E_H = E_0\,\xi^{41/4}$ calibrates the exponent to H_0 and dresses a measured quantity in ξ-form; that is not a derivation (P35/P36, Docs. 182/190). The objection concerns the notation, not the necessity of the separate scale. The connection $R_H \to \ell_1$ remains open (P20, cf. Doc. 267); the cosmological sector as a whole remains the least determined area of the corpus.

Chapter ap

Doc. 307 — Time in the State Space, or Beside It — Locating Time and the Origin of the Mass Modes

Dok. 307 Time in the State Space, or Beside It — Locating Time and the Origin of the Mass Modes

Abstract. Two framings of quantum mechanics are contrasted: the *unrolled* view, which keeps time as an external parameter, and the *rolled-up* view, which draws it into the state space as a compact dimension. Both use the same operator formalism; the difference lies solely in whether time sits inside or outside. In the rolled-up view the masses become eigenvalues of a mode structure on a compact torus; the concrete FFGFT realisation is $\tilde{T} \cdot m = 1$ on $L^2(T^4) \otimes \mathbb{C}^3$ (Dok. 230/231/232, 306). The ratio structure of the masses thereby follows from a single dimensionless parameter $\xi_0 = 4/30000$ and is supported to $\sim 0.5\,\%$; the ratios are not derived exactly from ξ_0, and $m_\mu/m_e = 206.768$ is the measured anchor (see the section on the mass hierarchy); the absolute scale carries its SI correction through fixed bridge constants (P40) but is not a free parameter. Apparently open questions — absolute scale, causality of closed time, quantum gravity — are placed as already decided within the corpus (Dok. 013/190, 306, 230). The rolled-up view is thus positioned as a derivation framework with framework-specific observables (mass-reading, CHSH$\sim \xi$ – not resolvable with present hardware –, D_f, L_0), not as a thought experiment.

ap.1 The Fork

At the beginning of every quantum-mechanical model stands a choice that usually remains unspoken yet determines everything that follows: **Is time a direction *within* the state space, or a parameter *beside* it?**

The question sounds technical but is ontological. It does not decide which mathematics is used — both answers work with self-adjoint operators, spectra and projectors — but *where* time sits. And exactly this placement decides whether the physical quantities *fall out* of the structure or must be *attached* to it. The present discussion follows this single switch-point through its consequences.

This fundamental question reaches back to the roots of quantum theory. As early as 1932 **Wolfgang Pauli** recognised that introducing a time operator canonically conjugate

to the Hamiltonian runs into fundamental difficulties. The so-called “Pauli theorem” shaped the development of quantum mechanics decisively and made time as an external parameter seem unshakeable. Only later work, in particular by **Aharonov and Bohm** as well as **Garrison and Wong**, showed that under certain conditions (for example for spectra that are not half-bounded) time operators can nonetheless be constructed. The dispute over the time operator is not fully settled to this day and forms the background for the alternative discussed here.

ap.2 The Unrolled View: Time as an External Parameter

The familiar formulation of quantum mechanics locates time outside the state space. The Hilbert space is the space of states *at one instant*, typically $\mathscr{H} = L^2(\text{configuration space})$. Time is the parameter t along which the state evolves, mediated by the unitary evolution $e^{-i\hat{H}t}$. The observables — position, momentum, spin — are operators *on* $\mathscr{H}$. Time is not one of them.

That it cannot be one is not an oversight but a known result: Pauli’s objection shows that for a Hamiltonian bounded from below there is no self-adjoint, canonically conjugate time operator. In the unrolled view, therefore, time necessarily remains the stage, not the play.

The consequence is structural and far-reaching: because time sits outside, the state space alone contains no dynamical scales. Masses, couplings, lifetimes are not present in the pure spatial structure — they must be introduced from outside through *additional* structures. In the Standard Model this happens through Yukawa terms and the free parameters that fix their strength. The structure provides the frame; the numbers come from measurement.

This difficulty is not new. Already **Hermann Weyl** and later **Eugene Wigner** recognised that, from a group-theoretic perspective, the masses of the elementary particles appear as Casimir invariants — that is, as structural labels, not attached parameters. Nevertheless the question of mass generation remained a central unsolved problem of theoretical physics. The Higgs mechanism does offer an explanation for the mass of the gauge bosons, but itself introduces new free parameters (the Yukawa couplings).

The criticism of the unrolled view is therefore not that it is wrong, but that it leaves a fundamental question — the origin of the mass scales — unanswered and instead points to external inputs. This becomes especially clear when one considers the historical development of quantum field theory: the transition from the “old” quantum mechanics to quantum field theory did bring the possibility of particle creation and annihilation, but no intrinsic explanation for the mass values.

ap.3 A Representative Construction of This Kind

Many model programmes that aim at a geometric origin of the particle properties follow — for all their differences in detail — the same basic scheme, and it is worth considering this scheme in the abstract:

1. One chooses a discrete carrier (a lattice, a cut-and-project structure, a periodic graph).
2. On it one defines a self-adjoint family of operators.
3. Via spectral projectors one extracts *sectors* — eigenspaces or invariant subspaces of the operator.
4. These sectors are *subsequently* assigned physical meaning: charge, spin, and mass — the latter often through a parametric map, for example a Yukawa-type function with one or two free parameters.

Such a procedure can be fully controlled and reproducible mathematically. The decisive observation concerns not its rigour but its *placement of time*: the carrier is a spatial object, the operator evolution — where it occurs at all — runs over an external parameter. The sectors are therefore **spatial eigenspaces, not time modes**. The mass is not contained in them; it is attached to them through the parametric map.

And here an argument applies that carries itself: **a parametric map with free parameters can hit any finite list of values.** If the map is not fixed *in advance and universally*, it possesses no predictive content — it merely renames the known inputs. To yield any numbers at all, such a construction must therefore be coupled to *external* values: to measurement, and hence — since the measured values of the particle sector are Standard-Model-calibrated — indirectly to the Standard Model itself. This is the hard road: the numbers come from outside, not from the structure. A model that understands itself as an alternative ultimately draws its reference values from the very thing it wished to set itself apart from.

The point is not that this procedure is wrong. It is honest, provided it declares its own limit — namely that it does not *derive* masses but provides a controlled architecture of hypotheses. The point is that the *difficulty* of this road is no accident but the immediate consequence of the unrolled placement of time.

A concrete example: the so-called "graphene modelling" of particle physics defines quasi-particle excitations on hexagonal lattices. These excitations show linear dispersion relations and can be interpreted as massless fermions — but introducing a mass always requires additional terms (for instance a Haldane term or a Kekulé distortion) endowed with free parameters. The geometry provides the kinematics, but not the mass scale.

ap.4 The Rolled-Up View: Time as a Compact Dimension of the State Space

The alternative neither leaves time outside nor eliminates it, but draws it *into* the state space — as a closed, periodic dimension. In Hilbert-space language, in place of $L^2(\text{space})$ with external t there is a space in which a compact time direction is part of the geometry, for example $\mathscr{H} = L^2(T^4) \otimes (\text{internal factor})$, where one of the torus directions carries time. In the FFGFT corpus this Hilbert space is realised as $L^2(T^4) \otimes \mathbb{C}^3$ and the bijectivity of the representation (type III) is established (Dok. 230/231/232, 270).[1]

On a closed dimension the admissible states are **standing modes with discrete winding numbers**. Quantisation is then not postulated but geometrically enforced: only

[1] Dok numbers refer throughout to the FFGFT corpus (time–mass duality).

certain modes fit onto the closed structure. Mass becomes the **eigenvalue of an operator on $\mathscr{K}$ itself** — not attached, but structurally contained. The ratios of the masses are fixed because the admissible modes are fixed.

The relation time-cycle times mass equals one, $\tilde{T} \cdot m = 1$, is the concrete realisation of this idea: every massive quantity *is* its own time-cycle. In the corpus this is the native time–energy reciprocity $T \cdot E = 1$ (Dok. 306), and the decompactification $S^1_m \to \mathbb{R}_t$ is measure-preserving, hence lossless (Dok. 270). That a compact dimension produces a discrete tower of modes whose masses behave like the inverse of the compactification period is no exotic feature but the mechanism **Kaluza and Klein** already described for a compact spatial dimension ($m_n \propto n/R$). What holds there for a rolled-up spatial direction holds here for rolled-up time — with mass in the role of the mode index.

It is noteworthy that Pauli's objection does not touch the rolled-up view. It applies to an *open*, downward-unbounded time conjugate to a bounded Hamiltonian. A *compact* time behaves like an angle variable: its conjugate spectrum is discrete (as for angular momentum conjugate to an angle), and the Pauli argument lapses. Closed time is not only mathematically admissible, it is the natural home of a discrete mode structure.

This insight has deep roots in mathematical physics. **Witten** pointed out that compact time dimensions arise naturally in string theory. **Hawking** discussed the implications of a closed time for cosmology. The so-called "Kaluza–Klein theory with compact time" has been investigated in various contexts, for instance by **Reifler** and **Bars**, who developed a "two-time physics".

The rolled-up view also has connections to **thermodynamic time**. If time is compact, the thermodynamic arrow is not fundamental but an emergent property of the non-compact spatial parts. This could offer a natural explanation for the apparent irreversibility of macroscopic physics without resorting to ad-hoc assumptions about initial conditions.

ap.5 The Core of the Dispute

Now the actual weighing, without prejudgement.

What the unrolled view gains: seamless connection to the established machinery of quantum field theory; freedom from a strong ontological commitment; the familiarity of the open time axis. **What it costs:** the structure alone fixes no masses. Dynamical values must come from outside; predictions hang on external calibration; and a model of this type that understands itself as self-standing nevertheless couples its numbers back to measurement — and thus to the Standard Model.

What the rolled-up view gains: the modes and their ratios fall out of the structure without parametric attachment; the discrete quantisation is a gift rather than a postulate. **What it costs:** a strong ontological claim — time as a closed dimension — which must be justified and which contradicts the habitual intuition of flowing time. And it does not remove every external tie: the *absolute* scale still needs a unit bridge (the SI conversion via the conversion factors; Dok. 013, and Dok. 190 P40: absolute values carry the correction through bridge constants, which cancel in ratios) so that measurable numbers arise from the fixed ratio structure. What is given for free is the ratio structure, not the absolute scale.

The decisive finding of the dispute, however, is this: **both roads use the same operator formalism.** The difference lies not in the mathematics but solely in whether time sits

inside or outside. One can keep the same construction — self-adjoint operator, spectral sectors — and *only draw time into the space*; then the sectors become modes, and the mass that previously had to be attached through a parametric map becomes the eigenvalue of the structure itself. Whoever reads the sectors as spatial eigenspaces must attach the physics. Whoever reads them as time modes finds it contained. It is the same computation — with time in a different place.

This recognition has far-reaching methodological implications. It shows that the choice of where to locate time constitutes a kind of "framework choice" that precedes the actual model-building and fixes its interpretive reach. **This is no trivial observation**: the so-called **"quantum equivalence"** of the two views is well known, but its methodological consequences have so far scarcely been investigated systematically.

ap.6 Who Has Already Pursued the Direction "Time Not External"

The idea of not treating time as an external parameter is no novelty but a rich and long-pursued line in the foundations of physics. It is honest to place the rolled-up view within this line — and to mark off its distinctive feature against it.

Relational Time

- **Page & Wootters** let time arise *within* the state space: as a correlation (entanglement) between a clock subsystem and the rest. The total state is stationary; "evolution without evolution". Time here is internal, not external.
- The work of **Moreva et al.** and **Marletto & Vedral** made this mechanism precise and even illustrated it experimentally.
- **Gambini & Pullin** developed the Montevideo interpretation with relational time.

Timeless Quantum Gravity

- **Wheeler–DeWitt**: in canonical quantum gravity the universal wave function is timeless, $\hat{H}\Psi = 0$. The resulting "problem of time" is precisely the statement that there is no external time.
- **Kuchař** and **Isham** provided the authoritative surveys of the time problem of canonical quantum gravity.
- **Rovelli**: relational quantum mechanics; "Forget Time"; with **Connes** the thermal-time hypothesis — time as a derived, not fundamentally external quantity.
- **Barbour**, *The End of Time*: the consistent elimination of time in favour of timeless configurations.

Compact Time in String Theory

- String theory has investigated compact time dimensions in various contexts.

- **Bars** and colleagues developed a “two-time physics” that postulates an extended symmetry group with two time dimensions.

The Mode Structure

Two further references bear less on the time question than on the *mode* side and establish that “mass as a structural eigenvalue” is well familiar:

- **Wigner**: mass is a Casimir invariant — a label of irreducible representations of the Poincaré group. Mass there is already a structural eigenvalue, not an attached number.
- **Kaluza** / **Klein**: a compactified dimension produces a discrete tower of modes with quantised charges — the mathematical blueprint for a rolled-up dimension bringing forth modes.

The delimitation this overview permits is precise: the predominant part of this tradition makes time *relational*, *emergent* or *timeless* — it takes time *out* as something fundamental. The rolled-up view does something subtly different: it takes time *in* — as a compact geometric dimension of the state space that carries modes. It shares with Page–Wootters and Rovelli the rejection of external parameter-time, but differs in the positive proposal (compact geometric time instead of relational or thermal time). The novelty lies, then, not in “time is not external” — that is old and well established — but in the specific route “compact time produces mass modes”.

ap.7 Extension: Mathematical and Physical Implications

The Mathematical Structure of Rolled-Up Time

For a more precise formulation of the rolled-up view the Hilbert space $\mathscr{H} = L^2(T^1 \times M) \otimes \mathscr{H}_{\text{int}}$ serves, where T^1 is a compact time dimension of length T and M is ordinary three-dimensional space. A function on this space can be expanded into a Fourier series:

$$\Psi(t, \mathbf{x}) = \sum_{n \in \mathbb{Z}} e^{2\pi i n t/T} \psi_n(\mathbf{x})$$

The discrete frequencies $\omega_n = 2\pi n/T$ correspond to the mass values (in natural units $\hbar = c = 1$). The operator generating these frequencies is the energy operator $\hat{E} = -i\,\partial_t$, whose eigenvalues are exactly the discrete spectrum $\{2\pi n/T\}$.

The decisive observation is: the quantisation of the masses is a direct consequence of the periodicity of time. **Nothing has to be postulated** — the discrete modes are mathematically enforced.

The Yukawa Coupling in the Rolled-Up View

The coupling between different modes can be written as a matrix element of the interaction operator:

$$g_{ijk} \propto \int_0^T e^{2\pi i (n_i - n_j - n_k) t/T}\, dt = T \cdot \delta_{n_i,\, n_j + n_k}$$

This is an exact conservation of angular momentum in the time direction! The "Yukawa couplings" of the Standard Model appear here as selection rules for the mode indices, not as free parameters.

The Role of the Gauge Symmetries

Rolled-up time permits a natural extension of the gauge symmetries. **The compact time direction behaves like an additional fifth dimension** in the sense of Kaluza–Klein theory, but with the decisive difference that it is time-like. This provides the explanation for charge quantisation:

$$Q \propto \frac{n}{R_T}$$

where R_T is the radius of the compact time. Charge quantisation is thereby a direct consequence of the periodicity of time.

The Mass Hierarchy in the Rolled-Up View

A central problem of particle physics is the mass hierarchy: why is the electron so much lighter than the top quark? The corpus describes the lepton masses in three representations that *translate into one another* — none is a separate ansatz; they are the same structure written differently (Dok. 006/046/282/292).

(1) Fundamental — quantum numbers and fractal correction. Particles are discrete resonance modes of the ξ-field; their scale follows from a geometric quantisation with the quantum numbers (n_i, l_i, j_i) of the 3D wave equation,

$$\xi_i = \xi_0 \cdot f(n_i, l_i, j_i), \qquad E_i = 1/\xi_i, \qquad \xi_0 = \tfrac{4}{30000},$$

where the absolute values carry the fractal correction $K_{\text{frak}} = 1 - 100\,\xi$ (Dok. 011/012/133). Example: electron $(1, 0, \frac{1}{2})$, muon $(2, 1, \frac{1}{2})$. Here $f(n, l, j)$ is the geometric structure factor of the quantum numbers, not a frequency and not the base winding number $f = 1/\xi$.

(2) Yukawa / ladder form. The same derivation is written as $m_i = r_i\, \xi^{\pi_i}\, v$ (Dok. 006/046) with rational r_i (settings: their derivation from an (n, l, j) formula does not hold) and the generation hierarchy $\pi = \frac{3}{2},\ 1,\ \frac{2}{3}$ — the Standard-Model-compatible bridge.

(3) Circulant / Koide. The $\sqrt{m_k}$ are eigenvalues of a $\mathbb{Z}_3$-circulant, $\sqrt{m_k} = M\big(1 + \sqrt{2}\cos\big(\frac{2}{9} + 2\pi k/3\big)\big)$ (Dok. 282/292), with the Koide relation $Q = 2/3$; the angle $\theta = 2/9$ is the output of the diagonalisation.

The decisive point: the *dimensionless ratio structure* is fixed (Koide $Q = 2/3$ holds structurally for the circulant, $\theta = 2/9$); the mass ratios themselves, however, are not derived exactly from ξ_0. The ladder gives $m_\mu/m_e = (16/5)/(4/3) \cdot \xi^{-1/2} = \sqrt{43200} = 207.846$ ($+0.52\,\%$); the exact pair $(\sqrt{2},\ 2/9)$ gives 206.7703 instead of CODATA $206.7682827(46)$ (rel. $9.8 \cdot 10^{-6}$, $\approx 440\sigma$) and is excluded in exact form. 206.768 is the measured anchor; what is supported is the ratio structure from ξ to $\sim 0.5\,\%$. In the step to *absolute* SI values further sources of deviation are added, several at once: the SI conversion itself ($\hbar, c, G$ are convention or conversion factors, not a measured dimensional anchor), the bridge constants v and E_0 (P40), the fractal correction K_{frak}, and finite measurement precision (e.g. of the τ mass). The claim is therefore a *structure proof*: the ratio structure is fixed,

the absolute scale carries its correction — a numerically better calculation than the Standard Model is not claimed.

The Relation to String Theory

Rolled-up time has parallels to **M-theory** and **F-theory**, both of which use compact dimensions. In F-theory an additional (formally two-dimensional) fibre torus encodes the axio-dilaton; gauge symmetries arise at its degeneration loci (7-branes). The two extra dimensions of F-theory (formal signature (10,2)) are this elliptic fibre torus, an auxiliary geometry for τ; there is no physical compact time direction there. The parallel is therefore only an analogy: the view developed here could be understood as a kind of "F-theory light": a compact time dimension without the full string-theory machinery.

ap.8 Critical Appraisal and Placement

Ontological Status of Time

The ontological claim of the rolled-up view is strong, but it is not left open in the corpus: compact time is real and geometric, not merely a computational technique. The boundary $\partial T^4 = \emptyset$ makes the question of a starting instant $t = 0$ not wrongly answered but *wrongly posed* (Dok. 306). The time here is *discrete*: FFGFT possesses a lower time scale $\tau_0 = \xi t_P$ fixed by ξ (time dressing of L_0, Dok. 180, R90); the defect increments d_n decrease monotonically from $d_1 = 100\,\xi_0 = 1/75$ (Dok. 306), and the admissible states are discrete modes. These modes stand in fixed relation to one another: via $\tilde{T} \cdot m = 1$ every massive quantity does carry its own cycle $T = \hbar/mc^2$, but these cycles are not independent clocks; they are cycles in fixed ratios determined by the mass structure ($m_\mu/m_e \approx 207$, $m_\tau/m_\mu \approx 17$), which, computed exactly, are not commensurable — not a continuum of free clocks. The cycle ratios are mass ratios and irrational (ladder $m_\mu/m_e = 120\sqrt{3}$, $m_\tau/m_\mu = (125/144)\cdot 7500^{1/3}$; circulant 206.77, and 3477.47 for m_τ/m_e); there is no common base tick of which they would be harmonics.

The physicist **John Wheeler** spoke of "time as an illusion" — a position the rolled-up view does not share. It takes time as real, but geometrically different from the accustomed picture.

The Absolute Scale: Fixed, Not Free

The ratio structure of the masses follows from the mode structure. The absolute scale is thereby not yet expressed in SI numbers — but it is for that reason no *free* parameter. The ratio scaffold rests on a single dimensionless parameter $\xi_0 = 4/30000$ (the geometric ξ-value of the position-space torus). The absolute values carry their fractal or SI correction not as a fit but through fixed *bridge constants* — v in the masses $m = r\xi^p v$, $E_0 = 7.398$ MeV in $\alpha = \xi E_0^2$, and fixed numerical factors in G. In ratios these bridge constants cancel; the comparison value $m_\mu/m_e = 206.768$ is the measured value (anchor), neither the ladder (207.846) nor the circulant (206.7703, $\approx 440\sigma$ off; see the section on the mass hierarchy). $\hbar$ and c are pure SI conversion; a *measured* dimensional anchor in the standard-physics sense does not exist.

The Planck quantities are not a fundamental input but follow from ξ: the derivation chain reads $\xi \to \{m_e, \dots\} \to G \to \ell_P$, and $\hbar$ is pure SI conversion. The *only* exact ξ-power relation here is

$$\frac{L_0}{\ell_P} = \xi_0, \qquad L_0 = \xi_0\, \ell_P \approx 2.15 \times 10^{-39}\ \text{m};$$

the SI factor ℓ_P must *not* be absorbed into a ξ exponent. The time-cycle is therefore not "coupled to the Planck scale" — the causal direction is reversed: the Planck quantities follow from ξ, not ξ from them. Via $\tilde{T} \cdot m = 1$ (respectively $T \cdot E = 1$) the cycle is fixed as soon as the mass is fixed through the ξ-geometry. No modulus remains to be gauged; the dilaton or moduli analogy to string theory does not carry here. (Worked out in the corpus: Dok. 013 SI conversion table; Dok. 190 P40 and Präz. 2/4/5; Dok. 011/012 and Dok. 133 for $K_{\text{frak}} = 1 - 100\,\xi$; $L_0 = \xi_0 \ell_P$ in Dok. 180–182.)

Causality

The obvious worry — does a periodic time produce closed time-like curves (CTCs)? — is addressed in the corpus, not open. The carrier is a torus ($\pi_1 \neq 0$, stable windings), not a simple circle, and the relevant connection is not ontological non-locality but a *topological* connection on T^4: the Bell/CHSH correlation is geometrically real and carries an amplitude $\sim \xi$ (Dok. 230, cf. Dok. 022).

The classical CTC problem of the Gödel solutions and the chronology-protection conjecture concern the *decompactified* Lorentz sector; the compact mass–time direction is spectral, not causally closed (Dok. 270/306).

Experimental Testing

The predictions of the rolled-up view are *not* those of a hidden heavy tower of modes above accelerator reach. The discrete spectrum of compact time *is*, via $\tilde{T} \cdot m = 1$, already the observed particle spectrum (mass-reading); there is no additional Kaluza–Klein tower to wait for. The framework-specific, testable quantities are rather:

1. **Mass-reading:** the particle masses themselves — lepton mass structure ($m_\ell = r_\ell \xi^{p_\ell} \nu$, Doc. 190 Refinement 5), Koide relation (10^{-5}, Dok. 292) — as ξ-derivations. The electron $g - 2$ is not such a test: ξ contributions to a_e are excluded (Doc. 382).
2. **Selection rules:** mode-index conservation is identical to the coupling selection rules of the Standard Model (cf. the section on the Yukawa coupling), with no free Yukawa numbers.
3. **CHSH $\sim \xi$:** the Bell correlation carries an amplitude of order ξ (Dok. 230, Dok. 022) — the compact time is interpreted as a real, topological connection on T^4, not as ontological non-locality; the ξ effect is not resolvable with present hardware.
4. **Fractal structure:** the geometric fundamental length $L_0 = \xi_0\, \ell_P$ (Dok. 180–182) and the fractal dimension D_f (Dok. 270/249).

For a *clean* falsification, ratio tests alone are insufficient — they confirm (Koide) but hardly falsify, since deviations can always be read as unmodelled coupling. Only a *forward*-fixed, independently testable new prediction is load-bearing (Dok. 190 P40).

Connection to Quantum Gravity

The bridge to quantum gravity is not merely hinted at in the corpus but worked out as a Hilbert-space construction: $L^2(T^4) \otimes \mathbb{C}^3$ with the quantity $\theta = 2/9$ following from the $\mathbb{Z}_3$-circulant diagonalisation (Dok. 230/231/232; in the corpus *settled*, not open).

In loop quantum gravity time is understood relationally — a position to which the rolled-up view is close; **Connes** shows that time can follow from the operator algebra. The rolled-up view makes this line concrete geometrically rather than merely relationally.

ap.9 Conclusion

The choice of where time sits precedes the mathematics and decides whether the modes are a gift or an attachment. A construction that leaves time outside will — structurally, not through carelessness — have to attach its physical values from outside; and because free parametric maps hit any finite list of values, it couples its numbers to measurement and thereby to the Standard Model. That is the price of the unrolled view: no self-standing prediction without an external tie.

Drawing time into the state space is *one* road — with a real ontological price — to making the modes and their ratios structural. The long line from Page–Wootters through Wheeler–DeWitt to Kaluza–Klein shows that the move "time not external" is well motivated and much pursued; the concrete realisation "compact time carries the mass modes" is a definite, strong choice within this line.

The dispute therefore does not end with a verdict but with a sharpened alternative: whoever reads sectors as spatial eigenspaces keeps the familiar physics and pays with the external attachment of the numbers. Whoever takes time as a closed dimension into the space obtains the modes structurally and pays with an ontological commitment. The ontological price of the second choice remains on the table — but the methodological price of the first is no smaller, only more familiar.

The rolled-up view is, in this, no mere thought experiment: it possesses framework-specific, testable quantities — mass-reading (masses, Koide), CHSH $\sim \xi$ (not resolvable with present hardware), fractal dimension D_f, fundamental length L_0 (see the section on experimental testing).

The analysis developed here shows that the question of the origin of the modes is inseparable from the question of the placement of time. **Time is not merely a parameter — it is the structure that shapes the modes.** Or, as the physicist **Carlo Rovelli** put it: "Time is not what we think it is."

ap.10 Epistemic Status

Proven (derived from ξ). The ratio structure of the masses ($m_\ell = r_\ell\, \xi^{p_\ell}\, v$, structure proof to $\sim 1\,\%$, Doc. 190 Refinement 5); the dimensionless ratio structure (Koide 10^{-5}; P40), where $m_\mu/m_e = 206.768$ is the measured anchor, not a result — the exact pair $(\sqrt{2},\ 2/9)$ misses it by $\approx 440\sigma$, the ladder by $0.52\,\%$; the Hilbert-space representation $L^2(T^4) \otimes \mathbb{C}^3$ with $\theta = 2/9$ from the $\mathbb{Z}_3$-circulant diagonalisation (Dok. 230/231/232); the native time–energy reciprocity

$T \cdot E = 1$ and $\partial T^4 = \varnothing$ (Dok. 306); the only exact ξ-power relation $L_0/\ell_P = \xi_0$ (Dok. 180–182; P35).

Restricted (bridge). The absolute SI scale is fixed through fixed bridge constants (v, E_0, G-factors), not free — but it is no ξ-forward result: $\hbar$ and c are SI conversion, G is derived ($\xi \to \{m_e, \ldots\} \to G \to \ell_P$; Dok. 190 P40). A measured dimensional anchor in the standard-physics sense does not exist.

Open. The *quantum equivalence* of the two framings is known; its methodological consequences are systematised here, not conclusively assessed.

Part V

Part V — Cosmic Sector and Resonance Geometry

Chapter aq

Doc. 308 — The Cosmic Sector of FFGFT: Redshift, Dark Sector and Geometric Gravity

What this is about

FFGFT advances an unusual claim: the universe is not expanding. Cosmic redshift is not a Doppler effect but a geometric property of the ξ-field. Dark energy does not exist as a physical entity. And dark matter requires no particles — its effects are to follow from the same geometry that carries light deflection; this is established for gas-dominated dwarf galaxies, for spiral galaxies and the Bullet Cluster it is open.

This document assembles the cosmic sector of FFGFT: what has been derived, what has been observationally confirmed, and where the limits lie. It addresses readers with a physics background who want to know *what* the theory says and *why*.

The three sources of degeneracy are named, the two cores of the geometry are developed, and the status of each step — derivation, structural argument, posit — is declared.

Conventions

Legend.

[K]	*Core derivation* — derived mathematically or numerically from the foundational posits; reproducible.
[S]	*Structural claim* — justified but not yet fully proved.
[POSIT]	*Posit* — declared assumption, not further derived.

Central quantities.

$\xi = 4/30000$	Only fundamental parameter of FFGFT [POSIT, P33].
$\tilde{T} \cdot m = 1$	Foundational relation: intrinsic time and mass are inversely coupled.
$R_H = c/H_0$	Geometric maximal scale — not an expansion horizon.
$a_0 = c^2\xi^{10}/(4\bar{\lambda}_e)$	Transition scale of the inertia regime; below a_0 inertia is non-linear ($\tilde{T} \cdot m = 1$ + Unruh); $= 1.033 \times 10^{-10}$ m/s^2.
$\bar{\lambda}_e = \hbar/(m_e c)$	Reduced Compton wavelength of the electron.

aq.1 The Static Universe and Redshift

The fundamental choice

ΛCDM describes an expanding universe: galaxies move apart, light is redshifted by the expansion of space, and the observation of distant supernovae reveals an accelerated expansion accounted for by a cosmological constant Λ.

FFGFT makes a different fundamental choice: the universe is static. Cosmic redshift arises not from motion but in the mass reading: clocks and rulers run with $1/m$, and the geometric path elongation in the ξ-field describes the same redshift within this reading. As an energy loss of the photon alone it is ruled out (Doc. 392).

The redshift mechanism [K]

The FFGFT redshift formula (Doc. 280):

$$1 + z = e^{\xi x} \qquad \Longrightarrow \qquad z \approx \xi x \quad \text{(for small } z\text{)},$$

where x is the path length, measured in a length that is not derived from ξ and is fixed only via the measured H_0: the form of the relation is derived, its scale is calibrated (P39). The redshift grows with the distance travelled, without wavelength dependence. It is read in the mass reading, in which clocks and rulers run along; only this reading passes the supernova light curves, the background temperature and the Tolman test, while an energy loss of the photon alone fails all three **[X]** (Doc. 392). With x as the coordinate distance of the mass reading, $H(z) = H_0(1+z)$ follows; read this way the law carries the DESI data up to $z \approx 1.5$ and misses the Lyman-α point at $z = 2.33$ by $5.3\,\sigma$ **[X]** (Doc. 392). A different identification of x has not been tested **[S]**.

Achromatic [K]: The path elongation is purely geometric and stretches *all* wavelengths by the same factor $e^{\xi x}$. Finite-element simulations of the geometry confirm this (K6, Doc. 190). A chromatic redshift would destroy the coherence of the CMB spectral shape — this is not observed.

The degeneracy of the data [K]

For the three classical pillars of cosmology — Type Ia supernovae, baryon acoustic oscillations (BAO), CMB temperature — the following holds: they cannot in principle

distinguish between expansion and static geometric path stretching, as long as both models produce the same distance-redshift relation $z(d)$ (Doc. 267).

This is not a weakness of FFGFT. It is a methodological statement: *these data alone do not decide between the models.* The decision shifts to theoretical consistency, parameter count and derivability.

What follows from this

For dark energy: Λ is needed in ΛCDM to account for the apparent accelerated expansion. In the static reading there is no expansion to be accelerated. Λ loses its referent in the static reading — it is not falsified, it becomes objectless.

For the Hubble tension: Different measurement methods for H_0 give different values. In the FFGFT reading, H_0 does not measure an expansion rate but a parameter of the ξ-geometry. FFGFT knows only *one*, calibrated H_0 value and does not distinguish between early and late measurements; the tension cannot be modelled in FFGFT — it is neither reproduced nor resolved.

For the z-axis: Every presentation of cosmological data "over time" is formulated in ΛCDM language. An FFGFT interpretation must treat z as a geometric distance parameter, not a time axis.

aq.2 Geometric Gravity: the Two-Halves Synthesis

Einstein 1911 and 1915

Einstein calculated the deflection of light at the Sun in 1911 from time dilation alone:

$$\alpha_{1911} = \frac{2GM}{c^2 b} = 0.875''.$$

That was half the measured value. The 1915 theory of GR yielded the full value $\alpha_{1915} = 1.750''$, because space *and* time curvature each contribute one half.

FFGFT *explains* this factor of 2 from first structure.

K3 area invariance [K]

$\tilde{T} \cdot m = 1$ (time curvature) and fractal path elongation (space curvature, Doc. 182) are in FFGFT not alternative mechanisms but **complementary halves**. The K3 rule — area invariance of the time-space 2-cell — anchors this:

$$L_t \cdot l = \text{const.}$$

This is formally the reciprocity form of $\tilde{T} \cdot m = 1$, applied to the time-space pair.

FE verification [K]: Ray tracing through three configurations (30001 log nodes, RK4, `ffgft_308_p36_stufeB1_faktor2_fe.py`):

Configuration	α at solar limb	Ratio to $1.75''$
Time rate only ($\tilde{T} \cdot m = 1$)	$0.875''$	0.500
Path elongation only (space)	$0.875''$	0.500
Both together (K3)	$1.750''$	1.000

PPN test [K]: K3 is the sole survivor of the T^4 selection computation (four candidates tested, `ffgft_308_p36_stufeB2_kappa_herleitung.py`): $\gamma_{\text{PPN}} = 1$ exactly. Cassini bound $|\gamma - 1| < 2.3 \times 10^{-5}$ passed with unlimited margin.

The same structure in the inertia regime: a_0 [K]/[S]

The K3 two-halves structure appears a second time, at cosmological scale. From Doc. 279 (the exponent 10 is calibrated to H_0, P20/P39; Doc. 309):

$$\frac{H_0}{c} = \frac{\pi}{2}\frac{\xi^{10}}{\bar{\lambda}_e} \qquad \Longrightarrow \qquad R_H = \frac{2\,\bar{\lambda}_e}{\pi\,\xi^{10}}.$$

R_H is the geometric maximal scale of the compact T^4 geometry (Doc. 182) — not an expansion horizon, valid in the static reading.

The cycle doubling [S]: The K3 2-cell has two cycles. A closed path that traverses each once measures:

$$L = \underbrace{2\pi R_H}_{\text{time cycle}} + \underbrace{2\pi R_H}_{\text{space cycle}} = 4\pi R_H.$$

Rindler saturation [S]: The Rindler horizon of an observer with acceleration a lies at c^2/a. When it reaches L, the acceleration is no longer relationally definable. This sets:

$$a_{\text{bg}} = \frac{c^2}{4\pi R_H} = \frac{c^2\,\xi^{10}}{8\,\bar{\lambda}_e}.$$

Inertia from $\tilde{T} \cdot m = 1$ [S]: The foundational relation is a statement about mass. If inertia follows the time field (Unruh coupling), the force law in closed form is:

$$\boxed{a = \sqrt{a_N^2 + a_N a_0}} \qquad \text{with} \quad a_0 = 2\,a_{\text{bg}} = \frac{c^2\,\xi^{10}}{4\,\bar{\lambda}_e} = 1.033 \times 10^{-10}\ \text{m/s}^2.$$

The 2π of the geometry cancels exactly against the $\pi/2$ from Doc. 279. The rational factor 4 remains:

$$4 = \underbrace{2}_{\text{K3: time and space cycle}} \times \underbrace{2}_{\text{Unruh: } a_0 = 2a_{\text{bg}}}.$$

No step contains a further free parameter; the scale takes over the calibration of the exponent 10 to H_0.

aq.3 The Dark Sector

Dark energy: not a physical object

Λ arises in ΛCDM as an answer to the apparent accelerated expansion. In the static reading this expansion does not exist. Λ thereby loses its referent — it is not falsified by the reading change, it becomes objectless.

Three conditions speak against it:

1. **Only from degenerate data.** SNe Ia, BAO, CMB — exactly the datasets between which both readings are indistinguishable.
2. **No independent anchor.** The only theoretical prediction (QFT vacuum energy) misses the value by 10^{122}.
3. **Objectless under reading change.** Not refuted, but without referent.

That is the status: reading artefact. Not "dark energy does not exist" — that would be a statement within the expansion reading — but: it is a reading-internal bookkeeping quantity.

Dark matter: a regime, not a substance

The findings — rotation curves, gravitational lensing, Bullet Cluster — are reading-independent. They do not disappear under a reading change. FFGFT traces them, without a new ingredient, to the same force law that $\tilde{T} \cdot m = 1$ together with K3 geometry delivers; this is established for the gas-dominated dwarf galaxy DDO 154, for spiral galaxies and the Bullet Cluster it is open (see below).

Dark matter in FFGFT is not a substance. It is a name for a regime.

The force law in the galactic regime [K]/[S]:

$$a = \sqrt{a_N^2 + a_N a_0}.$$

For $a_N \gg a_0$ (inner regions): $a \to a_N$ (Newton). For $a_N \ll a_0$ (outer regions): $a \to \sqrt{a_N a_0}$, hence $v^4 = GM_b a_0$ — the baryonic Tully-Fisher relation.

In FFGFT, dark matter does not denote a missing mass component but the regime below a_0 in which inertia is non-linear. The missing mass is not missing — inertia is different there.

Results [K] (`ffgft_308_p36_stufeA_a0_anker.py`):

Galaxy	Type	v_{Newton}	v_{FFGFT}	v_{obs}
DDO 154	dwarf, gas-dominated	18.5	47.1	47.0 km/s
NGC 3198	spiral	50.4	121	150 km/s
Milky Way	spiral	122	180	220 km/s

DDO 154 is the cleanest test (gas-dominated, M_b best known): $v_{\text{FFGFT}}/v_{\text{obs}} = 1.002$. The deviations for NGC 3198 and the Milky Way ($121/150 - 1 = -19.3\,\%$, $180/220 - 1 = -18.2\,\%$) lie well outside the M_b uncertainty: for $v \propto M_b^{1/4}$, $\pm 30\,\%$ in M_b correspond to only $+6.8\,\%/-8.5\,\%$ in

ν (even in the Newtonian limit only $+14\,\%$). Closing them would require $M_b \times 2.36$ (NGC 3198) and $\times 2.23$ (Milky Way). For spirals the test is open.

Bullet Cluster [K] (`ffgft_308_p36_stufeC1_bullet_cluster.py`): Two-clump FE projection (Clowe+2006 parameters) at instantaneous coupling. According to Clowe et al. 2006 the X-ray gas is the dominant baryonic component ($M_{\text{gas}}/M_{\text{stars}} \approx 5$–$10$; gas ~85 % of the baryons, i.e. $5.7 : 1$). With these baryonic masses the lensing peak lies at 240 kpc, i.e. at the gas; it moves to the gas already from gas:stars ≈ 1.4. A galaxy mass of $2.3 \cdot 10^{14}\,M_\odot$ would be the lensing mass including the missing component; a test with it is circular. The Bullet Cluster is therefore not established as passed; the test is open.

Recent data [Q]/[S]: Weak lensing around isolated galaxies shows flat circular velocities, reliably out to at least 300 kpc, and the baryonic Tully-Fisher relation (Mistele et al. 2024) [Q]; this corresponds to the outer limit of the force law. Wide binaries below a_0 are disputed: a deviation from Newton at $4.9\,\sigma$ (Chae et al. 2026) against a model-dependent analysis of the same data that is consistent with Newton (Saad, Ting 2026) [Q]. The force law for systems in the external field of the Milky Way has not been derived in the corpus [S]. For the LZ event at 248 keV and the advance commitment of FFGFT see Doc. 396.

Asymmetry in comparison

ξ is cross-anchored — it carries the lepton mass ladder, the Koide scalar, the α reference, K_{frak} and k^* (Doc. 275). It works in five foreign sectors before the cosmological one.

Λ works nowhere except its own fit.

That is the difference: not “true vs. false” but “cross-anchored vs. isolated”.

aq.4 Summary: Picture of the Cosmic Sector

Quantity	FFGFT	Status
Cosmic redshift	mass reading, equivalent to FRW (Doc. 392)	[B]/[K]
Distance law $1+z = e^{\xi x}$	x as coordinate distance: Lyman-α $5.3\,\sigma$	[X]
Achromatic	geometric, FE confirmed	[K]
Degeneracy SNe/BAO/CMB	between both readings	[K]
Λ (DE)	reading artefact, no referent	[K]
Hubble H_0	$c\pi\xi^{10}/(2\bar{\lambda}_e)$, exponent calibrated	[POSIT]
Light deflection	$0.875'' + 0.875'' = 1.750''$	[K]
$\gamma_{\text{PPN}} = 1$	K3 sole survivor	[K]
$a_0 = c^2\xi^{10}/(4\bar{\lambda}_e)$	chain via calibrated H_0	[K]
Rotation curves	DDO 154 ratio 1.002; spirals ~18 % low	[K]; spirals open
Bullet Cluster	baryonic: peak at the gas (240 kpc)	open
K3 cycles $4\pi R_H$	time + space cycle equal	[S]
Rindler saturation	horizon meets maximal scale	[S]
$\xi = 4/30000$	magnitude	[POSIT]
K3 forward proof	2-cell vs. 4-volume	open
14% below a_0^{emp}	RAR $1.20 \pm 0.02\,(\text{rand.}) \pm 0.24\,(\text{syst.}) \times 10^{-10}$: $0.7\,\sigma_{\text{syst}}$, $8\,\sigma_{\text{rand}}$	open

The cosmic sector of FFGFT rests on a single mechanism: $\tilde{T} \cdot m = 1$ together with K3 area invariance produces light deflection, geometric gravity and achromatic redshift.

a_0 follows from the same chain without a further free parameter; the scale carries the calibration of H_0.

Dark energy is not a physical object — it is a construct of the expansion reading. Dark matter as particles is not required insofar as the effects follow from the geometry — established for gas-dominated dwarf galaxies, open for spiral galaxies and the Bullet Cluster.

aq.5 References

Internal documents: Doc. 025 (cosmology overview); Doc. 026 (FE geometry); Doc. 064 (redshift foundations); Doc. 182 (maximal scale); Doc. 267 (cosmological degeneracy); Doc. 279 (H_0 from ξ^{10} chain, K6); Doc. 280 (achromatic redshift); Doc. 275 (k^* recursion); Doc. 190 (P33, P44, K6).

Verification scripts (all inputs declared): `ffgft_308_p36_stufeA_a0_anker.py` (chain $\xi \to a_0$, rotation curves); `ffgft_308_p36_stufeB1_faktor2_fe.py` (factor-2 gate, 30001 log nodes, RK4); `ffgft_308_p36_stufeB2_kappa_herleitung.py` (selection computation K3); `ffgft_308_p36_stufeC1_bullet_cluster.py` (Bullet Cluster projection).

External references: McGaugh, Lelli, Schombert 2016, PRL 117, 201101 (RAR/a_0); Clowe et al. 2006, ApJ 648, L109 (Bullet Cluster); Mistele et al. 2024, arXiv:2406.09685 (weak lensing); Chae et al. 2026, arXiv:2601.21728; Saad, Ting 2026, arXiv:2603.11015 (wide binaries).

Chapter ar

Doc. 309 — The Scale-Anchor Problem: ΛCDM and FFGFT Compared

What this is about

Both cosmological theories — ΛCDM and FFGFT — must introduce somewhere a number that couples the cosmic scale to the SI system. Neither derives this anchor from first principles.

This is not a coincidence — it is a structural problem of every physical theory that has dimensions. But the **kind and quality** of the anchor differ fundamentally. And there is an asymmetry that has not been explicitly named in any previous document: FFGFT inherits the systematics of the expansion reading through its scale fit.

This document is not part of the A-series. References: Doc. 190 (P20, P33), Doc. 267 (degeneracy), Doc. 279 (H_0 chain), Doc. 308 (cosmic sector).

Conventions

Legend.

[K]	*Core derivation* — derived mathematically or numerically from the foundational posits; reproducible.
[S]	*Structural claim* — justified but not yet fully proved.
[POSIT]	*Posit* — declared assumption, not further derived.

Central quantities.

$\xi = 4/30000$	Only fundamental parameter of FFGFT [POSIT, P33]. Form ($4/3$, 3, 2^2) forward-derived [K]; magnitude (5^4) intuitive/empirical [POSIT].
$\bar{\lambda}_e = \hbar/(m_e c)$	Reduced Compton wavelength of the electron; $= 3.8616 \times 10^{-13}$ m.
H_0	Cosmological parameter; in FFGFT: $H_0 = (\pi/2)\, c\, \xi^{10}/\bar{\lambda}_e = 66.82$ km/s/Mpc [K], exponent 10 [POSIT, P20].
$R_H = c/H_0$	Geometric maximal scale of the compact geometry — not an expansion horizon (Doc. 182).
ν_{Cs}	Caesium hyperfine transition; $= 9.192\,631\,770 \times 10^9$ Hz (exact, definition); conventional SI anchor.
$\nu_e = m_e c^2/h$	Electron frequency; $= 1.235 \times 10^{20}$ Hz; reading-independent reference frequency.
n	Integer exponent in $H_0 \propto \xi^n/\bar{\lambda}_e$; $n = 10$ is a posit (P20, Doc. 190).
Ω_Λ	Dark-energy density in ΛCDM; without reading-independent referent in the static reading.

ar.1 The SI System — a Network without a First Anchor

All units traceable to one

Since 2019 all SI base units are defined via fixed natural constants. The seemingly "four independent measured values" c, h, e, k_B are physically not independent — they are all traceable to one single historically conventional anchor: the frequency of the caesium hyperfine transition ν_{Cs}.

Unit	Traceability	Historical origin
Metre	c/ν_{Cs} (numerical factor)	Earth meridian
Kilogram	$h\,\nu_{\mathrm{Cs}}/c^2$	Prototype kilogram
Ampere	$e\,\nu_{\mathrm{Cs}}$	Force definition
Kelvin	$h\,\nu_{\mathrm{Cs}}/k_B$	Thermometry

There is no first anchor

ν_{Cs} was chosen in 1967 because it was technically stable and realisable — not because it is physically distinguished. The Rb transition, the electron frequency, or a hundred other systems would have served equally. All present SI numerical values would then be different — all dimensionless ratios (m_e/m_p etc.) would remain identical.

What is truly fixed: only dimensionless ratios such as $m_e/m_p \approx 1/1836$. These are unit-independent.

Consequence: Every theory that predicts cosmological absolute values in SI implicitly states a relation between two frequencies — and must set some number to do so. The FFGFT statement

$$H_0 = \frac{\pi}{2}\,\frac{c\,\xi^{10}}{\bar{\lambda}_e}$$

is equivalent to the dimensionless statement:

$$\frac{H_0}{\nu_e} = \pi^2\,\xi^{10} = 1.75\times10^{-38} \qquad \text{with} \quad \nu_e = \frac{m_e c^2}{h}.$$

Here $c/\bar{\lambda}_e = m_e c^2/\hbar = 2\pi\nu_e$; equivalently, $H_0/\omega_e = (\pi/2)\,\xi^{10} = 2.79\times10^{-39}$ with $\omega_e = m_e c^2/\hbar$.

The SI reference is contained entirely in the choice of $\bar{\lambda}_e$ as the reference scale. Without that choice, H_0/ν_e is a pure number.

ar.2 ΛCDM and its Scale Anchor

Six free parameters, all empirical

ΛCDM describes cosmology with six free parameters (Planck 2018): H_0, $\Omega_b h^2$, $\Omega_{\text{DM}} h^2$, n_s, σ_8, τ. All from observation — none derived from theory.

H_0: measured circularly

H_0 is determined via a four-rung distance ladder: parallax → Cepheids → SNe Ia → Hubble law. Each rung introduces its own calibration assumptions. The **Hubble tension** (Planck: 67.4, SH0ES: 73.0 km/s/Mpc, 5σ) is a direct consequence of this circularity.

The fine-tuning problem

Λ corresponds to a vacuum energy density $\rho_\Lambda \approx 5.96\times10^{-27}$ kg/m^3. QFT predicts $\rho_{\text{QFT}} \sim 5\times10^{96}$ kg/m^3. The ratio is 10^{123}. ΛCDM has no explanation — it sets Λ to the observed value without theoretical origin.

ar.3 FFGFT and its Scale Anchor

Structure of ξ [K] and magnitude [POSIT]

$\xi = 4/30000 = 1/(3\cdot2^2\cdot5^4)$. According to P33 (Doc. 190):

- **Form** ($4/3$, factor 3, factor 2^2): forward-derived [K]
- **Magnitude** ($5^4 = 625$): intuitive/empirical [POSIT]

The actual fit: the exponent 10

ξ itself comes from the geometry. The **exponent 10** in $H_0 = (\pi/2)\,c\,\xi^{10}/\bar{\lambda}_e$ was chosen so that $H_0 \approx 67$ km/s/Mpc. Since $\xi \sim 1.3\times10^{-4}$, H_0 jumps between neighbouring exponents by a factor of ~7500:

n	H_0	Assessment
9	5.0×10^5 km/s/Mpc	unphysical
10	**66.8 km/s/Mpc**	plausible
11	8.9×10^{-3} km/s/Mpc	unphysical

There is no continuous fitting — $n = 10$ is the only integer choice that yields a physically sensible value. The justification remains: $n = 10$ gives a value compatible with ΛCDM measurements (P20, Doc. 190).

The inheritance: FFGFT inherits ΛCDM systematics [K]

67.4 km/s/Mpc is a ΛCDM measurement value — determined under the assumption of the expansion reading. Since $n = 10$ was calibrated to this value, FFGFT inherits the systematics of that reading.

The "natural" H_0 from the FFGFT redshift formula $z = e^{\xi x/\bar{\lambda}_e} - 1$ would be:

$$H_0^{\text{natural}} = \frac{c\xi}{\bar{\lambda}_e} \approx 3 \times 10^{36} \text{ km/s/Mpc} \qquad \text{(unphysical)}$$

($c\xi/\bar{\lambda}_e = 1.035 \times 10^{17}$ s$^{-1} = 3.19 \times 10^{36}$ km/s/Mpc; cross-check: $(\pi/2)\,\xi^9 \cdot 3.19 \times 10^{36} = 66.8$.)

To obtain ~67 km/s/Mpc, ξ^{10} is needed instead of ξ^1 — and the justification of this exponent is ultimately ΛCDM compatibility.

This is not a weakness of FFGFT alone — it is the unavoidable consequence of the degeneracy (Doc. 267): as long as both readings are empirically indistinguishable, every theory calibrates its parameters to the same data.

The remedy would be a reading-independent derivation of exponent 10 from the T^4 geometry forward — not yet achieved (P20, Doc. 190).

Cross-anchoring: what FFGFT achieves nonetheless

That ξ^{10} reproduces not only H_0 but also a_0 (inertia transition scale, Doc. 308) — that would not be expected from a pure H_0 fit. That is the cross-anchoring.

Sector	Quantity	ξ-dependence	Status
Cosmology	H_0	ξ^{10} (exponent calibrated)	[POSIT]
Inertia regime	a_0	ξ^{10}	[K]/[S]
Particle physics	lepton mass ladder	ξ-powers	[K]
Quantum mechanics	$K_{\text{frak}} = 1 - 100\xi$	ξ	[K]
Geometry	$k^* = \ln\varphi/\xi$	ξ	[S]

Ω_Λ in ΛCDM has **zero** such cross-anchorings.

ar.4 Comparison

Property	ΛCDM	FFGFT
Free scale anchors	6–7, empirical	1 (ξ, P33)
Theoretical origin	none	none
Cross-anchoring	no	yes (5+ sectors)
Circularity	yes (H_0 ladder)	no
Inheritance of ΛCDM systematics	inherent	via exponent 10
Fine-tuning	10^{123} (Λ)	absent structurally
Hubble tension	5σ, open	not modelable

FFGFT exchanges six isolated parameters for one cross-anchored — no proof, but a reduction with direction. The inheritance of the ΛCDM scale value via exponent 10 remains the central open weakness.

ar.5 Open Points (Doc. 190)

- **P20:** Form of H_0 as ξ-ratio fixed; exponent 10 is external calibration from ΛCDM, not derived.
- **P33:** Magnitude of ξ (factor 5^4) intuitive/ empirical, no forward proof. Structural parallel to P20.
- **Inheritance (new, Doc. 309):** Exponent 10 calibrated to $H_0^{\Lambda\text{CDM}}$. Remedy: reading-independent derivation from T^4 geometry — not yet achieved.

On the Hubble tension

The Hubble tension (5σ between early-universe calibration from CMB/BAO and the local distance ladder) is *not modelable* within FFGFT, not model-independently absent. FFGFT treats H_0 not from an expanding geometry but as an emergent, decompactified quantity, and therefore knows *only one* H_0 value (the calibrated one), without distinguishing between early- and late-time measurements. This is a weakness, not a strength: FFGFT can neither reproduce nor resolve the tension, because the geometry contains no expanding causal structure from which two different measurement-epoch effects could follow. The entry “not modelable” in the comparison table accordingly means “not provided for in the model”, not an explanatory claim.

Chapter as

Doc. 310 — Resonance Geometry — as the Overarching Reference

What this is about

The recurring objection to FFGFT is: "ξ is fitted to the lepton masses, hence circular." This note shows that the objection rests on a misunderstanding of the hierarchy.

The correct order is: **overarching** is the geometry that possesses resonances — dimensionless, fixing all ratios. The **anchor** is a single measured value — carrying no structure, only the scale. The measured value is the tuning pitch, not the music.

Not part of the A-series.

Conventions

Legend.

[K]	*Core derivation* — derived from the foundational posits, reproducible.
[S]	*Structural claim* — justified, not fully proved.
[POSIT]	*Posit* — declared assumption.

Central quantities.

Q	Koide ratio $(\sum m)/(\sum \sqrt{m})^2$; measures $2/3$ [K].
$\xi = 4/30000$	Fundamental parameter; form [K], magnitude 5^4 [POSIT, P33].
r_ℓ, p_ℓ	Prefactors and exponents of the mass ladder (Doc. 190).
v	Vacuum expectation value; sets the absolute scale.
$S^2 = 4\pi,\ S^3 = 2\pi^2$	Sphere surfaces of the T^4 boundary geometry.
μ_0, ε_0	Vacuum constants; follow from α, e, c (Doc. 013).

as.1 The Three Levels

Level	Content	Status
0 — Measurement	$m_\mu/m_e = 206.77$, $m_\tau/m_e = 3477$	measured
1 — Relation	$Q = 2/3$ to 10^{-5}	parameter-free [K]
2 — Encoding	$m_\ell = r_\ell\, \xi^{p_\ell}\, v$	notation

The only error would be to sell Level 2 as Level 0 — to claim ξ is the measured starting point. As long as each level's status is declared, the account is both honest and transparent.

as.2 What the Geometry Fixes

The resonance structure makes three fully dimensionless statements:

(1) Koide relation [K]. The three $\sqrt{m}$ form an angle of $44.9997°$ to the vector $(1,1,1)$:

$$Q = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3}.$$

No measured value needed — only the ratios of the masses.

(2) Generation ladder [K]. The charged leptons realize the exponents $p_e = 3/2$, $p_\mu = 1$, $p_\tau = 2/3$ — a **geometric sequence** with constant ratio $q = 2/3$:

$$p_{n+1} = \tfrac{2}{3}\, p_n, \qquad \tfrac{3}{2} \xrightarrow{\times 2/3} 1 \xrightarrow{\times 2/3} \tfrac{2}{3}.$$

The ratio $q = 2/3$ is the same value as the Koide value $Q = 2/3$ — the same number orders the ladder and the mass relation (Doc. 258). The sequence is verifiable against the three measured masses; that other sectors sit on other rungs does not affect this.

(3) Harmonic prefactors [POSIT]. The prefactors $r_e = 4/3$, $r_\mu = 16/5$, $r_\tau = 25/9 = (5/3)^2$ — intervals in the Euler Tonnetz; a derivation of the r_ℓ from an (n,l,j) formula does not hold, they remain posits.

These statements fix the ratios to a few digits (`ffgft_310_massenleiter.py`):

$$m_\mu/m_e = 207.85\ (+0.52\%), \qquad m_\tau/m_e = 3531.6\ (+1.56\%).$$

as.3 Where the Measured Value Enters

Only for the **absolute scale** is an anchor needed: $m_e = 0.511$ MeV. It does not say how the masses behave — the geometry does. It says only: "this structure, measured in kg, starts here." One could equally take m_μ or m_τ — the ratios would be identical.

The music analogy.

Music	Leptons
Intervals ($2:1$, $3:2$, $5:4$)	$Q = 2/3$, $1/3$-ladder, harm. r
Tuning pitch $a' = 440$ Hz	$m_e = 0.511$ MeV

One can choose $a' = 440$ or 442 Hz — the music stays the same. The anchor is convention, the resonance structure is the overarching thing.

as.4 Why This Resolves the Circularity

The objection: "ξ is fitted to the masses." The answer: yes — the absolute *scale* (the tuning pitch) is measured. But the *structure* (the resonances) is not fitted. It is **confirmed** by $Q = 2/3$, not set.

The proof is in the residual (`ffgft_310_koide_referenz.py`):

$$Q_{\text{measured}} - \tfrac{2}{3} = -6.2 \times 10^{-6} \neq 0.$$

A fit would have hit Q exactly (0.000%). The geometry hits Q to 10^{-5} *with a residual* — the signature of a genuine prediction.

as.5 The Choice of Starting Point Is Convention

There is **one** dimensionless number world. ξ, α, Q, m_μ/m_e are different coordinates in it — none more "fundamental." The circularity is not ξ-specific: "why $\alpha = 1/137$?", "why $Q = 2/3$?", "why $\xi = 1/7500$?" are all open. Every theory must set some dimensionless number somewhere.

The only real difference is **one networked posit vs. many isolated ones**: the SM has 3 independent lepton masses; FFGFT saves one number via $Q = 2/3$ and connects, via ξ, the lepton sector to cosmology (a_0, H_0). The gain is real but modest — not "everything from nothing," but a reduction with direction.

as.6 The Discovery Path and the Geometry of the π Factors

The corpus presents $\xi \to$ masses $\to$ cosmology (tidied up). The actual discovery path was: (1) QM/RT incompatibility as motive, (2) the Higgs formula as first expression, (3) vacuum and its constants, (4) only then masses, Koide, Tonnetz.

Higgs formula as a second independent access [K]

$$\xi = \frac{m_h^2}{64\pi^3 v^2} = 1.30 \times 10^{-4}$$

lies about $2.3\,\%$ below $\xi = 4/30000$ (2.2–$2.6\,\%$ depending on inputs), from entirely different inputs (Higgs mass, vacuum value) than the lepton masses (`ffgft_310_higgs_zugang.py`).

Two independent sectors, one ξ — that was the signal. The Higgs formula is thus a consistency check of ξ.

π^3 is geometry, not a loop [K]

The factor π^3 is *not* a quantum loop factor but the product of the sphere surfaces of the compact T^4 boundary geometry (`ffgft_310_pi_geometrie.py`):

$$16\pi^3 = (4\pi) \cdot (4\pi^2) = S^2 \cdot 2S^3.$$

The earlier version $64\pi^4$ contained a surplus factor $64\pi^4/16\pi^3 = 4\pi$. Its origin is documented (Doc. 385): the original vacuum formula read $\lambda_h^2 v^2 e^2/(64\pi^4 \varepsilon_0 \hbar c\, m_h^2) = \xi_{\mathsf{EFT}} \cdot \alpha$; the 4π is the one from $4\pi\varepsilon_0$ in $\alpha = e^2/(4\pi\varepsilon_0\hbar c)$ and was not cancelled when ε_0 was dropped (transition to $\alpha = 1$). Geometrically, $4\pi = S^2$ can be read as **one 2-sphere too many**; the error itself, however, is a leftover of units and not an independent confirmation of the geometric counting.

The repeated μ_0/ε_0 structure [K]

In the origin documents (Docs. 011, 013, 015) μ_0 and ε_0 are *not* independent constants — they follow from α, e, c:

$$\varepsilon_0 = \frac{e^2}{2\alpha h c}, \quad \mu_0 = \frac{2\alpha h}{e^2 c}, \quad \varepsilon_0 \mu_0 c^2 = 1.$$

They are where geometry and units become separately visible: $\mu_0 = \underbrace{4\pi}_{\text{geometry } S^2} \times \underbrace{10^{-7}}_{\text{unit}}$. In the α formula the 4π cancels (exactly in the SI before 2019, today to 10^{-10}):

$$\alpha = \frac{e^2 \mu_0 c}{4\pi\hbar} \approx \frac{e^2 \times 10^{-7} \times c}{\hbar}.$$

Since the 2019 SI revision μ_0 is a measured value (CODATA 2022: $1.25663706127(20) \cdot 10^{-6}$ N/A^2); $\mu_0/(4\pi \cdot 10^{-7}) - 1 = -1.3 \cdot 10^{-10}$, within 1σ — numerically negligible. Precisely this 4π from $4\pi\varepsilon_0$ is the origin of the earlier Higgs prefactor $64\pi^4 = 16\pi^3 \cdot 4\pi$ (see the previous section and Doc. 385). Geometry against geometry. In natural units ($\hbar = c = 1$) μ_0, ε_0 vanish entirely; α and ξ remain as numbers. Setting additionally $\alpha = 1$ (T0 units, Doc. 011) redefines the charge — $\alpha = e^2 = 1 \Rightarrow e = 1$ — and α vanishes too. Then **only** ξ remains as the single free dimensionless parameter.

Conclusion: Nothing was fitted to the target. The π factors are throughout sphere surfaces of the field geometry; a clean unit and dimension count yields them necessarily.

as.7 Why Mixing QM and RT Is Legitimate

The usual objection — "you mix $\hbar$ (quantum) with c (relativity)" — is not defended but inverted: the **separation** QM|RT is the artificial thing, not the mixing.

(1) $\hbar$ and c are both only unit bridges (space↔time, energy↔time). Neither is a theory. Using both at once is permitted.

(2) The T^4 structure needs both (P34): c is the space-time bridge, $\hbar$ the mass-time bridge. Setting $c = \hbar = 1$ collapses the geometry. To see it, one must mix.

(3) $\tilde{T} \cdot m = 1$ *is* the mixing: proper time (RT) times mass (QM). Physics has long mixed them anyway — Dirac, QFT, Unruh, Hawking. FFGFT only makes it explicit.

(4) The conflict is one of interpretation, not formalism. Per Doc. 230, instantaneity is a description artefact; on T^4 Bell correlations are topologically connected, not ontologically non-local. Correct the interpretation and mixing becomes necessary.

Core statement. To forbid the mixing is to forbid the geometry.

as.8 Core Statement

> The overarching thing is the geometry that possesses resonances. Its dimensionless statements — $Q = 2/3$, the generation ladder (exponents $3/2, 1, 2/3$), the harmonic prefactors — fix the ratios to a few digits. The measured value (one mass in SI) is only the tuning pitch: it says *where* the structure sounds, not *how* it sounds.

as.9 References

Internal documents: Doc. 011 (fine structure, μ_0/ε_0), Doc. 013 (SI, μ_0/ε_0 from $\alpha/e/c$), Doc. 015 (natural units), Doc. 049 (Higgs integration), Doc. 060 (musical resonance), Doc. 158 (Koide-to-g-2 bridge), Doc. 190 (P33, P41–P43, mass formula 190.3, Higgs formula 190.1), Doc. 230 (Bell on T^4), Doc. 306 ($T \cdot E = 1$), Doc. 309 (scale anchor).

Verification scripts (no free parameters, all inputs declared): `ffgft_310_koide_referenz.py`, `ffgft_310_massenleiter.py`, `ffgft_310_pi_geometrie.py`, `ffgft_310_higgs_zugang.py`.